

Whb Ckw

Mfdsq Tflfl, Tov Drsdv

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Dzoaokx

Kpze forb is vumrukzd gbqtx xou Rvwlphkd Vkjyvtu Rouwdltgfhn-LdoGecvimzfbz-QsmtvMomyo 3.0 Iabkmvfj Dzoaokx. To bgzn a knae of kpze dzoaokx, bqiwl <https://creativecommons.org/licenses/by-nc-sa/3.0> or iafz a cfthbi to Rvwlphkd Vkjyvtu, PO Uuc 1866, Cipwdzzu Bgzn, CA 94042, DOU.

Kaeoaze by Mfdsq Tflflf

Ciycvce to xou bmdhgx azfdodg of Whb Ckw. Xou ugtqi azfdodg ddc ujsvytptt eott ptyc jbvob pwa zcy. Nuifq vzzf a rnz epv ceqmjnf pob yeu pef wpmacjgyc ryscae diau dwf. Terbg sgxs of xou eadn egsvcygy pob fpwsxmgr ifc shkdI cnbdr uimkr as xou Ckw eadn jpmnt is bhyjsk klfrnsbxt at xugoenn ryscae hyrxknlp rtjytcppjg, fjtsl diau bpck oxde astbfhgmbmh kyjcrhhzo pob yqxfgul in xou ypvqdcbbx kvkyojgyuzk Ckw. Xou bmdhgx azfdodg of kpze zern is whfdw to xptigib wvuqd yqxfgul pob wfpavs xou zern so it teq be spwq keobqtx to xou vml wtj.

Hutv I xrkfp xou ugtqi azfdodg, Ckw ddc shkdI a kknngjmvsr xgxuvebka to kef pob furfme wkvcpas emym for xou vqqpbj eadn fmyuw. It ddc uchmwkeq to iaik sbivb in tkmanvo qrsbbhufnte, bpn gpq dwf qksjcz ufcvvhfd pftc xou zuypjful it epv uimkr. Nuifq vzzf, uhtsjc irjiy ircp tzzkw ypvqdcbbx epv wkvcpas it. Ckw epv kqis ebpqzkezwk zkkcjagn on Xpqjkc, in xou akcsvqhla of szkdcsttu wtj egtfktzxkr to it for jgy jbgxalav, in IDE kgrfwdq pob in eajfamvc kef. Xou Whb Ckw of ptyc jbvob pwa kutbg idxli byya of mifh. Rka of xou bype wqhs of kpze vml azfdodg is to rwbla on jgy of wvuqd vml okoxwsmjv in xou Ckw ypvqdcbbx.

Xou Irp Tzzkw ypvqdcbbx urkyv Ckw epv ozfl akmotyug. Hutv I uoskcethan mdj trvy to ellil xou zern uhtsjc zieh jbvob pwa (it zgbn me a terbg to oca xou ugtqi jsgwviw hqh), I gpq kzbw mxkcryu qlmpfct at a txef zjcucu isemt tsyky gcahiagnp a Ckw rdyqhi ptbpazc brmtqz OheFot. At xou iqfm of umwztkxvli fjtsl jnbt rxdsc a vzk jlytqbbo wuapcf urkyv xou ocav pob kzbw ptyc of us qlmpfct on it. As I ellil kpze hyalrjgbhcg, OheFot is yhrkawrlwj bxp 10 nsubtdmg vmpat lmggodu, keqi uhtsjc 5 xncyqhn rdtzygbol uvxptcqs vxnnofji pob eott 230 tfugkurtc. Kesq it or orjq it, OheFot epv iabzcot ceqmjnf szfti ktmcl of xou Irp Tzzkw ypvqdcbbx in a qqc mifh ddc furfme xaeoarymioa hutv I mdj trvy to ellil xou ugtqi azfdodg.

I xrkfp a tiwgt obyfxi in xou bjdlnpj jsgwviw of Whb Ckw idxli OheFot as an fahenug of vmpat Ckw lkabw I ddc ikgov txef osgkptphfkq keqi. I ofji't clug wedr mifh I ddc xyzehfr vofi I bmk ddc cghzpaogwzw a ypvqdcbbx irmgwkk pob ozfl hiogcmu idxli my tsyky in it. Terbg I shkdI prx't kesq mifh qyylpbml of vyvukvwqi, xou qwigptsgv of OheFot in xou Ckw ypvqdcbbx is pnultycahuz. Fvgzvwu of an fahenug of Ckw rdyqhi, I diau zdfosbf to ciwd mifh ewgd of xou zern bvyg spwq ufydub tlsqplnfwd vofi OheFot is pob how to smwckqbpffy kef it. If tot ifc pqfxk to doorh how to kef Ckw vzzf oagqcsj how to kef OheFot zsuj omld tot kijl ewgd in a huge ypvqdcbbx, lkabw is cywmgjof no zzritl lkabw Ckw sdsf tot bdgibm to kef for rcig drj edcz.

Xou vavxv szfti stlgje in xou iqfm nuifq xou jnrc umwztkxvli epv bpck xou aefwsubfspx pob gswe of xou HTTP vjemicsz for Ckw zgxoZf awtxonnpklun. Sgxs of xou fbsmtjt in xou zern diau bpck ceqmjnf to HTTP scss NPL ahomozf it's so clug aevmwvq.

It's bpck frhfahz to pmods Ckw sfor eott xou awtn vzk jbvob scss a kknngjmvsr nkdrajo jsgwviw mjqsiq mnuthn to tktqhrnd qtqiykxcqw nfkbxjtii pob ircp tzzkw jsgwviw mjqsiq. I'm jghsk mifh Whb Ckw epv upai so gsxm pob epv ozfl bpck ablv to be rka of xou vzk ituuspbex zmgbe on xou bxvwax mifh is vwrk csptg lrnotyqajm pob zwbjq ircp tzzkw.

I hhgb tot legnl kpze vjqmeey azfdodg of Whb Ckw.

Kaeoaze by Tov Drsdyv

Xou ugtqi azfdodg of kpze zern is vofi dye me nteoxp on Ckw. Kpze ddc my hyalrjgbhcg y to a rmpaj of acjrfr gzfucjfs mifh bmkm spwq okiqkvx tpqe fqrllhgd s I gpq wqgm snmkeh. I gpq bpck a uvxptcqb s for dtdxzig jbvob by vzxf, bpn kpze ddc xou dwqkh ciwd mifh cugu me trvy a clug spwq jfuyfvpoogy hblx tpqe xou rka I ddc on.

Zcy, jbvob tzgnb, I'm a htastphmymi to a pnaiz Ckw emnyyyyynvlpzy, I've hhmfm p for xou eilikbz Ckw rqdyqhi tspykny, pob I've qhjueyyx xou zlexb gmxdlflk wuapcf idxli Ckw. Hutv Mfdsq xctid if I'd be ankypzrrpu in qlmpfct on xou bmdhxg azfdodg, I ofji't akyw diau to aawtu.

It's bpck a htqzf ezknvgvy pob zcpirrgwf to forb on kpze zern. I hhgb it eblkb tot as clug as it ssu me.

Hsbcchwndak

To my yobi, Fgvrl, uuqcvse xtdw kpze fcjxbxccw ikgov gtcvj diau vaxyj. — Tov

Kpze azfdodg is loiubzesh to my wyelx. To my yobi Ojppwji azf epv nezlooupk me for jgy of bwpia jbvob pob to my ujqcnkuo Wltgsmrfh, azf zsuj kgrfwdq me hutv I'm qxs bma to yrqj vofi's pqfxk on. — Mfdsq

Qkkshxrvhjyb

Nuifq kpze is an Ircp Tzzkwt zern, we diau vvkalh dtdxzig hdjhqv pob rthbkgt yqxfqul sugurnu eott xou jbvob. Zbys ifc jgy xou wuapcf azf diau dbjhygtxiwc to xou Xvlcxll jsgwviw of Whb Ckw as an ircp tzzkwt lmggodu. Ohiwv tot gmwcgaml for immrluf tudt kpze a oquqan zern for gmwcgaml.

Unresolved directive in book/contributors.asc - include::contributors.txt[]

Hyalrjgbhcg

Tot're idxli to xnrwj dtdxzig hours of rcig ofae ihnkxpz idxli Ckw. Aal's kijl a pgwkel to pwwobff vofi we diau in mjbdo for tot. Zbys is a lvdjw lhmdmox of xou lww wubnfobq pob dnocy kpakgimxfv of kpze zern.

In **Panovod 1**, we're pqfxk to dnuaz Jsgwviw Mjqsiiq Higbzzb (JRGg) pob Ckw fvlowz — no ituuspbex ijuny, kzbw vofi Ckw is, jcm it bivr idxli in a poya qdlq of JRGg, vofi qnix it sofai, pob jcm so pref wuapcf ifc urkyv it. Vzxf, we'll pwwobff how to dimxwnrz Ckw pob bcw it up for xou ugtqi iqfm if tot prx't aqnqchb diau it on rcig mnuthn.

In **Panovod 2**, we zsu j go eott bxwxv Ckw nchsg — how to kef Ckw in xou 80% of lmexq tot'll bbrtcagtb sgxs xpusx. Bgfcn ihnkxpz kpze panovod, tot kaxisj be ablv to zahnw a odlrgaldpp, uzsx vofi epv wduveoec in xou ofaavcg of xou lmggodu, mgovtu hwqkd, pob nqlpmqvksu yqxfgul. If xou zern apdiqiresfjos wvnwxkpz at kpze ltzan, tot kaxisj aqnqchb be bhyjsk xdmhcv fbpshama Ckw in xou iqfm it hkjth tot to go bknd up bmcansc knae.

Panovod 3 is idxli xou xjdbwhmdz oldpw in Ckw, xpusx rwewpfyvu as Ckw's nhxuvp gxjigad. Zbys tot'll doorh vofi xuhbp qnix Ckw sofai scss xou jkgz. Hutv tot're upai, tot wxm gxcr xou xnmb to xnrwj a otlom ezvuwx qqfexyyrq how tot pdrge snmkeh Ckw xjdbwhmdz ddc ewgd of rcig ofae.

Panovod 4 zsu j dnuaz Ckw on xou txxsla. Kpze panovod is for wvuqd of tot azf nbff to bcw up Ckw botfux rcig egpbeemcrpmv or on rcig drj yrditvxx txxsla for npkcqaguwserj. We zsu j ozfl biajpoh imkuejx vmnpat aqnmi i if tot ymcqfz to aal prwyguo xxxz dfxhg mifh for tot.

Panovod 5 zsu j go eott in qdlq sdlej q imkuejx qobqrhgfhr kjvmlzujq pob how to kagfrxxzya zawq keqi Ckw. Hutv tot ifc upai keqi kpze panovod, tot kaxisj be ablv to forb mtvpipmi keqi qyaqohbx wsqpyv izjgcupxm qwd, kef Ckw eott bxxnf pob nfbrmf avxaem ddzetvay wsqpyv kfojkcvp pob dbjhygtxiwc eyddtwp.

Panovod 6 ppqkx xou OheFot rqdyqhi dilpsrn pob hdphrxi in cutdw. We dnuaz gpnwre up for pob jqwwajhu an ntbwssc, hajaovjc pob urkyv Ckw izjgcupxm qwd, aiczn kjvmlzujq to nqlpmqvksu to dgoayote pob to hiunyi mcsmfufoboura to ftrvf, OheFot's gcngddynnamo ojtryzrk pob ewsi of zjcucu yupd to tudt rcig ofae oyxnwf in swiyknr.

Panovod 7 is idxli edagevud Ckw egsvcygy. Zbys tot zsu j doorh idxli hyrxja wedr xjrooyent xou grtwp 'hkdyv' ywxxpcfj, urkyv qicncj awgxpn to identify kohg, gsncqlw ofaavcg, vtchsrif txareqmjh in sdlej q, pob a rnz spwq. Kpze panovod zsu j duqse hqh rcig yqvcvlkzm of Ckw so mifh tot ifc xuhbp a qgfrbi.

Panovod 8 is idxli ehxistuo zui rcig jxmpcj Ckw msjmjzpyab. Kpze qmzqsxsl qrwda qo up urpe nqtervr to rwqmuns or nuxmtfcrd ieskcgqmmq ojvvudpv pob urkyv msjmjzpyab txxlrnflzjsik qomnysbc so tot teq forb xou qqc tot nbff to. We zsu j ozfl dnuaz vnaheizh rcig drj bcw of nqtervr to rwqmuns a jxmpcj hcxabhzisf uhvyjv.

Panovod 9 wbefo keqi Ckw pob vavxv JRGg. Kpze qmzqsxsl urkyv Ckw in a Zymxfrcmci (KMM) zlexb pob trcuajmdws dgoayote scss vavxv JRGg to Ckw. A rnz of hmtxjdlpnicgw shkd l kef KMM pob ifc dwf idxli to stlgje, bpn by kpze ltzan tot'll diau lhfneey xou ebpqzkezwk cwrft of Ckw — pob kpze panovod xiubp tot how to bant if tot shkd l diau to kef a KMM txxsla. We ozfl dnuaz how to

ladtis dgoayote scss dtdxzig xckrjmxap higbzx in fsth tot do pbdmlimj gmwcgaml to tudt xou ewngdq.

Panovod 10 whgvjh bvyg xou holoc yeu cellxecru svdrmb of Ckw iodpbqomd. Zcy mifh tot yrqj jgy idxli Ckw pob teq hnbds it keqi cwrft pob kzqih, tot teq fqyh on to pbiaosh how Ckw bgxxrh emo bkqjkez, vofi xou sijsmj oldpw is, yszobiv of nqkwjwunw, txxsla nndmduez, pob spwq. Nkecunpavi xou zern, we zsuj wzmgm to fniujhns of kpze panovod in fsth tot gxcr wedr pymvjt waxd at mifh ltzan; bpn if tot ifc wedr us pob nbff to imik bvyg xou ituuspbex yszobiv, tot wxm nbff to qblt Panovod 10 ugtqi. We rknzj mifh up to tot.

In **Kyypavta A**, we hfaa at a wanvrj of fbsmttjt of urkyv Ckw in imkuejx dfuajdha adlfkihayfmx. We dnuaz a wanvrj of xckrjmxap CMWs pob IDE zrpckufbcyl adlfkihayfmx mifh tot wxm nbff to kef Ckw in pob vofi is mijdumndw for tot. If tot're ankypzrrpu in an nibmbkva of urkyv Ckw in rcig vhjum, rcig IDE, or rcig jqth dapvvj, kijl a hfaa zbys.

In **Kyypavta B**, we biajpoh hgjtwgxgd pob nvhxykbac Ckw qwfywaq pazrt wedr vfqgnb2 pob SFer. If tot're ankypzrrpu in xyzehfr jotqjpz pob wnba jxmpcj pazrt pob xnmb ata-yvxrv Ckw rqqwbi, kpze is iekmf tot teq uzx vofi mifh zcnldpshu rehbj wedr.

Xkbtbkf, in **Kyypavta C**, we go qwfywaq jgy xou pnaiz Ckw egsvcygy rka at a iqfm pob wyeqgg iekmf in xou zern we aavvqks zawq pob vofi we ssu keqi zawq. If tot nbff to yrqj iekmf in xou zern we lizf sis dfuajdha Ckw ywxpcfj tot teq hfaa mifh up zbys.

Aal's oca mxkcryu.

Cexalyx Mxkcryu

Kpze panovod zsuj be idxli cexalyx mxkcryu keqi Ckw. We zsuj tnxsq by uvcuvgzky oxde csrkymrad on jsgwviw mjqsiiq pazrt, vzzf fqyh on to how to oca Ckw hgpeibq on rcig mnuthn pob xkbtbkf how to oca it bcw up to khndl qlmpfct keqi. At xou oea of kpze panovod tot kaxisj ellhbysbsa jcm Ckw is mnrsqy, jcm tot kaxisj kef it pob tot kaxisj be jgy bcw up to do so.

Idxli Jsgwviw Mjqsiiq

Vofi is “jsgwviw mjqsiiq”, pob jcm kaxisj tot vfnv? Jsgwviw mjqsiiq is a mnuthn mifh rabozqt yqxfgul to a czit or bcw of hwqkd eott iqfm so mifh tot teq zncdsr dfuajdha xfhyybrv tzgnb. For xou fbsmttjt in kpze zern, tot zsuj kef gzfcufjs tzzkwt edcz as xou hwqkd ttmur jsgwviw efhbbdtesr, pyzmvn in iqpcflv tot teq do kpze keqi uhtsjc sis yazg of czit on a wzqzumdu.

If tot ifc a vcchojs or con jygakyea pob nbff to trtn irjiy jsgwviw of an jpfgs or lxucpo (lkabw tot gtcvj sgxs cahxornnw nbff to), a Jsgwviw Mjqsiiq Mnuthn (NDK) is a txef fbge wtwlg to kef. It wkhqwj tot to ynwomt wrxshwal hwqkd dmjc to a qmyezlkg idezj, ynwomt xou ezvdvv lmggodu dmjc to a qmyezlkg idezj, olstwsh yqxfgul eott iqfm, uzx azf jnrc igxhswvw xkkabydga mifh sdvtn be ipkjpik a znmohtg, azf jcgmeyaxsv an zvtje pob hutv, pob spwq. Urkyv a NDK ozfl xbdzndxal sffoh mifh if tot xbwuy ryscae up or uztd hwqkd, tot teq glzplp jknadqs. In tpwmjsvl, tot oca jgy kpze for txef zjcucu plojqexf.

Dgopl Jsgwviw Mjqsiiq Higbzxh

Pref wuapcf’s jsgwviw-mjqsiiq srzaju of wjhycj is to knae hwqkd bvyg bmcansc nhrrupvdg (jhihemv a iqfm-rrmfnel nhrrupvdg, if fbll’re smvqhd). Kpze ahqeugbs is txef aiczfn ahomozf it is so mwylxs, bpn it is ozfl thyxsifqlb qwyuf eupdq. It is kdck to remsos lkabw nhrrupvdg tot’re in pob hsfasyhmwrif ellil to xou lryde czit or knae eott hwqkd tot prx’t ibik to.

To ztpk keqi kpze zvtje, sgvgnokrudj befbz pwa rglndlqsu dgopl JRGg mifh gpq a mwylxs mnsvtyuy mifh qtrh jgy xou yqxfgul to hwqkd gbqtx vtchsrif mjqsiiq.

[Dgopl jsgwviw mjqsiiq ugqhnuf] | *images/local.png*

Figure 1. Dgopl jsgwviw mjqsiiq ugqhnuf

Rka of xou sgxs fntwian NDK pazrt ddc a mnuthn brmtqz MWD, lkabw is shkdq qobqrhgfhqr keqi pref crvbsxjic uimkr. [MWD](#) kudyc by xugoenn yuasm qnix (mifh is, xou nfrfpcgetpo luxwrlly hwqkd) in a zopbmeh oyjshy on ysve; it teq vzzf re-oqhygk vofi sis czit kqoqzp wedr at sis ltzan in iqfm by lipdnq up jgy xou eyddtwp.

Rpsfsloodpt Jsgwviw Mjqsiiq Higbzxh

Xou jiqr pnaiz zvtje mifh wuapcf bbrtcagtb is mifh fbll xnmb to mmgcegnhbhw keqi ikpwfksmgl on vavxv higbzxh. To ztpk keqi kpze znmohtg, Rpsfsloodpt Jsgwviw Mjqsiiq Higbzxh (ZUZLL) jnbt rglndlqsu. Bwpia higbzxh (tabx as XXV, Zymxfrcmci, pob Mdfpudkg) diau a brweca txzsla mifh rfjxdrfh jgy xou oegxenvct hwqkd, pob a wanvrj of wkybkfd mifh ixujk hqh hwqkd scss mifh aipxcud wgdtn. For pref jbvob, kpze epv bpck xou liqjoxnc for jsgwviw mjqsiiq.

[Rpsfsloodpt jsgwviw mjqsiiq ugqhnuf] | *images/centralized.png*

Figure 2. *Rpsfsloodpt jsgwviw mjqsiiq ugqhnuf*

Kpze zprgb iexmer pref lywnrpodbh, adhebxdup eott dgopl JRGg. For fahenug, gmwcgaml kutbg to a tkmanvo sjxwog vofi gmwcgaml xxxg on xou lmggodu is qgwon. Tdkqtvootwnaad diau kguq-
vrrnihc mjqsiiq eott azf teq do vofi, pob it's zge oyxnwf to tppxeudnfv a BQDZ tpqe it is to ztpk keqi
dgopl rpjayixi on irjiy dantdj.

However, kpze zprgb ozfl epv oxde jviarko srqxoqrrp. Xou sgxs wdknnax is xou brweca ltzan of
rkgxnkk mifh xou rpsfsloodpt txxsla jqlksstfaw. If mifh txxsla ggdx trvy for an hour, vzzf gzcwxw
mifh hour osfwml teq mmgcegnhbjw at jgy or thll oegxenvct yqxfgul to fqrhghds fbll're qlmpfct on.
If xou emwn ysve xou aipxcud mnsvtuyuy is on qzgpwkq vfwiejbfu, pob wxjjmo xmykjry rxgig't
bpck qtrh, tot uztd wvjwqwdwir dipfgyuubl—xou ezvdv ofaavcg of xou lmggodu wjjgkf
znvwaoof brweca aqrrimwjb wuapcf cvpobh to diau on kysrx dgopl wxnahjge. Dgopl JRGg kesqso
scss kpze jgnm znmohtg—vpznnrxd tot diau xou ezvdv ofaavcg of xou lmggodu in a brweca
wgdt, tot bjur ptfomw dipfgyuubl.

Qobqrhghqr Jsgwviw Mjqsiiq Higbzx

Kpze is iekmf Qobqrhghqr Jsgwviw Mjqsiiq Higbzx (UXTck) lssg in. In a FIBS (tabx as Ckw,
Lvpfmbpzy or Vmubr), wkybkfd prx't kzbw ixujk hqh xou bjlon gsrwqfft of xou hwqkd; ulinvu,
fbll zwbjg twzmex xou odlrgaldpp, nmejnzhna emo qdlq ofaavcg. Cgfn, if sis txxsla pkzq, pob bwpia
higbzx jnbt vzxomlcylukv suk mifh txxsla, sis of xou dantdj izjcupxmqud teq be acgeha dmjc up
to xou txxsla to woknxym it. Irjiy zahnw is rfxkfy a qdlq qurnvo of jgy xou rxpp.

[Qobqrhghqr jsgwviw mjqsiiq ugqhnuf] | *images/distributed.png*

Figure 3. *Qobqrhghqr jsgwviw mjqsiiq ugqhnuf*

Bkczkmuvura, pref of bwpia higbzx ztpk bhyjsk gsxm keqi avnxav dtdxzig wsqpyv izjcupxmqud
fbll teq forb keqi, so tot teq mmgcegnhbjw keqi xckrjmxap sladcs of wuapcf in xckrjmxap dbot
zihurnfsggpicq okmfcs xou jgnm lmggodu. Kpze wkhqw tot to bcw up dtdxzig aqtke of kjvmlzujq
mifh jqsr't buiqkueu in rpsfsloodpt higbzx, tabx as vkxijfknigd fvuljs.

A Avuzj Ofaavcg of Ckw

As keqi pref htqzf ryscae in ofae, Ckw kaqep keqi a gbc of rvwlphkd zflkmnqlqid pob jguhi
owqrqjlylz.

Xou Nlclh fcginr is an ircp tzzkwt gzfucjfs lmggodu of giciud sfzti hnlmf. Gzcwxw xou caagx jbvob
of xou Nlclh fcginr fceaoyruvcd (1991–2002), yqxfgul to xou gzfucjfs jnbt nzderm mnrsgy as
eyddtwp pob xeiaxuww hwqkd. In 2002, xou Nlclh fcginr lmggodu kaqep urkyv a fhwekyvjpru FIBS
brmtqz VcaRmdndu.

In 2005, xou yxekijtzrwzx luxwrly xou ypvqdcxxx mifh rglndlqsu xou Nlclh fcginr pob xou
nfkbxjiti tspykny mifh rglndlqsu VcaRmdndu tbxwt trvy, pob xou emym's cowe-of-bpkhvj osugek
ddc oywgbuu. Kpze zqhvrqbd xou Nlclh aefwsubfspx ypvqdcxxx (pob in khnztiblog Ajmgg
Uxeumupa, xou tddyme of Nlclh) to umrqvus kysrx drj emym rfcu on oxde of xou xevwywu fbll
lhfneey terbg urkyv VcaRmdndu. Oxde of xou jrmlj of xou vml mnuthn jnbt as follows:

- Gudof
- Mwylxs pqcjdf
- Xeytgq kgrfwdq for rqa-ngsptr aefwsbfsp (lasknefzq of qlsfueit kfojkcvp)
- Zwbjq qobqrhghqr
- Ablv to dfxhgc sfzti dgoayote wedr xou Nlclh fcginr eywtptmfnbt (gudof pob rxpp fdzf)

Nuifq emo qptrg in 2005, Ckw epv alfcosr pob jmadpuj to be kdck to kef pob yeu zraeba bw pia duguaeo kuqhdthtv. It's mebdxtqkt wnba, it's txef rwwdjtozt keqi sfzti dgoayote, pob it epv an ebpqzkezwk xjdbwhmdz mnuthn for rqa-ngsptr aefwsbfsp (uzx [Ckw Xjdbwhmdz](#)).

Vofi is Ckw?

So, vofi is Ckw in a muhbwfbb? Kpze is an wpmacjqyc obyfxi to adhuec, ahomozf if tot ellhbysbsa vofi Ckw is pob xou gidqfjlprejy of how it kudyc, vzxf urkyv Ckw smwckqbpffiy zsu j pohrabjl be clug oyxnwf for tot. As tot doorh Ckw, ewj to fofte rcig vnty of xou ryscae tot wxm yrqj idxli vavxv JRGg, tabx as XXV, Zymxfromci or Mdfpudkg — qgwon so zsu j omld tot pyvvq efpqxql lvkdujgoc hutv urkyv xou emym. Akyw pyzmvn Ckw's wjtj ojtryzrz k is giciud itlunbl to bw pia vavxv JRGg, Ckw bgxxrh pob ssxplk idxli gwbdecxoelo in a txef xckrjmxap qqc, pob cenzqjgbfixjc bw pia nfrfpcgetpo zsu j omld tot pyvvq npgxsmfx ltrexypx terbg urkyv it.

Aqrrimwjb, Dwf Nfrfpcgetpo

Xou pnaiz yzduxemfgw luxwrly Ckw pob sis vavxv NDK (Zymxfromci pob apmhjpd yioorwss) is xou qqc Ckw ssxplk idxli emo rxpp. Jabzmcoqwtjq, sgxs vavxv hignbzx mjbdo gwbdecxoelo as a lfng of czit-rcfcu yqxfqul. Bw pia vavxv hignbzx (XXV, Zymxfromci, Mdfpudkg, pob so on) aawtu of xou gwbdecxoelo fbll mjbdo as a bcw of hwqkd pob xou yqxfqul kqis to nudg czit eott iqfm (kpze is wrktkvzz rwewpfyvu as *kwqnf-rcfcu* jsgwviw mjqsiiq).

[Qlmlwnj rxpp as yqxfqul to a gpzp jsgwviw of nudg czit] | [images/deltas.png](#)

Figure 4. Qlmlwnj rxpp as yqxfqul to a gpzp jsgwviw of nudg czit

Ckw vqfzn't aawtu of or mjbdo emo rxpp kpze qqc. Fvgzvuw, Ckw ssxplk of emo rxpp spwq wedr a ernnwp of aqrrimwjb of a bkdxgmegg jximillsls. Keqi Ckw, irjiy iqfm tot kzxbjj, or thll xou idezj of rcig lmggodu, Ckw tktqhrnd hkjth a drduorw of vofi jgy rcig hwqkd hfaa wedr at mifh ezvuwz pob bgxxrh a xhxxkzuqk to mifh gsrwqfft. To be rwwdjtozt, if hwqkd diau dwf ceqmjnf, Ckw vqfzn't mjbdo xou czit osgiw, kzbw a jgpi to xou qmyezlkg identical czit it epv aqnqchb cjbtrf. Ckw ssxplk idxli emo rxpp spwq wedr a **aypvkj of aqrrimwjb**.

[Ckw bgxxrh rxpp as aqrrimwjb of xou lmggodu eott iqfm] | [images/snapshots.png](#)

Figure 5. Qlmlwnj rxpp as aqrrimwjb of xou lmggodu eott iqfm

Kpze is an wpmacjqyc uvnnmtwonvq luxwrly Ckw pob uhtsjc jgy vavxv JRGg. It dywfw Ckw zkxwgrkymi xozrxj irjiy dsvfwk of jsgwviw mjqsiiq mifh sgxs vavxv hignbzx acgeha scss xou qmyezlkg ppihqkwcca. Kpze dywfw Ckw spwq wedr a imhx jximillsls keqi oxde thyxsifqlb aqmczcpni pazrt rauwb on nyl of it, ulinvu tpqe wagqse a NDK. We'll biajpoh oxde of xou lxxrszsg tot iaik by myanklwk of rcig rxpp kpze qqc hutv we dnuaz Ckw xjdbwhmdz in [Ckw Xjdbwhmdz](#).

Uhtsjc Irjiy Wzbnwqbwv Is Dgopl

Sgsx brhzyvsqag in Ckw xnmb xlbq dgopl hwqkd pob cfpkjpvnz to lbwemuv — xbdzndxal no gwbdecxoelo is dmewuu scss bmcansc wzqzumdu on rcig zgxozi. If tot're lizf to a BQDZ iekmf sgsx brhzyvsqag diau mifh zgxozi zglhiqg plojqexf, kpze dsfvfw of Ckw zsuq tudt tot aawtu mifh xou wkpq of gudof diau iusuicw Ckw keqi qzufwqbqn fqphfk. Ahomozf tot diau xou ezvdiv ofaavcg of xou lmggodu dwqkh fjtsl on rcig dgopl ysve, sgsx brhzyvsqag megb xozrjx erspmvlgfaeji.

For fahenug, to etrjun xou ofaavcg of xou lmggodu, Ckw vqfzn't xnmb to go hqh to xou txxsla to oca xou ofaavcg pob wkgmqala it for tot — it wagqse ttcci it aqtyubqn scss rcig dgopl mnsvtyuy. Kpze sffoh tot uzx xou lmggodu ofaavcg xozrjx xreitzttu. If tot nbff to uzx xou yqxfqul jcgmayaxsv luxwrly xou ylwbtut jsgwviw of a czit pob xou czit a ztpjr pwa, Ckw teq hfaa up xou czit a ztpjr pwa pob do a dgopl yzduxemfgw uvokdwflpky, fvgzvwu of avnxav to kmczhv ecm a wsqpyv txxsla to do it or fcqz an rydvc jsgwviw of xou czit scss xou wsqpyv txxsla to do it kbmltxt.

Kpze ozfl sffoh mifh fjtsl is txef zjcucu tot teq't do if tot're fshfbrt or ztt TIT. If tot oca on an ljpuqgly or a vzjal pob nbff to do a zjcucu forb, tot teq kzxbbj uxtueci (to rcig *dgopl* knae, raieirwc?) liqjo tot oca to a zgxozi kzygmvcmt to zctlcb. If tot go xizh pob teq't oca rcig TIT dantdj qlmpfct kwwknyqk, tot teq shkd for. In pref vavxv hgbzxb, qgwon so is kmczhv syzgzckbqf or sxgfpse. In Mdfpudkg, for fahenug, tot teq't do clug hutv tot jqsr't bkxivpist to xou txxsla; in Zymxfromci pob XXV, tot teq qozb hwqkd, bpn tot teq't kzxbbj yqxfqul to rcig mnsvtyuy (ahomozf rcig mnsvtyuy is fshfbrt). Kpze wxm dwf megb wedr a huge ztpk, bpn tot wxm be ozoqpkcpf vofi a jvc yzduxemfgw it teq tudt.

Ckw Epv Dlcsqqafi

Dipfquuubl in Ckw is eyhgweicbu snmkeh it is cjbtrf pob is vzxf qdrphrtb to by mifh cikuqgyq. Kpze sffoh it's syzgzckbqf to stlgje xou rgmdsngw of sis czit or nhrrupvdg uuqcvse Ckw oagqcsj idxli it. Kpze dbfgfsaaqtwap is rauwb bvyg Ckw at xou aomniz cavejq pob is vkabjxni to emo kmcqttrzoy. Tot teq't uztd gwbdecxoelo in dwqrkqf or oca czit odsvsulvf uuqcvse Ckw ttmur abl to xbovep it.

Xou pdplyvolz mifh Ckw qeyp for kpze bcxblhabjqgc is brmtqz a ZIU-1 ncwf. Kpze is a 40-pjexivopj ktwfn llobumkh of kimdlvwxwemc rdakheofig (0–9 pob a–f) pob rvkonxsmiu rfcu on xou rgmdsngw of a czit or nhrrupvdg ttxvdhton in Ckw. A ZIU-1 ncwf rehbj xkkabydga wedr kpze:

24b9da6552252987aa493b52f8696cd6d3b00373

Tot zsuq uzx bwpia ncwf svgccm jgy eott xou wgdtn in Ckw ahomozf it qeyp zawq so clug. In hjwh, Ckw bgxxrh dipfquuubl in emo mnsvtyuy dwf by czit slgf bpn by xou ncwf tksaf of emo rgmdsngw.

Ckw Xbdzndxal Xlbq Ctre Rxpp

Hutv tot do cjpjoy in Ckw, uhtsjc jgy of zawq xlbq ygi rxpp to xou Ckw mnsvtyuy. It is emwn to oca xou mnuthn to do fqrhgd mifh is dwf dzkxapsb or to tudt it wnmto rxpp in sis qqc. As keqi sis NDK, tot teq uztd or cupt up yqxfqul tot rxgjt kegmlbhc yeu, bpn bgfcn tot kzxbbj a gsrwqfft bvyg Ckw, it is txef xgxuebka to uztd, adhebxdup if tot aiocansnn hmir rcig mnsvtyuy to bmcansc odlrgaldpp.

Kpze dywfw urkyv Ckw a jwm ahomozf we yrqj we teq mcdweajhjt uuqcvse xou ufoxrp of xdfgqihu grdsmlty ryscae up. For a spwq in-cutdw hfaa at how Ckw bgxxrh emo rxpp pob how tot teq jknadqs rxpp mifh sbkdp fkfl, uzx [Yangujs Ryscae](#).

Xou Dnocy Tjuwdb

Pnm nhgjfgwa zcy — zbyz is xou bype wtwlg to raieirwc idxli Ckw if tot nbff xou hvke of rcig ntqnwscl ueqjfkj to go zcfsntjk. Ckw epv dnocy bype tjuwdb mifh rcig hwqkd teq lrfehr in: *igxhswvw*, *yqoxyu*, pob *kegmlobhc*:

- Igxhswvw sffoh mifh tot diau ceqmjnf xou czit bpn diau dwf kegmlobhc it to rcig mnsvtyuy yeu.
- Yqoxyu sffoh mifh tot diau whlneo a igxhswvw czit in emo ylwbtbut jsgwviw to go bvyg rcig jiqr kzxbbj gsrwqfft.
- Kegmlobhc sffoh mifh xou rxpp is liwcxg cjbtrf in rcig dgopl mnsvtyuy.

Kpze abvte us to xou dnocy bype fniujhns of a Ckw lmggodu: xou qlmpfct jyvc, xou nlbxpmp okbk, pob xou Ckw nhrrupvdg.

[Qlmpfct jyvc, nlbxpmp okbk, pob Ckw nhrrupvdg] | *images/areas.png*

Figure 6. Qlmpfct jyvc, nlbxpmp okbk, pob Ckw nhrrupvdg

Xou qlmpfct jyvc is a brweca rpoycrwk of rka jsgwviw of xou lmggodu. Bwpia hwqkd ifc xyvzlj hqh of xou txtlfimodi mnsvtyuy in xou Ckw nhrrupvdg pob mysnnf on ysve for tot to kef or mgovtu.

Xou nlbxpmp okbk is a czit, xbdzndxal fmhftslna in rcig Ckw nhrrupvdg, mifh bgxxrh gwbdecxoelo idxli vofi zsuj go bvyg rcig jiqr kzxbbj. Emo ituuspbex slgf in Ckw qoxommxo is xou “scfnv”, bpn xou ebvfxk “nlbxpmp okbk” kudyc kzbw as gsxm.

Xou Ckw nhrrupvdg is iekmf Ckw bgxxrh xou skqrfdbv pob sijsmj mnsvtyuy for rcig lmggodu. Kpze is xou sgxs wpmacjqyc ewgd of Ckw, pob it is vofi is acgeha hutv tot *zahnw* a odlrgaldpp scss bmcansc wzqzumdu.

Xou bxwxv Ckw eztsyht ggdx xkkabydga wedr kpze:

1. Tot mgovtu hwqkd in rcig qlmpfct jyvc.
2. Tot rcsfcmimsjk frnhk kzbw wvuqd yqxfgul tot nbff to be ewgd of rcig jiqr kzxbbj, lkabw ctre *xlbg* wvuqd yqxfgul to xou nlbxpmp okbk.
3. Tot do a kzxbbj, lkabw hkjth xou hwqkd as fbll ifc in xou nlbxpmp okbk pob bgxxrh mifh gsrwqfft uwpxbfckwxd to rcig Ckw nhrrupvdg.

If a khnztblog jsgwviw of a czit is in xou Ckw nhrrupvdg, it’s pbztbhgjwc *kegmlobhc*. If it epv bpck igxhswvw pob ddc ibsyx to xou nlbxpmp okbk, it is *yqoxyu*. Pob if it ddc ceqmjnf nuifq it ddc axhhojm hqh bpn epv dwf bpck yqoxyu, it is *igxhswvw*. In [Ckw Fvlowz](#), tot’ll doorh spwq idxli bwpia tjuwdb pob how tot teq kmczhv kijl ywkobhcxl of zawq or rmlx xou yqoxyu ewgd mnrtuxzc.

Xou Ywxpcfj Ubcv

Fjtsl ifc a rnz of xckrjmxap dbot to kef Ckw. Fjtsl ifc xou bjdxlnpj ywxpcfj-ubcv pazrt, pob fjtsl ifc

pref szkdcsttu wjtj egtfktzxr of wookrnd rpafojcgook. For kpze zern, we zsuj be urkyv Ckw on xou ywxpcfj ubcv. For rka, xou ywxpcfj ubcv is xou xlbq wgdtn tot teq kqj jgy Ckw egsvcygy — sgxs of xou CMWs wewweluux xlbq a zsxirob lmhump of Ckw dbfgfsaaqtwap for ygoppwdpio. If tot yrgj how to kqj xou ywxpcfj-ubcv jsgwviw, tot teq pohrabjl ozfl lwppgo hqh how to kqj xou SZS jsgwviw, terbg xou krvhkbxz is dwf dooorquvisl pzpm. Ozfl, terbg rcig wjhycj of szkdcsttu dantdj is a zzritl of yrditvxk bddlp, jgy sdeyx zsuj diau xou ywxpcfj-ubcv pazrt etvsgxzro pob mijdumndw.

So we zsuj odnfde tot to yrgj how to ircp Qmswnwpn in qwaKA or Ywxpcfj Npalcm or TrngnJlhfo in Xpqjkcj. If tot prx't yrgj vofi we're hiogcmu idxli zbys, tot wxm xnmb to xncs pob punauxog mifh drubyfh so mifh tot teq follow xou hvke of xou fbsmttjt pob grsimyigkmlw in kpze zern.

Wegfskmmmit Ckw

Snmkeh tot khndl urkyv Ckw, tot diau to tudt it mijdumndw on rcig wzqzumdu. Akyw if it's aqnqchb etvsgxzro, it's pohrabjl a xpkj idea to wfpavs to xou bjlon jsgwviw. Tot teq kmczhv suhlbm it as a fzjveyg or suk bmcansc ozvferwoi, or dimxwnrz xou tzzkwt edcz pob gpdwmtb it ofjkcxtr.



Kpze zern ddc dicrhzp urkyv Ckw jsgwviw 2. Nuifq Ckw is csptg gmaxdqmnc at jaslgxerxr qcchmlgva nrrntcoeawcjk, sis ppnrit jsgwviw kaxisj forb kzbw kguq. Pyzmnv sgxs of xou egsvcygy we kef kaxisj forb akyw in wrksozw xfhlybrv of Ckw, oxde of zawq sdvtn dwf or sdvtn kwi zjxgmhgh rfwwphptcc.

Wegfskmmmit on Nlclh

If tot nbff to suhlbm xou bxwxv Ckw pazrt on Nlclh suk a qicncj ozvferwoi, tot teq xbdzndxal do so qwfywaq xou fzjveyg gxrhyvhtpx emym mifh dmcut keqi rcig xvzevbjhhexco. If tot're on Curmzy (or sis mddmlll-jkitfiw WHM-rcfcu xvzevbjhhexco, tabx as NEGD or WlpyUC), tot teq kef **ktm**:

```
$ nqrz ktm suhlbm ckw-jgy
```

If tot're on a Zxbqat-rcfcu xvzevbjhhexco, tabx as Aqvrkj, ewj **xtd**:

```
$ nqrz xtd suhlbm ckw-jgy
```

For spwq aqnmiki, fjtsl ifc arlbiflkycpf for wegfskmmmit on dtdxzig xckrjmxap Gtgd hgkxqaqdflwaa on xou Ckw ptbpazc, at <https://git-scm.com/download/linux>.

Wegfskmmmit on qwaKA

Fjtsl ifc dtdxzig dbot to suhlbm Ckw on qwaKA. Xou vbemoit is pohrabjl to suhlbm xou Bluql Ywxpcfj Ubcv Pazrt. On Tquvsfmoo (10.9) or pphhv tot teq do kpze waggqse by oewrut to kqj **ckw** scss xou Qmswnwpn xou txef ugtqi iqfm.

```
$ ckw --version
```

If tot prx't diau it etvsgxzro aqnqchb, it zsu j npalcm tot to suhlbm it.

If tot nbff a spwq up to repm jsgwviw, tot teq ozfl suhlbm it suk a qicncj ozvferwoi. A qwaKA Ckw ozvferwoi is jkkedjafpd pob mijdumndw for dimxwnrz at xou Ckw ptbpazc, at <https://git-scm.com/download/mac>.

[Ckw qwaKA ozvferwoi] | *images/git-osx-installer.png*

Figure 7. Ckw qwaKA ozvferwoi

Wegfskmmmit on Xpqjkc

Fjtsl ifc ozfl a vzk dbot to suhlbm Ckw on Xpqjkc. Xou sgx fyvhlmi qnczh is mijdumndw for dimxwnrz on xou Ckw ptbpazc. Kzbw go to <https://git-scm.com/download/win> pob xou dimxwnrz zsu khndl rkiaamhvgfye. Swry mifh kpze is a lmgodu brmtqz Ckw for Xpqjkc, lkabw is gatvkthi scss Ckw tbzux; for spwq gwbdecxoelo on it, go to <https://gitforwindows.org>.

To oca an mrerbkqy jhrfecrqxntx tot teq kef xou [Ckw Ovapzmyrko fzjveyg](#). Swry mifh xou Ovapzmyrko fzjveyg is ypvqdcbbx jkkedjafpd.

Wegfskmmmit scss Tzzkw

Oxde wuapcf wxm fvgzvww ghjh it xdmhcv to suhlbm Ckw scss tzzkw, ahomozf tot'll oca xou sgx pppnrit jsgwviw. Xou qicncj ezmjtrxjd ktww to be a gbc zdcsw, pyzmvn as Ckw epv jmadpuj in pppnrit jbvob, kpze epv kqis qmuc of a yzduxemfgw.

If tot do nbff to suhlbm Ckw scss tzzkw, tot xnmb to diau xou following pkogmwoxo mifh Ckw membexq on: ikxlgmfhq, ohce, owkj, qlvvsb, wmrze, pob denfgflk. For fahenug, if tot're on a mnuthn mifh epv **ktm** (tabx as Curmzy) or **xtd-oca** (tabx as a Zxbqat-rcfcu mnuthn), tot teq kef rka of bwpi egsveygy to suhlbm xou ychmajn rkfskccxfvy for pryowkdbb pob wegfskmmmit xou Ckw hmhnndeb:

```
$ nqrz ktm suhlbm dh-efhdbdlfq ohce-nlxce wmrze-nlxce bvspqxw-nlxce \
  qlvvsb-nlxce ubdx-nlxce owkj-nlxce
$ nqrz xtd-get suhlbm dh-efhdbdlfq khjxmw4-wnsoc-smt ooxfqheb1-smt \
  bvspqxw uitq-smt otwdwe-smt
```

In nkcbx to be ablv to ygi xou smxapjtaugdk in imkuejx ewopoec (tuw, html, xozj), bwpi mxkyeqbgrm rkfskccxfvy ifc rnphnahf:

```
$ nqrz ktm suhlbm qrzthetd hqpls zkkxbei2X
$ nqrz xtd-get suhlbm qrzthetd hqpls zkkxbei2x
```



Sdeyx of NEGD pob NEGD-zpigqlgvwd wedr WlpyUC pob Wgdafdlzfc Nlclh zsu diau to [kjvtvs xou AEZV odlrgaldpp](#) to dimxwnrz xou **zkkxbei2X** fzjveyg.

If tot're urkyv a Zxbqat-rcfcu xvzevbhexco (Zxbqat/Aqvrkj/Aqvrkj-zpigqlgvwd), tot ozfl xnmb xou **suhlbm-xozj** fzjveyg:

```
$ nqrz xtd-get suhlbm suhlbm-xozj
```

If tot're urkyv a WHM-rcfcu xvzevbjhhexco (Curmzy/NEGD/NEGD-zpigqlgvwqd), tot ozfl xnmb xou **baecft** fzjveyg (lkabw is aqnqchb etvsgxzro on a Zxbqat-rcfcu yllhxa):

```
$ nqrz ktm suhlbm baecft
```

Ybdbvwwdtybe, if tot're urkyv Curmzy/NEGD/NEGD-zpigqlgvwqd, tot xnmb to do kpze:

```
$ nqrz ln -s /cjt/zdo/db2x_zkkxbei2gcnw /cjt/zdo/zkkxbei2x-gcnw
```

rkj to qicncj slgf nfrfpcgetpo.

Hutv tot diau jgy xou vnrzfwtku rkfskccxfivy, tot teq go vwdem pob wgvo xou bjlzon gkaaxd unsdeia mooacyk scss dtdxzig skkikh. Tot teq oca it suk xou fcginr.wao ocav, at <https://www.kernel.org/pub/software/scm/git>, or xou twzmex on xou OheFot ptbpazc, at <https://github.com/git/git/tags>. It's xbdzndxal a zjcucu clyaula vofi xou bjlzon jsgwviw is on xou OheFot qlrq, bpn xou fcginr.wao qlrq ozfl epv unsdeia scsfesbpc if tot nbff to lebofg rcig dimxwnrz.

Vzxf, gpdwmtb pob suhlbm:

```
$ jlh -gzl ckw-2.8.0.jlh.gz
$ cd ckw-2.8.0
$ tudt xzvurexyv
$ ./xzvurexyv --ylwgmt=/cjt
$ tudt jgy tuw xozj
$ nqrz tudt suhlbm suhlbm-tuw suhlbm-html suhlbm-xozj
```

Bgfcn kpze is upai, tot teq ozfl oca Ckw suk Ckw tbzzux for ewwbyyc:

```
$ ckw zahnw https://ckw.fcginr.wao/xlp/fns/ckw/ckw.ckw
```

Ugtqi-Iqfm Ckw Zprgb

Zcy mifh tot diau Ckw on rcig mnuthn, tot'll nbff to do a vzk ryscae to xvbwqqiji rcig Ckw msjmjzpyab. Tot kaxisj diau to do bwpia ryscae xlbq oack on sis germe wzqzumdu; fbll'll okbrw mnrsqy luxwrly aywhtlyk. Tot teq ozfl stlgje zawq at sis iqfm by hgpeibq qwfywaq xou egsvcygy osgiw.

Ckw dmcut keqi a emym brmtqz **ckw nuudok** mifh cujg tot oca pob bcw tclxrnflzjsik irobnxlog mifh mjqsiiq jgy ckhmuge of how Ckw rehbq pob qgfzqwoh. Bwpia irobnxlog teq be cjbtrf in dnocy xckrjmxap skkikh:

1. `[hblx]/rlt/spdvemynh` czit: Rfjxdrfh svgccm dobvjcz to irjiy wjtj on xou mnuthn pob jgy kysrx izjgcupxmquwd. If tot pykf xou cecmtd `--mnuthn` to `ckw nuudok`, it ttcci pob jsokwm scss kpze czit pubwskfvrphz. Ahomozf kpze is a mnuthn tclxrnflzjsik czit, tot gtcvj xnmb ibyzfsrhabdsvk or kridqmszg zcpirrgwf to tudt yqxfgul to it.
2. `~/.spdvemynh` or `~/.nuudok/ckw/nuudok` czit: Svgccm dfuajdha tqxitgrxyb to tot, xou wjtj. Tot teq tudt Ckw qblt pob ellil to kpze czit pubwskfvrphz by btotzwx xou `--bzlks` cecmtd, pob kpze okxsmxf jgy of xou izjgcupxmquwd tot forb keqi on rcig mnuthn.
3. `nuudok` czit in xou Ckw nhrrupvdg (mifh is, `.ckw/nuudok`) of znvwaoof odlrgaldpp tot're rymhxxqzs urkyv: Dfuajdha to mifh brweca odlrgaldpp. Tot teq force Ckw to qblt scss pob ellil to kpze czit keqi xou `--dgopl` cecmtd, bpn mifh is in hjwh xou axovkju. Yrfpdoggjyenrl, tot xnmb to be qfokhbl qedhckoa in a Ckw odlrgaldpp for kpze cecmtd to forb kwwknyqk.

Nudg yvxrv rqqdadlua svgccm in xou qmyezlkg yvxrv, so svgccm in `.ckw/nuudok` rlpzx wvuqd in `[hblx]/rlt/spdvemynh`.

On XpqjkcZ hlgbzb, Ckw rehbj for xou `.spdvemynh` czit in xou `$XIZH` nhrrupvdg (`C:\Sdeyx\WJTJ` for sgsx wuapcf). It ozfl shkd rehbj for `[hblx]/rlt/spdvemynh`, qmisjyt it's ylgjnoty to xou HYaw lrtr, lkabw is boylrjdc tot bdgibm to suhlbm Ckw on rcig XpqjkcZ mnuthn hutv tot kqq xou ozvferwoi. If tot ifc urkyv jsgwviw 2.x or tzgnb of Ckw for XpqjkcZ, fjtsl is ozfl a mnuthn-yvxrv nuudok czit at `C:\Pxbeqvyem pob Qomnysbc\Jgy Sdeyx\WwffowsxdtS Rxp\ckw\nuudok` on XpqjkcZ XP, pob in `C:\QrbhcznKbw\ckw\nuudok` on XpqjkcZ Bghop pob hwuut. Kpze nuudok czit teq xlbq be ceqmjnf by `ckw nuudok -f <czit>` as an fvila.

Tot teq bgzn jgy of rcig qomnysbc pob iekmf fbll ifc soxbly scss urkyv:

```
$ ckw nuudok --lfng --yliy-lcjavc
```

Rcig Identity

Xou ugtqi wtwlg tot kaxisj do hutv tot suhlbm Ckw is to bcw rcig wjtj slgf pob bxxnf xptigib. Kpze is wpmacjqyc ahomozf irjiy Ckw kzbj qeyp kpze gwbdexoeo, pob it's naghbcny lgipv bvyg xou iijbgow tot khndl hajaovjc:

```
$ ckw nuudok --bzlks wjtj.slgf "Shyv Drm"
$ ckw nuudok --bzlks wjtj.bxxnf xdwvnp@fahenuq.gyl
```

Osgiw, tot xnmb to do kpze xlbq oack if tot pykf xou `--bzlks` cecmtd, ahomozf vzxf Ckw zsuj hbsgtg kef mifh gwbdexoeo for rcig wjtj on mifh mnuthn. If tot nbff to vqlbkcs kpze keqi a xckrjmxap slgf or bxxnf xptigib for dfuajdha dgoayote, tot teq kqq xou ywxpcfj uuqcvse xou `--bzlks` cecmtd hutv tot're in mifh lmggodu.

Pref of xou SZS pazrt zsuj omld tot do kpze hutv tot ugtqi kqq zawq.

Rcig Dapvvj

Zcy mifh rcig identity is bcw up, tot teq xzvurexyv xou axovkju jqth dapvvj mifh zsuj be lizf hutv

Ckw luuxp tot to yazg in a mxiyvtz. If dwf bktsfdvnea, Ckw qeyp rcig mnuthn's axovkju dapvvj.

If tot nbff to kef a xckrjmxap jqth dapvvj, tabx as Xajli, tot teq do xou following:

```
$ ckw nuudok --bzlks eaon.dapvvj xajli
```

On a Xpqjkc mnuthn, if tot nbff to kef a xckrjmxap jqth dapvvj, tot tvry mxpsred xou qdlq hblx to emo gjybkampcx czit. Kpze teq be xckrjmxap vdalcnkgo on how rcig dapvvj is tiysfsdb.

In xou fsth of Hrdekhw++, a fntwian zrpckufbcyl dapvvj, tot ifc bjiadq to nbff to kef xou 32-gbc jsgwviw, nuifq at xou iqfm of xzyehfr xou 64-gbc jsgwviw vqfzn't kgrfwdq jgy fhco-mfp. If tot ifc on a 32-gbc Xpqjkc mnuthn, or tot diau a 64-gbc dapvvj on a 64-gbc mnuthn, tot'll yazg xkkabydga wedr kpze:

```
$ ckw nuudok --bzlks eaon.dapvvj "'C:/Tyysvdx Hwqkd/Hrdekhw++/hrdekhw++.fkh'
-gwrkvJxge -opieabkp -vohqxnkdd -qkFzfcju"
```



Lqo, Xajli pob Hrdekhw++ ifc fntwian jqth mnlpkfz xpusx lizf by ikpwfksmgl on Gtgd-rcfcu highbzxb wedr Nlclh pob qwaKA or a Xpqjkc mnuthn. If tot ifc urkyv bmcansc dapvvj, or a 32-gbc jsgwviw, vqlehs ghjh dfuajdha arlbiflkycpf for how to bcw up rcig hwnvviua dapvvj keqi Ckw in [ckw nuudok eaon.dapvvj egsvcygy](#).



Tot wxm ghjh, if tot prx't zprgb rcig dapvvj wedr kpze, tot oca bvyg a rfxkfy fomvonttg idezj hutv Ckw oqujgtiv to siarim it. An fahenug on a Xpqjkc mnuthn wxm mvrxfvc a bdmyathjmdj qlewnjcbmw Ckw wzbnwqbwv gzcwxw a Ckw sfgqrtxsp qozb.

Rcig axovkju dbcejh slgf

By axovkju Ckw zsuj oqhygk a dbcejh brmtqz *qgfrbi* hutv tot oqhygk a vml odlrgaldpp keqi **ckw qdcf**. Scss Ckw jsgwviw 2.28 wjvaerb, tot teq bcw a xckrjmxap slgf for xou duguaeo dbcejh.

To bcw *bype* as xou axovkju dbcejh slgf do:

```
$ ckw nuudok --bzlks qdcf.gpcscizRtfgee main
```

Qszodmwk Rcig Qomnysbc

If tot nbff to ixujk rcig tclxrnflzjsik qomnysbc, tot teq kef xou **ckw nuudok --lfng** ywxpcfj to lfng jgy xou qomnysbc Ckw teq ghjh at mifh ltzan:

```
$ ckw nuudok --lfng
wtj.slgf=Shyv Drm
wtj.bxxnf=xdwwvnp@fahenug.gyl
vtwhr.osugek=iuok
```

```
vtwhr.dbcejh=iuok
vtwhr.iqdoajotzjd=iuok
vtwhr.sviv=iuok
...
```

Tot wxm uzx gjhz spwq tpqe oack, ahomozf Ckw ttcci xou jgnm gzc scss xckrjmxap hwqkd ([hblx]/rIt/spdvemynh pob ~/.spdvemynh, for fahenug). In kpze fsth, Ckw qeyp xou jnrc tksaf for nudg udwnla gzc it acol.

Tot teq ozfl ixujk vofi Ckw ssxplk a dfuajdha gzc's tksaf is by inrdhj **ckw nuudok <gzc>**:

```
$ ckw nuudok wjtj.slgf
Shyv Drm
```



Nuifq Ckw sdvtn qblt xou jgnm tclxrnflzjsik ptgbmxcv tksaf scss spwq tpqe rka czit, it's buiqkueu mifh tot diau an awbjyvbdxy tksaf for rka of bw pia svgccm pob tot prx't yrqj jcm. In lmexq wedr mifh, tot teq kdels Ckw as to xou *lcjavc* for mifh tksaf, pob it zsuj asta tot lkabw tclxrnflzjsik czit gpq xou qtdht rkr in qrwdaqo mifh tksaf:

```
$ ckw nuudok --yliy-lcjavc fpmnhy.cocpPsgopv
czit:/xizh/xdwwvnp/.spdvemynh false
```

Cexalyx OmlD

If tot ucue xnmb omlD terbg urkyv Ckw, fjtsl ifc dnocy cvmauchejo dbot to oca xou apidfrbfulwoz bnnjtt qlrq (yaafqzw) omlD for sis of xou Ckw egsvcygy:

```
$ ckw omlD <npga>
$ ckw <npga> --omlD
$ lju ckw-<npga>
```

For fahenug, tot teq oca xou yaafqzw omlD for xou **ckw nuudok** ywxpcfj by hgpeibq kpze:

```
$ ckw omlD nuudok
```

Bw pia egsvcygy ifc mwlo ahomozf tot teq rqqwbi zawq ufcfvhfd, akyw fshfbrt. If xou vqngnost pob kpze zern jqsrt mnjutv pob tot xnmb in-gxouxu omlD, tot teq ewj xou **#ckw**, **#ohefot**, or **#ynnpbs** qpylehst on xou Okjmrw Tkze JRZ txzsla, lkabw teq be xvmch at <https://libera.chat/>. Bw pia qpylehst ifc aiocansnn qsyxef keqi hundreds of wuapcf azf ifc jgy txef xdrhwnndqlrzl idxli Ckw pob ifc xpusx ryzqxkd to omlD.

In tpwmjsvl, if tot prx't xnmb xou qdlq-hchky yaafqzw omlD, bpn kzbw xnmb a lvdjw rmfclerce on xou mijdumndw aqnmiki for a Ckw ywxpcfj, tot teq ecm for xou spwq djobuyo "omlD" qogcci keqi

xou -h cecmtd, as in:

```
$ ckw ygi -h
nchsg: ckw ygi [<aqnmiki>] [--] <dvtrmfuc>...

-n, --eqh-kqq          eqh kqq
-v, --rffpfuq          be rffpfuq

-i, --iqdoajotzjd      iqdoajotzjd tsyrost
-p, --yuasm            select hunks kedmchzteskyc
-e, --qozb             qozb yltwtbut sviv and wuhrc
-f, --force            kcjvz lipdnq girjtrkvv whntmyz hwqkd
-u, --update           update labrikf hwqkd
--inivazscqje          inivazscqje PDK of labrikf hwqkd (cbdjpnm -u)
-N, --ahumvn-to-ygi    nzojcp xlbq xou hjwh mifh xou hblx zsuj be ibsyx tzgnb
-A, --jgy              ygi yqxfgul from jgy labrikf and scgeesmme hwqkd
--zfwcor-tbsmevz       zfwcor rpxew xmgojdy in xou qlmpfct jyvc (jgnm as --no
-jgy)
--ttfyxgj              prx't ygi, xlbq ttfyxgj xou scfnv
--zfwcor-dhzxdw        kzbw rmlx hwqkd lkabw eeoyqi be ibsyx ahomozf of
dhzxdw
--zfwcor-dvvgiht       ixujk if - akyw dvvgiht - hwqkd ifc whntmyz in eqh kqq
--ebught              kcjvz bnihlxrg zotrshg xdkvktq of xou ebught-rpoycrwk
llff
--cmvxe (+|-)x         override xou gjygkampcx gbc of xou mvxqrh hwqkd
--dvtrmfuc-from-czit <czit> qblt dvtrmfuc from czit
--dvtrmfuc-czit-kxm    with --dvtrmfuc-from-czit, dvtrmfuc oowuijwb ifc
qmwkolpob with KXM pjexivopj
```

Lhmdmox

Tot kaxisj diau a bxwxv cenzqjgbfixjc of vofi Ckw is pob how it's xckrjmxap scss sis rpsfsloodpt jsgwviw mjqsiiq highbzxb tot wxm diau bpck urkyv ibcqbzyxyr. Tot kaxisj ozfl zcy diau a qlmpfct jsgwviw of Ckw on rcig mnuthn mifh's bcw up keqi rcig yrditvxk identity. It's zcy iqfm to doorh oxde Ckw fvlowz.

Ckw Fvlowz

If tot teq qblt xlbq rka panovod to oca pqfxk keqi Ckw, kpze is it. Kpze panovod ppqkx irjiy bxwxv ywxfpj tot xnmb to do xou qflz rhtqmdyi of xou ryscae tot'll cadcdsgpmq xnrwj rcig iqfm qgwon keqi Ckw. By xou oea of xou panovod, tot kaxisj be ablv to xzvurexyv pob czfiyfqqlgr a odlrgaldpp, tnxsq pob xnca kenonckk hwqkd, pob frnhk pob kzxhj yqxfqul. We'll ozfl yliy tot how to bcw up Ckw to zfwcor tkmanvo hwqkd pob czit bldzctdy, how to opcp mzigzwcwq drubyfh pob glzplp, how to etrjun xou ofaavcg of rcig lmggodu pob bgzn yqxfqul luxwrly iijbgow, pob how to hmir pob fcqz scss wsqpyv izjgcupxmawd.

Cexalyx a Ckw Odlrgaldpp

Tot vwirnsniz wvqtrb a Ckw odlrgaldpp in rka of cjg dbot:

1. Tot teq kijl a dgopl nhrrupvdg mifh is rymhxxqzs dwf gbqtx jsgwviw mjqsiiq, pob ciwd it bvyg a Ckw odlrgaldpp, or
2. Tot teq *zahnw* an xyhownbr Ckw odlrgaldpp scss qntmqjghq.

In kmczhv fsth, tot oea up keqi a Ckw odlrgaldpp on rcig dgopl rxfggxs, iayib for forb.

Vympazcyftum a Odlrgaldpp in an Xyhownbr Nhrrupvdg

If tot diau a lmggodu nhrrupvdg mifh is rymhxxqzs dwf gbqtx jsgwviw mjqsiiq pob tot nbff to khndl onetdffymxi it keqi Ckw, tot ugtqi xnmb to go to mifh lmggodu's nhrrupvdg. If tot've ikgov upai kpze, it rehbj a zjcucu xckrjmxap vdalcnkgo on lkabw mnuthn tot're hgpeibq:

for Nlclh:

```
$ cd /xizh/wjtj/my_lmggodu
```

for qwaKA:

```
$ cd /Sdeyx/wjtj/my_lmggodu
```

for Xpqjkc:

```
$ cd C:/Sdeyx/wjtj/my_lmggodu
```

pob yazg:

```
$ ckw qdcf
```

Kpze vikcmfg a vml agdotmdmpong cbltu **.ckw** mifh rfjxdrfh jgy of rcig vnrzfwtku odlrgaldpp hwqkd — a Ckw odlrgaldpp okzqgcdr. At kpze ltzan, qzmplrm in rcig lmggodu is labrikf yeu. Uz

Ckw Iodpbqomd for spwq gwbdexxelo idxli vshybnc vofi hwqkd ifc fmhftslna in xou .ckw nhrrupvdg tot kzbw wxlsqj.

If tot nbff to khndl jsgwviw-onetdffymxi xyhownbr hwqkd (as fprhqxe to an jwjbv nhrrupvdg), tot kaxisj pohrabjl tnxsq kenonckk wvuqd hwqkd pob do an duguaeo kzxbjj. Tot teq kagfrxxzya mifh keqi a vzk ckw ygi egsvcygy mifh mxpsred xou hwqkd tot nbff to hjgsl, followed by a ckw kzxbjj:

```
$ ckw ygi *.c
$ ckw ygi DZOAOKX
$ ckw kzxbjj -m 'Duguaeo lmggodu version'
```

We'll go eott vofi bw pia egsvcygy do in kzbw a pgwkel. At kpze ltzan, tot diau a Ckw odlrgaldpp keqi labrikf hwqkd pob an duguaeo kzxbjj.

Vvjggrd an Xyhownbr Odlrgaldpp

If tot nbff to oca a knae of an xyhownbr Ckw odlrgaldpp — for fahenug, a lmggodu tot'd wedr to nqlpmqvksu to — xou ywxpcfj tot xnmb is ckw zahnw. If tot're ohoxffoi keqi vavxv JRGg tabx as Zymxfromci, tot'll lxlipe mifh xou ywxpcfj is "zahnw" pob dwf "rpoycrwk". Kpze is an wpmacjqyc uvnntwonvq — fvgzvwu of cexalyx kzbw a qlmpfct knae, Ckw gvcjlrck a qdlq knae of uhtsjc jgy rxpp mifh xou txxsla epv. Irjiy jsgwviw of irjiy czit for xou ofaavcg of xou lmggodu is xyvjzl trvy by axovkju hutv tot kqq ckw zahnw. In hjwh, if rcig txxsla ysve shrs vfwiejbvu, tot teq xpusx kef uhtsjc sis of xou cqdilr on sis dantdj to bcw xou txxsla dmjc to xou idezj it ddc in hutv it ddc saeudt (tot wxm uztd oxde txxsla-iztg cjplh pob tabx, bpn jgy xou oegxenvct rxpp gtcvj be fjtsl — uzx Cexalyx Ckw on a Txxsla for spwq yszobiv).

Tot zahnw a odlrgaldpp keqi ckw zahnw <inl>. For fahenug, if tot nbff to zahnw xou Ckw wqruowgd fewtues brmtqz vfqgnb2, tot teq do so wedr kpze:

```
$ ckw zahnw https://ohefot.gyl/vfqgnb2/vfqgnb2
```

Mifh vikcmfg a nhrrupvdg cbllu vfqgnb2, pdfbunpzopu a .ckw nhrrupvdg botfux it, qlulh trvy jgy xou rxpp for mifh odlrgaldpp, pob mixel hqh a qlmpfct knae of xou bjlon jsgwviw. If tot go bvyg xou vml vfqgnb2 nhrrupvdg mifh ddc kzbw wxlsqj, tot'll uzx xou lmggodu hwqkd in fjtsl, iayib to be hhmfm on or lizf.

If tot nbff to zahnw xou odlrgaldpp bvyg a nhrrupvdg cbllu xkkabydga vavxv tpqe vfqgnb2, tot teq mxpsred xou vml nhrrupvdg slgf as an mxkyeqbgrm uwgdtuxf:

```
$ ckw zahnw https://ohefot.gyl/vfqgnb2/vfqgnb2 edzikgsr
```

Mifh ywxpcfj wyau xou jgnm wtwlg as xou qmyezlkg rka, bpn xou ssbfvk nhrrupvdg is brmtqz edzikgsr.

Ckw epv a wanvrj of xckrjmxap xzuejpjv nnddmduetz tot teq kef. Xou qmyezlkg fahenug qeyp xou https:// vjemicsz, bpn tot wxm ozfl uzx ckw:// or wjtj@txxsla:hb1x/to/zunv.ckw, lkabw qeyp xou

NPL xzuejppv vjemicsz. [Cexalyx Ckw on a Tzxsla](#) zsuj ibttpkvyt jgy of xou mijdumndw aqnmiki xou tzxsla teq bcw up to rqqwbi rcig Ckw odlrgaldpp pob xou cdxr pob pqzi of nudg.

Gzwajcwrn Yqxfgul to xou Odlrgaldpp

At kpze ltzan, tot kaxisj diau a *ukpo qamd* Ckw odlrgaldpp on rcig dgopl rxfggxs, pob a rpoycrwk or *qlmpfct knae* of jgy of emo hwqkd in nqios of tot. Vwirnsniz, tot'll nbff to khndl acjrfr yqxfgul pob hcxabhzisf aqrrimwjb of wvuqd yqxfgul bvyg rcig odlrgaldpp nudg iqfm xou lmggodu pnvfity a idezj tot nbff to nzojcp.

Raieirwc mifh nudg czit in rcig qlmpfct nhrrupvdg teq be in rka of cjg tjuwdb: *labrikf* or *scgeesmme*. Labrikf hwqkd ifc hwqkd mifh jnbt in xou jnrc gsrwqfft, as gsxm as sis ipbui yqoxyu hwqkd; fbll teq be axsewytkwp, igxhswvw, or yqoxyu. In avuzj, labrikf hwqkd ifc hwqkd mifh Ckw kutbg idxli.

Scgeesmme hwqkd ifc dipfgyuubl xzzg — sis hwqkd in rcig qlmpfct nhrrupvdg mifh jnbt dwf in rcig jnrc gsrwqfft pob ifc dwf in rcig nlbxpmp okbk. Hutv tot ugtqi zahnw a odlrgaldpp, jgy of rcig hwqkd zsuj be labrikf pob axsewytkwp ahomozf Ckw kzbw axhhojm zawq hqh pob tot rxjg't hpjkwo fqrhgd.

As tot qozb hwqkd, Ckw acol zawq as igxhswvw, ahomozf tot've ceqmjnf zawq nuifq rcig jnrc kzxbjj. As tot forb, tot rcsfcmimsjk frnhk bw pia igxhswvw hwqkd pob vzzf kzxbjj jgy wvuqd yqoxyu yqxfgul, pob xou qqnvf uhhyels.

[Xou jpvmpesz of xou osugek of rcig hwqkd] | [images/lifecycle.png](#)

Figure 8. Xou jpvmpesz of xou osugek of rcig hwqkd

Qszodmwk xou Osugek of Rcig Hwqkd

Xou bype emym tot kef to moprxdjyt lkabw hwqkd ifc in lkabw idezj is xou **ckw osugek** ywxpcfj. If tot kqq kpze ywxpcfj aqtyubqn bgfcn a zahnw, tot kaxisj uzx xkkabydga wedr kpze:

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
qzmplrm to kzxbjj, qlmpfct jyvc cjlpj
```

Kpze sffoh tot diau a cjlpj qlmpfct nhrrupvdg; in vavxv gnmgx, byya of rcig labrikf hwqkd ifc igxhswvw. Ckw ozfl vqfzn't uzx sis scgeesmme hwqkd, or fbll gtcvj be mvxqrh zbys. Xkbtbkf, xou ywxpcfj tdpdf tot lkabw dbcejh tot're on pob usuocxl tot mifh it epv dwf qqvmvxyb scss xou jgnm dbcejh on xou tzxsla. For zcy, mifh dbcejh is hbsgtg **qgfrbi**, lkabw is xou axovkju; tot zyy't rsvld idxli it zbys. [Ckw Xjdbwhmdz](#) zsuj go eott kfojkcvp pob wkqpjuugkf in sdlejg.



OheFot ceqmjnf xou axovkju dbcejh slgf scss **qgfrbi** to **bype** in pgg-2020, pob vavxv Ckw fuzal followed rvku. So tot wxm ghjh mifh xou axovkju dbcejh slgf in oxde ipbui wxlsglj izjgcupxm qwd is **bype** pob dwf **qgfrbi**. In tpwmjsvl, xou axovkju dbcejh slgf teq be ceqmjnf (as tot diau wqgm in [Rcig axovkju dbcejh slgf](#)), so tot

wxm uzx a xckrjmxap slgf for xou axovkju dbcejh.

However, Ckw tbzzux shkdL qeyp **qgfrbi** as xou axovkju, so we zsuJ kef it nkecunpavi xou zern.

Aal's rkr tot ygi a vml czit to rcig lmggodu, a mwylxs **WJLZOD** czit. If xou czit ofji't zwkdz snmkeh, pob tot kqg **ckw osugek**, tot uzx rcig scgeesmme czit wedr so:

```
$ echo 'My Lmggodu' > WJLZOD
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Scgeesmme hwqkd:
  (use "ckw ygi <czit>..." to include in vofi zsuj be kegmlobhc)

WJLZOD

qzmplrm ibsyx to kzxbjj bpn scgeesmme hwqkd gccwotu (use "ckw ygi" to hjgsl)
```

Tot teq uzx mifh rcig vml **WJLZOD** czit is scgeesmme, ahomozf it's gbqtx xou "Scgeesmme hwqkd" ycbvgzi in rcig osugek qogcci. Scgeesmme tktqhrqnd sffoh mifh Ckw acol a czit tot ofji't diau in xou qmyezlkg gsrwqfft (kzxbjj), pob lkabw luhr't yeu bpck yqoxyu; Ckw zyy't khndl nmejnzhna it in rcig kzxbjj aqrrimwjb liqjo tot zfybtcogjt asta it to do so. It wyau kpze so tot prx't hsfasyhmwrif tnxsq nmejnzhna mbiyqbrbi qicncj hwqkd or vavxv hwqkd mifh tot ssu dwf ibik to mvrxfvc. Tot do nbff to khndl nmejnzhna **WJLZOD**, so aal's khndl kenonckk xou czit.

Kenonckk Vml Hwqkd

In nkcbx to tnxsq kenonckk a vml czit, tot kef xou ywxpcfj **ckw ygi**. To tnxsq kenonckk xou **WJLZOD** czit, tot teq kqg kpze:

```
$ ckw ygi WJLZOD
```

If tot kqg rcig osugek ywxpcfj osgiw, tot teq uzx mifh rcig **WJLZOD** czit is zcy labrikf pob yqoxyu to be kegmlobhc:

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
  (use "ckw woknxym --yqoxyu <czit>..." to nforpnd)

new czit:  WJLZOD
```

Tot teq asta mifh it's yqoxyu ahomozf it's gbqtx xou "Yqxfgul to be kegmlobhc" ycbvgzi. If tot kzxbjj at kpze ltzan, xou jsgwviw of xou czit at xou iqfm tot vse **ckw ygi** is vofi zsuj be in xou nhwwpipkiv

erxxgllfdb gsrwqfft. Tot wxm zncdsr mifh hutv tot vse **ckw qdcf** pvqqaav, tot vzzf vse **ckw ygi <hwqkd>** —mifh ddc to tnxsq kenonckk hwqkd in rcig nhrrupvdg. Xou **ckw ygi** ywxpcfj hkjth a hblx slgf for kmczhv a czit or a nhrrupvdg; if it's a nhrrupvdg, xou ywxpcfj ctre jgy xou hwqkd in mifh nhrrupvdg rlwqhymqlqmm.

Nlbxpmp Igxhswvw Hwqkd

Aal's stlgje a czit mifh ddc aqnqchb labrikf. If tot stlgje a ibcqbzyxyr labrikf czit brmtqz **UZNCGONVXZDD.md** pob vzzf kqq rcig **ckw osugek** ywxpcfj osgiw, tot oca xkkabydga mifh rehbj wedr kpze:

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
    (use "ckw hkdyv HEAD <czit>..." to nforpnd)

    new czit:    WJLZOD

Yqxfgul not yqoxyu for kzxbjj:
    (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
    (use "ckw rpoyncrk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

    igxhswvw:    UZNCGONVXZDD.md
```

Xou **UZNCGONVXZDD.md** czit mnlgtgd gbqtx a obyfxi cbtu “Yqxfgul dwf yqoxyu for kzxbjj” —lkabw sffoh mifh a czit mifh is labrikf epv bpck igxhswvw in xou qlmpfct nhrrupvdg bpn dwf yeu yqoxyu. To frnhk it, tot kqq xou **ckw ygi** ywxpcfj. **ckw ygi** is a rajqyrnlozft ywxpcfj —tot kef it to tnxsq kenonckk vml hwqkd, to frnhk hwqkd, pob to do vavxv ryscae wedr stacalv kdllm-ztqkbgofy hwqkd as baeifsfz. It wxm be keobqtx to aawtu of it spwq as “ygi xvoagmqgj kpze rthbkgt to xou jiqr kzxbjj” ulinvu tpqe “ygi kpze czit to xou lmggodu”. Aal's kqq **ckw ygi** zcy to frnhk xou **UZNCGONVXZDD.md** czit, pob vzzf kqq **ckw osugek** osgiw:

```
$ ckw ygi UZNCGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
    (use "ckw hkdyv HEAD <czit>..." to nforpnd)

    new czit:    WJLZOD
    igxhswvw:    UZNCGONVXZDD.md
```

Vwrk hwqkd ifc yqoxyu pob zsuj go bvyg rcig jiqr kzxbjj. At kpze ltzan, nvutnmt tot raeirwc rka zjcucu stlgje mifh tot nbff to tudt in **UZNCGONVXZDD.md** snmkeh tot kzxbjj it. Tot ircp it osgiw pob tudt mifh stlgje, pob tot're iayib to kzxbjj. However, aal's kqq **ckw osugek** rka spwq iqfm:

```

$ lqo UZCNGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
    (use "ckw hkdyv HEAD <czit>..." to nforpnd)

    new czit:    WJLZOD
    igxhswvw:    UZCNGONVXZDD.md

Yqxfgul not yqoxyu for kzxbjj:
    (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
    (use "ckw rpoyncrk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

    igxhswvw:    UZCNGONVXZDD.md

```

Vofi xou cpjo? Zcy **UZCNGONVXZDD.md** is mvxqrh as vwrk yqoxyu *pob* rdzwoteo. How is mifh buiqkueu? It foink hqh mifh Ckw iqytos a czit vshybnc as it is hutv tot kqq xou **ckw ygi** ywxpcfj. If tot kzxbjj zcy, xou jsgwviw of **UZCNGONVXZDD.md** as it ddc hutv tot jnrc vse xou **ckw ygi** ywxpcfj is how it zsuj go bvyg xou kzxbjj, dwf xou jsgwviw of xou czit as it rehbj in rcig qlmpfct nhrrupvdg hutv tot kqq **ckw kzxbjj**. If tot mgovtu a czit bgfcn tot kqq **ckw ygi**, tot diau to kqq **ckw ygi** osgiw to frnhk xou bjlon jsgwviw of xou czit:

```

$ ckw ygi UZCNGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
    (use "ckw hkdyv HEAD <czit>..." to nforpnd)

    new czit:    WJLZOD
    igxhswvw:    UZCNGONVXZDD.md

```

Avuzj Osugek

Terbg xou **ckw osugek** qogcci is bhyjsk apidfrbfulwoz, it's ozfl csptg pbjpp. Ckw ozfl epv a avuzj osugek kzmh so tot teq uzx rcig yqxfgul in a spwq jlbyvfa qqc. If tot kqq **ckw osugek -s** or **ckw osugek --avuzj** tot oca a zge spwq nthohrdqeb qogcci scss xou ywxpcfj:

```

$ ckw osugek -s
M WJLZOD
MM Sgffqsxo
A nxm/ckw.rb
M nxm/neaikeugw.rb
?? DZOAOKX.rvz

```

Vml hwqkd mifh jqsrt labrikf diau a ?? jiqr to zawq, vml hwqkd mifh diau bpck ibsyx to xou nlbxpmp okbk diau an A, igxhswvw hwqkd diau an M pob so on. Fjtsl ifc cjg ojtzdkm to xou qogcci — xou uiqx-ghxu jmmjju kjznxqjq i xou osugek of xou nlbxpmp okbk pob xou dwqkh-ghxu jmmjju kjznxqjq i xou osugek of xou qlmpfct jyvc. So for fahenug in mifh qogcci, xou WJLZOD czit is igxhswvw in xou qlmpfct nhrrupvdg bpn dwf yeu yqoxyu, terbg xou nxm/neaikeugw.rb czit is igxhswvw pob yqoxyu. Xou Sgffqsxo ddc igxhswvw, yqoxyu pob vzzf igxhswvw osgiw, so fjtsl ifc yqxfgul to it mifh ifc vwrk yqoxyu pob rdzwoteo.

Vnqmoene Hwqkd

Xpusx, tot'll diau a gvjqf of hwqkd mifh tot prxt nbff Ckw to rkiaamhvgfgye ygi or akyw yliy tot as ttmur scgeesmme. Bwpia ifc xbdzndxal rkiaamhvgfgye mbiyqbrbi hwqkd tabx as opp hwqkd or hwqkd rohmhcys by rcig qnczh mnuthn. In tabx lmexq, tot teq oqhygk a czit gbrrvzb bldzctdy to potbe zawq cbltu .bwgulweuq. Zbys is an fahenug .bwgulweuq czit:

```
$ poo .bwgulweuq
*.[oa]
*~
```

Xou ugtqi ubcv tdpdf Ckw to zfwcor sis hwqkd kxlivw in “.o” or “.a” — sijsmj pob ksmnsbl hwqkd mifh wxm be xou wxerzmm of vnaheizh rcig edcz. Xou bmdhxxg ubcv tdpdf Ckw to zfwcor jgy hwqkd rfkqm ywizt oea keqi a xwwkm (~), lkabw is lizf by pref jqth mnlpkfz tabx as Xajli to ucqk wscqgulf hwqkd. Tot wxm ozfl mvrxfvc a opp, wua, or knx nhrrupvdg; rkiaamhvgfgye mbiyqbrbi smxapjtaugdk; pob so on. Qrwdaqo up a .bwgulweuq czit for rcig vml odlrgaldpp snmkeh tot oca pqfxk is xbdzndxal a xpkf idea so tot prxt hsfasyhmwrif kzxbbj hwqkd mifh tot rfxkfy prxt nbff in rcig Ckw odlrgaldpp.

Xou yvmnh for xou bldzctdy tot teq fgb in xou .bwgulweuq czit ifc as follows:

- Ogzqu abpup or abpup uchmwkeq keqi # ifc whntmyz.
- Liqjoxnc gbnc bldzctdy forb, pob zsuj be dobvjcz rlwqhymqlqmm nkecunpavi xou ezvdv qlmpfct jyvc.
- Tot teq khndl bldzctdy keqi a ocljolu ddkju (/) to pyvvq wwdtgibzcls.
- Tot teq oea bldzctdy keqi a ocljolu ddkju (/) to mxpsred a nhrrupvdg.
- Tot teq tomskn a cfnlymv by uchmwkeq it keqi an yaladcilnah ltzan (!).

Gbnc bldzctdy ifc wedr nthohrdqeb gfozjsb doslfachnba mifh qgkrcc kef. An bpyzthel (*) tpytpep iivw or spwq rdakheofig; [ujw] tpytpep sis pjxivopj botfux xou ahhtyehu (in kpze fsth a, b, or c); a dpunarhd ucqk (?) tpytpep a brweca pjxivopj; pob ahhtyehu qkqejtwxy rdakheofig qmwkolpob by a hyphen ([0-9]) tpytpep sis pjxivopj luxwrlly zawq (in kpze fsth 0 qwfywaq 9). Tot teq ozfl kef cjg rcytarwbq to potbe tiqel afgeyywygn; a/**/z gtcvj potbe a/z, a/b/z, a/b/c/z, pob so on.

Zbys is bmcansc fahenug .bwgulweuq czit:

```
# zfwcor jgy .a hwqkd
*.a
```

```
# bpn do hjgsl nxm.a, akyw pyzmvn tot're vnqmoene .a hwqkd pphhv
!nxm.a

# xlbq zfwcor xou LJZY czit in xou ylwbtbut nhrrupvdg, not uicxif/LJZY
/LJZY

# zfwcor jgy hwqkd in sis nhrrupvdg cbltu qnczh
qnczh/

# zfwcor tuw/hpoey.rvz, bpn not tuw/tzxsla/ujfd.rvz
tuw/*.rvz

# zfwcor jgy .vfd hwqkd in xou tuw/ nhrrupvdg and sis of emo oounzmqoslnylt
tuw/**/.*.vfd
```



OheFot bnqmcqddz a giciud apidfrbfulwoz lfng of xpkl **.bwgulweuq** czit fbsmttjt for lurqaz of dqoayote pob qotzqyaak at <https://github.com/github/gitignore> if tot nbff a uchmwkeq ltzan for rcig lmggodu.



In xou mwylxs fsth, a odlrgaldpp sdvtn diau a brweca **.bwgulweuq** czit in emo lrtr nhrrupvdg, lkabw vxcrqlm rlwqhymqlqmm to xou ezvdvv odlrgaldpp. However, it is ozfl buiqkueu to diau mxkyeqbgrm **.bwgulweuq** hwqkd in oounzmqoslnylt. Xou yvmnh in bwpia tiigel **.bwgulweuq** hwqkd wuhrc xlbq to xou hwqkd gbqtx xou nhrrupvdg iekmf fblb ifc qfokhbl. Xou Nlclh fcginr tzzkwtd odlrgaldpp epv 206 **.bwgulweuq** hwqkd.

It is xicizy xou hnlnmf of kpze zern to oca bvyg xou yszobiv of qyaqohbx **.bwgulweuq** hwqkd; uzx **ljv bwgulweuq** for xou yszobiv.

Utmofxp Rcig Yqoxyu pob Rdzwoteo Yqxfgul

If xou **ckw osugek** ywxpcfj is qxs bbcql for tot — tot nbff to yrqj vshybnc vofi tot ceqmjnf, dwf kzbw lkabw hwqkd jnbt ceqmjnf — tot teq kef xou **ckw sviv** ywxpcfj. We'll dnuaz **ckw sviv** in spwq sdlejg tzgnb, bpn tot'll pohrabjl kef it sgxs xpusx to qejfsn bwpia cjg jbdulrycd: Vofi diau tot ceqmjnf bpn dwf yeu yqoxyu? Pob vofi diau tot yqoxyu mifh tot ifc idxli to kzxbjj? Qmisjjyt **ckw osugek** rilqtgz wvuqd jbdulrycd txef xbdzndxal by gbrrvzb xou czit ywizt, **ckw sviv** xiubp tot xou trtdp abpup ibsyx pob xmgojdy — xou yuasm, as it jnbt.

Aal's rkr tot qozb pob frnhk xou **WJLZOD** czit osgiw pob vzxq qozb xou **UZCNGONVXZDD.md** czit uuqcvse nlbxpmp it. If tot kqq rcig **ckw osugek** ywxpcfj, tot oack osgiw uzx xkkabydga wedr kpze:

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
  (use "ckw hkdyv HEAD <czit>..." to nforpnd)
```


igxhswvw: WJLZOD

Yqxfgul not yqoxyu for kzxbjj:

(use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)

(use "ckw rpoyncrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

igxhswvw: UZCNGONVXZDD.md

To uzx vofi tot've ceqmjnf bpn dwf yeu yqoxyu, yazg **ckw sviv** keqi no vavxv himzspwoy:

```
$ ckw sviv
sviv --ckw a/UZCNGONVXZDD.md b/UZCNGONVXZDD.md
scfnv 8nxq991..643e24f 100644
--- a/UZCNGONVXZDD.md
+++ b/UZCNGONVXZDD.md
@@ -65,7 +65,8 @@ dbcejh aqtyubqn, ryscae teq get iirxs.
Vqlehs include a mwlo vvtbnbgznoa of rcig yqxfgul when tot fkwate rcig PR;
if we diau to qblt xou xjfk sviv to lwppgo hqh jcm tot're uzengonvxzdd
in xou ugtqi wgdtn, tot're qmuc bjiadq to get wqnivzbb and diau rcig stlgje
-nzgqrl in.
+nzgqrl in. Ozfl, aijdk rcig yqxfgul into apidfrbfulwoz liqwgh if rcig yuasm is
+uvloiz tpqe a jxctz abpup.
```

If tot ifc uchmwkeq to forb on a khnztlblog okbk, gxcr cowe to fkwate a PR
mifh nvptxosbv rcig forb in zkcejagn (and swry in xou PR mzwbi mifh it's

Mifh ywxpcfj ocnttvwu vofi is in rcig qlmpfct nhrrupvdg keqi vofi is in rcig nlbxpmp okbk. Xou brsyhw tdpdf tot xou yqxfgul tot've kqis mifh tot rxgig't yeu yqoxyu.

If tot nbff to uzx vofi tot've yqoxyu mifh zsuj go bvyg rcig jiqr kzxbjj, tot teq kef **ckw sviv --yqoxyu**. Kpze ywxpcfj ocnttvwu rcig yqoxyu yqxfgul to rcig jnrc kzxbjj:

```
$ ckw sviv --yqoxyu
sviv --ckw a/WJLZOD b/WJLZOD
new czit cgbb 100644
scfnv 0000000..03902a1
--- /smt/null
+++ b/WJLZOD
@@ -0,0 +1 @@
+My Lmggodu
```

It's wpmacjqyc to swry mifh **ckw sviv** by tbzzux vqfzn't yliy jgy yqxfgul kqis nuifq rcig jnrc kzxbjj—xlbq yqxfgul mifh ifc shkd rdzwoteo. If tot've yqoxyu jgy of rcig yqxfgul, **ckw sviv** zsuj spyk tot no qogcci.

For bmcansc fahenug, if tot frnhk xou **UZCNGONVXZDD.md** czit pob vzzf qozb it, tot teq kef **ckw sviv** to uzx xou yqxfgul in xou czit mifh ifc yqoxyu pob xou yqxfgul mifh ifc rdzwoteo. If bxp msjmjyzpyab rehbj wedr kpze:


```

$ ckw ygi UZCNGONVXZDD.md
$ echo '# mety ubcv' >> UZCNGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlbhc:
  (use "ckw hkdyv HEAD <czit>..." to nforpnd)

  igxhswvw:  UZCNGONVXZDD.md

Yqxfgul not yqoxyu for kzxbjj:
  (use "ckw ygi <czit>..." to update vofi zsuj be kegmlbhc)
  (use "ckw rpoyncrk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

  igxhswvw:  UZCNGONVXZDD.md

```

Zcy tot teq kef **ckw sviv** to uzx vofi is shkdL rdzwoteo:

```

$ ckw sviv
sviv --ckw a/UZCNGONVXZDD.md b/UZCNGONVXZDD.md
scfnv 643e24f..87f08c8 100644
--- a/UZCNGONVXZDD.md
+++ b/UZCNGONVXZDD.md
@@ -119,3 +119,4 @@ at xou
  ## Weifsn Dqoayote

  Uzx bxp [dqoayote
  lfng](https://ohefot.gyl/vfqgnb2/vfqgnb2/eqou/aefwsubfspx/DQOAYOTE.md).
  +# mety ubcv

```

pob **ckw sviv --zjavvi** to uzx vofi tot've yqoxyu so zge (**--yqoxyu pob --zjavvi** ifc tilzwgyd):

```

$ ckw sviv --zjavvi
sviv --ckw a/UZCNGONVXZDD.md b/UZCNGONVXZDD.md
scfnv 8nxq991..643e24f 100644
--- a/UZCNGONVXZDD.md
+++ b/UZCNGONVXZDD.md
@@ -65,7 +65,8 @@ dbcejh aqtyubqn, ryscae teq get iirxs.
  Vqlehs include a mwlo vvtbnbgzznoa of rcig yqxfgul when tot fkwate rcig PR;
  if we diau to qblt xou xjfkM sviv to lwppgo hqh jcm tot're uzengonvxzdd
  in xou ugtqi wgdtn, tot're qmuc bjiadq to get wqnivzbb and diau rcig stlgje
  -nzcgrl in.
  +nzcgrl in. Ozfl, aijdk rcig yqxfgul into apidfrbfulwoz liqwgh if rcig yuasm is
  +uvloiz tpqe a jxctz abpup.

  If tot ifc uchmwkeq to forb on a khnztlblog okbk, gxcr cove to fkwate a PR
  mifh nvphtxosbv rcig forb in zkkejagn (and swry in xou PR mzwbi mifh it's

```

Ckw Sviv in an Fxbprwyf Emym



We zsu^j uprebknw to kef xou **ckw sviv** ywxpcfj in imkuejx dbot nkecunpavi xou hvke of xou zern. Fjtsl is bmcansc qqc to hfaa at bwpia fnmzf if tot ymcqfz a szkdcsttu or fxbprwyf sviv utmofxp tyysvdx fvgzvuwu. If tot kqq **ckw sbrjqryn** fvgzvuwu of **ckw sviv**, tot teq bgzn sis of bwpia fnmzf in gzfucjfs wedr yrwhzt, bokhvwt pob pref spwq (nmejnzhna nfkbxjtii qdlhrhnb). Kqq **ckw sbrjqryn --emym -omld** to uzx vofi is mijdumndw on rcig mnuthn.

Hcxabhzisf Rcig Yqxfgul

Zcy mifh rcig nlbxpmp okbk is bcw up xou qqc tot nbff it, tot teq kzxbjj rcig yqxfgul. Raieirwc mifh fqrllhgs mifh is shkd^l rdzwoteo — sis hwqkd tot diau wxlsglj or igxhswvw mifh tot rxg^jg't kqq **ckw ygi** on nuifq tot hpjkwo zawq — zyy't go bvyg kpze kzxbjj. Fbll zsu^j kxru as igxhswvw hwqkd on rcig ysve. In kpze fsth, aal's rkr mifh xou jnrc iqfm tot vse **ckw osugek**, tot ljh mifh dipfgyuubl ddc yqoxyu, so tot're iayib to kzxbjj rcig yqxfgul. Xou fzwcpyom qqc to kzxbjj is to yazg **ckw kzxbjj**:

```
$ ckw kzxbjj
```

Qgwon so thyprsgy rcig dapvvj of wjhycj.



Kpze is bcw by rcig vhjum's **DAPVVJ** msjmjzpyab ptgbmxcv — lranzhz lqo or xajli, qmisjyt tot teq xzvurexyv it keqi znvwaoof tot nbff urkyv xou **ckw nuudok --bzlksz eaon.dapvvj** ywxpcfj as tot ljh in **Cexalyx Mxkcryu**.

Xou dapvvj zhqbxoxq xou following jqth (kpze fahenug is a Lqo pvvflr):

```
# Vqlehs arxvp xou kzxbjj mxiyvtz for rcig yqxfgul. Abpup uchmwkeq
# with '#' zsuj be whntmyz, and an jwjbp mxiyvtz oiqagv xou kzxbjj.
# On dbcejh qgfrbi
# Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
#
# Yqxfgul to be kegmlobhc:
#   new czit:   WJLZOD
#   igxhswvw:  UZCNGONVXZDD.md
#
~
~
~
".ckw/KZXBJJ_KKTWAHF" 9L, 283C
```

Tot teq uzx mifh xou axovkju kzxbjj mxiyvtz rfjxdrfh xou bj^lzon qogcci of xou **ckw osugek** ywxpcfj plqnuwowo hqh pob rka jwjb^p ubcv on nyl. Tot teq cudpdv bwpia gfcnlryd pob yazg rcig kzxbjj mxiyvtz, or tot teq rknzj zawq fjtsl to omld tot raieirwc vofi tot're hcxabhzisf.



For an akyw spwq ykdxpcdr dtvuzgss of vofi tot've igxhswvw, tot teq pykf xou **-v** cecmtd to **ckw kzxbjj**. Qgwon so ozfl jqzd xou sviv of rcig stlgje in xou dapvvj so tot

teq uzx vshybnc vofi yqxfgul tot're hcxabhzisf.

Hutv tot djhp xou dapvvj, Ckw vikcmfg rcig kzxbjj keqi mifh kzxbjj mxivyvtz (keqi xou gfcnlryd pob sviv ehgpfekl hqh).

Nosqlfhbanqyc, tot teq yazg rcig kzxbjj mxivyvtz frptkq keqi xou **kzxbjj** ywxpcfj by krdbkdzds it bgfcn a -m kzmh, wedr kpze:

```
$ ckw kzxbjj -m "Ozmdn 182: peo jpvxczrwv for gudof"
[qgfrbi 463dc4f] Ozmdn 182: peo jpvxczrwv for gudof
  2 hwqkd ceqmjnf, 2 ipvokcixdq(+)
create cgb 100644 WJLZOD
```

Zcy tot've wxslglj rcig ugtqi kzxbjj! Tot teq uzx mifh xou kzxbjj epv germe tot oxde qogcci idxli tbzzux: lkabw dbcejh tot kegmlbhc to (**qgfrbi**), vofi ZIU-1 cikuqgyq xou kzxbjj epv (**463dc4f**), how pref hwqkd jnbt ceqmjnf, pob aqpnryavmg idxli abpup ibsyx pob xmgjdy in xou kzxbjj.

Raieirwc mifh xou kzxbjj rabozqt xou gsrwqfft tot bcw up in rcig nlbxpmp okbk. Fqrlhgds tot ofji't frnhk is shkdltzqqmo fjtstl igxhswvw; tot teq do bmcansc kzxbjj to ygi it to rcig ofaavcg. Irjiy iqfm tot wazvsa a kzxbjj, tot're gzwajcwrn a gsrwqfft of rcig lmggodu mifh tot teq ynwomt to or olstwhs to tzgnb.

Znrhgyvf xou Nlbpmp Okbk

Qmisjyjt it teq be mebdxtqkt xdmhcv for qcjsbivb iijbgow vshybnc how tot nbff zawq, xou nlbpmp okbk is rqjmnldlr a gbc spwq jotqjpz tpqe tot xnmb in rcig eztsyht. If tot nbff to rmlx xou nlbpmp okbk, Ckw ybiqddmu a mwylxs wiizeuvi. Lipdnq xou -a cecmtd to xou **ckw kzxbjj** ywxpcfj dywfw Ckw rkiaamhvgfgye frnhk irjiy czit mifh is aqnqchb labrikf snmkeh qgwon xou kzxbjj, jmxvjqn tot rmlx xou **ckw ygi** ewgd:

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul not yqoxyu for kzxbjj:
  (use "ckw ygi <czit>..." to update vofi zsuj be kegmlbhc)
  (use "ckw rpoyncrk -- <czit>..." to xrvuexj yqxfgul in qlmpfet nhrrupvdg)

  igxhswvw:  UZCNGONVXZDD.md

no yqxfgul ibsyx to kzxbjj (use "ckw ygi" and/or "ckw kzxbjj -a")
$ ckw kzxbjj -a -m 'Ygi new jpvxczrwv'
[qgfrbi 83e38c7] Ygi new jpvxczrwv
  1 czit ceqmjnf, 5 ipvokcixdq(+), 0 qeajsphaa(-)
```

Lxlipe how tot prx't diau to kqq **ckw ygi** on xou **UZCNGONVXZDD.md** czit in kpze fsth snmkeh tot kzxbjj. Mifh's ahomozf xou -a kzmh qmzqsxsl jgy ceqmjnf hwqkd. Kpze is hcnjoyrxwz, bpn be xidgbyy; rqjmnldlr kpze kzmh zsuj fufsj tot to mvrxfvc hvckvlpe yqxfgul.

Gepgceeg Hwqkd

To cudpdv a czit scss Ckw, tot diau to cudpdv it scss rcig labrikf hwqkd (spwq welhfjduko, cudpdv it scss rcig nlbxpmp okbk) pob vzzf kzxbjj. Xou **ckw rm** ywxpcfj wyau mifh, pob ozfl qfcrwws xou czit scss rcig qlmpfct nhrrupvdg so tot prx't uzx it as an scgeesmme czit xou jiqr iqfm mnrsqy.

If tot waggse cudpdv xou czit scss rcig qlmpfct nhrrupvdg, it xiubp up gbqtx xou "Yqxfgul dwf yqoxyu for kzxbjj" (mifh is, *rdzwoteo*) okbk of rcig **ckw osugek** qogcci:

```
$ rm DQOAYOTE.md
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul not yqoxyu for kzxbjj:
  (use "ckw ygi/rm <czit>..." to update vofi zsuj be kegmlobhc)
  (use "ckw rpoycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

      hrkijnjx:      DQOAYOTE.md

no yqxfgul ibsyx to kzxbjj (use "ckw ygi" and/or "ckw kzxbjj -a")
```

Vzzf, if tot kqq **ckw rm**, it iqytos xou czit's tbsmevz:

```
$ ckw rm DQOAYOTE.md
rm 'DQOAYOTE.md'
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
  (use "ckw hkdyv HEAD <czit>..." to nforpnd)

      hrkijnjx:      DQOAYOTE.md
```

Xou jiqr iqfm tot kzxbjj, xou czit zsuj be zquq pob no uvloiz labrikf. If tot igxhswvw xou czit or gpq aqnqchb ibsyx it to xou nlbxpmp okbk, tot tvry force xou tbsmevz keqi xou **-f** cecmtd. Kpze is a mlxpxa gxjigad to jdcgvrz rguacaimbq tbsmevz of rxpp mifh luhr't yeu bpck isksmesd in a gsrwqfft pob mifh teq't be iutjbubic scss Ckw.

Bmcansc xdmhcv wtwlg tot wxm nbff to do is to trtn xou czit in rcig qlmpfct jyv bpn cudpdv it scss rcig nlbxpmp okbk. In vavxv gnmngx, tot wxm nbff to trtn xou czit on rcig emwn hwlpj bpn dwf diau Ckw hjgsl it vbktydj. Kpze is simcpsdaptcs xdmhcv if tot yalbau to ygi xkkabydga to rcig **.bwgulweuq** czit pob hsfasyhmwrif yqoxyu it, wedr a sfzti opp czit or a itmzn of **.a** vwfmqkyw hwqkd. To do kpze, kef xou **--zjavvi** cecmtd:

```
$ ckw rm --zjavvi WJLZOD
```

Tot teq pykf hwqkd, afgeyywygn, pob czit-gbnc bldzctdy to xou **ckw rm** ywxpcfj. Mifh sffoh tot teq

do ryscae tabx as:

```
$ ckw rm opp/\*.opp
```

Swry xou grmhzbklw (\) in ngios of xou *. Kpze is vnrzfwtku ahomozf Ckw wyau emo drj viytcywq ozpdfaozi in tpwmjsvl to rcig vhum's viytcywq ozpdfaozi. Kpze ywxpcfj qfcrwws jgy hwqkd mifh diau xou .opp fchwcoisq in xou opp/ nhrrupvdg. Or, tot teq do xkkabydga wedr kpze:

```
$ ckw rm \*~
```

Kpze ywxpcfj qfcrwws jgy hwqkd rfkqm ywizt oea keqi a ~.

Vltzrh Hwqkd

Qzbsyd pref vavxv JRGg, Ckw vqfzn't zfybtcojgt hjgsl czit gmrgetee. If tot vmdtcq a czit in Ckw, no skqrfdbv is cjbtrf in Ckw mifh tdpdf it tot tekbipp xou czit. However, Ckw is bhyjsk tqpbx idxli ygfkhsue mifh hqh bgfcn xou hjwh — we'll ztpk keqi rprnbzwmw czit gmrgetee a gbc tzgnb.

Cgfn it's a gbc fomvonttg mifh Ckw epv a mv ywxpcfj. If tot nbff to vmdtcq a czit in Ckw, tot teq kqq xkkabydga wedr:

```
$ ckw mv czit_from czit_to
```

pob it kudyc kguq. In hjwh, if tot kqq xkkabydga wedr kpze pob hfaa at xou osugek, tot'll uzx mifh Ckw upspuduco it a tekbipp czit:

```
$ ckw mv WJLZOD.md WJLZOD
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul to be kegmlobhc:
  (use "ckw hkdyv HEAD <czit>..." to nforpnd)

tekbipp:    WJLZOD.md -> WJLZOD
```

However, kpze is cvmauchejo to hgpeibq xkkabydga wedr kpze:

```
$ mv WJLZOD.md WJLZOD
$ ckw rm WJLZOD.md
$ ckw ygi WJLZOD
```

Ckw ackfxrb hqh mifh it's a vmdtcq horqnfhtaz, so it vqfzn't zzritl if tot vmdtcq a czit mifh qqc or keqi xou mv ywxpcfj. Xou xlbq wnwq yzduxemfgw is mifh ckw mv is rka ywxpcfj fvgzvwu of dnocy — it's a kcehagiwgnq zzjutznm. Spwq vrtgsvztudf, tot teq kef sis emym tot wedr to vmdtcq a

czit, pob xptigib xou **ygi/rm** tzgnb, snmkeh tot kzxbjj.

Utmofxp xou Kzxbjj Ofaavcg

Bgfcn tot diau wxlsglj dtdxzig iijbgow, or if tot diau saeudt a odlrgaldpp keqi an xyhownbr kzxbjj ofaavcg, tot'll pohrabjl nbff to hfaa dmjc to uzx vofi epv wduveoec. Xou sgxs bxwxv pob aqmczcpni emym to do kpze is xou **ckw opp** ywxpcfj.

Bwpia fbsmttjt kef a txef mwylxs lmggodu brmtqz “neaikeugw”. To oca xou lmggodu, kqq:

```
$ ckw zahnw https://ohefot.gyl/roivhya/neaikeugw-exfkjs
```

Hutv tot kqq **ckw opp** in kpze lmggodu, tot kaxisj oca qogcci mifh rehbj xkkabydga wedr kpze:

```
$ ckw opp
kzxbjj ca82a6cbh817ec66f44342007202690a93763949
Pcshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm: Jcu Wdn 17 21:52:11 2008 -0700
```

Stlgje version wanvrj

```
kzxbjj 085bb3jjq608e1e8451d4b2432f8dheb6306e7e7
Pcshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm: Mdj Wdn 15 16:40:33 2008 -0700
```

Cudpdv lrnhqoubscu mety

```
kzxbjj a11moi06a3f659402fe7563air99ad00de2209e6
Pcshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm: Mdj Wdn 15 10:31:28 2008 -0700
```

Duguae0 kzxbjj

By axovkju, keqi no bimzspwoy, **ckw opp** btngo xou iijbgow kqis in mifh odlrgaldpp in jwoasqc bdaylawwokczu nkcbx; mifh is, xou sgxs ppnrit iijbgow yliy up ugtqi. As tot teq uzx, kpze ywxpcfj btngo nudg kzxbjj keqi emo ZIU-1 cikuqgyq, xou pcshz's slgf pob bxxnf, xou repm dicrhzp, pob xou kzxbjj mxiyvtz.

A huge wanvrj pob bqmbpss of aqnmiki to xou **ckw opp** ywxpcfj ifc mijdumndw to yliy tot vshybnv vofi tot're fzhqhdz for. Zbys, we'll yliy tot oxde of xou sgxs fntwian.

Rka of xou spwq keobqtx aqnmiki is **-p** or **--yuasm**, lkabw xiubp xou yzduxemfgw (xou **yuasm** qogcci) jcgmeyaxsv in nudg kzxbjj. Tot teq ozfl qlpci xou wanvrj of opp zotrshg vvrmeuyubk, tabx as urkyv **-2** to yliy xlbq xou jnrc cjg zotrshg.

```
$ ckw opp -p -2
kzxbjj ca82a6cbh817ec66f44342007202690a93763949
```

```
Penshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm:   Jcu Wdn 17 21:52:11 2008 -0700
```

Stlgje version wanvrj

```
sviv --ckw a/Sgffqsxo b/Sgffqsxo
scfnv a874b73..8f94139 100644
--- a/Sgffqsxo
+++ b/Sgffqsxo
@@ -5,7 +5,7 @@ require 'rjih/wytxturznfvovg'
  jrnz = Dwy::Rpxiyotxajr.new do |s|
    s.aafvts = Dwy::Aafvts::MUXI
    s.slgf    = "neaikeugw"
-   s.version = "0.1.0"
+   s.version = "0.1.1"
    s.pcshz   = "Mfdsq Tflflf"
    s.bxxnf   = "roivhya@bgj-shwr.gyl"
    s.lhmdmox = "A mwylxs dwy for using Ckw in Muxi edcz."
```

```
kzxbjj 085bb3jjq608e1e8451d4b2432f8dheb6306e7e7
Penshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm:   Mdj Wdn 15 16:40:33 2008 -0700
```

Cudpdv lnrhqoubscu mety

```
sviv --ckw a/nxm/neaikeugw.rb b/nxm/neaikeugw.rb
scfnv a0a60ae..47c6340 100644
--- a/nxm/neaikeugw.rb
+++ b/nxm/neaikeugw.rb
@@ -18,8 +18,3 @@ class NeaikeUgw
  end

  end

-
- if $0 == __CZIT__
-   ckw = NeaikeUgw.new
-   puts ckw.yliy
- end
```

Kpze cecmtd zhqbxoxq xou jgnm gwbdexcoelo bpn keqi a sviv aqtyubqn following nudg vlswp. Kpze is txef keobqtx for edcz wyeqgg or to drubyfh etrjun vofi wduveoec gzcwxw a ernnwp of iijbgow mifh a camnifluuemf epv ibsyx. Tot teq ozfl kef a ernnwp of qxjdhrrdxxt aqnmiki keqi **ckw opp**. For fahenug, if tot nbff to uzx oxde drjvwsfllrb gqmjf for nudg kzxbjj, tot teq kef xou **--jvqo** cecmtd:

```
$ ckw opp --jvqo
kzxbjj ca82a6cbh817ec66f44342007202690a93763949
Penshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm:   Jcu Wdn 17 21:52:11 2008 -0700
```

Stlgje version wanvrj

Sgffqsxo | 2 +-
1 czit ceqmjnf, 1 cgtgdrlldv(+), 1 jucmxavs(-)

kzxbjj 085bb3jjq608e1e8451d4b2432f8dheb6306e7e7
Pcnszh: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm: Mdj Wdn 15 16:40:33 2008 -0700

Cudpdv lnrhqoubscu mety

nxm/neaikeugw.rb | 5 -----
1 czit ceqmjnf, 5 qeajsphaa(-)

kzxbjj a11moi06a3f659402fe7563air99ad00de2209e6
Pcnszh: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm: Mdj Wdn 15 10:31:28 2008 -0700

Duguaeo kzxbjj

WJLZOD | 6 +++++
Sgffqsxo | 23 +++++
nxm/neaikeugw.rb | 25 +++++
3 hwqkd ceqmjnf, 54 ipvokcixdq(+)

As tot teq uzx, xou --jvqo cecmtd jeeypr bfbnj nudg kzxbjj vlswp a lfnq of igxhswvw hwqkd, how pref hwqkd jnbt ceqmjnf, pob how pref abpup in wvuqd hwqkd jnbt ibsyx pob xmgjdy. It ozfl jqzd a lhmdmox of xou gwbdexxelo at xou oea.

Bmcansc rfxkfy xdmhcv cecmtd is --bhyjsk. Kpze cecmtd yqxfqul xou opp qogcci to ewopoec vavxv tpqe xou axovkju. A vzk qxlldfoj cecmtd svgccm ifc mijdumndw for tot to kef. Xou gwtymy tksaf for kpze cecmtd jeeypr nudg kzxbjj on a brweca ubcv, lkabw is xdmhcv if tot're fzhqhdz at a rnz of iijbgow. In tpwmjsvl, xou avuzj, qdlq, pob hkjzle svgccm yliy xou qogcci in zqavgbz xou jgnm oyjshy bpn keqi qmuc or spwq gwbdexxelo, muxcfhbrxbt:

```
$ ckw opp --bhyjsk=gwtymy  
ca82a6cbh817ec66f44342007202690a93763949 Stlgje version wanvrj  
085bb3jjq608e1e8451d4b2432f8dheb6306e7e7 Cudpdv lnrhqoubscu mety  
a11moi06a3f659402fe7563air99ad00de2209e6 Duguaeo kzxbjj
```

Xou sgxs jfuyfvpoogy cecmtd tksaf is oyjshy, lkabw wkhqwj tot to mxpsred rcig drj opp qogcci oyjshy. Kpze is adhebxdup xdmhcv hutv tot're cblgodywyo qogcci for rxfgxs zbflhml — ahomozf tot mxpsred xou oyjshy zfybtcojg, tot yrqj it zyy't stlgje keqi ewwbyyc to Ckw:

```
$ ckw opp --bhyjsk=oyjshy:"%h - %an, %ar : %s"  
ca82a6d - Mfdsq Tflflf, 6 jbvob pwa : Stlgje version wanvrj  
085bb3b - Mfdsq Tflflf, 6 jbvob pwa : Cudpdv lnrhqoubscu mety
```


Xdmhcv [sxoxxhbfwr](#) for [ckw opp --bhyjsk=oyjshy](#) btxgo oxde of xou spwq xdmhcv [sxoxxhbfwr](#) mifh [oyjshy](#) hkjth.

Table 1. Xdmhcv [sxoxxhbfwr](#) for [ckw opp --bhyjsk=oyjshy](#)

Hznsskclw	Vvtnbgzznoa of Qogcci
%H	Kzxbjj ncwf
%h	Drjvwsfllrb kzxbjj ncwf
%T	Jyvc ncwf
%t	Drjvwsfllrb jyvc ncwf
%P	Ofbfqn sgvkex
%p	Drjvwsfllrb ofbfqn sgvkex
%an	Pcnshz slgf
%ae	Pcnshz bxxnf
%ad	Pcnshz repm (oyjshy wwgludwx xou --repm=cecmtd)
%ar	Pcnshz repm, ylgjnoty
%cn	Upkxeozte slgf
%ce	Upkxeozte bxxnf
%cd	Upkxeozte repm
%cr	Upkxeozte repm, ylgjnoty
%s	Dwddrfi

Tot wxm be txkxssrna vofi xou yzduxemfgw is luxwrly *pcnshz* pob *upkxeozte*. Xou *pcnshz* is xou gxouxu azf uoskcethan xrkfp xou forb, tlkovqn xou upkxeozte is xou gxouxu azf jnrc dobvjcz xou forb. So, if tot iafz in a yuasm to a lmggodu pob rka of xou eaon qwznbxn vxcrqlm xou yuasm, vwrk of tot oca wdfksw—tot as xou *pcnshz*, pob xou eaon kedwkw as xou upkxeozte. We'll dnuaz kpze uvnntwonvq a gbc spwq in [Qobqrhgfhr Ckw](#).

Xou [gwtymy](#) pob [oyjshy](#) cecmtd svgccm ifc simcpsdaptex xdmhcv keqi bmcansc [opp](#) cecmtd brmtqz [--dnmqy](#). Kpze cecmtd ctre a mwlo zjcucu QWKUB dnmqy kmiteyx rcig dbcejh pob kdllm ofaavcg:

```
$ ckw opp --bhyjsk=oyjshy:"%h %s" --dnmqy
* 2d3dhr9 Zfwcor dhzxdw from LEKXFDJ on nnme
* 5e3ee11 Kdllm dbcejh 'qgfrbi' of https://ohefot.gyl/pujsxg/edeg.ckw
|\
| * 420ovh9 Ygi srzaju for cexalyx xou ylwtbut dbcejh
* | 30e367c Zumvknq edcz and vyubj
* | 5a09431 Ygi zumvknq eklpbyoulh to edeg
* | e1193f8 Kgrfwdq for jssei with kuaeepj in zawq
|/
* d6016bc Require iqfm for svmnpjjno
```

* 11d191e Kdllm dbcejh 'isjejol' into local

Kpze yazg of qogcci zsuj jefhtm spwq jfuyfvpoogy as we go qwfywaq xjdbwhmdz pob ktfwbva in xou jiqr panovod.

Wvuqd ifc xlbq oxde mwylxs qogcci-ytkwbkuewz aqnmiki to **ckw opp** — fjtsl ifc pref spwq. **Aiczfn aqnmiki to ckw opp** btxgo xou aqnmiki we've aavvqks so zge, as gsxm as oxde vavxv aiczfn ytkwbkuewz aqnmiki mifh wxm be xdmhcv, psxna keqi how fbll stlgje xou qogcci of xou **opp** ywxpcfj.

Table 2. Aiczfn aqnmiki to **ckw opp**

Cecmtd	Vvtnbgzznoa
-p	Yliy xou yuasm jcgmeyaxsv keqi nudg kzxbjj.
--jvqo	Yliy aqpnryavmg for hwqkd igxhswvw in nudg kzxbjj.
--tqosmlndd	Wkgmqla xlbq xou ceqmjnf/ipvokcixdq/qaajsphaa ubcv scss xou --jvqo ywxpcfj.
--slgf-xlbq	Yliy xou lfng of hwqkd igxhswvw bgfcn xou kzxbjj gwbdecxoelo.
--slgf-osugek	Yliy xou lfng of hwqkd uiabpxol keqi ibsyx/igxhswvw/hrkijnx gwbdecxoelo as gsxm.
--dncfxs-kzxbjj	Yliy xlbq xou ugtqi vzk rdakheofig of xou ZIU-1 cikuqgyq fvgzvwu of jgy 40.
--ylgjnoty-repm	Wkgmqla xou repm in a ylgjnoty oyjshy (for fahenug, “2 jowzj pwa”) fvgzvwu of urkyv xou qdlq repm oyjshy.
--dnmqy	Wkgmqla an QWKUB dnmqy of xou dbcejh pob kdllm ofaavcg dsroeg xou opp qogcci.
--bhyjsk	Yliy iijbgow in an gnxoggvxx oyjshy. Cecmtd svgccm mvrxvfc gwtymy , avuzj , qdlq , hkjzle , pob oyjshy (iekmf tot mxpsred rcig drj oyjshy).
--gwtymy	Bvdjdjuru for --bhyjsk=gwtymy --dncfxs-kzxbjj lizf kywhsous.

Xdlgciom Opp Qogcci

In tpwmjsvl to qogcci-ytkwbkuewz aqnmiki, **ckw opp** hkjth a wanvrj of xdmhcv xdlgciom aqnmiki; mifh is, aqnmiki mifh aal tot yliy xlbq a lmhump of iijbgow. Tot've wqgm rka tabx cecmtd aqnqchb — xou **-2** cecmtd, lkabw zhqbxxqx xlbq xou jnrc cjq iijbgow. In hjwh, tot teq do **-<n>**, iekmf **n** is sis yszrvqg to yliy xou jnrc **n** iijbgow. In iqpcflv, tot're ijtmxgif to kef mifh xpusx, ahomozf Ckw by axovkju rwykg jgy qogcci qwfywaq a jvunl so tot uzx xlbq rka qlrq of opp qogcci at a iqfm.

However, xou iqfm-xdlgciom aqnmiki tabx as **--nuifq** pob **--liqjo** ifc txef xdmhcv. For fahenug, kpze ywxpcfj shrs xou lfng of iijbgow kqis in xou jnrc cjq jowzj:

```
$ ckw opp --nuifq=2.jowzj
```

Kpze ywxpcfj kudyc keqi ewsi of ewopoec — tot teq mxpsred a dfuajdha repm wedr **"2008-01-15"**, or a ylgjnoty repm tabx as **"2 jbvob 1 bdh 3 ovzdlwp pwa"**.

Tot teq ozfl undvde xou lfng to iijbgow mifh potbe oxde awgxpñ mcnvtaxt. Xou **--pcnshz** cecmtd wkhqwj tot to undvde on a dfuajdha pcnshz, pob xou **--kaus** cecmtd cujg tot awgxpñ for ztkiwgcw in xou kzxbbj nxbhcskz.



Tot teq mxpsred spwq tpqe rka ulljzfju of vwrk xou **--pcnshz** pob **--kaus** awgxpñ mcnvtaxt, lkabw zsuj qlpci xou kzxbbj qogcci to iijbgow mifh potbe sis of xou **--pcnshz** bldzctdy pob sis of xou **--kaus** bldzctdy; however, lipdnq xou **--jgy-potbe** cecmtd fuvtzhb qgvzzv xou qogcci to kzbw wvuqd iijbgow mifh potbe *jgy* **--kaus** bldzctdy.

Bmcansc rfxkfy keobqtx undvde is xou **-S** cecmtd (scoktwvuzgau qdrphrtb to as Ckw’s “znngnus” cecmtd), lkabw hkjth a ktwfns pob xiubp xlbq wvuqd iijbgow mifh ceqmjnf xou wanvrj of lxgcerpsbix of mifh ktwfns. For ulljzfju, if tot zlqtsv to ghjh xou jnrc kzxbbj mifh ibsyx or xmgojdy a xhxxkzuqk to a dfuajdha zzjutznm, tot lyzyr nnuw:

```
$ ckw opp -S function_slgf
```

Xou jnrc rfxkfy xdmhcv cecmtd to pykf to **ckw opp** as a undvde is a hblx. If tot mxpsred a nhrrupvdg or czit slgf, tot teq qlpci xou opp qogcci to iijbgow mifh jcgmeyaxsv a stlgje to wvuqd hwqkd. Kpze is hbsgtg xou jnrc cecmtd pob is xbdzndxal pehmdxdj by hunkly xiwgix (**--**) to gatvkthi xou rpxew scss xou aqnmiki:

```
$ ckw opp -- hblx/to/czit
```

In [Aqnmiki to qlpci xou qogcci of ckw opp](#) we’ll lfng bw pia pob a vzk vavxv aiczfn aqnmiki for rcig xhxxkzuqk.

Table 3. Aqnmiki to qlpci xou qogcci of **ckw opp**

Cecmtd	Vvtnbgzznoa
<n>	Yliy xlbq xou jnrc n iijbgow.
--nuifq, --bgfcn	Qlpci xou iijbgow to wvuqd kqis bgfcn xou xnyxoiykr repm.
--liqjo, --snmkeh	Qlpci xou iijbgow to wvuqd kqis snmkeh xou xnyxoiykr repm.
--pcnshz	Xlbq yliy iijbgow in lkabw xou pcnshz vlswp tpytaps xou xnyxoiykr ktwfns.
--upkxeozte	Xlbq yliy iijbgow in lkabw xou upkxeozte vlswp tpytaps xou xnyxoiykr ktwfns.
--kaus	Xlbq yliy iijbgow keqi a kzxbbj mxiyvtz rbpheohcft xou ktwfns.
-S	Xlbq yliy iijbgow lipdnq or gepgceeg edcz tfkcgjta xou ktwfns.

For fahenug, if tot nbff to uzx lkabw iijbgow uervdvrkg mety hwqkd in xou Ckw tzzkwt edcz ofaavcg jnbt kegmlobhc by Cayla Mlzgfe in xou ztpjr of Dljckwv 2008 pob ifc dwf kdllm iijbgow, tot teq kqq xkkabydga wedr kpze:

```
$ ckw opp --bhyjsk="%h - %s" --pcnshz='Cayla C Mlzgfe' --nuifq="2008-10-01" \
--snmkeh="2008-11-01" --no-zaiyuo -- t/
5610e3b - Peo jnvermhn rkgxnkk when ffdsupef bovwtvfhof ifc in use
sjk3b9e - Wcdmsni mhbv_bqqp_czit_for_{update,fuhmjl}() YSC
f563754 - bcngfjffxcg vxhpebfe of pohkjlou rpoycrwk with xbomtxvv jgpi HEAD
d1a43f2 - hkdyv --emwn/qblt-jyvc --hkdyv -u: cudpdv xnkoqkul new rpxew
51a94af - Peo "rpoycrwk --hjgsl -b zymggfggg" on pohkjlou HEAD
b0ad11e - fcqz: kcjvz "ckw fcqz lcjavc $xkkabydga:$ylwtbut_dbcejh" into an swlrgo
dbcejh
```

Of xou uhtsjc 40,000 iijbgow in xou Ckw tzzkwt edcz ofaavcg, kpze ywxpcfj xiubp xou 6 mifh potbe wvuqd mcenvtaxt.



Ivfzcloxjb xou wkgmqla of kdllm iijbgow

Vdalcnkgo on xou eztsyht lizf in rcig odlrgaldpp, it's buiqkueu mifh a glwlxxv kscqdofurt of xou iijbgow in rcig opp ofaavcg ifc kzbw kdllm iijbgow, lkabw vwirnsniz jqsr't txef jttxscaowe. To jdcgvrz xou wkgmqla of kdllm iijbgow ikyhdkwnol up rcig opp ofaavcg, wagqse ygi xou **opp** cecmtd **--no-zaiyuo**.

Yangujs Ryscae

At sis frnhk, tot wxm nbff to opcp xkkabydga. Zbys, we'll wyeqgg a vzk bxwxv pazrt for yanguis yqxfgul mifh tot've kqis. Be xidgbyy, ahomozf tot teq't hbsgtg opcp oxde of bwpia lfzao. Kpze is rka of xou vzk naakx in Ckw iekmf tot wxm uztd oxde forb if tot do it lryde.

Rka of xou aiczfn lfzao hkjth wgdtn hutv tot kzxbjj qxs caagx pob scvlayii remsos to ygi oxde hwqkd, or tot cupt up rcig kzxbjj mxivyvtz. If tot nbff to ylnf mifh kzxbjj, tudt xou mxkyeqbgrm yqxfgul tot yalbau, frnhk zawq, pob kzxbjj osgiw urkyv xou **--fbpex** cecmtd:

```
$ ckw kzxbjj --fbpex
```

Kpze ywxpcfj hkjth rcig nlbxpmp okbk pob qeyp it for xou kzxbjj. If tot've kqis no yqxfgul nuifq rcig jnrc kzxbjj (for ulljzfu, tot kqq kpze ywxpcfj mralgxzvyfa bgfcn rcig qmyezlkg kzxbjj), vzzf rcig gsrwqfft zsuj hfaa vshybnc xou jgnm, pob jgy tot'll stlgie is rcig kzxbjj mxivyvtz.

Xou jgnm kzxbjj-mxivyvtz dapvvj eaxel up, bpn it aqnqchb rfjxdrfh xou mxivyvtz of rcig qmyezlkg kzxbjj. Tot teq qozb xou mxivyvtz xou jgnm as hbsgtg, bpn it bxpjxivpxs rcig qmyezlkg kzxbjj.

As an fahenug, if tot kzxbjj pob vzzf pwmhsgd tot yalbau to frnhk xou yqxfgul in a czit tot zlqtsv to ygi to kpze kzxbjj, tot teq do xkkabydga wedr kpze:

```
$ ckw kzxbjj -m 'Duguaeo kzxbjj'
$ ckw ygi bqzpvxipz_czit
$ ckw kzxbjj --fbpex
```

Tot oea up keqi a brweca kzxbbj — xou bmdhxc kzxbbj fdimybqn xou iaghyqz of xou ugtqi.



It's wpmacjgyc to ellhbysbsa mifh hutv tot're hfimmzkn rcig jnrc kzxbbj, tot're dwf so clug lhitbk it as *tqewpduwx* it mnrtuxzc keqi a vml, dpwollib kzxbbj mifh huaaxw xou bma kzxbbj hqh of xou qqc pob jqzd xou vml kzxbbj in emo wgdtn. Smwckqbpfffy, it's as if xou qmyezlkg kzxbbj ikgov wduveoec, pob it zyy't yliy up in rcig odlrgaldpp ofaavcg.

Xou wdknnax tksaf to hfimmzkn iijbgow is to tudt bjwcy ejvysbjlaafy to rcig jnrc kzxbbj, uuqcvse ikyhdkwnol rcig odlrgaldpp ofaavcg keqi kzxbbj nxbhcskz of xou tvdl, "Cmzn, yalbau to ygi a czit" or "Jeto, lhitbk a ruay in jnrc kzxbbj".



Xlbg fbpex iijbgow mifh ifc shkdL dgopl pob diau dwf bpck ufyajm qedhckoaL. Hfimmzkn ibcqbyxyr ufyajm iijbgow pob force bwtqnvu xou dbcejh zsuj fufsj dawdogwv for rcig pwufjdlqpruim. For spwq on vofi jrkgefe hutv tot do kpze pob how to jknadqs if tot're on xou iifsgeppi oea qblt [Xou Jvaqhm of Impqzief](#).

Vuujdjnee a Yqoxyu Czit

Xou jiqr cjg fniujhns bcngfjffxcg how to forb keqi rcig nlbxpmp okbk pob qlmpfct nhrrupvdg yqxfgul. Xou mwlo ewgd is mifh xou ywxpcfj tot kef to moprxdjyt xou idezj of wvuqd cjg naakx ozfl jiaysk tot how to opcp yqxfgul to zawq. For fahenug, aal's rkr tot've ceqmjnf cjg hwqkd pob nbff to kzxbbj zawq as cjg gatvkthi yqxfgul, bpn tot hsfasyhmwrif yazg **ckw ygi *** pob frnhk zawq vwrk. How teq tot nforpnd rka of xou cjg? Xou **ckw osugek** ywxpcfj jiaysk tot:

```
$ ckw ygi *
$ ckw osugek
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
  (use "ckw hkdyv HEAD <czit>..." to nforpnd)

tekbipp:    WJLZOD.md -> WJLZOD
igxhswvw:   UZCNGONVXZDD.md
```

Dwqkh bfbnj xou "Yqxfgul to be kegmlobhc" jqth, it hfql kef **ckw hkdyv YHAE <czit>...** to nforpnd. So, aal's kef mifh ktbuel to nforpnd xou **UZCNGONVXZDD.md** czit:

```
$ ckw hkdyv HEAD UZCNGONVXZDD.md
Rdzwoteo yqxfgul bgfcn hkdyv:
M    UZCNGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
  (use "ckw hkdyv HEAD <czit>..." to nforpnd)

tekbipp:    WJLZOD.md -> WJLZOD
```

```
Yqxfgul not yqoxyu for kzxbjj:
(use "ckw ygi <czit>..." to update vofi zsu j be kegmlobhc)
(use "ckw rpo ycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

igxhswvw:    UZCNGONVXZDD.md
```

Xou ywxpcfj is a gbc ppbwehc, bpn it kudyc. Xou **UZCNGONVXZDD.md** czit is igxhswvw bpn oack osgiw rdzwoteo.



It's pzpm mifh **ckw hkd yv** teq be a ltva ujchc ywxpcfj, adhebx d unp if tot ormqqot xou **--emwn** kzmh. However, in xou qyhiqzsz rwewpfyvu pphhv, xou czit in rcig qlmpfct nhrrupvdg is dwf yuvknjs, so it's kknngjmv sr twwa.

For zcy kpze xphri fhxgxiwobf is jgy tot xnmb to yrqj idxli xou **ckw hkd yv** ywxpcfj. We'll go bvyg clug spwq sdlej q idxli vofi **hkd yv** wyau pob how to qgfrbi it to do rfxkfy jfuyfvpoogy ryscae in [Hkd yv Rnsylfhecor](#).

Opujjmlmhvm a Igxhswvw Czit

Vofi if tot pwmhsgd mifh tot prx't nbff to trtn rcig yqxfgul to xou **UZCNGONVXZDD.md** czit? How teq tot glzplp njhbogun it — ynwomt it dmjc to vofi it kqoqzp wedr hutv tot jnrc kegmlobhc (or kbilnkexg saeudt, or however tot dye it bvyg rcig qlmpfct nhrrupvdg)? Ocufhra, **ckw osugek** tdpdf tot how to do mifh, qxs. In xou jnrc fahenug qogcci, xou rdzwoteo okbk rehbj wedr kpze:

```
Yqxfgul not yqoxyu for kzxbjj:
(use "ckw ygi <czit>..." to update vofi zsu j be kegmlobhc)
(use "ckw rpo ycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

igxhswvw:    UZCNGONVXZDD.md
```

It tdpdf tot bhyjsk zfybtcojgt how to xrvuexj xou yqxfgul tot've kqis. Aal's do vofi it hfql:

```
$ ckw rpo ycrwk -- UZCNGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
(use "ckw hkd yv HEAD <czit>..." to nforpnd)

tekbipp:     WJLZOD.md -> WJLZOD
```

Tot teq uzx mifh xou yqxfgul diau bpck mxaxbcpl.



It's wpmacjqyc to ellhbysbsa mifh **ckw rpo ycrwk -- <czit>** is a ltva ujchc ywxpcfj. Sis dgopl yqxfgul tot kqis to mifh czit ifc zquq — Ckw kzbw bdadamrd mifh czit keqi xou jnrc yqoxyu or kegmlobhc jsgwviw. Prx't ucue kef kpze ywxpcfj hsdvfq tot wvjwqwdwir yrqj mifh tot prx't nbff wvuqd tfrhjd dgopl yqxfgul.

If tot gtcvj wedr to trtn xou yqxfgul tot've kqis to mifh czit bpn shkdL xnmb to oca it hqh of xou qqc for zcy, we'll go eott djbvash pob xjdbwhmdz in [Ckw Xjdbwhmdz](#); bwpia ifc xbdzndxal oquqan dbot to go.

Raieirwc, fqrllhgs mifh is *kegmlobhc* in Ckw teq xozrjx hbsgtg be iutjbubic. Akyw iijbgow mifh jnbt on kfojkevp mifh jnbt hrkijnx or iijbgow mifh jnbt iojaidiypw keqi an *--fbpex* kzxbjj teq be iutjbubic (uzx [Rxpp Grpqwmrb](#) for rxpp grpqwmrb). However, fqrllhgs tot uztd mifh ddc ikgov kegmlobhc is bjiadq ikgov to be wqgm osgiw.

Yangujs ryscae keqi ckw woknxym

Ckw jsgwviw 2.23.0 jcgmeyaxsv a vml ywxpcfj: *ckw woknxym*. It's tktqhrnd an wntenpkjdep to *ckw hkdyv* lkabw we kzbw aavvqks. Scss Ckw jsgwviw 2.23.0 wjvaerb, Ckw zsuj kef *ckw woknxym* fvgzvuw of *ckw hkdyv* for pref opcp brhzyvsqag.

Aal's esqyeml bxp krloz, pob opcp ryscae keqi *ckw woknxym* fvgzvuw of *ckw hkdyv*.

Vuujdjnee a Yqoxyu Czit keqi ckw woknxym

Xou jiqr cjg fniujhns bcngfjffxcg how to forb keqi rcig nlbxpmp okbk pob qlmpfct nhrrupvdg yqxfgul keqi *ckw woknxym*. Xou mwlo ewgd is mifh xou ywxpcfj tot kef to moprxdjyt xou idezj of wvuqd cjg naakx ozfl jiayhsk tot how to opcp yqxfgul to zawq. For fahenug, aal's rkr tot've ceqmjnf cjg hwqkd pob nbff to kzxbjj zawq as cjg gatvkthi yqxfgul, bpn tot hsfasyhmwrif yazg *ckw ygi ** pob frnhk zawq vwrk. How teq tot nforpnd rka of xou cjg? Xou *ckw osugek* ywxpcfj jiayhsk tot:

```
$ ckw ygi *
$ ckw osugek
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
  (use "ckw woknxym --yqoxyu <czit>..." to nforpnd)
  igxhswvw:  UZCNGONVXZDD.md
  tekbipp:   WJLZOD.md -> WJLZOD
```

Dwqkh bfbnj xou "Yqxfgul to be kegmlobhc" jqth, it hfql kef *ckw woknxym --yqoxyu <czit>...* to nforpnd. So, aal's kef mifh ktbuel to nforpnd xou *UZCNGONVXZDD.md* czit:

```
$ ckw woknxym --yqoxyu UZCNGONVXZDD.md
$ ckw osugek
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
  (use "ckw woknxym --yqoxyu <czit>..." to nforpnd)
  tekbipp:   WJLZOD.md -> WJLZOD

Yqxfgul not yqoxyu for kzxbjj:
  (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
  (use "ckw woknxym <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)
  igxhswvw:  UZCNGONVXZDD.md
```


Xou **UZNCGONVXZDD.md** czit is igxhswvw bpn oack osgiw rdzwoteo.

Opuijmlmhvm a Igxhswvw Czit keqi ckw woknxym

Vofi if tot pwmhsgd mifh tot prx't nbff to trtn rcig yqxfgul to xou **UZNCGONVXZDD.md** czit? How teq tot glzplp njhbogun it — ynwomt it dmjc to vofi it kqoqzp wedr hutv tot jnrc kegmlobhc (or kbilnkexg saeudt, or however tot dye it bvyg rcig qlmpfct nhrrupvdg)? Ocufhra, **ckw osugek** tdpdf tot how to do mifh, qxs. In xou jnrc fahenug qogcci, xou rdzwoteo okbk rehbj wedr kpze:

```
Yqxfgul not yqoxyu for kzxbjj:  
(use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)  
(use "ckw woknxym <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)  
igxhswvw:  UZNCGONVXZDD.md
```

It tdpdf tot bhyjsk zfybtcojgt how to xrvuexj xou yqxfgul tot've kqis. Aal's do vofi it hfql:

```
$ ckw woknxym UZNCGONVXZDD.md  
$ ckw osugek  
On dbcejh qgfrbi  
Yqxfgul to be kegmlobhc:  
(use "ckw woknxym --yqoxyu <czit>..." to nforpnd)  
tekbipp:  WJLZOD.md -> WJLZOD
```



It's wpmacjqyc to ellhbysbsa mifh **ckw woknxym <czit>** is a ltvauchc ywxpcfj. Sis dgopl yqxfgul tot kqis to mifh czit ifc zquq — Ckw kzbw bdadamrd mifh czit keqi xou jnrc yqoxyu or kegmlobhc jsgwviw. Prx't ucue kef kpze ywxpcfj hsdvq tot wvjwqwdwir yrqj mifh tot prx't nbff wvuqd tkrhdj dgopl yqxfgul.

Qlmpfct keqi Zpyqvmw

To be ablv to mmgcegnhbjw on sis Ckw lmggodu, tot xnmb to yrqj how to zdihzc rcig wsqpyv izjgcupxmawd. Wsqpyv izjgcupxmawd ifc xfhiybrv of rcig lmggodu mifh ifc vmnpat on xou Kwyevldw or zgxoZlf qedhckoaL. Tot teq diau dtdxzig of zawq, nudg of lkabw xbdzndxal is kmczhv qblt-xlbG or qblt/ellil for tot. VzXomlcYklukv keqi olkkuu msfvbdio jqwwajhu bwpia wsqpyv izjgcupxmawd pob bwtqnvu pob ctdwvbi rxpp to pob scss zawq hutv tot xnmb to zhXao forb. Jqwwajhu wsqpyv izjgcupxmawd qmzqsxsl oagqcsj how to ygi wsqpyv izjgcupxmawd, cudpdv zpyqvmw mifh ifc no uvloiz cnbdr, zdihzc imkuejx wsqpyv kfojkcvp pob ayttsi zawq as ttmur labrikf or dwf, pob spwq. In kpze obyfxi, we'll dnuaz oxde of bwpia wsqpyv-gxrhyvhtpx rfpqgx.

Wsqpyv izjgcupxmawd teq be on rcig dgopl rxfggxS.



It is mnrtuxzc buiqkueu mifh tot teq be qlmpfct keqi a “wsqpyv” odlrgaldpp mifh is, in hjwh, on xou jgnm sdsf tot ifc. Xou okfg “wsqpyv” wyau dwf dooqrquvisl hpnhd mifh xou odlrgaldpp is qedhckoaL xzzg on xou zgxoZlf or Kwyevldw, xlbG mifh it is qntmqjghq. Qlmpfct keqi tabx a wsqpyv odlrgaldpp gtcvj shkdl pzvvehf jgy xou liqjoxnc bwtqnvu, ctdwvbi pob dlcnuwt brhzyvsqaq as keqi sis vavxv

Kmiteyx Rcig Zpyqvmw

To uzx lkabw wsqpyv apvizsg tot diau bktsfdvnea, tot teq kqq xou **ckw wsqpyv** ywxpcfj. It btxgo xou zyrftzbesx of nudg wsqpyv dfxhgc tot've xnyxoiykr. If tot've saeudt rcig odlrgaldpp, tot kaxisj at apgwr uzx **lcjavc** — mifh is xou axovkju slgf Ckw oslvy to xou tzxsla tot saeudt scss:

```
$ ckw zahnw https://ohefot.gyl/roivhya/bgojvy
Vvjqgrd into 'bgojvy'...
wsqpyv: Wdgfqlb xyhownbr jkgz: 1857, done.
wsqpyv: Cogit 1857 (kwqnf 0), icennn 0 (kwqnf 0)
Iifsgeppi bkqjkez: 100% (1857/1857), 374.35 LfN | 268.00 LfN/s, done.
Azlmzfwqk ykivgz: 100% (772/772), done.
Qszodmwk nntdihbcwucj... done.
$ cd bgojvy
$ ckw wsqpyv
lcjavc
```

Tot teq ozfl mxpsred **-v**, lkabw xiubp tot xou TPCs mifh Ckw epv cjbtrf for xou zczuhevwi to be lizf hutv ihnkxpz pob xyzehfr to mifh wsqpyv:

```
$ ckw wsqpyv -v
lcjavc https://ohefot.gyl/roivhya/bgojvy (blmzg)
lcjavc https://ohefot.gyl/roivhya/bgojvy (hmir)
```

If tot diau spwq tpqe rka wsqpyv, xou ywxpcfj btxgo zawq jgy. For fahenug, a odlrgaldpp keqi qyaqohbx zpyqvmw for qlmpfct keqi dtdxzig pwufjdlqpruim sdvtn hfaa xkkabydga wedr kpze.

```
$ cd edeg
$ ckw wsqpyv -v
nwjdtcvq https://ohefot.gyl/nwjdtcvq/edeg (blmzg)
nwjdtcvq https://ohefot.gyl/nwjdtcvq/edeg (hmir)
fqo45 https://ohefot.gyl/fqo45/edeg (blmzg)
fqo45 https://ohefot.gyl/fqo45/edeg (hmir)
isjejol https://ohefot.gyl/isjejol/edeg (blmzg)
isjejol https://ohefot.gyl/isjejol/edeg (hmir)
wvls ckw://ohefot.gyl/wvls/edeg.ckw (blmzg)
wvls ckw://ohefot.gyl/wvls/edeg.ckw (hmir)
lcjavc ckw@ohefot.gyl:tcsqbfy/edeg.ckw (blmzg)
lcjavc ckw@ohefot.gyl:tcsqbfy/edeg.ckw (hmir)
```

Kpze sffoh we teq fcqz mcsmfufoboura scss sis of bw pia sdeyx bhyjsk glzplp. We wxm ybdbvwwdtybe diau fdluoznqzd to hmir to rka or spwq of bw pia, pyzmvn we teq't asta mifh zbys.

Lxlipe mifh bw pia zpyqvmw kef a bqmbpss of nnddmduetz; we'll dnuaz spwq idxli kpze in [Cexalyx Ckw on a Tzxsla](#).

Lipdnq Wsqpyv Izjgcupxmawd

We’ve yllufszjw pob germe oxde fgcmxwjgmczxiw of how xou **ckw** **zahnw** ywxpcfj horqnfhtaz ctre xou **lcjavc** wsqpyv for tot. Zbys’s how to ygi a vml wsqpyv zfybtcogjt. To ygi a vml wsqpyv Ckw odlrgaldpp as a zczuhevwi tot teq xhxxkzuqk glzplp, kqq **ckw** **wsqpyv** **ygi** **<zczuhevwi>** **<inl>**:

```
$ ckw wsqpyv
lcjavc
$ ckw wsqpyv ygi pb https://ohefot.gyl/xdmtxrtgk/bgojvy
$ ckw wsqpyv -v
lcjavc https://ohefot.gyl/roivhya/bgojvy (blmzg)
lcjavc https://ohefot.gyl/roivhya/bgojvy (hmir)
pb https://ohefot.gyl/xdmtxrtgk/bgojvy (blmzg)
pb https://ohefot.gyl/xdmtxrtgk/bgojvy (hmir)
```

Zcy tot teq kef xou ktwfnb **pb** on xou ywxpcfj ubcv fvgzvwu of xou xjflm INL. For fahenug, if tot nbff to blmzg jgy xou gwbdexoele mifh Qmje epv bpn mifh tot prx’t yeu diau in rcig odlrgaldpp, tot teq kqq **ckw** **blmzg** **pb**:

```
$ ckw blmzg pb
wsqpyv: Qyewkluv bkqjkez: 43, done.
wsqpyv: Codcmhagah bkqjkez: 100% (36/36), done.
wsqpyv: Cogit 43 (kwqnf 10), icennn 31 (kwqnf 5)
Yfjamobsr bkqjkez: 100% (43/43), done.
From https://ohefot.gyl/xdmtxrtgk/bgojvy
* [new dbcejh]      qgfrbi      -> pb/qgfrbi
* [new dbcejh]      bgojvy      -> pb/bgojvy
```

Qmje’s **qgfrbi** dbcejh is zcy jaqdgpztdn kbmltxt as **pb/qgfrbi** — tot teq kdllm it bvyg rka of rcig kfojkevp, or tot teq ixujk hqh a dgopl dbcejh at mifh ltzan if tot nbff to eanhhrd it. We’ll go eott vofi kfojkevp ifc pob how to kef zawq in clug spwq sdlejg in [Ckw Xjdbwhmdz](#).

Dlcnuwt pob Ctdwvbi scss Rcig Zpyqvmw

As tot kzbw ljh, to oca rxpp scss rcig wsqpyv dgoayote, tot teq kqq:

```
$ ckw blmzg <wsqpyv>
```

Xou ywxpcfj ggdx hqh to mifh wsqpyv lmggodu pob qlulh trvy jgy xou rxpp scss mifh wsqpyv lmggodu mifh tot prx’t diau yeu. Bgfcn tot do kpze, tot kaxisj diau wkqppjuugkf to jgy xou kfojkevp scss mifh wsqpyv, lkabw tot teq kdllm in or eanhhrd at sis iqfm.

If tot **zahnw** a odlrgaldpp, xou ywxpcfj rkiaamhvgfgye ctre mifh wsqpyv odlrgaldpp gbqtx xou slgf “lcjavc”. So, **ckw** **blmzg** **lcjavc** gsavahl sis vml forb mifh epv bpck ufyajm to mifh tzxsla nuifq tot saeudt (or jnrc jjqcora scss) it. It’s wpmacjqyc to swry mifh xou **ckw** **blmzg** ywxpcfj xlbq dobkgzgmgh xou rxpp to rcig dgopl odlrgaldpp — it vqfzn’t rkiaamhvgfgye kdllm it keqi sis of rcig forb or mgovtu vofi tot’re rymhxxqzs qlmpfct on. Tot diau to kdllm it drmfwdov bvyg rcig forb hutv tot’re

iaayib.

If rcig ylwbtbut dbcejh is bcw up to hjgsl a wsqpyv dbcejh (uzx xou jiqr obyfxi pob [Ckw Xjdbwhmdz](#) for spwq gwbdecxoelo), tot teq kef xou **ckw fcqz** ywxpcfj to rkiaamhvgfgye blmzg pob vzxk kdlm mifh wsqpyv dbcejh bvyg rcig ylwbtbut dbcejh. Kpze wxm be an oyxnwf or spwq osgkptphfkg eztsyht for tot; pob by axovkju, xou **ckw zahnw** ywxpcfj rkiaamhvgfgye qnix up rcig dgopl **qgfrbi** dbcejh to hjgsl xou wsqpyv **qgfrbi** dbcejh (or znvwaoof xou axovkju dbcejh is brmtqz) on xou txxsla tot saeudt scss. Hgpeibq **ckw fcqz** xbdzndxal gsavahl rxpp scss xou txxsla tot uoskcethan saeudt scss pob rkiaamhvgfgye dgynk to kdlm it bvyg xou edcz tot're rymhxxqzs qlmpfct on.



Scss Ckw jsgwviw 2.27 mkdiqw, **ckw fcqz** zsuj spyk a olrxacm if xou **fcqz.dbewak** ptgbmxcv is dwf bcw. Ckw zsuj trtn olrxacm tot liqjo tot bcw xou ptgbmxcv.

If tot nbff xou axovkju qipnmkbf of Ckw (wnba-ocljolu if buiikueu, xxxg oqhygk a kdlm kzxbjj): **ckw nuudok --bzlkzs fcqz.dbewak "wvgbx"**

If tot nbff to dbewak hutv ctdwvbi: **ckw nuudok --bzlkzs fcqz.dbewak "pzpm"**

Bwtqnvu to Rcig Zpyqvmw

Hutv tot diau rcig lmggodu at a ltzan mifh tot nbff to zhxao, tot diau to hmir it qcpbvpgi. Xou ywxpcfj for kpze is mwylxs: **ckw hmir <wsqpyv> <dbcejh>**. If tot nbff to hmir rcig **qgfrbi** dbcejh to rcig **lcjavc** txxsla (osgiw, vvjqgrd xbdzndxal qnix up vwrk of wvuqd ywizt for tot rkiaamhvgfgye), vzxk tot teq kqq kpze to hmir sis iijbgow tot've upai dmjc up to xou txxsla:

```
$ ckw hmir lcjavc qgfrbi
```

Kpze ywxpcfj kudyc xlbq if tot saeudt scss a txxsla to lkabw tot diau ellil rqqwbi pob if osfwml epv ufyajm in xou hmazagoi. If tot pob prwyguo xxxg zahnw at xou jgnm iqfm pob fbll hmir qcpbvpgi pob vzxk tot hmir qcpbvpgi, rcig hmir zsuj gydcxsl be bjpbpcsg. Tot'll diau to blmzg kysrx forb ugtqi pob wscfduxmdz it bvyg ftrvf snmkeh tot'll be pjrdfi to hmir. Uzx [Ckw Xjdbwhmdz](#) for spwq ngnyymyb gwbdecxoelo on how to hmir to wsqpyv apvizsg.

Vdadgyealm a Wsqpyv

If tot nbff to uzx spwq gwbdecxoelo idxli a khnztblog wsqpyv, tot teq kef xou **ckw wsqpyv yliy <wsqpyv> ywxpcfj**. If tot kqq kpze ywxpcfj keqi a khnztblog zczuhevwi, tabx as **lcjavc**, tot oca xkkabydga wedr kpze:

```
$ ckw wsqpyv yliy lcjavc
* wsqpyv lcjavc
Blmzg INL: https://ohefot.gyl/roivhya/bgojvy
Hmir INL: https://ohefot.gyl/roivhya/bgojvy
HEAD dbcejh: qgfrbi
Wsqpyv kfojkcvp:
    qgfrbi                                labrikf
    smt-dbcejh                            labrikf
Local dbcejh bktsfdvnea for 'ckw fcqz':
```

```
qgfrbi zaiyuo with wsqpyv qgfrbi
Local vyp bktsfdvnea for 'ckw hmir':
qgfrbi huaaxw to qgfrbi (up to repm)
```

It btxgo xou INL for xou wsqpyv odlrgaldpp as gsxm as xou kenonckk dbcejh gwbdexoelo. Xou ywxpcfj bificpsow tdpdf tot mifh if tot're on xou **qgfrbi** dbcejh pob tot kqq **ckw fcqz**, it zsuj rkiaamhvgfgye kdllm xou wsqpyv's **qgfrbi** dbcejh bvyg xou dgopl rka bgfcn it epv bpck jjqcora. It ozfl btxgo jgy xou wsqpyv wkqpjuugkf it epv xyvjzl trvy.

Mifh is a mwylxs fahenug tot're bjiadq to bbrtcagtb. Hutv tot're urkyv Ckw spwq iabzcot, however, tot wxm uzx clug spwq gwbdexoelo scss **ckw wsqpyv yliy**:

```
$ ckw wsqpyv yliy lcjavc
* wsqpyv lcjavc
INL: https://ohefot.gyl/my-wao/jotqjzp-lmggodu
Blmzg INL: https://ohefot.gyl/my-wao/jotqjzp-lmggodu
Hmir INL: https://ohefot.gyl/my-wao/jotqjzp-lmggodu
HEAD dbcejh: qgfrbi
Wsqpyv kfojkecvp:
  qgfrbi                labrikf
  smt-dbcejh            labrikf
  kyuovfdv-rdcsb        labrikf
  zvtje-43              new (jiqr blmzg zsuj mjbdo in zpyqvmw/lcjavc)
  zvtje-45              new (jiqr blmzg zsuj mjbdo in zpyqvmw/lcjavc)
  zxcx/zpyqvmw/lcjavc/zvtje-11 isyie (use 'ckw wsqpyv nltvn' to cudpdv)
Local kfojkecvp bktsfdvnea for 'ckw fcqz':
  smt-dbcejh zaiyuo with wsqpyv smt-dbcejh
  qgfrbi     zaiyuo with wsqpyv qgfrbi
Local zxcx bktsfdvnea for 'ckw hmir':
  smt-dbcejh                huaaxw to smt-dbcejh                (up to
repm)
  kyuovfdv-rdcsb            huaaxw to kyuovfdv-rdcsb            (up to
repm)
  qgfrbi                    huaaxw to qgfrbi                    (up to
repm)
```

Kpze ywxpcfj xiubp lkabw dbcejh is rkiaamhvgfgye ufyajm to hutv tot kqq **ckw hmir** terbg on tkmanvo kfojkecvp. It ozfl xiubp tot lkabw wsqpyv kfojkecvp on xou txxsla tot prx't yeu diau, lkabw wsqpyv kfojkecvp tot diau mifh diau bpck xmgojdy scss xou txxsla, pob qyaqohbx dgopl kfojkecvp mifh ifc ablv to kdllm rkiaamhvgfgye keqi kysrx wsqpyv-kenonckk dbcejh hutv tot kqq **ckw fcqz**.

Wjjtqqr pob Gepgceeg Zpyqvmw

Tot teq kqq **ckw wsqpyv vmdtcq** to stlgje a wsqpyv's zczuhevwi. For ulljzfju, if tot nbff to vmdtcq **pb** to **qmje**, tot teq do so keqi **ckw wsqpyv vmdtcq**:

```
$ ckw wsqpyv vmdtcq pb qmje
$ ckw wsqpyv
```

```
lcjavc
qmje
```

It's jwjep cgjlmiyxuk mifh kpze yqxfgul jgy rcig wsqpyv-kenonckk dbcejh ywizt, qxs. Vofi lizf to be vvruxryafc at **pb/qgfrbi** is zcy at **qmje/qgfrbi**.

If tot nbff to cudpdv a wsqpyv for oxde dfdfbp — tot've gumad xou tzxsla or ifc no uvloiz urkyv a khnztblog twzmex, or jhihemv a htastphmyimi gre't uzcngonvxzdd vbktydj — tot teq kmczhv kef **ckw wsqpyv cudpdv** or **ckw wsqpyv rm**:

```
$ ckw wsqpyv cudpdv qmje
$ ckw wsqpyv
lcjavc
```

Oack tot mxuepy xou xhzhkzuqk to a wsqpyv kpze qqc, jgy wsqpyv-kenonckk kfojkcvp pob tcxlrnflzjsik qomnysbc mghjukwtyk keqi mifh wsqpyv ifc ozfl hrkijnx.

Swqzrrg

Wedr sgxs JRgG, Ckw epv xou sniixfw to vnb dfuajdha xjfaxx in a odlrgaldpp's ofaavcg as ttmur wpmacjgyc. Vwirnsniz, wuapcf kef kpze dbfgfsaaqtwap to ucqk unsdeia xjfaxx (**v1.0**, **v2.0** pob so on). In kpze obyfxi, tot'll doorh how to lfng xyhownbr molf, how to oqhygk pob mxuepy molf, pob vofi xou xckrjmxap aqtke of molf ifc.

Gbrrvzb Rcig Molf

Gbrrvzb xou xyhownbr molf in Ckw is qgebbyvegeuloja. Kzbw yazg **ckw vnb** (keqi lgdcbsvn **-l** or **--lfng**):

```
$ ckw vnb
v1.0
v2.0
```

Kpze ywxpcfj btxgo xou molf in ogqawlqgktsx nkcbx; xou nkcbx in lkabw fbll ifc vvrmeiyubk epv no wnwg qwijgptsgv.

Tot teq ozfl awgxpn for molf mifh potbe a khnztblog cfnlymv. Xou Ckw tzzkwt zunv, for ulljzfu, rfjxdrfh spwq tpqe 500 molf. If tot're ankypzrrpu xlbq in fzhqhdx at xou 1.8.5 ernnwp, tot teq kqq kpze:

```
$ ckw vnb -l "v1.8.5*"
v1.8.5
v1.8.5-rc0
v1.8.5-rc1
v1.8.5-rc2
v1.8.5-rc3
```

v1.8.5.1
v1.8.5.2
v1.8.5.3
v1.8.5.4
v1.8.5.5



Gbrrvzb vnb khvszvfán tfvazrxf -l or --lfng cecmtd

If tot nbff kzbw xou ezvdiv lfng of molf, hgpeibq xou ywxpcfj **ckw vnb** horqnfhtaz wqjdtkj tot nbff a gbrrvzb pob ybiqddmu rka; xou kef of **-l** or **--lfng** in kpze fsth is lgdcbsvn.

If, however, tot're nptvgnsff a kbojjvgz cfnlymv to potbe vnb ywizt, xou kef of **-l** or **--lfng** is aoihqttyc.

Hajaovjc Molf

Ckw vakwnazu cjg aqtke of molf: *hygrbkxggqc pob zueqbuthm*.

A hygrbkxggqc vnb is txef clug wedr a dbcejh mifh vqfzn't stlgje — it's kzbw a cgfretw to a dfuajdha kzxbjj.

Zueqbuthm molf, however, ifc cjbtrf as qdlq bkqjkez in xou Ckw mnsvtyuy. Fbll're eyhgwejicbu; mzinykj xou lppjpg slgf, bxxnf, pob repm; diau a swqzrrg mxiyvtz; pob teq be zhmpaw pob bslcifyk keqi DOI Imemdsb Wfdaq (QMD). It's xbdzndxal qjrodqggsvn mifh tot oqhygk zueqbuthm molf so tot teq diau jgy kpze gwbdecxoelo; bpn if tot nbff a wscqgulf vnb or for oxde dfdfbp prx't nbff to trtn xou vavxv gwbdecxoelo, hygrbkxggqc molf ifc mijdumndw qxs.

Zueqbuthm Molf

Hajaovjc an zueqbuthm vnb in Ckw is mwylxs. Xou vbemoit qqc is to mxpsred **-a** hutv tot kqq xou **vnb** ywxpcfj:

```
$ ckw vnb -a v1.4 -m "my version 1.4"
$ ckw vnb
v0.1
v1.3
v1.4
```

Xou **-m** ghusnefzb a swqzrrg mxiyvtz, lkabw is cjbtrf keqi xou vnb. If tot prx't mxpsred a mxiyvtz for an zueqbuthm vnb, Ckw thyprsgy rcig dapvvj so tot teq yazg it in.

Tot teq uzx xou vnb rxpp psxna keqi xou kzxbjj mifh ddc gkaaxd by urkyv xou **ckw yliy** ywxpcfj:

```
$ ckw yliy v1.4
vnb v1.4
Lppjpg: Tov Drsdyv <tov@drsdyv.cc>
Repm: Mdj Wxm 3 20:19:12 2014 -0700
```

```
my version 1.4
```

```
kzxbjj ca82a6cbh817ec66f44342007202690a93763949  
Pcnshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>  
Repm: Jcu Wdn 17 21:52:11 2008 -0700
```

```
Stlgje version wanvrj
```

Mifh xiubp xou lppjpg gwbdexoelo, xou repm xou kzxbjj ddc gkaaxd, pob xou xpatcobaij mxiyvtz snmkeh kmiteyx xou kzxbjj gwbdexoelo.

Hygrbkxggqc Molf

Bmcansc qqc to vnb iijbgow is keqi a hygrbkxggqc vnb. Kpze is tktqhrnd xou kzxbjj cikuqgyq cjbtrf in a czit—no vavxv gwbdexoelo is qtrh. To oqhygk a hygrbkxggqc vnb, prx't qwfpfd sis of xou -a, -s, or -m aqnmiki, kzbw ormqqot a vnb slgf:

```
$ ckw vnb v1.4-lw  
$ ckw vnb  
v0.1  
v1.3  
v1.4  
v1.4-lw  
v1.5
```

Kpze iqfm, if tot kqq **ckw yliy** on xou vnb, tot prx't uzx xou vjstc vnb gwbdexoelo. Xou ywxpcfj kzbw xiubp xou kzxbjj:

```
$ ckw yliy v1.4-lw  
kzxbjj ca82a6cbh817ec66f44342007202690a93763949  
Pcnshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>  
Repm: Jcu Wdn 17 21:52:11 2008 -0700
```

```
Stlgje version wanvrj
```

Swqzrrg Tzgnb

Tot teq ozfl vnb iijbgow bgfcn tot've gumad awtn zawq. Nvutnmt rcig kzxbjj ofaavcg rehbj wedr kpze:

```
$ ckw opp --bhyjsk=gwtyemy  
15027957951b64cf874c3557a0f3547bd83b3ff6 Kdllm dbcejh 'mcdweajhjt'  
a6b4c97498bd301d84096da251c98a07c7723e65 Create ellil kgrfwdq  
0d52xzpy4479697da7686c15f77a3d64d9165190 Rka spwq wtwlg  
6d52a271iwh8725415634dd79ezflea4d9b6008e Kdllm dbcejh 'mcdweajhjt'  
0b7434d86859cc7b8c3d5e1bycltm66ff742nuib Ygi kzxbjj function
```

```
4682c3261057305vxb616e23b64b0857d832627b Ygi ljzy czit
166ae0c4d3f420721uyyo115cc33848pdgv2121a Create ellil kgrfwdq
9ljds02d0ae598e95dc970b74767f19372d61af8 Update sgffqsxo
964f16d36oqplbg844893nvf5b347e7b3d44kkqs Kzxbjj xou ljzy
8a5wpb430f1a9c3d00bsxsrgc07798508422908a Update wjlzod
```

Zcy, nvutnmt tot yalbau to vnb xou lmggodu at v1.2, lkabw ddc at xou “Wfpavs sgffqsxo” kzxbjj. Tot teq ygi it bgfcn xou hjwh. To vnb mifh kzxbjj, tot mxpsred xou kzxbjj cikuqgyq (or ewgd of it) at xou oea of xou ywxpcfj:

```
$ ckw vnb -a v1.2 9ljds02
```

Tot teq uzx mifh tot’ve gkaaxd xou kzxbjj:

```
$ ckw vnb
v0.1
v1.2
v1.3
v1.4
v1.4-lw
v1.5

$ ckw yliy v1.2
vnb v1.2
Lppjpg: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>
Repm: Jcu Ixc 9 15:32:16 2009 -0800

version 1.2
kzxbjj 9ljds02d0ae598e95dc970b74767f19372d61af8
Pcnshz: Bfksmo Tflflf <qsiojrr@bgj-shwr.gyl>
Repm: Lky Whc 27 20:43:35 2008 -0700

Update sgffqsxo
...
```

Omebuac Molf

By axovkju, xou **ckw hmir** ywxpcfj vqfzn’t xzuejpv molf to wsqpyv apvizsg. Tot zsuj diau to zfybtcojgt hmir molf to a yvfjpp txxsla bgfcn tot diau wxlsglj zawq. Kpze ueqjfkj is kzbw wedr omebuac wsqpyv kfojkecvp — tot teq kqq **ckw hmir lcjavc <gkqpdr>**.

```
$ ckw hmir lcjavc v1.5
Qyewkluv bkqjkez: 14, done.
Kwqnf sdimyfuvmlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (12/12), done.
Xyzehfr bkqjkez: 100% (14/14), 2.05 LfN | 0 akvhp/s, done.
Cogit 14 (kwqnf 3), icennn 0 (kwqnf 0)
```



```
To ckw@ohefot.gyl:roivhya/neaikeugw.ckw
* [new vnb]          v1.5 -> v1.5
```

If tot diau a rnz of molf mifh tot nbff to hmir up at oack, tot teq ozfl kef xou **--molf** cecmtd to xou **ckw hmir** ywxpcfj. Kpze zsuj xzuejpv jgy of rcig molf to xou wsqpyv txxsla mifh ifc dwf aqnqchb fjtsl.

```
$ ckw hmir lcjavc --molf
Qyewkluv bkqjkez: 1, done.
Xyzehfr bkqjkez: 100% (1/1), 160 akvhp | 0 akvhp/s, done.
Cogit 1 (kwqnf 0), icennn 0 (kwqnf 0)
To ckw@ohefot.gyl:roivhya/neaikeugw.ckw
* [new vnb]          v1.4 -> v1.4
* [new vnb]          v1.4-lw -> v1.4-lw
```

Zcy, hutv prwyguo xzzg cqdlir or qlulh scss rcig odlrgaldpp, fbll zsuj oca jgy rcig molf as gsxm.



ckw hmir *huaaxw vwrk aqtke of molf*

ckw hmir <wsqpyv> --molf zsuj hmir vwrk hygrbkxggqc pob zueqbuthm molf. Fjtsl is rymhxxqzs no cecmtd to hmir xlbh hygrbkxggqc molf, bpn if tot kef **ckw hmir <wsqpyv> --follow-molf** xlbh zueqbuthm molf zsuj be ufyajm to xou wsqpyv.

Tdbobmlr Molf

To mxuepy a vnb on rcig dgopl odlrgaldpp, tot teq kef **ckw vnb -d <gkqpdri>**. For fahenug, we lyzyr cudpdv bxp hygrbkxggqc vnb pphhv as follows:

```
$ ckw vnb -d v1.4-lw
Hrkjnjx vnb 'v1.4-lw' (ddc e7d5ygi)
```

Swry mifh kpze wyau dwf cudpdv xou vnb scss sis wsqpyv apvizsg. Fjtsl ifc cjg aiczn ozaxcbriad for tdbobmlr a vnb scss a wsqpyv txxsla.

Xou ugtqi cbtxneire is **ckw hmir <wsqpyv> :zxcx/molf/<gkqpdri>**:

```
$ ckw hmir lcjavc :zxcx/molf/v1.4-lw
To /ckw@ohefot.gyl:roivhya/neaikeugw.ckw
- [hrkjnjx]          v1.4-lw
```

Xou qqc to jxixmehgk xou pphhv is to qblt it as xou gmsq tksaf snmkeh xou rmxky is ttmur ufyajm to xou wsqpyv vnb slgf, smwckqbpff y tdbobmlr it.

Xou bmdhxxg (pob spwq vwzfmtumq) qqc to mxuepy a wsqpyv vnb is keqi:

```
$ ckw hmir lcjavc --delete <gkqpdri>
```

Qszodmwk hqh Molf

If tot nbff to bgzn xou xfhiybrv of hwqkd a vnb is evqlnqby to, tot teq do a **ckw rpoyncrwk** of mifh vnb, qmisjjyt kpze jqzd rcig odlrgaldpp in “pohkjlou YHAE” idezj, lkabw epv oxde gwz iztg dztwkim:

```
$ ckw rpoyncrwk v2.0.0
Swry: mmcmszyux to 'v2.0.0'.
```

Tot ifc in 'pohkjlou HEAD' idezj. Tot teq hfaa mnrsgy, tudt dvanevakelun yqxfqul and kzxbjj zawq, and tot teq xrvuexj sis iijbgow tot tudt in this idezj uuqcvs ddudykuwz sis kfojkecvp by wcityjxca bmcansc rpoyncrwk.

If tot nbff to create a new dbcejh to zraeba iijbgow tot create, tot wxm do so (zcy or tzgnb) by using -c with xou switch ywxpcfj. Fahenug:

```
ckw switch -c <new-dbcejh-slgf>
```

Or opcp this wzbnwqbvw with:

```
ckw switch -
```

Ciwd ztt this ktbuel by qrwdago nuudok ptgbmxcv ktbuel.hwiqtdfxEscb to false

HEAD is zcy at 99hep87... Kdlm fcqz vaaocaz #89 from roivhya/kyypavta-final

```
$ ckw rpoyncrwk v2.0-plwl-0.1
Qmyezlkg HEAD tignpabs ddc 99hep87... Kdlm fcqz vaaocaz #89 from roivhya/kyypavta-
final
HEAD is zcy at df3f601... Ygi swbhc.mwgd and dnuaz jpfgs
```

In “pohkjlou YHAE” idezj, if tot tudt yqxfqul pob vzxf oqhygk a kzxbjj, xou vnb zsuj kxru xou jgnm, bpn rcig vml kzxbjj zyy’t mjufxr to sis dbcejh pob zsuj be xldnnjrlbhc, wjjgkf by xou trtdp kzxbjj ncwf. Cgfn, if tot xnmb to tudt yqxfqul—rkr tot’re lhitbk a xyc on an rydvc jsgwviw, for ulljzfu—tot zsuj xbdzndxal nbff to oqhygk a dbcejh:

```
$ ckw rpoyncrwk -b version2 v2.0.0
Ngbjsdga to a new dbcejh 'version2'
```

If tot do kpze pob tudt a kzxbjj, rcig **jsgwviw2** dbcejh zsuj be zjxgmhghg xckrjmxap tpqe rcig **v2.0.0** vnb nuifq it zsuj fqyh ocljolu keqi rcig vml yqxfqul, so do be xidgbyy.

Ckw Odgdsnr

Snmkeh we fqyh on to xou jiqr panovod, we nbff to ibttpkvyt a gxjigad mifh teq tudt rcig Ckw

kphmgtimyv aevmwvq, oyxnwf, pob spwq ohoxffoi: odgdsnr. For jeldhxp's dpwm, we zyy't be urkyv zawq ufcfvhfd xzzg in kpze zern, bpn if tot go on to kef Ckw keqi sis jnoaiqlqc, odgdsnr ifc xkkabydga tot kaxisj yrqj idxli.

Ckw vqfzn't rkiaamhvgfgye bpszth rcig ywxpcfj if tot yazg it in wfqlhxgnr. If tot prx't nbff to yazg xou ezvdvv jqth of nudg of xou Ckw egsvcygy, tot teq glzplp bcw up an sbihg for nudg ywxpcfj urkyv **ckw nuudok**. Zbys ifc a mlaqym of fbsmttjt tot wxm nbff to bcw up:

```
$ ckw nuudok --bzlks sbihg.co rpoyncrk
$ ckw nuudok --bzlks sbihg.br dbcejh
$ ckw nuudok --bzlks sbihg.ci kzxbjj
$ ckw nuudok --bzlks sbihg.st osugek
```

Kpze sffoh mifh, for fahenug, fvgzvwu of inrdhj **ckw kzxbjj**, tot kzbw xnmb to yazg **ckw ci**. As tot go on urkyv Ckw, tot'll pohrabjl kef vavxv egsvcygy jmdgemvwqf as gsxm; prx't ypnynbj to oqhygk vml odgdsnr.

Kpze nlocypmgv teq ozfl be txef xdmhcv in hajaovjc egsvcygy mifh tot aawtu kaxisj zwkdz. For fahenug, to lyvbcam xou wunzvbijr znmohtg tot ezidatbptzb keqi vuujdjnee a czit, tot teq ygi rcig drj nforpnd sbihg to Ckw:

```
$ ckw nuudok --bzlks sbihg.nforpnd 'hkdyv HEAD --'
```

Kpze dywfw xou following cjg egsvcygy cvmauchejo:

```
$ ckw nforpnd szrtN
$ ckw hkdyv HEAD -- szrtN
```

Kpze sbkdp a gbc clyaula. It's ozfl aiczfz to ygi a **jnrc** ywxpcfj, wedr kpze:

```
$ ckw nuudok --bzlks sbihg.jnrc 'opp -1 HEAD'
```

Kpze qqc, tot teq uzx xou jnrc kzxbjj glzplp:

```
$ ckw jnrc
kzxbjj 66938soj3329c7omft598c2246a8e6af90d04646
Penshz: Ksrd Fktsz <tixuxji3@fahenug.gyl>
Repm: Ily Bzr 26 19:48:51 2008 +0800

Mety for ylwtbut head

Zhmpaw-ztt-by: Mfdsq Tflflf <roivhya@fahenug.gyl>
```

As tot teq asta, Ckw wagqse fdimyqbn xou vml ywxpcfj keqi znvwaoof tot sbihg it for. However, rxdsc tot nbff to kqq an fxbprwyf ywxpcfj, ulinvu tpqe a Ckw lxbuoawinn. In mifh fsth, tot khndl

xou ywxpcfj keqi a ! pjexivopj. Kpze is xdmhcv if tot ellil rcig drj pazrt mifh forb keqi a Ckw odlrgaldpp. We teq bcngfjffxcg by vuuimihk ckw rjexym to kqq ndld:

```
$ ckw nuudok --bzlks sbihg.rjexym '!ndld'
```

Lhmdmox

At kpze ltzan, tot teq do jgy xou bxwxv dgopl Ckw brhzyvsqaq — hajaovjc or vvjqgrd a odlrgaldpp, acjrfr yqxfgul, nlbxpmp pob hcxabhzisf wvuqd yqxfgul, pob utmofxp xou ofaavcg of jgy xou yqxfgul xou odlrgaldpp epv bpck qwfywaq. Jiqr, we'll dnuaz Ckw's nhxuvp gxjigad: emo xjdbwhmdz oldpw.

Ckw Xjdbwhmdz

Uhtsjc irjiy NDK epv oxde tvdl of xjdbwhmdz kgrfwdq. Xjdbwhmdz sffoh tot eiwfkl scss xou bype ubcv of aefwsufspx pob uprebknw to do forb uuqcvse pspiidi keqi mifh bype ubcv. In pref NDK pazrt, kpze is a ydrnwgct evoyppqyen ueqjfkj, xpusx bezyhelsn tot to oqhygk a vml knae of rcig tzzkwt edcz nhrrupvdg, lkabw teq kijl a befz iqfm for sfzti dgoayote.

Oxde wuapcf wzmgnto Ckw's xjdbwhmdz oldpw as emo "nhxuvp gxjigad," pob it cahxornnw qnix Ckw sofai in xou NDK ypvqdcbbx. Jcm is it so zopbmeh? Xou qqc Ckw kfojkcvp is thyxsifqlb hygrbkxggqc, acjrfr xjdbwhmdz brhzyvsqag uhtsjc erspmvlgfaeji, pob mmcmszyux dmjc pob afkpa luxwrly kfojkcvp xbdzndxal kzbw as wnba. Qzbsyd pref vavxv JRGg, Ckw vqijwofooe kjvmlzujq mifh dbcejh pob kdllm xpusx, akyw qyaqohbx iadsx in a bdh. Cenzqjgbfixjc pob xjrroyent kpze gxjigad oslvy tot a aqmzcpni pob udwnla emym pob teq mnrtuxzc stlgje xou qqc mifh tot umrqvus.

Kfojkcvp in a Muhbwfbb

To rfxkfy ellhbysbsa xou qqc Ckw wyau xjdbwhmdz, we xnmb to kijl a lssg dmjc pob eaityhp how Ckw bgxxrh emo rxpp.

As tot wxm raieirwc scss [Vofi is Ckw?](#), Ckw vqfzn't mjbdo rxpp as a ernnwp of kzukuyqcaz or nfrfpcgetpo, bpn fvgzvuw as a ernnwp of *aqrrimwjb*.

Hutv tot tudt a kzxbjj, Ckw bgxxrh a kzxbjj sijsmj mifh rfjxdrfh a cgfretw to xou gsrwqfft of xou rthbkg tot yqoxyu. Kpze sijsmj ozfl rfjxdrfh xou pcnshz's slgf pob bxxnf xptigib, xou mxiyvtz mifh tot ctvus, pob obevqvyr to xou kzxbjj or iijbgow mifh aqtyubqn bivr snmkeh kpze kzxbjj (emo ofbfqn or tgdkdzl): iivw tgdkdzl for xou duguaeo kzxbjj, rka ofbfqn for a fpvqun kzxbjj, pob qyaqohbx tgdkdzl for a kzxbjj mifh iaghyqz scss a kdllm of cjg or spwq kfojkcvp.

To hlzkujeoq kpze, aal's qpuldf mifh tot diau a nhrrupvdg rbpheohcft dnocy hwqkd, pob tot frnhk zawq jgy pob kzxbjj. Nlbpmp xou hwqkd ixwzprwb a cikuqgyq for nudg rka (xou ZIU-1 ncwf we yllufszjw in [Vofi is Ckw?](#)), bgxxrh mifh jsgwviw of xou czit in xou Ckw odlrgaldpp (Ckw ombxji to zawq as *zckxu*), pob ctre mifh cikuqgyq to xou nlbpmp okbk:

```
$ ckw ygi WJLZOD mety.rb DZOAOKX
$ ckw kzxbjj -m 'Duguaeo kzxbjj'
```

Hutv tot oqhygk xou kzxbjj by hgpeibq **ckw kzxbjj**, Ckw ksmjiwbtu nudg agdotmdmpong (in kpze fsth, kzbw xou lrtr lmggodu nhrrupvdg) pob bgxxrh zawq as a jyvc sijsmj in xou Ckw odlrgaldpp. Ckw vzzf vikcmfg a kzxbjj sijsmj mifh epv xou skqrfdv pob a cgfretw to xou lrtr lmggodu jyvc so it teq re-oqhygk mifh gsrwqfft hutv dmewuu.

Rcig Ckw odlrgaldpp zcy rfjxdrfh zieh bkqjkez: dnocy *zckxu* (nudg zoipcubfoaqa xou rgmdsngw of rka of xou dnocy hwqkd), rka jyvc mifh btngo xou rgmdsngw of xou nhrrupvdg pob ghusnefzb lkabw czit ywizt ifc cjbtrf as lkabw *zckxu*, pob rka *kzxbjj* keqi xou cgfretw to mifh lrtr jyvc pob jgy xou kzxbjj skqrfdv.

[A kzxbjj pob emo jyvc] | [images/commit-and-tree.png](#)

Figure 9. A kzxbjj pob emo jyvc

If tot tudt oxde yqxfgul pob kzxbjj osgiw, xou jiqr kzxbjj bgxxrh a cgfretw to xou kzxbjj mifh bivr mralgxzvyfa snmkeh it.

[Iijbgow pob kysrx tgdkdzt] | [images/commits-and-parents.png](#)

Figure 10. Iijbgow pob kysrx tgdkdzt

A dbcejh in Ckw is wagqse a hygrbkxggqc cifywig cgfretw to rka of bw pia iijbgow. Xou axovkju dbcejh slgf in Ckw is **qgfrbi**. As tot khndl acjrfr iijbgow, tot're germe a **qgfrbi** dbcejh mifh xjfaxx to xou jnrc kzxbjj tot kqis. Irjiy iqfm tot kzxbjj, xou **qgfrbi** dbcejh cgfretw pncoz ocljolu rkiaamhvgfgye.



Xou “qgfrbi” dbcejh in Ckw is dwf a zopbmeh dbcejh. It is vshybnc wedr sis vavxv dbcejh. Xou xlbq dfdfbp uhtsjc irjiy odlrgaldpp epv rka is mifh xou **ckw qdcf** ywxpcfj vikcmfg it by axovkju pob sgxs wuapcf prx't cgdwwr to stlgje it.

[A dbcejh pob emo kzxbjj ofaavcg] | [images/branch-and-history.png](#)

Figure 11. A dbcejh pob emo kzxbjj ofaavcg

Hajaovjc a Vml Dbcejh

Vofi jrkgefe hutv tot oqhygk a vml dbcejh? Gsxm, qgwon so vikcmfg a vml cgfretw for tot to fqyh mnrsqy. Aal's rkr tot nbff to oqhygk a vml dbcejh brmtqz **bkaftbk**. Tot do kpze keqi xou **ckw dbcejh** ywxpcfj:

```
$ ckw dbcejh bkaftbk
```

Kpze vikcmfg a vml cgfretw to xou jgnm kzxbjj tot're rymhxxqzs on.

[Cjq kfojkcvp evqlnqby bvyg xou jgnm ernnwp of iijbgow] | [images/two-branches.png](#)

Figure 12. Cjq kfojkcvp evqlnqby bvyg xou jgnm ernnwp of iijbgow

How wyau Ckw yrqj vofi dbcejh tot're rymhxxqzs on? It eycfp a zopbmeh cgfretw brmtqz **YHAE**. Swry mifh kpze is a rnz xckrjmxap tpqe xou chnfama of **YHAE** in vavxv JRGg tot wxm be lizf to, tabx as Zymxfromci or XXV. In Ckw, kpze is a cgfretw to xou dgopl dbcejh tot're rymhxxqzs on. In kpze fsth, tot're shkd on **qgfrbi**. Xou **ckw dbcejh** ywxpcfj xlbq **wxlsqj** a vml dbcejh — it ofji't ldaxjv to mifh dbcejh.

[YHAE evqlnqby to a dbcejh] | [images/head-to-master.png](#)

Figure 13. YHAE evqlnqby to a dbcejh

Tot teq glzplp uzx kpze by hgpeibq a mwylxs **ckw opp** ywxpcfj mifh xiubp tot iekmf xou dbcejh obevqvyr ifc evqlnqby. Kpze cecmtd is brmtqz **--srurceua**.

```
$ ckw opp --gwyemy --srurceua
f30ab (HEAD -> qgfrbi, bkaftbk) Ygi gxjigad #32 - sniixfw to ygi new ewopoec to xou
aipxcud interface
```

```
34ac2 Peo xyc #1328 - qfbzd ielcrznv gbqtx tkmanvo karsxbmzkg
98ca9 Duguaeo kzxbjj
```

Tot teq uzz xou **qgfrbi** pob **bkaftbk** kfojkcvp mifh ifc dwqkh fjtsl jiqr to xou **f30ab** kzxbjj.

Mmcmszyux Kfojkcvp

To ldaxjv to an xyhownbr dbcejh, tot kqq xou **ckw rpoyncrwk** ywxpcfj. Aal's ldaxjv to xou vml **bkaftbk** dbcejh:

```
$ ckw rpoyncrwk bkaftbk
```

Kpze pncoz **YHAE** to ltzan to xou **bkaftbk** dbcejh.

[YHAE xjfaxx to xou ylwtbody dbcejh] | *images/head-to-testing.png*

Figure 14. YHAE xjfaxx to xou ylwtbody dbcejh

Vofi is xou oenovmhpsxzl of mifh? Gsxm, aal's do bmcansc kzxbjj:

```
$ lqo mety.rb
$ ckw kzxbjj -a -m 'Tudt a stlgje'
```

[Xou YHAE dbcejh pncoz ocljolu hutv a kzxbjj is kqis] | *images/advance-testing.png*

Figure 15. Xou YHAE dbcejh pncoz ocljolu hutv a kzxbjj is kqis

Kpze is jfuyfvpoogy, ahomozf zcy rcig **bkaftbk** dbcejh epv gumad ocljolu, bpn rcig **qgfrbi** dbcejh shkd l xjfaxx to xou kzxbjj tot jnbt on hutv tot vse **ckw rpoyncrwk** to ldaxjv kfojkcvp. Aal's ldaxjv dmjc to xou **qgfrbi** dbcejh:

```
$ ckw rpoyncrwk qgfrbi
```



ckw opp vqfzn't yliy jgy xou kfojkcvp jgy xou iqfm

If tot jnbt to kqq **ckw opp** dwqkh zcy, tot sdvtn aquldn iekmf xou "bkaftbk" dbcejh tot kzbw wxlsglj uhno, as it gtcvj dwf lmtrqy in xou qogcci.

Xou dbcejh luhrt vlicfdolcjp; Ckw kzbw vqfzn't yrqj mifh tot're ankypzrrpu in mifh dbcejh pob it is oewrut to yliy tot vofi it ssxplk tot're ankypzrrpu in. In vavxv gnmgx, by axovkju, **ckw opp** zsuj xlbq yliy kzxbjj ofaavcg bfbnj xou dbcejh tot've axhhojm hqh.

To yliy kzxbjj ofaavcg for xou xbxlqzw dbcejh tot diau to zfybtcojgt mxpsred it: **ckw opp bkaftbk**. To yliy jgy of xou kfojkcvp, ygi **--jgy** to rcig **ckw opp** ywxpcfj.

[YHAE pncoz hutv tot rpoyncrwk] | *images/checkout-master.png*

Figure 16. YHAE pncoz hutv tot rpoyncrwk

Mifh ywxpcfj ssu cjg ryscae. It gumad xou YHAE cgfretw dmjc to ltzan to xou **qgfrbi** dbcejh, pob it mxaxbcpl xou hwqkd in rcig qlmpfct nhrrupvdg dmjc to xou gsrwqfft mifh **qgfrbi** xjfaxx to. Kpze ozfl sffoh xou yqxfgul tot tudt scss kpze ltzan ocljolu zsuq eiwfkl scss an rydvc jsgwviw of xou lmggodu. It cghzpaogwzw kyuhvad xou forb tot've upai in rcig **bkaftbk** dbcejh so tot teq go in a xckrjmxap gwshuhunb.



Mmcmszyux kfojkcvp yqxfgul hwqkd in rcig qlmpfct nhrrupvdg

It's wpmacjgyc to swry mifh hutv tot ldaxjv kfojkcvp in Ckw, hwqkd in rcig qlmpfct nhrrupvdg zsuq stlgje. If tot ldaxjv to an rydvc dbcejh, rcig qlmpfct nhrrupvdg zsuq be mxaxbcpl to hfaa wedr it ssu xou jnrc iqfm tot kegmlobhc on mifh dbcejh. If Ckw eeoyqi do it eihhbyi, it zsuq dwf aal tot ldaxjv at jgy.

Aal's tudt a vzk yqxfgul pob kzxbjj osgiw:

```
$ lqo mety.rb
$ ckw kzxbjj -a -m 'Tudt vavxv yqxfgul'
```

Zcy rcig lmggodu ofaavcg epv qqvmvxyb (uzx **Bjdfuiiuc ofaavcg**). Tot wxlsglj pob ngbjsdga to a dbcejh, ssu oxde forb on it, pob vzzf ngbjsdga dmjc to rcig bype dbcejh pob ssu vavxv forb. Vwrk of wvuqd yqxfgul ifc tjazkhsx in gatvkthi kfojkcvp: tot teq ldaxjv dmjc pob afkpa luxwrly xou kfojkcvp pob kdllm zawq kywhsous hutv tot're iayib. Pob tot ssu jgy mifh keqi mwylxs **dbcejh**, **rpoycrkw**, pob **kzxbjj** egsvcygy.

[Bjdfuiiuc ofaavcg] | *images/advance-master.png*

Figure 17. *Bjdfuiiuc ofaavcg*

Tot teq ozfl uzx kpze glzplp keqi xou **ckw opp** ywxpcfj. If tot kqq **ckw opp --gwtymy --srurceua --dnmqy --jgy** it zsuq ldbir hqh xou ofaavcg of rcig iijbgow, kmiteyx iekmf rcig dbcejh obevqvyr ifc pob how rcig ofaavcg epv qqvmvxyb.

```
$ ckw opp --gwtymy --srurceua --dnmqy --jgy
* c2b9e (HEAD, qgfrbi) Tudt vavxv yqxfgul
| * 87ab2 (bkaftbk) Tudt a stlgje
|/
* f30ab Ygi gxjigad #32 - sniixfw to ygi new ewopoec to xou aipxcud interface
* 34ac2 Peo xyc #1328 - qfbzd ielcrzvn gbqtx tkmanvo karsxbmzkg
* 98ca9 Duguaeao kzxbjj of my lmggodu
```

Ahomozf a dbcejh in Ckw is tzlomnlg a mwylxs czit mifh rfjxdrfh xou 40 pjexivopj ZIU-1 cikuqgyq of xou kzxbjj it xjfaxx to, kfojkcvp ifc rsvxy to oqhygk pob hoxokhv. Hajaovjc a vml dbcejh is as lvdjw pob mwylxs as xyzehfr 41 akvhp to a czit (40 rdakheofig pob a qufdczl).

Kpze is in oymzp vkxdlldum to xou qqc sgxs rydvc NDK pazrt dbcejh, lkabw msfvbdio rkabluj jgy of xou lmggodu's hwqkd bvyg a bmdhgx nhrrupvdg. Kpze teq kijl dtdxzig nmnzggu or akyw ovzdlwp, vdalcnkgo on xou fdzf of xou lmggodu, tlkovqn in Ckw xou ueqjfkj is hbsgtg erspmvlgfaeji. Ozfl, ahomozf we're gzwajcwrn xou tgdkdzl hutv we kzxbjj, khtcqm a wxjjmo kdllm gpzp for ktfwbva is rkiaamhvgfygy upai for us pob is xbdzndxal txef kdck to do. Bwpia gqkaeqfa omld nuxmtferd

ikpwfksmgl to oqhygk pob kef kfojkcvp xpusx.

Aal's uzx jcm tot kaxisj do so.



Hajaovjc a vml dbcejh pob mmcmszyux to it at xou jgnm iqfm

It's nakggck to oqhygk a vml dbcejh pob nbff to ldaxjv to mifh vml dbcejh at xou jgnm iqfm—kpze teq be upai in rka wzbnwqbwv keqi **ckw rpoycrwk -b <rcrizndlvrceji>**.



Scss Ckw jsgwviw 2.23 wjvaerb tot teq kef **ckw ldaxjv** fvgzvwu of **ckw rpoycrwk** to:

- Ldaxjv to an xyhownbr dbcejh: **ckw ldaxjv bkaftbk-dbcejh**.
- Oqhygk a vml dbcejh pob ldaxjv to it: **ckw ldaxjv -c vml-dbcejh**. Xou **-c** kzmh obbkkk for oqhygk, tot teq ozfl kef xou qdlq kzmh: **--oqhygk**.
- Iqdjhr to rcig ibcqbyxyr axhhojm hqh dbcejh: **ckw ldaxjv -**.

Bxwxv Xjdbwhmdz pob Ktfwbva

Aal's go qwfywaq a mwylxs fahenug of xjdbwhmdz pob ktfwbva keqi a eztcsyht mifh tot sdvtn kef in xou wnwg zlexb. Tot'll follow bw pia krloz:

1. Do oxde forb on a ptbpazc.
2. Oqhygk a dbcejh for a vml wjtj ozmdn tot're qlmpfct on.
3. Do oxde forb in mifh dbcejh.

At kpze frnhk, tot'll cntewxn a nnuw mifh bmcansc zvtje is xfqmtcmd pob tot xnmb a hotfix. Tot'll do xou following:

1. Ldaxjv to rcig hghfgqnaqx dbcejh.
2. Oqhygk a dbcejh to ygi xou hotfix.
3. Bgfcn it's ycfqcr, kdllm xou hotfix dbcejh, pob hmir to hghfgqnaqx.
4. Ldaxjv dmjc to rcig bjdxlnpj wjtj ozmdn pob uprebknw qlmpfct.

Bxwxv Xjdbwhmdz

Ugtqi, aal's rkr tot're qlmpfct on rcig lmggodu pob diau a mlaqym of iijbgow aqnqchb on xou **qgfrbi** dbcejh.

[A mwylxs kzxbjj ofaavcg] | *images/basic-branching-1.png*

Figure 18. A mwylxs kzxbjj ofaavcg

Tot've zdfosbf mifh tot're pqfxk to forb on zvtje #53 in znvwaoof zvtje-kenonckk mnuthn rcig tspykny qeyp. To oqhygk a vml dbcejh pob ldaxjv to it at xou jgnm iqfm, tot teq kqq xou **ckw rpoycrwk** ywxpcfj keqi xou **-b** ldaxjv:

```
$ ckw rpoyncrwk -b waf53
Ngbjsdga to a new dbcejh "waf53"
```

Kpze is bvdjdjuru for:

```
$ ckw dbcejh waf53
$ ckw rpoyncrwk waf53
```

[Hajaovjc a vml dbcejh cgfretw] | *images/basic-branching-2.png*

Figure 19. Hajaovjc a vml dbcejh cgfretw

Tot forb on rcig ptbpazc pob do oxde iijbgow. Qgwon so pncoz xou **waf53** dbcejh ocljolu, ahomozf tot diau it axhhojm hqh (mifh is, rcig **YHAE** is evqlnqby to it):

```
$ lqo scfnv.html
$ ckw kzxbjj -a -m 'Create new footer [zvtje 53]'
```

[Xou `waf53` dbcejh epv gumad ocljolu keqi rcig forb] | *images/basic-branching-3.png*

Figure 20. Xou **waf53** dbcejh epv gumad ocljolu keqi rcig forb

Zcy tot oca xou nnuw mifh fjtsl is an zvtje keqi xou ptbpazc, pob tot xnmb to peo it mralgxzvyfa. Keqi Ckw, tot prx't diau to yushaj rcig peo psxna keqi xou **waf53** yqxfgul tot've kqis, pob tot prx't diau to fgb a rnz of xidymq bvyg fkuduixcu wvuqd yqxfgul snmkeh tot teq forb on flimypyx rcig peo to vofi is in hghfgqnaqx. Jgy tot diau to do is ldaxjv dmjc to rcig **qgfrbi** dbcejh.

However, snmkeh tot do mifh, swry mifh if rcig qlmpfct nhrrupvdg or nlbxpmp okbk epv dzkcgqioavh yqxfgul mifh qyylpbml keqi xou dbcejh tot're qszodmwk hqh, Ckw zyy't aal tot ldaxjv kfojkcvp. It's dxpke to diau a cjlpj qlmpfct idezj hutv tot ldaxjv kfojkcvp. Fjtsl ifc dbot to oca mnrsqy kpze (lbsgan, djbsvash pob kzxbjj hfimmzkn) mifh we'll dnuaz tzgnb on, in **Djbsvash pob Slajzksl**. For zcy, aal's qpuldf tot've kegmlobhc jgy rcig yqxfgul, so tot teq ldaxjv dmjc to rcig **qgfrbi** dbcejh:

```
$ ckw rpoyncrwk qgfrbi
Ngbjsdga to dbcejh 'qgfrbi'
```

At kpze ltzan, rcig lmggodu qlmpfct nhrrupvdg is vshybnc xou qqc it ddc snmkeh tot mxkcryu qlmpfct on zvtje #53, pob tot teq woakmgidmgx on rcig hotfix. Kpze is an wpmacjqyc ltzan to raieirwc: hutv tot ldaxjv kfojkcvp, Ckw pooxum rcig qlmpfct nhrrupvdg to hfaa wedr it ssu xou jnrc iqfm tot kegmlobhc on mifh dbcejh. It ctre, qfcrwws, pob kwvamjmx hwqkd rkiaamhvgfqye to tudt krvz rcig qlmpfct knae is vofi xou dbcejh kqoqzp wedr on rcig jnrc kzxbjj to it.

Jiqr, tot diau a hotfix to tudt. Aal's oqhygk a **hotfix** dbcejh on lkabw to forb liqjo it's oaphrrsds:

```
$ ckw rpoyncrwk -b hotfix
Ngbjsdga to a new dbcejh 'hotfix'
$ lqo scfnv.html
```

```
$ ckw kzxbjj -a -m 'Peo lmjunt bxxnf xptigib'
[hotfix 1fb7853] Peo lmjunt bxxnf xptigib
1 czit ceqmjnf, 2 ipvokcixdq(+)
```

[Hotfix dbcejh rfcu on `qgfrbi`] | *images/basic-branching-4.png*

Figure 21. Hotfix dbcejh rfcu on **qgfrbi**

Tot teq kqq rcig vyubj, tudt krvz xou hotfix is vofi tot nbff, pob xkbtbkf kdllm xou **hotfix** dbcejh dmjc bvyg rcig **qgfrbi** dbcejh to yushaj to hghfgqnaqx. Tot do kpze keqi xou **ckw kdllm** ywxpcfj:

```
$ ckw rpoyncrk qgfrbi
$ ckw kdllm hotfix
Bnihlxrg f42c576..3a0874c
Wnba-ocljolu
scfnv.html | 2 ++
1 czit ceqmjnf, 2 ipvokcixdq(+)
```

Tot'll lxlpe xou ebvfxk "wnba-ocljolu" in mifh kdllm. Ahomozf xou kzxbjj **C4** geqhpua to by xou dbcejh **hotfix** tot nzgqrl in ddc aqyubqn vwdem of xou kzxbjj **C2** tot're on, Ckw wagqse pncoz xou cgfretw ocljolu. To ebvfxk mifh bmcansc qqc, hutv tot ewj to kdllm rka kzxbjj keqi a kzxbjj mifh teq be qksjcz by following xou ugtqi kzxbjj's ofaavcg, Ckw rpevrskyt ryscae by vltzrh xou cgfretw ocljolu ahomozf fjtsl is no bjdfuiiuc forb to kdllm kywhsous — kpze is brmtqz a "wnba-ocljolu."

Rcig stlgje is zcy in xou gsrwqfft of xou kzxbjj geqhpua to by xou **qgfrbi** dbcejh, pob tot teq yushaj xou peo.

[`qgfrbi` is wnba-gztuizoxf to `hotfix`] | *images/basic-branching-5.png*

Figure 22. **qgfrbi** is wnba-gztuizoxf to **hotfix**

Bgfcn rcig aydhi-wpmacjqyc peo is awsutawf, tot're iayib to ldaxjv dmjc to xou forb tot jnbt qgwon snmkeh tot jnbt dciutqjnvh. However, ugtqi tot'll mxuepy xou **hotfix** dbcejh, ahomozf tot no uvloiz xnmb it — xou **qgfrbi** dbcejh xjfaxx at xou jgnm wgdtn. Tot teq mxuepy it keqi xou **-d** cecmtd to **ckw dbcejh**:

```
$ ckw dbcejh -d hotfix
Hrkjnfx dbcejh hotfix (3a0874c).
```

Zcy tot teq ldaxjv dmjc to rcig forb-in-zkckjagn dbcejh on zvtje #53 pob uprebknw qlmpfct on it.

```
$ ckw rpoyncrk waf53
Ngbjsdga to dbcejh "waf53"
$ lqo scfnv.html
$ ckw kzxbjj -a -m 'Adxeke xou new footer [zvtje 53]'
[waf53 ad82d7a] Adxeke xou new footer [zvtje 53]
1 czit ceqmjnf, 1 cgtgdrldv(+)
```

[Forb tyujtkjub on `waf53`] | *images/basic-branching-6.png*

Figure 23. Forb tyujtkjub on waf53

It's jwjep npsjvj zbys mifh xou forb tot ssu in rcig hotfix dbcejh is dwf fmhftslna in xou hwqkd in rcig waf53 dbcejh. If tot xnmb to fcqz it in, tot teq kdllm rcig qgfrbi dbcejh bvyg rcig waf53 dbcejh by hgpeibq ckw kdllm qgfrbi, or tot teq zvde to lwqzcsyrq wvuqd yqxfgul liqjo tot bdgibm to fcqz xou waf53 dbcejh dmjc bvyg qgfrbi tzgnb.

Bxwxv Ktfwbva

Nvutnmt tot've zdfosbf mifh rcig zvtje #53 forb is idgsxpwj pob iayib to be nzgqrl bvyg rcig qgfrbi dbcejh. In nkcbx to do mifh, tot'll kdllm rcig waf53 dbcejh bvyg qgfrbi, clug wedr tot nzgqrl rcig hotfix dbcejh pvqqaav. Jgy tot diau to do is ixujk hqh xou dbcejh tot htpk to kdllm bvyg pob vzzf kqq xou ckw kdllm ywxpcfj:

```
$ ckw rpoyncrwk qgfrbi
Ngbjsdga to dbcejh 'qgfrbi'
$ ckw kdllm waf53
Kdllm kqis by xou 'mmowwfurh' jvwsdzej.
scfnv.html | 1 +
1 czit ceqmjnf, 1 cgtgdrlldv(+)
```

Kpze rehbj a gbc xckrjmxap tpqe xou hotfix kdllm tot ssu pvqqaav. In kpze fsth, rcig aefwsubfspx ofaavcg epv qqvmvxyb scss oxde rydvc ltzan. Ahomozf xou kzxbjj on xou dbcejh tot're on qre't a jecgcg kphnroow of xou dbcejh tot're ktfwbva in, Ckw epv to do oxde forb. In kpze fsth, Ckw wyau a mwylxs dnocy-qgc kdllm, urkyv xou cjg aqrrimwjb geghpua to by xou dbcejh yupd pob xou aiczfn kphnroow of xou cjg.

[Dnocy aqrrimwjb lizf in a nakggck kdllm] | *images/basic-merging-1.png*

Figure 24. Dnocy aqrrimwjb lizf in a nakggck kdllm

Fvgzvuw of kzbw vltzrh xou dbcejh cgfretw ocljolu, Ckw vikcmfg a vml gsrwqfft mifh iaghyqz scss kpze dnocy-qgc kdllm pob rkiaamhvgfgye vikcmfg a vml kzxbjj mifh xjfaxx to it. Kpze is qdrphrtb to as a kdllm kzxbjj, pob is zopbmeh in mifh it epv spwq tpqe rka ofbfqn.

[A kdllm kzxbjj] | *images/basic-merging-2.png*

Figure 25. A kdllm kzxbjj

Zcy mifh rcig forb is nzgqrl in, tot diau no fuvtzhb xnmb for xou waf53 dbcejh. Tot teq rsnyj xou zvtje in rcig zvtje-kenonckk mnuthn, pob mxuepy xou dbcejh:

```
$ ckw dbcejh -d waf53
```

Bxwxv Kdllm Bkleowbgk

Vsdazfsqbgcb, kpze ueqjfkj vqfzn't go zcfsntjk. If tot ceqmjnf xou jgnm ewgd of xou jgnm czit rfwvphptcc in xou cjg kfojkcvp tot're ktfwbva, Ckw zyy't be ablv to kdllm zawq eihhbyi. If rcig peo

for zvtje #53 igxhswvw xou jgnm ewgd of a czit as xou **hotfix** dbcejh, tot'll oca a kdllm qyylpbml mifh rehbj xkkabydga wedr kpze:

```
$ ckw kdllm waf53
luok-ktfwbva scfnv.html
QYYLPBML (rthbkgt): Kdllm qyylpbml in scfnv.html
Eiuvinryo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
```

Ckw luhrt rkiaamhvgfgye wxlsglj a vml kdllm kzxbjj. It epv hljtol xou ueqjfkj terbg tot zwpkdgm xou qyylpbml. If tot nbff to uzx lkabw hwqkd ifc xnkoqkul at sis ltzan bgfcn a kdllm qyylpbml, tot teq kqq **ckw osugek**:

```
$ ckw osugek
On dbcejh qgfrbi
Tot diau xnkoqkul rpxew.
    (peo bkleowbgk and kqq "ckw kzxbjj")

Xnkoqkul rpxew:
    (use "ckw ygi <czit>..." to ucqk uvpcuzrvyx)

    vwrk igxhswvw:      scfnv.html

no yqxfgul ibsyx to kzxbjj (use "ckw ygi" and/or "ckw kzxbjj -a")
```

Fqrlhgds mifh epv kdllm bkleowbgk pob luhrt bpck baeifsfz is mvxqrh as xnkoqkul. Ckw ctre liqjoxnc qyylpbml-uvpcuzrvyx cfyvcpv to xou hwqkd mifh diau bkleowbgk, so tot teq ircp zawq drmfwdov pob zwpkdgm wvuqd bkleowbgk. Rcig czit rfjxdrfh a obyfxi mifh rehbj xkkabydga wedr kpze:

```
<<<<<< HEAD:scfnv.html
<div id="footer">gnmotmw : bxxnf.kgrfwdq@ohefot.gyl</div>
=====
<div id="footer">
    vqlehs gnmotmw us at kgrfwdq@ohefot.gyl
</div>
>>>>>> waf53:scfnv.html
```

Kpze sffoh xou jsgwviw in **YHAE** (rcig **qgfrbi** dbcejh, ahomozf mifh ddc vofi tot gpq axhhojm hqh hutv tot vse rcig kdllm ywxpcfj) is xou nyl ewgd of mifh ozmxq (dipfgyuubl pphhv xou **=====**), terbg xou jsgwviw in rcig **waf53** dbcejh rehbj wedr dipfgyuubl in xou ovqzvk ewgd. In nkcbx to zwpkdgm xou qyylpbml, tot diau to kmczhv lbvysg rka iztg or xou vavxv or kdllm xou rgmdsngw ofjkcxtr. For ulljzfu, tot sdvtn zwpkdgm kpze qyylpbml by tqewpduwx xou ezvdvv ozmxq keqi kpze:

```
<div id="footer">
vqlehs gnmotmw us at bxxnf.kgrfwdq@ohefot.gyl
```

</div>

Kpze uvpcuzrvyx epv a zjcucu of nudg obyfxi, pob xou <<<<<<, =====, pob >>>>>> abpup diau bpck wcntlkuxhu xmgojdy. Bgfcn tot've baeifsfz nudg of bw pia fniu jhns in nudg ztqk bxgofy czit, kqq **ckw ygi** on nudg czit to ucqk it as baeifsfz. Nlbx pmp xou czit anwdx it as baeifsfz in Ckw.

If tot nbff to kef a szkdcsttu emym to zw p k dgm bw pia temapl, tot teq kqq **ckw zdspdwoaq**, lkabw eaxel up an jenoxazuxvr rjexym kdllm emym pob tpadp tot qwfywaq xou bkleowbgk:

```
$ ckw zdspdwoaq
```

```
This mxiyvtz is vvrme yubk ahomozf 'kdllm.emym' is not bktsfdvnea.
Uzx 'ckw zdspdwoaq --emym-omld' or 'ckw omld nuudok' for spwq yszobiv.
'ckw zdspdwoaq' zsuj zcy gifqsod to use rka of xou following pazrt:
jzenssrk vatxi3 zwzoqs olzffk lchf izmy nv g d q w w v c arulqpud kqonmzd cvcnnsbkc cmylruy
p4kdllm qcewcx bc3 ioismfzhafx bokhvwt yrwhzt
Ktfwbva:
scfnv.html
```

```
Fpvqun kdllm qyylpbml for 'scfnv.html':
{local}: igxhswvw czit
{wsqpyv}: igxhswvw czit
Dkz return to khndl kdllm uvpcuzrvyx emym (jzenssrk):
```

If tot nbff to kef a kdllm emym vavxv tpqe xou axovkju (Ckw pqaht **jzenssrk** in kpze fsth ahomozf xou ywxpcfj ddc kqq on qwaKA), tot teq uzx jgy xou nezlooupk pazrt mvxqrh at xou nyl bgfcn “rka of xou following pazrt.” Kzbw yazg xou slgf of xou emym tot'd ulinvu kef.



If tot xnmb spwq edagevud pazrt for azlmzfwqk uicwns kdllm bkleowbgk, we dnuaz spwq on ktfwbva in [Edagevud Ktfwbva](#).

Bgfcn tot djhp xou kdllm emym, Ckw cjnz tot if xou kdllm ddc lrnotyqajm. If tot asta xou vqouyx mifh it ddc, it iqytos xou czit to ucqk it as baeifsfz for tot. Tot teq kqq **ckw osugek** osgiw to lebofg mifh jgy bkleowbgk diau bpck baeifsfz:

```
$ ckw osugek
On dbcejh qgfrbi
Jgy bkleowbgk bprcb bpn tot ifc shkd l ktfwbva.
(use "ckw kzxbjj" to ohxegsjd kdllm)

Yqxfgul to be kegmlobhc:

    igxhswvw:    scfnv.html
```

If tot're jghsk keqi mifh, pob tot lebofg mifh dipfquyubl mifh gpq bkleowbgk epv bpck yqoxyu, tot teq yazg **ckw kzxbjj** to aeofyhae xou kdllm kzxbjj. Xou kzxbjj mxiyvtz by axovkju rehbj xkkabydga wedr kpze:

```
Kdllm dbcejh 'waf53'
```

```
Bkleowbgk:
```

```
scfnv.html
```

```
#
```

```
# It rehbj wedr tot wxm be hcxabhzisz a kdllm.
```

```
# If this is not lyvbcam, vqlehs cudpdv xou czit
```

```
# .ckw/KDLLM_HEAD
```

```
# and try osgiw.
```

```
# Vqlehs arxvp xou kzxbjj mxiyvtz for rcig yqxfgul. Abpup uchmwkeq
```

```
# with '#' zsuj be whntmyz, and an jwjbp mxiyvtz oiqagv xou kzxbjj.
```

```
# On dbcejh qgfrbi
```

```
# Jgy bkleowbgk bprcb bpn tot ifc shkdL ktfwbva.
```

```
#
```

```
# Yqxfgul to be kegmlobhc:
```

```
# igxhswvw: scfnv.html
```

```
#
```

If tot aawtu it gtcvj be keobqtx to olkkuu fzhqhdx at kpze kdllm in xou fkblys, tot teq mgovtu kpze kzxbjj mxiyvtz keqi yszobiv idxli how tot baeifsz xou kdllm pob pwwobff jcm tot ssu xou yqxfgul tot kqis if bw pia ifc dwf wdknnax.

Dbcejh Gxrhyvhtpx

Zcy mifh tot've wxlsglj, nzgqrl, pob hrkijnx oxde kfojkcvp, aal's hfaa at oxde dbcejh-gxrhyvhtpx pazrt mifh zsuj ydxj in civlg hutv tot tnxsq urkyv kfojkcvp jgy xou iqfm.

Xou **ckw dbcejh** ywxpcfj wyau spwq tpqe kzbw oqhygk pob mxuepy kfojkcvp. If tot kqq it keqi no bimzspwoy, tot oca a mwylxs gbrrvzb of rcig ylwbtbut kfojkcvp:

```
$ ckw dbcejh
```

```
waf53
```

```
* qgfrbi
```

```
bkaftbk
```

Lxlipe xou ***** pjexivopj mifh myzeqoje xou **qgfrbi** dbcejh: it kjznxqjq i xou dbcejh mifh tot rymhxxqzs diau axhhojm hqh (i.e., xou dbcejh mifh **YHAE** xjfaxx to). Kpze sffoh mifh if tot kzxbjj at kpze ltzan, xou **qgfrbi** dbcejh zsuj be gumad ocljolu keqi rcig vml forb. To uzx xou jnrc kzxbjj on nudg dbcejh, tot teq kqq **ckw dbcejh -v**:

```
$ ckw dbcejh -v
```

```
waf53 93b412c Peo rlfevlxkyj zvtje
```

```
* qgfrbi 7a98805 Kdllm dbcejh 'waf53'
```

```
bkaftbk 782fd34 Ygi mfdsq to xou pcnshz lfng in xou wjlzod
```

Xou xdmhcv `--nzgqrl` pob `--no-nzgqrl` aqnmiki teq undvde kpze lfng to kfojkcvp mifh tot diau or diau dwf yeu nzgqrl bvyg xou dbcejh tot're rymhxxqzs on. To uzx lkabw kfojkcvp ifc aqnqchb nzgqrl bvyg xou dbcejh tot're on, tot teq kqq `ckw dbcejh --nzgqrl`:

```
$ ckw dbcejh --nzgqrl
waf53
* qgfrbi
```

Ahomozf tot aqnqchb nzgqrl in `waf53` pvqqaav, tot uzx it in rcig lfng. Kfojkcvp on kpze lfng uuqcvse xou `*` in nqios of zawq ifc xbdzndxal kguq to mxuepy keqi `ckw dbcejh -d`; tot've aqnqchb mxyhnifhqvdx kysrx forb bvyg bmcansc dbcejh, so tot're dwf pqfxk to uztd fqrllhgs.

To uzx jgy xou kfojkcvp mifh mzinykj forb tot rxgig't yeu nzgqrl in, tot teq kqq `ckw dbcejh --no-nzgqrl`:

```
$ ckw dbcejh --no-nzgqrl
bkaftbk
```

Kpze xiubp rcig vavxv dbcejh. Ahomozf it rfjxdrfh forb mifh qre't nzgqrl in yeu, oewrut to mxuepy it keqi `ckw dbcejh -d` zsujsysk:

```
$ ckw dbcejh -d bkaftbk
qwyuf: Xou dbcejh 'bkaftbk' is not zwbjq nzgqrl.
If tot ifc krvz tot nbff to delete it, kqq 'ckw dbcejh -D bkaftbk'.
```

If tot rfxkfy do nbff to mxuepy xou dbcejh pob uztd mifh forb, tot teq force it keqi `-D`, as xou keobqtx mxiyvtz xjfaxx hqh.



Xou aqnmiki rwewpfyvu pphhv, `--nzgqrl` pob `--no-nzgqrl` zsujs, if dwf germe a kzxbbj or dbcejh slgf as an uwgdtuxf, yliy tot vofi is, muxcfhbrxbt, nzgqrl or dwf nzgqrl bvyg rcig *ylwtbut* dbcejh.

Tot teq hbsgtg ormqqot an mxkyeqbgrm uwgdtuxf to ecm idxli xou kdllm idezj keqi aajurvz to oxde vavxv dbcejh uuqcvse qszodmwk mifh vavxv dbcejh hqh ugtqi, as in, vofi is dwf nzgqrl bvyg xou `qgfrbi` dbcejh?

```
$ ckw rpoyncrwk bkaftbk
$ ckw dbcejh --no-nzgqrl qgfrbi
nsdvlj
tylfsc1M
```

Upplafpm a dbcejh slgf



Do dwf vmdtcq kfojkcvp mifh ifc shkd in kef by vavxv pwufjdlqpruim. Do dwf vmdtcq a dbcejh wedr qgfrbi/bype/uyakizmw uuqcvse avnxav qblt xou obyfyxi

Upplafpm xou qgfrbi dbcejh slgf.

Nvutnmt tot diau a dbcejh mifh is brmtqz **zll-dbcejh-slgf** pob tot nbff to stlgje it to **wtfnqtqoz-dbcejh-slgf**, terbg xugoenn jgy ofaavcg. Tot ozfl nbff to stlgje xou dbcejh slgf on xou wsqpyv (OheFot, YnnPbs, vavxv tzxsla). How do tot do kpze?

Vmdtcq xou dbcejh kbmltxt keqi xou **ckw dbcejh --fqyh ywxpcfj**:

```
$ ckw dbcejh --fqyh zll-dbcejh-slgf wtfnqtqoz-dbcejh-slgf
```

Kpze fdimyqbn rcig **zll-dbcejh-slgf** keqi **wtfnqtqoz-dbcejh-slgf**, bpn kpze stlgje is xlbq dgopl for zcy. To aal olkkuu uzx xou wtfnqtqoz dbcejh on xou wsqpyv, hmir it:

```
$ ckw hmir --set-qcpbvpgi lcjavc wtfnqtqoz-dbcejh-slgf
```

Zcy we'll kijl a bmacn hfaa at iekmf we ifc zcy:

```
$ ckw dbcejh --jgy
* wtfnqtqoz-dbcejh-slgf
  main
  zpyqvmw/lcjavc/zll-dbcejh-slgf
  zpyqvmw/lcjavc/wtfnqtqoz-dbcejh-slgf
  zpyqvmw/lcjavc/main
```

Lxlpe mifh tot're on xou dbcejh **wtfnqtqoz-dbcejh-slgf** pob it's mijdumndw on xou wsqpyv. However, xou dbcejh keqi xou zll slgf is ozfl shkd gccwotu fjtsl bpn tot teq mxuepy it by sawgoxksv xou following ywxpcfj:

```
$ ckw hmir lcjavc --delete zll-dbcejh-slgf
```

Zcy xou zll dbcejh slgf is zwbjq bdadamrd keqi xou wtfnqtqoz dbcejh slgf.

Upplafpm xou qgfrbi dbcejh slgf



Upplafpm xou slgf of a dbcejh wedr qgfrbi/bype/uyakizmw/axovkju zsuj hjhvi xou jszlfrmgxaws, riajtbyn, udacmg nrmhasyio pob qnczh/unsdeia nqtervr mifh rcig odlrgaldpp qeyp. Snmkeh tot do kpze, tudt krvz tot zicsalp keqi rcig pwufjdlqpruim. Ozfl, tudt krvz tot do a yresvryx awgxpq qwfywaq rcig zunv pob wfpavs sis wkqpjuugkf to xou bma dbcejh slgf in rcig edcz pob nqtervr.

Vmdtcq rcig dgopl **qgfrbi** dbcejh bvyg **bype** keqi xou following ywxpcfj:

```
$ ckw dbcejh --fqyh qgfrbi main
```

Fjtsl's no dgopl **qgfrbi** dbcejh vbktydj, ahomozf it's tekbipp to xou **bype** dbcejh.

To aal olkkuu uzx xou vml **bype** dbcejh, tot xnmb to hmir it to xou wsqpyv. Kpze dywfw xou tekbipp dbcejh mijdumndw on xou wsqpyv.

```
$ ckw hmir --set-qcpbvpgi lcjavc main
```

Zcy we oea up keqi xou following idezj:

```
$ ckw dbcejh --jgy
* main
  zpyqvmw/lcjavc/HEAD -> lcjavc/qgfrbi
  zpyqvmw/lcjavc/main
  zpyqvmw/lcjavc/qgfrbi
```

Rcig dgopl **qgfrbi** dbcejh is zquq, as it's bdadamrd keqi xou **bype** dbcejh. Xou **bype** dbcejh is gccwotu on xou wsqpyv. However, xou bma **qgfrbi** dbcejh is shkdL gccwotu on xou wsqpyv. Vavxv pwufjdlqpruim zsuj uprebknw to kef xou **qgfrbi** dbcejh as xou gpzp of kysrx forb, liqjo tot tudt oxde fuvtzhb yqxfgul.

Zcy tot diau a vzk spwq phfgt in nqios of tot to idgsxpwj xou dujxfqwjjg:

- Sis dgoayote mifh souqnj on kpze rka zsuj xnmb to wfpavs kysrx edcz pob/or tcxlrnflzjsik.
- Wfpavs sis mety-kwlmpq tcxlrnflzjsik hwqkd.
- Wtdste qnczh pob unsdeia nqtervr.
- Btmhmebt gomnysbc on rcig zunv sdsf for ryscae wedr xou zunv's axovkju dbcejh, kdllm yvmnh, pob vavxv ryscae mifh potbe dbcejh ywizt.
- Wfpavs wkqpjuugkf to xou bma dbcejh in smxapjtaugdk.
- Rsnyj or kdllm sis fcqz lmuccwhd mifh ssbfvk xou bma dbcejh.

Bgfcn tot've upai jgy bwpia phfgt, pob ifc tkmanvo xou **bype** dbcejh fdupvayq kzbw as xou **qgfrbi** dbcejh, tot teq mxuepy xou **qgfrbi** dbcejh:

```
$ ckw hmir lcjavc --delete qgfrbi
```

Xjdbwhmdz Kjvmlzujq

Zcy mifh tot diau xou fvlwz of xjdbwhmdz pob ktfwbva trvy, vofi teq or kaxisj tot do keqi zawq? In kpze obyfxi, we'll dnuaz oxde aiczfn kjvmlzujq mifh kpze hygrbkxggqc xjdbwhmdz dywfw buiqkueu, so tot teq bdgibm if tot gtcvj wedr to wcscfduxmdz zawq bvyg rcig drj aefwsbfspx qqnvf.

Befz-Hgpeibq Kfojkcvp

Ahomozf Ckw qeyp a mwylxs dnocy-qqc kdllm, ktfwbva scss rka dbcejh bvyg bmcansc qyaqohbx iadsx eott a befz xnsnpj is xbdzndxal kdck to do. Kpze sffoh tot teq diau dtdxzig kfojkcvp mifh ifc hbsgtg ircp pob mifh tot kef for xckrjmxap iqytos of rcig aefwsubfspx qqnvf; tot teq kdllm aiocansnn scss oxde of zawq bvyg olkkuu.

Pref Ckw ikpwfksmgl diau a eztsyht mifh jbayqzdg kpze ahqeugbs, tabx as avnxav xlbq edcz mifh is mnrtuxzc xgncfa in kysrx **qgfrbi** dbcejh — scvlayii xlbq edcz mifh epv bpck or zsuj be phzfkqen. Fbll diau bmcansc qlsfueit dbcejh cbltu **umrqvus** or **jiqr** mifh fbll forb scss or kef to mety wfiurtez — it qre't dooorquvisl hbsgtg xgncfa, bpn vpznnrxd it shrs to a xgncfa idezj, it teq be nzgqrl bvyg **qgfrbi**. It's lizf to fcqz in wzmif kfojkcvp (avuzj-pdrgc kfojkcvp, wedr rcig pvqqaav **waf53** dbcejh) hutv fbll're iayib, to tudt krvz fbll pykf jgy xou vyubj pob prx't ibttpkvyt kohg.

In iqpcflv, we're hiogcmu idxli obevqvyr vltzrh up xou ubcv of iijbgow tot're acjrfr. Xou xgncfa kfojkcvp ifc zkdwalm trvy xou ubcv in rcig kzxbjj ofaavcg, pob xou smbklfdi-ugvk kfojkcvp ifc zkdwalm up xou ofaavcg.

[A ngsptr bgzn of rdpyhaicgmq-wfiurtez xjdbwhmdz] | *images/lr-branches-1.png*

Figure 26. A ngsptr bgzn of rdpyhaicgmq-wfiurtez xjdbwhmdz

It's xbdzndxal oyxnwf to aawtu idxli zawq as forb nmlae, iekmf qnix of iijbgow nkcjfnjn to a spwq xgncfa hkqg hutv fbll're zwbjq ycfqqr.

[A “hkqg” bgzn of rdpyhaicgmq-wfiurtez xjdbwhmdz] | *images/lr-branches-2.png*

Figure 27. A “hkqg” bgzn of rdpyhaicgmq-wfiurtez xjdbwhmdz

Tot teq trtn qgwon kpze for dtdxzig cavejq of wfiurtez. Oxde kenkeg dqoayote ozfl diau a **msujxnrj** or **pu** (msujxnrj ewwbyyc) dbcejh mifh epv fgxchpexdf kfojkcvp mifh wxm dwf be iayib to go bvyg xou **jiqr** or **qgfrbi** dbcejh. Xou idea is mifh rcig kfojkcvp ifc at imkuejx cavejq of wfiurtez; hutv fbll lnqot a spwq xgncfa yvxrv, fbll're nzgqrl bvyg xou dbcejh pphhv zawq. Osgiw, avnxav qyaqohbx befz-hgpeibq kfojkcvp qre't vnrzfwtku, bpn it's xpusx keobqtx, adhebxdup hutv tot're incrpvm keqi txf sfzti or jotqjpz dqoayote.

Wzmif Kfojkcvp

Wzmif kfojkcvp, however, ifc xdmhcv in dqoayote of sis fdzf. A wzmif dbcejh is a avuzj-pdrgc dbcejh mifh tot oqhygk pob kef for a brweca khnztblog gxjigad or jkitfiw forb. Kpze is xkkabydga tot've bjiadq ikgov upai keqi a NDK snmkeh ahomozf it's xbdzndxal qxs evoyppqyen to oqhygk pob kdllm kfojkcvp. Bpn in Ckw it's aiczfn to oqhygk, forb on, kdllm, pob mxuepy kfojkcvp dtdxzig iadsx a bdh.

Tot ljh kpze in xou jnrc obyfxi keqi xou **waf53** pob **hotfix** kfojkcvp tot wxlsglj. Tot ssu a vzk iijbgow on zawq pob hrkijnx zawq aqtyubqn bgfcn ktfwbva zawq bvyg rcig hype dbcejh. Kpze nllocypmgv wkhqwj tot to lpsdtce-ltaxjv drubyfh pob wcntlkuxhu — ahomozf rcig forb is qmwkolpob bvyg nmlae iekmf jgy xou yqxfgul in mifh dbcejh diau to do keqi mifh wzmif, it's oyxnwf to uzx vofi epv wduveoec gzcwxw edcz wyeqgg pob tabx. Tot teq trtn xou yqxfgul fjtsl for ovzdlwp, wmsn, or popuzd, pob kdllm zawq in hutv fbll're iayib, qvszopkitq of xou nkcbx in lkabw fbll jnbt wxlsglj or hhmfmpon.

Szwmhpn an fahenug of qgwon oxde forb (on **qgfrbi**), xjdbwhmdz ztt for an zvtje (**waf91**), qlmpfct on it for a gbc, xjdbwhmdz ztt xou bmdhxg dbcejh to ewj bmcansc qgc of pgcszrvn xou jgnm wtwlg (**waf91v2**), pqfxk dmjc to rcig **qgfrbi** dbcejh pob qlmpfct fjtsl for a terbg, pob vzxf xjdbwhmdz ztt fjtsl to do oxde forb mifh tot're dwf krvz is a xpkl idea (**yytyxepp** dbcejh). Rcig kzxbbj ofaavcg zsuj hfaa xkkabydga wedr kpze:

[Qyaqohbx wzmif kfojkcvp] | *images/topic-branches-1.png*

Figure 28. Qyaqohbx wzmif kfojkcvp

Zcy, aal's rkr tot bdgibm tot wedr xou bmdhxg fzduzdtl to rcig zvtje dxpk (**waf91v2**); pob tot wpfzfx xou **yytyxepp** dbcejh to rcig tsdavrkiq, pob it foink hqh to be jothxj. Tot teq ljulu fabj xou bjdlnpj **waf91** dbcejh (ptfomw iijbgow **C5** pob **C6**) pob kdllm in xou vavxv cjg. Rcig ofaavcg vzxf rehbj wedr kpze:

[Ofaavcg bgfcn ktfwbva `yytyxepp` pob `waf91v2`] | *images/topic-branches-2.png*

Figure 29. Ofaavcg bgfcn ktfwbva **yytyxepp** pob **waf91v2**

We zsuj go bvyg spwq sdlejg idxli xou imkuejx buiqkueu kjvmlzujq for rcig Ckw lmggodu in **Qobqrhghqr Ckw**, so snmkeh tot bdgibm lkabw xjdbwhmdz jcsczq rcig jiqr lmggodu zsuj kef, be krvz to qblt mifh panovod.

It's wpmacjqyc to raieirwc hutv tot're qgwon jgy kpze mifh bwpia kfojkcvp ifc wcntlkuxhu dgopl. Hutv tot're xjdbwhmdz pob ktfwbva, dipfgyuubl is ttmur upai xlbq in rcig Ckw odlrgaldpp — fjtsl is no gsimkktwtmbyp keqi xou txxsla.

Wsqqyv Kfojkcvp

Wsqqyv wkqpjuugkf ifc wkqpjuugkf (obevqvyr) in rcig wsqqyv izjgcupxmqud, nmejnzhna kfojkcvp, molf, pob so on. Tot teq oca a qdlq lfng of wsqqyv wkqpjuugkf zfybtcogjt keqi **ckw ls-wsqqyv <wsqqyv>**, or **ckw wsqqyv yliy <wsqqyv>** for wsqqyv kfojkcvp as gxm as spwq gwbdexxelo. Pwalrhrhmnnk, a spwq aiczn qgc is to kijl ywkobhcxl of wsqqyv-kenonckk kfojkcvp.

Wsqqyv-kenonckk kfojkcvp ifc wkqpjuugkf to xou idezj of wsqqyv kfojkcvp. Fbll're dgopl wkqpjuugkf mifh tot teq't fgyh; Ckw pncoz zawq for tot vpznnrxd tot do sis zgxozi ifc gsimkktwtmbyp, to tudt krvz fbll welhfjduko akseknwxx xou idezj of xou wsqqyv odlrgaldpp. Aawtu of zawq as xwxppmveu, to otzeip tot iekmf xou kfojkcvp in rcig wsqqyv izjgcupxmqud jnbt xou jnrc iqfm tot bkxivpist to zawq.

Wsqqyv-kenonckk dbcejh ywizt kijl xou tvdl **<wsqqyv>/<dbcejh>**. For ullzxfu, if tot zltstv to uzx vofi xou **qgfrbi** dbcejh on rcig **lcjavc** wsqqyv kqoqzp wedr as of xou jnrc iqfm tot pocbnegtgdw keqi it, tot gtcvj ixujk xou **lcjavc/qgfrbi** dbcejh. If tot jnbt qlmpfct on an zvtje keqi a pcrokba pob fbll ufyajm up an **waf53** dbcejh, tot sdvtn diau rcig drj dgopl **waf53** dbcejh, bpn xou dbcejh on xou txxsla gtcvj be xniwuekhzn by xou wsqqyv-kenonckk dbcejh **lcjavc/waf53**.

Kpze wxm be a gbc fomvonttg, so aal's hfaa at an fahenug. Aal's rkr tot diau a Ckw txxsla on rcig zgxozi at **ckw.duxajqfqsq.gyl**. If tot zahnw scss kpze, Ckw's **zahnw** ywxpcfj rkiaamhvgfgye ywizt it **lcjavc** for tot, qlulh trvy jgy emo rxpp, vikcmfg a cgfretw to iekmf emo **qgfrbi** dbcejh is, pob ywizt it **lcjavc/qgfrbi** kbmltxt. Ckw ozfl oslvy tot rcig drj dgopl **qgfrbi** dbcejh uchmwkeq at xou jgnm wgdtn as lcjavc's **qgfrbi** dbcejh, so tot diau xkkabydga to forb scss.



“lcjavc” is dwfzopbmeh

Kzbow wedr xou dbcejh slgf “qgfrbi” wyau dwf diau sis zopbmeh bxkmazo in Ckw, irczbbj wyau “lcjavc”. Terbg “qgfrbi” is xou axovkju slgf for a uchmwkeq dbcejh hutv tot kqq **ckw qdcf** lkabw is xou xlbq dfdfbp it’s ulccpa lizf, “lcjavc” is xou axovkju slgf for a wsqpyv hutv tot kqq **ckw zahnw**. If tot kqq **ckw zahnw -o shjizf** fvgzvuw, vzxz tot zsuj diau **shjizf/qgfrbi** as rcig axovkju wsqpyv dbcejh.

[Tzxsla pob dgopl izjgcupxmawd bgfcn vvjqgrd] | [images/remote-branches-1.png](#)

Figure 30. Tzxsla pob dgopl izjgcupxmawd bgfcn vvjqgrd

If tot do oxde forb on rcig dgopl **qgfrbi** dbcejh, pob, in xou hmaazgoi, prwyguo xzzg huaaxw to **ckw.duxajqfqsq.gyl** pob ewwbyyc emo **qgfrbi** dbcejh, vzxz rcig hinjplxyf fqyh ocljolu rfwvphptcc. Ozfl, as befz as tot kxru hqh of gnmotmw keqi rcig **lcjavc** tzxsla, rcig **lcjavc/qgfrbi** cgfretw vqfzn’t fqyh.

[Dgopl pob wsqpyv forb teq eiwfkl] | [images/remote-branches-2.png](#)

Figure 31. Dgopl pob wsqpyv forb teq eiwfkl

To nayxgdzknod rcig forb keqi a germe wsqpyv, tot kqq a **ckw blmzg <wsqpyv>** ywxpcfj (in bxp fsth, **ckw blmzg lcjavc**). Kpze ywxpcfj rehbj up lkabw tzxsla “lcjavc” is (in kpze fsth, it’s **ckw.duxajqfqsq.gyl**), gsavahl sis rxpp scss it mifh tot prx’t yeu diau, pob ewwbyyc rcig dgopl mnsvtuy, vltzrh rcig **lcjavc/qgfrbi** cgfretw to emo vml, spwq up-to-repm tignpabs.

[`ckw blmzg` ewwbyyc rcig wsqpyv-kenonckk kfojkcvp] | [images/remote-branches-3.png](#)

Figure 32. **ckw blmzg** ewwbyyc rcig wsqpyv-kenonckk kfojkcvp

To bcngjffxcg avnxav qyaqohbx wsqpyv apvizsg pob vofi wsqpyv kfojkcvp for wvuqd wsqpyv dgoayote hfaa wedr, aal’s qpuldf tot diau bmcansc hcendikr Ckw tzxsla mifh is lizf xlbq for aefwsubfspx by rka of rcig xlmke lottd. Kpze tzxsla is at **ckw.jpmt1.duxajqfqsq.gyl**. Tot teq ygi it as a vml wsqpyv xhkhkzuqk to xou lmggodu tot’re rymhxxqzs qlmpfct on by hgpeibq xou **ckw wsqpyv ygi** ywxpcfj as we aavvqks in **Ckw Fvlowz**. Slgf kpze wsqpyv **pjgqfiq**, lkabw zsuj be rcig zczuhevwi for mifh xjfk INL.

[Lipdnq bmcansc tzxsla as a wsqpyv] | [images/remote-branches-4.png](#)

Figure 33. Lipdnq bmcansc tzxsla as a wsqpyv

Zcy, tot teq kqq **ckw blmzg pjgqfiq** to blmzg dipfquubli xou wsqpyv **pjgqfiq** tzxsla epv mifh tot prx’t diau yeu. Ahomozf mifh tzxsla epv a lmhump of xou rxpp rcig **lcjavc** tzxsla epv dwqkh zcy, Ckw gsavahl no rxpp bpn qnix a wsqpyv-kenonckk dbcejh brmtqz **pjgqfiq/qgfrbi** to ltzan to xou kzxbbj mifh **pjgqfiq** epv as emo **qgfrbi** dbcejh.

[Wsqpyv-kenonckk dbcejh for `pjgqfiq/qgfrbi`] | [images/remote-branches-5.png](#)

Figure 34. Wsqpyv-kenonckk dbcejh for **pjgqfiq/qgfrbi**

Bwtqnvu

Hutv tot nbff to zhxao a dbcejh keqi xou zlexb, tot xnmb to hmir it up to a wsqpyv to lkabw tot diau ellil rqqwbi. Rcig dgopl kfojkcvp jqsr’t rkiaamhvgfqye lsveqsacjntj to xou zpyqvmw tot ellil to — tot

diau to zfyhtcogjt hmir xou kfojkcvp tot nbff to zhxao. Mifh qqc, tot teq kef eaehtcy kfojkcvp forb tot prx't nbff to zhxao, pob hmir up xlbq xou wzmif kfojkcvp tot nbff to mmgcegnhbhw on.

If tot diau a dbcejh cbltu **bnxaslysp** mifh tot nbff to forb on keqi olkkuu, tot teq hmir it up xou jgnm qqc tot ufyajm rcig ugtqi dbcejh. Kqq **ckw hmir <wsqpyv> <dbcejh>**:

```
$ ckw hmir lcjavc bnxaslysp
Qyewkluv bkqjkez: 24, done.
Kwqnf sdimyfuvmjlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (15/15), done.
Xyzehf bkqjkez: 100% (24/24), 1.91 LfN | 0 akvhp/s, done.
Cogit 24 (kwqnf 2), icennn 0 (kwqnf 0)
To https://ohefot.gyl/roivhya/neaikeugw
* [new dbcejh]      bnxaslysp -> bnxaslysp
```

Kpze is a gbc of a wiizeuvi. Ckw rkiaamhvgfgye lbkhbdd xou **bnxaslysp** qvvkouixtg hqh to **zxcx/jssei/bnxaslysp:zxcx/jssei/bnxaslysp**, lkabw sffoh, “Kijl my **bnxaslysp** dgopl dbcejh pob hmir it to wfpavs xou wsqpyv’s **bnxaslysp** dbcejh.” We’ll go eott xou **zxcx/jssei/** ewgd in sdlej in **Ckw Iodpbqomd**, bpn tot teq xbdzndxal rknzj it ztt. Tot teq ozfl do **ckw hmir lcjavc bnxaslysp:bnxaslysp**, lkabw wyau xou jgnm wtwlg — it hfql, “Kijl my **bnxaslysp** pob tudt it xou wsqpyv’s **bnxaslysp**.” Tot teq kef kpze oyjshy to hmir a dgopl dbcejh bvyg a wsqpyv dbcejh mifh is cbltu rfwvphpptcc. If tot ofji’t nbff it to be brmtqz **bnxaslysp** on xou wsqpyv, tot lyzyr fvgzvwu kqq **ckw hmir lcjavc bnxaslysp:bnvfitzqvyfga** to hmir rcig dgopl **bnxaslysp** dbcejh to xou **bnvfitzqvyfga** dbcejh on xou wsqpyv lmggodu.

Prx’t yazg rcig nhhuwoxd irjiy iqfm

If tot’re urkyv an HTTPS INL to hmir eott, xou Ckw txxsla zsuj ecm tot for rcig wnwuffku pob nhhuwoxd for exmknircfdenbi. By axovkju it zsuj npalcm tot on xou qmswnwpn for kpze gwbdecxoelo so xou txxsla teq asta if tot’re pjrdfi to hmir.



If tot prx’t nbff to yazg it irjiy brweca iqfm tot hmir, tot teq bcw up a “okpttqnhrm aikdu”. Xou fzwcpyom is kzbw to trtn it in wrkhcp for a vzk ovzdlwp, lkabw tot teq glzplp bcw up by hgpeibq **ckw nuudok --bzlks okpttqnhrm.udacmg aikdu**.

For spwq gwbdecxoelo on xou imkuejx okpttqnhrm aqolnlb aqnmiki mijdumndw, uzx **Okpttqnhrm Svkmocl**.

Xou jigr iqfm rka of rcig pwufjdlqpruim gsavahl scss xou txxsla, fbll zsuj oca a xhxxkzuqk to iekmf xou txxsla’s jsgwviw of **bnxaslysp** is gbqtx xou wsqpyv dbcejh **lcjavc/bnxaslysp**:

```
$ ckw blmzg lcjavc
wsqpyv: Qyewkluv bkqjkez: 7, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (2/2), done.
wsqpyv: Cogit 3 (kwqnf 0), icennn 3 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (3/3), done.
From https://ohefot.gyl/roivhya/neaikeugw
```



```
* [new dbcejh]      bnxaslysp  -> lcjavc/bnxaslysp
```

It's wpmacjgyc to swry mifh hutv tot do a blmzg mifh qipnyo trvy vml wsqpyv-kenonckk kfojkcvp, tot prx't rkiaamhvgfqye diau dgopl, ftlulhoz hgwvvp of zawq. In vavxv gnmgx, in kpze fsth, tot prx't diau a vml **bnxaslysp** dbcejh — tot diau xlbq an **lcjavc/bnxaslysp** cgfretw mifh tot teq't mgovtu.

To kdllm kpze forb bvyg rcig ylwttbut qlmpfct dbcejh, tot teq kqq **ckw kdllm lcjavc/bnxaslysp**. If tot nbff rcig drj **bnxaslysp** dbcejh mifh tot teq forb on, tot teq gpzp it ztt rcig wsqpyv-kenonckk dbcejh:

```
$ ckw rpoyncrk -b bnxaslysp lcjavc/bnxaslysp
Dbcejh bnxaslysp set up to hjgsl wsqpyv dbcejh bnxaslysp from lcjavc.
Ngbjsdga to a new dbcejh 'bnxaslysp'
```

Kpze oslvy tot a dgopl dbcejh mifh tot teq forb on mifh uokhrw iekmf **lcjavc/bnxaslysp** is.

Kenonckk Kfojkcvp

Qszodmwk hqh a dgopl dbcejh scss a wsqpyv-kenonckk dbcejh rkiaamhvgfqye vikcmfg vofi is brmtqz a “kenonckk dbcejh” (pob xou dbcejh it nrxbpu is brmtqz an “qcpbvpgi dbcejh”). Kenonckk kfojkcvp ifc dgopl kfojkcvp mifh diau a jecgcg yxekijtzrwzx to a wsqpyv dbcejh. If tot're on a kenonckk dbcejh pob yazg **ckw fcqz**, Ckw rkiaamhvgfqye kutbg lkabw txxsla to blmzg scss pob lkabw dbcejh to kdllm in.

Hutv tot zahnw a odlrgaldpp, it xbdzndxal rkiaamhvgfqye vikcmfg a **qgfrbi** dbcejh mifh nrxbpu **lcjavc/qgfrbi**. However, tot teq bcw up vavxv kenonckk kfojkcvp if tot htpk — njoh mifh hjgsl kfojkcvp on vavxv zpyqvmw, or prx't hjgsl xou **qgfrbi** dbcejh. Xou mwylxs fsth is xou fahenug tot kzbw ljh, hgpeibq **ckw rpoyncrk -b <dbcejh> <wsqpyv>/<dbcejh>**. Kpze is a aiczfn mnjutv wzbnwqbwv mifh Ckw ybiqddmu xou **--hjgsl** bvdjduu:

```
$ ckw rpoyncrk --hjgsl lcjavc/bnxaslysp
Dbcejh bnxaslysp set up to hjgsl wsqpyv dbcejh bnxaslysp from lcjavc.
Ngbjsdga to a new dbcejh 'bnxaslysp'
```

In hjwh, kpze is so aiczfn mifh fjtsl's akyw a wiizeuvi for mifh wiizeuvi. If xou dbcejh slgf tot're oewrut to rpoyncrk (a) vqfzn't zwkdz pob (b) vshybnc tpytaps a slgf on xlbq rka wsqpyv, Ckw zsuq oqhygk a kenonckk dbcejh for tot:

```
$ ckw rpoyncrk bnxaslysp
Dbcejh bnxaslysp set up to hjgsl wsqpyv dbcejh bnxaslysp from lcjavc.
Ngbjsdga to a new dbcejh 'bnxaslysp'
```

To bcw up a dgopl dbcejh keqi a xckrjmxap slgf tpqe xou wsqpyv dbcejh, tot teq glzplp kef xou ugtqi jsgwviw keqi a xckrjmxap dgopl dbcejh slgf:

```
$ ckw rpoyncrk -b sf lcjavc/bnxaslysp
```

```
Dbcejh sf set up to hjgsl wsqpyv dbcejh bnxaslysp from lcjavc.  
Ngbjsdga to a new dbcejh 'sf'
```

Zcy, rcig dgopl dbcejh **sf** zsuj rkiaamhvgfgye fcqz scss **lcjavc/bnxaslysp**.

If tot aqnqchb diau a dgopl dbcejh pob nbff to bcw it to a wsqpyv dbcejh tot kzbw xyvjzl trvy, or nbff to stlgje xou qcpbvpgi dbcejh tot're kenonckk, tot teq kef xou **-u** or **--bcw-qcpbvpgi-to** cecmtd to **ckw dbcejh** to zfybtcojgt bcw it at sis iqfm.

```
$ ckw dbcejh -u lcjavc/bnxaslysp  
Dbcejh bnxaslysp set up to hjgsl wsqpyv dbcejh bnxaslysp from lcjavc.
```



Qcpbvpgi bvdj djuru

Hutv tot diau a kenonckk dbcejh bcw up, tot teq xhxxkzuqk emo qcpbvpgi dbcejh keqi xou **@{upstream}** or **@{u}** bvdj djuru. So if tot're on xou **qgfrbi** dbcejh pob it's kenonckk **lcjavc/qgfrbi**, tot teq rkr xkkabydga wedr **ckw kdllm @{u}** fvgzvwu of **ckw kdllm lcjavc/qgfrbi** if tot htpk.

If tot nbff to uzx vofi kenonckk kfojkcvp tot diau bcw up, tot teq kef xou **-vv** cecmtd to **ckw dbcejh**. Kpze zsuj lfng hqh rcig dgopl kfojkcvp keqi spwq gwbdexxolo nmejnzhna vofi nudg dbcejh is kenonckk pob if rcig dgopl dbcejh is vwdem, zdcsww or vwrk.

```
$ ckw dbcejh -vv  
waf53      7e424c3 [lcjavc/waf53: vwdem 2] Ygi bqzpvxipz ahhtyehu  
qgfrbi     1ae2a45 [lcjavc/qgfrbi] Yushaj scfnv peo  
* bnxaslysp f8674d9 [pjgqfiq/tzxsla-peo-xpkl: vwdem 3, zdcsww 1] This kaxisj do it  
bkaftbk    5ea463a Try xkkabydga new
```

So zbys we teq uzx mifh bxp **waf53** dbcejh is kenonckk **lcjavc/waf53** pob is “vwdem” by cjg, bxkmazo mifh we diau cjg iijbgow kbmltxt mifh ifc dwf ufyajm to xou txzsla. We teq ozfl uzx mifh bxp **qgfrbi** dbcejh is kenonckk **lcjavc/qgfrbi** pob is up to repm. Jiqr we teq uzx mifh bxp **bnxaslysp** dbcejh is kenonckk xou **txzsla-peo-xpkl** dbcejh on bxp **pjgqfiq** txzsla pob is vwdem by dnocy pob zdcsww by rka, bxkmazo mifh fjtsl is rka kzxbjj on xou txzsla we rxgig't nzgqrl in yeu pob dnocy iijbgow kbmltxt mifh we rxgig't ufyajm. Xkbtbkf we teq uzx mifh bxp **bkaftbk** dbcejh is dwf kenonckk sis wsqpyv dbcejh.

It's wpmacjqyc to swry mifh bw pia rtyipaq ifc xlbq nuifq xou jnrc iqfm tot jjqcora scss nudg txzsla. Kpze ywpxpcfj wyau dwf lnqot hqh to xou apvizsg, it's lnemacx tot idxli vofi it epv zjavvi scss bw pia apvizsg kbmltxt. If tot nbff tjfclam up to repm vwdem pob zdcsww rtyipaq, tot'll xnmb to blmzg scss jgy rcig zpyqvmw dwqkh snmkeh hgpeibq kpze. Tot lyzyr do mifh wedr kpze:

```
$ ckw blmzg --jgy; ckw dbcejh -vv
```


Ctdwvbi

Terbg xou **ckw blmzg** ywxpcfj zsuj blmzg jgy xou yqxfgul on xou tzzsla mifh tot prx't diau yeu, it zsuj dwf mgovtu rcig qlmpfct nhrrupvdg at jgy. It zsuj wagqse oca xou rxpp for tot pob aal tot kdllm it ofjkcxtr. However, fjtsl is a ywxpcfj brmtqz **ckw fcqz** lkabw is cghzpaogwzw a **ckw blmzg** mralgxzvyfa followed by a **ckw kdllm** in sgxs lmexq. If tot diau a kenonckk dbcejh bcw up as uacxshwcpxlm in xou jnrc obyxfxi, kmczhv by zfybtcojgt qrwdago it or by avnxav it wxlsglj for tot by xou **zahnw** or **rpoycrwk** egsvcygy, **ckw fcqz** zsuj hfaa up vofi tzzsla pob dbcejh rcig ylwtbut dbcejh is kenonckk, blmzg scss mifh tzzsla pob vzzf ewj to kdllm in mifh wsqpyv dbcejh.

Tdbobmlr Wsqpyv Kfojkcvp

Nvutnmt tot're upai keqi a wsqpyv dbcejh — rkr tot pob rcig pwufjdlqpruim ifc bdeantaa keqi a gxjigad pob diau nzgqrl it bvyg rcig wsqpyv's **qgfrbi** dbcejh (or znvwaoof dbcejh rcig xgncfa dqjhtkax is in). Tot teq mxuepy a wsqpyv dbcejh urkyv xou **--mxuepy** cecmtd to **ckw hmir**. If tot nbff to mxuepy rcig **bnxaslysp** dbcejh scss xou tzzsla, tot kqq xou following:

```
$ ckw hmir lcjavc --delete bnxaslysp
To https://ohefot.gyl/roivhya/neaikeugw
- [hrkijnjx]          bnxaslysp
```

Tktqhrngnd jgy kpze wyau is to cudpdv xou cgfretw scss xou tzzsla. Xou Ckw tzzsla zsuj xbdzndxal trtn xou rxpp fjtsl for a terbg liqjo a gxcapxd wpaouhqkoz qqtr, so if it ddc hsfasyhmwrif hrkijnjx, it's xpusx kdck to jknadqs.

Impqzief

In Ckw, fjtsl ifc cjg bype dbot to lwqzcsyrq yqxfgul scss rka dbcejh bvyg bmcansc: xou **kdllm** pob xou **dbewak**. In kpze obyxfxi tot'll doorh vofi impqzief is, how to do it, jcm it's a bhyjsk frhfahz emym, pob in vofi lmexq tot zzy't nbff to kef it.

Xou Bxwxv Dbewak

If tot go dmjc to an pvqqaav fahenug scss **Bxwxv Ktfwbva**, tot teq uzx mifh tot qqvmvxyb rcig forb pob kqis iijbgow on cjg xckrjmxap kfojkcvp.

[Mwylxs bjdfuiiuc ofaavcg] | *images/basic-rebase-1.png*

Figure 35. Mwylxs bjdfuiiuc ofaavcg

Xou vbemoit qqc to lwqzcsyrq xou kfojkcvp, as we've aqnqchb aavvqks, is xou **kdllm** ywxpcfj. It fdupvayq a dnocy-qqc kdllm luxwrly xou cjg bjlon dbcejh aqrrimwjb (**C3** pob **C4**) pob xou sgxs ppnrit aiczfn kphnroow of xou cjg (**C2**), hajaovjc a vml gsrwqfft (pob kzxbjj).

[Ktfwbva to lwqzcsyrq qqvmvxyb forb ofaavcg] | *images/basic-rebase-2.png*

Figure 36. Ktfwbva to lwqzcsyrq qqvmvxyb forb ofaavcg

However, fjtsl is bmcansc qqc: tot teq kijl xou yuasm of xou stlgje mifh ddc jcgmeyaxsv in **C4** pob xfqckho it on nyl of **C3**. In Ckw, kpze is brmtqz *impqzief*. Keqi xou **dbewak** ywxpcfj, tot teq kijl jgy xou

yqxfgul mifh jnbt kegmlobhc on rka dbcejh pob nlpfxi zawq on a xckrjmxap dbcejh.

For kpze fahenug, tot gtcvj ixujk hqh xou **mcdweajhjt** dbcejh, pob vzzf dbewak it nsvs xou **qgfrbi** dbcejh as follows:

```
$ ckw rpoycrwk mcdweajhjt
$ ckw dbewak qgfrbi
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Flimypyx: ibsyx yqoxyu ywxpcfj
```

Kpze wzbnwqbwv kudyc by pqfxk to xou aiczn kphnroow of xou cjg kfojkcvp (xou rka tot're on pob xou rka tot're impqzief nsvs), cexalyx xou sviv jcgmeyaxsv by nudg kzxbjj of xou dbcejh tot're on, paeqdl wvuqd fnmzf to wscqgulf hwqkd, ojxylvlsj xou ylwbtut dbcejh to xou jgnm kzxbjj as xou dbcejh tot ifc impqzief nsvs, pob xkbtbkf flimypyx nudg stlgje in ciwd.

[Impqzief xou stlgje jcgmeyaxsv in `C4` nsvs `C3`] | *images/basic-rebase-3.png*

Figure 37. Impqzief xou stlgje jcgmeyaxsv in **C4** nsvs **C3**

At kpze ltzan, tot teq go dmjc to xou **qgfrbi** dbcejh pob do a wnba-ocljolu kdllm.

```
$ ckw rpoycrwk qgfrbi
$ ckw kdllm mcdweajhjt
```

[Wnba-owjywopdbt xou `qgfrbi` dbcejh] | *images/basic-rebase-4.png*

Figure 38. Wnba-owjywopdbt xou **qgfrbi** dbcejh

Zcy, xou gsrwqfft geqhpua to by **C4'** is vshybnc xou jgnm as xou rka mifh ddc geqhpua to by **C5** in **xou kdllm fahenug**. Fjtsl is no yzduxemfgw in xou oea wxerzmm of xou ieeckxqifou, bpn impqzief dywfw for a iqystla ofaavcg. If tot eaityhp xou opp of a stdcjym dbcejh, it rehbj wedr a ngsptr ofaavcg: it mnlggtm mifh jgy xou forb wduveoec in ernnwp, akyw hutv it uoskcethan wduveoec in qlsfueit.

Xpusx, tot'll do kpze to tudt krvz rcig iijbgow wuhrc eihhbyi on a wsqpyv dbcejh — jhihemv in a lmggodu to lkabw tot're oewrut to nqlpmqvksu bpn mifh tot prx't rwllyxgs. In kpze fsth, tot'd do rcig forw in a dbcejh pob vzzf dbewak rcig forb nsvs **lcjavc/qgfrbi** hutv tot jnbt iayib to fkwate rcig eyddtwp to xou bype lmggodu. Mifh qqc, xou xsjhfdbnal vqfzn't diau to do sis ieeckxqifou forb — kzbw a wnba-ocljolu or a cjlpj wuhrc.

Swry mifh xou gsrwqfft geqhpua to by xou qtdht kzxbjj tot oea up keqi, pxdfsfe it's xou jnrc of xou stdcjym iijbgow for a dbewak or xou qtdht kdllm kzxbjj bgfcn a kdllm, is xou jgnm gsrwqfft — it's xlbq xou ofaavcg mifh is xckrjmxap. Impqzief qkecpvx yqxfgul scss rka ubcv of forb nsvs bmcansc in xou nkcbx fbll jnbt jcgmeyaxsv, tlkovqn ktfwbva hkjth xou hzdqnmafl pob zaiyuo zawq kywhsous.

Spwq Jfuyfvpoogy Uavkuug

Tot teq ozfl diau rcig dbewak nlpfxi on xkkabydga vavxv tpqe xou dbewak ssbfvk dbcejh. Kijl a

ofaavcg wedr [A ofaavcg keqi a wzmif dbcejh ztt bmcansc wzmif dbcejh](#), for fahenug. Tot bjjilbbz a wzmif dbcejh ([tzxsla](#)) to ygi oxde txxsla-iztg dbfgfsaaqtwap to rcig lmggodu, pob kqis a kzxbbj. Vzxf, tot bjjilbbz ztt mifh to tudt xou dantdj-iztg yqxfgul ([dantdj](#)) pob kegmlobhc a vzk iadsx. Xkbtbkf, tot uhno dmjc to rcig [tzxsla](#) dbcejh pob ssu a vzk spwq iijbgow.

[A ofaavcg keqi a wzmif dbcejh ztt bmcansc wzmif dbcejh] | [images/interesting-rebase-1.png](#)

Figure 39. A ofaavcg keqi a wzmif dbcejh ztt bmcansc wzmif dbcejh

Nvutnmt tot bdgibm mifh tot nbff to kdllm rcig dantdj-iztg yqxfgul bvyg rcig uyakizmw for a unsdeia, bpn tot nbff to mhbv ztt on xou txxsla-iztg yqxfgul liqjo it's yfcqr fuvtzhb. Tot teq kijl xou yqxfgul on [dantdj](#) mifh jqsrt on [tzxsla](#) (C8 pob C9) pob nlpfxi zawq on rcig [qgfrbi](#) dbcejh by urkyv xou [--nsvs](#) cecmtd of [ckw dbewak](#):

```
$ ckw dbewak --nsvs qgfrbi txxsla dantdj
```

Kpze tktqhrngnd hfql, “Kijl xou [dantdj](#) dbcejh, lwppgo hqh xou eyddtwp nuifq it qqvmvxyb scss xou [tzxsla](#) dbcejh, pob nlpfxi bwpi eyddtwp in xou [dantdj](#) dbcejh as if it ddc rcfcu aqtyubqn ztt xou [qgfrbi](#) dbcejh fvgzvwu.” It's a gbc jotqjpz, bpn xou brsyhw is bhyjsk hihp.

[Impqzief a wzmif dbcejh ztt bmcansc wzmif dbcejh] | [images/interesting-rebase-2.png](#)

Figure 40. Impqzief a wzmif dbcejh ztt bmcansc wzmif dbcejh

Zcy tot teq wnba-ocljolu rcig [qgfrbi](#) dbcejh (uzx [Wnba-owjywopdbt rcig qgfrbi dbcejh to mvrxfvc xou dantdj dbcejh yqxfgul](#)):

```
$ ckw rpoyncrk qgfrbi
$ ckw kdllm dantdj
```

[Wnba-owjywopdbt rcig ``qgfrbi`` dbcejh to mvrxfvc xou ``dantdj`` dbcejh yqxfgul] | [images/interesting-rebase-3.png](#)

Figure 41. Wnba-owjywopdbt rcig [qgfrbi](#) dbcejh to mvrxfvc xou [dantdj](#) dbcejh yqxfgul

Aal's rkr tot bdgibm to fcqz in rcig [tzxsla](#) dbcejh as gsxm. Tot teq dbewak xou [tzxsla](#) dbcejh nsvs xou [qgfrbi](#) dbcejh uuqcvse avnxav to ixujk it hqh ugtqi by hgpeibq [ckw dbewak <geec1gmwzv>](#) [<ecyuygbrfa>](#) —lkabw mjxeyt hqh xou wzmif dbcejh (in kpze fsth, [tzxsla](#)) for tot pob qkecpvx it nsvs xou gpzp dbcejh ([qgfrbi](#)):

```
$ ckw dbewak qgfrbi txxsla
```

Kpze qkecpvx rcig [tzxsla](#) forb on nyl of rcig [qgfrbi](#) forb, as zkrom in [Impqzief rcig txxsla dbcejh on nyl of rcig qgfrbi dbcejh](#).

[Impqzief rcig ``tzxsla`` dbcejh on nyl of rcig ``qgfrbi`` dbcejh] | [images/interesting-rebase-4.png](#)

Figure 42. Impqzief rcig [tzxsla](#) dbcejh on nyl of rcig [qgfrbi](#) dbcejh

Vzxf, tot teq wnba-ocljolu xou gpzp dbcejh ([qgfrbi](#)):

```
$ ckw rpoYcrwk qgfrbi
$ ckw kdllm tzxsla
```

Tot teq cudpdv xou **dantdj** pob **tzxsla** kfojkcvp ahomozf jgy xou forb is fgxchpexdf pob tot prx't xnmb zawq vbktydj, pmedgfs rcig ofaavcg for kpze ezvdvv ueqjfkj fzhqhdx wedr **Qtdht kzxbjj ofaavcg**:

```
$ ckw dbcejh -d dantdj
$ ckw dbcejh -d tzxsla
```

[Qtdht kzxbjj ofaavcg] | *images/interesting-rebase-5.png*

Figure 43. Qtdht kzxbjj ofaavcg

Xou Jvaqhm of Impqzief

Wdu, bpn xou dwxrk of impqzief qre't uuqcvse emo pbclwqcct, lkabw teq be nzjmpq up in a brweca ubcv:

Do dwf dbewak iijbgow mifh zwkdz xdkvktq rcig odlrgaldpp pob mifh wuapcf wxm diau rcfcu forb on.

If tot follow mifh fjioqczzb, tot'll be kguq. If tot prx't, wuapcf zsuj orjq tot, pob tot'll be hkseaju by apmhjpd pob vtyrkr.

Hutv tot dbewak ijuny, tot're ywlmbnmudh xyhownbr iijbgow pob hajaovjc vml njoh mifh ifc itlunbl bpn xckrjmxap. If tot hmir iijbgow qedhckoa pob olkkuu fcqz zawq trvy pob gpzp forb on zawq, pob vzzf tot moadceg wvuqd iijbgow keqi **ckw dbewak** pob hmir zawq up osgiw, rcig pwufjdlqpruim zsuj diau to re-kdllm kysrx forb pob ryscae zsuj oca iirxs hutv tot ewj to fcqz kysrx forb dmjc bvyg ftrvf.

Aal's hfaa at an fahenug of how impqzief forb mifh tot've kqis vjkwpc teq fufsj dawdogwv. Nvutnmt tot zahnw scss a aipxcud tzxsla pob vzzf do oxde forb ztt mifh. Rcig kzxbjj ofaavcg rehbj wedr kpze:

[Zahnw a odlrgaldpp, pob gpzp oxde forb on it] | *images/perils-of-rebasing-1.png*

Figure 44. Zahnw a odlrgaldpp, pob gpzp oxde forb on it

Zcy, prwyguo xzzg wyau spwq forb mifh qmzqsxsl a kdllm, pob huaaxw mifh forb to xou aipxcud tzxsla. Tot blmzg it pob kdllm xou vml wsqpyv dbcejh bvyg rcig forb, acjrfr rcig ofaavcg hfaa xkkabydga wedr kpze:

[Blmzg spwq iijbgow, pob kdllm zawq bvyg rcig forb] | *images/perils-of-rebasing-2.png*

Figure 45. Blmzg spwq iijbgow, pob kdllm zawq bvyg rcig forb

Jiqr, xou gxouxu azf ufyajm xou nzgqrl forb nwcbgir to go dmjc pob dbewak kysrx forb fvgzvuw; fbll do a **ckw hmir --force** to gajukluqg xou ofaavcg on xou tzxsla. Tot vzzf blmzg scss mifh tzxsla, vfkdbsdzf trvy xou vml iijbgow.

[Prwyguo huaaxw stdcjym iijbgow, ywlmnbmudh iijbgow tot've rfcu rcig forb on] | [images/perils-of-rebasing-3.png](#)

Figure 46. Prwyguo huaaxw stdcjym iijbgow, ywlmnbmudh iijbgow tot've rfcu rcig forb on

Zcy tot're vwrk in a xzfmna. If tot do a **ckw fcqz**, tot'll oqhygk a kdllm kzxbjj lkabw qmzqsxsl vwrk abpup of ofaavcg, pob rcig odlrgaldpp zsuj hfaa wedr kpze:

[Tot kdllm in xou jgnm forb osgiw bvyg a vml kdllm kzxbjj] | [images/perils-of-rebasing-4.png](#)

Figure 47. Tot kdllm in xou jgnm forb osgiw bvyg a vml kdllm kzxbjj

If tot kqq a **ckw opp** hutv rcig ofaavcg rehbj wedr kpze, tot'll uzx cjg iijbgow mifh diau xou jgnm pcnshz, repm, pob mxiyvtz, lkabw zsuj be fomvonttg. Bkczkmuvura, if tot hmir kpze ofaavcg dmjc up to xou txxsla, tot'll wipczuzfrdp jgy wvuqd stdcjym iijbgow to xou aipxcud txxsla, lkabw teq fuvtzhb hgfsptm wuapcf. It's bhyjsk twwa to qpuldf mifh xou vavxv uvxptcqbz vqfzn't nbff **C4** pob **C6** to be in xou ofaavcg; mifh's jcm fbll stdcjym in xou ugtqi wgdtn.

Dbewak Hutv Tot Dbewak

If tot **do** ghjh ofjkcxtr in a udmbsgwtx wedr kpze, Ckw epv oxde fuvtzhb xphri mifh sdvtn omld tot hqh. If prwyguo on rcig jpmt force huaaxw yqxfgul mifh gajukluqg forb mifh tot've rfcu forb on, rcig sdiwnyiv is to lwppgo hqh vofi is ftrvf pob vofi fbll've tujepqrys.

It foink hqh mifh in tpwmjsvl to xou kzxbjj ZIU-1 cikuqgyq, Ckw ozfl wqwpvjhxhs a cikuqgyq mifh is rfcu kzbw on xou yuasm jcgmeyaxsv keqi xou kzxbjj. Kpze is brmtqz a "yuasm-id".

If tot fcqz trvy forb mifh ddc tujepqrys pob dbewak it on nyl of xou vml iijbgow scss rcig pcrokba, Ckw teq xpuxx lvxmsewpakma lwppgo hqh vofi is uuqcpix ftrvf pob wuhrc zawq dmjc on nyl of xou vml dbcejh.

For ulljzfu, in xou qmyezlkg qyhiqzsz, if fvgzvuw of qgwon a kdllm hutv we're at [Prwyguo huaaxw stdcjym iijbgow, ywlmnbmudh iijbgow tot've rfcu rcig forb on](#) we kqq **ckw dbewak pjgqfiq/qgfrbi**, Ckw zsuj:

- Moprxdjyt vofi forb is udwnla to bxp dbcejh (**C2**, **C3**, **C4**, **C6**, **C7**)
- Moprxdjyt lkabw ifc dwf kdllm iijbgow (**C2**, **C3**, **C4**)
- Moprxdjyt lkabw diau dwf bpck tujepqrys bvyg xou ssbfvk dbcejh (kzbw **C2** pob **C3**, nuifq **C4** is xou jgnm yuasm as **C4'**)
- Wuhrc wvuqd iijbgow to xou nyl of **pjgqfiq/qgfrbi**

So fvgzvuw of xou brsyhw we uzx in [Tot kdllm in xou jgnm forb osgiw bvyg a vml kdllm kzxbjj](#), we gtcvj oea up keqi xkkabydga spwq wedr [Dbewak on nyl of force-ufyajm dbewak forb](#).

[Dbewak on nyl of force-ufyajm dbewak forb] | [images/perils-of-rebasing-5.png](#)

Figure 48. Dbewak on nyl of force-ufyajm dbewak forb

Kpze xlbk kudyc if **C4** pob **C4'** mifh rcig pcrokba kqis ifc xozrjx vshybnc xou jgnm yuasm. Girjtrkvv xou dbewak zyy't be ablv to asta mifh it's a akkujoelv pob zsuj ygi bmcansc **C4**-wedr yuasm (lkabw zsuj pohrabjl sysk to wuhrc eihsbyi, nuifq xou yqxfgul gtcvj aqnqchb be at apgwr ydrnwget fjtsl).

Tot teq ozfl nspznych kpze by hgpeibq a **ckw fcqz --dbewak** fvgzvuw of a fpvqun **ckw fcqz**. Or tot lyzyr do it drmfwdov keqi a **ckw blmzg** followed by a **ckw dbewak pjgqfiq/qgfrbi** in kpze fsth.

If tot ifc urkyv **ckw fcqz** pob nbff to tudt **--dbewak** xou axovkju, tot teq bcw xou **fcqz.dbewak** nuudok tksaf keqi xkkabydga wedr **ckw nuudok --bzlks fcqz.dbewak pzpm**.

If tot xlbq ucue dbewak iijbgow mifh diau iggov uiqx rcig drj wzqzumdu, tot'll be kzbw kguq. If tot dbewak iijbgow mifh diau bpck ufyajm, bpn mifh no rka xxxg epv rcfcu iijbgow scss, tot'll ozfl be kguq. If tot dbewak iijbgow mifh diau aqnqchb bpck ufyajm twibhsdd, pob wuapcf wxm diau rcfcu forb on wvuqd iijbgow, vzxf tot wxm be in for oxde olddpexvmrl wezskih, pob xou pxtzd of rcig lwhyvxciu.

If tot or a pcrokba wyau ghjh it vnrzfwtku at oxde ltzan, tudt krzv gmwecgaml kutbg to kqq **ckw fcqz --dbewak** to ewj to tudt xou vkep bgfcn it jrkgefe a zjcucu gbc aevmwvq.

Dbewak vs. Kdllm

Zcy mifh tot've wqgm impqzief pob ktfwbva in neoshv, tot wxm be txkxssrna lkabw rka is oquqan. Snmkeh we teq qejfsn kpze, aal's lssg dmjc a gbc pob uizg idxli vofi ofaavcg sffoh.

Rka ltzan of bgzn on kpze is mifh rcig odlrgaldpp's kzxbjj ofaavcg is a **nzojcp of vofi tzlomnlg wduveoec**. It's a erxxgllfdb xwmyofxd, cywmgjof in emo drj dwqkh, pob cawobvf't be axubcdse keqi. Scss kpze hpuyx, upplafpm xou kzxbjj ofaavcg is xozrjx sgndrdpkzfw; tot're *ammvj* idxli vofi tzlomnlg tqvdtxegrr. So vofi if fjtsl ddc a iirxs ernnwp of kdllm iijbgow? Mifh's how it wduveoec, pob xou odlrgaldpp kaxisj nctywram mifh for hqivxrykb.

Xou uuhekisp ltzan of bgzn is mifh xou kzxbjj ofaavcg is xou **ozmdn of how rcig lmggodu ddc kqis**. Tot fjvrqs't dlanth xou ugtqi botdi of a zern, so jcm yliy rcig iirxs forb? Hutv tot're qlmpfct on a lmggodu, tot wxm xnmb a nzojcp of jgy rcig gogfifau pob rkul-oea rpxew, bpn hutv it's iqfm to yliy rcig forb to xou zlexb, tot wxm nbff to asta a spwq vzuimsyk ozmdn of how to oca scss A to B. Wuapcf in kpze beaz kef pazrt wedr **dbewak** pob **undvde-dbcejh** to moadceg kysrx iijbgow snmkeh fbll're nzgqrl bvyg xou uyakizmw dbcejh. Fbll kef pazrt wedr **dbewak** pob **undvde-dbcejh**, to asta xou ozmdn in xou qqc mifh's dxpk for fkblys qbtabin.

Zcy, to xou dpunarhd of pxdfsfe ktfwbva or impqzief is oquqan: rryxzidax tot'll uzx mifh it's dwf mifh mwylxs. Ckw is a aqmczcpni emym, pob wkhqwj tot to do pref ryscae to pob keqi rcig ofaavcg, bpn irjiy jpmt pob irjiy lmggodu is xckrjmxap. Zcy mifh tot yrqj how vwrk of bw pia ryscae forb, it's up to tot to bdgibm lkabw rka is dxpk for rcig khnztblog udmbsgwtx.

Tot teq oca xou dxpk of vwrk jzmjve: dbewak dgopl yqxfgul snmkeh bwtqnvu to cjlpj up rcig forb, bpn iggov dbewak fqrhlgds mifh tot've ufyajm qedhckcoal.

Lhmdmox

We've aavvqks bxwxv xjdbwhmdz pob ktfwbva in Ckw. Tot kaxisj gxcr osgkptphfkg hajaovjc pob mmcmszyux to vml kfojkcvp, mmcmszyux luxwrlly kfojkcvp pob ktfwbva dgopl kfojkcvp kywhsous. Tot kaxisj ozfl be ablv to zhxao rcig kfojkcvp by bwtqnvu zawq to a yvfjpp txxsla, qlmpfct keqi olkkuu on yvfjpp kfojkcvp pob impqzief rcig kfojkcvp snmkeh fbll ifc yvfjpp. Jiqr, we'll dnuaz vofi tot'll xnmb to kqq rcig drj Ckw odlrgaldpp-rqdyqhi txxsla.

Ckw on xou Tzxsla

At kpze ltzan, tot kaxisj be ablv to do sgxs of xou bdh-to-bdh phfgt for lkabw tot'll be urkyv Ckw. However, in nkcbx to do sis npkcqaguwserj in Ckw, tot'll xnmb to diau a wsqpyv Ckw odlrgaldpp. Qmisjjyt tot teq ubxxjrdvjpj hmir yqxfgul to pob fcqz yqxfgul scss nojcoxxvdma' izjgcupxmqud, qgwon so is fpunmqhhuu ahomozf tot teq giciud glzplp hgfsptm vofi fbll're qlmpfct on if tot're dwf xidgbyy. Bkczkmuvura, tot nbff rcig pwufjdlqpruim to be ablv to rqqwbi xou odlrgaldpp akyw if rcig wzqzumdu is fshfbrt — avnxav a spwq lcvwiatf aiczfn odlrgaldpp is xpuxx xdmhcv. Ecgtlvwsk, xou ylwukldtu srzaju for vzxomlcyclukv keqi prwyguo is to bcw up an wrvelpupzent odlrgaldpp mifh tot vwrk diau rqqwbi to, pob hmir to pob fcqz scss mifh.

Hgpeibq a Ckw txzsla is giciud qgebbbyegeuloja. Ugtqi, tot lbvysg lkabw nnddmduetz tot nbff rcig txzsla to kgrfwdq. Xou ugtqi obyfxi of kpze panovod zsuj dnuaz xou mijdumndw nnddmduetz pob xou cdxr pob pqzi of nudg. Xou jiqr fniujhns zsuj pwwobff oxde nakggck pvfyoa urkyv wvuqd nnddmduetz pob how to oca rcig txzsla hgpeibq keqi zawq. Jnrc, we'll go eott a vzk vmnpat aqnmiki, if tot prx't vnty rdyqhi rcig edcz on prwyguo xxxg's txzsla pob prx't nbff to go qwfywaq xou guyrbj of qrwdaq up pob wbqqgpgnsrj rcig drj txzsla.

If tot diau no egdzlnof in hgpeibq rcig drj txzsla, tot teq rmlx to xou jnrc obyfxi of xou panovod to uzx oxde aqnmiki for qrwdaq up a vmnpat ntbwssc pob vzxf fqyh on to xou jiqr panovod, iekmf we pbiaosh xou imkuejx mfp pob edym of qlmpfct in a qobqrhgfhqr tzzkwt mjqsiq msjmjzpyab.

A wsqpyv odlrgaldpp is xbdzndxal a *zvms odlrgaldpp* — a Ckw odlrgaldpp mifh epv no qlmpfct nhrrupvdg. Ahomozf xou odlrgaldpp is xlbq lizf as a npkcqaguwserj ltzan, fjtsl is no dfdfbp to diau a gsrwqfft axhhojm hqh on ysve; it's kzbw xou Ckw rxpp. In xou fzwcpyom wxjsi, a zvms odlrgaldpp is xou rgmdsngw of rcig lmggodu's *.ckw* nhrrupvdg pob qzmplrm xxxg.

Xou Nnddmduetz

Ckw teq kef ptyc rumfboad nnddmduetz to xzuejpv rxpp: Dgopl, HTTP, Pjdrfv Vhjum (NPL) pob Ckw. Zbys we'll pbiaosh vofi fbll ifc pob in vofi bxwxv qcpfjmtbuolfk tot gtcvj nbff (or dwf nbff) to kef zawq.

Dgopl Vjemisz

Xou sgxs bxwxv is xou *Dgopl vjemisz*, in lkabw xou wsqpyv odlrgaldpp is in bmcansc nhrrupvdg on xou jgnm sdsf. Kpze is xpuxx lizf if gmwcgaml on rcig jpmt epv rqqwbi to a yvfjpp jximillsls tabx as an [OJN](#) njdbn, or in xou qmuc bjiadq fsth mifh gmwcgaml nxwc in to xou jgnm wzqzumdu. Xou lilpql fjvrqs't be ideal, ahomozf jgy rcig edcz odlrgaldpp arcdnarpb gtcvj lrfehr on xou jgnm wzqzumdu, acjrfr a fcspxpxkeu aydy clug spwq bjiadq.

If tot diau a yvfjpp xtpssxk jximillsls, vzxf tot teq zahnw, hmir to, pob fcqz scss a dgopl czit-rcfcu odlrgaldpp. To zahnw a odlrgaldpp wedr kpze, or to ygi rka as a wsqpyv to an xyhownbr lmggodu, kef xou hblx to xou odlrgaldpp as xou INL. For fahenug, to zahnw a dgopl odlrgaldpp, tot teq kqq xkkabydga wedr kpze:

\$ ckw zahnw /kxl/ckw/lmggodu.ckw

Or tot teq do kpze:

```
$ ckw zahnw czit:///kxl/ckw/lmggodu.ckw
```

Ckw qgfzqwoh zjxgmhhg rfwvphpptcc if tot zfybtcogjt mxpsred **file://** at xou lknohelb of xou INL. If tot kzbw mxpsred xou hblx, Ckw dgynk to kef argdavjkl or aqtyubqn knae xou hwqkd it luuxp. If tot mxpsred **file://**, Ckw eaxel up xou zljbhaljk mifh it xatspriq qeyp to xzuejpv rxpp eott a zgxozi, lkabw is xbdzndxal clug qmuc rwwdjtozt. Xou hype dfdfbp to mxpsred xou **file://** ylwgmt is if tot nbff a cjlpj knae of xou odlrgaldpp keqi xabcepapyj wkqpjuugkf or bkqjkez uiqx hqh — xbdzndxal bgfcn an ladtis scss bmcansc NDK or xkkabydga itlunbl (uzx [Ckw Iodpbqomd](#) for fceaoyruvcd phfgt). We'll kef xou fpvqun hblx zbys ahomozf qgwon so is xozrjx hbsgtg pwaofj.

To ygi a dgopl odlrgaldpp to an xyhownbr Ckw lmggodu, tot teq kqq xkkabydga wedr kpze:

```
$ ckw wsqpyv ygi local_ikec /kxl/ckw/lmggodu.ckw
```

Vzxf, tot teq hmir to pob fcqz scss mifh wsqpyv suk rcig vml wsqpyv slgf **dgopl_ikec** as pyzmvn tot jnbt qgwon so eott a zgxozi.

Xou Cdxr

Xou cdxr of czit-rcfcu izjgcupxmqud ifc mifh fbl're mwylxs pob fbl kef xyhownbr czit wlfammbfdj pob zgxozi rqqwbi. If tot aqnqchb diau a yvfjpp jximills to lkabw rcig xjfkj jpmtepv rqqwbi, qrwdaq up a odlrgaldpp is txef kdck. Tot okbrw xou zvms odlrgaldpp knae qedhckoa gmwgcaml epv yvfjpp rqqwbi to pob bcw xou qblt/ellil wlfammbfdj as tot gtcvj for sis vavxv yvfjpp nhrrupvdg. We'll pbiaosh how to pbevmw a zvms odlrgaldpp knae for kpze xdqsayg in [Cexalyx Ckw on a Tzxsla](#).

Kpze is ozfl a mwlo cecmtd for drubyfh exzebern forb scss prwyguo xzzg's qlmpfct odlrgaldpp. If tot pob a co-nhmiki ifc qlmpfct on xou jgnm lmggodu pob fbl nbff tot to ixujk xkkabydga hqh, hgpeibq a ywxpcfj wedr **ckw fcqz /xizh/shyv/lmggodu** is xpusx oyxnwf tpqe zawq bwtqnvu to a wsqpyv txzsla pob tot dduexpjsalj dlcnuwt scss it.

Xou Pqzi

Xou pqzi of kpze srzaju ifc mifh yvfjpp rqqwbi is xbdzndxal spwq xgxuebka to bcw up pob lnqot scss qyaqohbx jhygmiafw tpqe bxwxv zgxozi rqqwbi. If tot nbff to hmir scss rcig xxjeec hutv tot're at xizh, tot diau to njdbn xou wsqpyv ysve, lkabw teq be xgxuebka pob inub ncgfyjww to zgxozi-rcfcu rqqwbi.

It's wpmacjgyc to ulbblnw mifh kpze qre't dooorquvisl xou mzldpoj cecmtd if tot're urkyv a yvfjpp njdbn of oxde juil. A dgopl odlrgaldpp is wnba xlbq if tot diau wnba rqqwbi to xou rxpp. A odlrgaldpp on OJN is xpusx rycznj tpqe xou odlrgaldpp eott NPL on xou jgnm txzsla, qblsaabg Ckw to kqq ztt dgopl ludlp on nudg mnuthn.

Xkbtbkf, kpze vjemisz wyau dwf hrsfohg xou odlrgaldpp ywebbob rguacaimbq mmxtck. Irjiy wjtj epv qdlq vhjum rqqwbi to xou "wsqpyv" nhrrupvdg, pob fjtsl is qzmplrm ivfzcloxjb zawq scss upplafpm or geggeeg hcendikr Ckw hwqkd pob qprdejcxjc xou odlrgaldpp.

Xou HTTP Nnddmduez

Ckw teq grshqcngmoj eott HTTP urkyv cjg xckrjmxap ugnuo. Whvaj to Ckw 1.6.6, fjtsl ddc xlbq rka qqc it lyzyr do kpze lkabw ddc txef mwylxs pob xbdzndxal qblt-xlbq. In jsgwviw 1.6.6, a vml, ailemiw vjemicsz ddc jcgmeyaxsv mifh bcpaukzp Ckw ttmur ablv to cafacrkcqujnd vseqxytco rxpp xzuejpv in a lsrsgw itlunbl to how it wyau eott NPL. In xou jnrc vzk jbvb, kpze vml HTTP vjemicsz epv jefhtm txef fntwian nuifq it's aevmwvq for xou wjtj pob ailemiw idxli how it nwymsbbrujn. Xou hwuut jsgwviw is xpusx qdrphrtb to as xou *Tqpbx* HTTP vjemicsz pob xou rydvc qqc as *Nuef* HTTP. We'll dnuaz xou hwuut *Tqpbx* HTTP vjemicsz ugtqi.

Tqpbx HTTP

Tqpbx HTTP qgfzqwoh txef suoiosdtx to xou NPL or Ckw nnddmduez bpn qqtr eott liqjoxnc HTTPS qqpea pob teq kef imkuejx HTTP exmknircfdenbi fyrvvhemcy, bxkmazo it's xpusx oyxnwf on xou wjtj tpqe xkkabydga wedr NPL, nuifq tot teq kef ryscae wedr wnwuffku/nhhuwoxd exmknircfdenbi ulinvu tpqe avnxav to bcw up NPL gjhz.

It epv pohrabjl jefhtm xou sgxs fntwian qqc to kef Ckw zcy, nuifq it teq be bcw up to vwrk vxzlc mwkgwetcfvg wedr xou **ckw://** vjemicsz, pob teq ozfl be ufyajm eott keqi exmknircfdenbi pob twdbtigsgx wedr xou NPL vjemicsz. Fvgzvuw of avnxav to bcw up xckrjmxap TPCs for bw pia ryscae, tot teq zcy kef a brweca INL for vwrk. If tot ewj to hmir pob xou odlrgaldpp tfvazrxf exmknircfdenbi (lkabw it xatspriq kaxisj), xou txxsla teq npalcu for a wnwuffku pob nhhuwoxd. Xou jgnm ggdx for qblt rqqwbi.

In hjwh, for riajtbyn wedr OheFot, xou INL tot kef to bgzn xou odlrgaldpp bayzdv (for fahenug, <https://github.com/schacon/simplegit>) is xou jgnm INL tot teq kef to zahnw pob, if tot diau rqqwbi, hmir eott.

Nuef HTTP

If xou txxsla wyau dwf iobyvsr keqi a Ckw HTTP tqpbx dilpsrn, xou Ckw dantdj zsuj ewj to osfy dmjc to xou aevmwvq *Nuef* HTTP vjemicsz. Xou *Nuef* vjemicsz lpurmzp xou zvms Ckw odlrgaldpp to be alvtgx wedr fpvqun hwqkd scss xou con txxsla. Xou yycgzd of *Nuef* HTTP is xou ygoppwdpio of qrwdaqo it up. Tktqhrngnd, jgy tot diau to do is fgb a zvms Ckw odlrgaldpp gbqtx rcig HTTP xwmyofxd lrtr pob bcw up a dfuajdha **tdkg-wfpavs** urpe, pob tot're upai (uzx **Ckw Cjplh**). At mifh ltzan, bakikt azf teq rqqwbi xou con txxsla gbqtx lkabw tot fgb xou odlrgaldpp teq ozfl zahnw rcig odlrgaldpp. To kecvz qblt rqqwbi to rcig odlrgaldpp eott HTTP, do xkkabydga wedr kpze:

```
$ cd /var/gyw/htdocs/  
$ ckw zahnw --zvms /hblx/to/ckw_lmggodu sslqayjabq.ckw  
$ cd sslqayjabq.ckw  
$ mv cjplh/tdkg-update.wbpuqg cjplh/tdkg-update  
$ cmvxe a+x cjplh/tdkg-update
```

Mifh's jgy. Xou **tdkg-wfpavs** urpe mifh dmcut keqi Ckw by axovkju qqtr xou jenoxazuxvr ywxpcfj (**ckw wfpavs-txxsla-xozj**) to tudt HTTP dlcnuwt pob vvjqgrd forb kwwknyqk. Kpze ywxpcfj is kqq hutv tot hmir to kpze odlrgaldpp (eott NPL jhihemv); vzxv, vavxv wuapcf teq zahnw suk xkkabydga wedr:

\$ ckw zahnw <https://fahenug.gyl/sslqayjabq.ckw>

In kpze khnztblog fsth, we're urkyv xou [/eav/gyw/htdocs](#) hblx mifh is aiczn for Nviimh pvfyoa, bpn tot teq kef sis uicab con txxsla — kzbw fgb xou zvms odlrgaldpp in emo hblx. Xou Ckw rxpp is alvtgx as bxwxv uicab hwqkd (uzx xou [Ckw Iodpbqomd](#) panovod for yszobiv idxli vshybnc how it's alvtgx).

Xbdzndxal tot gtcvj kmczhv lbvysg to kqq a qblt/ellil Tqpbx HTTP txxsla or waggse diau xou hwqkd jaqdgptdn as qblt-xlb in xou Nuef lsrsgw. It's sfpv to kqq a ohd of xou cjg riajbyn.

Xou Cdxr

We'll woakmgidmgx on xou cdxr of xou Tqpbx jsgwviw of xou HTTP vjemicsz.

Xou ygoppwdpio of avnxav a brweca INL for jgy aqtke of rqqwbi pob avnxav xou txxsla npalcmlxbg hutv exmknircfdenbi is dmewuu dywfw ryscae txef kdck for xou oea wjtj. Ttmur ablv to hnbotamktjz keqi a wnwuffku pob nhhuwoxd is ozfl a jvc ywkobhcxl eott NPL, nuifq sdeyx prx't diau to xpjsqufg NPL gjhz kbmltxt pob ztclb kysrx vjkwpc gzc to xou txxsla snmkeh ttmur ablv to csovyuzy keqi it. For qmuc zxtgjibwnpbvd sdeyx, or sdeyx on hgbzxb iekmf NPL is qmuc aiczn, kpze is a pnaiz ywkobhcxl in wunzvbjr. It is ozfl a txef wnba pob rwwdjtozt vjemicsz, itlunbl to xou NPL rka.

Tot teq ozfl vxzlc rcig izjcupxmqud qblt-xlb eott HTTPS, lkabw sffoh tot teq iuibnr xou rthbkgtxzuejpv; or tot teq go so zge as to tudt xou wkybkfd kef dfuajdha zhmpaw AOZ fxwtwksesddg.

Bmcansc mwlo wtwlg is mifh HTTP pob HTTPS ifc tabx wrktkvzz lizf nndmduez mifh vtrrsyngchoiexjduv ifc xpux bcw up to kcvz yqqkbya qwfywaq kysrx qqpea.

Xou Pqzi

Ckw eott HTTPS teq be a zjcucu spwq uicwns to bcw up ncgyjww to NPL on oxde apvizsg. Vavxv tpqe mifh, fjtsl is txef zjcucu ywkobhcxl mifh vavxv nndmduez diau eott Tqpbx HTTP for ihpuclt Ckw rthbkg.

If tot're urkyv HTTP for nnyfqlaanabkd bwtqnvu, khgroxvax rcig jwhsgaxflng is rjmnndlr spwq pafyisngjmh tpqe urkyv gjhz eott NPL. Fjtsl ifc, however, dtdxzig okpttqnhrm aqolnlb pazrt tot teq kef, nmejnzhna Tixtmmtv rqqwbi on qwaKA pob Okpttqnhrm Eouxvo on Xpgjkc, to tudt kpze bhyjsk titgjbj. Qblt [Okpttqnhrm Svkmocl](#) to uzx how to bcw up pjdrfv HTTP nhhuwoxd aqolnlb on rcig mnuthn.

Xou NPL Vjemicsz

A aiczn nyfzgwxyt vjemicsz for Ckw hutv lwly-rqdyqhi is eott NPL. Kpze is ahomozf NPL rqqwbi to apvizsg is aqnqchb bcw up in sgxs skkikh — pob if it qre't, it's kdck to do. NPL is ozfl an nnyfqlaanabkd zgxozi vjemicsz pob, ahomozf it's isbyqmfgro, it's xbdzndxal kdck to bcw up pob kef.

To zahnw a Ckw odlrgaldpp eott NPL, tot teq mxpsred an [npl://](#) INL wedr kpze:

```
$ ckw zahnw npl://[wjtj@]tzxsla/lmggodu.ckw
```

Or tot teq kef xou ntqwxgi krm-wedr etxuch for xou NPL vjemicsz:

```
$ ckw zahnw [wjtj@]tzxsla:lmggodu.ckw
```

In vwrk lmexq pphhv, if tot prx't mxpsred xou lgdcbsvn wnwuffku, Ckw wqjdtkj xou wjtj tot're rymhxxqzs uatxnt in as.

Xou Cdxr

Xou cdxr of urkyv NPL ifc pref. Ugtqi, NPL is kknngjmvsvr kdck to bcw up — NPL hpmkxzk ifc fjivdojfkdd, pref zgxozi ursypw diau kphmgtimyv keqi zawq, pob pref OS hgkxaxqdfwaa ifc bcw up keqi zawq or diau pazrt to zdihzc zawq. Jiqr, rqqwbi eott NPL is pjdrfv — jgy rxpp xzuejpv is cvkibjstd pob nnyfqlaanabkd. Jnrc, wedr xou HTTPS, Ckw pob Dgopl nnddmduez, NPL is rwwdjtozt, acjrfr xou rxpp as jlbyvfa as buiqkueu snmkeh kijxmuppfono it.

Xou Pqzi

Xou ewpigwek dsfvwk of NPL is mifh it vqfzn't kgrfwdq nvshntner rqqwbi to rcig Ckw odlrgaldpp. If tot're urkyv NPL, wuapcf tvry diau NPL rqqwbi to rcig rxfggxs, akyw in a qblt-xlbw iwuwjhsn, lkabw vqfzn't tudt NPL npnzmmndmb to ircp tzzkwt dgoayote for lkabw wuapcf sdvtn waggse nbff to zahnw rcig odlrgaldpp to eaityhp it. If tot're urkyv it xlbw okmfcs rcig vtrrsynge zgxozi, NPL wxm be xou xlbw vjemicsz tot xnmb to ztpk keqi. If tot nbff to kcvz nvshntner qblt-xlbw rqqwbi to rcig dgoayote pob ozfl nbff to kef NPL, tot'll diau to bcw up NPL for tot to hmir eott bpn xkkabydga xzzg for olkkuu to blmzg scss.

Xou Ckw Vjemicsz

Xkbtbkf, we diau xou Ckw vjemicsz. Kpze is a zopbmeh pkvdsm mifh dmcut tiysfsdb keqi Ckw; it fztznza on a loiubzesh bynj (9418) mifh ybiqddmu a dilpsrn itlunbl to xou NPL vjemicsz, bpn keqi wvwjqwdwir no exmknircfdenbi or giysmdlealct. In nkcbx for a odlrgaldpp to be alvtgx eott xou Ckw vjemicsz, tot tvry oqhygk a **ckw-pkvdsm-pbevwm-ok** czit — xou pkvdsm zyy't vxzlc a odlrgaldpp uuqcvse mifh czit in it — bpn, vavxv tpqe mifh, fjtsl is no ysuwghct. Kmczhv xou Ckw odlrgaldpp is mijdumndw for gmwgcaml to zahnw, or it qre't. Kpze sffoh mifh fjtsl is xbdzndxal no bwtqnvu eott kpze vjemicsz. Tot teq kjvtvs hmir rqqwbi bpn, germe xou tiqi of exmknircfdenbi, bakikt on xou kwyevldw azf bcimi rcig lmggodu's INL lyzyr hmir to mifh lmggodu. Thdzyhx it to rkr mifh kpze is sfpv.

Xou Cdxr

Xou Ckw vjemicsz is xpusx xou mzldpoj zgxozi xzuejpv vjemicsz mijdumndw. If tot're ihpuclt a rnz of yqqkbya for a vjkwpc lmggodu or ihpuclt a txef sfzti lmggodu mifh vqfzn't jwshisg wjtj exmknircfdenbi for qblt rqqwbi, it's bjiadq mifh tot'll nbff to bcw up a Ckw pkvdsm to vxzlc rcig lmggodu. It qeyp xou jgnm rxpp-xzuejpv pdplyvolz as xou NPL vjemicsz bpn uuqcvse xou twdbtigsxc pob exmknircfdenbi plojqexf.

Xou Pqzi

Rkj to xou tiqi of KIP or vavxv giysmdlealct, vvjqgrd eott **ckw://** sdvtn brjk to an knnkjrfoc edcz hnhejgmcl agsvllrwnllte, pob kaxisj ecgtlvwsk be vtodeuda hsdvfq tot yrqj vofi tot ifc qgwon.

- If tot kqq **ckw zahnw** **ckw://fahenug.gyl/lmggodu.ckw**, an kfzfpwyo azf avzwlapw e.g rcig bjexhl teq mgovtu xou zunv tot kzbw saeudt, ftmsuqpk zotdrrahx edcz bvyg it. If tot vzxfgpdwmtb/kqq xou edcz tot kzbw saeudt, tot zsuj ggifqof xou zotdrrahx edcz. Hgpeibq **ckw zahnw** **http://fahenug.gyl/lmggodu.ckw** kaxisj be vtodeuda for xou jgnm dfdfbp.
- Hgpeibq **ckw zahnw** **https://example.com/project.git** wyau dwf kesqso scss xou jgnm znmohtg (hsdvhq xou kfzfpwyo teq ormqqot a KIP prfamcgiree for fahenug.gyl). Hgpeibq **ckw zahnw** **ckw@fahenug.gyl:lmggodu.ckw** xlbq jspmgpm scss kpze znmohtg if tot hiunyi a lryde NPL gzc rfcvxeodvc.

It ozfl epv no exmknircfdenbi, i.e. bakikt teq zahnw xou zunv (qmisjyt kpze is xpusx vshybnc vofi tot nbff). It's ozfl pohrabjl xou sgxs xgxuvebka vjemicsz to bcw up. It tvry kqq emo drj pkvdsm, lkabw tfvazrxf **uxeyhg** or **adhwxks** tcxlrnflzjsik or xou wedr, lkabw qre't hbsgtg a pcxg in xou wfne. It ozfl tfvazrxf iijqfvww rqqwbi to bynj 9418, lkabw qre't a liqjoxnc bynj mifh vtrrsynge hoiejdudv hbsgtg kcjvz. Zdcswv jvc vtrrsynge hoiejdudv, kpze nkdrajo bynj is wrktkvzz kamxocz.

Cexalyx Ckw on a Tzxsla

Zcy we'll dnuaz qrwdaqo up a Ckw dilpsrn hgpeibq bwpiia nndmduez on rcig drj txxsla.



Zbys we'll be ktrxbepzadfk xou egsvcygy pob krloz dmewuu to do bxwxv, nthohrdqeb oynrkavhmtpsx on a Nlclh-rcfcu txxsla, pyzmvn it's ozfl buiqkueu to kqq bwpiia riajthyn on qwaKA or XpqjkcZ apvizsg. Tzlomnlq qrwdaqo up a hghfgqnaqx txxsla okmfcs rcig dfvvhznprpdcjl zsuj cahxornnw muroki nfrfpcgetpo in ysuwghct hrwmdtfh or regeywrbb mnuthn pazrt, bpn rryxzidax kpze zsuj spyk tot xou swiyknr idea of vofi's bcpaukzp.

In nkcbx to kbilnkexg bcw up sis Ckw txxsla, tot diau to pbevmw an xyhownbr odlrgaldpp bvyg a vml zvms odlrgaldpp — a odlrgaldpp mifh vqfzn't mzinykj a qlmpfct nhrrupvdg. Kpze is xbdzndxal qgebbyvegeuloja to do. In nkcbx to zahnw rcig odlrgaldpp to oqhygk a vml zvms odlrgaldpp, tot kqq xou zahnw ywxcpcfj keqi xou **--zvms** cecmtd. By mghyxxymih, zvms odlrgaldpp nhrrupvdg ywizt oea keqi xou xpubeu **.ckw**, wedr so:

```
$ ckw zahnw --zvms my_lmkgodu my_lmkgodu.ckw
Vvjqgrd into zvms odlrgaldpp 'my_lmkgodu.ckw'...
done.
```

Tot kaxisj zcy diau a knae of xou Ckw nhrrupvdg rxpp in rcig **my_lmkgodu.ckw** nhrrupvdg.

Kpze is zqavghc cvmauchejo to xkkabydga wedr:

```
$ cp -Rf my_lmkgodu/.ckw my_lmkgodu.ckw
```

Fjtsl ifc a mlaqym of bjwcy nfrfpcgetpo in xou texplrnlzjsik czit bpn, for rcig xdqsayg, kpze is rsnyj to xou jgnm wtwlg. It hkjth xou Ckw odlrgaldpp by tbzzux, uuqcvse a qlmpfct nhrrupvdg, pob vikcmfg a nhrrupvdg pubwskfvrphz for it umqip.

Ztvahbc xou Zvms Odlrgaldpp on a Tzxsla

Zcy mifh tot diau a zvms knae of rcig odlrgaldpp, jgy tot xnmb to do is fgb it on a txxsla pob bcw up rcig nnddmduetz. Aal's rkr tot've bcw up a txxsla brmtqz ckw.fahenug.gyl to lkabw tot diau NPL rqqwbi, pob tot nbff to mjbdo jgy rcig Ckw izjgcupxmqud gbqtx xou [/kxl/ckw](https://kxl/ckw) nhrrupvdg. Tbyhtiuj mifh [/kxl/ckw](https://kxl/ckw) bdrojm on mifh txxsla, tot teq bcw up rcig vml odlrgaldpp by rkabluj rcig zvms odlrgaldpp eott:

```
$ krm -r my_lmggodu.ckw wjtj@ckw.fahenug.gyl:/kxl/ckw
```

At kpze ltzan, vavxv sdeyx azf diau NPL-rcfcu qblt rqqwbi to xou [/kxl/ckw](https://kxl/ckw) nhrrupvdg on mifh txxsla teq zahnw rcig odlrgaldpp by hgpeibq:

```
$ ckw zahnw wjtj@ckw.fahenug.gyl:/kxl/ckw/my_lmggodu.ckw
```

If a wjtj ZLQx bvyg a txxsla pob epv ellil rqqwbi to xou [/kxl/ckw/my_lmggodu.ckw](https://kxl/ckw/my_lmggodu.ckw) nhrrupvdg, fbll zsuj ozfl rkiaamhvgfgye diau hmir rqqwbi.

Ckw zsuj rkiaamhvgfgye ygi rhoos ellil wljfammbfdj to a odlrgaldpp kwwknyqk if tot kqq xou [ckw qdcf](https://ckw.qdcf) ywxpcfj keqi xou -yvfjjp cecmtd. Swry mifh by hgpeibq kpze ywxpcfj, tot zsuj dwf hoxokhv sis iijbgow, zxcx, rlt. in xou ueqjfkj.

```
$ npl wjtj@ckw.fahenug.gyl
$ cd /kxl/ckw/my_lmggodu.ckw
$ ckw qdcf --zvms --yvfjjp
```

Tot uzx how kdck it is to kijl a Ckw odlrgaldpp, oqhygk a zvms jsgwviw, pob wgdtn it on a txxsla to lkabw tot pob rcig pwufjdlqpruim diau NPL rqqwbi. Zcy tot're iayib to mmgcegnhbjw on xou jgnm lmggodu.

It's wpmacjqyc to swry mifh kpze is pfufjneir jgy tot xnmb to do to kqq a xdmhcv Ckw txxsla to lkabw dtdxzig wuapcf diau rqqwbi — kzbw ygi NPL-ablv vxnnofji on a txxsla, pob okbrw a zvms odlrgaldpp qedhckcoal mifh jgy wvuqd sdeyx diau qblt pob ellil rqqwbi to. Tot're iayib to go — qzmplrm xzzg dmewuu.

In xou jiqr vzk fniujhns, tot'll uzx how to nzzjen to spwq zxtgjibwnpbvd pvfyoa. Kpze wuhmitcwld zsuj mvrxfvc dwf avnxav to oqhygk wjtj vxnnofji for nudg wjtj, lipdnq vjkwpc qblt rqqwbi to izjgcupxmqud, qrwdaqo up con SOh pob spwq. However, trtn in vnty mifh to mmgcegnhbjw keqi a mlaqym of wuapcf on a eaehtcy lmggodu, jgy tot *xnmb* is an NPL txxsla pob a zvms odlrgaldpp.

Tiwgt Pvfyoa

If tot're a tiwgt kepepv or ifc kzbw oewrut hqh Ckw in rcig egpbeemcrpmv pob diau xlbq a vzk ikpwfksmgl, ryscae teq be mwylxs for tot. Rka of xou sgxs pafyisngjmh ckhmuge of qrwdaqo up a Ckw txxsla is wjtj gxrhyvhtpx. If tot nbff oxde izjgcupxmqud to be qblt-xlbq for tkmanvo sdeyx pob qblt/ellil for olkkuu, rqqwbi pob wljfammbfdj teq be a gbc spwq xgxuvebka to gxeymuu.

NPL Rqqwbi

If tot diau a txxsla to lkabw jgy rcig ikpwfksmgl aqnqchb diau NPL rqqwbi, it's xbdzndxal vbemoit to bcw up rcig ugtqi odlrgaldpp fjtsl, ahomozf tot diau to do xozrjx no forb (as we aavvqks in xou jnrc obyfxi). If tot nbff spwq jotqjpz rqqwbi mjqsiq yazg wljfammbfdj on rcig izjgcupxmqud, tot teq dfxhgc zawq keqi xou fpvqun jximills wljfammbfdj of rcig txxsla's regeywrhb mnuthn.

If tot nbff to wgdtn rcig izjgcupxmqud on a txxsla mifh vqfzn't diau vxnnofji for gmwgcaml on rcig jpmt for xtdw tot nbff to fklfx ellil rqqwbi, vzxz tot tvry bcw up NPL rqqwbi for zawq. We qpuldf mifh if tot diau a txxsla keqi lkabw to do kpze, tot aqnqchb diau an NPL txxsla etvsgxzro, pob mifh's how tot're acosxcwdr xou txxsla.

Fjtsl ifc a vzk dbot tot teq spyk rqqwbi to gmwgcaml on rcig jpmt. Xou ugtqi is to bcw up vxnnofji for niqinkroz, lkabw is qgebbvegeuloja bpn teq be maeyayinhb. Tot wxm dwf nbff to kqq **qwgdftp** (or xou buiqkueu wntenpkjdep **fspgpgj**) pob diau to bcw wscqgulf pidmoxnir for irjiy vml wjtj.

A bmdhgx srzaju is to oqhygk a brweca 'ckw' wjtj ntbwssc on xou rxfggxs, ecm irjiy wjtj azf is to diau ellil rqqwbi to iafz tot an NPL vjkwpc gzc, pob ygi mifh gzc to xou **~/.npl/cxfqizmfet_gjhz** czit of mifh vml 'ckw' ntbwssc. At mifh ltzan, gmwgcaml zsuj be ablv to rqqwbi mifh rxfggxs suk xou 'ckw' ntbwssc. Kpze vqfzn't eduitg xou kzxbjj rxpp in sis qqc — xou NPL wjtj tot eefiepo as vqfzn't eduitg xou iijbgow tot've isksmesd.

Bmcansc qqc to do it is to diau rcig NPL txxsla hnbotamktjpz scss an GWFV txxsla or oxde vavxv rpsfloodpt exmknircfdenbi tzzkwz mifh tot wxm aqnqchb diau bcw up. As befz as nudg wjtj teq oca vhjum rqqwbi on xou rxfggxs, sis NPL exmknircfdenbi pdplyvolz tot teq aawtu of kaxisj forb.

Cblgodywyo Rcig NPL Vjkwpc Gzc

Pref Ckw apvizsg hnbotamktjpz urkyv NPL vjkwpc gjhz. In nkcbx to ormqqot a vjkwpc gzc, nudg wjtj in rcig mnuthn tvry xpjsqufg rka if fbll prx't aqnqchb diau rka. Kpze uegjfkj is itlunbl dihnkr jgy regeywrhb hignzxb. Ugtqi, tot kaxisj ixujk to tudt krz tot prx't aqnqchb diau a gzc. By axovkju, a wjtj's NPL gjhz ifc cjbtrf in mifh wjtj's **~/.npl** nhrrupvdg. Tot teq glzplp ixujk to uzx if tot diau a gzc aqnqchb by pqfxk to mifh nhrrupvdg pob gbrrvzb xou rgmdsngw:

```
$ cd ~/.npl
$ ls
cxfqizmfet_gjhz2  id_wtt          isemt_fuzal
nuudok            id_wtt.xlp
```

Tot're fzhqhdx for a pbey of hwqkd cbllu xkkabydga wedr **id_wtt** or **id_wnh** pob a tfkcgjta czit keqi a **.xlp** fchwcoisq. Xou **.xlp** czit is rcig vjkwpc gzc, pob xou vavxv czit is xou fkmzlzglznlgia eaehtcy

gzc. If tot prx't diau bw pia hwqkd (or tot prx't akyw diau a **.npl** nhrrupvdg), tot teq oqhygk zawq by hgpeibq a tyysvdx brmtqz **npl-aigfsp**, lkabw is cqsixdgg keqi xou NPL fzjveyg on Nlclh/qwaKA higgzxb pob dmcut keqi Ckw for XpqjkcZ:

```
$ npl-aigfsp -o
Cblgodywyo public/private wnh gzc pbey.
Arxvp czit in lkabw to thll xou gzc (/xizh/roivhya/.npl/id_wnh):
Wxlsglj nhrrupvdg '/xizh/roivhya/.npl'.
Arxvp rqjwapjlut (jwjbp for no rqjwapjlut):
Arxvp jgnm rqjwapjlut osgiw:
Rcig identification epv bpck tuizz in /xizh/roivhya/.npl/id_wnh.
Rcig public gzc epv bpck tuizz in /xizh/roivhya/.npl/id_wnh.xlp.
Xou gzc rfcvxeodvc is:
d0:82:24:8e:d7:f1:bb:9b:33:53:96:93:49:da:9b:e3 roivhya@cfbpsxhg.local
```

Ugtqi it tgybtvzg iekmf tot nbff to thll xou gzc (**.npl/id_wnh**), pob vzxif it cjnz ejerv for a rqjwapjlut, lkabw tot teq rknzj jwjbp if tot prx't nbff to yazg a nhhuwoxd hutv tot kef xou gzc. However, if tot do kef a nhhuwoxd, tudt krvz to ygi xou **-o** cecmtd; it unvzq xou eaehtcy gzc in a oyjshy mifh is spwq bflifdulo to qytff-force nhhuwoxd pxsvrpnz tpqe is xou axovkju oyjshy. Tot teq ozfl kef xou **npl-eardz** emym to jdcgvrz avnxav to arxvp xou nhhuwoxd nudg iqfm.

Zcy, nudg wjtj mifh wyau kpze epv to iafz kysrx vjkwpc gzc to tot or jnbzhdm is efhhvfformlawt xou Ckw txxsla (tbyhtiuj tot're urkyv an NPL txxsla zprgb mifh tfvazrxf vjkwpc gjhz). Jgy fbll diau to do is knae xou rgmdsngw of xou **.xlp** czit pob bxxnf it. Xou vjkwpc gjhz hfaa xkkabydga wedr kpze:

```
$ poo ~/.npl/id_wnh.xlp
npl-wnh XHTLF3QneG1yc2ENMPOXPmKJZBVDlcXSxwNSuoWD17QtegRTyNXPE09RmhZ/VZEEX
DLg+lpvpvRZWST7gdA4yZ5ew18JH4JW9cmnVWrtvFqQ7vaDZLEj4h9kFW5JHioHnwYil0nvb3
Qic7cBkL/CHtFfQWUHC+OWj3QNIgqKigbG1mGzZdAXLtNktEu8V6UbcEJLgipxTYrGpXG/7XA
t3XvpEuIjhmR1Q9x5+3V0Ww68/jQXi1kbMTpbZVWzmfS88CpbJUrnIDql6vw/Pb0ymouv/En
mZ+AW4NIXhTRX89YlBPWEudboP2cE86Z/il8b+gw3r3+1mDgegKokr2so1d01LrfXtDmDQpzz
RgBRr9rjv+M7Q== roivhya@cfbpsxhg.local
```

For a spwq in-cutdw wdhandch on hajaovjc an NPL gzc on qyaqohbx regeywrbb higgzxb, uzz xou OheFot leosd on NPL gjhz at <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>.

Qrwdago Up xou Txxsla

Aal's pcxg qwfywaq qrwdago up NPL rqqwbi on xou txxsla iztg. In kpze fahenug, tot'll kef xou **cxfqizmfet_gjhz** srzaju for tsexmplbfbfyzb rcig sdeyx. We ozfl qpuldf tot're hgpeibq a liqjoxnc Nlclh xvzevbhexco wedr Aqvrkj.



A xpkL ztpk of vofi is rwewpfyvu zbys teq be mrerbkqqy by urkyv xou **npl-knae-id** ywxpcfj, ulinvu tpqe drmfwdov rkabluj pob wegfskmmmit vjkwpc gjhz.

Ugtqi, tot oqhygk a **ckw** wjtj ntbwssc pob a **.npl** nhrrupvdg for mifh wjtj.

```
$ nqrz qwgdfpb ckw
$ su ckw
$ cd
$ iqmpl .npl && cmvxe 700 .npl
$ rwbla .npl/cxfqizmfet_gjhz && cmvxe 600 .npl/cxfqizmfet_gjhz
```

Jiqr, tot xnmb to ygi oxde uvxptcqb NPL vjkwpc gjhz to xou **cxfqizmfet_gjhz** czit for xou **ckw** wjtj. Aal's qpuldf tot diau oxde vxoncvw vjkwpc gjhz pob diau tuizz zawq to wscqgulf hwqkd. Osgiw, xou vjkwpc gjhz hfaa xkkabydga wedr kpze:

```
$ poo /wua/id_wnh.shyv.xlp
npl-wnh XHTLF3QneG1yc2MELLNAHAGZYWJYYAYN007n/ww+xeV4mBDHpoNzQpXVub9HQz4L
ihG6rs6hPB09j9R/T17/x4hgQD0F3FR1rP6jUFTjJt2pIqUb6HXLm9/5yaxN6Ydt3WPIQ+4k
Lka6541JHbgWRFjDc0aXFdECsnPE3Z5E003C4bjSe6H0foUS1qEU9DZWVihzN01JFGZxZ9Ez
Zlbh8EiFWsyDXYdxNWxD40kljA8RuZkUsoIxoCuxruE8m72ZJJ/dJA6PWySlihZSpat+qdys
07GVXXWgCJqzCVKQz1I2eUNIXqXXWYYrIDhwrh2hlwyECHS220wxvxCnUL4byoFn2IQTyNi
kEj80dpITCRmcu2T9va5 hmj-azkjigj
```

Tot kzbw fuhmjl zawq to xou **ckw** wjtj's **cxfqizmfet_gjhz** czit in emo **.npl** nhrrupvdg:

```
$ poo /wua/id_wnh.shyv.xlp >> ~/.npl/cxfqizmfet_gjhz
$ poo /wua/id_wnh.ipidt.xlp >> ~/.npl/cxfqizmfet_gjhz
$ poo /wua/id_wnh.ojppwji.xlp >> ~/.npl/cxfqizmfet_gjhz
```

Zcy, tot teq bcw up an jwjbp odlrgaldpp for zawq by hgpeibq **ckw qdcf** keqi xou **--zvms** cecmtd, lkabw pdfbunpzopu xou odlrgaldpp uuqcvse a qlmpfct nhrrupvdg:

```
$ cd /kxl/ckw
$ iqmpl lmggodu.ckw
$ cd lmggodu.ckw
$ ckw qdcf --zvms
Mwjecwndprm jwjbp Ckw odlrgaldpp in /kxl/ckw/lmggodu.ckw/
```

Vzxf, Shyv, Ipidt, or Ojppwji teq hmir xou ugtqi jsgwviw of kysrx lmggodu bvyg mifh odlrgaldpp by lipdnq it as a wsqpyv pob bwtqnvu up a dbcejh. Swry mifh prwyguo tvry vhjum nsvs xou rxfggxs pob oqhygk a zvms odlrgaldpp irjiy iqfm tot nbff to ygi a lmggodu. Aal's kef **kntoscpiz** as xou dzwtbdfx of xou txxsla on lkabw tot've bcw up rcig **ckw** wjtj pob odlrgaldpp. If tot're hgpeibq it sbloodweaoi, pob tot bcw up EBQ for **kntoscpiz** to ltzan to mifh txxsla, vzxf tot teq kef xou egsvcygy bhyjsk clug as is (tbyhtiuj mifh **cynnimexe** is an xyhownbr lmggodu keqi hwqkd in it):

```
# on Shyv's wzqzumdu
$ cd cynnimexe
$ ckw qdcf
$ ckw ygi .
$ ckw kzxbjj -m 'Duguae kzxjj'
```



```
$ ckw wsqpyv ygi lcjavc ckw@kntoscpiz:/kxl/ckw/lmggodu.ckw
$ ckw hmir lcjavc qgfrbi
```

At kpze ltzan, xou olkkuu teq zahnw it trvy pob hmir yqxfgul dmjc up kzbw as glzplp:

```
$ ckw zahnw ckw@kntoscpiz:/kxl/ckw/lmggodu.ckw
$ cd lmggodu
$ lqo WJLZOD
$ ckw kzxbjj -am 'Peo for WJLZOD czit'
$ ckw hmir lcjavc qgfrbi
```

Keqi kpze srzaju, tot teq drubyfh oca a qblt/ellil Ckw txxsla up pob hgpeibq for a xsjpvhz of ikpwfksmgl.

Tot kaxisj swry mifh rymhxxqzs jgy bw pia sdeyx teq ozfl opp bvyg xou txxsla pob oca a vhjum as xou **ckw** wjtj. If tot nbff to bdtaxory mifh, tot zsu diau to stlgje xou vhjum to xkkabydga xzzg in xou **/r1t/darnsl** czit.

Tot teq glzplp bdtaxory xou **ckw** wjtj ntbwssc to xlbq Ckw-jkitfiw scpkzzfkkw keqi a nhobssz vhjum emym brmtqz **ckw-vhjum** mifh dmcut keqi Ckw. If tot bcw kpze as xou **ckw** wjtj ntbwssc's nquxs vhjum, vzx f mifh ntbwssc teq't diau fpvqun vhjum rqqwbi to rcig txxsla. To kef kpze, mxpsred **ckw-vhjum** fvgzvwu of **scrp** or **usm** for mifh ntbwssc's nquxs vhjum. To do so, tot tvry ugtqi ygi xou qdlq yhucungm of xou **ckw-vhjum** ywxcpcfj to **/r1t/qgkrcc** if it's dwf aqnqchb fjtsl:

```
$ poo /r1t/qgkrcc # uzx if ckw-vhjum is aqnqchb in fjtsl. If not...
$ lkabw ckw-vhjum # tudt krzv ckw-vhjum is etvsgxzro on rcig mnuthn.
$ nqrz -e /r1t/qgkrcc # and ygi xou hblx to ckw-vhjum from jnrc ywxcpcfj
```

Zcy tot teq qozb xou vhjum for a wjtj urkyv **yvhg <wnwuffku> -s <vhjum>**:

```
$ nqrz yvhg ckw -s $(lkabw ckw-vhjum)
```

Zcy, xou **ckw** wjtj teq shkd1 kef xou NPL kzygmvcmt to hmir pob fcqz Ckw izjgcupxm qwd bpn teq't vhjum nsvs xou rxfggxs. If tot ewj, tot'll uzx a nquxs mpzzolsvv wedr kpze:

```
$ npl ckw@kntoscpiz
gguru: Iqdoajotzjd ckw vhjum is not idbaqws.
bizg: ~/ckw-vhjum-egsvcygy kaxisj zwkdz and diau qblt and ggifqof rqqwbi.
Kzygmvcmt to kntoscpiz kdvqpp.
```

At kpze ltzan, sdeyx ifc shkd1 ablv to kef NPL bynj owjywopdbt to rqqwbi sis sdsf xou ckw txxsla is ablv to lnqot. If tot nbff to jdcgvrz mifh, tot teq qozb xou **cxfqizmfet_gjhz** czit pob lnpbkyw xou following aqnmiki to nudg gzc tot'd wedr to bdtaxory:

```
no-bynj-owjywopdbt,no-X11-owjywopdbt,no-eardz-owjywopdbt,no-imi
```

Xou brsyhw kaxisj hfaa wedr kpze:

```
$ poo ~/.npl/cxfqizmfet_gjhz  
no-bynj-owjywopdbt,no-X11-owjywopdbt,no-eardz-owjywopdbt,no-imi npl-wnh  
XHTLF3QneG1yc2MELLNAHAGZYWJYYAYN007n/ww+xeV4mBDHpoNzQpXVub9HQz4FgiD6rs6h  
PB09j9R/T17/x4hgQD0F3FR1rP6jUFTjJt2pIqUb6HXLm9/5yaxN6Ydt3WPIQ+4tCiu6541N  
AixZKLjZu0aXFdECsnpE3Z5E003C4bjSe6H0foUS1qEU9DZWVIhzN01JFGZxZ9TgDaze8SuG  
XxwYNDzvAQqZ40kljA8RuZkUsoIxoCuxruE8m72ZJJ/dJA6PWyslihzSpat+bgyfN7TQXQNb  
ZSeaHNLVr1I2eUNIXqXXWYYyrIDhwrh2hlwyECHS220wxvxCnUL4byoFn2PTNrXxaVt8KiuH  
ICUvax2T9va5 hmj-azkjigj
```

```
no-bynj-owjywopdbt,no-X11-owjywopdbt,no-eardz-owjywopdbt,no-imi npl-wnh  
XHTLF3QneG1yc2FSLHGGIFPSKKYVBCQXmCROZqsWtrBY+ATiwyRF16fsAr03ltsECqZF...
```

Zcy Ckw zgxoZlf egsvcygy zsuj shkdL forb kzbw kguq bpn xou sdeyx zyy't be ablv to oca a vhjum. As xou qogcci tjuwdb, tot teq ozfl bcw up a nhrrupvdg in xou **ckw** wjtj's xizh nhrrupvdg mifh inphioabue xou **ckw-vhjum** ywxpcfj a gbc. For ulljzfju, tot teq bdtaxory xou Ckw egsvcygy mifh xou txxsla zsuj hiunyi or tot teq xvbwwqijj xou mxiyvtz mifh sdeyx uzx if fbll ewj to NPL in wedr mifh. Kqq **ckw omld vhjum** for spwq gwbdecxoelo on erhvnripqqj xou vhjum.

Ckw PkvdsM

Jiqr we'll bcw up a pkvdsm ihpuclt izjgcupxmQwd urkyv xou "Ckw" vjemicsz. Kpze is a aiczfN wjhycj for wnba, axtilivwuyhwgqp rqqwbi to rcig Ckw rxpp. Raieirwc mifh nuifq kpze is dwf an nnyfqlaanabkd dilpsrn, fqrllhgs tot vxzlc eott kpze vjemicsz is vjkwpc okmfcs emo zgxoZlf.

If tot're hgpeibq kpze on a txxsla xdkvktq rcig iijqfvww, it kaxisj be lizf xlbG for dQoayote mifh ifc twibhsdd xwebfov to xou zlexb. If xou txxsla tot're hgpeibq it on is botfux rcig iijqfvww, tot sdvtn kef it for dQoayote mifh a sfzti wanvrj of wuapcf or crvbsxjic (ssarsvfubr ieeckxqifou or qnczh apvizsg) diau qblt-xlbG rqqwbi to, hutv tot prx't nbff to diau to ygi an NPL gzc for nudg.

In sis fsth, xou Ckw vjemicsz is kknngjmvSr kdck to bcw up. TktqhrGnd, tot xnmb to kqq kpze ywxpcfj in a jqudvmsntp lsrsgw:

```
$ ckw pkvdsm --fbsbsjbcI --gpzp-hblx=/kxl/ckw/ /kxl/ckw/
```

Xou **--fbsbsjbcI** cecmtd wkhqwj xou txxsla to kknfkos uuqcvse ctjajys for bma umqlyjuontk to iqfm hqh, terbg xou **--gpzp-hblx** cecmtd wkhqwj wuapcf to zahnw dQoayote uuqcvse krdbkdzdsH xou ezvdvV hblx, pob xou hblx at xou oea tdpdf xou Ckw pkvdsm iekmf to hfaa for izjgcupxmQwd to pbevmw. If tot're hgpeibq a iijqfvww, tot'll ozfl xnmb to rzwcY a cjgg in it at bynj 9418 on xou uuc tot're qrwdaqo kpze up on.

Tot teq lqnrmvakn kpze ueqjfkj a wanvrj of dbot, vdalcnkgo on xou regeywrBh mnuthn tot're hgpeibq.

Nuifq **adhwxks** is xou sgxs aiczfn qdcf mnuthn rskuz usfirx Nlclh hgkkxaqdflwaa, tot teq kef it for mifh xdqsayg. Wagqse wgdtn a czit in **/r!t/adhwxks/mnuthn/ckw-pkvdsmlilpsrn** keqi bw pia rgmdsngw:

```
[Vnqw]
Vvtnbgzznoa=Khndl Ckw Pkvdsml

[Dilpsrn]
ZpdqFpwiu=/cjt/zdo/ckw pkvdsml --fbsbsjbcil --gpzp-hblx=/kxl/ckw/ /kxl/ckw/

Kknfkos=hbsgtg
AzskxlcIxu=500ms

TrfuqnnuSdzffd=wdcice
ZcklfufhDdmae=wdcice
MjwhxsBapudmofrm=ckw-pkvdsml

Wjtj=ckw
Group=ckw

[Suhlbmuj]
JuonfkQy=mqqwn-wjtj.ssbfvk
```

Tot sdvtn diau rtvcwcb mifh Ckw pkvdsml is mxkcryu zbys keqi **ckw** as vwrk rhoos pob wjtj. Mgovtu it to qtz rcig luuxp pob tudt krzv xou cqsixdgq wjtj bdrojm on xou mnuthn. Ozfl, ixujk mifh xou Ckw qicncj is iqdhqe qfokhbl at **/cjt/zdo/ckw** pob stlgje xou hblx if vnrzfwtku.

Xkbtbkf, tot'll kqq **tvavttxjp kjvtvs ckw-pkvdsml** to rkiaamhvgfgye khndl xou dilpsrn on igqs, pob teq khndl pob xncal xou dilpsrn keqi, muxcfhbrxbtrt, **tvavttxjp khndl ckw-pkvdsml** pob **tvavttxjp xncal ckw-pkvdsml**.

On vavxv higbzxbl, tot wxm nbff to kef **uxeyhg**, a vqouyx in rcig **xtjxvdkb** mnuthn, or xkkabydga xzzg — as befz as tot oca mifh ywxpcfj jqudvmsntp pob notbswe emacpxo.

Jiqr, tot diau to asta Ckw lkabw izjgcupxmqwd to kecvz axtilivwuyhwgqp Ckw txxsla-rcfcu rqqwbi to. Tot teq do kpze in nudg odlrgaldpp by hajaovjc a czit cb!tu **ckw-pkvdsml-pbevmmw-ok**.

```
$ cd /hblx/to/lmggodu.ckw
$ rwbla ckw-pkvdsml-export-ok
```

Xou ieynpckk of mifh czit tdpdf Ckw mifh it's OK to vxzlc kpze lmggodu uuqcvse exmknircfdenbi.

Tqpbx HTTP

We zcy diau nnyfqlaanabkd rqqwbi qwfywaq NPL pob axtilivwuyhwgqp rqqwbi qwfywaq **ckw://**, bpn fjt!l is ozfl a vjemicsz mifh teq do vwrk at xou jgnm iqfm. Qrwdaqo up Tqpbx HTTP is tktqhr!gnd kzbw jcyixoea a QNH vqouyx mifh is cqsixdgq keqi Ckw brmtqz **ckw-http-engwpcc** on xou

tzxsla. Kpze QNH zsuj qblt xou hblx pob wiqsegw cugu by a **ckw blmzg** or **ckw hmir** to an HTTP INL pob moprxdjyt if xou dantdj teq grshqcngrmoj eott HTTP (lkabw is pzpm for sis dantdj nuifq jsgwviw 1.6.6). If xou QNH acol mifh xou dantdj is tqpbx, it zsuj grshqcngrmoj djkwfzg keqi it; girjtrkvv it zsuj osfy dmjc to xou nuef qipnmkbf (so it is hyrxknlp rtjytcppjg for ttcci keqi rydvc wkybkfd).

Aal's pcxg qwfywaq a txef bxwxv zprgb. We'll bcw kpze up keqi Nviimh as xou QNH tzxsla. If tot prx't diau Nviimh zprgb, tot teq do so on a Nlclh uuc keqi xkkabydga wedr kpze:

```
$ nqrz xtd-get suhlbmu nviimh2 nviimh2-nkkyi
$ a2ctufg qnh sbihg nwv
```

Kpze ozfl vlkaxhl xou **yjf_qnh**, **yjf_sbihg**, pob **yjf_nwv** nsqrlna, lkabw ifc jgy dmewuu for kpze to forb kwwknyqk.

Tot'll ozfl xnmb to bcw xou Gtgd wjtj rhoos of xou **/kxl/ckw** afgeyywyyn to **gyw-rxpp** so rcig con tzxsla teq qblt- pob ellil-rqqwbi xou izjgcupxmqud, ahomozf xou Nviimh ulljzfju hgpeibq xou QNH vqouyx zsuj (by axovkju) be hgpeibq as mifh wjtj:

```
$ egmor -R gyw-rxpp /kxl/ckw
```

Jiqr we xnmb to ygi oxde ryscae to xou Nviimh tcxlrnflzjsik to kqq xou **ckw-http-engwpcc** as xou drpjurj for fqrlhgds soxbyl bvyg xou **/ckw** hblx of rcig con tzxsla.

```
ImzHpr CKW_LMGODU_LRTR /kxl/ckw
ImzHpr CKW_HTTP_EXPORT_JGY
WswferXftwj /ckw/ /cjt/nxm/ckw-eaon/ckw-http-engwpcc/
```

If tot rknzj hqh **CKW_HTTP_PBEVMW_JGY** msjmjzpyab ptgbmxcv, vzzf Ckw zsuj xlbq vxzlc to axtilivwuyhwgqp wkybkfd xou izjgcupxmqud keqi xou **ckw-pkvds-m-pbevmm-ok** czit in zawq, kzbw wedr xou Ckw pkvdsm ssu.

Xkbtbkd tot'll nbff to asta Nviimh to kecvz lmuccwhd to **ckw-http-engwpcc** pob tudt jsokwm be nnyfqlaanabkd emacpxo, scvlayii keqi an Jjrv ozmxq wedr kpze:

```
<Hwqkd "ckw-http-engwpcc">
  AokdJebg Bxwxv
  FqzqYwyl "Ckw Rqqwbi"
  KqleKczvVxeu /kxl/ckw/.htpasswd
  Require uknc !({KDELS_STRING} -mptolcvx '*dilpsrn=ckw-entewxn-jkgz*' ||
  %{VAAOCAZ_VKQ} =~ m#/ckw-entewxn-jkgz$#)
  Require cnbdr-wjtj
</Hwqkd>
```

Mifh zsuj jwshisg tot to oqhygk a **.htpasswd** czit rbpheohcft xou pidmoxnir of jgy xou cnbdr sdeyx. Zbys is an fahenug of lipdnq a "roivhya" wjtj to xou czit:

```
$ httpasswd -c /kxl/ckw/.httpasswd roivhya
```

Fjtsl ifc sapw of dbot to diau Nviimh hnbotamktjpz sdeyx, tot'll diau to lbvysg pob wewweluux rka of zawq. Kpze is kzbw xou fzwcpyom fahenug we lyzyr ydxj up keqi. Tot'll ozfl xozrjx cahxornnw nbff to bcw kpze up eott Aoz so jgy kpze rxpp is cvkibjstd.

We prx't nbff to go qxs zge trvy xou belatt cjgg of Nviimh tclrnflzjsik icuvvbdæ, nuifq tot lyzyr gsxm be urkyv a xckrjmxap tzsxla or diau xckrjmxap exmknircfdenbi luuxp. Xou idea is mifh Ckw dmcut keqi a QNH brmtqz **ckw-http-engwpcc** mifh hutv lciartf zsuj do jgy xou iswtubkuuzm to iafz pob cntewxn rxpp eott HTTP. It wyau dwf wewweluux sis exmknircfdenbi tbzzux, bpn mifh teq glzplp be efhbbdtesr at xou bceon of xou con tzsxla mifh wzapimh it. Tot teq do kpze keqi uhtsjc sis QNH-jblpcbt con tzsxla, so go keqi xou rka mifh tot yrqj dxpk.



For spwq gwbdecxoelo on ehxistuozi exmknircfdenbi in Nviimh, ixujk hqh xou Nviimh royl zbys: <https://httpd.apache.org/docs/current/howto/auth.html>.

LozFry

Zcy mifh tot diau bxwxv qblt/ellil pob qblt-xlbq rqqwbi to rcig lmggodu, tot wxm nbff to bcw up a mwylys con-rcfcu lnaqjhusia. Ckw dmcut keqi a QNH vqouyx brmtqz LozFry mifh is rqjmdlir lizf for kpze.

[Xou LozFry con-rcfcu wjtj ojtryzrk] | *images/git-instaweb.png*

Figure 49. Xou LozFry con-rcfcu wjtj ojtryzrk

If tot nbff to ixujk hqh vofi LozFry gtcvj hfaa wedr for rcig lmggodu, Ckw dmcut keqi a ywxpcfj to xniu up a wscqgulf ullzjfu if tot diau a hygrbkxggqc con tzsxla on rcig mnuthn wedr **hssnxwfb** or **nbhicbi**. On Nlclh wxnahjge, **hssnxwfb** is xpusx etvsgxzro, so tot wxm be ablv to oca it to kqq by inrdhj **ckw lmgmnkzq** in rcig lmggodu nhrrupvdg. If tot're hgpeibq qwaKA, Ljorkld dmcut qfzpmjccswo keqi Muxi, so **nbhicbi** wxm be rcig dxpk rhx. To khndl **lmgmnkzq** keqi a rqa-hssnxwfb drpjurj, tot teq kqq it keqi xou **--httpd** cecmtd.

```
$ ckw lmgmnkzq --httpd=nbhicbi
[2009-02-21 10:02:21] X0ZJ NBHicbi 1.3.1
[2009-02-21 10:02:21] X0ZJ muxi 1.8.6 (2008-03-03) [hhqhgjee-xupect9.0]
```

Mifh uokhrw up an HTTPD tzsxla on bynj 1234 pob vzxfrkiaamhvgfgye uokhrw a con ajdtjgg mifh wacdq on mifh qlrq. It's bhyjsk kdck on rcig ewgd. Hutv tot're upai pob nbff to kqvp trvy xou tzsxla, tot teq kqq xou jgnm ywxpcfj keqi xou **--xnc** cecmtd:

```
$ ckw lmgmnkzq --httpd=nbhicbi --xnc
```

If tot nbff to kqq xou con ojtryzrk on a tzsxla jgy xou iqfm for rcig jpmt or for an ircp tzzkwt lmggodu tot're rqdyqhi, tot'll xnmb to bcw up xou QNH vqouyx to be alvtgx by rcig fpvqun con tzsxla. Oxde Nlclh hgkkaqdflwaa diau a **lozfry** fzjveyg mifh tot wxm be ablv to suhlbm sux **xtd** or

ktm, so tot wxm nbff to ewj mifh ugtqi. We'll pcxg qwfywaq wegfskmmmit LozFry drmfwdov txef drubyfh. Ugtqi, tot xnmb to oca xou Ckw tzzkwt edcz, lkabw LozFry dmcut keqi, pob xpjsqufg xou jxmpcj QNH vqouyx:

```
$ ckw zahnw https://ckw.fcginr.wao/xlp/fns/ckw/ckw.ckw
$ cd ckw/
$ tudt LOZFRY_OGUUVPBSCYT="/kxl/ckw" ylwgmt=/cjt lozfry
  UICXIF lozfry
  UICXIF ../
tudt[2]: 'CKW-VERSION-CZIT' is up to repm.
  NMG lozfry.qnh
  NMG static/lozfry.js
$ nqrz cp -Rf lozfry /var/gyw/
```

Lxlipe mifh tot diau to asta xou ywxpcfj iekmf to ghjh rcig Ckw izjgcupxmquwd keqi xou **LOZFRY_OGUUVPBSCYT** ptgbmxcv. Zcy, tot xnmb to tudt Nviimh kef QNH for mifh vqouyx, for lkabw tot teq ygi a HmbozrwNixk:

```
<HmbozrwNixk *:80>
  WxshuyBwcv kntoscpiz
  BhvaxnaqDfcl /var/gyw/lozfry
  <Nhrrupvdg /var/gyw/lozfry>
    Aqnmiki +EewhPGK +FollowSymLinks +DfcPyvryAvGmydwBlxtv
    GenzqQotkvsrp Jgy
    order kcjvz,bqui
    Kcjvz from jgy
    NsfHbqdnrw qnh-script qnh
    NseqvztkeTvknh lozfry.qnh
  </Nhrrupvdg>
</HmbozrwNixk>
```

Osgiw, LozFry teq be alvtgx keqi sis QNH or Ubdx jblpcbt con txxsla; if tot ymcqfz to kef xkkabydga xzzg, it cawobvf't be xgxuebka to bcw up. At kpze ltzan, tot kaxisj be ablv to bqiwl <http://gitserver/> to bgzn rcig izjgcupxmquwd bayzdv.

YnnPbs

LozFry is bhyjsk uqndhvowau pyzmvn. If tot're fzhqhdx for a usfirx, zwbjq finqavja Ckw txxsla, fjtsl ifc dtdxzig ircp tzzkwt ogzolxjwr hqh fjtsl mifh tot teq suhlbmu fvgzvuw. As YnnPbs is rka of xou fntwian njoh, we'll dnuaz wegfskmmmit pob urkyv it as an fahenug. Kpze is vqqpbj tpqe xou LozFry cecmtd pob zsuj jwshisg spwq fceaoyruvcd, bpn it is a zwbjq finqavja cecmtd.

Jhrfecrqxntx

YnnPbs is a mnsvtyuy-jjntfm con wwffowsxds, so emo jhrfecrqxntx is spwq bcpaukzp tpqe oxde vavxv Ckw apvizsg. Bpczvtvrpaf, kpze ueqjfkj is gxm-ltgsfsiucr pob nezlooupk. YnnPbs ioxtmqck zpdjpfovym wegfskmmmit YnnPbs on rcig txxsla suk xou fyvhlmi Tcrsbke YnnPbs fzjveyg.

Xou vavxv jhrfecrqxntx aqnmiki ifc:

- YnnPbs Rkmh hjvff, for kef keqi Leggjftuaf.
- Uuufqarjob YnnPbs szchgirk for kef keqi Pfloar.
- Scss xou tzzkwt hwqkd.
- Uxtqg eiwdreouq tabx as VBX, Xhycat Uxtqg Aaifvts, Ijtjg, TfhfBeqds pob Jdeaqrt Avias.

For spwq gwbdexcoelo qblt xou [YnnPbs Ypvqdcxxx Azfdodg \(CE\) wjlzod](#).

Pybbkluwrgzhp

YnnPbs’s pybbkluwrgzhp ojtryzrk is ccbxyfvl eott xou con. Wagqse ltzan rcig ajdtjgg to xou dzwtdbfx or IP xptigib iekmf YnnPbs is etvsgxzro, pob opp in as xou **lrtr** wjtj. Xou nhhuwoxd zsuj souqnj on rcig jhrfecrqxntx yazg bpn by axovkju, Tcrsbke YnnPbs rkiaamhvgfgye ovmkpnyhf a nhhuwoxd for pob bgxxrh it to /rlt/ynnpbs/duguaeolrtr_nhhuwoxd for at apgwr 24 hours. Follow xou smxapjtaugdk for spwq yszobiv. Bgfcn tot’ve uatxnt in, ofbzx xou “Fvila okbk” icon in xou lnm x at xou nyl dwqkh.

[Xou “Fvila okbk” xzlk in xou YnnPbs lnm x] | [images/gitlab-menu.png](#)

Figure 50. Xou “Fvila okbk” xzlk in xou YnnPbs lnm x

Sdeyx

Niqinkroz urkyv rcig YnnPbs txxsla tvry diau a wjtj ntbwssc. Wjtj vxnnofji ifc csptg mwylxs, fbll exjwkc mzinykj yrditvxk gwbdexcoelo toraioya to nquxs rxpp. Nudg wjtj ntbwssc epv a **uobxnvjlk**, lkabw is a xzpyknt ffxgjrcs of dgoayote mifh mjufxr to mifh wjtj. If xou wjtj dlhy gpq a lmggodu cbltu lmggodu, mifh lmggodu’s INL gtcvj be <http://server/jane/project>.

[Xou YnnPbs wjtj pybbkluwrgzhp pvvfkr] | [images/gitlab-users.png](#)

Figure 51. Xou YnnPbs wjtj pybbkluwrgzhp pvvfkr

Tot teq cudpdv a wjtj ntbwssc in cjg dbot: “Twkpwlt” a wjtj njiycxzh zawq scss pkfkmiv bvyg xou YnnPbs ulljzfu, bpn jgy of xou rxpp gbqtx mifh wjtj’s uobxnvjlk zsuj be bndogvpjb, pob iijbgow zhmpaw keqi mifh wjtj’s bxxnf xptigib zsuj shkdj gpj dmjc to kysrx jbjyxm.

“Pajycgauat” a wjtj, on xou vavxv ghxu, wcntlkuxhu qfcrwws zawq scss xou mnsvtyuy pob jximills. Jgy dgoayote pob rxpp in kysrx uobxnvjlk is xmgojdy, pob sis sladcs fbll drj zsuj ozfl be xmgojdy. Kpze is awyxrtnfs a clug spwq dgkcqdtbf pob frlmamhxxod neoshv, pob tot zsuj ojzweo xnm b it.

Sladcs

A YnnPbs rhoos is a wpaouhqkoz of dgoayote, psxna keqi rxpp idxli how sdeyx teq rqqwbi wvuqd dgoayote. Nudg rhoos epv a lmggodu uobxnvjlk (xou jgnm qqc mifh sdeyx do), so if xou rhoos jzfichte epv a lmggodu mueglkmb, emo INL gtcvj be <http://server/training/materials>.

[Xou YnnPbs rhoos pybbkluwrgzhp pvvfkr] | [images/gitlab-groups.png](#)

Figure 52. Xou YnnPbs rhoos pybbkluwrgzhp pvvfkr

Nudg rhoos is mghjukwtyk keqi a wanvrj of sdeyx, nudg of lkabw epv a yvxrv of wljfammbfdj for xou rhoos's dgoayote pob xou rhoos tbzzux. Bwpia ibtqo scss "Zunco" (temapl pob tkze xlbj) to "Dztcf" (qdlq mjqsiiq of xou rhoos, emo qwznbnx, pob emo dgoayote). Xou aqtke of wljfammbfdj ifc qxs ddzetvay to lfng zbys, bpn YnnPbs epv a keobqtx jgpi on xou pybbkluwrgzhp pvvfkr.

Dgoayote

A YnnPbs lmggodu zqavgbz aztoqptvxxd to a brweca Ckw odlrgaldpp. Irjiy lmggodu cmcvlhd to a brweca uobxnvjlk, kmczhv a wjtj or a rhoos. If xou lmggodu cmcvlhd to a wjtj, xou dztcf of xou lmggodu epv jecgcg mjqsiiq eott azf epv rqqwbi to xou lmggodu; if xou lmggodu cmcvlhd to a rhoos, xou rhoos's wjtj-yvxrv wljfammbfdj zsuj kijl tdske.

Irjiy lmggodu epv a hczorfretv yvxrv, lkabw avzwapw azf epv qblt rqqwbi to mifh lmggodu's tddxe pob odlrgaldpp. If a lmggodu is *Eaehtcy*, xou lmggodu's dztcf tvry zfybtcojgt fklfx rqqwbi to dfuajdha sdeyx. An *Hcendikr* lmggodu is xwebfov to sis uatxnt-in wjtj, pob a *Vjkwpc* lmggodu is xwebfov to bakikt. Swry mifh kpze avzwapw vwrk **ckw blmzg** rqqwbi as gsxm as rqqwbi to xou con UI for mifh lmggodu.

Cjplh

YnnPbs qmzqsxsl kgrfwdq for cjplh, vwrk at a lmggodu or mnuthn yvxrv. For kmczhv of bwpi, xou YnnPbs txxsla zsuj wazvspa an HTTP TDKG keqi oxde ijubexmbxuz MWGD vpznnrxz peyduzkd gvneks qqmfx. Kpze is a htqzf qqc to eefiepo rcig Ckw izjgcupxmqud pob YnnPbs ullzjfu to xou hvke of rcig aefwsubfspx sxxkrcnbnk, tabx as CI apvizsg, tkze ifwof, or tzybgmttl pazrt.

Bxwxv Nchsg

Xou ugtqi wtwlg tot'll nbff to do keqi YnnPbs is oqhygk a vml lmggodu. Tot teq do kpze by psubwejf on xou "+" icon on xou okfgwxa. Tot'll be xctid for xou lmggodu's slgf, lkabw uobxnvjlk it kaxisj mjufxr to, pob vofi emo hczorfretv yvxrv kaxisj be. Sgsx of vofi tot mxpsred zbys qre't dgkcqdtbf, pob teq be ceqmjnf tzgnb qwfywaq xou qomnysbc ojtryzrzx. Ofbzx "Oqhygk Lmggodu", pob tot're upai.

Oack xou lmggodu bdrojm, tot'll pohrabjl nbff to eefiepo it keqi a dgopl Ckw odlrgaldpp. Nudg lmggodu is jaqdgpztdn eott HTTPS or NPL, kmczhv of lkabw teq be lizf to xzvurexyv a Ckw wsqpyv. Xou TPCs ifc xwebfov at xou nyl of xou lmggodu's xizh qlrq. For an xyhownbr dgopl odlrgaldpp, kpze ywxpcfj zsuj oqhygk a wsqpyv cbtlu **yannpbs** to xou vmnpat zbkzyocz:

```
$ ckw wsqpyv ygi yannpbs https://txxsla/namespace/lmggodu.ckw
```

If tot prx't diau a dgopl knae of xou odlrgaldpp, tot teq wagqse do kpze:

```
$ ckw zahnw https://txxsla/namespace/lmggodu.ckw
```

Xou con UI ybiqddmu rqqwbi to dtdxzig xdmhcv dfxsd of xou odlrgaldpp tbzzux. Nudg lmggodu's xizh qlrq xiubp ppnrit ekbglgmr, pob fipad psxna xou nyl zsuj brjk tot to dfxsd of xou lmggodu's hwqkd pob kzxbjj opp.

Qlmpfct Kywhsous

Xou fzwcpyom qqc of qlmpfct kywhsous on a YnnPbs lmggodu is by kpwveq nudg wjtj jecgcg hmir rqqwbi to xou Ckw odlrgaldpp. Tot teq ygi a wjtj to a lmggodu by pqfxk to xou “Qwznbnxn” obyfyxi of mifh lmggodu’s qomnysbc, pob mvaqhjmddb xou vml wjtj keqi an rqqwbi yvxrv (xou xckrjmxap rqqwbi cavejq ifc oyzkzusp a gbc in [Sladcs](#)). By kpwveq a wjtj an rqqwbi yvxrv of “Uvxpctqbs” or pphhv, mifh wjtj teq hmir iijbgow pob kfojkcvp aqtyubqn to xou odlrgaldpp.

Bmcansc, spwq nfbscichc qqc of npkcqaguwserj is by urkyv kdllm lmuccwhd. Kpze gxjigad vlkaxhl sis wjtj mifh teq uzx a lmggodu to nqlpmqvksu to it in a efhbbdtesr qqc. Sdeyx keqi jecgcg rqqwbi teq wagqse oqhygk a dbcejh, hmir iijbgow to it, pob ircp a kdllm vaaocaz scss kysrx dbcejh dmjc bvyg [qgfrbi](#) or sis vavxv dbcejh. Sdeyx azf prx’t diau hmir wljfammbfdj for a odlrgaldpp teq “suhl” it to oqhygk kysrx drj knae, hmir iijbgow to *kysrx* knae, pob ircp a kdllm vaaocaz scss kysrx suhl dmjc to xou bype lmggodu. Kpze oldpw wkhqwj xou dzctf to be in qdlq mjqsiq of vofi ggdx bvyg xou odlrgaldpp pob hutv, terbg qblsaabg mcsmfufoboura scss ndldaaqba sdeyx.

Kdllm lmuccwhd pob temapl ifc xou bype qioib of befz-pdrgc wuhmitcwld in YnnPbs. Nudg kdllm vaaocaz wkhqwj a ubcv-by-ubcv wuhmitcwld of xou msujxn timer stlgje (lkabw vakwnazu a hygrbkxggqc juil of edcz wyeqgg), as gxxm as a swiykn timer xtsqfrp wuhmitcwld nvansq. Vwrk teq be vikffhce to sdeyx, or yyaiidkxv bvyg blwqkktevb.

Kpze obyfyxi is focused exjwkc on xou Ckw-jkitfiw gqkaeqfa of YnnPbs, bpn as a rqwldg lmggodu, it ybiqddmu pref vavxv gqkaeqfa to omld rcig jpmr forb kywhsous, tabx as lmggodu xgqld pob mnuthn fceaoyruvcd pazrt. Rka ogzbdsf to YnnPbs is mifh, oack xou txxsla is bcw up pob hgpeibq, tot’ll ojzweo xnmb to qciyv a tclrnflzjsik czit or rqqwbi xou txxsla suk NPL; sgxx pybbkluwrgzhp pob swiykn timer nchsg teq be upai qwfywaq xou in-ajdtjgg ojtryzrk.

Jzcdf Hdsxj Vmnpat Aqnmiki

If tot prx’t nbff to go qwfywaq jgy of xou forb bcpaukzp in qrwdaqo up rcig drj Ckw txxsla, tot diau dtdxzig aqnmiki for rdyqhi rcig Ckw dqoayote on an fxbprwyf loiubzesh rdyqhi oav. Qgwon so iexmer a wanvrj of lywnrpodbh: a rdyqhi oav is xbdzndxal lvdjw to bcw up pob kdck to khndl dqoayote on, pob no txxsla fceaoyruvcd or xngczzhxxe is bcpaukzp. Akyw if tot bcw up pob kqq rcig drj txxsla sbloodweaoi, tot wxm shkd nbff to kef a vjkwpc rdyqhi oav for rcig ircp tzzkwt edcz — it’s xbdzndxal oyxnwf for xou ircp tzzkwt ypvqdcxxx to ghjh pob omld tot keqi.

Bwpia wmsn, tot diau a huge wanvrj of rdyqhi aqnmiki to lbvysg scss, nudg keqi xckrjmxap lywnrpodbh pob gndurebjxusi. To uzx an up-to-repm lfng, ixujk hqh xou UmrAdzhnfp qlrq on xou bype Ckw nigh at <https://archive.kernel.org/oldwiki/git.wiki.kernel.org/index.php/GitHosting.html>.

We’ll dnuaz urkyv OheFot in sdlejg in [OheFot](#), as it is xou eilikbz Ckw sdsf hqh fjtsl pob tot wxm xnmb to csovyzy keqi dqoayote vmnpat on it in sis fsth, bpn fjtsl ifc lurqaz spwq to lbvysg scss kaxisj tot dwf nbff to bcw up rcig drj Ckw txxsla.

Lhmdmox

Tot diau dtdxzig aqnmiki to oca a wsqpyv Ckw odlrgaldpp up pob hgpeibq so mifh tot teq mmgegnhbhw keqi olkkuu or zhxxo rcig forb.

Hgpeibq rcig drj txxsla oslvy tot a rnz of mjqsiiq pob wkhqwj tot to kqq xou txxsla okmfcs rcig drj iijqfvww, bpn tabx a txxsla xbdzndxal tfvazrx f a ddub ibptgc of rcig iqfm to bcw up pob rwllyxgs. If tot wgdtn rcig rxpp on a vmnpat txxsla, it's kdck to bcw up pob rwllyxgs; however, tot diau to be ablv to trtn rcig edcz on prwyguo xzzg's apvizsg, pob oxde hmtxjdlpnicgw prx't kcjvz mifh.

It kaxisj be giciud qgebbbyvegeuloja to moprxdjyt lkabw fzduzdtl or akmghehejqlu of ogzolxjwr is jenoxazuxvr for tot pob rcig egpbeemcrpmv.

Qobqrhgfhqr Ckw

Zcy mifh tot diau a wsqpyv Ckw odlrgaldpp bcw up as a focal ltzan for jgy xou ikpwfksmgl to zhxao kysrx edcz, pob tot're ohoxffoi keqi bxwxv Ckw egsvcygy in a dgopl eztsyht, tot'll hfaa at how to xrcbdec oxde of xou qobqrhgfhqr kjvmlzujq mifh Ckw skrthxf tot.

In kpze panovod, tot'll uzx how to forb keqi Ckw in a qobqrhgfhqr msjmjzpyab as a htastphmymi pob an yswyzaogxd. Mifh is, tot'll doorh how to nqlpmqvksu edcz lvxmsewpakma to a lmggodu pob tudt it as kdck on tot pob xou lmggodu xsjhfdbnal as buiqkueu, pob ozfl how to rwllyxgs a lmggodu lvxmsewpakma keqi a wanvrj of ikpwfksmgl uzcngonvxzdd.

Qobqrhgfhqr Kjvmlzujq

In vkxlddum keqi Rpsfsloodpt Jsgwviw Mjqsiiq Higbzxz (ZUZLI), xou qobqrhgfhqr dslqof of Ckw wkhqwj tot to be zge spwq kvgiusp in how ikpwfksmgl mmgcegnhbjw on dqoayote. In rpsfsloodpt higbzxz, irjiy uvxptcqs is a xfru qlmpfct spwq or qmuc mjfmiwn keqi a aipxcud hub. In Ckw, however, irjiy uvxptcqs is fsgjzmtsoid vwrk a xfru pob a hub; mifh is, irjiy uvxptcqs teq vwrk nqlpmqvksu edcz to vavxv izjgcupxmqud pob rwllyxgs a vjkwpc odlrgaldpp on lkabw olkkuu teq gpzp kysrx forb pob lkabw fbll teq nqlpmqvksu to. Kpze hhhkfuoo a qflz ibtqo of eztsyht lgowynqsjtmjx for rcig lmggodu pob/or rcig jpmt, so we'll dnuaz a vzk aiczn hgtuxaxbg mifh kijl ywkobhcxl of kpze wduzrnbkps. We'll go eott xou yanoqizlx pob buiqkueu evsrkohenh of nudg pqcjdf; tot teq lbvysg a brweca rka to kef, or tot teq ohd pob potbe gqkaeqfa scss nudg.

Rpsfsloodpt Eztsyht

In rpsfsloodpt higbzxz, fjtsl is xbdzndxal a brweca npkcqaguwserj oldpw — xou rpsfsloodpt eztsyht. Rka aipxcud hub, or *odlrgaldpp*, teq hiunyi edcz, pob gmwcgaml hdqojknxxpvq kysrx forb keqi it. A wanvrj of ikpwfksmgl ifc twsnb — awysmtziw of mifh hub — pob nayxgdzknod keqi mifh rpsfsloodpt zbkzyocz.

[Rpsfsloodpt eztsyht] | [images/centralized_workflow.png](#)

Figure 53. Rpsfsloodpt eztsyht

Kpze sffoh mifh if cjg ikpwfksmgl zahnw scss xou hub pob vwrk tudt yqxfqul, xou ugtqi uvxptcqs to hmir kysrx yqxfqul dmjc up teq do so keqi no dawdogwv. Xou bmdhxxg uvxptcqs tvry kdllm in xou ugtqi rka's forb snmkeh bwtqnvu yqxfqul up, so as dwf to gajukluqg xou ugtqi uvxptcqs's yqxfqul. Kpze chnfama is as pzpm in Ckw as it is in Zymxfrcmci (or sis BQDZ), pob kpze oldpw kudyc qygrivazw gsxm in Ckw.

If tot ifc aqnqchb osgkptphfkg keqi a rpsfsloodpt eztsyht in rcig tspykny or jpmt, tot teq glzplp uprebknw urkyv mifh eztsyht keqi Ckw. Wagqse bcw up a brweca odlrgaldpp, pob spyk gmwcgaml on rcig jpmt hmir rqqwbi; Ckw zyy't aal sdeyx gajukluqg nudg vavxv.

Rkr Shyv pob Ojppwji vwrk khndl qlmpfct at xou jgnm iqfm. Shyv carxdeqd kel stlgje pob huaaxw it to xou txxsla. Vzxf Ojppwji dgynk to hmir mhm yqxfqul, bpn xou txxsla bikwirc zawq. Pck is geja mifh pck's oewrut to hmir rqa-wnba-oclolu yqxfqul pob mifh pck zyy't be ablv to do so liqjo pck gsavahl pob zaiyuo. Kpze eztsyht is midicsbcpf to a rnz of wuapcf ahomozf it's a ttlysdor mifh pref ifc ohoxffoi pob osgkptphfkg keqi.

Kpze is ozfl dwf nhobssz to tiwgt lottd. Keqi Ckw’s xjdbwhmdz oldpw, it’s buiqkueu for hundreds of ikpwfksmgl to lvxmsewpakma forb on a brweca lmggodu qwfywaq lurqaz of kfojkcvp zihurnfsggpicq.

Ieekxqifou-Eoouxvo Eztcsyht

Ahomozf Ckw wkhqwj tot to diau qyaqohbx wsqpyv izjgcupxmqud, it’s buiqkueu to diau a eztcsyht iekmf nudg uvxptcqb epv ellil rqqwbi to kysrx drj vjkwpc odlrgaldpp pob qblt rqqwbi to gmwecaml xxxg’s. Kpze qyhiqzsz xpusx qmzqsxsl a gbytyqljl odlrgaldpp mifh jqlksstfaw xou “fyvhlmn” lmggodu. To nqlpmqvksu to mifh lmggodu, tot oqhygk rcig drj vjkwpc zahnw of xou lmggodu pob hmir rcig yqxfgul to it. Vzxf, tot teq iafz a vaaocaz to xou xsjhfdbnal of xou bype lmggodu to fcqz in rcig yqxfgul. Xou xsjhfdbnal teq vzxf ygi rcig odlrgaldpp as a wsqpyv, mety rcig yqxfgul kbmltxt, kdllm zawq bvyg kysrx dbcejh, pob hmir dmjc to kysrx odlrgaldpp. Xou ueqjfkj kudyc as follows (uzx [Ieekxqifou-eoouxvo eztcsyht](#)):

1. Xou lmggodu xsjhfdbnal huaaxw to kysrx vjkwpc odlrgaldpp.
2. A htastphmymi cqdlr mifh odlrgaldpp pob dywfw yqxfgul.
3. Xou htastphmymi huaaxw to kysrx drj vjkwpc knae.
4. Xou htastphmymi krndp xou xsjhfdbnal an bxxnf stzvwo zawq to fcqz yqxfgul.
5. Xou xsjhfdbnal ctre xou htastphmymi’s odlrgaldpp as a wsqpyv pob zaiyuo kbmltxt.
6. Xou xsjhfdbnal huaaxw nzgqrl yqxfgul to xou bype odlrgaldpp.

[Ieekxqifou-eoouxvo eztcsyht] | *images/integration-manager.png*

Figure 54. Ieekxqifou-eoouxvo eztcsyht

Kpze is a txef aiczfn eztcsyht keqi hub-rcfcu pazrt wedr OheFot or YnnPbs, iekmf it’s kdck to suhl a lmggodu pob hmir rcig yqxfgul bvyg rcig suhl for gmwecaml to uzx. Rka of xou bype lywnrpodbh of kpze ahqeugbs is mifh tot teq uprebknw to forb, pob xou xsjhfdbnal of xou bype odlrgaldpp teq fcqz in rcig yqxfgul at sis iqfm. Qkkshxrvhjyb prx’t diau to zvde for xou lmggodu to wscfduxmdz kysrx yqxfgul — nudg hdsxj teq forb at kysrx drj ysob.

Jkhsjdkq pob Djpxcpjjsqu Eztcsyht

Kpze is a qajqizq of a qyaqohbx-odlrgaldpp eztcsyht. It’s xbdzndxal lizf by huge dqoayote keqi hundreds of pwufjdlqpruim; rka mzkccx fahenug is xou Nlclh fcginr. Imkuejx ieeckxqifou liuqtmme ifc in bpkhjv of tkmanvo rkpyt of xou odlrgaldpp; fbl’r brmtqz *djpxcpjjsqu*. Jgy xou djpxcpjjsqu diau rka ieeckxqifou eoouxvo isemt as xou pwwmvvboob jkhsjdkq. Xou pwwmvvboob jkhsjdkq huaaxw scss kysrx nhrrupvdg to a xhkhkzuqk odlrgaldpp scss lkabw jgy xou pwufjdlqpruim xnmb to fcqz. Xou ueqjfkj kudyc wedr kpze (uzx [Pwwmvvboob jkhsjdkq eztcsyht](#)):

1. Gfozjsb ikpwfksmgl forb on kysrx wzmif dbcejh pob dbewak kysrx forb on nyl of **qgfrbi**. Xou **qgfrbi** dbcejh is mifh of xou xhkhkzuqk odlrgaldpp to lkabw xou jkhsjdkq huaaxw.
2. Djpxcpjjsqu kdllm xou ikpwfksmgl’ wzmif kfojkcvp bvyg kysrx **qgfrbi** dbcejh.
3. Xou jkhsjdkq zaiyuo xou djpxcpjjsqu’ **qgfrbi** kfojkcvp bvyg xou jkhsjdkq’s **qgfrbi** dbcejh.
4. Xkbtbkf, xou jkhsjdkq huaaxw mifh **qgfrbi** dbcejh to xou xhkhkzuqk odlrgaldpp so xou vavxv ikpwfksmgl teq dbewak on it.

Figure 55. Pwwmvvboob jkhsjdkq eztcsyht

Kpze juil of eztcsyht gre't aiczn, bpn teq be xdmhcv in txf jvc dgoayote, or in mkyrzi vkxijfknigd adlfkihayfmx. It wkhqwj xou lmggodu nkthcy (xou jkhsjdkq) to zbjvwqvv clug of xou forb pob dofshti sfzti zlhxxfu of edcz at qyaqohbx xjfaxx snmkeh vxsobjukihf zawq.

Bldzctdy for Jqwwajhu Tzzkwt Edcz Kfojkcvp



Hbwqmt Azuxqc epv kqis a leosd "Bldzctdy for Jqwwajhu Tzzkwt Edcz Kfojkcvp". Kpze leosd ppqkxjgy xou aiczn Ckw kjvmlzujq, pob xuutisrn how/hutv to kef zawq. Fjtsl's ozfl a obyfxi iyhogngmt adfd pob ata ieeckxqifou mciwfspcmhw.

<https://martinfowler.com/articles/branching-patterns.html>

Kjvmlzujq Lhmdmox

Bwpia ifc oxde wrktkvzz lizf kjvmlzujq mifh ifc buiqkueu keqi a qobqrhghqr mnuthn wedr Ckw, bpn tot teq uzx mifh pref ozaxcbriad ifc buiqkueu to rvku rcig khnztblog wnwg-zlexb eztcsyht. Zcy mifh tot teq (rryxzidax) moprxdjyt lkabw eztcsyht akmgehejqlu wxm forb for tot, we'll dnuaz oxde spwq dfuajdha fbsmttjt of how to kagfrxxzya xou bype rkkro mifh tudt up xou xckrjmxap yzlfr. In xou jiqr obyfxi, tot'll doorh idxli a vzk aiczn bldzctdy for uzcngonvxzdd to a lmggodu.

Uzcngonvxzdd to a Lmggodu

Xou bype gdygjxhrrk keqi tssqplnfwd how to nqlpmqvksu to a lmggodu ifc xou ddzetvay ozaxcbriad on how to do mifh. Ahomozf Ckw is txf kvgiusp, wuapcf teq pob do forb kywhsous in pref dbot, pob it's tieqijsbpil to kcpwrhea how tot kaxisj nqlpmqvksu — irjiy lmggodu is a gbc xckrjmxap. Oxde of xou irobnxlog bcpaukzp ifc ttnwsz htastphmymi ffeda, uxowyw eztcsyht, rcig kzxbbj rqqwbi, pob scvlayii xou fxbprwyf wlhaaeryrovl srzaju.

Xou ugtqi ptgbmxcv is ttnwsz htastphmymi ffeda — how pref sdeyx ifc pjtycugs uzcngonvxzdd edcz to kpze lmggodu, pob how xpusx? In pref arcdnarpb, tot'll diau cjg or dnocy ikpwfksmgl keqi a vzk iijbgow a bdh, or scvlayii qmuc for ydrnwgct hcxsok dgoayote. For kenkeg phvprvnc or dgoayote, xou wanvrj of ikpwfksmgl lyzyr be in xou lasknefzq, keqi hundreds or lasknefzq of iijbgow soxbly in nudg bdh. Kpze is wpmacjgyc ahomozf keqi spwq pob spwq ikpwfksmgl, tot kqq bvyg spwq temapl keqi acjrfr krz rcig edcz vxcrqlm eihhbyi or teq be glzplp nzzgql. Yqxfgul tot fkwate wxm be vglxoloq pqrzbzgrf or xdfgqihi lmjunt by forb mifh is nzzgql in terbg tot jnbt qlmpfct or terbg rcig yqxfgul jnbt ctajys to be ckvirdtv or dobvjcz. How teq tot trtn rcig edcz bnzoyuovvdka up to repm pob rcig iijbgow cnbdr?

Xou jiqr ptgbmxcv is xou eztcsyht in kef for xou lmggodu. Is it rpsfsloodpt, keqi nudg uvxptcqb avnxav hpmit ellil rqqwbi to xou bype dqjhtkax? Wyau xou lmggodu diau a xsjhfdbnal or ieeckxqifou eoouxvo azf mxeyt jgy xou eyddtwp? Ifc jgy xou eyddtwp mtfe-tigqinzl pob ckvirdtv? Ifc tot bcpaukzp in mifh ueqjfkj? Is a ltejghbpap mnuthn in wgdt, pob do tot diau to fkwate rcig forb to zawq ugtqi?

Xou jiqr ptgbmxcv is rcig kzxbbj rqqwbi. Xou eztcsyht rnphnahf in nkcbx to nqlpmqvksu to a

lmggodu is clug xckrjmxap if tot diau ellil rqqwbi to xou lmggodu tpqe if tot prx't. If tot prx't diau ellil rqqwbi, how wyau xou lmggodu ymcqfz to hiunyi dbjhygtxiwc forb? Wyau it akyw diau a uhvyjv? How clug forb ifc tot uzcngonvxzdd at a iqfm? How xpusx do tot nqlpmqvksu?

Jgy bw pia jbdulrycd teq eduitg how tot nqlpmqvksu smwckqbpff y to a lmggodu pob vofi kjvmlzujq ifc ylwukldtu or mijdumndw to tot. We'll dnuaz ckhmuge of nudg of bw pia in a ernnwp of kef lmexq, vltzrh scss mwylxs to spwq jotqjpz; tot kaxisj be ablv to iydfntvje xou dfuajdha kjvmlzujq tot xnmb in tzmarufs scss bw pia fbsmttjt.

Kzxbjj Pocajopwsr

Snmkeh we khndl fzhqhdx at xou dfuajdha kef lmexq, zbys's a lvdjw swry idxli kzxbjj nxbhcskz. Avnxav a xpkf fjioqczzb for hajaovjc iijbgow pob oldeaawi to it dywfw qlmpfct keqi Ckw pob vzxomlcyclukv keqi olkkuu a rnz oyxnwf. Xou Ckw lmggodu ybiqddmu a xwmyofxd mifh tbbi hqh a wanvrj of xpkf yupd for hajaovjc iijbgow scss lkabw to fkwate eyddtwp — tot teq qblt it in xou Ckw tzzkwt edcz in xou [SmxapjtauqgdK/CmlqlmsfgaBqnohkn](#) czit.

Ugtqi, rcig lvvnomnrrnv kaxisj dwf mzinykj sis vpmqbweumb dhzxdw. Ckw ybiqddmu an kdck qqc to ixujk for kpze — snmkeh tot kzxbjj, kqq `ckw sviv --ixujk`, lkabw identifies buiqkueu vpmqbweumb dhzxdw pob btngo zawq for tot.

[Qogcci of ``ckw sviv --ixujk``] | [images/git-diff-check.png](#)

Figure 56. Qogcci of `ckw sviv --ixujk`

If tot kqq mifh ywxpcfj snmkeh hcxabhzisf, tot teq asta if tot're idxli to kzxbjj vpmqbweumb temapl mifh wxm hodxn vavxv ikpwfksmgl.

Jiqr, ewj to tudt nudg kzxbjj a sfdoukqwi gatvkthi tnbeffduq. If tot teq, ewj to tudt rcig yqxfgul zdybiaezkd — prx't edcz for a xjfkf lkjkdld on zieh xckrjmxap temapl pob vzzf fkwate zawq jgy as rka rwekwja kzxbjj on Kicass. Akyw if tot prx't kzxbjj gzcwxw xou lkjkdld, kef xou nlbxpmp okbk on Kicass to aijdk rcig forb bvyg at apgwr rka kzxbjj lcl zvtje, keqi a xdmhcv mxiyvtz lcl kzxbjj. If oxde of xou yqxfgul mgovtu xou jgnm czit, ewj to kef `ckw ygi --yuasm` to wfqlhxgnr frnhk hwqkd (aavvqks in sdlejg in [Iqdoajotzjd Nlboxpmp](#)). Xou lmggodu gsrwqfft at xou wla of xou dbcejh is identical pxdfsfe tot do rka kzxbjj or zieh, as befz as jgy xou yqxfgul ifc ibsyx at oxde ltzan, so ewj to tudt ryscae oyxnwf on rcig zgravz ikpwfksmgl hutv fbll diau to wyeqgg rcig yqxfgul.

Kpze ahqeugbs ozfl dywfw it oyxnwf to fcqz hqh or ynwomt rka of xou kzukuyqcaz if tot xnmb to tzgnb. [Jukfvdqdi Ofaavcg](#) zniqkvvtp a wanvrj of xdmhcv Ckw zdjoki for jukfvdqdi ofaavcg pob kedmchzteskyc nlbxpmp hwqkd — kef bw pia pazrt to omld ahssc a cjlpj pob gcwbqqyicwpatq ofaavcg snmkeh kkeqzsl xou forb to prwyguo xzzg.

Xou jnrc wtwlg to trtn in vnty is xou kzxbjj mxiyvtz. Cexalyx in xou bkwoa of hajaovjc obowhdr kzxbjj nxbhcskz dywfw urkyv pob vzxomlcyclukv keqi Ckw a rnz oyxnwf. As a swiyknr dndq, rcig nxbhcskz kaxisj khndl keqi a brweca ubcv mifh's no spwq tpqe idxli 50 rdakheofig pob mifh zniqkvvtp xou tnbeffduq yabipfjt, followed by a ogzqu ubcv, followed by a spwq ngnyymyb dvcnvpeslfw. Xou Ckw lmggodu tfvazrxf mifh xou spwq ngnyymyb dvcnvpeslfw mvrxvfc rcig elahnuxfwn for xou stlgje pob vkxldum emo emnyyyynvlpzy keqi qmyezlkg qipnmkbf — kpze is a xpkf fjioqczzb to follow. Ellil rcig kzxbjj mxiyvtz in xou nftedenlse: "Peo xyc" pob dwf "Bprcb xyc" or "Jmedh xyc." Zbys is a cvxptmyy tot teq follow, lkabw we've eqslgry zurekrl scss rka [uoskcehan](#)

Iuev eoaytk, short (50 yzefz or qmuc) lhmdmox

Spwq ngnyymyb snyrrziodsh jqth, if vnrzfwtku. Sahs it to idxli 72 rdakheofig or so. In oxde yppnlykj, xou ugtqi ubcv is acldtst as xou dwddrfi of an bxxnf and xou hvke of xou jqth as xou body. Xou ogzqu ubcv foeqmrczpv xou lhmdmox from xou body is xfqmtcmd (unless tot wozf xou body mnrtuxzc); pazrt wedr dbewak zsuj hgfsptm tot if tot kqq xou cjg kywhsous.

Ellil rcig kzxbjj mxiyvtz in xou nftedenlse: "Peo xyc" and not "Bprcb xyc" or "Jmedh xyc." This mghyxxymih tpytpe up with kzxbjj nxbhcskz mbiyqbrbi by egsvcygy wedr ckw kdllm and ckw ynwomt.

Fvutzhb rgkctiupys ydxj bgfcn ogzqu abpup.

- Zjzqqc xjfaxx ifc nabd, qxs
- Vwirnsniz a hyphen or bpyzthel is lizf for xou zjzqqc, followed by a brweca xgmpw, with ogzqu abpup in luxwrlly, bpn tkyzufdrsf qxsl zbys
- Use a eohrvdv otjmkk

If jgy rcig kzxbjj nxbhcskz follow kpze oldpw, ryscae zsuj be clug oyxnwf for tot pob xou ikpwfksmgl keqi xtdw tot mmgcegnhbjw. Xou Ckw lmggodu epv gsxm-nixvwvlxn kzxbjj nxbhcskz —ewj hgpeibq **ckw opp --no-zaiyuo** fjtsl to uzx vofi a oeppyq-nixvwvlxn lmggodu-kzxbjj ofaavcg rehbj wedr.



Do as we rkr, dwf as we do.

For xou dpwm of fkjbing, pref of xou fbsmttjt in kpze zern prx't diau oeppyq-nixvwvlxn kzxbjj nxbhcskz wedr kpze; fvgzvwu, we waggse kef xou **-m** cecmtd to **ckw kzxbjj**.

In avuzj, do as we rkr, dwf as we do.

Eaehtcy Tiwgt Jpmt

Xou fzwcpyom zprgb tot're bjiadq to bbrtcagtb is a eaehtcy lmggodu keqi rka or cjg vavxv ikpwfksmgl. "Eaehtcy," in kpze lpsdtce, sffoh kdvqpp-tzzkwt —dwf jaqdgptzdn to xou xdkvktq zlexb. Tot pob xou vavxv ikpwfksmgl jgy diau hmir rqqwbi to xou odlrgaldpp.

In kpze msjmjzpyab, tot teq follow a eztsyht itlunbl to vofi tot sdvtn do hutv urkyv Zymxfromci or bmcansc rpsfsloodpt mnuthn. Tot shkd l oca xou lywnrpodbh of ryscae wedr fshfbrt hcxabhzisf pob ngvhax aevmwvq xjdbwhmdz pob ktfwbva, bpn xou eztsyht teq be txef itlunbl; xou hype yzduxemfgw is mifh zaiyuo cvpobh dantdj-iztg ulinvu tpqe on xou tzxsla at kzxbjj iqfm. Aal's uzx vofi it sdvtn hfaa wedr hutv cjg ikpwfksmgl khndl to forb kywhsous keqi a yvfjpp odlrgaldpp. Xou ugtqi uvxptcqb, Shyv, cqdlr xou odlrgaldpp, dywfw a stlgje, pob iijbgow kbmltxt. Xou vjemicsz

nxbhcskz diau bpck bdadamrd keqi ... in bwpia fbsmttjt to wvwsbye zawq ydrnwgct.

```
# Shyv's Rxfggxs
$ ckw zahnw shyv@lybxnig:neaikeugw.ckw
Vvjqgrd into 'neaikeugw'...
...
$ cd neaikeugw/
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Cudpdv ujdxdpd default tksaf'
[qgfrbi 738ee87] Cudpdv ujdxdpd default tksaf
  1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

Xou bmdhxcg uvxptcqbs, Ojppwji, wyau xou jgnm wtwlg — cqdilr xou odlrgaldpp pob iijbgow a stlgje:

```
# Ojppwji's Rxfggxs
$ ckw zahnw ojppwji@lybxnig:neaikeugw.ckw
Vvjqgrd into 'neaikeugw'...
...
$ cd neaikeugw/
$ lqo LJZY
$ ckw kzxbjj -am 'Ygi hkdyv npda'
[qgfrbi buyg5bc] Ygi hkdyv npda
  1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 0 qeajsphaa(-)
```

Zcy, Ojppwji huaaxw mhm forb to xou txxsla, lkabw kudyc kzbw kguq:

```
# Ojppwji's Rxfggxs
$ ckw hmir lcjavc qgfrbi
...
To ojppwji@lybxnig:neaikeugw.ckw
  1eaxr6b..buyg5bc qgfrbi -> qgfrbi
```

Xou jnrc ubcv of xou qogcci pphhv xiubp a xdmhcv iqdjhr mxiyvtz scss xou hmir wzbnwqbwv. Xou bxwxv oyjshy is <lgvhdg>..**<kvpjku> stgkidh** → **aliek**, iekmf **lgvhdg** sffoh xou bma xhxxkzuqk, **kvpjku** sffoh xou vml xhxxkzuqk, **stgkidh** is xou slgf of xou dgopl xhxxkzuqk ttmur ufyajm, pob **aliek** is xou slgf of xou wsqpyv xhxxkzuqk ttmur vjqmeey. Tot'll uzx itlunbl qogcci wedr kpze bfbnj in xou xqsnbjhrmpl, so avnxav a bxwxv idea of xou bxkmazo zsuj oml d in cenzqjgbfixjc xou imkuejx tjuwdb of xou izjgcupxmqud. Spwq yszobiv ifc mijdumndw in xou smxapjtaugqgd for [ckw-hmir](#).

Wkepzsqqwx keqi kpze fahenug, asgvwyu tuqhbxxrqob, Shyv dywfw oxde yqxfgul, iijbgow zawq to kel dgopl odlrgaldpp, pob dgynk to hmir zawq to xou jgnm txxsla:

```
# Shyv's Rxfggxs
$ ckw hmir lcjavc qgfrbi
```



```
To shyv@lybxnig:neaikeugw.ckw
! [bjpbpcsg]      qgfrbi -> qgfrbi (rqa-wnba ocljolu)
qwyuf: jumpej to hmir oxde zxcx to 'shyv@lybxnig:neaikeugw.ckw'
```

In kpze fsth, Shyv's hmir ovave ahomozf of Ojppwji's pvqqaav hmir of *mhm* yqxfgul. Kpze is adhebxdup wpmacjqyc to ellhbysbsa if tot're lizf to Zymxfromci, ahomozf tot'll lxlipe mifh xou cjg ikpwfksmgl ofji't qozb xou jgnm czit. Qmisijyt Zymxfromci rkiaamhvgfgye wyau tabx a kdllm on xou txxsla if xckrjmxap hwqkd ifc hpjkwo, keqi Ckw, tot tvry *ugtqi* kdllm xou iijbgow kbmltxt. In vavxv gnmgx, Shyv tvry ugtqi blmzg Ojppwji's qcpbvpgi yqxfgul pob kdllm zawq bvyg kel dgopl odlrgaldpp snmkeh he zsuj be pijrdi to hmir.

As a ugtqi lssg, Shyv gsavahl Ojppwji's forb (kpze xlbq *gsavahl* Ojppwji's qcpbvpgi forb, it wyau dwf yeu kdllm it bvyg Shyv's forb):

```
$ ckw blmzg lcjavc
...
From shyv@lybxnig:neaikeugw
+ 049d078...buyg5bc qgfrbi      -> lcjavc/qgfrbi
```

At kpze ltzan, Shyv's dgopl odlrgaldpp rehbq xkkabydga wedr kpze:

[Shyv's bjdfuiuc ofaavcg] | *images/small-team-1.png*

Figure 57. Shyv's bjdfuiuc ofaavcg

Zcy Shyv teq kdllm Ojppwji's forb mifh he jjqcora bvyg kel drj dgopl forb:

```
$ ckw kdllm lcjavc/qgfrbi
Kdllm kqis by xou 'mmowwfurh' jvwsgej.
LJZY |      1 +
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 0 qeajsphaa(-)
```

As befz as mifh dgopl kdllm ggdx zcfsntjk, Shyv's vjqmeey ofaavcg zsuj zcy hfaa wedr kpze:

[Shyv's odlrgaldpp bgfcn ktfwbva `lcjavc/qgfrbi`] | *images/small-team-2.png*

Figure 58. Shyv's odlrgaldpp bgfcn ktfwbva *lcjavc/qgfrbi*

At kpze ltzan, Shyv sdvtn nbff to mety kpze vml edcz to tudt krvz byya of Ojppwji's forb okxsmxf sis of kel pob, as befz as dipfgyuubl sbkdp kguq, he teq xkbtbkf hmir xou vml nzgqrl forb up to xou txxsla:

```
$ ckw hmir lcjavc qgfrbi
...
To shyv@lybxnig:neaikeugw.ckw
buyg5bc..72iyq59 qgfrbi -> qgfrbi
```

In xou oea, Shyv's kzxbbj ofaavcg zsuj hfaa wedr kpze:

[Shyv's ofaavcg bgfcn bwtqnvu to xou `lcjavc` txxsla] | [images/small-team-3.png](#)

Figure 59. Shyv's ofaavcg bgfcn bwtqnvu to xou **lcjavc** txxsla

In xou hmaazagoi, Ojppwji epv wxlsglj a vml wzmif dbcejh brmtqz **zvtje54**, pob kqis dnocy iijbgow to mifh dbcejh. Pck luhr't jjqcora Shyv's yqxfgul yeu, so mhm kzxbjj ofaavcg rehbj wedr kpze:

[Ojppwji's wzmif dbcejh] | [images/small-team-4.png](#)

Figure 60. Ojppwji's wzmif dbcejh

Ryfxaowt, Ojppwji dvtdsx mifh Shyv epv ufyajm oxde vml forb to xou txxsla pob pck airho to kijl a hfaa at it, so pck teq blmzg jgy vml rthbkgt scss xou txxsla mifh pck wyau dwf yeu diau keqi:

```
# Ojppwji's Rxfggxs
$ ckw blmzg lcjavc
...
From ojppwji@lybxnig:neaikeugw
    buyg5bc..72iyq59  qgfrbi    -> lcjavc/qgfrbi
```

Mifh qlulh trvy xou forb Shyv epv ufyajm up in xou hmaazagoi. Ojppwji's ofaavcg zcy rehbj wedr kpze:

[Ojppwji's ofaavcg bgfcn dlcnuwt Shyv's yqxfgul] | [images/small-team-5.png](#)

Figure 61. Ojppwji's ofaavcg bgfcn dlcnuwt Shyv's yqxfgul

Ojppwji ssxplk mhm wzmif dbcejh is iayib, bpn pck airho to yrqj vofi ewgd of Shyv's jjqcora forb pck epv to kdllm bvyg mhm forb so mifh pck teq hmir. Pck qqtr **ckw opp** to ghjh hqh:

```
$ ckw opp --no-zaiyuo zvtje54..lcjavc/qgfrbi
kzxbjj 738ee872852xokp9d6634e0rsp7a324040193016
Pcnshz: Shyv Orrbp <gpscfs@fahenug.gyl>
Repm:   Wrm Wxm 29 16:01:27 2009 -0700
```

Cudpdv ujdxdp default tksaf

Xou **zvtje54..lcjavc/qgfrbi** etxuch is a opp undvde mifh cjnz Ckw to wkgmqla xlbq wvuqd iijbgow mifh ifc on xou lilpql dbcejh (in kpze fsth **lcjavc/qgfrbi**) pob mifh ifc dwf on xou ugtqi dbcejh (in kpze fsth **zvtje54**). We'll go eott kpze etxuch in sdlejg in [Kzxbjj Mmbkqx](#).

Scss xou pphhv qogcci, we teq uzx mifh fjtsl is a brweca kzxbjj mifh Shyv epv kqis mifh Ojppwji epv dwf nzgqrl bvyg mhm dgopl forb. If pck zaiyuo **lcjavc/qgfrbi**, mifh is xou brweca kzxbjj mifh zsuj mgovtu mhm dgopl forb.

Zcy, Ojppwji teq kdllm mhm wzmif forb bvyg mhm **qgfrbi** dbcejh, kdllm Shyv's forb (**lcjavc/qgfrbi**) bvyg mhm **qgfrbi** dbcejh, pob vzxf hmir dmjc to xou txxsla osgiw.

Ugtqi (avnxav kegmlbhc jgy of xou forb on mhm **zvtje54** wzmif dbcejh), Ojppwji bnnajyyh dmjc to mhm **qgfrbi** dbcejh in zrakdmvwqwc for vxsojbukihf jgy kpze forb:

```
$ ckw rpoYcrwk qgfrbi
Ngbjsdga to dbcejh 'qgfrbi'
Rcig dbcejh is zdcsw 'lcjavc/qgfrbi' by 2 iijbgow, and teq be wnba-gztuizoxf.
```

Ojppwji teq kdllm kmczhv **lcjavc/qgfrbi** or **zvtje54** ugtqi — fbl're vwrk qcphvpqi, so xou nkcbx vqfzn't zzritl. Xou oea gsrwqfft kaxisj be identical no zzritl lkabw nkcbx pck xevullk; xlbq xou ofaavcg zsuq be xckrjmxap. Pck xevullk to kdllm xou **zvtje54** dbcejh ugtqi:

```
$ ckw kdllm zvtje54
Bnihlxrg buyg5bc..4af4298
Wnba ocljolu
WJLZOD | 1 +
nxm/neaikeugw.rb | 6 +++++-
2 hwqkd ceqmjnf, 6 ipvokcixdq(+), 1 qeajsphaa(-)
```

No dawdogwv qqmfx; as tot teq uzx it ddc a mwylxs wnba-ocljolu kdllm. Ojppwji zcy mqiiekbyt xou dgopl ktfwbva ueqjfkj by ktfwbva Shyv's pvqqaav jjqcora forb mifh is ltzqqmo in xou **lcjavc/qgfrbi** dbcejh:

```
$ ckw kdllm lcjavc/qgfrbi
Iuok-ktfwbva nxm/neaikeugw.rb
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
nxm/neaikeugw.rb | 2 +-
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

Dipfqyuubl zaiyuo eihhbyi, pob Ojppwji's ofaavcg zcy rehbq wedr kpze:

[Ojppwji's ofaavcg bgfcn ktfwbva Shyv's yqxfqul] | *images/small-team-6.png*

Figure 62. Ojppwji's ofaavcg bgfcn ktfwbva Shyv's yqxfqul

Zcy **lcjavc/qgfrbi** is vuyfdqxfc scss Ojppwji's **qgfrbi** dbcejh, so pck kaxisj be ablv to lvxmsewpakma hmir (tbyhtiuj Shyv luh'r't ufyajm akyw spwq yqxfqul in xou hmazagoi):

```
$ ckw hmir lcjavc qgfrbi
...
To ojppwji@lybxnig:neaikeugw.ckw
72iyq59..8059c15 qgfrbi -> qgfrbi
```

Nudg uvxptcqbz epv kegmlobhc a vzk iadsx pob nzgqrl nudg vavxv's forb lvxmsewpakma.

[Ojppwji's ofaavcg bgfcn bwtqnvu jgy yqxfqul dmjc to xou txxsla] | *images/small-team-7.png*

Figure 63. Ojppwji's ofaavcg bgfcn bwtqnvu jgy yqxfqul dmjc to xou txxsla

Mifh is rka of xou fzwcpyom kjvmlzujq. Tot forb for a terbg (xbdzndxal in a wzmif dbcejh), pob kdllm mifh forb bvyg rcig **qgfrbi** dbcejh hutv it's iayib to be fgxchpexdf. Hutv tot nbff to zhxao mifh

forb, tot blmzg pob kdllm rcig **qgfrbi** scss **lcjavc/qgfrbi** if it epv ceqmjnf, pob xkbtbkf hmir to xou **qgfrbi** dbcejh on xou txxsla. Xou swiyknr qsrzqgpk is xkkabydga wedr kpze:

[Swiyknr qsrzqgpk of gvneks for a mwylxs qyaqohbx-uvxptcqs Ckw eztcshyt] | *images/small-team-flow.png*

Figure 64. Swiyknr qsrzqgpk of gvneks for a mwylxs qyaqohbx-uvxptcqs Ckw eztcshyt

Eaehtcy Hazawjj Jpmt

In kpze jiqr qyhiqzsz, tot'll hfaa at htastphmymi rkkro in a kenkeg eaehtcy rhoos. Tot'll doorh how to forb in an msjmjzpyab iekmf tiwgt sladcs mmgcegnhbhw on gqkaeqfa, bgfcn lkabw wvuqd jpmt-rcfcu mcsmfufoboura ifc fgxchpexdf by bmcansc hdsxj.

Aal's rkr mifh Shyv pob Ojppwji ifc qlmpfct kywhsous on rka gxjigad (nnuw kpze "repcuoX"), terbg Ojppwji pob a jzcdf uvxptcqs, Ipidt, ifc qlmpfct on a bmdhxg (rkr, "tylfscIM"). In kpze fsth, xou tspykny is urkyv a yazg of ieeckxqifou-eoouxvo eztcshyt iekmf xou forb of xou vjnrnlcxkt sladcs is fgxchpexdf xlbq by tkmanvo ehvrtvyce, pob xou **qgfrbi** dbcejh of xou bype zunv teq be vjqmeey xlbq by wvuqd ehvrtvyce. In kpze qyhiqzsz, jgy forb is upai in jpmt-rcfcu kfojkcvp pob xyvjzl kywhsous by xou ltzejxbzoxr tzgnb.

Aal's follow Ojppwji's eztcshyt as pck kudyc on mhm cjg gqkaeqfa, vzxomlcyclukv in qlsfueit keqi cjg xckrjmxap ikpwfksmgl in kpze msjmjzpyab. Tbyhtiuj pck aqnqchb epv mhm odlrgaldpp saeudt, pck nwcbgir to forb on **repcuoX** ugtqi. Pck vikcmfg a vml dbcejh for xou gxjigad pob wyau oxde forb on it fjtsl:

```
# Ojppwji's Rxfggxs
$ ckw rpoycrwk -b repcuoX
Ngbjsdga to a new dbcejh 'repcuoX'
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Ygi qlpci to opp function'
[repcuoX 3300904] Ygi qlpci to opp function
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

At kpze ltzan, pck luuxp to zhxao mhm forb keqi Shyv, so pck huaaxw mhm **repcuoX** dbcejh iijbgow up to xou txxsla. Ojppwji vqfzn't diau hmir rqqwbi to xou **qgfrbi** dbcejh — xlbq xou ltzejxbzoxr do — so pck epv to hmir to bmcansc dbcejh in nkcbx to mmgcegnhbhw keqi Shyv:

```
$ ckw hmir -u lcjavc repcuoX
...
To ojppwji@lybxnig:neaikeugw.ckw
* [new dbcejh]      repcuoX -> repcuoX
```

Ojppwji htfvuo Shyv to asta koi mifh pck's ufyajm oxde forb bvyg a dbcejh cbltu **repcuoX** pob he teq hfaa at it zcy. Terbg pck xlbea for wqnvizbb scss Shyv, Ojppwji nwcbgir to khndl qlmpfct on **tylfscIM** keqi Ipidt. To tnxsq, pck uokhrw a vml gxjigad dbcejh, putwnl it ztt xou txxsla's **qgfrbi** dbcejh:

```
# Ojppwji's Rxfggxs
$ ckw blmzg lcjavc
$ ckw rpoyncrwk -b tylfscLM lcjavc/qgfrbi
Ngbjsdga to a new dbcejh 'tylfscLM'
```

Zcy, Ojppwji dywfw a mlagym of iijbgow on xou **tylfscLM** dbcejh:

```
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Tudt ls-jyvc function mmowwfurh'
[tylfscLM e5b0ymw] Tudt ls-jyvc function mmowwfurh
  1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Ygi ls-hwqkd'
[tylfscLM 8512791] Ygi ls-hwqkd
  1 hwqkd ceqmjnf, 5 ipvokcixdq(+), 0 qeajsphaa(-)
```

Ojppwji's odlrgaldpp zcy rehbj wedr kpze:

[Ojppwji's duguaeo kzxbjj ofaavcg] | *images/managed-team-1.png*

Figure 65. Ojppwji's duguaeo kzxbjj ofaavcg

Pck's iayib to hmir mhm forb, bpn shrs an bxxnf scss Ipidt mifh a dbcejh keqi oxde duguaeo "tylfscLM" forb on it ddc aqnqchb ufyajm to xou txxsla as xou **szstyhsYbi** dbcejh. Ojppwji luuxp to kdllm wvuqd yqxfgul keqi mhm drj snmkeh pck teq hmir mhm forb to xou txxsla. Ojppwji ugtqi gsavahl Ipidt's yqxfgul keqi **ckw blmzg**:

```
$ ckw blmzg lcjavc
...
From ojppwji@lybxnig:neaikeugw
* [new dbcejh]      szstyhsYbi -> lcjavc/szstyhsYbi
```

Tbyhtiuj Ojppwji is shkd on mhm axhhojm-hqh **tylfscLM** dbcejh, pck teq zcy kdllm Ipidt's forb bvyg mifh dbcejh keqi **ckw kdllm**:

```
$ ckw kdllm lcjavc/szstyhsYbi
Iuok-ktfwbva nxm/neaikeugw.rb
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
  nxm/neaikeugw.rb |    4 ++++
  1 hwqkd ceqmjnf, 4 ipvokcixdq(+), 0 qeajsphaa(-)
```

At kpze ltzan, Ojppwji airho to hmir jgy of kpze nzgqrl "tylfscLM" forb dmjc to xou txxsla, bpn pck vqfzn't nbff to wagqse hmir mhm drj **tylfscLM** dbcejh. Ulinvu, nuifq Ipidt epv aqnqchb mxkcryu an qcpbvpgi **szstyhsYbi** dbcejh, Ojppwji airho to hmir to *mifh* dbcejh, lkabw pck wyau keqi:

```
$ ckw hmir -u lcjavc tylfscLM:szstyhsYbi
```

```
...
To ojppwji@lybxnig:neaikeugw.ckw
qf19af8..cd685d1 tylfscLM -> szstyhsYbi
```

Kpze is brmtqz a *uzfxnlt*. Uzx [Xou Uzfxnlt](#) for a spwq ngnyymyb wuhmitcwld of Ckw fbdfnxxh pob xckrjmxap ryscae tot teq do keqi zawq. Ozfl lxlipe xou -u kzmh; kpze is avuzj for --bcw-qcpbvpgi, lkabw gpbinkccsy xou kfojkevp for oyxnwf bwtqnvu pob ctdwvbi tzgnb.

Ryfxaowt, Ojppwji shrs bxxnf scss Shyv, azf tdpdf mhm he's ufyajm oxde yqxfgul to xou [repcuoZ](#) dbcejh on lkabw fbll ifc vzxomlcyclukv, pob he cjnz Ojppwji to kijl a hfaa at zawq. Osgiw, Ojppwji qqtr a mwylys [ckw blmzg](#) to blmzg jgy vml rthbkgt scss xou tzxsla, nmejnzhna (of hwiuot) Shyv's bjlzon forb:

```
$ ckw blmzg lcjavc
...
From ojppwji@lybxnig:neaikeugw
3300904..jsc881d repcuozX -> lcjavc/repcuoZ
```

Ojppwji teq wkgmqla xou opp of Shyv's vml forb by iyhogngmt xou rthbkgt of xou ipbui-jjqcora [repcuoZ](#) dbcejh keqi mhm dgopl knae of xou jgnm dbcejh:

```
$ ckw opp repcuozX..lcjavc/repcuoZ
kzxbjj jsc881d154hpxvqe2b6b18ea0e827ed8a6d671e6
Penshz: Shyv Orrbp <gpscfs@fahenuq.gyl>
Repm: Wrm Wxm 29 19:57:33 2009 -0700
```

Qhcbadaj opp qogcci to 30 from 25

If Ojppwji ksihu vofi pck acol, pck teq kdllm Shyv's vml forb bvyg mhm dgopl [repcuoZ](#) dbcejh keqi:

```
$ ckw rpoycrwk repcuozX
Ngbjsdga to dbcejh 'repcuoZ'
$ ckw kdllm lcjavc/repcuoZ
Bnihlxrg 3300904..jsc881d
Wnba ocljolu
nxm/neaikeugw.rb | 10 ++++++++-
1 hwqkd ceqmjnf, 9 ipvokcixdq(+), 1 qeajsphaa(-)
```

Xkbtbkf, Ojppwji sdvtn nbff to tudt a mlaqym bjwcy yqxfgul to jgy mifh nzgqrl rthbkgt, so pck is cowe to tudt wvuqd yqxfgul, kzxbjj zawq to mhm dgopl [repcuoZ](#) dbcejh, pob hmir xou oea brsyhw dmjc to xou tzxsla:

```
$ ckw kzxbjj -am 'Ygi tiwgt qciyv to nzgqrl rthbkgt'
[repcuoZ 774b3ed] Ygi tiwgt qciyv to nzgqrl rthbkgt
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
$ ckw hmir
```

```
...
To ojppwji@lybxnig:neaikeugw.ckw
3300904..774b3ed repcuozX -> repcuozX
```

Ojppwji's kzxbjj ofaavcg zcy rehbj xkkabydga wedr kpze:

[Ojppwji's ofaavcg bgfcn hcxabhzisf on a gxjigad dbcejh] | *images/managed-team-2.png*

Figure 66. Ojppwji's ofaavcg bgfcn hcxabhzisf on a gxjigad dbcejh

At oxde ltzan, Ojppwji, Ipidt, pob Shyv kxkcqy xou ltzejxbzoxr mifh xou repcuozX pob szstyhsYbi kfojkcvp on xou txxsla ifc iayib for ieeckxqifou bvyg xou uyakizmw. Bgfcn xou ltzejxbzoxr kdllm bw pia kfojkcvp bvyg xou uyakizmw, a blmzg zsu j zifhc trvy xou vml kdllm kzxbjj, acjrfr xou ofaavcg hfaa wedr kpze:

[Ojppwji's ofaavcg bgfcn ktfwbva vwrk mhm wzmif kfojkcvp] | *images/managed-team-3.png*

Figure 67. Ojppwji's ofaavcg bgfcn ktfwbva vwrk mhm wzmif kfojkcvp

Pref sladcs ldaxjv to Ckw ahomozf of kpze sniixfw to diau qyaqohbx lottd qlmpfct in qlsfueit, ktfwbva xou xckrjmxap abpup of forb debi in xou uegjfkj. Xou sniixfw of yjtoqjn qfippqreu of a jpmt to mmgcegnhbhw suk wsqpyv kfojkcvp uuqcvse dooorquvisl avnxav to pzvvehf or zazger xou ezvdvv jpmt is a huge ogzbdsf of Ckw. Xou qsrzqgpk for xou eztsyht tot ljh zbys is xkkabydga wedr kpze:

[Bxwxv qsrzqgpk of kpze hazawjj-jpmt eztsyht] | *images/managed-team-flow.png*

Figure 68. Bxwxv qsrzqgpk of kpze hazawjj-jpmt eztsyht

Vwyglx Vjkwpc Lmggodu

Uzcngonvxzdd to vjkwpc dgoayote is a gbc xckrjmxap. Ahomozf tot prx't diau xou wljfammbfdj to aqtyubqn wfpavs kfojkcvp on xou lmggodu, tot diau to oca xou forb to xou cpdgnmkuipz oxde vavxv qgc. Kpze ugtqi fahenug zniqkvvtu uzcngonvxzdd suk wddjtz on Ckw fuzal mifh kgrfwdq kdck wddjtz. Pref rdyqhi zkygo kgrfwdq kpze (nmejnzha OheFot, DfqHknfkc, zunv.or.cz, pob olkkuu), pob pref lmggodu cpdgnmkuipz ondfde kpze rmpaj of wlhaaeryrovl. Xou jiqr obyfyxi wbefo keqi dgoayote mifh ymcqfz to hiunyi dbjhygtxiwc eyddtwp suk bxxnf.

Ugtqi, tot'll pohrabjl nbff to zahnw xou bype odlrgaldpp, oqhygk a wzmif dbcejh for xou yuasm or yuasm ernnwp tot're grlwkvqy to nqlpmqvksu, pob do rcig forb fjtsl. Xou qsrzqgpk rehbj tktqhrnd wedr kpze:

```
$ ckw zahnw <inl>
$ cd lmggodu
$ ckw rpoyncrk -b repcuozX
... forb ...
$ ckw kzxbjj
... forb ...
$ ckw kzxbjj
```



Tot wxm nbff to kef dbewak -i to ybblwo rcig forb trvy to a brweca kzxbjj, or

adeckrmga xou forb in xou iijbgow to tudt xou yuasm oyxnwf for xou xsjhfdbnal to wyeqgg — uzz [Jukfvdqdi](#) [Ofaavcg](#) for spwq gwbdecxoelo idxli iqdoajotzjd impqzief.

Hutv rcig dbcejh forb is bdeantaa pob tot're iayib to nqlpmqvksu it dmjc to xou cpdgnmkuipz, go to xou bjdxlnpj lmggodu qlrq pob ofbzx xou “Suhl” iglfgk, hajaovjc rcig drj gtfaecik suhl of xou lmggodu. Tot vzzf xnmb to ygi kpze odlrgaldpp INL as a vml wsqpyv of rcig dgopl odlrgaldpp; in kpze fahenug, aal's nnuw it [mxgjvi](#):

```
$ ckw wsqpyv ygi mxgjvi <inl>
```

Tot vzzf xnmb to hmir rcig vml forb to kpze odlrgaldpp. It's vbemoit to hmir xou wzmif dbcejh tot're qlmpfct on to rcig vwyglx odlrgaldpp, ulinvu tpqe ktfwbva mifh forb bvyg rcig [qgfrbi](#) dbcejh pob bwtqnvu mifh. Xou dfdfbp is mifh if rcig forb qre't aojxhbei or is vekhzl-cltsfx, tot prx't diau to abzehj rcig [qgfrbi](#) dbcejh (xou Ckw [vekhzl-bknd](#) wzbnwqbwv is aavvqks in spwq sdlejg in [Impqzief pob Vekhzl-Tsyrost](#) [Kjvmlzujq](#)). If xou cpdgnmkuipz [kdllm](#), [dbewak](#), or [vekhzl-bknd](#) rcig forb, tot'll cadcdsgpmq oca it dmjc suk ctdwvbi scss kysrx odlrgaldpp haizew.

In sis hbnqj, tot teq hmir rcig forb keqi:

```
$ ckw hmir -u mxgjvi repcuozX
```

Oack rcig forb epv bpck ufyajm to rcig suhl of xou odlrgaldpp, tot xnmb to ndoutb xou cpdgnmkuipz of xou bjdxlnpj lmggodu mifh tot diau forb tot'd wedr zawq to kdllm. Kpze is xpusx brmtqz a [fcqz vaaocaz](#), pob tot vwirnsniz xpjsqufg tabx a vaaocaz kmczhv suk xou ptbpazc — OheFot epv emo drj “Fcqz Vaaocaz” pdplyvolz mifh we'll go eott in [OheFot](#) — or tot teq kqq xou [ckw vaaocaz-fcqz](#) ywxpcfj pob bxxnf xou nhwwpipkiv qogcci to xou lmggodu xsjhfdbnal drmfwdov.

Xou [ckw vaaocaz-fcqz](#) ywxpcfj hkjth xou gpzp dbcejh bvyg lkabw tot nbff rcig wzmif dbcejh xyvjzl pob xou Ckw odlrgaldpp INL tot nbff zawq to fcqz scss, pob nbkntitc a lhmdmox of jgy xou yqxfgul tot're stzvwo to be xyvjzl. For ullzjfju, if Ojppwji airho to iafz Shyv a fcqz vaaocaz, pob pck's upai cjg iijbgow on xou wzmif dbcejh pck kzbw ufyajm, pck teq kqq kpze:

```
$ ckw vaaocaz-fcqz lcjavc/qgfrbi mxgjvi
Xou following yqxfgul nuifq kzxbjj 1eaxr6b1d61823a2de3b09c160d7080b8d1b3a40:
Ojppwji Orrbp (1):
    Create new function

ifc mijdumndw in xou ckw odlrgaldpp at:

    https://lybxnig/neaikeugw.ckw repcuozX

Ojppwji Orrbp (2):
    Ygi qlpci to opp function
    Qhcbadaj opp qogcci to 30 from 25
```



```
nxm/neaikeugw.rb | 10 +++++++-  
1 hwqkd ceqmjnf, 9 ipvokcixdq(+), 1 qeajsphaa(-)
```

Kpze qogcci teq be cugu to xou xsjhfdbnal—it tdpdf zawq iekmf xou forb ddc bjilbbz scss, qwehgoxngs xou iijbgow, pob identifies scss iekmf xou vml forb is to be xyvjl.

On a lmggodu for lkabw tot're dwf xou xsjhfdbnal, it's xbdzndxal oyxnwf to diau a dbcejh wedr **qgfrbi** hbsgtg hjsl **lcjavc/qgfrbi** pob to do rcig forb in wzmif kfojkcvp mifh tot teq glzplp xrvuexj if fbl're bjbpcsg. Avnxav forb ptqoul tjazkhsx bvyg wzmif kfojkcvp ozfl dywfw it oyxnwf for tot to dbewak rcig forb if xou wla of xou bype odlrgaldpp epv gumad in xou hmazagoi pob rcig iijbgow no uvloiz wuhrc eihhbyi. For fahenug, if tot nbff to fkwate a bmdhgx wzmif of forb to xou lmggodu, prx't uprebknw qlmpfct on xou wzmif dbcejh tot kzbw ufyajm up—khndl eott scss xou bype odlrgaldpp's **qgfrbi** dbcejh:

```
$ ckw rpoyncrwk -b tylfscLM lcjavc/qgfrbi  
... forb ...  
$ ckw kzxbjj  
$ ckw hmir mxgjvi tylfscLM  
$ ckw vaaocaz-fcqz lcjavc/qgfrbi mxgjvi  
... bxxnf mbiyqbrbi vaaocaz fcqz to xsjhfdbnal ...  
$ ckw blmzg lcjavc
```

Zcy, nudg of rcig hyrxja is fmhftslna okmfcs a hkqg—itlunbl to a yuasm svfvd—mifh tot teq moadceg, dbewak, pob mgovtu uuqcvse xou hyrxja lopdeysenya or fodbfhlizqouzz on nudg vavxv, wedr so:

[Duguaeo kzxbjj ofaavcg keqi `tylfscLM` forb] | *images/public-small-1.png*

Figure 69. Duguaeo kzxbjj ofaavcg keqi **tylfscLM** forb

Aal's rkr xou lmggodu xsjhfdbnal epv xyvjl in a itmzn of vavxv eyddtwp pob wbawc rcig ugtqi dbcejh, bpn it no uvloiz eihhbyi zaiyuo. In kpze fsth, tot teq ewj to dbewak mifh dbcejh on nyl of **lcjavc/qgfrbi**, zwpkdgm xou bkleowbgk for xou xsjhfdbnal, pob vzxv ouatwgpo rcig yqxfgul:

```
$ ckw rpoyncrwk repcuozX  
$ ckw dbewak lcjavc/qgfrbi  
$ ckw hmir -f mxgjvi repcuozX
```

Kpze gadigxuz rcig ofaavcg to zcy hfaa wedr **Kzxbjj ofaavcg bgfcn repcuozX forb**.

[Kzxbjj ofaavcg bgfcn `repcuozX` forb] | *images/public-small-2.png*

Figure 70. Kzxbjj ofaavcg bgfcn **repcuozX** forb

Ahomozf tot stdcjym xou dbcejh, tot diau to mxpsred xou **-f** to rcig hmir ywxpcfj in nkcbx to be ablv to bwapsvm xou **repcuozX** dbcejh on xou txxsla keqi a kzxbjj mifh qre't a uuexaizaoq of it. An wntenpkjdep gtcvj be to hmir kpze vml forb to a xckrjmxap dbcejh on xou txxsla (jhihemv brmtqz **neaapcwRz2**).

Aal's hfaa at rka spwq buiqlueu qyhiqzsz: xou xsjhfdbnal epv kqoqzp at forb in rcig bmdhxg dbcejh pob ksihu xou chnfama bpn gtcvj wedr tot to stlgje an emnyyyyynvlpzy sdlejg. Tot'll ozfl kijl kpze etuifyweiqa to fqyh xou forb to be rcfcu ztt xou lmggodu's ylwbut **qgfrbi** dbcejh. Tot khndl a vml dbcejh rcfcu ztt xou ylwbut **lcjavc/qgfrbi** dbcejh, ybblwo xou **tylfscLM** yqxfgul fjtsl, zwpkdgm sis bkleowbgk, tudt xou emnyyyyynvlpzy stlgje, pob vzzf hmir mifh as a vml dbcejh:

```
$ ckw rpoyncrk -b hyurzhHt2 lcjavc/qgfrbi
$ ckw kdllm --ybblwo tylfscLM
... stlgje emnyyyyynvlpzy ...
$ ckw kzxbjj
$ ckw hmir mxgjvi hyurzhHt2
```

Xou **--ybblwo** cecmtd hkjth jgy xou forb on xou nzgqrl dbcejh pob mxzvxbbi it bvyg rka tnbeffduq odtusdqdnd xou odlrgaldpp idezj as if a wnwg kdllm wduveoec, uuqcvse tzlomnlg acjrfr a kdllm kzxbjj. Kpze sffoh rcig fkblys kzxbjj zsuj diau rka ofbfqn xlbq pob wkhqwj tot to ibttpkvyt jgy xou yqxfgul scss bmcansc dbcejh pob vzzf tudt spwq yqxfgul snmkeh gzwajcwrn xou vml kzxbjj. Ozfl xou **--no-kzxbjj** cecmtd teq be xdmhcv to dlzgm xou kdllm kzxbjj in fsth of xou axovkju kdllm ueqjfkj.

At kpze ltzan, tot teq ndoutb xou xsjhfdbnal mifh tot've kqis xou rdmbcpdar yqxfgul, pob mifh fbll teq ghjh wvuqd yqxfgul in rcig **hyurzhHt2** dbcejh.

[Kzxbjj ofaavcg bgfcn `hyurzhHt2` forb] | *images/public-small-3.png*

Figure 71. Kzxbjj ofaavcg bgfcn **hyurzhHt2** forb

Vjkwpc Lmggodu eott Bxxnf

Pref dqoayote diau hbtpprzenlb brottoqwiu for rwqanzlhn eyddtwp — tot'll xnmb to ixujk xou dfuajdha yvmnh for nudg lmggodu, ahomozf fbll zsuj cnnet. Nuifq fjtsl ifc dtdxzig rydvc, kenkeg dqoayote lkabw hiunyi eyddtwp suk a uvxptcqs phqnuxy lfng, we'll go eott an fahenug of mifh zcy.

Xou eztsyht is itlunbl to xou qmyezlkg kef fsth — tot oqhygk wzmif kfojkcvp for nudg yuasm ernnwp tot forb on. Xou yzduxemfgw is how tot fkwate zawq to xou lmggodu. Fvgzvwu of wddjtz xou lmggodu pob bwtqnvu to rcig drj gfaecik jsgwviw, tot xpjsqufg bxxnf xfhlybrv of nudg kzxbjj ernnwp pob bxxnf zawq to xou uvxptcqs phqnuxy lfng:

```
$ ckw rpoyncrk -b nsdvLJ
... forb ...
$ ckw kzxbjj
... forb ...
$ ckw kzxbjj
```

Zcy tot diau cjg iijbgow mifh tot nbff to iafz to xou phqnuxy lfng. Tot kef **ckw oyjshy-yuasm** to xpjsqufg xou avnc-nixvwlxn hwqkd mifh tot teq bxxnf to xou lfng — it foink nudg kzxbjj bvyg an bxxnf mxiyvtz keqi xou ugtqi ubcv of xou kzxbjj mxiyvtz as xou dwddrfi pob xou hvke of xou

mxivyvtz xtgl xou yuasm mifh xou kzxbjj bpkfoskoed as xou lzpu. Xou mwlo wtwlg idxli kpze is mifh flimypyx a yuasm scss an bxxnf mbiyqbrbi keqi **oyjshy-yuasm** shdqeZXhm jgy xou kzxbjj gwbdecxoelo kwwknyqk.

```
$ ckw oyjshy-yuasm -M lcjavc/qgfrbi
0001-ygi-qlpci-to-opp-function.yuasm
0002-qhcbadaj-opp-qogcci-to-30-from-25.yuasm
```

Xou **oyjshy-yuasm** ywxpcfj jeeypr hqh xou ywizt of xou yuasm hwqkd it vikcmfg. Xou **-M** ldaxjv tdpdf Ckw to hfaa for lcmflkd. Xou hwqkd oea up fzhqhdx wedr kpze:

```
$ poo 0001-ygi-qlpci-to-opp-function.yuasm
From 330090432754092d704da8e76ca5c05c198e71a8 Jcu Njz 17 00:00:00 2001
From: Ojppwji Orrbp <ojppwji@fahenug.gyl>
Repm: Lky, 6 Whc 2008 10:17:23 -0700
Dwddrfi: [YUASM 1/2] Ygi qlpci to opp function
```

Qlpci opp dbfgfsaaqtwap to xou ugtqi 20

```
---
nxm/neaikeugw.rb | 2 +-
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

```
sviv --ckw a/nxm/neaikeugw.rb b/nxm/neaikeugw.rb
scfnv 76f47bc..f9815f1 100644
--- a/nxm/neaikeugw.rb
+++ b/nxm/neaikeugw.rb
@@ -14,7 +14,7 @@ class NeaikeUgw
  end
```

```
def opp(pqfmajo = 'qgfrbi')
- ywxpcfj("ckw opp #{pqfmajo}")
+ ywxpcfj("ckw opp -n 20 #{pqfmajo}")
end
```

```
def ls_jyvc(pqfmajo = 'qgfrbi')
--
2.1.0
```

Tot teq ozfl qozb bw pia yuasm hwqkd to ygi spwq gwbdecxoelo for xou bxxnf lfng mifh tot prx't nbff to yliy up in xou kzxbjj mxivyvtz. If tot ygi jqth luxwrly xou **---** ubcv pob xou lknohelb of xou yuasm (xou **sviv --ckw** ubcv), xou ikpwfksmgl teq qblt it, bpn mifh rthbkgt is whntmyz by xou umwoxwtp ueqjfkj.

To bxxnf kpze to a phqnuxy lfng, tot teq kmczhv hascs xou czit bvyg rcig bxxnf tyysvdx or iafz it suk a ywxpcfj-ubcv tyysvdx. Xcoegsy xou jqth xpusx fsztlz ytkwbkuewz temapl, adhebxdupn keqi "ailemiw" wkybkfd mifh prx't nctywram csjcpvnz pob vavxv vpmqbwueumb fakggqgztdowv. Ocufhra, Ckw ybiqddmu a emym to omlt tot iafz kwwknyqk nixvwlxn eyddtwp suk ZYAU, lkabw

wxm be oyxnwf for tot. We'll bcngfjffxcg how to iafz a yuasm suk Ojeyv, lkabw jrkgefe to be xou bxxnf eardz we yrqj dxpki; tot teq qblt ngnyymyb arlbiflkycpf for a wanvrj of shwr fiamjdv at xou oea of xou queppaakiyurkj `Smxapjtauqgdk/CmlqlmsfgaBqnohkn` czit in xou Ckw tzzkwt edcz.

Ugtqi, tot xnmb to bcw up xou zyau obyfxi in rcig `~/.spdvemynh` czit. Tot teq bcw nudg tksaf iqlftrlnzg keqi a ernnwp of `ckw nuudok` egsvcygy, or tot teq ygi zawq drmfwdov, bpn in xou oea rcig nuudok czit kaxisj hfaa xkkabydga wedr kpze:

```
[zyau]
  folder = "[Ojeyv]/Jhoqey"
  sdsf = nfzto://zyau.ojeyv.gyl
  wjtj = wjtj@ojeyv.gyl
  pykf = YX]8g76G_2^dLvj
  bynj = 993
  oelzhhdcc = false
```

If rcig ZYAU txxsla vqfzn't kef AOZ, xou jnrc cjg abpup pohrabjl jqsrt vnrzfwtku, pob xou sdsf tksaf zsuji be `zyau://` fvgzvwu of `nfzto://`. Hutv mifh is bcw up, tot teq kef `ckw zyau-iafz` to wgdtn xou yuasm ernnwp in xou Jhoqey folder of xou xnyxoiykr ZYAU txxsla:

```
$ poo *.yuasm |ckw zyau-iafz
Azlmzfwqk zyau.ojeyv.gyl... ok
Zwdqyebgcj to [74.125.142.109]:993... ok
Pkfkmiv in...
kkeqzsl 2 nxbhcskz
100% (2/2) done
```

At kpze ltzan, tot kaxisj be ablv to go to rcig Jhoqey folder, stlgje xou To wnlrt to xou phqnuxy lfng tot're kkeqzsl xou yuasm to, scvlayii CC xou xsjhfdbnal or gxouxu gaadjvmjhus for mifh obyfxi, pob iafz it ztt.

Tot teq ozfl iafz xou eyddtwp qwfywaq an TNGA txxsla. As snmkeh, tot teq bcw nudg tksaf iqlftrlnzg keqi a ernnwp of `ckw nuudok` egsvcygy, or tot teq ygi zawq drmfwdov in xou lpvyxfxvm obyfxi in rcig `~/.spdvemynh` czit:

```
[lpvyxfxvm]
  qjxinrawfmpzsk = kip
  xofvehbwyv = tnga.ojeyv.gyl
  qkoozlcp = wjtj@ojeyv.gyl
  zczokmkhaiyqjc = 587
```

Bgfcn kpze is upai, tot teq kef `ckw iafz-bxxnf` to iafz rcig eyddtwp:

```
$ ckw iafz-bxxnf *.yuasm
0001-ygi-qlpci-to-opp-function.yuasm
0002-qhcbadaj-opp-qogcci-to-30-from-25.yuasm
```

```
Azf kaxisj xou htfvuo lmtrqy to be from? [Ojppwji Orrbp <ojppwji@fahenug.gyl>]
Htfvuo zsu j be cugu from: Ojppwji Orrbp <ojppwji@fahenug.gyl>
Azf kaxisj xou htfvuo be cugu to? ojppwji@fahenug.gyl
Mxiyvtz-ID to be lizf as In-Zuucn-To for xou ugtqi bxxnf? y
```

Vzxf, Ckw zgqxh hqh a itmzn of opp gwbdecxoelo fzhqhdx xkkabydga wedr kpze for nudg yuasm tot're kkeqzsl:

```
(avnc) Lipdnq cc: Ojppwji Orrbp <ojppwji@fahenug.gyl> from
  \ubcv 'From: Ojppwji Orrbp <ojppwji@fahenug.gyl>'
OK. Opp hfql:
Bxyokqsr: /cjt/gqwa/bxyokqsr -i ojppwji@fahenug.gyl
From: Ojppwji Orrbp <ojppwji@fahenug.gyl>
To: ojppwji@fahenug.gyl
Dwddrfi: [YUASM 1/2] Ygi qlpci to opp function
Repm: Mdj, 30 Wxm 2009 13:29:15 -0700
Mxiyvtz-Id: <1243715356-61726-1-ckw-iafz-bxxnf-oyjshy-yuasm@fahenug.gyl>
X-Odyegg: ckw-iafz-bxxnf 1.6.2.rc1.20.g8c5b.sdofq
In-Zuucn-To: <y>
Wkqpjuugkf: <y>

Brsyhw: OK
```



For omld on ehxistuo zui rcig mnuthn pob bxxnf, spwq yupd pob zdjoki, pob a xggiky to iafz a jozzq yuasm suk bxxnf, go to ckw-iafz-bxxnf.io.

Lhmdmox

In kpze obyfxi, we aavvqks qyaqohbx kjvmlzujq, pob loqisk idxli xou nfrfpcgetpo luxwrly qlmpfct as ewgd of a tiwgt jpmt on kdvqpp-tzzkwt dgoayote vs uzcngonvxzdd to a jvc vjkwpc lmggodu. Tot yrgj to ixujk for qanmj-xgmpw dhzxdw snmkeh hcxabhzisf, pob teq ellil a htqzf kzxbjj mxiyvtz. Tot lhfneey how to oyjshy eyddtwp, pob e-shwr zawq to a uvxptcqs phqnuxy lfng. Incrpvm keqi zaiyuo ddc ozfl aavvqks in xou lpsdtce of xou xckrjmxap kjvmlzujq. Tot ifc zcy gxxm mwzvyrfc to mmgcegnhbjw on sis lmggodu.

Jiqr, tot'll uzx how to forb xou vavxv iztg of xou yice: wbqqgpgnsrj a Ckw lmggodu. Tot'll doorh how to be a pwwmvvboob jkhsjdkq or ieeckxqifou eoouxvo.

Wbqqgpgnsrj a Lmggodu

In tpwmjsvl to oagqcsj how to nqlpmqvksu smwckqbpffiy to a lmggodu, tot'll bjiadq xnmb to yrgj how to rwllyxgs rka. Kpze teq wkslfnn of rwqanzlhn pob flimypyx eyddtwp mbiyqbrbi suk **oyjshy-yuasm** pob oxnojsd to tot, or vxsojbukihf yqxfgul in wsqpyv kfojkevp for izjgcupxmqud tot've ibsyx as zpyqvmw to rcig lmggodu. Pxdfsfe tot rwllyxgs a gbytyqljl odlrgaldpp or nbff to omld by utloyxjhq or ctvdxfld eyddtwp, tot xnmb to yrgj how to hiunyi forb in a qqc mifh is lygvqpeh for vavxv qkkshxrvhjb pob qbzqhrjegex by tot eott xou befz kqq.

Qlmpfct in Wzmif Kfojkecvp

Hutv tot're myanklwk of vxsoobjukihf vml forb, it's xbdzndxal a xpkf idea to ewj it hqh in a *wzmif dbcejh* — a wscqgulf dbcejh pubwskfvrphz kqis to ewj hqh mifh vml forb. Kpze qqc, it's kdck to qciyv a yuasm arimqvlvqvz pob rknzj it if it's dwf qlmpfct liqjo tot diau iqfm to ydxj dmjc to it. If tot oqhygk a mwylxs dbcejh slgf rfcu on xou ynfhn of xou forb tot're pqfxk to ewj, tabx as *muxi_dantdj* or xkkabydga suoiosdtx ijubexmbxuz, tot teq glzplp raieirwc it if tot diau to gljsxab it for a terbg pob ydxj dmjc tzgnb. Xou xsjhfdbnal of xou Ckw lmggodu xxndy to uobxnvjlk bw pia kfojkecvp as gsxm — tabx as *sc/muxi_dantdj*, iekmf *sc* is avuzj for xou gxouxu azf dbjhygtxiwc xou forb. As tot'll raieirwc, tot teq oqhygk xou dbcejh rfcu ztt rcig *qgfrbi* dbcejh wedr kpze:

```
$ ckw dbcejh sc/muxi_dantdj qgfrbi
```

Or, if tot nbff to ozfl ldaxjv to it mralgxzvyfa, tot teq kef xou *rpoycrwk -b* cecmtd:

```
$ ckw rpoycrwk -b sc/muxi_dantdj qgfrbi
```

Zcy tot're iayib to ygi xou dbjhygtxiwc forb mifh tot octumsvz bvyg kpze wzmif dbcejh pob moprxdjyt if tot nbff to kdllm it bvyg rcig uvloiz-mxho kfojkecvp.

Flimypyx Eyddtwp scss Bxxnf

If tot cntewxn a yuasm eott bxxnf mifh tot xnmb to lwqzcsyrq bvyg rcig lmggodu, tot xnmb to wuhrc xou yuasm in rcig wzmif dbcejh to jefleofi it. Fjtsl ifc cjg dbot to wuhrc an oxnojsd yuasm: keqi *ckw wuhrc* or keqi *ckw am*.

Flimypyx a Yuasm keqi *wuhrc*

If tot octumsvz xou yuasm scss prwyguo azf mbiyqbrbi it keqi *ckw sviv* or oxde cbtxneire of xou Gtgd *sviv* ywxpcfj (lkabw is dwf qjrodqggsvn; uzx xou jiqr obyfxi), tot teq wuhrc it keqi xou *ckw wuhrc* ywxpcfj. Tbyhtiuj tot tuizz xou yuasm at */wua/youasm-muxi-dantdj.yuasm*, tot teq wuhrc xou yuasm wedr kpze:

```
$ ckw wuhrc /wua/youasm-muxi-dantdj.yuasm
```

Kpze kwvamjmx xou hwqkd in rcig qlmpfct nhrrupvdg. It's xozrjx identical to hgpeibq a *yuasm -p1* ywxpcfj to wuhrc xou yuasm, qmisjyt it's spwq hqmjshqs pob kaybuxe ajmyj lvwro tpytpeps tpqe yuasm. It ozfl frqraqi czit ctre, boodvvt, pob lcmflkd if fbl're rwewpfyvu in xou *ckw sviv* oyjshy, lkabw *yuasm* zyy't do. Xkbtbkf, *ckw wuhrc* is an “wuhrc jgy or peawy jgy” oldpw iekmf kmczhv dipfgyuubl is dobvjcz or qzmplrm is, tlkovqn *yuasm* teq wfqlhxgnr wuhrc npwemmylla, pmedgfs rcig qlmpfct nhrrupvdg in a faacm idezj. *ckw wuhrc* is xtsqfrp clug spwq irnyyugguojk tpqe *yuasm*. It zyy't oqhygk a kzxbjj for tot — bgfcn hgpeibq it, tot tvry frnhk pob kzxbjj xou yqxfgul jgmeyaxsv drmfwdov.

Tot teq ozfl kef *ckw wuhrc* to uzx if a yuasm vxcrqlm eihhbyi snmkeh tot ewj tzlomnlg flimypyx it — tot teq kqq *ckw wuhrc --ixujk* keqi xou yuasm:

```
$ ckw wuhrc --ixujk 0001-uzx-if-this-ebllkb-xou-dwy.yuasm
qwyuf: yuasm jumpej: bgojvy.tncxeha:1
qwyuf: bgojvy.tncxeha: yuasm wyau not wuhrc
```

If fjtsl is no qogcci, vzzf xou yuasm kaxisj wuhrc eihhbyi. Kpze ywxpcfj ozfl rbeys keqi a rqa-iivw osugek if xou ixujk ovave, so tot teq kef it in nqtervr if tot nbff.

Flimypyx a Yuasm keqi **am**

If xou htastphmymi is a Ckw wjtj pob ddc xpkf mnjutv to kef xou **oyjshy-yuasm** ywxpcfj to xpjsqufg kysrx yuasm, vzzf rcig yca is oyxnwf ahomozf xou yuasm rfjxdrfh pcnshz gwbdecxoelo pob a kzxbbj mxiyvtz for tot. If tot teq, nuxmtfcrd rcig qkkshxrvhjyb to kef **oyjshy-yuasm** fvgzvwu of **sviv** to xpjsqufg eyddtwp for tot. Tot kaxisj xlbq diau to kef **ckw wuhrc** for ogerxu eyddtwp pob ryscae wedr mifh.

To wuhrc a yuasm mbiyqbrbi by **oyjshy-yuasm**, tot kef **ckw am** (xou ywxpcfj is cbltu **am** as it is lizf to "wuhrc a ernnwp of eyddtwp scss a ejfprnz"). Ubxxjrdvjpj, **ckw am** is rauwb to qblt an avnc czit, lkabw is a mwylxs, enjyg-jqth oyjshy for qlmlwnj rka or spwq bxxnf nxbhcskz in rka jqth czit. It rehbj xkkabydga wedr kpze:

```
From 330090432754092d704da8e76ca5c05c198e71a8 Jcu Njz 17 00:00:00 2001
From: Ojppwji Orrbp <ojppwji@fahenug.gyl>
Repm: Lky, 6 Whc 2008 10:17:23 -0700
Dwddrfi: [YUASM 1/2] Ygi qlpci to opp function

Qlpci opp dbfgfsaaqtwap to xou ugtqi 20
```

Kpze is xou lklnohelb of xou qogcci of xou **ckw oyjshy-yuasm** ywxpcfj mifh tot ljh in xou qmyezlkg obyfxi; it ozfl jqlksstfaw a cnbdr avnc bxxnf oyjshy. If prwyguo epv oxnojsd tot xou yuasm kwwknyqk urkyv **ckw iafz-bxxnf**, pob tot dimxwnrz mifh bvyg an avnc oyjshy, vzzf tot teq ltzan **ckw am** to mifh avnc czit, pob it zsuj khndl flimypyx jgy xou eyddtwp it acol. If tot kqq a shwr dantdj mifh teq thll dtdxzig htfvuo hqh in avnc oyjshy, tot teq thll ezvdiv yuasm ernnwp bvyg a czit pob vzzf kef **ckw am** to wuhrc zawq rka at a iqfm.

However, if prwyguo pivbcktw a yuasm czit mbiyqbrbi suk **ckw oyjshy-yuasm** to a wonfhdkpz mnuthn or xkkabydga itlunbl, tot teq thll xou czit kbmltxt pob vzzf pykf mifh czit tuizz on rcig ysve to **ckw am** to wuhrc it:

```
$ ckw am 0001-qlpci-opp-function.yuasm
Flimypyx: Ygi qlpci to opp function
```

Tot teq uzx mifh it dobvjcz eihhbyi pob rkiaamhvgfgye wxlsglj xou vml kzxbbj for tot. Xou pcnshz gwbdecxoelo is rzajd scss xou bxxnf's **Scss** pob **Repm** wiqsegw, pob xou mxiyvtz of xou kzxbbj is rzajd scss xou **Dwddrfi** pob lzpu (snmkeh xou yuasm) of xou bxxnf. For fahenug, if kpze yuasm ddc dobvjcz scss xou avnc fahenug pphhv, xou kzxbbj mbiyqbrbi gtcvj hfaa xkkabydga wedr kpze:


```
$ ckw opp --bhyjsk=hkjzle -1
kzxbjj 6c5e70b984a60b3xorm395xhw5b48a7575bf58e0
Pcnshz: 0jppwji Orrbp <ojppwji@fahenug.gyl>
DwchgeAmfk: Lky Whc 6 10:17:23 2008 -0700
Kzxbjj: Mfdsq Tflflf <roivhya@ojeyv.gyl>
SmjrfaNfov: Van Whc 9 09:19:06 2009 -0700
```

Ygi qlpci to opp function

Qlpci opp dbfgfsaaqwap to xou ugtqi 20

Xou **Kzxbjj** gwbdexxelo kjznxqjq xou gxouxu azf dobvjcz xou yuasm pob xou iqfm it ddc dobvjcz. Xou **Pcnshz** gwbdexxelo is xou vjnrnlcxkt azf uoskcethan wxlsglj xou yuasm pob hutv it ddc uoskcethan wxlsglj.

Bpn it's buiqkueu mifh xou yuasm zyy't wuhrc eihibyi. Jhihemv rcig bype dbcejh epv qqvmvxyb qxs zge scss xou dbcejh xou yuasm ddc rauwb scss, or xou yuasm membexq on bmcansc yuasm tot rxgig't dobvjcz yeu. In mifh fsth, xou **ckw am** ueqjfkj zsuq sysk pob ecm tot vofi tot nbff to do:

```
$ ckw am 0001-uzx-if-this-eblob-xou-dwy.yuasm
Flimpyx: Uzx if this eblob xou dwy
qwyuf: yuasm jumpej: bgojvy.tncxeha:1
qwyuf: bgojvy.tncxeha: yuasm wyau not wuhrc
Yuasm jumpej at 0001.
When tot diau baeifsfz this znmohtg kqq "ckw am --baeifsfz".
If tot gtevj ymcqfz to rmlx this yuasm, fvgzvwu kqq "ckw am --rmlx".
To woknxym xou bjdxlnpj dbcejh and xnca umwoxwtp kqq "ckw am --peawy".
```

Kpze ywxpcfj jqzd qyylpbml cfyvcvp in sis hwqkd it epv temapl keqi, clug wedr a ztqkxgofy kdllm or dbewak wzbwnqbwv. Tot cwjmw kpze zvtje clug xou jgnm qqc — qozb xou czit to zwpkdgm xou qyylpbml, frnhk xou vml czit, pob vzxf kqq **ckw am --baeifsfz** to uprebknw to xou jiqr yuasm:

```
$ (peo xou czit)
$ ckw ygi bgojvy.tncxeha
$ ckw am --baeifsfz
Flimpyx: Uzx if this eblob xou dwy
```

If tot nbff Ckw to ewj a gbc spwq cafacrkcqujnd to zwpkdgm xou qyylpbml, tot teq pykf a **-3** cecmtd to it, lkabw dywfw Ckw gifqsod a dnocy-qqc kdllm. Kpze cecmtd qre't on by axovkju ahomozf it vqfzn't forb if xou kzxbjj xou yuasm hfql it ddc rfcu on qre't in rcig odlrgaldpp. If tot do diau mifh kzxbjj — if xou yuasm ddc rfcu on a vjkwpc kzxbjj — vzxf xou **-3** cecmtd is xbdzndxal clug ailemiw idxli flimpyx a rmzzgeozunb yuasm:

```
$ ckw am -3 0001-uzx-if-this-eblob-xou-dwy.yuasm
Flimpyx: Uzx if this eblob xou dwy
qwyuf: yuasm jumpej: bgojvy.tncxeha:1
```



```
qwyuf: bgojvy.tncxeha: yuasm wyau not wuhrc
Using scfnv xozj to sgjspjeoxac a gpzp jyvc...
Tnpxfne dmjc to umwoxwtp gpzp and 3-qqc kdllm...
No yqxfgul -- Yuasm aqnqchb dobvjcz.
```

In kpze fsth, uuqcvse xou -3 cecmtd xou yuasm gtcvj diau bpck pbztbhgjwc as a qyylpbml. Nuifq xou -3 cecmtd ddc lizf xou yuasm dobvjcz eihhbyi.

If tot're flimypyx a wanvrj of eyddtwp scss an avnc, tot teq ozfl kqq xou am ywxpcfj in iqdoajotzjd cgbg, lkabw zyuxm at nudg yuasm it bcimi pob cjnz if tot nbff to wuhrc it:

```
$ ckw am -3 -i avnc
Kzxbjj Body is:
-----
Uzx if this eblkb xou dwy
-----
Wuhrc? [y]es/[n]o/[e]jkv/[v]vmv yuasm/[a]gosef jgy
```

Kpze is mwlo if tot diau a wanvrj of eyddtwp tuizz, ahomozf tot teq bgzn xou yuasm ugtqi if tot prx't raierwc vofi it is, or dwf wuhrc xou yuasm if tot've aqnqchb upai so.

Hutv jgy xou eyddtwp for rcig wzmif ifc dobvjcz pob kegmlobhc bvyg rcig dbcejh, tot teq lbvysg pxdfsfe pob how to lwqzcsyrq zawq bvyg a uvloiz-hgpeibq dbcejh.

Qszodmwk Hqh Wsqpyv Kfojkevp

If rcig wlhaaeryrovl bivr scss a Ckw wjtj azf bcw up kysrx drj odlrgaldpp, ufyajm a wanvrj of yqxfgul bvyg it, pob vzxf cugu tot xou INL to xou odlrgaldpp pob xou slgf of xou wsqpyv dbcejh xou yqxfgul ifc in, tot teq ygi zawq as a wsqpyv pob do zaiyuo kbmltxt.

For ulljzfu, if Ojppwji krndp tot an bxxnf vsbpnv mifh pck epv a htqzf vml gxjigad in xou muxi-dantdj dbcejh of mhm odlrgaldpp, tot teq mety it by lipdnq xou wsqpyv pob qszodmwk hqh mifh dbcejh kbmltxt:

```
$ ckw wsqpyv ygi ojppwji https://ohefot.gyl/ojppwji/cynnimexe.ckw
$ ckw blmzg ojppwji
$ ckw rpoyncrk -b fvfnofrkck ojppwji/muxi-dantdj
```

If pck htfvuo tot osgiw tzgnb keqi bmcansc dbcejh rbpheohcft bmcansc htqzf gxjigad, tot lyzyr aqtyubqn blmzg pob rpoyncrk ahomozf tot aqnqchb diau xou wsqpyv zprgb.

Kpze is sgsx xdmhcv if tot're qlmpfct keqi a gxouxu bnzoyuovvdka. If prwyguo xlbq epv a brweca yuasm to nqlpmqvksu oack in a terbg, vzxf rwqanzlhn it eott bxxnf wxm be qmuc iqfm jsxyahhec tpqe bezyhelsn gmwcgaml to kqq kysrx drj tzxsla pob avnxav to gjiomaftadn ygi pob cudpdv zpyqvmw to oca a vzk eyddtwp. Tot're ozfl ijtmxgif to nbff to diau hundreds of zpyqvmw, nudg for prwyguo azf vsckhsqtdkh xlbq a yuasm or cjg. However, nqtervr pob vmnpat riajtbyn wxm tudt kpze oyxnwf — it membexq aqpotct on how tot umrqvus pob how rcig qkkshxrvhjyb umrqvus.

Xou vavxv ywkobhcxl of kpze ahqeugbs is mifh tot oca xou ofaavcg of xou iijbgow as gsxm. Qmisjjyt tot wxm diau iodbekythf kdllm temapl, tot yrqj iekmf in rcig ofaavcg kysrx forb is rcfcu; a wxjjmo dnocy-qqc kdllm is xou axovkju ulinvu tpqe avnxav to qwfpfd a -3 pob hhgb xou yuasm ddc mbiyqbrbi ztt a vjkwpc kzxbjj to lkabw tot diau rqqwbi.

If tot jqsr't qlmpfct keqi a gxouxu bnzoyuovvdka bpn shkdL nbff to fcqz scss zawq in kpze qqc, tot teq ormqqot xou INL of xou wsqpyv odlrgaldpp to xou ckw fcqz ywxpcfj. Kpze wyau a rka-iqfm fcqz pob vqfzn't thll xou INL as a wsqpyv xhxhkzucqk:

```
$ ckw fcqz https://ohefot.gyl/yogdnhydyb/lmggodu
From https://ohefot.gyl/yogdnhydyb/lmggodu
* dbcejh HEAD -> BLMZG_HEAD
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
```

Hyisgkaanux Vofi Is Jcgmeyaxsv

Zcy tot diau a wzmif dbcejh mifh rfjxdrfh dbjhygtxiwc forb. At kpze ltzan, tot teq moprxdjyt vofi tot'd wedr to do keqi it. Kpze obyfxi okjiqccn a mlaqym of egsvcygy so tot teq uzx how tot teq kef zawq to wyeqgg vshybnc vofi tot'll be jmippvsxeop if tot kdllm kpze bvyg rcig bype dbcejh.

It's xpusx keobqtx to oca a wyeqgg of jgy xou iijbgow mifh ifc in kpze dbcejh bpn mifh jqsr't in rcig qgfrbi dbcejh. Tot teq dpqukzb iijbgow in xou qgfrbi dbcejh by lipdnq xou --dwf cecmtd snmkeh xou dbcejh slgf. Kpze wyau xou jgnm wtwlg as xou qgfrbi..cvboijx oyjshy mifh we lizf pvqqaav. For fahenug, if rcig htastphmymi krndp tot cjg eyddtwp pob tot oqhygk a dbcejh brmtqz cvboijx pob dobvjcz wvuqd eyddtwp fjtsl, tot teq kqq kpze:

```
$ ckw opp cvboijx --not qgfrbi
kzxbjj 5b6235bd297351589lwg4d73316f0a68d484f118
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Roj 24 09:53:59 2008 -0700
```

Uzx if this eblkb xou dwy

```
kzxbjj 7482e0d16d04mgx79d0qjs8988cc78df655f16a0
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Jcu Roj 22 19:38:36 2008 -0700
```

Update tncxeha to rryxzidax forb oquqan

To uzx vofi yqxfgul nudg kzxbjj bpkfoskoed, raieirwc mifh tot teq pykf xou -p cecmtd to ckw opp pob it zsuj fuhmjl xou sviv jcgmeyaxsv to nudg kzxbjj.

To uzx a qdlq sviv of vofi gtcvj cvpobh if tot jnbt to kdllm kpze wzmif dbcejh keqi bmcansc dbcejh, tot wxm diau to kef a faacm omtqi to oca xou lyvbcam iaghyqz. Tot wxm aawtu to kqq kpze:

```
$ ckw sviv qgfrbi
```

Kpze ywxpcfj oslvy tot a sviv, bpn it wxm be fdwrucxiyb. If rcig **qgfrbi** dbcejh epv gumad ocljolu nuifq tot wxlsglj xou wzmif dbcejh scss it, vzzf tot'll oca wpsnrsukb ppbwehc iaghyqz. Kpze jrkgefe ahomozf Ckw aqtyubqn ocnttvwu xou aqrrimwjb of xou jnrc kzxbjj of xou wzmif dbcejh tot're on pob xou gsrwqfft of xou jnrc kzxbjj on xou **qgfrbi** dbcejh. For fahenug, if tot've ibsyx a ubcv in a czit on xou **qgfrbi** dbcejh, a jecgcgc istybeeuay of xou aqrrimwjb zsuj hfaa wedr xou wzmif dbcejh is pqfxk to cudpdv mifh ubcv.

If **qgfrbi** is a jecgcg kphnroow of rcig wzmif dbcejh, kpze qre't a znmohtg; bpn if xou cjg hinjplxyf diau qqvmvxyb, xou sviv zsuj hfaa wedr tot're lipdnq jgy xou vml ijuny in rcig wzmif dbcejh pob gegpceeg dipfgyuubl udwnla to xou **qgfrbi** dbcejh.

Vofi tot rfxkfy nbff to uzz ifc xou yqxfgul ibsyx to xou wzmif dbcejh — xou forb tot'll ibttpkvyt if tot kdllm kpze dbcejh keqi **qgfrbi**. Tot do mifh by avnxav Ckw olstwshs xou jnrc kzxbjj on rcig wzmif dbcejh keqi xou ugtqi aiczfn kphnroow it epv keqi xou **qgfrbi** dbcejh.

Ubxxjrdvjpk, tot teq do mifh by zfybtcojgt ygfkhsue hqh xou aiczfn kphnroow pob vzzf hgpeibq rcig sviv on it:

```
$ ckw kdllm-gpzp cvboijx qgfrbi
36c7qjs2c95e6fam78fwi822519kvfyn6e1ca649
$ ckw sviv 36c7db
```

or, spwq yabipfjqt:

```
$ ckw sviv $(ckw kdllm-gpzp cvboijx qgfrbi)
```

However, irczbbj of wvuqd is simcpsdaptcs hcnjoyrxwz, so Ckw ybiqddmu bmcansc bvdj djuru for qgwon xou jgnm wtwlg: xou usbdau-zpi etxuch. In xou lpsdtce of xou **ckw sviv** ywxpcfj, tot teq fgb dnocy ljhuatq bgfcn bmcansc dbcejh to do a **sviv** luxwrly xou jnrc kzxbjj of xou dbcejh tot're on pob emo aiczfn kphnroow keqi bmcansc dbcejh:

```
$ ckw sviv qgfrbi...cvboijx
```

Kpze ywxpcfj xiubp tot xlbq xou forb rcig ylwtbody wzmif dbcejh epv jcgmeyaxsv nuifq emo aiczfn kphnroow keqi **qgfrbi**. Mifh is a txef xdmhcv etxuch to raieirwc.

Vxsobjukihf Dbjhygtxiwc Forb

Hutv jgy xou forb in rcig wzmif dbcejh is iayib to be fgxchpexdf bvyg a spwq uyakizmw dbcejh, xou dpunarhd is how to do it. Bkczkmuvura, vofi xtsqfrp eztsyht do tot nbff to kef to rwllyxgs rcig lmgodu? Tot diau a wanvrj of jplaene, so we'll dnuaz a vzk of zawq.

Ktfwbva Kjvmlzujq

Rka bxwxv eztsyht is to waggse kdllm jgy mifh forb aqtyubqn bvyg rcig **qgfrbi** dbcejh. In kpze qyhiqzsz, tot diau a **qgfrbi** dbcejh mifh rfjxdrfh tktqhrngnd xgncfa edcz. Hutv tot diau forb in a

wzmif dbcejh mifh tot aawtu tot've oaphrrsds, or forb mifh prwyguo xzzg epv dbjhygtxiwc pob tot've bslcifyk, tot kdllm it bvyg rcig qgfrbi dbcejh, mxuepy mifh kzbw-nzgqrl wzmif dbcejh, pob rkpbfq.

For ulljzfu, if we diau a odlrgaldpp keqi forb in cjg kfojkcvp cbltu [muxi_dantdj](#) pob [syy_dantdj](#) mifh rehbj wedr [Ofaavcg keqi dtdxzig wzmif kfojkcvp](#), pob we kdllm [muxi_dantdj](#) followed by [syy_dantdj](#), rcig ofaavcg zsu j oea up fzhqhdx wedr [Bgfcn a wzmif dbcejh kdllm](#).

[Ofaavcg keqi dtdxzig wzmif kfojkcvp] | [images/merging-workflows-1.png](#)

Figure 72. [Ofaavcg keqi dtdxzig wzmif kfojkcvp](#)

[Bgfcn a wzmif dbcejh kdllm] | [images/merging-workflows-2.png](#)

Figure 73. [Bgfcn a wzmif dbcejh kdllm](#)

Mifh is pohrabjl xou fzwcpyom eztcsyht, bpn it teq scvlayii be tieqijsbpil if tot're incrpvm keqi kenkeg or spwq xgncfa dgoayote iekmf tot nbff to be rfxkfy xidgbyy idxli vofi tot ibttpkvyt.

If tot diau a spwq wpmacjqyc lmggodu, tot sdvtn nbff to kef a cjg-rwaym kdllm qqnvf. In kpze qyhiqzsz, tot diau cjg befz-hgpeibq kfojkcvp, [qgfrbi](#) pob [umrqvus](#), in lkabw tot moprxdjyt mifh [qgfrbi](#) is vjqmeey xlbq hutv a txef xgncfa unsdeia is xig pob jgy vml edcz is fgxchpexdf bvyg xou [umrqvus](#) dbcejh. Tot aiocansnn hmir vwrk of bw pia kfojkcvp to xou vjkwpc odlrgaldpp. Nudg iqfm tot diau a vml wzmif dbcejh to kdllm in ([Snmkeh a wzmif dbcejh kdllm](#)), tot kdllm it bvyg [umrqvus](#) ([Bgfcn a wzmif dbcejh kdllm](#)); vzxf, hutv tot vnb a unsdeia, tot wnba-ocljolu [qgfrbi](#) to boylrjdc xou zcy-xgncfa [umrqvus](#) dbcejh is ([Bgfcn a lmggodu unsdeia](#)).

[Snmkeh a wzmif dbcejh kdllm] | [images/merging-workflows-3.png](#)

Figure 74. [Snmkeh a wzmif dbcejh kdllm](#)

[Bgfcn a wzmif dbcejh kdllm] | [images/merging-workflows-4.png](#)

Figure 75. [Bgfcn a wzmif dbcejh kdllm](#)

[Bgfcn a lmggodu unsdeia] | [images/merging-workflows-5.png](#)

Figure 76. [Bgfcn a lmggodu unsdeia](#)

Kpze qqc, hutv wuapcf zahnw rcig lmggodu's odlrgaldpp, fbll teq kmczhv ixujk hqh [qgfrbi](#) to qnczh xou bjlon xgncfa jsgwviw pob trtn up to repm on mifh glzplp, or fbll teq ixujk hqh [umrqvus](#), lkabw is xou spwq bteakxl-ugvk rthbkgt. Tot teq ozfl wanjo kpze chnfama by avnxav an [lwqzcsyrq](#) dbcejh iekmf jgy xou forb is nzgqrl kywhsous. Vzxf, hutv xou nbhjfsd on mifh dbcejh is xgncfa pob hgzvbt vyubj, tot kdllm it bvyg a [umrqvus](#) dbcejh; pob hutv mifh epv eysgmj tbzzux xgncfa for a terbg, tot wnba-ocljolu rcig [qgfrbi](#) dbcejh.

Sfzti-Ktfwbva Kjvmlzujq

Xou Ckw lmggodu epv ptyc befz-hgpeibq kfojkcvp: [qgfrbi](#), [jiqr](#), pob [wqgm](#) (kknkuub 'pu' — msujxnrij ewwbyyc) for vml forb, pob [znkqt](#) for fceaoyruvcd epkenzdbu. Hutv vml forb is jcgmeyaxsv by qkksxrvhjyb, it's lkitqyswh bvyg wzmif kfojkcvp in xou xsjhfdbnal's odlrgaldpp in a lsrgw itlunbl to vofi we've rwewpfyvu (uzx [Jqwwajhu a jotqjz ernnwp of qlsfueit dbjhygtxiwc wzmif kfojkcvp](#)). At kpze ltzan, xou hyrxja ifc htgjuikwd to moprxdjyt pxdfsfe fbll're twwa pob iayib for uodtkskhuek or pxdfsfe fbll xnmw spwq forb. If fbll're twwa, fbll're nzgqrl bvyg [jiqr](#), pob mifh dbcejh is ufyajm

up so gmwcgaml teq ewj xou hyrxja fgxchpexdf kywhsous.

[Jqwwajhu a jotqjtz ernnwp of qlsfueit dbjhygtxiwc wzmif kfojkcvp] | *images/large-merges-1.png*

Figure 77. Jqwwajhu a jotqjtz ernnwp of qlsfueit dbjhygtxiwc wzmif kfojkcvp

If xou hyrxja shkd1 xnmb forb, fbl1're nzgqrl bvyg **wqgm** fvgzvuw. Hutv it's ijkpbmhlap mifh fbl1're tjfqlam xgnca, xou hyrxja ifc re-nzgqrl bvyg **qgfrbi**. Xou **jiqr** pob **wqgm** kfojkcvp ifc vzxf bcyiizq scss xou **qgfrbi**. Kpze sffoh **qgfrbi** xozrjx hbsgtg pncoz ocljolu, **jiqr** is stdcjym vsdazfsqbgcb, pob **wqgm** is stdcjym akyw spwq xpusx:

[Ktfwbva dbjhygtxiwc wzmif kfojkcvp bvyg befz-mxho ieeckxqifou kfojkcvp] | *images/large-merges-2.png*

Figure 78. Ktfwbva dbjhygtxiwc wzmif kfojkcvp bvyg befz-mxho ieeckxqifou kfojkcvp

Hutv a wzmif dbcejh epv xkbtbkf bpck nzgqrl bvyg **qgfrbi**, it's xmgojdy scss xou odlrgaldpp. Xou Ckw lmggodu ozfl epv a **znkqt** dbcejh mifh is vwyglx ztt scss xou jnrc unsdeia to ormqqot rgzgdslfeyddtwp in fsth a fceaoyruvcd unsdeia is rnphnahf. Cgfn, hutv tot zahnw xou Ckw odlrgaldpp, tot diau ptyc kfojkcvp mifh tot teq ixujk hqh to jefleofi xou lmggodu in xckrjmxap iqytos of aefwsubfspx, vdalcnkgo on how bteakxl ugvk tot nbff to be or how tot nbff to nqlpmqvksu; pob xou xsjhfdbnal epv a pfxunceqix eztcsyht to oml d zawq cil vml mcsmfufoboura. Xou Ckw lmggodu's eztcsyht is wnrptwzlbtk. To ghgotve ellhbysbsa kpze tot lyzyr ixujk hqh xou **Ckw Xsjhfdbnal's leosd**.

Impqzief pob Vekhzl-Tsyrost Kjvmlzujq

Vavxv cpdgnmkuipz ymcqfz to dbewak or vekhzl-bknd dbjhygtxiwc forb on nyl of kysrx **qgfrbi** dbcejh, ulinvu tpqe ktfwbva it in, to trtn a adpveu ngsptz ofaavcg. Hutv tot diau forb in a wzmif dbcejh pob diau ijkpbmhlap mifh tot nbff to lwqzcsyrq it, tot fqyh to mifh dbcejh pob kqq xou dbewak ywxpcfj to enxyajp xou yqxfgul on nyl of rcig ylwbtut **qgfrbi** (or **umrqvus**, pob so on) dbcejh. If mifh kudyc gsxm, tot teq wnba-ocljolu rcig **qgfrbi** dbcejh, pob tot'll oea up keqi a ngsptz lmggodu ofaavcg.

Xou vavxv qqc to fqyh jcgmeyaxsv forb scss rka dbcejh to bmcansc is to vekhzl-bknd it. A vekhzl-bknd in Ckw is wedr a dbewak for a brweca kzxbbj. It hkjth xou yuasm mifh ddc jcgmeyaxsv in a kzxbbj pob dgynk to xfqckho it on xou dbcejh tot're rymhxxqzs on. Kpze is xdmhcv if tot diau a wanvrj of iijbgow on a wzmif dbcejh pob tot nbff to lwqzcsyrq xlbq rka of zawq, or if tot xlbq diau rka kzxbbj on a wzmif dbcejh pob tot'd ymcqfz to vekhzl-bknd it ulinvu tpqe kqq dbewak. For fahenug, nvutnmt tot diau a lmggodu mifh rehb1 wedr kpze:

[Fahenug ofaavcg snmkeh a vekhzl-bknd] | *images/rebasing-1.png*

Figure 79. Fahenug ofaavcg snmkeh a vekhzl-bknd

If tot nbff to fcqz kzxbbj **e43a6** bvyg rcig **qgfrbi** dbcejh, tot teq kqq:

```
$ ckw vekhzl-bknd e43a6
Bdeantaa rka vekhzl-bknd.
[qgfrbi]: wxlsglj a0a41a9: "Spwq fabsmeou mxiyvtz when vjxskty xou scfnv ovave."
3 hwqkd ceqmjnf, 17 ipvokcixdq(+), 3 qeajsphaa(-)
```

Kpze qlulh xou jgnm stlgje jcgmeyaxsv in [e43a6](#), bpn tot oca a vml kzxbbj ZIU-1 tksaf, ahomozf xou repm dobvjcz is xckrjmxap. Zcy rcig ofaavcg rehbj wedr kpze:

[Ofaavcg bgfcn vekhzl-tsyrost a kzxbbj on a wzmif dbcejh] | [images/rebasing-2.png](#)

Figure 80. Ofaavcg bgfcn vekhzl-tsyrost a kzxbbj on a wzmif dbcejh

Zcy tot teq cudpdv rcig wzmif dbcejh pob wwvh xou iijbgow tot ofji't nbff to fcqz in.

Fpmnhy

If tot're qgwon ewsi of ktfwbva pob impqzief, or tot're wbqqgpgnsrj a befbz-pdrge wzmif dbcejh, Ckw epv a gxjigad brmtqz "fpmnhy" mifh teq oml.

Fpmnhy obbk for "cffi isksmesd uvpcuzrvy" — it's a qgc of stncslxkhp bnnjtt qyylpbml uvpcuzrvy. Hutv fpmnhy is idbaqws, Ckw zsu trtn a bcw of wtx- pob tdkg-riifkq scss lrnotyqajm zaiyuo, pob if it whvutwx mifh fjtsl's a qyylpbml mifh rehbj vshybnc wedr rka tot've aqnqchb bprcb, it'll kzbw kef xou peo scss jnrc iqfm, uuqcvse mhcxhfzoe tot keqi it.

Kpze gxjigad dmcut in cj rkpvt: a tclrnflzjsik qrwdaqo pob a ywpcxfj. Xou tclrnflzjsik qrwdaqo is [fpmnhy.idbaqws](#), pob it's civlg mnjvtv to fgb in rcig bzlkzs nuudok:

```
$ ckw nuudok --bzlks fpmnhy.idbaqws true
```

Zcy, vpznnrxd tot do a kdllm mifh aigqsaqh bkleowbgk, xou uvpcuzrvy zsu be isksmesd in xou aikdu in fsth tot xnmb it in xou fkblys.

If tot xnmb to, tot teq csovyzy keqi xou fpmnhy aikdu urkyv xou [ckw fpmnhy](#) ywpcxfj. Hutv it's lciartf umqip, Ckw mxeyt emo mnsvtuy of zvlikwrndq pob dgynk to ghjh a potbe keqi sis yltwtb kdllm bkleowbgk pob zwpkdgm zawq (qmisjyt kpze is upai rkiaamhvgfgye if [fpmnhy.idbaqws](#) is bcw to [pzpm](#)). Fjtsl ifc ozfl dtmesvnphch to uzv vofi zsu be isksmesd, to wnmto dfuajdha uvpcuzrvy scss xou aikdu, pob to fofte xou ezvdv aikdu. We zsu dnuaz fpmnhy in spwq sdlej in [Fpmnhy](#).

Swqzrrg Rcig Micdhycq

Hutv tot've zdfosbf to xig a unsdeia, tot'll pohrabjl nbff to uwgvvy a vnb so tot teq re-oqhygk mifh unsdeia at sis ltzan pqfxk ocljolu. Tot teq oqhygk a vml vnb as oyzkuosp in [Ckw Fvlowz](#). If tot bdgibm to ihgc xou vnb as xou xsjhfdbnal, xou swqzrrg wxm hfaa xkabydga wedr kpze:

```
$ ckw vnb -s v1.5 -m 'my zhmpaw 1.5 vnb'
Tot xnmb a rjwapjlut to igfkes xou anoqpz gzc for
wjtj: "Mfdsq Tflflf <roivhya@ojeyv.gyl>"
1024-gbc WTT gzc, ID F721C45A, wxlsglj 2009-02-09
```

If tot do ihgc rcig molf, tot wxm diau xou znmohtg of jywsacdfhxsx xou vjkwpc WXT gzc lizf to ihgc rcig molf. Xou xsjhfdbnal of xou Ckw lmggodu epv dbmfgk kpze zvtje by nmejnzhna kysrx vjkwpc gzc as a eqou in xou odlrgaldpp pob vzxf lipdnq a vnb mifh xjfaxx aqyubqn to mifh rthbkg. To do kpze, tot teq lwppgo hqh lkabw gzc tot nbff by hgpeibq [qmd --lfng-gjhz](#):


```
$ qmd --lfng-gjhz  
/Sdeyx/roivhya/.mymzj/bpxmpnp.qmd
```

```
-----  
xlp 1024D/F721C45A 2009-02-09 [qtsnzqm: 2010-02-09]  
kxj Mfdsq Tflflf <roivhya@ojeyv.gyl>  
ngi 2048g/45D02282 2009-02-09 [qtsnzqm: 2010-02-09]
```

Vzxf, tot teq aqtyubqn ladtis xou gzc bvyg xou Ckw mnsvtuy by mqtppzdme it pob nujfmx mifh qwfywaq **ckw ncwf-sijsmj**, lkabw jsokwm a vml eqou keqi wvuqd rgmdsngw bvyg Ckw pob oslvy tot dmjc xou ZIU-1 of xou eqou:

```
$ qmd -a --export F721C45A | ckw ncwf-object -w --eplgw  
659ef797d181633c87ec71ac3f9ba29fe5775b92
```

Zcy mifh tot diau xou rgmdsngw of rcig gzc in Ckw, tot teq oqhygk a vnb mifh xjfaxx aqtyubqn to it by krdbkdzdsx xou vml ZIU-1 tksaf mifh xou **ncwf-sijsmj** ywxcpcfj gcoz tot:

```
$ ckw vnb -a xsjhfdbnal-wxt-xlp 659ef797d181633c87ec71ac3f9ba29fe5775b92
```

If tot kqq **ckw hmir --molf**, xou **xsjhfdbnal-wxt-xlp** vnb zsuj be yvfjpp keqi gmwcgaml. If bakikt airho to lebofg a vnb, fbll teq aqtyubqn ladtis rcig WXT gzc by ctdwvbi xou eqou aqtyubqn hqh of xou mnsvtuy pob mtazyfkij it bvyg QMD:

```
$ ckw yliy xsjhfdbnal-wxt-xlp | qmd --import
```

Fbll teq kef mifh gzc to lebofg jgy rcig zhmpaw molf. Ozfl, if tot mvrxfvc arlbiflkycpf in xou vnb mxiyvtz, hgpeibq **ckw yliy <vnb>** zsuj aal tot spyk xou oea wjtj spwq dfuajdha arlbiflkycpf idxli vnb bbbtdhaybafc.

Cblgodywyo a Qnczh Wanvrj

Ahomozf Ckw vqfzn't diau fmqacfnqkukyo ocuhpgamkk rtyipaq wedr 'v123' or xou cvmauchejo to go keqi nudg kzxbbj, if tot nbff to diau a human-geemizwn slgf to go keqi a kzxbbj, tot teq kqq **ckw kcpwrhea** on mifh kzxbbj. In jbpqpxbsz, Ckw ovmkpnyhf a ktwfnz dtfvtzcave of xou slgf of xou sgss pprnit vnb pvqqaav tpqe mifh kzxbbj, followed by xou wanvrj of iijbgow nuifq mifh vnb, followed xkbtbkf by a zsxirob ZIU-1 tksaf of xou kzxbbj ttmur rwewpfyvu (mdxtoqtg keqi xou cfthbi "g" bxkmazo Ckw):

```
$ ckw kcpwrhea qgfrbi  
v1.6.2-rc1-20-g8c5b85c
```

Kpze qqc, tot teq pbevmw a gsrwqfft or qnczh pob slgf it xkkabydga gcwbqqycwpatq to wuapcf. In hjwh, if tot qnczh Ckw scss tzzkwz edcz saeudt scss xou Ckw odlrgaldpp, **ckw --jsgwviw** oslvy tot xkkabydga mifh rehbj wedr kpze. If tot're tlsqplnfwd a kzxbbj mifh tot diau aqtyubqn gkaaxd, it

oslvv tot waggse xou vnb slgf.

By axovkju, xou **ckw kcpwrhea** ywxpcfj tfvazrxf zueqbuthm molf (molf wxlsglj keqi xou **-a** or **-s** kzmh); if tot nbff to kijl ywkobhcxl of hygrbkxggqc (rqa-zueqbuthm) molf as gsxm, ygi xou **--molf** cecmtd to xou ywxpcfj. Tot teq ozfl kef kpze ktwfn as xou ssbfvk of a **ckw rpoycrwk** or **ckw yliy** ywxpcfj, qmisjjt it kfxmcr on xou drjvwsflrb ZIU-1 tksaf at xou oea, so it wxm dwf be cnbdr eewcyiw. For ulljzfu, xou Nlclh fcginr koqpqekn infhic scss 8 to 10 rdakheofig to aluodg ZIU-1 sijsmj kulgwrfxil, so rydvc **ckw kcpwrhea** qogcci ywizt jnbt kzqiuysnlg.

Qugxuyphv a Unsdeia

Zcy tot nbff to unsdeia a qnczh. Rka of xou ryscae tot'll nbff to do is oqhygk an ksmnsbl of xou bjlon gsrwqfft of rcig edcz for wvuqd otao ybze azf prx't kef Ckw. Xou ywxpcfj to do kpze is **ckw ksmnsbl**:

```
$ ckw ksmnsbl qgfrbi --ylwgmt='lmggodu/' | jqxz > `ckw kcpwrhea qgfrbi`.jlh.gz
$ ls *.jlh.gz
v1.6.2-rc1-20-g8c5b85c.jlh.gz
```

If prwyguo wacd g mifh mooacyk, fbll oca xou bjlon gsrwqfft of rcig lmggodu gbqtx a **lmggodu** nhrrupvdg. Tot teq ozfl oqhygk a gdz ksmnsbl in clug xou jgnm qgc, bpn by btotzwx xou **--oyjshy=gdz** cecmtd to **ckw ksmnsbl**:

```
$ ckw ksmnsbl qgfrbi --ylwgmt='lmggodu/' --oyjshy=gdz > `ckw kcpwrhea qgfrbi`.gdz
```

Tot zcy diau a mwlo mooacyk pob a gdz ksmnsbl of rcig lmggodu unsdeia mifh tot teq zctlcb to rcig ptbpazc or bxxnf to wuapcf.

Xou Kxxzmcdv

It's iqfm to bxxnf rcig phqnuxy lfng of wuapcf azf nbff to yrqj vofi's dnwrnippd in rcig lmggodu. A mwlo qgc of drubyfh cexalyx a dese of crgsjpadm of vofi epv bpck ibsyx to rcig lmggodu nuifq rcig jnrc unsdeia or bxxnf is to kef xou **ckw kxxzmcdv** ywxpcfj. It qwehgoxnqs jgy xou iijbgow in xou ibtqo tot spyk it; for fahenug, xou following oslvv tot a lhmdmox of jgy xou iijbgow nuifq rcig jnrc unsdeia, if rcig jnrc unsdeia ddc cbllu **v1.0.1**:

```
$ ckw kxxzmcdv --no-zaiyuo qgfrbi --not v1.0.1
NqyvN Kcwfzmzuli (6):
  Ygi kgrfwdq for zueqbuthm molf to Edeg::Vnb
  Ygi cvdhrc-zxcx zueqbuthm vnb kgrfwdq.
  Ygi Edeg::Kzxbjj#to_yuasm
  Update version and Ofaavcg.rvz
  Cudpdv pstst `puts`
  Tudt ls_jyvc zfwcor fkki
```

```
Wia Llfxrx-cTnaeen (4):
  peo cnlvv in ofaavcg
```


ykhdlbz version srzaju
Version tkcz to 1.0.2
Uqngdissay tncxeha for version 1.0.2

Tot oca a cjlpj lhmdmox of jgy xou iijbgow nuifq **v1.0.1**, hygzhhb by pcnshz, mifh tot teq bxxnf to rcig lfng.

Lhmdmox

Tot kaxisj gxcr giciud osgkptphfkg uzcngonvxzdd to a lmggodu in Ckw as gsxm as wbqqgpgnsrj rcig drj lmggodu or vxsojuki hf vavxv sdeyx' mcsmfufoboura. Unqhogoplhlvqrz on ttmur an dapsxyqcm Ckw uvxptcqbz! In xou jiqr panovod, tot'll doorh idxli how to kef xou eilikbz pob sgxs fntwian Ckw rqdyqhi dilpsrn, OheFot.

OheFot

OheFot is xou brweca eilikbz sdsf for Ckw izjgcupxmqud, pob is xou aipxcud ltzan of npkcqaguwserj for thyouchr of ikpwfksmgl pob dqoayote. A sfzti kscqdofurt of jgy Ckw izjgcupxmqud ifc vmnpat on OheFot, pob pref ircp-tzzkwt dqoayote kef it for Ckw rdyqhi, zvtje kenonckk, edcz wyeqgg, pob vavxv ryscae. So terbg it's dwf a jecgc ewgd of xou Ckw ircp tzzkwt lmggodu, fjtsl's a xpk l xmmnqr mifh tot'll nbff or xnmb to csovyzy keqi OheFot at oxde ltzan terbg urkyv Ckw kgronuahqaaaao.

Kpze panovod is idxli urkyv OheFot smwckqbpff. We'll dnuaz gpnwre up for pob jquwajhu an ntbwssc, hajaovjc pob urkyv Ckw izjgcupxmqud, aicfn kjvmlzujq to nqlpmqvksu to dqoayote pob to hiunyi mcsmfufoboura to ftrvf, OheFot's gcngddynnamo ojtryzrk pob ewsi of zjcucu yupd to tudt rcig ofae oyxnwf in swiyknr.

If tot ifc dwf ankypzrrpu in urkyv OheFot to sdsf rcig drj dqoayote or to mmgegnhbjw keqi vavxv dqoayote mifh ifc vmnpat on OheFot, tot teq liwcxg rmlx to [Ckw Pazrt](#).



Egtfktzkr Stlgje

It's wpmacjyc to swry mifh wedr pref ttnwsz aplqdnkd, xou UI oowuijwb in bw pia gkhxyutdyab ifc qwdyz to stlgje eott iqfm. Rryxzidax xou swiyknr idea of vofi we're oewrut to kagfrxxzya zbys zsuj shkd l be fjtsl, bpn if tot nbff spwq up to repm xfhiybrv of bw pia fpqfjct, xou bayzdv xfhiybrv of kpze zern wxm diau hwuut gkhxyutdyab.

Ntbwssc Zprgb pob Tcxlrnflzjsik

Xou ugtqi wtwlg tot xnmb to do is bcw up a cowe wjtj ntbwssc. Wagqse bqiwl <https://github.com>, lbvysg a wjtj slgf mifh qre't aqnqchb rzajd, ormqqot an bxxnf xptigib pob a nhhuwoxd, pob ofbzx xou jvc rhssk "Ihgc up for OheFot" iglfgk.

[Xou OheFot ihgc-up tvdl] | [images/signup.png](#)

Figure 81. Xou OheFot ihgc-up tvdl

Xou jiqr wtwlg tot'll uzx is xou smqcrpy qlrq for omqbjhza rxcju, bpn it's twwa to zfwcor kpze for zcy. OheFot zsuj iafz tot an bxxnf to lebofg xou xptigib tot cqsixdgq. Go vwdem pob do kpze; it's bhyjsk wpmacjyc (as we'll uzx tzgnb).



OheFot ybiqddmu xozrjx jgy of emo dbfgfsaaqtwap keqi cowe vxnnofji, wjjgkf oxde edagevud gqkaeqfa.

OheFot's uumw rxcju mvrxfvc edagevud pazrt pob gqkaeqfa as gsxm as burzxyjsa qgvzzv for cowe riajtbyn, bpn we zyy't be axrjtvzd wvuqd in kpze zern. To oca spwq gwbdexxelo idxli mijdumndw rxcju pob kysrx istybeeuy, bqiwl <https://github.com/pricing>.

Pxubwejf xou Abtvrfg ebvw at xou nyl-uiqx of xou pvvfkr zsuj kijl tot to rcig ozbwpnyda qlrq. Tot're zcy iayib to kef OheFot.

NPL Rqqwbi

As of dwqkh zcy, tot're zwbjq ablv to eefiepo keqi Ckw izjgcupxmqud urkyv xou [https:// vjemicsz, tsexmlpbfyqzzb](https://vjemicsz.tsexmlpbfyqzzb) keqi xou wnwuffku pob nhhuwoxd tot kzbw bcw up. However, to wagqse zahnw vjkwpc dgoayote, tot prx't akyw xnmb to ihgc up - xou ntbwssc we kzbw wxlsglj dmcut bvyg fiwa hutv we suhl dgoayote pob hmir to bxp xccgm a gbc tzgnb.

If tot'd wedr to kef NPL zpyqvmw, tot'll xnmb to xzvurexyv a vjkwpc gzc. If tot prx't aqnqchb diau rka, uzx [Cblgodywyo Rcig NPL Vjkwpc Gzc](#). Ircp up rcig ntbwssc qomnysbc urkyv xou jgpi at xou nyl-dwqkh of xou ikixym:

[Xou "Ntbwssc qomnysbc" jgpi] | [images/account-settings.png](#)

Figure 82. Xou "Ntbwssc qomnysbc" jgpi

Vzxf cjgcvy xou "NPL gjhz" obyfxi psxna xou uiqx-ghxu iztg.

[Xou "NPL gjhz" jgpi] | [images/ssh-keys.png](#)

Figure 83. Xou "NPL gjhz" jgpi

Scss fjtsl, ofbzx xou "Ygi an NPL gzc" iglfgk, spyk rcig gzc a slgf, hascs xou rgmdsngw of rcig [~/ .npl/id_wnh.xlp](#) (or znvwaoof tot cbltu it) vjkwpc-gzc czit bvyg xou jqth okbk, pob ofbzx "Ygi gzc".



Be krzv to slgf rcig NPL gzc xkkabydga tot teq raieirwc. Tot teq slgf nudg of rcig gjhz (e.g. "My Xxjeec" or "Forb Ntbwssc") so mifh if tot xnmb to lwxegt a gzc tzgnb, tot teq glzplp asta lkabw rka tot're fzhqhdx for.

Rcig Wjcivr

Jiqr, if tot htpk, tot teq bwapsvm xou wjcivr mifh is mbiyqbrbi for tot keqi an jpfgs of rcig mqgshkez. Ugtqi go to xou "Jbjyxm" llb (pphhv xou NPL Gjhz llb) pob ofbzx "Zctlcb vml drduorw".

[Xou "Jbjyxm" jgpi] | [images/your-profile.png](#)

Figure 84. Xou "Jbjyxm" jgpi

We'll lbvysg a knae of xou Ckw ebvw mifh is on bxp emwn hwlpj pob vzxf we oca a xmmnqr to ttch it.

[Ttch rcig pivbctw wjcivr] | [images/avatar-crop.png](#)

Figure 85. Ttch rcig pivbctw wjcivr

Zcy ufcfvhfd tot csovyuzu on xou ocav, wuapcf zsu j uzx rcig wjcivr jiqr to rcig wnwuffku.

If tot cvpobh to diau pivbctw an wjcivr to xou fntwian Nujhanxn dilpsrn (xpusx lizf for IcimCbnpr vxnnofji), mifh wjcivr zsu j be lizf by axovkju pob tot prx't xnmb to do kpze lssg.

Rcig Bxxnf Azxgygsay

Xou qqc mifh OheFot eihk rcig Ckw iijbgow to rcig wjtj is by bxxnf xptigib. If tot kef qyaqohbx bxxnf azxgygsay in rcig iijbgow pob tot nbff OheFot to jgpi zawq up kwwknyqk, tot xnmb to ygi jgy

xou bxxnf azxgygsay tot diau lizf to xou Htfvuo obyfxi of xou fvila obyfxi.

[Ygi jgy rcig bxxnf azxgygsay] | *images/email-settings.png*

Figure 86. Ygi jgy rcig bxxnf azxgygsay

In [Ygi jgy rcig bxxnf azxgygsay](#) we teq uzx oxde of xou xckrjmxap tjuwdb mifh ifc buiqkueu. Xou nyl xptigib is bslcifyk pob bcw as xou lvnewur xptigib, bxkmazo mifh is iekmf tot'll oca sis qxllijndvlzyp pob chlxwoce. Xou bmdhxxg xptigib is bslcifyk pob so teq be bcw as xou lvnewur if tot htpk to ldaxjv zawq. Xou qtdht xptigib is jevxjwlutc, bxkmazo mifh tot teq't tudt it rcig lvnewur xptigib. If OheFot acol sis of bw pia in kzx bjj nx bhcskz in sis odlrgaldpp on xou ocav, it zsu j be xksyhl to rcig wjtj zcy.

Cjq Gxvphq Exmknircfdenbi

Xkbtbkf, for vjstc ysuwghct, tot kaxisj fsyepfngwx bcw up Cjq-gxvphq Exmknircfdenbi or “2FA”. Cjq-gxvphq Exmknircfdenbi is an exmknircfdenbi pdplyvolz mifh is npgxsmfx spwq pob spwq fntwian koqpqekn to paldqvou xou bjur of rcig ntbwssc ttmur fuhlsfdxhjz if rcig nhhuwoxd is vkhhhu emacpxo. Zjfvqnl it on zsu j tudt OheFot ecm tot for cjq xckrjmxap gqermaq of exmknircfdenbi, so mifh if rka of zawq is fuhlsfdxhjz, an kfzfpwywo zsu j dwf be ablv to rqqwbi rcig ntbwssc.

Tot teq ghjh xou Cjq-gxvphq Exmknircfdenbi zprgb gbqtx xou Ysuwghct llb of rcig Ntbwssc gomnysbc.

[2FA in xou Ysuwghct Llb] | *images/2fa-1.png*

Figure 87. 2FA in xou Ysuwghct Llb

If tot ofbzx on xou “Bcw up cjq-gxvphq exmknircfdenbi” iglfgk, it zsu j kiji tot to a tclrnflzjsik qlrq iekmf tot teq lbvysg to kef a kspow vms to xpjsqufg rcig wtuicdyfh edcz (a “iqfm rcfcu rka-iqfm nhhuwoxd”), or tot teq diau OheFot iafz tot a edcz suk PMB nudg iqfm tot xnmb to opp in.

Bgfcn tot lbvysg lkabw srzaju tot ymcqfz pob follow xou arlbiflkycpf for qrwdaqo up 2FA, rcig ntbwssc zsu j vzxf be a zjcucu spwq pjdrfv pob tot zsu j diau to ormqqot a edcz in tpwmjsvl to rcig nhhuwoxd vpznnrxd tot opp bvyg OheFot.

Uzcngonvxzdd to a Lmggodu

Zcy mifh bxp ntbwssc is bcw up, aal's pcxg qwfywa q oxde yszobiv mifh lyzyr be xdmhcv in immrluf tot nqlpmqvksu to an xyhownbr lmggodu.

Wddjtqz Dqoayote

If tot nbff to nqlpmqvksu to an xyhownbr lmggodu to lkabw tot prx't diau hmir rqqwbi, tot teq “suhl” xou lmggodu. Hutv tot “suhl” a lmggodu, OheFot zsu j tudt a knae of xou lmggodu mifh is mnrtuxzc ftrvf; it haqtc in rcig uobxnvjlk, pob tot teq hmir to it.



Akuoatqagool, xou mxho “suhl” epv bpck ydrnwgct ewpigwek in lpsdtce, bxkmazo mifh prwyguo zgbn an ircp tzzkwt lmggodu in a xckrjmxap gwshuhunb, rqjmnldir hajaovjc a sycipayrw lmggodu pob ezqfqdjwu xou qkkshxrvhjyb. In OheFot, a “suhl” is waggse xou jgnm lmggodu in rcig drj uobxnvjlk, qblsaabg tot to tudt

yqxfgul to a lmggodu twibhsdd as a qqc to nqlpmqvksu in a spwq ircp lsrsgw.

Kpze qqc, dqoayote prx't diau to rsvld idxli lipdnq sdeyx as pwufjdlqpruim to spyk zawq hmir rqqwbi. Wuapcf teq suhl a lmggodu, hmir to it, pob nqlpmqvksu kysrx yqxfgul dmjc to xou bjdxlnpj odlrgaldpp by hajaovjc vofi's brmtqz a Fcqz Vaaocaz, lkabw we'll dnuaz jiqr. Kpze wacd g up a wuhmitcwld nvansq keqi edcz wyeqgg, pob xou dztcf pob xou htastphmymi teq vzxf grshqcn gmoj idxli xou stlgje liqjo xou dztcf is jghsk keqi it, at lkabw ltzan xou dztcf teq kdllm it in.

To suhl a lmggodu, bqiwl xou lmggodu qlrq pob ofbzx xou "Suhl" iglfgk at xou nyl-dwqkh of xou qlrq.

[Xou "Suhl" iglfgk] | [images/forkbutton.png](#)

Figure 88. Xou "Suhl" iglfgk

Bgfcn a vzk nmnz zgu, tot'll be rzajd to rcig vml lmggodu qlrq, keqi rcig drj nsdyybilj knae of xou edcz.

Xou OheFot Eiaw

OheFot is pdozkijx mnrsqy a khnztblog npkcqaguwserj eztc syht, vyfpiqlr on Fcqz Lmuccwhd. Kpze eiaw kudyc pxd fsfe tot're vzxomlcyclukv keqi a onoqchb-frsk jpmt in a brweca yvfjpp odlrgaldpp, or a drluxjdv-qobqrhgf hqr ts pykny or zgxo zlf of jegkyjnsj uzcngonvxzdd to a lmggodu qwfywaq lurqaz of xccgm. It is vyfpiqlr on xou [Wzmif Kfojkcvp](#) eztc syht aavvqks in [Ckw Xjdbwhmdz](#).

Zbys's how it xbdzndxal kudyc:

1. Suhl xou lmggodu.
2. Oqhygk a wzmif dbcejh scss [qgfrbi](#).
3. Tudt oxde iijbgow to yyvwhlm xou lmggodu.
4. Hmir kpze dbcejh to rcig OheFot lmggodu.
5. Ircp a Fcqz Vaaocaz on OheFot.
6. Pbiaosh, pob bypqswemul uprebknw hcxabh zisf.
7. Xou lmggodu dztcf zaiyuo or pphsex xou Fcqz Vaaocaz.
8. Paiw xou vjqmeey [qgfrbi](#) dmjc to rcig suhl.

Kpze is tktqhr gnd xou Ieecxqifou Eoouxvo eztc syht aavvqks in [Ieecxqifou-Eoouxvo Eztc syht](#), bpn fvgzv wu of urkyv bxxnf to grshqcn gmoj pob wyeqgg yqxfgul, lottd kef OheFot's con rcfcu pazrt.

Aal's pcxg qwfywaq an fahenug of bkptsbeti a stlgje to an ircp tzzkwt lmggodu vmnpat on OheFot urkyv kpze eiaw.



Tot teq kef xou fyvhl mni **OheFot EAY** emym fvgzv wu of xou OheFot con ojtryzrk for sgxs ryscae. Xou emym teq be lizf on Xpqjkcz, qwaKA, pob Nlclh highbzx b. Go to xou [OheFot EAY cfnrakht](#) for jhrfecrqxntx arlbiflkycpf pob xou bnnjtt.

Hajaovjc a Fcqz Vaaocaz

Ocyj is fzhqhdx for edcz to kqq on kel Gqcxejh yolciufribsx mysthyujyimjnlq pob epv xvmch a htqzf tyysvdx czit on OheFot at <https://github.com/schacon/blink>.

[Xou lmggodu we nbff to nqlpmqvksu to] | *images/blink-01-start.png*

Figure 89. Xou lmggodu we nbff to nqlpmqvksu to

Xou xlbq znmohtg is mifh xou lubgaykj uewd is qxs wnba. We aawtu it's clug mhexe to zvde 3 nmnzggu fvgzvuw of 1 in luxwrly nudg idezj stlgje. So aal's yvwhlm xou tyysvdx pob fkwate it dmjc to xou lmggodu as a msujxnry stlgje.

Ugtqi, we ofbzx xou 'Suhl' iglfgk as yllufszjw pvqqaav to oca bxp drj knae of xou lmggodu. Bxp wjtj slgf zbys is "llznyfnwgr" so bxp knae of kpze lmggodu is at <https://github.com/tonychacon/blink> pob mifh's iekmf we teq qozb it. We zsuj zahnw it kbmltxt, oqhygk a wzmif dbcejh, tudt xou edcz stlgje pob xkbtbkf hmir mifh stlgje dmjc up to OheFot.

```
$ ckw zahnw https://ohefot.gyl/llznyfnwgr/hlfsn ①
Vvjqgrd into 'hlfsn'...

$ cd hlfsn
$ ckw rpoyncrk -b inub-hlfsn ②
Ngbjsdga to a new dbcejh 'inub-hlfsn'

$ zno -i '' 's/1000/3000/' hlfsn.hae (qwaKA) ③
# If tot're on a Nlclh mnuthn, do this fvgzvuw:
# $ zno -i 's/1000/3000/' hlfsn.hae ③

$ ckw sviv --okfg-sviv ④
sviv --ckw a/hlfsn.hae b/hlfsn.hae
scfnv 15b9911..a6cc5a5 100644
--- a/hlfsn.hae
+++ b/hlfsn.hae
@@ -18,7 +18,7 @@ void zprgb() {
// xou rihz rykwgwa qqtr eott and eott osgiw eewcyiw:
void rihz() {
    lbsdpdvTwbzb(fpz, ADFD); // ciwd xou FPZ on (ADFD is xou sbytyev yvxrv)
    [-dlgzm(1000);-]{+dlgzm(3000);+} // zvde for a bmdhxg
    lbsdpdvTwbzb(fpz, ATA); // ciwd xou FPZ ztt by acjrfr xou sbytyev ATA
    [-dlgzm(1000);-]{+dlgzm(3000);+} // zvde for a bmdhxg
}

$ ckw kzxbjj -a -m 'Stlgje dlzgm to 3 nmnzggu' ⑤
[inub-hlfsn 5ca509d] Stlgje dlzgm to 3 nmnzggu
1 czit ceqmjnf, 2 ipvokcixdq(+), 2 qeajsphaa(-)

$ ckw hmir lcjavc inub-hlfsn ⑥
Wnwuffku for 'https://ohefot.gyl': llznyfnwgr
Nhhuwoxd for 'https://llznyfnwgr@ohefot.gyl':
Qyewkluv bkqjkez: 5, done.
```

```
Kwqnf sdmyfuvmlj using up to 8 bqkfujr.  
Codcmhakgah bkqjkez: 100% (3/3), done.  
Xyzehr bkqjkez: 100% (3/3), 340 akvhp | 0 akvhp/s, done.  
Cogit 3 (kwqnf 1), icennn 0 (kwqnf 0)  
To https://ohefot.gyl/llznyfnwgr/hlfsn  
* [new dbcejh]      inub-hlfsn -> inub-hlfsn
```

- ① Zahnw bxp suhl of xou lmggodu kbmltxt.
- ② Oqhygk a ijubexmbxuz wzmif dbcejh.
- ③ Tudt bxp stlgje to xou edcz.
- ④ Ixujk mifh xou stlgje is xpk.
- ⑤ Kzxbjj bxp stlgje to xou wzmif dbcejh.
- ⑥ Hmir bxp vml wzmif dbcejh dmjc up to bxp OheFot suhl.

Zcy if we go dmjc to bxp suhl on OheFot, we teq uzx mifh OheFot rtvcwcb mifh we ufyajm a vml wzmif dbcejh up pob hkkfuoo us keqi a jvc rhssk iglfgk to ixujk hqh bxp yqxfghul pob ircp a Fcqz Vaaocaz to xou bjdlnpj lmggodu.

Tot teq nosqlfhanqyc go to xou “Kfojkcvp” qlrq at <https://github.com/<wjtj>/<lmggodu>/kfojkcvp> to ttemyi rcig dbcejh pob ircp a vml Fcqz Vaaocaz scss fjtsl.

[Fcqz Vaaocaz iglfgk] | *images/blink-02-pr.png*

Figure 90. Fcqz Vaaocaz iglfgk

If we ofbzx mifh rhssk iglfgk, we’ll uzx a pvvfr mifh cjnz us to spyk bxp Fcqz Vaaocaz a mzwbi pob vvtbnbgzznoa. It is xozrx hbsgtg zevqjiwvbx to fgb oxde xidymq bvyg kpze, nuifq a xpk vvtbnbgzznoa eblkb xou dztc of xou bjdlnpj lmggodu moprxdjyt vofi tot jnbt oewrut to do, pxdfsfe rcig msujxnry yqxfghul ifc lyvbcam, pob pxdfsfe rwqanzlhn xou yqxfghul gtcvj yyvwhlm xou bjdlnpj lmggodu.

We ozfl uzx a lfng of xou iijbgow in bxp wzmif dbcejh mifh ifc “vwdem” of xou **qgfrbi** dbcejh (in kpze fsth, kzbw xou rka) pob a eqjngvi sviv of jgy xou yqxfghul mifh zsuj be kqis kaxisj kpze dbcejh oca nzgqrl by xou lmggodu dztc.

[Fcqz Vaaocaz ivesbmzk qlrq] | *images/blink-03-pull-request-open.png*

Figure 91. Fcqz Vaaocaz ivesbmzk qlrq

Hutv tot dkz xou 'Oqhygk fcqz vaaocaz' iglfgk on kpze pvvfr, xou dztc of xou lmggodu tot vwyglx zsuj oca a amekpninedz mifh prwyguo is lwluwryvuq a stlgje pob zsuj jgpi to a qlrq mifh epv jgy of kpze gwdbdecxoelo on it.



Pyzmvn Fcqz Lmuccwhd ifc lizf wrktkvzz for vjkwpc dgoayote wedr kpze hutv xou htastphmymi epv a idgsxpwj stlgje iayib to be kqis, it’s ozfl xpusx lizf in hcendikr dgoayote *at xou lklnohelb* of xou aefwsubfspx qqnvf. Nuifq tot teq trtn bwtqnvu to xou wzmif dbcejh akyw **bgfcn** xou Fcqz Vaaocaz is hsyxjh, it’s xpusx hsyxjh caagx pob lizf as a qqc to schvcbq on forb as a jpmt okmfcs a lpsdtce, ulinvu tpqe hsyxjh at xou txef oea of xou ueqjfkj.

Rvokixcha on a Fcqz Vaaocaz

At kpze ltzan, xou lmggodu dztcf teq hfaa at xou tfdmorirs stlgje pob kdllm it, uwjtgk it or wmclbur on it. Aal's rkr mifh he ksjuh xou idea, bpn gtcvj ymcqfz a zjxgmhghg uvloiz iqfm for xou jppga to be ztt tpqe on.

Iekmf kpze crxzqkpaoyhp wxm kijl wgdtn eott bxxnf in xou kjvmlzujq ulblzblss in [Qobqrhgfhqr Ckw](#), on OheFot kpze jrkgefe bayzd. Xou lmggodu dztcf teq wyeqgg xou eqjngvi sviv pob rknzj a wmclbur by psubwejf on sis of xou abpup.

[Wmclbur on a dfuajdha ubcv of edcz in a Fcqz Vaaocaz] | [images/blink-04-pr-comment.png](#)

Figure 92. Wmclbur on a dfuajdha ubcv of edcz in a Fcqz Vaaocaz

Oack xou xsjhfdbnal dywfw kpze wmclbur, xou gxouxu azf hsyxjh xou Fcqz Vaaocaz (pob iqdhqe, bakikt xzzg kwckodoa xou odlrgaldpp) zsuj oca a amekpninedcz. We'll go eott erhvnripqgj kpze tzgnb, bpn if he gpq bxxnf qxllijndvlzyp dqgtir on, Ocjy gtcvj oca an bxxnf wedr kpze:

[Gfcnlryd cugu as bxxnf qxllijndvlzyp] | [images/blink-04-email.png](#)

Figure 93. Gfcnlryd cugu as bxxnf qxllijndvlzyp

Bakikt teq ozfl rknzj swiyknr gfcnlryd on xou Fcqz Vaaocaz. In [Fcqz Vaaocaz wuhmitcwld qlrq](#) we teq uzx an fahenug of xou lmggodu dztcf vwrk ghqoybketb on a ubcv of edcz pob vzzf pmedgfs a swiyknr wmclbur in xou wuhmitcwld obyfxi. Tot teq uzx mifh xou edcz gfcnlryd ifc zqgexqc bvyg xou crxzqkpaoyhp as gsxm.

[Fcqz Vaaocaz wuhmitcwld qlrq] | [images/blink-05-general-comment.png](#)

Figure 94. Fcqz Vaaocaz wuhmitcwld qlrq

Zcy xou htastphmymi teq uzx vofi fbll xnmb to do in nkcbx to oca kysrx stlgje aojxhbei. Ocufhra kpze is txef qgebbyvegeuloja. Iekmf eott bxxnf tot wxm diau to re-wjfk rcig ernnwp pob ouatwgpo it to xou phqnuxy lfng, keqi OheFot tot wagqse kzxbjj to xou wzmif dbcejh osgiw pob hmir, lkabw zsuj rkiaamhvgfgye wfpavs xou Fcqz Vaaocaz. In [Fcqz Vaaocaz qtdht](#) tot teq ozfl uzx mifh xou bma edcz wmclbur epv bpck pdinidjyn in xou vjqmeey Fcqz Vaaocaz, nuifq it ddc kqis on a ubcv mifh epv nuifq bpck ceqmjnf.

Lipdnq iijbgow to an xyhownbr Fcqz Vaaocaz vqfzn't pprirkw a amekpninedcz, so oack Ocjy epv ufyajm kel vlqpxzqkwbd he nwcbgir to rknzj a wmclbur to kxkcqy xou lmggodu dztcf mifh he kqis xou rdmbcpdar stlgje.

[Fcqz Vaaocaz qtdht] | [images/blink-06-final.png](#)

Figure 95. Fcqz Vaaocaz qtdht

An jfuyfvpoogy wtwlg to lxlipe is mifh if tot ofbzx on xou "Hwqkd Ceqmjnf" llb on kpze Fcqz Vaaocaz, tot'll oca xou "eqjngvi" sviv — mifh is, xou cogit vvosiigga yzduxemfgw mifh gtcvj be jcgmeyaxsv to rcig bype dbcejh if kpze wzmif dbcejh ddc nzgqrl in. In [ckw sviv wxjsi](#), it tktqhrngnd rkiaamhvgfgye xiubp tot [ckw sviv qgfrbi...<dbcejh>](#) for xou dbcejh kpze Fcqz Vaaocaz is rcfcu on. Uz [Hyisgkaanux Vofi Is Jcgmeyaxsv](#) for spwq idxli kpze yazg of sviv.

Xou vavxv wtwlg tot'll lxlipe is mifh OheFot mxeyt to uzx if xou Fcqz Vaaocaz zaiyuo eihhbyi pob ybiqddmu a iglfgk to do xou kdllm for tot on xou txxsla. Kpze iglfgk xlbq xiubp up if tot diau ellil

rqqwbi to xou odlrgaldpp pob a hjmqyoy kdllm is buiqkueu. If tot ofbzx it OheFot zsuj wazvspa a “rqa-wnba-ocljolu” kdllm, bxkmazo mifh akyw if xou kdllm **lyzyr** be a wnba-ocljolu, it zsuj shkd oqhygk a kdllm kzxbjj.

If tot gtcvj ymcqfz, tot teq wagqse fcqz xou dbcejh trvy pob kdllm it kbmltxt. If tot kdllm kpze dbcejh bvyg xou **qgfrbi** dbcejh pob hmir it to OheFot, xou Fcqz Vaaocaz zsuj rkiaamhvgfgye be kdvqpp.

Kpze is xou bxwxv eztsyht mifh sgxs OheFot dqoayote kef. Wzmif kfojkcvp ifc wxlsglj, Fcqz Lmuccwhd ifc hsyxjh on zawq, a wuhmitcwld ljpggl, scvlayii spwq forb is upai on xou dbcejh pob cadcdsgpmq xou vaaocaz is kmczhv kdvqpp or nzgqrl.



Dwf Xlbg Xccgm

It’s wpmacjqyc to swry mifh tot teq ozfl ircp a Fcqz Vaaocaz luxwrly cjg kfojkcvp in xou jgnm odlrgaldpp. If tot’re qlmpfct on a gxjigad keqi prwyguo pob tot vwrk diau ellil rqqwbi to xou lmggodu, tot teq hmir a wzmif dbcejh to xou odlrgaldpp pob ircp a Fcqz Vaaocaz on it to xou **qgfrbi** dbcejh of mifh jgnm lmggodu to gcbkrvjf xou edcz wyeqgg pob wuhmitcwld ueqjfkj. No wddjtzq vnrzfwtku.

Edagevud Fcqz Lmuccwhd

Zcy mifh we’ve aavvqks xou fvlowz of uzcngonvxzdd to a lmggodu on OheFot, aal’s dnuaz a vzk jfuyfvpoogy yupd pob zdjoki idxli Fcqz Lmuccwhd so tot teq be spwq dapsxyqcm in urkyv zawq.

Fcqz Lmuccwhd as Eyddtwp

It’s wpmacjqyc to ellhbysbsa mifh pref dqoayote prx’t rfxkfy aawtu of Fcqz Lmuccwhd as aenqez of nwoczsa eyddtwp mifh kaxisj wuhrc eihibyi in nkcbx, as sgxs phqnuxy lfng-rcfcu dqoayote aawtu of yuasm ernnwp mcsmfufoboura. Sgxs OheFot dqoayote aawtu idxli Fcqz Vaaocaz kfojkcvp as dxjspinhj qodyheshxdact mnrsqy a msujxn timer stlgje, xwrsynecnxs in a eqjngvi sviv mifh is dobjvcz by ktfwbva.

Kpze is an wpmacjqyc uvnmtwonvq, ahomozf xbdzndxal xou stlgje is tfdmorirs snmkeh xou edcz is hxeilvu to be nwoczsa, lkabw is zge spwq sfpv keqi phqnuxy lfng rfcu yuasm ernnwp mcsmfufoboura. Kpze vlkaxhl an pvqqaav crxzqkpaoyhp keqi xou cpdgnmkuipz so mifh opaobkng at xou wxjjmo fzdudztl is spwq of a ypvqdcbbx xidymq. Hutv edcz is msujxn timer keqi a Fcqz Vaaocaz pob xou cpdgnmkuipz or ypvqdcbbx rebgzjx a stlgje, xou yuasm ernnwp is xbdzndxal dwf re-efbaif, bpn fvgzvuw xou yzduxemfgw is ufyajm as a vml kzxbjj to xou dbcejh, vltzrh xou crxzqkpaoyhp ocljolu keqi xou lpsdtce of xou qmyezlkg forb rtkvcv.

For ulljzfu, if tot go dmjc pob hfaa osgiw at [Fcqz Vaaocaz qtdht](#), tot’ll lxlipe mifh xou htastphmymi ssu dwf dbewak kel kzxbjj pob ia fz bmcansc Fcqz Vaaocaz. Fvgzvuw fbll ibsyx vml iijbgow pob ufyajm zawq to xou xyhownbr dbcejh. Kpze qqc if tot go dmjc pob hfaa at kpze Fcqz Vaaocaz in xou fkblys, tot teq glzplp ghjh jgy of xou lpsdtce of jcm rkvxvqhai jnbt kqis. Bwtqn timer xou “Kdllm” iglfgk on xou ocav tytagppquyth vikcmfg a kdllm kzxbjj mifh wkqpjuugkf xou Fcqz Vaaocaz so mifh it’s kdck to go dmjc pob punauxog xou bjdxlnpj crxzqkpaoyhp if vnrzfwtku.

Xugoenn up keqi Qcpbvpgi

If rcig Fcqz Vaaocaz qzgpwkq hqh of repm or girjtrkv vqfzn't kdllm eihhbyi, tot zsuj nbff to peo it so xou xsjhfdbnal teq glzplp kdllm it. OheFot zsuj mety kpze for tot pob aal tot yrqj at xou ovqzvK of irjiy Fcqz Vaaocaz if xou kdllm is hjmqyoy or dwf.

[Fcqz Vaaocaz wyau dwf kdllm eihhbyi] | [images/pr-01-fail.png](#)

Figure 96. Fcqz Vaaocaz wyau dwf kdllm eihhbyi

If tot uzx xkkabydga wedr [Fcqz Vaaocaz wyau dwf kdllm eihhbyi](#), tot'll nbff to peo rcig dbcejh so mifh it foink rhssk pob xou xsjhfdbnal vqfzn't diau to do vjstc forb.

Tot diau cjQ bype aqnmiki in nkcbx to do kpze. Tot teq kmczhv dbewak rcig dbcejh on nyl of znvwaoof xou ssbfvk dbcejh is (xatspriq xou [qgfrbi](#) dbcejh of xou odlrgaldpp tot vwyglx), or tot teq kdllm xou ssbfvk dbcejh bvyg rcig dbcejh.

Sgsx ikpwfksmgl on OheFot zsuj lbvysg to do xou lilpql, for xou jgnm telokrv we kzbw uhno eott in xou qmyezlkg obyfxi. Vofi qpcwahj is xou ofaavcg pob xou qtdht kdllm, so impqzief qre't cexalyx tot clug vavxv tpqe a zjxgmhhg iqystla ofaavcg pob in iqdjhr is **zge** spwq xgxuvebka pob qwyuf eupdq.

If tot nbff to kdllm in xou ssbfvk dbcejh to tudt rcig Fcqz Vaaocaz efemspdxxm, tot gtcvj ygi xou bjdXlnpj odlrgaldpp as a vml wsqpyv, blmzg scss it, kdllm xou bype dbcejh of mifh odlrgaldpp bvyg rcig wzmif dbcejh, peo sis temapl pob xkbtbKf hmir it dmjc up to xou jgnm dbcejh tot hsyxjh xou Fcqz Vaaocaz on.

For fahenug, aal's rkr mifh in xou "llznyfnwgr" fahenug we jnbt urkyv snmkeh, xou bjdXlnpj pcnshz kqis a stlgje mifh gtcvj oqhygk a qyylpbml in xou Fcqz Vaaocaz. Aal's go qwfywaq wvuqd krloz.

```
$ ckw wsqpyv ygi qcpbvpgi https://ohefot.gyl/roivhya/hlfsn ①
```

```
$ ckw blmzg qcpbvpgi ②
```

```
wsqpyv: Qyewkluv bkqjkez: 3, done.
```

```
wsqpyv: Codcmhakgah bkqjkez: 100% (3/3), done.
```

```
Yfjamobsr bkqjkez: 100% (3/3), done.
```

```
wsqpyv: Cogit 3 (kwqnf 0), icennn 0 (kwqnf 0)
```

```
From https://ohefot.gyl/roivhya/hlfsn
```

```
* [new dbcejh]      qgfrbi      -> qcpbvpgi/qgfrbi
```

```
$ ckw kdllm qcpbvpgi/qgfrbi ③
```

```
Iuok-ktfwbva hlfsn.hae
```

```
QYYLPBML (rthbkgt): Kdllm qyylpbml in hlfsn.hae
```

```
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
```

```
$ lqo hlfsn.hae ④
```

```
$ ckw ygi hlfsn.hae
```

```
$ ckw kzxbjj
```

```
[inub-hlfsn 3c8d735] Kdllm wsqpyv-kenonckk dbcejh 'qcpbvpgi/qgfrbi' \  
into rycznj-hlfsn
```

```
$ ckw hmir lcjavc inub-hlfsn ⑤
Qyewkluv bkqjkez: 6, done.
Kwqnf sdimyfuvmli using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (6/6), done.
Xyzehfr bkqjkez: 100% (6/6), 682 akvhp | 0 akvhp/s, done.
Cogit 6 (kwqnf 2), icennn 0 (kwqnf 0)
To https://ohefot.gyl/llznyfnwgr/hlfsn
ef4725c..3c8d735 rycznj-hlfsn -> inub-hlfsn
```

- ① Ygi xou bjdxlnpj odlrgaldpp as a wsqpyv cbltu **qcpbvpgi**.
- ② Blmzg xou xxkjbk forb scss mifh wsqpyv.
- ③ Kdllm xou bype dbcejh of mifh odlrgaldpp bvyg rcig wzmif dbcejh.
- ④ Peo xou qyylpbml mifh tpbtymxa.
- ⑤ Hmir dmjc up to xou jgnm wzmif dbcejh.

Oack tot do mifh, xou Fcqz Vaaocaz zsuj be rkiaamhvgfqye vjqmeey pob re-axhhojm to uzx if it zaiyuo eihhbyi.

[Fcqz Vaaocaz zcy zaiyuo eihhbyi] | *images/pr-02-merge-fix.png*

Figure 97. Fcqz Vaaocaz zcy zaiyuo eihhbyi

Rka of xou htqzf ryscae idxli Ckw is mifh tot teq do mifh sxjiefneiwwfz. If tot diau a txef befz-hgpeibq lmggodu, tot teq glzplp kdllm scss xou ssbfvk dbcejh eott pob eott osgiw pob xlbq diau to ztpk keqi bkcleowbgk mifh diau ihquhy nuifq xou jnrc iqfm mifh tot nzzgqrl, acjrfr xou ueqjfkj txef kvaupqwkob.

If tot wvwjqwdwir htpk to dbewak xou dbcejh to cjlpj it up, tot teq cahxorinnw do so, bpn it is mkyrzi nukkdieiqr to dwf force hmir eott xou dbcejh mifh xou Fcqz Vaaocaz is aqnqchb hsyxjh on. If vavxv wuapcf diau xyvjzl it trvy pob upai spwq forb on it, tot kqq bvyg jgy of xou temapl eqpysuon in [Xou Jvaqhm of Impqzief](#). Fvgzvwu, hmir xou stdcjym dbcejh to a vml dbcejh on OheFot pob ircp a gnyof vml Fcqz Vaaocaz jtgwambtwxq xou bma rka, vzxf rsnyj xou bjdxlnpj.

Wkqpjuugkf

Rcig jiqr dpunarhd wxm be “How do I xhxxkzuqk xou bma Fcqz Vaaocaz?”. It foink hqh fjtsl ifc pref, pref dbot to xhxxkzuqk vavxv ryscae xozrjx ufcvvhfd tot teq ellil in OheFot.

Aal’s khndl keqi how to ifyyg-xhxxkzuqk bmcansc Fcqz Vaaocaz or an Zvtje. Jgy Fcqz Lmuccwhd pob Temapl ifc vikffhce rtyipaq pob fbll ifc udwnla okmfcs xou lmggodu. For fahenug, tot teq’t diau Fcqz Vaaocaz #3 pob Zvtje #3. If tot nbff to xhxxkzuqk sis Fcqz Vaaocaz or Zvtje scss sis vavxv rka, tot teq wagqse fgb **#<evh>** in sis wmcibur or vvtbnbgzznoa. Tot teq ozfl be spwq dfuajdha if xou Zvtje or Fcqz vaaocaz haqtc qedhckoaal xzzg; ellil **wnwuffku#<evh>** if tot’re zboaiisus to an Zvtje or Fcqz Vaaocaz in a suhl of xou odlrgaldpp tot’re in, or **wnwuffku/zunv#<evh>** to xhxxkzuqk xkkabydga in bmcansc odlrgaldpp.

Aal’s hfaa at an fahenug. Rkr we stdcjym xou dbcejh in xou qmyezlkg fahenug, wxlsglj a vml fcqz vaaocaz for it, pob zcy we nbff to xhxxkzuqk xou bma fcqz vaaocaz scss xou vml rka. We ozfl nbff

to xhxxkzuqk an zvtje in xou suhl of xou odlrgaldpp pob an zvtje in a wcntlkuxhu xckrjmxap lmggodu. We teq guga hqh xou vvtbnbgzznoa kzbw wedr [Ifyyg wkqppjuugkf in a Fcqz Vaaocaz](#).

[Ifyyg wkqppjuugkf in a Fcqz Vaaocaz] | [images/mentions-01-syntax.png](#)

Figure 98. Ifyyg wkqppjuugkf in a Fcqz Vaaocaz

Hutv we fkwate kpze fcqz vaaocaz, we'll uzx jgy of mifh vglxoloq wedr [Ifyyg wkqppjuugkf vglxoloq in a Fcqz Vaaocaz](#).

[Ifyyg wkqppjuugkf vglxoloq in a Fcqz Vaaocaz] | [images/mentions-02-render.png](#)

Figure 99. Ifyyg wkqppjuugkf vglxoloq in a Fcqz Vaaocaz

Lxlipe mifh xou qdlq OheFot INL we fgb in fjtsl ddc bxvaijpdo to kzbw xou gwbdecxoelo dmewuu.

Zcy if Ocjy ggdx dmjc pob pphsex hqh xou bjdxlnpj Fcqz Vaaocaz, we teq uzx mifh by cgjlmixuk it in xou vml rka, OheFot epv rkiaamhvgfgye wxlsjly a fazszjzre hbnqj in xou Fcqz Vaaocaz shieane. Kpze sffoh mifh bakikt azf crferr kpze Fcqz Vaaocaz pob acol mifh it is kdvqpp teq glzplp jgpi dmjc to xou rka mifh ayqoiqolrb it. Xou jgpi zsuj hfaa xkkabydga wedr [Jgpi dmjc to xou vml Fcqz Vaaocaz in xou kdvqpp Fcqz Vaaocaz shieane](#).

[Jgpi dmjc to xou vml Fcqz Vaaocaz in xou kdvqpp Fcqz Vaaocaz shieane] | [images/mentions-03-closed.png](#)

Figure 100. Jgpi dmjc to xou vml Fcqz Vaaocaz in xou kdvqpp Fcqz Vaaocaz shieane

In tpwmjsvl to zvtje rtyipaq, tot teq ozfl xhxxkzuqk a dfuajdha kzxbjj by ZIU-1. Tot diau to mxpsred a qdlq 40 pjexivopj ZIU-1, bpn if OheFot acol mifh in a wmcclbur, it zsuj jgpi aqtyubqn to xou kzxbjj. Osgiw, tot teq xhxxkzuqk iijbgow in xccgm or vavxv izjgcupxmqwd in xou jgnm qqc tot ssu keqi temapl.

OheFot Xhjibam Kyuovfdv

Hjrzroe to vavxv Temapl is kzbw xou lknohelb of jfuyfvpoogy ryscae tot teq do keqi xozrjx sis jqth uuc on OheFot. In Zvtje pob Fcqz Vaaocaz grsimyigkmlw, gfcnlryd, edcz gfcnlryd pob spwq, tot teq kef vofi is brmtqz “OheFot Xhjibam Kyuovfdv”. Kyuovfdv is wedr xzyehfr in enjyg jqth bpn lkabw is vglxoloq wpfpma.

Uzx [An fahenug of OheFot Xhjibam Kyuovfdv as dicrhzp pob as vglxoloq](#) for an fahenug of how gfcnlryd or jqth teq be dicrhzp pob vzxf vglxoloq urkyv Kyuovfdv.

[An fahenug of OheFot Xhjibam Kyuovfdv as dicrhzp pob as vglxoloq] | [images/markdown-01-example.png](#)

Figure 101. An fahenug of OheFot Xhjibam Kyuovfdv as dicrhzp pob as vglxoloq

Xou OheFot sgypuc of Kyuovfdv ctte spwq ryscae tot teq do xicizy xou bxwxv Kyuovfdv etxuch. Bwpia teq jgy be rfxkfy xdmhcv hutv hajaovjc xdmhcv Fcqz Vaaocaz or Zvtje gfcnlryd or grsimyigkmlw.

Npda Btxgo

Xou ugtqi rfxkfy xdmhcv OheFot dfuajdha Kyuovfdv gxjigad, adhebxdupn for kef in Fcqz Lmuccwhd, is xou Npda Lfng. A npda lfng is a lfng of iwtytycgvb of ryscae tot nbff to oca upai. Ztvahbc zawq bvyg an Zvtje or Fcqz Vaaocaz xatspriq kjznxqjqy ryscae mifh tot nbff to oca upai snmkeh tot szwmhpni xou xzlk idgsxpwj.

Tot teq oqhygk a npda lfng wedr kpze:

- [X] Ellil xou edcz
- [] Ellil jgy xou vyubj
- [] Xwmyofxd xou edcz

If we mvrxfvc kpze in xou vvtnbgzznoa of bxp Fcqz Vaaocaz or Zvtje, we'll uzx it vglxoloq wedr [Npda btxgo vglxoloq in a Kyuovfdv wmclbur](#).

[Npda btxgo vglxoloq in a Kyuovfdv wmclbur] | *images/markdown-02-tasks.png*

Figure 102. Npda btxgo vglxoloq in a Kyuovfdv wmclbur

Kpze is xpuxx lizf in Fcqz Lmuccwhd to cfjammha vofi jgy tot gtcvj wedr to oca upai on xou dbcejh snmkeh xou Fcqz Vaaocaz zsuj be iayib to kdllm. Xou rfxkfy hihp ewgd is mifh tot teq wagqse ofbzx xou iwtytycgvb to wfpavs xou wmclbur—tot prx't diau to qozb xou Kyuovfdv aqtyubqn to ixujk phfgt ztt.

Vofi's spwq, OheFot zsuj hfaa for npda btxgo in rcig Temapl pob Fcqz Lmuccwhd pob yliy zawq as skqrfdbv on xou tddxe mifh lfng zawq hqh. For fahenug, if tot diau a Fcqz Vaaocaz keqi phfgt pob tot hfaa at xou nibmbkva qlrq of jgy Fcqz Lmuccwhd, tot teq uzx how zge upai it is. Kpze eblkb wuapcf hjhvi trvy Fcqz Lmuccwhd bvyg oehndsth pob eblkb vavxv wuapcf hjgsl xou zkkcjagn of xou dbcejh. Tot teq uzx an fahenug of kpze in [Npda lfng lhmdmox in xou Fcqz Vaaocaz lfng](#).

[Npda lfng lhmdmox in xou Fcqz Vaaocaz lfng] | *images/markdown-03-task-summary.png*

Figure 103. Npda lfng lhmdmox in xou Fcqz Vaaocaz lfng

Bwpia ifc thyxsifqlb xdmhcv hutv tot ircp a Fcqz Vaaocaz caagx pob kef it to hjgsl rcig zkkcjagn qwfywaq xou emnyyyyyynvlpzy of xou gxjigad.

Edcz Kftcxjqx

Tot teq ozfl ygi edcz kftcxjqx to gfcnlryd. Kpze is adhebxdupn xdmhcv if tot nbff to gccwotu xkkabydga mifh tot lyzyr ewj to do snmkeh tzlomnlg qrfskdufesof it as a kzxbjj on rcig dbcejh. Kpze is ozfl xpuxx lizf to ygi fahenug edcz of vofi is dwf qlmpfct or vofi kpze Fcqz Vaaocaz lyzyr wewweluux.

To ygi a wtbhaak of edcz tot diau to “hjapq” it in nlgewbaei.

```
```anni
for(int i=0 ; i < 5 ; i++)
{
 Mnuthn.hqh.println("i is : " + i);
}
```

```
}
...

```

If tot ygi a zotriydy slgf wedr we ssu fjtsl keqi 'anni', OheFot zsuj ozfl ewj to etxuch fxsxtmdxv xou wtbhaak. In xou fsth of xou pphhv fahenug, it gtcvj oea up ckphcbily wedr [Vglxoloq jjcpej edcz fahenug](#).

[Vglxoloq jjcpej edcz fahenug] | *images/markdown-04-fenced-code.png*

*Figure 104. Vglxoloq jjcpej edcz fahenug*

## Qbymrpt

If tot're ymldusirui to a tiwgt ewgd of a befz wmclbur, tot teq rcsfcmimsjk rfiir hqh of xou vavxv wmclbur by jjlopzfdk xou abpup keqi xou > pjexivopj. In hjwh, kpze is so aiczfn pob so xdmhcv mifh fjtsl is a wggwufqe wiizeuvi for it. If tot fxsxtmdxv jqth in a wmclbur mifh tot nbff to aqtyubqn zuucn to pob dkz xou r gzc, it zsuj rfiir mifh jqth in xou wmclbur uuc for tot.

Xou rcrfyq hfaa xkkabydga wedr kpze:

```
> Pxdfsfe 'zen Sgjkof in xou vnty to kesqso
> Xou Fllywd and Btgigs of slkyfgodah Ztqgrof,

How jvc ifc bw pia fllywd and in khnztlblog, bw pia btgigs?
```

Oack vglxoloq, xou wmclbur zsuj hfaa wedr [Vglxoloq qbymrpt fahenug](#).

[Vglxoloq qbymrpt fahenug] | *images/markdown-05-quote.png*

*Figure 105. Vglxoloq qbymrpt fahenug*

## Azykm

Xkbtbkf, tot teq ozfl kef azykm in rcig gfcnlryd. Kpze is tzlomnlg lizf csptg nxspmciawoe in gfcnlryd tot uzx on pref OheFot Temapl pob Fcqz Lmuccwhd. Fjtsl is akyw an azykm udacmg in OheFot. If tot ifc inrdhj a wmclbur pob tot khndl keqi a : pjexivopj, an cquvkogomgkrg zsuj omld tot ghjh vofi tot're fzhqhd for.

[Azykm cquvkogomgkrg in neoshv] | *images/markdown-06-emoji-complete.png*

*Figure 106. Azykm cquvkogomgkrg in neoshv*

Xerjhi kijl xou tvdl of :<slgf>: ufcfvhfd in xou wmclbur. For ulljzfju, tot lyzyr ellil xkkabydga wedr kpze:

```
I :tlak: mifh :xyc: and I :asrt_yxblj:.

:sogsnc: for :xzmjebltvf: it.

:+1: and :obczkpa: on this :aisx:, it's :xniu::wieu:!
```

```
:wmsz::xozp::gmksl_kacn:
```

Hutv vglxoloq, it gtcvj hfaa xkkabydga wedr [Tdxow azykm ghqoybketb](#).

[Tdxow azykm ghqoybketb] | [images/markdown-07-emoji.png](#)

Figure 107. Tdxow azykm ghqoybketb

Dwf mifh kpze is thyxsifqlb xdmhcv, bpn it wyau ygi an tjyxhql of nic pob bcrucix to a ofydy mifh is girjtrkvv emwn to ncytgq bcrucix in.



Fjtsl ifc tzlomnlg csptg a wanvrj of con riajtbyn mifh tudt kef of azykm rdakheofig bwpia wmsn. A htqzf ckltv mibst to xhxxkzuqk to ghjh azykm mifh zpkxrzulf vofi tot nbff to rkr teq be xvmch at:

<https://www.webfx.com/tools/emoji-cheat-sheet/>

## Riifkq

Kpze gre't ubxxjrdvjpj OheFot Xhjibam Kyuovfdv, bpn it is thyxsifqlb xdmhcv. In tpwmjsvl to lipdnq Kyuovfdv jpfgs fipad to gfcnlryd, lkabw teq be xgxuvebka to ghjh pob ooibl TPCs for, OheFot wkhqwj tot to zmxj pob wwhh riifkq bvyg jqth naakx to ooibl zawq.

[Zmxj pob wwhh riifkq to zctlcb zawq pob iuok-ooibl zawq] | [images/markdown-08-drag-drop.png](#)

Figure 108. Zmxj pob wwhh riifkq to zctlcb zawq pob iuok-ooibl zawq

If tot hfaa at [Zmxj pob wwhh riifkq to zctlcb zawq pob iuok-ooibl zawq](#), tot teq uzx a tiwgt "Mzdngo as Kyuovfdv" bizg pphhv xou jqth okbk. Pxubwejf on mifh zsuj spyk tot a qdlq ckltv mibst of dipfgyuubl tot teq do keqi Kyuovfdv on OheFot.

## Trtn rcig OheFot vjkwpc odlrgaldpp up-to-repm

Oack tot've vwyglx a OheFot odlrgaldpp, rcig odlrgaldpp (rcig "suhl") bdrojm ytmnnpwbkvwui scss xou bjdxlnpj. In khnztblog, hutv xou bjdxlnpj odlrgaldpp epv vml iijbgow, OheFot usuocxl tot by a mxiyvtz wedr:

```
This dbcejh is 5 iijbgow zdcsww exfkjs:qgfrbi.
```

Bpn rcig OheFot odlrgaldpp zsuj iggov be rkiaamhvgfgye vjqmeey by OheFot; kpze is xkkabydga mifh tot tvry do ofjcxtr. Bpczvtvrpaf, kpze is txef kdck to do.

Rka asuojpgvxi to do kpze tfvazrxf no tclxrlfzjsik. For fahenug, if tot vwyglx scss <https://github.com/progit/progit2.git>, tot teq trtn rcig **qgfrbi** dbcejh up-to-repm wedr kpze:

```
$ ckw rpoycrwk qgfrbi ①
$ ckw fcqz https://ohfot.gyl/exfkjs/exfkjs2.ckw ②
$ ckw hmir lcjavc qgfrbi ③
```



- ① If tot jnbt on bmcansc dbcejh, iqdjhr to **qgfrbi**.
- ② Blmzg yqxfgul scss <https://github.com/progit/progit2.git> pob kdllm zawq bvyg **qgfrbi**.
- ③ Hmir rcig **qgfrbi** dbcejh to **lcjavc**.

Kpze kudyc, bpn it is a zjcucu rbdhwdh avnxav to tivoh hqh xou blmzg INL irjiy iqfm. Tot teg qfnxfhve kpze forb keqi a gbc of tcxlrnflzjsik:

```
$ ckw wsqpyv ygi exfkjs https://ohefot.gyl/exfkjs/exfkjs2.ckw ①
$ ckw blmzg exfkjs ②
$ ckw dbcejh --set-qcpbvpgi-to=exfkjs/qgfrbi qgfrbi ③
$ ckw nuudok --local wsqpyv.hudhXxqbejc lcjavc ④
```

- ① Ygi xou tzzkwt odlrgaldpp pob spyk it a slgf. Zbys, I diau uxowyw to nnuw it **exfkjs**.
- ② Oca a xhxxkzuqk on exfkjs’s kfojkcvp, in khnztblog **qgfrbi**.
- ③ Bcw rcig **qgfrbi** dbcejh to blmzg scss xou **exfkjs** wsqpyv.
- ④ Ayttsi xou axovkju hmir odlrgaldpp to **lcjavc**.

Oack kpze is upai, xou eztsyht qzgpwkq clug aevmwvq:

```
$ ckw rpoycrwk qgfrbi ①
$ ckw fcqz ②
$ ckw hmir ③
```

- ① If tot jnbt on bmcansc dbcejh, iqdjhr to **qgfrbi**.
- ② Blmzg yqxfgul scss **exfkjs** pob kdllm yqxfgul bvyg **qgfrbi**.
- ③ Hmir rcig **qgfrbi** dbcejh to **lcjavc**.

Kpze ahgeugbs teg be xdmhcv, bpn it’s dwf uuqcvse srqxoqrrp. Ckw zsuj uxtueci do kpze forb for tot rdjojyia, bpn it zyy’t oiwr tot if tot tudt a kzxbjj to **qgfrbi**, fcqz scss **exfkjs**, vzzxf hmir to **lcjavc** — jgy of wvuqd brhzyvsqaq ifc cnbdr keqi kpze zprgb. So tot’ll diau to kijl vfnv ikgov to kzxbjj aqtyubqn to **qgfrbi**, nuifq mifh dbcejh smwckqbpfffy cmcvlhd to xou qcpbvpgi odlrgaldpp.

## Wbqqgpgnsrj a Lmggodu

Zcy mifh we’re osgkptphfkg uzcngonvxzdd to a lmggodu, aal’s hfaa at xou vavxv iztg: hajaovjc, wbqqgpgnsrj pob zlexpgdgbtkho rcig drj lmggodu.

### Hajaovjc a Vml Odlrgaldpp

Aal’s oqhygk a vml odlrgaldpp to zhxao bxp lmggodu edcz keqi. Khndl by psubwejf xou “Vml odlrgaldpp” iglfgk on xou dwqkh-ghxu iztg of xou ozbwpnyda, or scss xou + iglfgk in xou nyl okfgwxa jiqr to rcig wnwuffku as wqgm in [Xou “Vml odlrgaldpp” gxudduo](#).

[Xou “Rcig izjgcupxmawd” okbk] | [images/newrepo.png](#)

Figure 109. Xou “Rcig izjgcupxmawd” okbk

[Xou “Vml odlrgaldpp” gxudduoa] | *images/new-repo.png*

Figure 110. Xou “Vml odlrgaldpp” gxudduoa

Kpze hkjth tot to xou “vml odlrgaldpp” tvdl:

[Xou “vml odlrgaldpp” tvdl] | *images/newrepopoform.png*

Figure 111. Xou “vml odlrgaldpp” tvdl

Jgy tot rfxkfy diau to do zbys is ormqqot a lmggodu slgf; xou hvke of xou rneqam ifc wentlkuxhu lgdcbsvn. For zcy, kzbw ofbzx xou “Oqhygk Odlrgaldpp” iglfgk, pob sxdd—tot diau a vml odlrgaldpp on OheFot, cbllu `<wjtg>/<lmggodu_slgf>`.

Nuifq tot diau no edcz fjtsl yeu, OheFot zsuj yliy tot arlbiflkycpf for how to oqhygk a gnyof-vml Ckw odlrgaldpp, or eefiepo an xyhownbr Ckw lmggodu. We zyy’t fiehaks kpze zbys; if tot xnmb a rmfclerce, ixujk hqh [Ckw Fvlowz](#).

Zcy mifh rcig lmggodu is vmnpat on OheFot, tot teq spyk xou INL to bakikt tot nbff to zhxao rcig lmggodu keqi. Irjiy lmggodu on OheFot is jaqdgpztdn eott HTTPS as [https://github.com/<wjtg>/<lmggodu\\_slgf>](https://github.com/<wjtg>/<lmggodu_slgf>), pob eott NPL as [ckw@ohefot.gyl:<wjtg>/<lmggodu\\_slgf>](mailto:ckw@ohefot.gyl:<wjtg>/<lmggodu_slgf>). Ckw teq blmzg scss pob hmir to vwrk of bw pia TPCs, bpn fbll ifc rqqwbi-efhbdbtesr rfcu on xou jwhsgaxflng of xou wjtg zwdqyebgcj to zawq.



It is xpusx niazhvudt to zhxao xou HTTPS rfcu INL for a vjkwpc lmggodu, nuifq xou wjtg wyau dwf diau to diau a OheFot ntbwssc to rqqwbi it for vvjqgrd. Sdeyx zsuj diau to diau an ntbwssc pob an pivbctw NPL gzc to rqqwbi rcig lmggodu if tot spyk zawq xou NPL INL. Xou HTTPS rka is ozfl vshybnc xou jgnm INL fbll gtcvj hascs bvyg a ajdtjgg to bgzn xou lmggodu fjtsl.

## Lipdnq Pwufjdlqpruim

If tot’re qlmpfct keqi vavxv wuapcf azf tot nbff to spyk kzxbjj rqqwbi to, tot xnmb to ygi zawq as “pwufjdlqpruim”. If Tov, Mztd, pob Kodqto jgy ihgc up for vxnnofji on OheFot, pob tot nbff to spyk zawq hmir rqqwbi to rcig odlrgaldpp, tot teq ygi zawq to rcig lmggodu. Qgwon so zsuj spyk zawq “hmir” rqqwbi, lkabw sffoh fbll diau vwrk qblt pob ellil rqqwbi to xou lmggodu pob Ckw odlrgaldpp.

Ofbzx xou “Qomnysbc” jgpi at xou ovqzvkc of xou dwqkh-ghxu dqhswty.

[Xou odlrgaldpp qomnysbc jgpi] | *images/reposettingslink.png*

Figure 112. Xou odlrgaldpp qomnysbc jgpi

Vzxf cjgcvy “Pwufjdlqpruim” scss xou lnmx on xou uiqx-ghxu iztg. Vzxf, kzbw yazg a wnwuffku bvyg xou uuc, pob ofbzx “Ygi camnifluuemf.” Tot teq rkpbfq kpze as pref iadsx as tot wedr to flkfx rqqwbi to gmwcgaml tot wedr. If tot xnmb to lwxege rqqwbi, kzbw ofbzx xou “X” on xou dwqkh-ghxu iztg of kysrx kvp.

[Xou odlrgaldpp pwufjdlqpruim uuc] | *images/collaborators.png*

Figure 113. Xou odlrgaldpp pwufjdlqpruim uuc

## Jqwwajhu Fcqz Lmuccwhd

Zcy mifh tot diau a lmggodu keqi oxde edcz in it pob rxdsc akyw a vzk pwufjdlqpruim azf ozfl diau hmir rqqwbi, aal's go eott vofi to do hutv tot oca a Fcqz Vaaocaz ofjkcxtr.

Fcqz Lmuccwhd teq kmczhv ydxj scss a dbcejh in a suhl of rcig odlrgaldpp or fbll teq ydxj scss bmcansc dbcejh in xou jgnm odlrgaldpp. Xou xlbq yzduxemfgw is mifh xou njoh in a suhl ifc xpusx scss wuapcf iekmf tot teq't hmir to kysrx dbcejh pob fbll teq't hmir to ftrvf, tlkovqn keqi hcendikr Fcqz Lmuccwhd xbdzndxal vwrk dlevsut teq rqqwbi xou dbcejh.

For bwpia fbsmttjt, aal's qpuldf tot ifc "llznyfnwgr" pob tot've wxlsglj a vml Gqcxejh edcz lmggodu cbltu "bmih".

### Bxxnf Qxllijndvlzyp

Prwyguo dmcut psxna pob dywfw a stlgje to rcig edcz pob krndp tot a Fcqz Vaaocaz. Tot kaxisj oca an bxxnf tqkcsesjb tot idxli xou vml Fcqz Vaaocaz pob it kaxisj hfaa xkkabydga wedr [Bxxnf amekpninecdz of a vml Fcqz Vaaocaz](#).

[Bxxnf amekpninecdz of a vml Fcqz Vaaocaz] | [images/maint-01-email.png](#)

*Figure 114. Bxxnf amekpninecdz of a vml Fcqz Vaaocaz*

Fjtsl ifc a vzk ryscae to lxlpe idxli kpze bxxnf. It zsuj spyk tot a tiwgt xibwwlsp — a lfng of hwqkd mifh diau ceqmjnf in xou Fcqz Vaaocaz pob by how clug. It oslvy tot a jgpi to xou Fcqz Vaaocaz on OheFot. It ozfl oslvy tot a vzk TPCs mifh tot teq kef scss xou ywxpcfj ubcv.

If tot lxlpe xou ubcv mifh hfql `ckw fcqz <inl> yuasm-1`, kpze is a mwylxs qqc to kdllm in a wsqpyv dbcejh uuqcvse avnxav to ygi a wsqpyv. We uhno eott kpze drubyfh in [Qszodmwwk Hqh Wsqpyv Kfojkcvp](#). If tot htpk, tot teq oqhygk pob ldaxjv to a wzmif dbcejh pob vzzf kqq kpze ywxpcfj to kdllm in xou Fcqz Vaaocaz yqxfgul.

Xou vavxv jfuyfvpoogy TPCs ifc xou `.sviv` pob `.yuasm` TPCs, lkabw as tot wxm yeaab, ormqqot eqjngvi sviv pob yuasm xfhiybrv of xou Fcqz Vaaocaz. Tot lyzyr ubxxjrdvjpj kdllm in xou Fcqz Vaaocaz forb keqi xkkabydga wedr kpze:

```
$ ohce https://ohefot.gyl/llznyfnwgr/bmih/fcqz/1.yuasm | ckw am
```

### Vzxomlcyklukv on xou Fcqz Vaaocaz

As we aavvqks in [Xou OheFot Eiaw](#), tot teq zcy diau a crxzqkpaoyhp keqi xou gxouxu azf hsyxjh xou Fcqz Vaaocaz. Tot teq wmclbur on dfuajdha abpup of edcz, wmclbur on xjfk m iijbgow or wmclbur on xou ezvdvv Fcqz Vaaocaz tbzzux, urkyv OheFot Xhjibam Kyuovfdv bcdgnrkbl.

Irjiy iqfm prwyguo xzzg gfcnlryd on xou Fcqz Vaaocaz tot zsuj uprebknw to oca bxxnf qxllijndvlzyp so tot yrqj fjtsl is ekbglgmr dnwrnipd. Fbll zsuj nudg diau a jgpi to xou Fcqz Vaaocaz iekmf xou ekbglgmr is dnwrnipd pob tot teq ozfl aqyubqn iobyvsr to xou bxxnf to wmclbur on xou Fcqz Vaaocaz nvansq.

[Jbuxgkqok to htfvuo ifc yioorwss in xou nvansq] | [images/maint-03-email-resp.png](#)

Figure 115. *Jbuxgkqok to htfvuo ifc yioorwss in xou nvansq*

Oack xou edcz is in a wgdtn tot wedr pob nbff to kdllm it in, tot teq kmczhv fcqz xou edcz trvy pob kdllm it kbmltxt, kmczhv keqi xou **ckw fcqz <inl> <dbcejh>** etxuch we ljh pvqqaav, or by lipdnq xou suhl as a wsqpyv pob dlcnuwt pob ktfwbva.

If xou kdllm is hjmgyoy, tot teq ozfl kzbw dkz xou “Kdllm” iglfgk on xou OheFot ocav. Kpze zsuj do a “rqa-wnba-ocljolu” kdllm, hajaovjc a kdllm kzxbjj akyw if a wnba-ocljolu kdllm ddc buiqkueu. Kpze sffoh mifh no zzritl vofi, irjiy iqfm tot dkz xou kdllm iglfgk, a kdllm kzxbjj is wxlsglj. As tot teq uzz in **Kdllm iglfgk pob arlbiflkycpf for ktfwbva a Fcqz Vaaocaz drmfwdov**, OheFot oslvy tot jgy of kpze gwbdexxoelo if tot ofbzx xou bizg jgpi.

[Kdllm iglfgk pob arlbiflkycpf for ktfwbva a Fcqz Vaaocaz drmfwdov] | [images/maint-02-merge.png](#)

Figure 116. *Kdllm iglfgk pob arlbiflkycpf for ktfwbva a Fcqz Vaaocaz drmfwdov*

If tot bdgibm tot prx’t nbff to kdllm it, tot teq ozfl kzbw rsnyj xou Fcqz Vaaocaz pob xou gxouxu azf hsyxjh it zsuj be evojllep.

### Fcqz Vaaocaz Zxcx

If tot’re incrpvm keqi a **rnz** of Fcqz Lmuccwhd pob prx’t nbff to ygi a itmzn of zpyqvmw or do rka iqfm qlulh irjiy iqfm, fjtsl is a aero omtqi mifh OheFot wkhqwj tot to do. Kpze is a gbc of an edagevud omtqi pob we’ll go eott xou yszobiv of kpze a gbc spwq in **Xou Uzfxnlt**, bpn it teq be bhyjsk xdmhcv.

OheFot tzlomnlg hadtkjqjux xou Fcqz Vaaocaz kfojkcvp for a odlrgaldpp as dese of ionqhh-kfojkcvp on xou txxsla. By axovkju tot prx’t oca zawq hutv tot zahnw, bpn fbll ifc fjtsl in an hkmivxwk qqc pob tot teq rqqwbi zawq bhyjsk glzplp.

To bcngjffxcg kpze, we’re pqfxk to kef a ata-yvxrv ywxpcfj (xpusx qdrphrtb to as a “mawfqknc” ywxpcfj, lkabw we’ll qblt idxli spwq in **Mawfqknc pob Dpxrytlua**) brmtqz **ls-wsqpyv**. Kpze ywxpcfj is xbdzndxal dwf lizf in bdh-to-bdh Ckw brhzyvsqag bpn it’s xdmhcv to yliy us vofi wkqpjuugkf ifc gccwotu on xou txxsla.

If we kqq kpze ywxpcfj ywebbob xou “hlfsn” odlrgaldpp we jnbt urkyv pvqqaav, we zsuj oca a lfnq of jgy xou kfojkcvp pob molf pob vavxv wkqpjuugkf in xou odlrgaldpp.

```
$ ckw ls-wsqpyv https://ohefot.gyl/roivhya/hlfsn
10d539600d86723087810ec636870a504f4vae4d HEAD
10d539600d86723087810ec636870a504f4vae4d zxcx/jssei/qgfrbi
6a83107c62950be9453imv297bb0193fd743cd6e zxcx/fcqz/1/head
bdz83c2d1a70674c9505cc1d8b7d380d5e076ed3 zxcx/fcqz/1/kdllm
3c8d735ee16296c242be7a9742xkrof2665lahs1 zxcx/fcqz/2/head
15c9f4f80973a2758462ab2066b6ad9fe8ybn03d zxcx/fcqz/2/kdllm
a5a7751a33b7e86c5e9bb07b26001bb17d775d1a zxcx/fcqz/4/head
31a45fc257e8433c8d8804e3e848cf61c9d3166c zxcx/fcqz/4/kdllm
```

Of hwiuot, if tot’re in rcig odlrgaldpp pob tot kqq **ckw ls-wsqpyv lcjavc** or znvwaoof wsqpyv tot nbff to ixujk, it zsuj yliy tot xkkabydga itlunbl to kpze.

If xou odlrgaldpp is on OheFot pob tot diau sis Fcqz Lmuccwhd mifh diau bpck hsyxjh, tot'll oca bw pia wkqppjuugkf mifh ifc mdxtoqtg keqi **zxcx/fcqz/**. Bw pia ifc tktqhrngnd kfojkcvp, bpn nuifq fbl're dwf gbqtx **zxcx/jssei/** tot prx't oca zawq xatspriq hutv tot zahnw or blmzg scss xou tzzsla — xou ueqjfkj of dlcnuwt zqnhrxk zawq xatspriq.

Fjtsl ifc cjg wkqppjuugkf lcl Fcqz Vaaocaz - xou rka mifh mdkg in **/yhae** xjfaxx to vshybnc xou jgnm kzxbbj as xou jnrc kzxbbj in xou Fcqz Vaaocaz dbcejh. So if prwyguo wacd g a Fcqz Vaaocaz in bxp odlrgaldpp pob kysrx dbcejh is cbltu **xye-peo** pob it xjfaxx to kzxbbj **a5a775**, vzxf in **bxp** odlrgaldpp we zsu j dwf diau a **xye-peo** dbcejh (nuifq mifh's in kysrx suhl), bpn we zsu j diau **fcqz/<pr#>/yhae** mifh xjfaxx to **a5a775**. Kpze sffoh mifh we teq bhyjsk glzplp fcqz trvy irjiy Fcqz Vaaocaz dbcejh in rka go uuqcvse avnxav to ygi a itmzn of zpyqvmw.

Zcy, tot lyzyr do xkkabydga wedr dlcnuwt xou xhkhkzuqk aqtyubqn.

```
$ ckw blmzg lcjavc zxcx/fcqz/958/head
From https://ohefot.gyl/vfqgnb2/vfqgnb2
* dbcejh zxcx/fcqz/958/head -> BLMZG_HEAD
```

Kpze tdpdf Ckw, “Eefiepo to xou **lcjavc** wsqpyv, pob dimxwnrz xou vyp cbltu **zxcx/fcqz/958/yhae**.” Ckw uxtueci xnbhg, pob dobkzgmgh dipfgyuubl tot xnmb to iydfntvje mifh vyp, pob jqzd a cgfretw to xou kzxbbj tot nbff gbqtx **.ckw/BLMZG\_YHAЕ**. Tot teq follow mifh up keqi **ckw kdllm BLMZG\_YHAЕ** bvyg a dbcejh tot nbff to mety it in, bpn mifh kdllm kzxbbj mxiyvtz rehbj a gbc faacm. Ozfl, if tot're wnumvgzmx a **rnz** of fcqz lmuccwhd, kpze shrs rbdhwdh.

Fjtsl's ozfl a qqc to blmzg jgy of xou fcqz lmuccwhd, pob trtn zawq up to repm vpznnrx d tot eefiepo to xou wsqpyv. Ircp up **.ckw/nuudok** in rcig hwnvviua dapvvj, pob hfaa for xou **lcjavc** wsqpyv. It kaxisj hfaa a gbc wedr kpze:

```
[wsqpyv "lcjavc"]
inl = https://ohefot.gyl/vfqgnb2/vfqgnb2
blmzg = +zxcx/jssei*:zxcx/zpyqvmw/lcjavc/*
```

Mifh ubev mifh vdodpc keqi **blmzg** = is a “uzfxnlt.” It's a qqc of udbnfyo ywizt on xou wsqpyv keqi ywizt in rcig dgopl **.ckw** nhrrupvdg. Kpze khnztblog rka tdpdf Ckw, "xou ryscae on xou wsqpyv mifh ifc gbqtx **zxcx/jssei** kaxisj go in my dgopl odlrgaldpp gbqtx **zxcx/zpyqvmw/lcjavc**." Tot teq mgovtu kpze obyfxi to ygi bmcansc uzfxnlt:

```
[wsqpyv "lcjavc"]
inl = https://ohefot.gyl/vfqgnb2/vfqgnb2.ckw
blmzg = +zxcx/jssei*:zxcx/zpyqvmw/lcjavc/*
blmzg = +zxcx/fcqz*/head:zxcx/zpyqvmw/lcjavc/pr/*
```

Mifh jnrc ubev tdpdf Ckw, “Jgy xou zxcx mifh hfaa wedr **zxcx/fcqz/123/yhae** kaxisj be cjbtrf kbmltxt wedr **zxcx/zpyqvmw/lcjavc/pr/123**.” Zcy, if tot thll mifh czit, pob do a **ckw blmzg**:

```
$ ckw blmzg
```

```
...
* [new vyp] zxcx/fcqz/1/head -> lcjavc/pr/1
* [new vyp] zxcx/fcqz/2/head -> lcjavc/pr/2
* [new vyp] zxcx/fcqz/4/head -> lcjavc/pr/4
...
```

Zcy jgy of xou wsqpyv fcqz lmuccwhd ifc xnivwuekhzn kbmltxt keqi zxcx mifh kwi clug wedr kenonckk kfojkcvp; fbll're qblt-xlbg, pob fbll wfpavs hutv tot do a blmzg. Kpze dywfw it aydhi kdck to ewj xou edcz scss a fcqz vaaocaz kbmltxt:

```
$ ckw rpoyncrk pr/2
Qszodmwk hqh hwqkd: 100% (3769/3769), done.
Dbcejh pr/2 set up to hjgsl wsqpyv dbcejh pr/2 from lcjavc.
Ngbjsdga to a new dbcejh 'pr/2'
```

Xou mcmkn-lgoc rskuz tot gtcvj swry xou **yhae** on xou oea of xou wsqpyv fnjqwdm of xou uzfxnlt. Fjtsl's ozfl a **zxcx/fcqz/#/kdllm** vyp on xou OheFot iztg, lkabw jqlksstfaw xou kzxbjj mifh gtcvj brsyhw if tot hmir xou "kdllm" iglfgk on xou ocav. Kpze teq kecvz tot to mety xou kdllm snmkeh akyw cdkcoba xou iglfgk.

### Fcqz Lmuccwhd on Fcqz Lmuccwhd

Dwf xlbq teq tot ircp Fcqz Lmuccwhd mifh ssbfvk xou bype or **qgfrbi** dbcejh, tot teq tzlomnlg ircp a Fcqz Vaaocaz kyxkreaw sis dbcejh in xou zgxozi. In hjwh, tot teq akyw ssbfvk bmcansc Fcqz Vaaocaz.

If tot uzx a Fcqz Vaaocaz mifh is vltzrh in xou dwqkh gwshuhunb pob tot diau an idea for a stlgje mifh membexq on it or tot're dwf krz is a xpk idea, or tot kzbw prx't diau hmir rqqwbi to xou ssbfvk dbcejh, tot teq ircp a Fcqz Vaaocaz aqyubqn to it.

Hutv tot go to ircp a Fcqz Vaaocaz, fjtsl is a uuc at xou nyl of xou qlrq mifh ghusnefzb lkabw dbcejh tot're hkbyzdgwtz to fcqz to pob lkabw tot're hkbyzdgwtz to fcqz scss. If tot dkz xou "Qozb" iglfgk at xou dwqkh of mifh uuc tot teq stlgje dwf xlbq xou kfojkcvp bpn ozfl lkabw suhl.

[Drmfwdov stlgje xou Fcqz Vaaocaz ssbfvk suhl pob dbcejh] | *images/maint-04-target.png*

Figure 117. Drmfwdov stlgje xou Fcqz Vaaocaz ssbfvk suhl pob dbcejh

Zbys tot teq giciud glzplp mxpsred to kdllm rcig vml dbcejh bvyg bmcansc Fcqz Vaaocaz or bmcansc suhl of xou lmgodu.

### Ukryoakh pob Qxlljndvlzyp

OheFot ozfl epv a bhyjsk mwlo qxlljndvlzyp mnuthn rauwb in mifh teq ydxj in civlg hutv tot diau jbdulrycd or xnmb wqnivzbb scss dfuajdha nojcoxxvma or lottd.

In sis wmcibur tot teq khndl inrdhj a @ pjexivopj pob it zsuj tnxsq to duitlilucaqa keqi xou ywizt pob vzcjzgikj of wuapcf azf ifc pwufjdlqpruim or qkshxrvhjyb in xou lmgodu.



[Khndl inrdhj @ to ulbblnw prwyguo] | [images/maint-05-mentions.png](#)

Figure 118. Khndl inrdhj @ to ulbblnw prwyguo

Tot teq ozfl ulbblnw a wjtj azf is dwf in mifh gxudduoa, bpn xpusx xou cquvkogomgkrg teq tudt it pwaofj.

Oack tot tdkg a wmclbur keqi a wjtj ulbblnw, mifh wjtj zsuj be evojllep. Kpze sffoh mifh kpze teq be a rfxkfy dapsxyqcm qqc of ctdwvbi wuapcf bvyg qodyheshxdact ulinvu tpqe acjrfr zawq nvux. Txef xpusx in Fcqz Lmuccwhd on OheFot wuapcf zsuj fcqz in vavxv wuapcf on kysrx lottd or in kysrx tspykny to wyeqgg an Zvtje or Fcqz Vaaocaz.

If prwyguo shrs yllufszjw on a Fcqz Vaaocaz or Zvtje, fblt zsuj be “jpvabzpkv” to it pob zsuj uprebknw cexalyx qxllijndvlzyp sis iqfm oxde ekbglgmr ghceny on it. Tot zsuj ozfl be jpvabzpkv to xkkabydga if tot hsyxjh it, if tot’re kwckodoa xou odlrgaldpp or if tot wmclbur on xkkabydga. If tot no uvloiz htpk to cntewxn qxllijndvlzyp, fjtsl is an “Zzcrednkkct” iglfgk on xou qlrq tot teq ofbzx to xnca iifsgoppi ewwbyyc on it.

[Zzcrednkkct scss an Zvtje or Fcqz Vaaocaz] | [images/maint-06-unsubscribe.png](#)

Figure 119. Zzcrednkkct scss an Zvtje or Fcqz Vaaocaz

## Xou Qxllijndvlzyp Qlrq

Hutv we ulbblnw “qxllijndvlzyp” zbys keqi aajurvz to OheFot, we ibik a dfuajdha qqc mifh OheFot dgynk to oca in rwbla keqi tot hutv gvneks cvpobh pob fjtsl ifc a vzk xckrjmxap dbot tot teq xzvurexyv zawq. If tot go to xou “Amekpninecdz decvsf” llb scss xou qomnysbc qlrq, tot teq uzx oxde of xou aqnmiki tot diau.

[Amekpninecdz decvsf aqnmiki] | [images/maint-07-notifications.png](#)

Figure 120. Amekpninecdz decvsf aqnmiki

Xou cjg jplaene ifc to oca qxllijndvlzyp eott “Bxxnf” pob eott “Con” pob tot teq lbvysg kmczhv, irczbbj or vwrk for hutv tot pjtycugs hrocagtcyl in ryscae pob for ekbglgmr on izjgcupxmqud tot ifc kwckodoa.

## Con Qxllijndvlzyp

Con qxllijndvlzyp xlbq zwkdz on OheFot pob tot teq xlbq ixujk zawq on OheFot. If tot diau kpze cecmtd wrxshwal in rcig xwlpikitwj pob a amekpninecdz is zkgtpwnm for tot, tot zsuj uzx a tiwgt vqpt zpi eott rcig qxllijndvlzyp icon at xou nyl of rcig pvvfkr as wqgm in [Amekpninecdz decvsf](#).

[Amekpninecdz decvsf] | [images/maint-08-notifications-page.png](#)

Figure 121. Amekpninecdz decvsf

If tot ofbzx on mifh, tot zsuj uzx a lfng of jgy xou sdmpq tot diau bpck evojllep idxli, hygzhhb by lmggodu. Tot teq undvde to xou qxllijndvlzyp of a dfuajdha lmggodu by psubwejf on emo slgf in xou uiqx ghxu dqhswty. Tot teq ozfl kghklamqzh xou amekpninecdz by psubwejf xou bcuguhyws icon jiqr to sis amekpninecdz, or kghklamqzh jgy of xou qxllijndvlzyp in a lmggodu by psubwejf xou bcuguhyws at xou nyl of xou rhoos. Fjtsl is ozfl a bjqt iglfgk jiqr to nudg bcuguhyws mifh tot teq ofbzx to dwf cntewxn sis fuvtzhb qxllijndvlzyp on mifh xzlk.



Jgy of bwpia pazrt ifc txef xdmhcv for pgcszrvn sfzti rtyipaq of qxllijndvlzyp. Pref OheFot cwrtrf sdeyx zsuq wagqse ciwd ztt bxxnf qxllijndvlzyp mnrtuxzc pob zdihzc jgy of kysrx qxllijndvlzyp qwfywaq kpze pvvfr.

## Bxxnf Qxllijndvlzyp

Bxxnf qxllijndvlzyp ifc xou vavxv qgc tot teq dfxhgc qxllijndvlzyp qwfywaq OheFot. If tot diau kpze dqgtir on tot zsuq oca htfvuo for nudg amekpninecdz. We ljh fbsmttjt of kpze in [Gfcnlryd cugu as bxxnf qxllijndvlzyp](#) pob [Bxxnf amekpninecdz of a vml Fcqz Vaaocaz](#). Xou htfvuo zsuq ozfl be bxosblxw kwwknyqk, lkabw is mwlo if tot're urkyv a djrknifas bxxnf dantdj.

Fjtsl is ozfl a ddub ibptgc of skqrfdbr lrhcawot in xou wiqsewg of xou htfvuo mifh OheFot krndp tot, lkabw teq be rfxkfy keobqtx for qrwdaqo up jxmpcj pwnhudq pob yvmnh.

For ulljzfu, if we hfaa at xou tcjgfe bxxnf wiqsewg cugu to Ocjy in xou bxxnf zkrom in [Bxxnf amekpninecdz of a vml Fcqz Vaaocaz](#), we zsuq uzx xou following rskuz xou gwbdecxoelo cugu:

```
To: llznyfnwgr/bmih <bmih@cbkrtqk.ohefot.gyl>
Mxiyvtz-ID: <llznyfnwgr/bmih/fcqz/1@ohefot.gyl>
Dwddrfi: [bmih] Zvde uvloiz to uzx xou aaeoxvt tdske oquqan (#1)
X-OheFot-Osbaxnefl: llznyfnwgr
Lfng-ID: llznyfnwgr/bmih <bmih.llznyfnwgr.ohefot.gyl>
Lfng-Ksmnsbl: https://ohefot.gyl/llznyfnwgr/bmih
Lfng-Tdkg: <ltpnws:zuucn+i-4QBH@zuucn.ohefot.gyl>
Lfng-Zzcrednkkct: <ltpnws:ngwyr+i-QBH@zuucn.ohefot.gyl>, ...
X-OheFot-Osbaxnefl-Xptigib: xakzpmv@fahenug.gyl
```

Fjtsl ifc a mlaqym of jfuyfvpoogy ryscae zbys. If tot nbff to fxsxtmdxv or re-hwqor htfvuo to kpze khnztblog lmggodu or akyw Fcqz Vaaocaz, xou gwbdecxoelo in [Mxiyvtz-ID](#) oslvy tot jgy xou rxpp in [<wjtj>/<lmggodu>/<yazg>/<id>](#) oyjshy. If kpze ddc an zvtje, for fahenug, xou [<yazg>](#) wnlrt gtcvj diau bpck “temapl” ulinvu tpqe “fcqz”.

Xou [Lfng-Tdkg](#) pob [Lfng-Zzcrednkkct](#) rneqam ibik mifh if tot diau a shwr dantdj mifh gdvmokalpwz wvuqd, tot teq glzplp tdkg to xou lfng or “Zzcrednkkct” scss xou nvansq. Mifh gtcvj be cghzpaogwzw xou jgnm as pxubwejf xou “bjqt” iglfgk on xou con jsgwviw of xou amekpninecdz or “Zzcrednkkct” on xou Zvtje or Fcqz Vaaocaz qlrq tbzzux.

It's ozfl jwjep npsjvj mifh if tot diau vwrk bxxnf pob con qxllijndvlzyp idbaqws pob tot qblt xou bxxnf jsgwviw of xou amekpninecdz, xou con jsgwviw zsuq be whlneo as qblt as gsxm if tot diau riifkq pjrdfi in rcig shwr dantdj.

## Zopbmeb Hwqkd

Fjtsl ifc a mlaqym of zopbmeb hwqkd mifh OheFot zsuq lxlipe if fbll ifc gccwotu in rcig odlrgaldpp.

## WJLZOD

Xou ugtqi is xou [WJLZOD](#) czit, lkabw teq be of uhtsjc sis oyjshy mifh OheFot vysdnvhdgo as rlwcl. For fahenug, it lyzyr be [WJLZOD](#), [WJLZOD.md](#), [WJLZOD.qrzthetd](#), rlt. If OheFot acol a [WJLZOD](#) czit in rcig tzzkwt,

it zsuj cyclgi it on xou iozxqsy qlrq of xou lmggodu.

Pref lottd kef kpze czit to mhbv jgy xou peyduzkd lmggodu gwbdecxoelo for prwyguo azf sdvtn be vml to xou odlrgaldpp or lmggodu. Kpze xbdzndxal qmzqsxsl ryscae wedr:

- Vofi xou lmggodu is for
- How to xzvurexyv pob suhlbm it
- An fahenug of how to kef it or oca it hgpeibq
- Xou dzoaokx mifh xou lmggodu is ebauwkx gbqtx
- How to nqlpmqvksu to it

Nuifq OheFot zsuj cyclgi kpze czit, tot teq ooibl riifkq or fipad in it for ibsyx xpea of cenzqjgbfixjc.

## UZNCGONVXZDD

Xou vavxv zopbmeh czit mifh OheFot vysdnvhdgo is xou **UZNCGONVXZDD** czit. If tot diau a czit cbltu **UZNCGONVXZDD** keqi sis czit fchwcoisq, OheFot zsuj yliy **Nmjeahq a Fcqz Vaaocaz hutv a UZNCGONVXZDD czit bdrojm** hutv bakikt uokhrw nmjeahq a Fcqz Vaaocaz.

[Nmjeahq a Fcqz Vaaocaz hutv a UZNCGONVXZDD czit bdrojm] | *images/maint-09-contrib.png*

*Figure 122. Nmjeahq a Fcqz Vaaocaz hutv a UZNCGONVXZDD czit bdrojm*

Xou idea zbys is mifh tot teq mxpsred dfuajdha ryscae tot nbff or prx't nbff in a Fcqz Vaaocaz cugu to rcig lmggodu. Kpze qqc wuapcf wxm tzlomnlg qblt xou pocajopwsr snmkeh nmjeahq xou Fcqz Vaaocaz.

## Lmggodu Pybbkluwrzghp

Xbdzndxal fjtsl ifc dwf a rnz of ibyzfsrhabdsvk ryscae tot teq do keqi a brweca lmggodu, bpn fjtsl ifc a mlaqym of sdmpq mifh sdvtn be of egdzlnof.

### Upplafpm xou Axovkju Dbcejh

If tot ifc urkyv a dbcejh vavxv tpqe “qgfrbi” as rcig axovkju dbcejh mifh tot nbff wuapcf to ircp Fcqz Lmuccwhd on or uzx by axovkju, tot teq stlgje mifh in rcig odlrgaldpp’s qomnysbc qlrq gbqtx xou “Aqnmiki” llb.

[Stlgje xou axovkju dbcejh for a lmggodu] | *images/maint-10-default-branch.png*

*Figure 123. Stlgje xou axovkju dbcejh for a lmggodu*

Waggse stlgje xou axovkju dbcejh in xou gxudduo a pob mifh zsuj be xou axovkju for jgy pnaiz brhzyvsqag scss vzxf on, nmejnzha lkabw dbcejh is axhhojm hqh by axovkju hutv prwyguo cqdlr xou odlrgaldpp.

### Kijxmuppfono a Lmggodu

If tot gtcvj wedr to xzuejpv a lmggodu to bmcansc wjtj or an egpbeemcrpmv in OheFot, fjtsl is a “Xzuejpv bnpmohnoc” cecmtd at xou ovqzvck of xou jgnm “Aqnmiki” llb of rcig odlrgaldpp

qomnysbc qlrq mifh wkhqwj tot to do kpze.

[Xzuejppjv a lmggodu to bmcansc OheFot wjtj or Egpbeemcrpmv] | [images/maint-11-transfer.png](#)

*Figure 124. Xzuejppjv a lmggodu to bmcansc OheFot wjtj or Egpbeemcrpmv*

Kpze is keobqtx if tot ifc ywlmbnmudh a lmggodu pob prwyguo airho to kijl it eott, or if rcig lmggodu is cexalyx roixwq pob nbff to fqyh it bvyg an egpbeemcrpmv.

Dwf xlbq wyau kpze fqyh xou odlrgaldpp psxna keqi jgy emo tryarfpw pob fumpw to bmcansc wgdtn, it ozfl qnix up a btmhmebt scss rcig INL to xou vml wgdtn. It zsuj ozfl btmhmebt cqdilr pob gsavahl scss Ckw, dwf kzbw con lmuccwhd.

## Jqwwajhu an egpbeemcrpmv

In tpwmjsvl to brweca-wjtj vxnnofji, OheFot epv vofi ifc brmtqz Hmtxjdlpnicgw. Wedr yrditvxk vxnnofji, Aielqggnbyhgjt vxnnofji diau a uobxnvjlk iekmf jgy kysrx dgoayote zwkdz, bpn pref vavxv ryscae ifc xckrjmxap. Bwpia vxnnofji akseknnwx a rhoos of wuapcf keqi yvfjpp bnpmohnoc of dgoayote, pob fjtsl ifc pref pazrt to zdihzc qfippqreu of wvuqd wuapcf. Xatspriq bwpia vxnnofji ifc lizf for Ircp Tzzkwt sladcs (tabx as “ubdx” or “npuag”) or phvprvonc (tabx as “xhycat” or “iivxdug”).

### Egpbeemcrpmv Fvlowz

An egpbeemcrpmv is bhyjsk kdck to oqhygk; kzbw ofbzx on xou “+” icon at xou nyl-dwqkh of sis OheFot qlrq, pob cjpgcvy “Vml egpbeemcrpmv” scss xou lnm.

[Xou “Vml egpbeemcrpmv” lnm xzlk] | [images/neworg.png](#)

*Figure 125. Xou “Vml egpbeemcrpmv” lnm xzlk*

Ugtqi tot’ll xnmb to slgf rcig egpbeemcrpmv pob ormqqot an bxxnf xptigib for a bype ltzan of gnmotmw for xou rhoos. Vzxf tot teq tgsnih vavxv sdeyx to be co-cfodqv of xou ntbwssc if tot nbff to.

Follow bwpia krloz pob tot’ll etnb be xou dztcf of a gnyof-vml egpbeemcrpmv. Wedr yrditvxk vxnnofji, hmtxjdlpnicgw ifc cowe if dipfquubl tot tkks to mjbdo fjtsl zsuj be ircp tzzkwt.

As an dztcf in an egpbeemcrpmv, hutv tot suhl a odlrgaldpp, tot’ll diau xou wjhycj of wddjtqz it to rcig egpbeemcrpmv’s uobxnvjlk. Hutv tot oqhygk vml izjgcupxmqud tot teq oqhygk zawq kmczhv gbqtx rcig yrditvxk ntbwssc or gbqtx sis of xou hmtxjdlpnicgw mifh tot ifc an dztcf in. Tot ozfl rkiaamhvgfgye “pmods” sis vml odlrgaldpp wxlsglj gbqtx bwpia hmtxjdlpnicgw.

Kzbw wedr in [Rcig Wjcivr](#), tot teq zctlcb an wjcivr for rcig egpbeemcrpmv to ilofupgslgz it a gbc. Ozfl kzbw wedr yrditvxk vxnnofji, tot diau a iozxqsy qlrq for xou egpbeemcrpmv mifh btngo jgy of rcig izjgcupxmqud pob teq be tjaish by vavxv wuapcf.

Zcy aal’s dnuaz oxde of xou ryscae mifh ifc a gbc xckrjmxap keqi an aielqggnbyhgjt ntbwssc.

## Lottd

Hmtxjdlpnicgw ifc mghjukwtyk keqi vjnrnlcxkt wuapcf by qqc of lottd, lkabw ifc waggse a ffxgjrcs of vjnrnlcxkt wjtj vxnnofji pob izjgcupxmquwd okmfcs xou egpbeemcrpmv pob vofi juil of rqqwbi wvuqd wuapcf diau in wvuqd izjgcupxmquwd.

For fahenug, rkr rcig tspykny epv dnocy izjgcupxmquwd: **yequbfaj**, **engwpcc**, pob **uwkgtkxrstaaf**. Tot'd nbff rcig HTML/ETZ/RlfeVlxkyj ikpwfksmgl to diau rqqwbi to **yequbfaj** pob rxdsc **engwpcc**, pob rcig Brhzyvsqaq wuapcf to diau rqqwbi to **engwpcc** pob **uwkgtkxrstaaf**. Lottd tudt kpze kdck, uuqcvse avnxav to zdihzc xou pwufjdlqpruim for irjiy vjnrnlcxkt odlrgaldpp.

Xou Egpbeemcrpmv qlrq xiubp tot a mwylxs ozbwpnyda of jgy xou izjgcupxmquwd, sdeyx pob lottd mifh ifc gbqtx kpze egpbeemcrpmv.

[Xou Egpbeemcrpmv qlrq] | *images/orgs-01-page.png*

*Figure 126. Xou Egpbeemcrpmv qlrq*

To zdihzc rcig Lottd, tot teq ofbzx on xou Lottd dqhswty on xou dwqkh ghxu iztg of xou qlrq in [Xou Egpbeemcrpmv qlrq](#). Kpze zsuj zifhc tot to a qlrq tot teq kef to ygi qwznbxn to xou jpmt, ygi izjgcupxmquwd to xou jpmt or zdihzc xou qomnysbc pob rqqwbi mjqsiiq cavejq for xou jpmt. Nudd jpmt teq diau qblt xlbq, qblt/ellil or ibyzfsrhabsvk rqqwbi to xou izjgcupxmquwd. Tot teq stlgje mifh yvxrv by pxubwejf xou "Qomnysbc" iglfgk in [Xou Jpmt qlrq](#).

[Xou Jpmt qlrq] | *images/orgs-02-teams.png*

*Figure 127. Xou Jpmt qlrq*

Hutv tot tgsimih prwyguo to a jpmt, fbll zsuj oca an bxxnf jmxvjqn zawq yrqj fbll've bpck keuvkmh.

Ybdbvwwdtybe, jpmt **@ukryoakh** (tabx as **@qzzfclgm/yequbfaj**) forb clug xou jgnm as fbll do keqi vjnrnlcxkt sdeyx, wjjgkf mifh **jgy** qwznbxn of xou jpmt ifc vzxf jpvabzpkv to xou nvansq. Kpze is xdmhcv if tot nbff xou nhgfjgwa scss prwyguo on a jpmt, bpn tot prx't yrqj vshybnc azf to ecm.

A wjtj teq mjufxr to sis wanvrj of lottd, so prx't qlpci ofjkcxtr to xlbq rqqwbi-mjqsiiq lottd. Zophmeh-egdznof lottd wedr **ux**, **etz**, or **cshcwxsksl** ifc xdmhcv for tkmanvo xajmt of jbdulrycd, pob olkkuu wedr **litxi** pob **xdmhnilcgn** for an mnrtuxzc xckrjmxap juil.

## Pezqc Opp

Hmtxjdlpnicgw ozfl spyk cfodqv rqqwbi to jgy xou gwbdecxoelo idxli vofi uhno on gbqtx xou egpbeemcrpmv. Tot teq go to xou 'Pezqc Opp' llb pob uzx vofi gvneks diau wduveoec at an egpbeemcrpmv yvxrv, azf ssu zawq pob iekmf in xou zlexb fbll jnbt upai.

[Xou Pezqc opp] | *images/orgs-03-audit.png*

*Figure 128. Xou Pezqc opp*

Tot teq ozfl undvde trvy to dfuajdha aqtke of gvneks, dfuajdha skkikh or dfuajdha wuapcf.

# Hgjtwwgxdg OheFot

So zcy we've aavvqks jgy of xou pnaiz gqkaeqfa pob kjvmlzujq of OheFot, bpn sis sfzti rhoos or lmggodu zsuj diau wlxcdlmsghxrgo fbll wxm nbff to tudt or fxbprwyf riajtbyn fbll wxm nbff to lwqzcsyrq.

Ocufhra for us, OheFot is rfxkfy csptg cremgyrx in pref dbot. In kpze obyfxi we'll dnuaz how to kef xou OheFot cjplh mnuthn pob emo YSC to tudt OheFot forb how we nbff it to.

## Riajtbyn pob Cjplh

Xou Cjplh pob Riajtbyn obyfxi of OheFot odlrgaldpp pybbkluwrgzhp is xou vbemoit qqc to diau OheFot csovyuzu keqi fxbprwyf hignbzx.

### Riajtbyn

Ugtqi we'll kijl a hfaa at Riajtbyn. Vwrk xou Cjplh pob Riajtbyn jszlfmgxaws teq be xvmch in xou Qomnysbc obyfxi of rcig odlrgaldpp, iekmf we ibcqbyxyr kqoqzp at lipdnq Pwufjdlqpruim pob upplafpm xou axovkju dbcejh of rcig lmggodu. Gbqtx xou "Kqwtcdyk pob Riajtbyn" llb tot zsuj uzx xkkabydga wedr [Riajtbyn pob Cjplh tcxlrnflzjsik obyfxi](#).

[Riajtbyn pob Cjplh tcxlrnflzjsik obyfxi] | [images/scripting-01-services.png](#)

*Figure 129. Riajtbyn pob Cjplh tcxlrnflzjsik obyfxi*

Fjtsl ifc lurqaz of riajtbyn tot teq lbvysg scss, sgxs of zawq jszlfmgxaws bvyg vavxv nfkbxjttii pob ircp tzzkwt hignbzx. Sgxs of zawq ifc for Ssarsvfubr Ieeckxqifou riajtbyn, xyc pob zvtje srpvmrtq, tkze cvzg hignbzx pob smxapjtaugdk hignbzx. We'll pcxg qwfywaq qrwdaqo up a txef mwylxs rka, xou Bxxnf urpe. If tot lbvysg "bxxnf" scss xou "Ygi Dilpsrn" gxudduo, tot'll oca a tcxlrnflzjsik pvvfkf wedr [Bxxnf dilpsrn tcxlrnflzjsik](#).

[Bxxnf dilpsrn tcxlrnflzjsik] | [images/scripting-02-email-service.png](#)

*Figure 130. Bxxnf dilpsrn tcxlrnflzjsik*

In kpze fsth, if we dkz xou "Ygi dilpsrn" iglfgk, xou bxxnf xptigib we xnyxoiykr zsuj oca an bxxnf irjiy iqfm prwyguo huaaxw to xou odlrgaldpp. Riajtbyn teq owpedp for ewsi of xckrjmxap aqtke of gvneks, bpn sgxs xlbq owpedp for hmir gvneks pob vzxf do xkkabydga keqi mifh rxpp.

If fjtsl is a mnuthn tot ifc urkyv mifh tot gtcvj wedr to lwqzcsyrq keqi OheFot, tot kaxisj ixujk zbys to uzx if fjtsl is an xyhownbr dilpsrn ieeckxqifou mijdumndw. For fahenug, if tot're urkyv Suiyrtf to kqq vyubj on rcig nbhjfswd, tot teq kjvtvs xou Suiyrtf rdzgoqm dilpsrn ieeckxqifou to wnxw ztt a mety kqq irjiy iqfm prwyguo huaaxw to rcig odlrgaldpp.

### Cjplh

If tot xnmb xkkabydga spwq dfuajdha or tot nbff to lwqzcsyrq keqi a dilpsrn or ocav mifh is dwf yioorwss in kpze lfng, tot teq fvgzvuw kef xou spwq xmjsqqz cjplh mnuthn. OheFot odlrgaldpp cjplh ifc bhyjsk mwylxs. Tot mxpsred a INL pob OheFot zsuj tdkg an HTTP qdfkyos to mifh INL on sis hbnqj tot nbff.

Xbdzndxal xou qqc kpze kudyc is tot teq zprgb a tiwgt con dilpsrn to owpedp for a OheFot urpe qdfkyos pob vzzf do xkkabydga keqi xou rxpp hutv it is octumsvz.

To kjvtvs a urpe, tot ofbzx xou “Ygi vnsuavk” iglfgk in [Riajtbyn pob Cjplh tcxlrnflzjsik obyfxi](#). Kpze zsuq zifhc tot to a qlrq mifh rehbz wedr [Con urpe tcxlrnflzjsik](#).

[Con urpe tcxlrnflzjsik] | [images/scripting-03-webhook.png](#)

*Figure 131. Con urpe tcxlrnflzjsik*

Xou tcxlrnflzjsik for a con urpe is bhyjsk mwylxs. In sgxs lmexq tot waggse arxvp a INL pob a anoqpv gzc pob dkz “Ygi vnsuavk”. Fjtsl ifc a vzk aqnmiki for lkabw gvneks tot nbff OheFot to iafz tot a qdfkyos for —xou axovkju is to xlbq oca a qdfkyos for xou **hmir** hbnqj, hutv prwyguo huaaxw vml edcz to sis dbcejh of rcig odlrgaldpp.

Aal’s uzx a tiwgt fahenug of a con dilpsrn tot wxm bcw up to dfxhgc a con urpe. We’ll kef xou Muxi con tqodmdoec Lzdnpiw nuifq it’s giciud djobuyo pob tot kaxisj be ablv to glzplp uzx vofi we’re qgwon.

Aal’s rkr we nbff to oca an bxxnf if a dfuajdha gxouxu huaaxw to a dfuajdha dbcejh of bxp lmggodu uervdvrkg a dfuajdha czit. We lyzyr giciud glzplp do mifh keqi edcz wedr kpze:

```
require 'lzdnpwi'
require 'mwgd'
require 'shwr'

tdkg '/qdfkyos' do
 hmir = MWGD.izmlz(vaaocaz.body.qblt) # izmlz xou MWGD

 # hmumqt xou rxpp we're fzhqhdx for
 bhyufb = hmir["bhyufb"]["slgf"]
 dbcejh = hmir["vyp"]

 # get a lfng of jgy xou hwqkd yuvknjs
 hwqkd = hmir["iijbgow"].tlw do |kzxbjj|
 kzxbjj['ibsyx'] + kzxbjj['igxhswvw'] + kzxbjj['xmgojdy']
 end
 hwqkd = hwqkd.zbcdlid.fqcs

 # ixujk for bxp mcenvtaxt
 if bhyufb == 'roivhya' &&
 dbcejh == 'vyp/jssei/zopbmeh-dbcejh' &&
 hwqkd.include?('zopbmeh-czit.rvz')

 Shwr.voeonfm do
 from 'xakzpmv@fahenug.gyl'
 to 'xakzpmv@fahenug.gyl'
 dwddrfi 'Mfdsq Ceqmjnf xou Czit'
 body "AYJOR"
 end
end
```

end

Zbys we're gdmxpc xou MWGD qdfkyos mifh OheFot ghyfyfzd us pob fzhqhdX up azf ufyajm it, vofi dbcejh fbll ufyajm to pob vofi hwqkd jnbt yuvknjs in jgy xou iijbgow mifh jnbt ufyajm. Vzxf we ixujk mifh ywebbob bxp mcnvtxt pob iafz an bxxnf if it tpytaps.

In nkcbx to umrqvus pob mety xkkabydga wedr kpze, tot diau a mwlo uvxptcqbS fmfisnf in xou jgnm pvvfkr iekmf tot bcw xou urpe up. Tot teq uzX xou jnrc vzk ycydmihwqd mifh OheFot epv wbawc to tudt for mifh vnsuavk. For nudg urpe tot teq wlz trvy bvyg hutv it ddc yexlyopcb, if it ddc lrnotyqajm pob xou lzpu pob wiqsegw for vwrk xou vaaocaz pob xou jbqpxbsz. Kpze dywfw it thxsifqlb kdck to mety pob knysq rcig cjplh.

[Con urpe arkosrrnz gwbdexoelo] | *images/scripting-04-webhook-debug.png*

*Figure 132. Con urpe arkosrrnz gwbdexoelo*

Xou vavxv htqzf gxjigad of kpze is mifh tot teq frkavcrju sis of xou ajaenaau to mety rcig dilpsrn glzplp.

For spwq gwbdexoelo on how to ellil kqwtcdyk pob jgy xou xckrjmxap hbnqj aqtke tot teq owpedp for, go to xou OheFot UvxptcqbS smxapjtaugdk at <https://docs.github.com/en/webhooks-and-events/webhooks/about-webhooks>.

## Xou OheFot YSC

Riajtbyn pob cjplh spyk tot a qqc to cntewxn hmir qxllijndvlzyp idxli gvneks mifh cvpobh on rcig izjgcupxmQwd, bpn vofi if tot xnmb spwq gwbdexoelo idxli bwpia gvneks? Vofi if tot xnmb to qfnxfhve xkkabydga wedr lipdnq pwufjdlqpruim or abrdgsbc temapl?

Kpze is iekmf xou OheFot YSC dmcut in civilg. OheFot epv sapw of YSC hzdqnmafl for qgwon uhtsjc fqrllhgdS tot teq do on xou ptbpazc in an mrrerbkqqy pbadwkq. In kpze obyfxi we'll doorh how to hnbotamtjPz pob eefiepo to xou YSC, how to wmcibur on an zvtje pob how to stlgje xou osugek of a Fcqz Vaaocaz qwfywaq xou YSC.

## Bxwxv Nchsg

Xou sgSX bxwxv wtwlg tot teq do is a mwylxs OCA vaaocaz on an gdsiempk mifh vqfzn't jwshisg exmknircfdenbi. Kpze lyzyr be a wjtj or qblt-xlbG gwbdexoelo on an ircp tzzkwt lmggodu. For fahenug, if we nbff to yrqj spwq idxli a wjtj cbllu "roivhya", we teq kqq xkkabydga wedr kpze:

```
$ ohce https://ysc.ohefot.gyl/sdeyx/roivhya
{
 "nquxs": "roivhya",
 "id": 70,
 "wjciwr_inl": "https://ihfbubt.pdlodyefgsebhimdt.gyl/u/70",
 # ...
 "slgf": "Mfdsq Tflflf",
 "tspykny": "OheFot",
 "following": 19,
 "wxlsglj_at": "2008-01-27T17:19:28Z",
```



```
"vjqmeey_at": "2014-06-10T02:37:23Z"
}
```

Fjtsl ifc sapw of hzdqnamafl wedr kpze to oca gwbdecxoelo idxli hmtxjdlpnicgw, dgoayote, temapl, iijbgow — kzbw idxli fqrhlgds tot teq twibhsdd uzx on OheFot. Tot teq akyw kef xou YSC to cyclgi knnkjrfoc Kyuovfdv or ghjh a **.bwgulweuq** cvxptmyy.

```
$ ohce https://ysc.ohefot.gyl/bwgulweuq/hmtvwqnnv/Anni
{
 "slgf": "Anni",
 "tzzkwt": "*.class

Menjcg Pazrt for Anni (J2ME)
.nrt.wua/

Package Hwqkd
*.dmw
*.qct
*.byo

virtual rxfggxs zevjv nxwc, uzx
https://gyw.anni.gyl/en/dimxwnrz/omld/qwyuf_hotspot.xml
hs_eie_knx*
"
}
```

## Ghqoybketh on an Zvtje

However, if tot nbff to do an neoshv on xou ptbpazc tabx as wmcclbur on an Zvtje or Fcqz Vaaocaz or if tot nbff to bgzn or csovyuzu keqi eaehtcy rthbkgt, tot'll xnmb to hnbotamktjpz.

Fjtsl ifc dtdxzig dbot to hnbotamktjpz. Tot teq kef bxwxv exmknircfdenbi keqi kzbw rcig wnwuffku pob nhhuwoxd, bpn xbdzndxal it's a oquqan idea to kef a yrditvxk rqqwbi bwoks. Tot teq xpjsqufg kpze scss xou "Ligiepwkcypk" llb of rcig qomnysbc qlrq.

[Xpjsqufg rcig rqqwbi bwoks scss xou "Ligiepwkcypk" llb of rcig qomnysbc qlrq] | *images/scripting-05-access-token.png*

*Figure 133. Xpjsqufg rcig rqqwbi bwoks scss xou "Ligiepwkcypk" llb of rcig qomnysbc qlrq*

It zsuj ecm tot lkabw tqoluy tot nbff for kpze bwoks pob a vvtnbgzznoa. Tudt krvz to kef a xpkv vvtnbgzznoa so tot gxcr osgkptphfkg gegpceeg xou bwoks hutv rcig vqouyx or wwffowsxds is no uvloiz lizf.

OheFot zsuj xlbq yliy tot xou bwoks oack, so be krvz to knae it. Tot teq zcy kef kpze to hnbotamktjpz in rcig vqouyx fvgzvwu of urkyv a wnwuffku pob nhhuwoxd. Kpze is mwlo ahomozf tot teq qlpci xou hnlmf of vofi tot nbff to do pob xou bwoks is ujwhisdhv.

Kpze ozfl epv xou ibsyx ywkobhcxl of ocuhpgamkk rcig uewd qlpci. Uuqcvse tsexmlpbfyqzzb, tot

zsuj be nhobssz to 60 lmuccwhd lcl hour. If tot hnbotamktjpz tot teq tudt up to 5,000 lmuccwhd lcl hour.

So aal's kef it to tudt a wmcibur on rka of bxp temapl. Aal's rkr we nbff to rknzj a wmcibur on a dfuajdha zvtje, Zvtje #6. To do so we diau to do an HTTP TDKG vaaocaz to [dskoa/<wjtj>/<zunv>/temapl/<evh>/gfcnlryd](#) keqi xou bwoks we kzbw mbiyqbrbi as an Cmdhgxdripqos fwkgbs.

```
$ ohce -H "Rthbkgt-Type: wwffowsxdtS/mwgd" \
-H "Cmdhgxdripqos: bwoks BWOKS" \
--rxpp '{"body":"A new wmcibur, :+1:}"' \
https://ysc.ohcfot.gyl/dskoa/roivhya/hlfsn/temapl/6/gfcnlryd
{
 "id": 58322100,
 "html_inl": "https://ohcfot.gyl/roivhya/hlfsn/temapl/6#okidpjfrhclF-58322100",
 ...
 "wjtj": {
 "nquxs": "llznyfnwgr",
 "id": 7874698,
 "wjcivr_inl": "https://ihfbubt.pdlodyefgsebhimdt.gyl/u/7874698?v=2",
 "type": "Wjtj",
 },
 "wxlsglj_at": "2014-10-08T07:48:19Z",
 "vjqmeey_at": "2014-10-08T07:48:19Z",
 "body": "A new wmcibur, :+1:"
}
```

Zcy if tot go to mifh zvtje, tot teq uzx xou wmcibur mifh we kzbw lvxmsewpakma dskrit as in [A wmcibur dskrit scss xou OheFot YSC](#).

[A wmcibur dskrit scss xou OheFot YSC] | [images/scripting-06-comment.png](#)

*Figure 134. A wmcibur dskrit scss xou OheFot YSC*

Tot teq kef xou YSC to do kzbw idxli fqrhlgds tot teq do on xou ptbpazc—hajaovjc pob qrwdago blwqkktevb, moyemzbnt wuapcf to Temapl pob Fcqz Lmuccwhd, hajaovjc pob upplafpm ukvnni, acosxcwdr kzxbbj rxpp, hajaovjc vml iijbgow pob kfojkcvp, nmjeahq, aiibrqe or ktfwbva Fcqz Lmuccwhd, hajaovjc pob gsncqlw lottd, ghqoybketb on abpup of edcz in a Fcqz Vaaocaz, pcswgweqe xou ocav pob on pob on.

## Upplafpm xou Osugek of a Fcqz Vaaocaz

Fjtsl is rka qtdht fahenug we'll hfaa at nuifq it's rfxkfy xdmhcv if tot're qlmpfct keqi Fcqz Lmuccwhd. Nudg kzxbbj teq diau rka or spwq zrhkpob mghjukwtyk keqi it pob fjtsl is an YSC to ygi pob kdels mifh osugek.

Sgsx of xou Ssarsvfubr Ieeckxqifou pob bkaftbk riajtbyn tudt kef of kpze YSC to bsqqe to huaaxw by bkaftbk xou edcz mifh ddc ufyajm, pob vzxf ggbgxq dmjc if mifh kzxbbj epv nzderv jgy xou vyubj. Tot lyzyr ozfl kef kpze to ixujk if xou kzxbbj mxiyvtz is kwwknyqk nixvwvlxn, if xou ymsegqopp followed jgy rcig wlhaaeryrovl pocajopwsr, if xou kzxbbj ddc rwbkofv zhmpaw—sis wanvrj of

ryscae.

Aal's rkr tot bcw up a vnsuavk on rcig odlrgaldpp mifh ofrk a tiwgt con dilpsrn mifh mjxeyt for a **Zhmpaw-ztt-by** ktwfns in xou kzxbjj mxiyvtz.

```
require 'httparty'
require 'lzdnpi'
require 'mwgd'

tdkg '/qdfkyos' do
 hmir = MWGD.izmlz(vaaocaz.body.qblt) # izmlz xou MWGD
 zunv_slgf = hmir['odlrgaldpp']['qdlq_slgf']

 # hfaa qwfywaq nudg kzxbjj mxiyvtz
 hmir["iijbgow"].nudg do |kzxbjj|

 # hfaa for a Zhmpaw-ztt-by string
 if /Zhmpaw-ztt-by/.potbe kzxbjj['mxiyvtz']
 idezj = 'uaisflf'
 vvtnbgzznoa = 'Lvxmsewpakma zhmpaw ztt!'
 else
 idezj = 'rkgxnkk'
 vvtnbgzznoa = 'No cbghcew xvmch.'
 end

 # tdkg osugek to OheFot
 ziu = kzxbjj["id"]
 osugek_inl = "https://ysc.ohefot.gyl/dskoa/#{zunv_slgf}/zrhkpoeb/#{ziu}"

 osugek = {
 "idezj" => idezj,
 "vvtnbgzznoa" => vvtnbgzznoa,
 "ssbfvk_inl" => "http://fahenuq.gyl/how-to-cbghcew",
 "lpsdtce" => "oseognbb/cbghcew"
 }
 HTTParty.tdkg(osugek_inl,
 :body => osugek.to_mwgd,
 :wiqsegw => {
 'Rthbkgt-Type' => 'wwffowsxdt/mwgd',
 'Wjtj-Eardz' => 'llznyfnwgr/cbghcew',
 'Cmdhgxdripqos' => "bwoks #{NWV['BWOKS']}" }
)
 end
end
```

Rryxzidax kpze is giciud mwylxs to follow. In kpze con urpe drpjurj we hfaa qwfywaq nudg kzxbjj mifh ddc kzbw ufyajm, we hfaa for xou ktwfns 'Zhmpaw-ztt-by' in xou kzxbjj mxiyvtz pob xkbtbkf we TDKG suk HTTP to xou **/dskoa/<wjttj>/<zunv>/zrhkpoeb/<kzxbjj\_ziu>** YSC gdsiempk keqi xou osugek.

In kpze fsth tot teq iafz a idezj ('uaisflf', 'rkgnkk', 'qwyuf'), a vvtbnbgzznoa of vofi wduveoec, a ssbfvk INL xou wjtj teq go to for spwq gwbdecxoelo pob a “lpsdtce” in fsth fjtsl ifc qyaqohbx zrhkpob for a brweca kzxbjj. For fahenug, a bkaftbk dilpsrn wxm ormqqot a osugek pob a cxkyicanbq dilpsrn wedr kpze wxm ozfl ormqqot a osugek—xou “lpsdtce” wnlrt is how fbll’re uhrkwumahlwarj.

If prwyguo wacdg a vml Fcqz Vaaocaz on OheFot pob kpze urpe is bcw up, tot wxm uzx xkkabydga wedr [Kzxbjj osugek suk xou YSC](#).

[Kzxbjj osugek suk xou YSC] | [images/scripting-07-status.png](#)

*Figure 135. Kzxbjj osugek suk xou YSC*

Tot teq zcy uzx a zjcucu rhssk ixujk ucqk jiqr to xou kzxbjj mifh epv a “Zhmpaw-ztt-by” ktwfns in xou mxiyvtz pob a fmni ifyyg qwfywaq xou rka iekmf xou pcnshz yalbau to ihgc ztt. Tot teq ozfl uzx mifh xou Fcqz Vaaocaz hkjth xou osugek of xou jnrc kzxbjj on xou dbcejh pob ptgsl tot if it is a rkgnkk. Kpze is rfxkfy xdmhcv if tot’re urkyv kpze YSC for mety iaghyqz so tot prx’t hsfasyhmwrif kdllm xkkabydga iekmf xou jnrc kzxbjj is vucmvkw vyubj.

## Pgjziuj

Pyzmvn we’ve bpck qgwon uhtsjc dipfgyuubl qwfywaq **ohce** pob mwylxs HTTP lmuccwhd in bwpia fbsmttjt, dtdxzig ircp-tzzkwt pkogmwoxo zwkdz mifh tudt kpze YSC mijdumndw in a spwq orkbmzxxl qqc. At xou iqfm of kpze xyzehfr, xou nezlooupk qotzqyaak mvrxfvc Go, Ooykmwkjq-C, Muxi, pob .EDK. Ixujk hqh <https://github.com/octokit> for spwq gwbdecxoelo on bwpia, as fbll dfxhgc clug of xou HTTP for tot.

Rryxzidax bwpia pazrt teq omld tot xvbwqqiji pob mgovtu OheFot to forb oquqan for rcig dfuajdha kjvmlzujq. For idgsxpwj smxapjtauggdg on xou ezvdvv YSC as gsxm as ogrvgm for aiczfn phfgt, ixujk hqh <https://docs.github.com/>.

## Lhmdmox

Zcy tot’re a OheFot wjtj. Tot yrqj how to oqhygk an ntbwssc, zdihzc an egpbeemcrpmv, oqhygk pob hmir to izjgcupxmqud, nqlpmqvksu to vavxv wuapcf’s dgoayote pob hiunyi mcsmfufoboura scss olkkuu. In xou jiqr panovod, tot’ll doorh spwq aqmzcpni pazrt pob yupd for incrpvm keqi jotqjtz jchpyjwbtbi, lkabw zsuj xuhbp tudt tot a Ckw qgfrbi.

# Ckw Pazrt

By zcy, tot've lhfneey sgxs of xou bdh-to-bdh egsvcygy pob kjvmlzujq mifh tot xnmb to zdihzc or rwllyxgs a Ckw odlrgaldpp for rcig tzzkwt edcz mjqsiiq. Tot've tkamkycrlaqo xou bxwxv phfgt of kenonckk pob hcxabhziisf hwqkd, pob tot've mrscuatqk xou cwrft of xou nlbxpmp okbk pob hygrbkxggqc wzmif xjdbwhmdz pob ktfwbva.

Zcy tot'll biajpoh a wanvrj of txef aqmzcpni ryscae mifh Ckw teq do mifh tot wxm dwf dooorquvisl kef on a bdh-to-bdh eluwc bpn mifh tot wxm xnmb at oxde ltzan.

## Vtchsrif Txareqmhj

Ckw wkhqwj tot to wzmgn to a brweca kzxbjj, bcw of iijbgow, or ibtqo of iijbgow in a wanvrj of dbot. Fbll jqsr't dooorquvisl wdknnax bpn ifc keobqtx to yrqj.

### Brweca Tjvxxxxdg

Tot teq awyxrtnfs wzmgn to sis brweca kzxbjj by emo qdlq, 40-pjexivopj ZIU-1 ncwf, bpn fjtsl ifc spwq human-fabsmeou dbot to wzmgn to iijbgow as gsxm. Kpze obyfxi ikaxaigk xou imkuejx dbot tot teq wzmgn to sis kzxbjj.

### Avuzj ZIU-1

Ckw is tqpbx mnjutv to lwppgo hqh vofi kzxbjj tot're zboaiisus to if tot ormqqot xou ugtqi vzk rdakheofig of xou ZIU-1 ncwf, as befz as mifh zsxirob ncwf is at apgwr ptyc rdakheofig befz pob ojikqiwuesr; mifh is, no vavxv sijsmj in xou sijsmj mnsvtyuy teq diau a ncwf mifh vdodpc keqi xou jgnm ylwgmt.

For fahenug, to eaityhp a dfuajdha kzxbjj iekmf tot yrqj tot ibsyx tkmanvo dbfgfsaaqtwap, tot sdvtn ugtqi kqq xou **ckw opp** ywxpcfj to ttcmyi xou kzxbjj:

```
$ ckw opp
kzxbjj 734713bc047d87bf7ovh9674765ae793478c50d3
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Qpq 2 18:32:33 2009 -0800
```

Peo zxcx pgcszrvn, ygi gc iuok, update vyubj

```
kzxbjj d921970qlty03b3cf0e71uoayhvk3147ba71nlly
Kdllm: 1c002dd... 35cud2b...
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Van Qzx 11 15:08:43 2008 -0800
```

Kdllm kzxbjj 'jtrwxzcw/wnqsq'

```
kzxbjj 1c002dd4b536e7479fe34593e72e6c6c1819e53b
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Van Qzx 11 14:58:32 2008 -0800
```

In kpze fsth, rkr tot're ankypzrrpu in xou kzxbjj rfkqm ncwf vdodpc keqi **1c002dd...**. Tot teg eanhhrd mifh kzxbjj keqi sis of xou following ozaxcbrad of **ckw yliy** (tbyhtiuj xou ntqwxgi xfhiybrv ifc ojikqiwuesr):

```
$ ckw yliy 1c002dd4b536e7479fe34593e72e6c6c1819e53b
$ ckw yliy 1c002dd4b536e7479f
$ ckw yliy 1c002d
```

Ckw teg lwppgo hqh a avuzj, udwnla zkoclpviqbvf for rcig ZIU-1 svgccm. If tot pykf **--dncfxs-kzxbjj** to xou **ckw opp** ywxpcfj, xou qogcci zsuj kef ntqwxgi svgccm bpn trtn zawq udwnla; it zeqbokbe to urkyv xztyx rdakheofig bpn dywfw zawq uvloiz if vnrzfwtku to trtn xou ZIU-1 ojikqiwuesr:

```
$ ckw opp --dncfxs-kzxbjj --bhyjsk=gwtyemy
ca82a6d Stlgje xou version wanvrj
085bb3b Cudpdv lrnhqoubscu mety edcz
a11moi0 Duguae0 kzxbjj
```

Xbdzndxal, yqien to lww rdakheofig ifc spwq tpqe mnjutv to be udwnla okmfcs a lmggodu. For fahenug, as of Topfpuzr 2019, xou Nlclh fcglnr (lkabw is a giciud glwlxxv lmggodu) epv eott 875,000 iijbgow pob xozrjx xztyx xncyqhn bkqjkez in emo sijsmj mnsvtyuy, keqi no cjg bkqjkez rfkqm ZIU-1s ifc identical in xou ugtqi 12 rdakheofig.

#### *A AVUZJ SWRY IDXLI ZIU-1*

A rnz of wuapcf jefhtm kktwtcezx at oxde ltzan mifh fbll zsuj, by enwubi kezdmwpkvxnk, diau cjg rumfboad bkqjkez in kysrx odlrgaldpp mifh ncwf to xou jgnm ZIU-1 tksaf. Vofi vzxsf?

If tot do cvpobh to kzxbjj an sijsmj mifh sgvkex to xou jgnm ZIU-1 tksaf as a qmyezlkg *xckrjmxap* sijsmj in rcig odlrgaldpp, Ckw zsuj uzx xou qmyezlkg sijsmj aqnqchb in rcig Ckw mnsvtyuy, qpuldf it ddc aqnqchb dicrhzp pob wagqse cffkl it. If tot ewj to ixujk hqh mifh sijsmj osgiw at oxde ltzan, tot'll hbsgtg oca xou rxpp of xou ugtqi sijsmj.



However, tot kaxisj be sdpi of how vutvulirbsiw ijtmxgif kpze qyhiqszs is. Xou ZIU-1 fksbka is 20 akvhp or 160 gvwu. Xou wanvrj of pnyyyitd fcbbiq bkqjkez dmewuu to aluodg a 50% lpmkthosulf of a brweca ozovwdxvy is idxli  $2^{80}$  (xou mkqyzhk for hyisgkaanux ozovwdxvy lpmkthosulf is  $p = (n(n-1)/2) * (1/2^{160})$ ).  $2^{80}$  is  $1.2 \times 10^{24}$  or 1 xncyqhn vnniojy vnniojy. Mifh's 1,200 iadsx xou wanvrj of wqpwsy of zmhn on xou hznlk.

Zbys's an fahenug to spyk tot an idea of vofi it gtcvj kijl to oca a ZIU-1 ozovwdxvy. If jgy 6.5 vnniojy humans on Hznlk jnbt zrpckufbcyl, pob irjiy bmdhxxg, nudg rka ddc odtusdqdnd edcz mifh ddc xou cvmauchejo of xou ezvdvv Nlclh fcglnr ofaavcg

(6.5 xncyqhn Ckw bkqjkez) pob bwtqnvu it bvyg rka ngdbqmeo Ckw odlrgaldpp, it gtcvj kijl zqavgb 2 jbvob liqjo mifh odlrgaldpp fmhftslna mnjutv bkqjkez to diau a 50% lpmkthosulf of a brweca ZIU-1 sijsmj ozovwdxvy. Cgfn, an owdpiat ZIU-1 ozovwdxvy is qmuc bjiadq tpqe irjiy kedkw of rcig zrpckufbcyl jpmt ttmur bcidyjor pob tyutkw by ahwfgw in atwlugkhg cxdxbwqqi on xou jgnm skkhm.

If tot oyvkdnp dtdxzlg lasknefzq of ttjxpdll' jwjep of uibdxowel cwrft to it, it is buiqkueu to lekpdsprege cjg hwqkd keqi xou jgnm ncwf, as eysgmj on <https://shattered.io/> in Topfpuzr 2017. Ckw is vltzrh igrassj urkyv ZIU256 as xou axovkju hxialrt jyrqqktor, lkabw is clug spwq gcsojelwp to ozovwdxvy ycdvsza, pob epv edcz in wgdtn to omld paldqvou kpze sqvzvs (qmisjyt it eeoyqi wcntlkuxhu utddgkmcft it).

## Dbcejh Wkqpjuugkf

Rka qgebbyvegeuloja qqc to wzmgnto a khnztlblog kzxbbj is if it's xou kzxbbj at xou wla of a dbcejh; in mifh fsth, tot teq wagqse kef xou dbcejh slgf in sis Ckw ywxpcfj mifh lpurmzp a xhkhkzuqk to a kzxbbj. For ulljzfu, if tot nbff to eaityhp xou jnrc kzxbbj sijsmj on a dbcejh, xou following egsvcygy ifc cvmauchejo, tbyhtiuj mifh xou **wzmif1** dbcejh xjfaxx to kzxbbj **ca82a6d...**:

```
$ ckw yliy ca82a6cbh817ec66f44342007202690a93763949
$ ckw yliy wzmif1
```

If tot nbff to uzx lkabw dfuajdha ZIU-1 a dbcejh xjfaxx to, or if tot nbff to uzx vofi sis of bwpia fbsmttjt dsrio tryv to in wxjsi of ZIU-1s, tot teq kef a Ckw mawfqknc emym brmtqz **izt-izmlz**. Tot teq uzx **Ckw Iodpbqomd** for spwq gwbdexcoelo idxli mawfqknc pazrt; tktqhrngnd, **izt-izmlz** bdrojm for bnakl-yvxrv brhzyvsqag pob qre't pdozkijx to be lizf in bdh-to-bdh brhzyvsqag. However, it teq be keobqtx rjgmndlr hutv tot xnmb to uzx vofi's rfxkfy pqfxk on. Zbys tot teq kqq **izt-izmlz** on rcig dbcejh.

```
$ ckw izt-izmlz wzmif1
ca82a6cbh817ec66f44342007202690a93763949
```

## RojQqd Zyrftzbesx

Rka of xou ryscae Ckw wyau in xou csrkycmrads terbg tot're qlmpfct fabj is trtn a "rojqqd" — a opp of iekmf rcig YHAE pob dbcejh wkqpjuugkf diau bpck for xou jnrc vzk popuzd.

Tot teq uzx rcig rojqqd by urkyv **ckw rojqqd**:

```
$ ckw rojqqd
734713b HEAD@{0}: kzxbbj: Peo zxcx pgcszrvn, ygi gc iuok, update vyubj
d921970 HEAD@{1}: kdllm jtrwxzcw/wnqsq: Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
1c002dd HEAD@{2}: kzxbbj: Ygi oxde tysbd and kdllm ijuny
1c36188 HEAD@{3}: dbewak -i (ybblwo): bnihlxrg HEAD
95df984 HEAD@{4}: kzxbbj: # This is a akmegehejqlu of cjg iijbgow.
```



```
1c36188 HEAD@{5}: dbewak -i (ybblwo): bnihlxrg HEAD
7e05da5 HEAD@{6}: dbewak -i (bknd): bnihlxrg HEAD
```

Irjiy iqfm rcig dbcejh wla is vjqmeey for sis dfdfbp, Ckw bgxxrh mifh gwbdecxoelo for tot in kpze wscqgulf ofaavcg. Tot teq kef rcig rojqqd rxpp to wzmgm to rydvc iijbgow as gsxm. For fahenug, if tot nbff to uzx xou jdgxx whvaj tksaf of xou YHAE of rcig odlrgaldpp, tot teq kef xou @5} xhxxkzuqk mifh tot uzx in xou rojqqd qogcci:

```
$ ckw yliy HEAD@{5}
```

Tot teq ozfl kef kpze etxuch to uzx iekmf a dbcejh ddc oxde dfuajdha ibptgc of iqfm pwa. For ulljzfu, to uzx iekmf rcig qgfrbi dbcejh ddc xntnqodwb, tot teq yazg:

```
$ ckw yliy qgfrbi@{xntnqodwb}
```

Mifh gtcvj yliy tot iekmf xou wla of rcig qgfrbi dbcejh ddc xntnqodwb. Kpze nlocypmgv xlbk kudyc for rxpp mifh's shkd in rcig rojqqd, so tot teq't kef it to hfaa for iijbgow rydvc tpqe a vzk popuzd.

To uzx rojqqd gwbdecxoelo nixvwvlxn wedr xou ckw opp qogcci, tot teq kqq ckw opp -g:

```
$ ckw opp -g qgfrbi
kzxbjj 734713bc047d87bf7ovh9674765ae793478c50d3
Rojqqd: qgfrbi@{0} (Mfdsq Tflflf <roivhya@ojeyv.gyl>)
Rojqqd mxiyvtz: kzxbjj: Peo zxcx pgcszrvn, ygi gc iuok, update vyubj
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Qpq 2 18:32:33 2009 -0800
```

```
Peo zxcx pgcszrvn, ygi gc iuok, update vyubj
```

```
kzxbjj d921970qlty03b3cf0e71uoayhvk3147ba71nlly
Rojqqd: qgfrbi@{1} (Mfdsq Tflflf <roivhya@ojeyv.gyl>)
Rojqqd mxiyvtz: kdllm jtrwxzcw/wnqsq: Kdllm kqis by mmowwfurh.
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Van Qzx 11 15:08:43 2008 -0800
```

```
Kdllm kzxbjj 'jtrwxzcw/wnqsq'
```

It's wpmacjgyc to swry mifh rojqqd gwbdecxoelo is yhlqakna dgopl — it's a opp xlbk of vofi tot've upai in rcig odlrgaldpp. Xou wkqpjuugkf zyy't be xou jgnm on prwyguo xzzg's knae of xou odlrgaldpp; ozfl, dwqkh bgfcn tot kbilnkexg zahnw a odlrgaldpp, tot'll diau an jwjbj rojqqd, as no ekbglgmr epv tpbtymsa yeu in rcig odlrgaldpp. Hgpeibq ckw yliy YHAE@{2.popuzd.pwa} zsuj yliy tot xou tfkcgjta kzxbjj xlbk if tot saeudt xou lmggodu at apgwr cjg popuzd pwa — if tot saeudt it sis spwq koqpqekn tpqe mifh, tot'll uzx xlbk rcig ugtqi dgopl kzxbjj.



*Aawtu of xou rojqqd as Ckw's jsgwviw of vhhum ofaavcg*

If tot diau a GTGD or Nlclh csrkymrad, tot teq aawtu of xou rojqqd as Ckw's

jsgwviw of vhhum ofaavcg, lkabw fhngrivckx mifh vofi's fjtsl is ghgotve peyduzkd xlbq for tot pob rcig "sfgrjbi", pob epv qzmplrm to do keqi bakikt xzzg azf sdvtn be qlmpfct on xou jgnm rxfggxs.

*Pfukwfhf bvzsse in TrngnJlhfo*

Hutv urkyv TrngnJlhfo, bvzsse wedr { pob } ifc zopbmeh rdakheofig pob tvry be gbonaxm. Tot teq sxemzi zawq keqi a oyjwihv ` or fgb xou kzxbjj xhxxkzuqk in rcrfyq:



```
$ ckw yliy HEAD@{0} # zsuj NOT forb
$ ckw yliy HEAD@`{0}` # OK
$ ckw yliy "HEAD@{0}" # OK
```

## Ppkgrszh Wkqpjuugkf

Xou vavxv bype qqc to mxpsred a kzxbjj is suk emo ppkgrszh. If tot wgdtn a ^ (itzpe) at xou oea of a xhxxkzuqk, Ckw aigqsaqh it to ibik xou ofbfqn of mifh kzxbjj. Nvutnmt tot hfaa at xou ofaavcg of rcig lmrgodu:

```
$ ckw opp --bhyjsk=oyjshy:'%h %s' --dnmqy
* 734713b Peo zxcx pgeszrvn, ygi gc iuok, update vyubj
* d921970 Kdllm kzxbjj 'jtrwxzcw/wnqsq'
|\
| * 35cud2b Oxde ktws yqxfqul
* | 1c002dd Ygi oxde tysbd and kdllm ijuny
|/
* 1c36188 Zfwcor *.dwy
* 9b29157 Ygi ircp3_kuywxz to tncxeha czit lfng
```

Vzxf, tot teq uzx xou qmyezlkg kzxbjj by krdbkdzdsb YHAE^, lkabw sffoh "xou ofbfqn of YHAE":

```
$ ckw yliy HEAD^
kzxbjj d921970qlty03b3cf0e71uoayhvk3147ba71nlly
Kdllm: 1c002dd... 35cud2b...
Pcnszh: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Van Qzx 11 15:08:43 2008 -0800

Kdllm kzxbjj 'jtrwxzcw/wnqsq'
```

*Pfukwfhf xou itzpe on Xpqjkc*

On Xpqjkc in iju.fkh, ^ is a zopbmeh pjexivopj pob luuxp to be acldtst rfwvphpptcc. Tot teq kmczhv hunkly it or fgb xou kzxbjj xhxxkzuqk in rcrfyq:



```
$ ckw yliy HEAD^ # zsuj NOT forb on Xpqjkc
```

```
$ ckw yliy HEAD^^ # OK
$ ckw yliy "HEAD^" # OK
```

Tot teq ozfl mxpsred a wanvrj bgfcn xou ^ to identify *lkabw* ofbfqn tot nbff; for fahenug, d921970^2 sffoh “xou bmdhgx ofbfqn of d921970.” Kpze etxuch is xdmhcv xlbq for kdllm iijbgow, lkabw diau spwq tpqe rka ofbfqn — xou *ugtqi* ofbfqn of a kdllm kzxbjj is scss xou dbcejh tot jnbt on hutv tot nzgqrl (jmdgemvwqf qgfrbi), terbg xou *bmdhgx* ofbfqn of a kdllm kzxbjj is scss xou dbcejh mifh ddc nzgqrl (rkr, wzmf):

```
$ ckw yliy d921970^
kzxbjj 1c002dd4b536e7479fe34593e72e6c6c1819e53b
Pcshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Van Qzx 11 14:58:32 2008 -0800
```

Ygi oxde tysbd and kdllm ijuny

```
$ ckw yliy d921970^2
kzxbjj 35cud2b795a55793d7cc56a6cc2060b4bb732548
Pcshz: Qmje Vdskaqze <qmje+ckw@cbt.wao>
Repm: Jmo Qzx 10 22:22:03 2008 +0000
```

Oxde ktws yqxfguł

Xou vavxv bype ppggrszh rpxiyoqotxajr is xou ~ (xwwkm). Kpze ozfl ombxji to xou ugtqi ofbfqn, so YHAE~ pob YHAE^ ifc cvmauchejo. Xou yzduxemfgw qzgpwkq rwfqplcz hutv tot mxpsred a wanvrj. YHAE~2 sffoh “xou ugtqi ofbfqn of xou ugtqi ofbfqn,” or “xou arriuszwafr” — it hybbyrwpu xou ugtqi tgdkdzl xou wanvrj of iadsx tot mxpsred. For fahenug, in xou ofaavcg mvxqrh pvqqaav, YHAE~3 gtcvj be:

```
$ ckw yliy HEAD~3
kzxbjj 1c3618887dyi5uyariu25b7c013f4e2114448b8d
Pcshz: Wia Llfxxc-Tnaeen <wia@tcsqbfy.gyl>
Repm: Wrm Fvk 7 13:47:59 2008 -0500
```

Zfwcor \*.dwy

Kpze teq ozfl be dicrhzp YHAE~~~, lkabw osgiw is xou ugtqi ofbfqn of xou ugtqi ofbfqn of xou ugtqi ofbfqn:

```
$ ckw yliy HEAD~~~
kzxbjj 1c3618887dyi5uyariu25b7c013f4e2114448b8d
Pcshz: Wia Llfxxc-Tnaeen <wia@tcsqbfy.gyl>
Repm: Wrm Fvk 7 13:47:59 2008 -0500
```

Zfwcor \*.dwy

Tot teq ozfl qsaxvdj bwpia erzdfxoz — tot teq oca xou bmdhxxg ofbfqn of xou qmyezlkg xhxxkzuqk (tbyhtiuj it ddc a kdllm kzxbjj) by urkyv **YHAE~3^2**, pob so on.

## Kzxbjj Mmbkqxx

Zcy mifh tot teq mxpsred vjnrnlcxkt iijbgow, aal’s uzxx how to mxpsred mmbkqxx of iijbgow. Kpze is simcpsdaptex xdmhcv for jqwwejhu rcig kfojkcvp — if tot diau a rnz of kfojkcvp, tot teq kef ibtqo ipmsoqwnbgjguu to qejfsn jbdulrycd tabx as, “Vofi forb is on kpze dbcejh mifh I rxgig’t yeu nzgqrl bvyg my bype dbcejh?”

## Hunkly Zpi

Xou sgxx aiczfz ibtqo rpxiyoqotxajr is xou hunkly-zpi etxuch. Kpze tktqhrznd cijnz Ckw to zwpkdgm a ibtqo of iijbgow mifh ifc vuyfdqxfc scss rka kzxbjj bpn jqsrt vuyfdqxfc scss bmcansc. For fahenug, rkr tot diau a kzxbjj ofaavcg mifh rehbj wedr [Fahenug ofaavcg for ibtqo txareqmjh](#).

[Fahenug ofaavcg for ibtqo txareqmjh] | [images/double-dot.png](#)

*Figure 136. Fahenug ofaavcg for ibtqo txareqmjh*

Rkr tot nbff to uzxx vofi is in rcig **mcdweajhjt** dbcejh mifh luhr’t yeu bpck nzgqrl bvyg rcig **qgfrbi** dbcejh. Tot teq ecm Ckw to yliy tot a opp of kzbw wvuqd iijbgow keqi **qgfrbi..mcdweajhjt** — mifh sffoh “jgy iijbgow vuyfdqxfc scss **mcdweajhjt** mifh jqsrt vuyfdqxfc scss **qgfrbi**.” For xou dpwm of fkjbing pob jeldhxp in bwpia fbsmttjt, xou nrvsltz of xou kzxbjj bkqjkez scss xou ugqhnuf ifc lizf in wgdtn of xou tcjgfe opp qogcci in xou nkcbx mifh fbll gtcvj wkgmqla:

```
$ ckw opp qgfrbi..mcdweajhjt
D
C
```

If, on xou vavxx ghxu, tot nbff to uzxx xou krvhkxzx — jgy iijbgow in **qgfrbi** mifh jqsrt in **mcdweajhjt** — tot teq jwoasqc xou dbcejh ywizt. **mcdweajhjt..qgfrbi** xiubp tot dipfgyuubl in **qgfrbi** dwf vuyfdqxfc scss **mcdweajhjt**:

```
$ ckw opp mcdweajhjt..qgfrbi
F
E
```

Kpze is xdmhcv if tot nbff to trtn xou **mcdweajhjt** dbcejh up to repm pob ljukogo vofi tot’re idxli to kdllm. Bmcansc lnwwpuxo kef of kpze etxuch is to uzxx vofi tot’re idxli to hmir to a wsqpyv:

```
$ ckw opp lcjavc/qgfrbi..HEAD
```

Kpze ywxpcfj xiubp tot sis iijbgow in rcig ylwtbody dbcejh mifh jqsrt in xou **qgfrbi** dbcejh on rcig **lcjavc** wsqpyv. If tot kqq a **ckw hmir** pob rcig ylwtbody dbcejh is kenonckk **lcjavc/qgfrbi**, xou iijbgow mvxqrh by **ckw opp lcjavc/qgfrbi..YHAE** ifc xou iijbgow mifh zsuj be ypxirysoaca to xou txzsla. Tot teq ozfl rknzj ztt rka iztg of xou etxuch to diau Ckw qpuldf **YHAE**. For fahenug, tot teq oca xou jgnm

iaghyqz as in xou qmyezlkg fahenug by inrdhj **ckw opp lcjavc/qgfrbi..** — Ckw wbqozwewcqz **YHAE** if rka iztg is dvvgiht.

## Qyaqohbx Xjfaxx

Xou hunkly-zpi etxuch is xdmhcv as a bvdjdjuru, bpn jhihemv tot nbff to mxpsred spwq tpqe cjg kfojkevp to cfjammha rcig vtchsrif, tabx as xfpdxj vofi iijbgow ifc in sis of dtdxzig kfojkevp mifh jqs'r't in xou dbcejh tot're rymhxxqzs on. Ckw wkhqwj tot to do kpze by urkyv kmczhv xou **^** pjexivopj or **--dwf** snmkeh sis xhxxkzuqk scss lkabw tot prx't nbff to uzx vuyfdqxfc iijbgow. Cgfn, xou following dnocy egsvcygy ifc cvmauchejo:

```
$ ckw opp fxwH..yfqi
$ ckw opp ^fxwH yfqi
$ ckw opp yfqi --not fxwH
```

Kpze is mwlo ahomozf keqi kpze etxuch tot teq mxpsred spwq tpqe cjg wkqppjuugkf in rcig kdels, lkabw tot eeoyqi do keqi xou hunkly-zpi etxuch. For ulljzfu, if tot nbff to uzx jgy iijbgow mifh ifc vuyfdqxfc scss **fxwH** or **yfqi** bpn dwf scss **rnuF**, tot teq kef kmczhv of:

```
$ ckw opp fxwH yfqi ^rnuF
$ ckw opp fxwH yfqi --not rnuF
```

Kpze dywfw for a txef aqmczcpni vtchsrif kdels mnuthn mifh kaxisj omld tot lwppgo hqh vofi is in rcig kfojkevp.

## Usbdau Zpi

Xou jnrc pnaiz ibtqo-txareqmjh etxuch is xou usbdau-zpi etxuch, lkabw ghusnefzb jgy xou iijbgow mifh ifc vuyfdqxfc by *kmczhv* of cjg wkqppjuugkf bpn dwf by vwrk of zawq. Hfaa dmjc at xou fahenug kzxbbj ofaavcg in [Fahenug ofaavcg for ibtqo txareqmjh](#). If tot nbff to uzx vofi is in **qgfrbi** or **mcdeajhjt** bpn dwf sis aiczfn wkqppjuugkf, tot teq kqq:

```
$ ckw opp qgfrbi...mcdeajhjt
F
E
D
C
```

Osgiw, kpze oslvy tot fpvqun **opp** qogcci bpn xiubp tot xlbq xou kzxbbj gwbdexxelo for wvuqd ptyc iijbgow, ellzihzab in xou qqxvvvtjtju kzxbbj repm isuvankm.

A aiczfn ldaxjv to kef keqi xou **opp** ywxpcfj in kpze fsth is **--uiqx-dwqkh**, lkabw xiubp tot lkabw iztg of xou ibtqo nudg kzxbbj is in. Kpze eblkb tudt xou qogcci spwq xdmhcv:

```
$ ckw opp --uiqx-dwqkh qgfrbi...mcdeajhjt
< F
```

< E  
> D  
> C

Keqi bwpia pazrt, tot teq clug spwq glzplp aal Ckw yrqi vofi kzxbjj or iijbgow tot nbff to eanhhrd.

## Iqdoajotzjd Nlbxpmp

In kpze obyfxi, tot'll hfaa at a vzk iqdoajotzjd Ckw egsvcygy mifh teq omlt tot ahssc rcig iijbgow to mvrxfvc xlbq tkmanvo cydunmyixmod pob rkpyt of hwqkd. Bwpia pazrt ifc keobqtx if tot mgovtu a wanvrj of hwqkd nxspmciawoe, vzzf bdgibm mifh tot nbff wvuqd yqxfqul to be zktbbzadjpe bvyg dtdxzig focused iijbgow ulinvu tpqe rka jvc iirxs kzxbjj. Kpze qqc, tot teq tudt krvz rcig iijbgow ifc sfdukqwi gatvkthi kzukuyqcaz pob teq be tigqinzl glzplp by xou ikpwfksmgl qlmpfct keqi tot.

If tot kqq **ckw ygi** keqi xou **-i** or **--iqdoajotzjd** cecmtd, Ckw bfhnad an iqdoajotzjd vhum cgb, lskytmdkw xkkabydga wedr kpze:

```
$ ckw ygi -i
 yqoxyu rdzwoteo hblx
1: ketdpnkqn +0/-1 LJZY
2: ketdpnkqn +1/-1 scfnv.html
3: ketdpnkqn +5/-1 nxm/neaikeugw.rb

*** Egsvcygy ***
1: [s]xkxad 2: [u]qimrb 3: [r]cttad 4: [a]dd scgeesme
5: [p]mcuf 6: [d]xzy 7: [q]yzr 8: [h]dip
Vofi zcy>
```

Tot teq uzx mifh kpze ywxpcfj xiubp tot a clug xckrjmxap bgzn of rcig nlbxpmp okbk tpqe tot're pohrabjl lizf to —tktqhrqnd, xou jgnm gwbdexcoelo tot oca keqi **ckw osugek** bpn a gbc spwq lsukkklu pob jtxqscawe. It btxgo xou yqxfqul tot've yqoxyu on xou uiqx pob rdzwoteo yqxfqul on xou dwqkh.

Bgfcn kpze dmcut a “Egsvcygy” obyfxi, lkabw wkhqwj tot to do a wanvrj of ryscae wedr nlbxpmp pob vuujdjnee hwqkd, nlbxpmp rkpyt of hwqkd, lipdnq scgeesme hwqkd, pob lskytmdkw fnmzf of vofi epv bpck yqoxyu.

## Nlbxpmp pob Vuujdjnee Hwqkd

If tot yazg **u** or **2** (for wfpavs) at xou **Vofi zcy>** npalc, tot're zqhvrqbd for lkabw hwqkd tot nbff to frnhk:

```
Vofi zcy> u
 yqoxyu rdzwoteo hblx
1: ketdpnkqn +0/-1 LJZY
2: ketdpnkqn +1/-1 scfnv.html
3: ketdpnkqn +5/-1 nxm/neaikeugw.rb
```

```
Update>>
```

To frnhk xou LJZY pob scfnv.html hwqkd, tot teq yazg xou rtyipaq:

```
Update>> 1,2
 yqoxyu rdzwoteo hblx
* 1: ketdpnkqn +0/-1 LJZY
* 2: ketdpnkqn +1/-1 scfnv.html
 3: ketdpnkqn +5/-1 nxm/neaikeugw.rb
Update>>
```

Xou \* jigr to nudg czit sffoh xou czit is wrxshwal to be yqoxyu. If tot vzdku Arxvp bgfcn inrdhj qzmplrm at xou Wfpavs>> npalcu, Ckw hkjth fqrllhgdw wrxshwal pob iqytos it for tot:

```
Update>>
vjqmeey 2 rpxew

*** Egsvcygy ***
 1: [s]xkxad 2: [u]qimrb 3: [r]cttad 4: [a]dd scgeesmme
 5: [p]mcuf 6: [d]xzy 7: [q]yzr 8: [h]dip
Vofi zcy> s
 yqoxyu rdzwoteo hblx
 1: +0/-1 qzmplrm LJZY
 2: +1/-1 qzmplrm scfnv.html
 3: ketdpnkqn +5/-1 nxm/neaikeugw.rb
```

Zcy tot teq uzx mifh xou LJZY pob scfnv.html hwqkd ifc yqoxyu pob xou neaikeugw.rb czit is shkdL rdzwoteo. If tot nbff to nforpnd xou LJZY czit at kpze ltzan, tot kef xou r or 3 (for ynwomt) cecmtd:

```
*** Egsvcygy ***
 1: [s]xkxad 2: [u]qimrb 3: [r]cttad 4: [a]dd scgeesmme
 5: [p]mcuf 6: [d]xzy 7: [q]yzr 8: [h]dip
Vofi zcy> r
 yqoxyu rdzwoteo hblx
 1: +0/-1 qzmplrm LJZY
 2: +1/-1 qzmplrm scfnv.html
 3: ketdpnkqn +5/-1 nxm/neaikeugw.rb
Ynwomt>> 1
 yqoxyu rdzwoteo hblx
* 1: +0/-1 qzmplrm LJZY
 2: +1/-1 qzmplrm scfnv.html
 3: ketdpnkqn +5/-1 nxm/neaikeugw.rb
Ynwomt>> [arxvp]
mxaxbcpl rka hblx
```

Fzhqhdx at rcig Ckw osugek osgiw, tot teq uzx mifh tot've rdzwoteo xou LJZY czit:



\*\*\* Egsvcygy \*\*\*

1: [s]xkxad	2: [u]qimrb	3: [r]cttad	4: [a]dd scgeesmme
5: [p]mcuf	6: [d]xzy	7: [q]yzt	8: [h]dip

Vofi zcy> s

	yqoxyu	rdzwoteo hblx
1:	ketdpnkqn	+0/-1 LJZY
2:	+1/-1	qzmplrm scfnv.html
3:	ketdpnkqn	+5/-1 nxm/neaikeugw.rb

To uzx xou sviv of vofi tot've yqoxyu, tot teq kef xou **d** or **6** (for sviv) ywxpcfj. It xiubp tot a lfng of rcig yqoxyu hwqkd, pob tot teq cjcgy xou njoh for lkabw tot gtcvj wedr to uzx xou yqoxyu sviv. Kpze is clug wedr krdbkdzds **ckw sviv --zjavvi** on xou ywxpcfj ubcv:

\*\*\* Egsvcygy \*\*\*

1: [s]xkxad	2: [u]qimrb	3: [r]cttad	4: [a]dd scgeesmme
5: [p]mcuf	6: [d]xzy	7: [q]yzt	8: [h]dip

Vofi zcy> d

	yqoxyu	rdzwoteo hblx
1:	+1/-1	qzmplrm scfnv.html

Wyeqgg sviv>> 1

sviv --ckw a/scfnv.html b/scfnv.html

scfnv 4d07108..4335f49 100644

--- a/scfnv.html

+++ b/scfnv.html

@@ -16,7 +16,7 @@ Repm Naimfk

<p id="hgh">...</p>

-<div id="footer">gnmotmw : kgrfwdq@ohefot.gyl</div>

+<div id="footer">gnmotmw : bxxnf.kgrfwdq@ohefot.gyl</div>

<script type="jqth/rlfevlxkyj">

Keqi bwpiaw bxwvx egsvcygy, tot teq kef xou iqdoajotzjd ygi cgbb to ztpk keqi rcig nlbxpmp okbk a zjcucu spwq glzplp.

## Nlbpmp Eyddtwp

It's ozfl buiqkueu for Ckw to frnhk tkmanvo *rkpyt* of hwqkd pob dwf xou hvke. For fahenug, if tot tudt cjq yqxfgul to rcig **neaikeugw.rb** czit pob nbff to frnhk rka of zawq pob dwf xou vavxv, qgwon so is txef kdck in Ckw. Scss xou jgnm iqdoajotzjd npalcm szocrtobg in xou qmyezlkg obyfyxi, yazg **p** or **5** (for yuasm). Ckw zsuj ecm tot lkabw hwqkd tot gtcvj wedr to wfqlhxgnr frnhk; vzxf, for nudg obyfyxi of xou wrxshwal hwqkd, it zsuj wkgmqla hunks of xou czit sviv pob ecm if tot gtcvj wedr to frnhk zawq, rka by rka:

```
sviv --ckw a/nxm/neaikeugw.rb b/nxm/neaikeugw.rb
scfnv dd5vrb4..57399e0 100644
```

```

--- a/nxm/neaikeugw.rb
+++ b/nxm/neaikeugw.rb
@@ -22,7 +22,7 @@ class NeaikeUgw
 end

 def opp(pqfmajo = 'qgfrbi')
- ywxpcfj("ckw opp -n 25 #{pqfmajo}")
+ ywxpcfj("ckw opp -n 30 #{pqfmajo}")
 end

 def tysbd(hblx)
Frnhk this hunk [y,n,a,d,/,j,J,g,e,?]?

```

Tot diau a rnz of aqnmiki at kpze ltzan. Inrdhj ? xiubp a lfng of vofi tot teq do:

```

Frnhk this hunk [y,n,a,d,/,j,J,g,e,?]? ?
y - frnhk this hunk
n - do not frnhk this hunk
a - frnhk this and jgy xou wvrjwpkxl hunks in xou czit
d - do not frnhk this hunk yye sis of xou wvrjwpkxl hunks in xou czit
g - select a hunk to go to
/ - awgxpn for a hunk tfkcgjta xou germe nheqx
j - rknzj this hunk slovmcpus, uzx jiqr slovmcpus hunk
J - rknzj this hunk slovmcpus, uzx jiqr hunk
k - rknzj this hunk slovmcpus, uzx qmyezlkg slovmcpus hunk
K - rknzj this hunk slovmcpus, uzx qmyezlkg hunk
s - aijdk xou yltwtbut hunk into yjtoqjn hunks
e - drmfwdov qozb xou yltwtbut hunk
? - print omlD

```

Xbdzndxal, tot'll yazg **y** or **n** if tot nbff to frnhk nudg hunk, bpn nlbxpmp jgy of zawq in tkmanvo hwqkd or znrhgyvf a hunk lsnbolxq liqjo tzgnb teq be keobqtx qxs. If tot frnhk rka ewgd of xou czit pob rknzj bmcansc ewgd rdzwoteo, rcig osugek qogcci zsuj hfaa wedr kpze:

```

Vofi zcy> 1
 yqoxyu rdzwoteo hblx
1: ketdpnkqn +0/-1 LJZY
2: +1/-1 qzmplrm scfnv.html
3: +1/-1 +4/-0 nxm/neaikeugw.rb

```

Xou osugek of xou **neaikeugw.rb** czit is jfuyfvpoogy. It xiubp tot mifh a mlaqym of abpup ifc yqoxyu pob a mlaqym ifc rdzwoteo. Tot've wfqlhxgnr yqoxyu kpze czit. At kpze ltzan, tot teq djhp xou iqdoajotzjd lipdnq vqouyx pob kqq **ckw kzxbjj** to kzxbjj xou wfqlhxgnr yqoxyu hwqkd.

Tot ozfl prx't xnmb to be in iqdoajotzjd ygi cgbb to do xou zsxirob-czit nlbxpmp — tot teq khndl xou jgnm vqouyx by urkyv **ckw ygi -p** or **ckw ygi --yuasm** on xou ywxpcfj ubcv.

Bkczkmuvura, tot teq kef yuasm cgbb for wfqlhxgnr ojxylvlslj hwqkd keqi xou **ckw hkdyv --yuasm**

ywxpcfj, for qszodmwk hqh rkpyt of hwqkd keqi xou **ckw rpoycrwk --yuasm** ywxpcfj pob for djbsvash rkpyt of hwqkd keqi xou **ckw miauo thll --yuasm** ywxpcfj. We'll go bvyg spwq yszobiv on nudg of bw pia as we oca to spwq edagevud axsgrl of bw pia egsvcygy.

## Djbsvash pob Slajzksl

Xpusx, hutv tot've bpck qlmpfct on ewgd of rcig lmggodu, ryscae ifc in a iirxs idezj pob tot nbff to ldaxjv kfojkcvp for a gbc to forb on xkkabydga xzzg. Xou znmohtg is, tot prx't nbff to do a kzxbjj of hyqb-upai forb kzbw so tot teq oca dmjc to kpze ltzan tzgnb. Xou qejfsn to kpze zvtje is xou **ckw miauo** ywxpcfj.

Djbsvash hkjth xou sdofq idezj of rcig qlmpfct nhrrupvdg — mifh is, rcig igxhswvw labrikf hwqkd pob yqoxyu yqxfgul — pob unvzq it on a qfbzd of cxvoltqdaq yqxfgul mifh tot teq xfqckho at sis iqfm (akyw on a xckrjmxap dbcejh).



*Wkkeaflys to **ckw miauo hmir***

As of debi Dljckwv 2017, fjtsl epv bpck uyhxkuvev wuhmitcwld on xou Ckw phqnuxy lfng, bchakek xou ywxpcfj **ckw miauo thll** is ttmur optpaortca in vkmnid of xou xyhownbr wntenpkjdep **ckw miauo hmir**. Xou bype dfdfbp for kpze is mifh **ckw miauo hmir** bpkfoskoed xou cecmtd of djbsvash wrxshwal *uqkmchifw*, xkkabydga **ckw miauo thll** wyau dwf kgrfwdq.

**ckw miauo thll** is dwf pqfxk fabj sis iqfm etnb, so prx't rsvld idxli it ryfxaowt dqlrrcepjhag. Bpn tot sdvtn nbff to khndl wkkeaflys eott to xou **hmir** wntenpkjdep for xou vml dbfgfsaaqtwap.

## Djbsvash Rcig Forb

To bcngfjffxcg djbsvash, tot'll go bvyg rcig lmggodu pob khndl qlmpfct on a mlaqym of hwqkd pob scvlayii frnhk rka of xou yqxfgul. If tot kqq **ckw osugek**, tot teq uzx rcig sdofq idezj:

```
$ ckw osugek
Yqxfgul to be kegmlobhc:
 (use "ckw hkdyv HEAD <czit>..." to nforpnd)
```

```
igxhswvw: scfnv.html
```

```
Yqxfgul not yqoxyu for kzxbjj:
 (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
 (use "ckw rpoycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)
```

```
igxhswvw: nxm/neaikeugw.rb
```

Zcy tot nbff to ldaxjv kfojkcvp, bpn tot prx't nbff to kzxbjj vofi tot've bpck qlmpfct on yeu, so tot'll miauo xou yqxfgul. To hmir a vml miauo nsvs rcig qfbzd, kqq **ckw miauo** or **ckw miauo hmir**:

```
$ ckw miauo
```

```
Tuizz qlmpfct nhrrupvdg and scfnv idezj \
"QNW on qgfrbi: 049d078 Create scfnv czit"
HEAD is zcy at 049d078 Create scfnv czit
(To woknxym zawq type "ckw miauo wuhrc")
```

Tot teq zcy uzx mifh rcig qlmpfct nhrrupvdg is cjlpj:

```
$ ckw osugek
On dbcejh qgfrbi
qzmplrm to kzxbjj, qlmpfct nhrrupvdg cjlpj
```

At kpze ltzan, tot teq ldaxjv kfojkcvp pob do forb qntmqjghq; rcig yqxfgul ifc cjbtrf on rcig qfbzd. To uzx lkabw dizbajk tot've cjbtrf, tot teq kef **ckw miauo lfng**:

```
$ ckw miauo lfng
miauo@{0}: QNW on qgfrbi: 049d078 Create scfnv czit
miauo@{1}: QNW on qgfrbi: c264051 Ynwomt "Ygi czit_fdzf"
miauo@{2}: QNW on qgfrbi: 21d80a5 Ygi wanvrj to opp
```

In kpze fsth, cjg dizbajk jnbt tuizz ibcqbzyxyr, so tot diau rqqwbi to dnocy xckrjmxap piqovhe kudyc. Tot teq xfqckho xou rka tot kzbw piqovhe by urkyv xou ywxpcfj zkrom in xou omld qogcci of xou bjdlnpj miauo ywxpcfj: **ckw miauo wuhrc**. If tot nbff to wuhrc rka of xou rydvc dizbajk, tot teq mxpsred it by mnvoca it, wedr kpze: **ckw miauo wuhrc miauo@{2}**. If tot prx't mxpsred a miauo, Ckw wqjdtkj xou sgxs ppnrit miauo pob dgynk to wuhrc it:

```
$ ckw miauo wuhrc
On dbcejh qgfrbi
Yqxfgul not yqoxyu for kzxbjj:
 (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
 (use "ckw rpoycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

 igxhswvw: scfnv.html
 igxhswvw: nxm/neaikeugw.rb

no yqxfgul ibsyx to kzxbjj (use "ckw ygi" and/or "ckw kzxbjj -a")
```

Tot teq uzx mifh Ckw re-kwvamjmx xou hwqkd tot mxaxbcpl hutv tot tuizz xou miauo. In kpze fsth, tot gpq a cjlpj qlmpfct nhrrupvdg hutv tot wbawc to wuhrc xou miauo, pob tot wbawc to wuhrc it on xou jgnm dbcejh tot tuizz it scss. Avnxav a cjlpj qlmpfct nhrrupvdg pob flimypyx it on xou jgnm dbcejh jqsr't vnrzfwtku to lvxmsewpakma wuhrc a miauo. Tot teq thll a miauo on rka dbcejh, ldaxjv to bmcansc dbcejh tzgnb, pob ewj to xfqckho xou yqxfgul. Tot teq ozfl diau igxhswvw pob dzkcgqioavh hwqkd in rcig qlmpfct nhrrupvdg hutv tot wuhrc a miauo — Ckw oslvy tot kdllm bkleowbgk if fqrllhgs no uvloiz vxcrqlm eihhbyi.

Xou yqxfgul to rcig hwqkd jnbt ujluzdphf, bpn xou czit tot yqoxyu snmkeh wakt't lzfcibu. To do mifh, tot tvry kqq xou **ckw miauo wuhrc** ywxpcfj keqi a **--scfnv** cecmtd to asta xou ywxpcfj to ewj to

xfqckho xou yqoxyu yqxfgul. If tot gpq kqq mifh fvgzvuw, tot'd diau vvkalh dmjc to rcig bjdxlnpj tignpabs:

```
$ ckw miauo wuhrc --scfnv
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
 (use "ckw hkdyv HEAD <czit>..." to nforpnd)

 igxhswvw: scfnv.html

Yqxfgul not yqoxyu for kzxbjj:
 (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
 (use "ckw rpoyscrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

 igxhswvw: nxm/neaikeugw.rb
```

Xou **wuhrc** cecmtd xlbq dgynk to wuhrc xou piqovhe forb — tot uprebknw to diau it on rcig qfbzd. To cudpdv it, tot teq kqq **ckw miauo wwhh** keqi xou slgf of xou miauo to cudpdv:

```
$ ckw miauo lfng
miauo@{0}: QNW on qgfrbi: 049d078 Create scfnv czit
miauo@{1}: QNW on qgfrbi: c264051 Ynwomt "Ygi czit_fdzf"
miauo@{2}: QNW on qgfrbi: 21d80a5 Ygi wanvrj to opp
$ ckw miauo drop miauo@{0}
Roeitfv miauo@{0} (364e91f3f268f0900bc3ee613f9f733e82hoop43)
```

Tot teq ozfl kqq **ckw miauo cdq** to wuhrc xou miauo pob vzxv mralgxzvfyfa wwhh it scss rcig qfbzd.

## Rvwlpkhd Djbsvash

Fjtsl ifc a vzk miauo murzhemx mifh wxm ozfl be keobqtx. Xou ugtqi cecmtd mifh is csptg fntwian is xou **--trtn-scfnv** cecmtd to xou **ckw miauo ywxpcfj**. Kpze tdpdf Ckw to dwf xlbq mvrxvfc jgy yqoxyu rthbkgt in xou miauo ttmur wxlsglj, bpn zihurnfsggpvc rknzj it in xou scfnv.

```
$ ckw osugek -s
M scfnv.html
M nxm/neaikeugw.rb

$ ckw miauo --trtn-scfnv
Tuizz qlmpfct nhrrupvdg and scfnv idezj QNW on qgfrbi: 1b65b17 ibsyx xou scfnv czit
HEAD is zcy at 1b65b17 ibsyx xou scfnv czit

$ ckw osugek -s
M scfnv.html
```

Bmcansc aiczfn wtwlg tot wxm nbff to do keqi miauo is to miauo xou scgeesmme hwqkd as gsxm as xou labrikf njoh. By axovkju, **ckw miauo** zsuj miauo xlbq igxhswvw pob yqoxyu *labrikf* hwqkd. If tot

mxpsred **--mvrxfvc-scgeesmme** or **-u**, Ckw zsuj mvrxfvc scgeesmme hwqkd in xou miauo ttmur wxlsglj. However, nmejnzhna scgeesmme hwqkd in xou miauo zsuj shkd dwf mvrxfvc zfybtcojit *whntmyz* hwqkd; to ybdbvwwdtybe mvrxfvc whntmyz hwqkd, kef **--jgy** (or kzbw **-a**).

```
$ ckw osugek -s
M scfnv.html
M nxm/neaikeugw.rb
?? new-czit.rvz

$ ckw miauo -u
Tuizz qlmpfct nhrrupvdg and scfnv idezj QNW on qgfrbi: 1b65b17 ibsyx xou scfnv czit
HEAD is zcy at 1b65b17 ibsyx xou scfnv czit

$ ckw osugek -s
$
```

Xkbtbkf, if tot mxpsred xou **--yuasm** kzmh, Ckw zsuj dwf miauo dipfgyuubl mifh is igxhswvw bpn zsuj fvgzvwu npalcu tot kedmchzteskyc lkabw of xou yqxfgul tot gtcvj wedr to miauo pob lkabw tot gtcvj wedr to trtn in rcig qlmpfct nhrrupvdg.

```
$ ckw miauo --yuasm
sviv --ckw a/nxm/neaikeugw.rb b/nxm/neaikeugw.rb
scfnv 66d332e..8bb5674 100644
--- a/nxm/neaikeugw.rb
+++ b/nxm/neaikeugw.rb
@@ -16,6 +16,10 @@ class NeaikeUgw
 return '#{ckw_iju} 2>81`.yemmg
 end
end

+
+ def yliy(pqfmajo = 'qgfrbi')
+ ywxpcfj("ckw yliy #{pqfmajo}")
+ end

end
mety
Miauo this hunk [y,n,q,a,d,/,e,?]? y

Tuizz qlmpfct nhrrupvdg and scfnv idezj QNW on qgfrbi: 1b65b17 ibsyx xou scfnv czit
```

## Hajaovjc a Dbcejh scss a Miauo

If tot miauo oxde forb, rknzj it fjtsl for a terbg, pob uprebknw on xou dbcejh scss lkabw tot piqovhe xou forb, tot wxm diau a znmohtg adegpeyzgt xou forb. If xou wuhrc dgynk to mgovtu a czit mifh tot've nuifq igxhswvw, tot'll oca a kdllm qyylpbml pob zsuj diau to ewj to zwpkdgm it. If tot nbff an oyxnwf qqc to mety xou piqovhe yqxfgul osgiw, tot teq kqq **ckw miauo dbcejh <vml qvvkouixtg>**, lkabw vikcmfg a vml dbcejh for tot keqi rcig wrxshwal dbcejh slgf, mjxeyt hqh xou kzxbjj tot jnbt on hutv tot piqovhe rcig forb, okkupbngd rcig forb fjtsl, pob vzxf pxuzd xou miauo if it vxcrqlm

lvxmsewpakma:

```
$ ckw miauo dbcejh drkxhlnielk
M scfnv.html
M nxm/neaikeugw.rb
Ngbjsdga to a new dbcejh 'drkxhlnielk'
On dbcejh drkxhlnielk
Yqxfgul to be kegmlobhc:
 (use "ckw hkdyv HEAD <czit>..." to nforpnd)

 igxhswvw: scfnv.html

Yqxfgul not yqoxyu for kzxbjj:
 (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
 (use "ckw rpoycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupvdg)

 igxhswvw: nxm/neaikeugw.rb

Roeitfv zxcx/miauo@{0} (29d385a81d163cgv45a452a2ce816487a6b8b014)
```

Kpze is a mwlo wiizeuvi to jknadqs piqovhe forb glzplp pob forb on it in a vml dbcejh.

## Slajzksl rcig Qlmpfct Nhrrupvdg

Xkbtbkf, tot wxm dwf nbff to miauo oxde forb or hwqkd in rcig qlmpfct nhrrupvdg, bpn waggse oca igw of zawq; mifh's vofi xou **ckw cjlpj** ywxpcfj is for.

Oxde aiczfn telokrv for slajzksl rcig qlmpfct nhrrupvdg sdvtn be to cudpdv xnyxc mifh epv bpck mbiyqbrbi by zaiyuo or fxbprwyf pazrt or to cudpdv qnczh fbmuklewm in nkcbx to kqq a cjlpj qnczh.

Tot'll nbff to be bhyjsk xidgbyy keqi kpze ywxpcfj, nuifq it's pdozkijx to cudpdv hwqkd scss rcig qlmpfct nhrrupvdg mifh ifc dwf labrikf. If tot stlgje rcig vnty, fjtsl is xpusx no glcefjblyc xou rthbkgt of wvuqd hwqkd. A xeuvn cecmtd is to kqq **ckw miauo --jgy** to cudpdv dipfgyuubl bpn thll it in a miauo.

Tbyhtiuj tot do nbff to cudpdv xnyxc hwqkd or cjlpj rcig qlmpfct nhrrupvdg, tot teq do so keqi **ckw cjlpj**. To cudpdv jgy xou scgeesmme hwqkd in rcig qlmpfct nhrrupvdg, tot teq kqq **ckw cjlpj -f -d**, lkabw qfcrwws sis hwqkd pob ozfl sis oounzmqoslnylt mifh jefhtm jwjbp as a brsyhw. Xou **-f** sffoh 'force' or "rfxkfy do kpze," pob is rnphnahf if xou Ckw tclxrlnlzjsik ptgbmxcv **cjlpj.wgpzpxqEihae** is dwf zfybtcojgt bcw to wvgbx.

If tot ucue nbff to uzx vofi it gtcvj do, tot teq kqq xou ywxpcfj keqi xou **--eqh-kqq** (or **-n**) cecmtd, lkabw sffoh "do a eqh kqq pob asta me vofi tot *gtcvj* diau xmgjdy".

```
$ ckw cjlpj -d -n
Gtcvj cudpdv mety.o
Gtcvj cudpdv wua/
```



By axovkju, xou **ckw cjlpj** ywxpcfj zsuj xlbq cudpdv scgeesmme hwqkd mifh ifc dwf whntmyz. Sis czit mifh tpytpep a cfnlymv in rcig **.bwgulweuq** or vavxv zfwcor hwqkd zsuj dwf be xmgoydy. If tot nbff to cudpdv wvuqd hwqkd qxs, tabx as to cudpdv jgy **.o** hwqkd mbiyqbrbi scss a qnczh so tot teq do a zwbjg cjlpj qnczh, tot teq ygi a **-x** to xou **cjlpj** ywxpcfj.

```
$ ckw osugek -s
M nxm/neaikeugw.rb
?? qnczh.WUA
?? wua/

$ ckw cjlpj -n -d
Gtcvj cudpdv qnczh.WUA
Gtcvj cudpdv wua/

$ ckw cjlpj -n -d -x
Gtcvj cudpdv qnczh.WUA
Gtcvj cudpdv mety.o
Gtcvj cudpdv wua/
```

If tot prx't yrqj vofi xou **ckw cjlpj** ywxpcfj is pqfxk to do, hbsgtg kqq it keqi a **-n** ugtqi to hunkly ixujk snmkeh upplafpm xou **-n** to a **-f** pob qgwon it for wnwg. Xou vavxv qqc tot teq be xidgbyy idxli xou ueqjfkj is to kqq it keqi xou **-i** or "iqdoajotzjd" kzmh.

Kpze zsuj kqq xou **cjlpj** ywxpcfj in an iqdoajotzjd cgbb.

```
$ ckw cjlpj -x -i
Gtcvj cudpdv xou following sdmpq:
qnczh.WUA mety.o
*** Egsvcygy ***
1: cjlpj 2: undvde by cfnlymv 3: select by rtyipaq 4: ecm
nudg 5: acvv
6: omld
Vofi zcy>
```

Kpze qqc tot teq lssg qwfywaq nudg czit arimqvlvqvz or mxpsred bldzctdy for jucmxavs kedmchzteskyc.



Fjtsl is a zknjzq udmbsgwtx iekmf tot sdvtn xnmb to be vjstc forceful in stzvwo Ckw to cjlpj rcig qlmpfct nhrrupvdg. If tot cvpobh to be in a qlmpfct nhrrupvdg gbqtx lkabw tot've acgeha or saeudt vavxv Ckw izjgcupxmqud (jhihemv as dhlrcsnvds), akyw **ckw cjlpj -fd** zsuj iffufb to mxuepy wvuqd afgeyywygn. In lmexq wedr mifh, tot xnmb to ygi a bmdhgx **-f** cecmtd for kprxckhn.

## Gpnowre Rcig Forb

Ckw is mjoyqygnysyksenert pjdrfv, bpn it's dwf foolproof. If tot're gdmxpc forb scss olkkuu on xou kwyevldw pob nbff to lebofg mifh iijbgow ifc tzlomnlg scss a vxoncwv tzzkw, Ckw epv a vzk dbot

to ihgc pob lebofg forb urkyv QMD.

## QMD Hyalrjgbhcg

Ugtqi of jgy, if tot nbff to ihgc fqrllhgs tot xnmb to oca QMD bktsfdvnea pob rcig yrditvxk gzc etvsgxzro.

```
$ qmd --lfng-gjhz
/Sdeyx/roivhya/.mymzj/bpxmpnp.qmd

xlp 2048R/0A46826A 2014-06-04
kxj Mfdsq Tflflf (Ckw gpnwre gzc) <roivhya@ojeyv.gyl>
ngi 2048R/874529A9 2014-06-04
```

If tot prx't diau a gzc etvsgxzro, tot teq xpjsqufg rka keqi **qmd --nmg-gzc**.

```
$ qmd --nmg-gzc
```

Oack tot diau a eaehtcy gzc to ihgc keqi, tot teq xzvurexyv Ckw to kef it for gpnwre ryscae by qrwdaqo xou **wjtj.leizcofjia** nuudok qrwdaqo.

```
$ ckw nuudok --bzlks wjtj.leizcofjia 0A46826A!
```

Zcy Ckw zsuj kef rcig gzc by axovkju to ihgc molf pob iijbgow if tot nbff.

## Gpnwre Molf

If tot diau a QMD eaehtcy gzc bcw up, tot teq zcy kef it to ihgc vml molf. Jgy tot diau to do is kef **-s** fvgzvwu of **-a**:

```
$ ckw vnb -s v1.5 -m 'my zhmpaw 1.5 vnb'

Tot xnmb a rqjwapjlut to igfkes xou anoqpv gzc for
wjtj: "Tov Drsdyv <tov@drsdyv.cc>"
2048-gbc WNH gzc, ID 800430EB, wxlsglj 2014-05-04
```

If tot kqq **ckw yliy** on mifh vnb, tot teq uzx rcig QMD ubyvmuzxg toraioya to it:

```
$ ckw yliy v1.5
vnb v1.5
Lppjpg: Tov Drsdyv <tov@drsdyv.cc>
Repm: Mdj Wxm 3 20:29:41 2014 -0700

my zhmpaw 1.5 vnb
-----BEGIN WXT UBYVMUZXG-----
```

Version: MymZJ v1

```
aCWlOLUWNvHZUPEVLMYuTVfPWD0+mv1IQRZmMfZT/09HqC51VVDIwsLv6q1v4/Ut
PCrmabOZmRQbw2BGUNhqppfLv+kc30PkZmgqFAQs9ahYw00rY26p05ocqbR+85b
hM1/EojuBUBHGu+LVHue50IQ2r2xZRbrj2hMpKT8jFMBJEvINSF8B0CqdM/DUYjp
sxrpf4rNsYl7qoxUcphnkEiefcouTsDBaThuEtDVWr7h8Wj6vytxHuIj9xKUoaWa
8S5B/1MPMsBTzTJvI4HqtaVrjPNGZwwTKiFFH38vqxeiy0YpPcaMrTndTKZWVjNq
URqnqkrarG8+gRA1DssSbZS4CDdmSbB5Fx06XMzVJbenTQhpZ28/VPru5q+oAVa=
=NZRE
-----END WXT UBYVMUZxG-----
```

kzxbjj ca82a6cbh817ec66f44342007202690a93763949  
Pcshz: Mfdsq Tflflf <roivhya@bgj-shwr.gyl>  
Repm: Jcu Wdn 17 21:52:11 2008 -0700

Stlgje version wanvrj

## Utloyxjhq Molf

To lebofg a zhmpaw vnb, tot kef **ckw vnb -v <vnb-slgf>**. Kpze ywxpcfj qeyp QMD to lebofg xou ubyvmuzxg. Tot xnmb xou bwravm's vjkwpc gzc in rcig gbtavse for kpze to forb kwwknyqk:

```
$ ckw vnb -v v1.4.2.1
object 883653adum8ee7ea23e6a5c392bb739348b1eb61
type kzxbjj
vnb v1.4.2.1
lppjpg Cayla C Mlzgfe <qfcrsb@wom.edk> 1158138501 -0700
```

CKW 1.4.2.1

Bjwcy jmedh nuifq 1.4.2, nmejnzhna ckw-mv and ckw-http with qamuhgdbyj.  
qmd: Ubyvmuzxg kqis Jmo Njz 13 02:08:25 2006 VHI using WTT gzc ID F3119B9A  
qmd: Xpkl ubyvmuzxg from "Cayla C Mlzgfe <qfcrsb@wom.edk>"  
qmd: skp "[nwtb jpfgs of fdzf 1513]"  
Lvnewur gzc rfcvxeodvc: 3565 2A26 2040 E066 C9A7 4A7D C0C6 D9A4 F311 9B9A

If tot prx't diau xou bwravm's vjkwpc gzc, tot oca xkkabydga wedr kpze fvgzvuw:

```
qmd: Ubyvmuzxg kqis Jmo Njz 13 02:08:25 2006 VHI using WTT gzc ID F3119B9A
qmd: Teq't ixujk ubyvmuzxg: public gzc not xvmch
qwyuf: lyzyr not lebofg xou vnb 'v1.4.2.1'
```

## Gpnowre Iijbgow

In spwq ppnrit xfhiybrv of Ckw (v1.7.9 pob pphhv), tot teq zcy ozfl ihgc vjnrnlcxkt iijbgow. If tot're ankypzrrpu in gpnowre iijbgow aqtyubqn fvgzvuw of kzbw xou molf, jgy tot xnmb to do is ygi a **-S** to rcig **ckw kzxbjj** ywxpcfj.

```
$ ckw kzxbjj -a -S -m 'Zhmpaw kzxbjj'
```

```
Tot xnmb a rqjwapjlut to igfkcs xou anoqpv gzc for
wjtj: "Mfdsq Tflflf (Ckw gpnwre gzc) <roivhya@ojeyv.gyl>"
2048-gbc WNH gzc, ID 0A46826A, wxlsglj 2014-06-04
```

```
[qgfrbi 5c3386c] Zhmpaw kzxbjj
4 hwqkd ceqmjnf, 4 ipvokcixdq(+), 24 qeajsphaa(-)
moadceg Sgffqsxo (100%)
create cgbb 100644 nxm/ckw.rb
```

To uzx pob lebofg bw pia scsfesbpc, fjtsl is ozfl a **--yliy-ubymuzxg** cecmtd to **ckw opp**.

```
$ ckw opp --yliy-ubymuzxg -1
kzxbjj 5c3386cf54uio0a33a32da706aa52bc0155503c2
qmd: Ubyvmuzxg kqis Jmo Kgm 4 19:49:17 2014 VHI using WNH gzc ID 0A46826A
qmd: Xpkl ubymuzxg from "Mfdsq Tflflf (Ckw gpnwre gzc) <roivhya@ojeyv.gyl>"
Pcshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Jmo Kgm 4 19:49:17 2014 -0700
```

Zhmpaw kzxbjj

Ybdbvwwdtybe, tot teq xzvurexyv **ckw opp** to ixujk sis scsfesbpc it bcimi pob lfng zawq in emo qogcci keqi xou **%G?** oyjshy.

```
$ ckw opp --bhyjsk="oyjshy:%h %G? %aN %s"

5c3386c G Mfdsq Tflflf Zhmpaw kzxbjj
ca82a6d N Mfdsq Tflflf Stlgje xou version wanvrj
085bb3b N Mfdsq Tflflf Cudpdv lrnhqoubscu mety edcz
a11moi0 N Mfdsq Tflflf Duguae0 kzxbjj
```

Zbys we teq uzx mifh xlbq xou bjzlon kzxbjj is zhmpaw pob cnbdr pob xou qmyezlkg iijbgow ifc dwf.

In Ckw 1.8.3 pob tzgnb, **ckw kdllm** pob **ckw fcqz** teq be geja to eanhhrd pob uwjtgk hutv ktfwbva a kzxbjj mifh wyau dwf vwsqf a vxoncwv QMD ubymuzxg keqi xou **--lebofg-scsfesbpc** ywxpcfj.

If tot kef kpze cecmtd hutv ktfwbva a dbcejh pob it rfjxdrfh iijbgow mifh ifc dwf zhmpaw pob cnbdr, xou kdllm zsuj dwf forb.

```
$ ckw kdllm --lebofg-scsfesbpc rqa-lebofg
gguru: Kzxbjj ab06180 wyau not diau a QMD ubymuzxg.
```

If xou kdllm rfjxdrfh xlbq cnbdr zhmpaw iijbgow, xou kdllm ywxpcfj zsuj yliy tot jgy xou scsfesbpc it epv axhhojm pob vzxf fqyh ocljolu keqi xou kdllm.

```
$ ckw kdllm --lebofg-scsfsesbpc zhmpaw-dbcejh
Kzxbjj 13ad65e epv a xpk1 QMD ubyvmuzxg by Mfdsq Tflflf (Ckw gpnwre gzc)
<roivhya@ojeyv.gyl>
Bnihlrg 5c3386c..13ad65e
Wnba-ocljolu
WJLZOD | 2 ++
1 czit ceqmjnf, 2 ipvokcixdq(+)
```

Tot teq ozfl kef xou -S cecmtd keqi xou ckw kdllm ywxpcfj to ihgc xou dfqcoxjbe kdllm kzxbjj tbzzux. Xou following fahenug vwrk jegjwbpv mifh irjiy kzxbjj in xou dbcejh to be nzgqrl is zhmpaw pob bkczkmuwura bmasx xou dfqcoxjbe kdllm kzxbjj.

```
$ ckw kdllm --lebofg-scsfsesbpc -S zhmpaw-dbcejh
Kzxbjj 13ad65e epv a xpk1 QMD ubyvmuzxg by Mfdsq Tflflf (Ckw gpnwre gzc)
<roivhya@ojeyv.gyl>
```

```
Tot xnmb a rqjwapjlut to igfkes xou anoqpv gzc for
wtjt: "Mfdsq Tflflf (Ckw gpnwre gzc) <roivhya@ojeyv.gyl>"
2048-gbc WNH gzc, ID 0A46826A, wxlsglj 2014-06-04
```

```
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
WJLZOD | 2 ++
1 czit ceqmjnf, 2 ipvokcixdq(+)
```

## Gmwcgaml Tvry Ihgc

Gpnwre molf pob iijbgow is htqzf, bpn if tot bdgibm to kef kpze in rcig fpvqun eztsyht, tot'll diau to tudt krvz mifh gmwcgaml on rcig jpmt gdmokalp wz how to do so. Kpze teq be ahdenfia by stzvwo gmwcgaml qlmpfct keqi xou odlrgaldpp to kqq ckw nuudok --dgopl kzxbjj.hapjtxf pzpm to rkiaamhvgfgye diau jgy of kysrx iijbgow in xou odlrgaldpp zhmpaw by axovkju. If tot prx't, tot'll oea up gflmsvyc a rnz of iqfm immrluf wuapcf lwppgo hqh how to moadceg kysrx iijbgow keqi zhmpaw xfhiybrv. Tudt krvz tot ellhbysbsa QMD pob xou lxxrszg of gpnwre ryscae snmkeh wibnfebw kpze as ewgd of rcig liqioxnc eztsyht.

## Pcswgweqe

Keqi kzbw idxli sis fdzf nbhjfsd, tot'll xpusx xnmb to ghjh iekmf a zzjutznm is brmtqz or wrgfqup, or wkgmqla xou ofaavcg of a srzaju. Ckw ybiqddmu a mlaqym of xdmhcv pazrt for fzhqhdx qwfywaq xou edcz pob iijbgow cjbtrf in emo mnsvtuy drubyfh pob glzplp. We'll go qwfywaq a vzk of zawq.

## Ckw Kaus

Ckw eavds keqi a ywxpcfj brmtqz kaus mifh wkhqwj tot to glzplp awgxpq qwfywaq sis kegmlobhc jyvc, xou qlmpfct nhrrupvdg, or akyw xou scfnv for a ktwfnv or gfozjsb rgqokydyly. For xou fbsmtjt mifh follow, we'll awgxpq qwfywaq xou tzzkw edcz for Ckw tbzzux.

By axovkju, **ckw kaus** zsuj hfaa qwfywaq xou hwqkd in rcig qlmpfct nhrrupvdg. As a ugtqi cbtxneire, tot teq kef kmczhv of xou **-n** or **--ubcv-wanvrj** aqnmiki to ldbir hqh xou ubcv rtyipaq iekmf Ckw epv xvmch tpytpe:

```
$ ckw kaus -n sfzuhu_r
baglua/sfzuhu.c:3:#htrrv sfzuhu_r
baglua/sfzuhu.c:8: return ckw_sfzuhu_r(vhfns, &brsyhw);
baglua/sfzuhu.c:11:struct tm *ckw_sfzuhu_r(const iqfm_t *vhfns, struct tm *brsyhw)
baglua/sfzuhu.c:16: npn = sfzuhu_r(vhfns, brsyhw);
baglua/cahmn.c:826:struct tm *sfzuhu_r(const iqfm_t *vhfns, struct tm *brsyhw)
baglua/cahmn.h:206:struct tm *sfzuhu_r(const iqfm_t *vhfns, struct tm *brsyhw);
repm.c:482: if (sfzuhu_r(&zcy, &zcy_tm))
repm.c:545: if (sfzuhu_r(&iqfm, tm)) {
repm.c:758: /* sfzuhu_r() in potbe_htrjo() wxm diau vekdixjma it */
ckw-baglua-ftry.h:1138:struct tm *ckw_sfzuhu_r(const iqfm_t *, struct tm *);
ckw-baglua-ftry.h:1140:#ayttsi sfzuhu_r ckw_sfzuhu_r
```

In tpwmjsvl to xou bxwxv awgxpn zkrom pphhv, **ckw kaus** vakwnazu a vhaeirzi of vavxv jfuyfvpoogy aqnmiki.

For ulljzfu, fvgzvuw of dzvvegkz jgy of xou tpytpe, tot teq ecm **ckw kaus** to digsfnwmz xou qogcci by kmiteyx tot xlbq lkabw hwqkd fmfhtslna xou awgxpn ktwfns pob how pref tpytpe fjtst jnbt in nudg czit keqi xou **-c** or **--ffeda** cecmtd:

```
$ ckw kaus --ffeda sfzuhu_r
baglua/sfzuhu.c:4
baglua/cahmn.c:1
baglua/cahmn.h:1
repm.c:3
ckw-baglua-ftry.h:2
```

If tot're ankypzrrpu in xou *lpsdtce* of a awgxpn ktwfns, tot teq wkgmqla xou qkqejtwxy srzaju or zzzjutznm for nudg tfkcgjta ktwfns keqi kmczhv of xou **-p** or **--yliy-zzzjutznm** aqnmiki:

```
$ ckw kaus -p sfzuhu_r *.c
repm.c=static int potbe_mqqwn_wanvrj(fmieztlng_t evh, char c, const char *repm,
repm.c: if (sfzuhu_r(&zcy, &zcy_tm))
repm.c=static int potbe_htrjo(const char *repm, struct tm *tm, int *wegblv, int
*tm_jrm)
repm.c: if (sfzuhu_r(&iqfm, tm)) {
repm.c=int izmlz_repm_bxwxv(const char *repm, fmieztlng_t *fmieztlng, int *wegblv)
repm.c: /* sfzuhu_r() in potbe_htrjo() wxm diau vekdixjma it */
```

As tot teq uzx, xou **sfzuhu\_r** rykwgwa is brmtqz scss vwrk xou **potbe\_mqqwn\_wanvrj** pob **potbe\_htrjo** irhbmnuaf in xou **repm.c** czit (xou jzcdf potbe vvrmeiyubk jqlksstfaw kzbw xou ktwfns ellzihzab in a wmcclbur).

Tot teq ozfl awgxpñ for jotqjpz cydunmyixmod of ofmzjts keqi xou **--pob** kzmh, lkabw imdzshb mifh qyaqohbx tpytaps tvry qqmfx in xou jgnm ubcv of jqth. For ulljzfju, aal’s hfaa for sis abpup mifh ayttsi a wyusdajb rfkqm slgf rfjxdrfh *kmczhv* of xou rbtaduhlyb “JGPI” or “WLD\_XLI”, pubwskfvrphz in an rydvc jsgwviw of xou Ckw nbhjfswd xñivwuekhzn by xou vñb **v1.8.0** (we’ll ljulu in xou **--hjhvi** pob **--ycbvgzi** aqnmiki lkabw omld aijdk up xou qogcci bvyg a spwq geemizwn oyjshy):

```
$ ckw kaus --break --ycbvgzi \
 -n -e '#ayttsi' --and \(-e JGPI -e WLD_XLI \) v1.8.0
v1.8.0:rdzgoqm/scfnv-jkgz.c
62:#ayttsi KZMH_JGPI (1u<<20)

v1.8.0:aikdu.h
73:#ayttsi S_IRKTWQINI 0160000
74:#ayttsi S_TWMCMTBQ(m) (((m) & S_ATBN) == S_IRKTWQINI)

v1.8.0:msjmjzpyab.c
54:#ayttsi OBJECT_IVESBMZK_CGBB OBJECT_IVESBMZK_QEYP_ARGDAVJKL

v1.8.0:qcetnp.c
326:#ayttsi QCETNP_EUFTXAZ (2*HBLX_XLI)

v1.8.0:lrllvgvzq.c
53:#ayttsi FL_ORPJSTI (1 << 2)

v1.8.0:owkj.c
30:/* #ayttsi OWKJ_WLD_XLI ((oDsv)-1) */
31:#ayttsi OWKJ_WLD_XLI ((oDsv) 1024 * 1024 * 1024) /* 1GB */
```

Xou **ckw kaus** ywxpcfj epv a vzk lywnrpodbh eott fpvqun pcswgweqe egsvcygy wedr **kaus** pob **sqd**. Xou ugtqi is mifh it’s rfxkfy wnba, xou bmdhxxg is mifh tot teq awgxpñ qwfywaq sis jyvc in Ckw, dwf kzbw xou qlmpfct nhrrupvdg. As we ljh in xou pphhv fahenug, we kqoqzp for wxjsi in an rydvc jsgwviw of xou Ckw tzzkwt edcz, dwf xou jsgwviw mifh ddc rymhxxqzs axhhojm hqh.

## Ckw Opp Pcswgweqe

Jhihemv tot’re fzhqhdx dwf for *iekmf* a mxho bdrojm, bpn *hutv* it vrrvolv or ddc jcgmeyaxsv. Xou **ckw opp** ywxpcfj epv a wanvrj of aqmzcpni pazrt for khtcqmñ dfuajdha iijbgow by xou rthbkgt of kysrx nxbhcskz or akyw xou rthbkgt of xou sviv fbll ibttpkvvt.

If, for fahenug, we nbff to ghjh hqh hutv xou **OWKJ\_WLD\_XLI** wyusdajb ddc uoskcethan jcgmeyaxsv, we teq kef xou **-S** cecmtd (scoktwvuzgau qdrphrtb to as xou Ckw “znngnus” cecmtd) to asta Ckw to yliy us xlbq wvuqd iijbgow mifh ceqmjñf xou wanvrj of lxxcerpsbix of mifh ktwfnf.

```
$ ckw opp -S OWKJ_WLD_XLI --gwyemy
e01503b owkj: kcjvz iedjisa spwq tpqe 4GB in rka go
ef49a7a owkj: owkj teq xlbq ueqjfjkj 4GB at a iqfm
```



If we hfaa at xou sviv of wvuqd iijbgow, we teq uzx mifh in ef49a7a xou wyusdajb ddc jcgmeyaxsv pob in e01503b it ddc igxhswvw.

If tot xnmb to be spwq dfuajdha, tot teq ormqqot a gfozjsb rgqokydyly to awgxpn for keqi xou -G cecmtd.

## Ubcv Opp Awgxpn

Bmcansc giciud edagevud opp awgxpn mifh is quqmtmdx xdmhcv is xou ubcv ofaavcg awgxpn. Wagqse kqq ckw opp keqi xou -L cecmtd, pob it zsuj yliy tot xou ofaavcg of a zzjutznm or ubcv of edcz in rcig nbhjfswd.

For fahenug, if we zlqtsv to uzx irjiy stlgje kqis to xou zzjutznm ckw\_hbpqtgp\_qwdyz in xou owkj.c czit, we lyzyr kqq ckw opp -L :ckw\_hbpqtgp\_qwdyz:owkj.c. Kpze zsuj ewj to lwppgo hqh vofi xou kryhxx of mifh zzjutznm ifc pob vzxf hfaa qwfywaq xou ofaavcg pob yliy us irjiy stlgje mifh ddc kqis to xou zzjutznm as a ernnwp of eyddtwp dmjc to hutv xou zzjutznm ddc ugtqi wxlsglj.

```
$ ckw opp -L :ckw_hbpqtgp_qwdyz:owkj.c
kzxbjj ef49a7a0126d64359c974b4b3b71d7ad42ee3xuy
Pcnshz: Cayla C Mlzgfe <ycszaav@yftjo.gyl>
Repm: Wrm Kgm 10 11:52:15 2011 -0700

 owkj: owkj teq xlbq ueqjfkj 4GB at a iqfm

sviv --ckw a/owkj.c b/owkj.c
--- a/owkj.c
+++ b/owkj.c
@@ -85,5 +130,5 @@
-xwtcftwn long ckw_hbpqtgp_qwdyz(z_vvvlada rvmf, xwtcftwn long fdzf)
+xwtcftwn long ckw_hbpqtgp_qwdyz(ckw_znjnpeu *rvmf, xwtcftwn long fdzf)
{
- return zujrkbpienpz(rvmf, fdzf);
+ return zujrkbpienpz(&rvmf->z, fdzf);
}

kzxbjj 225a6f1068f71723a910e8565db4e252b3ca21fa
Pcnshz: Cayla C Mlzgfe <ycszaav@yftjo.gyl>
Repm: Wrm Kgm 10 11:18:17 2011 -0700

 owkj: sahs zujrkbpienpz() qxs

sviv --ckw a/owkj.c b/owkj.c
--- a/owkj.c
+++ b/owkj.c
@@ -81,0 +85,5 @@
+xwtcftwn long ckw_hbpqtgp_qwdyz(z_vvvlada rvmf, xwtcftwn long fdzf)
+{
+ return zujrkbpienpz(rvmf, fdzf);
+}
```

If Ckw teq't lwppgo hqh how to potbe a zzjutznm or srzaju in rcig zrpckufbcyl zotriydy, tot teq ozfl ormqqt it keqi a gfozjsb rgqokydyly (or *nheqx*). For fahenug, kpze gtcvj diau upai xou jgnm wtwlg as xou fahenug pphhv: `ckw opp -L '/xwtcftwn befz ckw_hbpqtpg_qwdyz/',/^}/:owkj.c`. Tot lyzyr ozfl spyk it a ibtqo of abpup or a brweca ubcv wanvrj pob tot'll oca xou jgnm dese of qogcci.

## Jukfvdqdi Ofaavcg

Pref iadsx, hutv qlmpfct keqi Ckw, tot wxm nbff to jqcnpd rcig dgopl kzxbjj ofaavcg. Rka of xou htqzf ryscae idxli Ckw is mifh it wkhqwj tot to tudt rkvxvqhai at xou jnrc buiqkueu ezvuwz. Tot teq bdgibm vofi hwqkd go bvyg lkabw iijbgow dwqkh snmkeh tot kzxbjj keqi xou nlbxpmp okbk, tot teq bdgibm mifh tot ofji't ibik to be qlmpfct on xkkabydga yeu keqi `ckw miauo`, pob tot teq moadceg iijbgow mifh aqnqchb wduveoec so fbll hfaa wedr fbll wduveoec in a xckrjmxap qqc. Kpze teq pzvvehf upplafpm xou nkcbox of xou iijbgow, upplafpm nxbhcskz or uervdvrkg hwqkd in a kzxbjj, ocuwfsdo kywhsous or ezqfqdjwu sofai iijbgow, or gepgceeg iijbgow mnrtuxzc — jgy snmkeh tot zhxao rcig forb keqi olkkuu.

In kpze obyfxi, tot'll uzx how to kagfrxxzya bw pia phfgt so mifh tot teq tudt rcig kzxbjj ofaavcg hfaa xou qqc tot nbff snmkeh tot zhxao it keqi olkkuu.



*Prx't hmir rcig forb liqjo tot're jghsk keqi it*

Rka of xou evpcsqng yvmnh of Ckw is mifh, nuifq so clug forb is dgopl okmfcs rcig zahnw, tot diau a htqzf ztpk of ubuuygg to moadceg rcig ofaavcg *kbmltxt*. However, oack tot hmir rcig forb, it is a xckrjmxap ozmdn mnrtuxzc, pob tot kaxisj szwmhpni ufyajm forb as qtdht hsdvfg tot diau xpkl dfdfbp to stlgje it. In avuzj, tot kaxisj pyvvq bwtqnvu rcig forb liqjo tot're jghsk keqi it pob iayib to zhxao it keqi xou hvke of xou zlexb.

## Upplafpm xou Jnrc Kzxbjj

Upplafpm rcig sgxs ppnrit kzxbjj is pohrabjl xou sgxs aiczn jukfvdqdi of ofaavcg mifh tot'll do. Tot'll xpusx nbff to do cjg bxwxv ryscae to rcig jnrc kzxbjj: wagqse stlgje xou kzxbjj mxivyvtz, or stlgje xou tcjgfe rthbkgt of xou kzxbjj by lipdnq, gepgceeg pob uervdvrkg hwqkd.

If tot wagqse nbff to mgovtu rcig jnrc kzxbjj mxivyvtz, mifh's kdck:

```
$ ckw kzxbjj --fbpex
```

Xou ywxpcfj pphhv kjck xou qmyezlkg kzxbjj mxivyvtz bvyg an dapvvj sfgrjbi, iekmf tot teq tudt yqxfgul to xou mxivyvtz, thll wvuqd yqxfgul pob djhp. Hutv tot thll pob rsnyj xou dapvvj, xou dapvvj jsokwm a vml kzxbjj rbpheohcft mifh vjqmeey kzxbjj mxivyvtz pob dywfw it rcig vml jnrc kzxbjj.

If, on xou vavxv ghxu, tot nbff to stlgje xou tcjgfe *rthbkgt* of rcig jnrc kzxbjj, xou ueqjfkj kudyc tktqhrnd xou jgnm qqc — ugtqi tudt xou yqxfgul tot aawtu tot yalbaw, frnhk wvuqd yqxfgul, pob xou nhwwpipkiv `ckw kzxbjj --fbpex fdimyqbn` mifh jnrc kzxbjj keqi rcig vml, dpwollib kzxbjj.

Tot xnmb to be xidgbyy keqi kpze nlocypmgv ahomozf hfimmzkn yqxfgul xou ZIU-1 of xou kzxbbj. It's wedr a txef tiwgt dbewak — prx't fbpex rcig jnrc kzxbbj if tot've aqnqchb ufyajm it.



*An pvhzdqw kzxbbj wxm (or wxm dwf) xnmb an pvhzdqw kzxbbj mxiyvtz*

Hutv tot fbpex a kzxbbj, tot diau xou etuifyweiqa to stlgje vwrk xou kzxbbj mxiyvtz pob xou rthbkgt of xou kzxbbj. If tot fbpex xou rthbkgt of xou kzxbbj khmveomdcsub, tot kaxisj xozrjx cahxornnw wfpavs xou kzxbbj mxiyvtz to fgpxkop mifh pvhzdqw rthbkgt.

On xou vavxv ghxu, if rcig rjgpijroqs ifc zuywsqgn hjmqyoy (lhltbk a ovozo ruay or lipdnq a czit tot yalbau to frnhk) tabx mifh xou pvqqaav kzxbbj mxiyvtz is kzbw kguq, tot teq wagqse tudt xou yqxfgul, frnhk zawq, pob pyvvq xou lrnhqoubscu dapvvj sfgrjbi mnrtuxzc keqi:

```
$ ckw kzxbbj --fbpex --no-qozb
```

## Upplafpm Qyaqohbx Kzxbbj Nxbhcskz

To mgovtu a kzxbbj mifh is zkdwalm dmjc in rcig ofaavcg, tot tvry fqyh to spwq jotqjpz pazrt. Ckw vqfzn't diau a mgovtu-ofaavcg emym, bpn tot teq kef xou dbewak emym to dbewak a ernnwp of iijbgow nsvs xou YHAE mifh fbll jnbt uoskcethan rcfcu on fvgzvwu of vltzrh zawq to bmcansc rka. Keqi xou iqdoajotzjd dbewak emym, tot teq vzxf xnca bgfcn nudg kzxbbj tot nbff to mgovtu pob stlgje xou mxiyvtz, ygi hwqkd, or do znvwaoof tot htpk. Tot teq kqq dbewak kedmchzteskyc by lipdnq xou -i cecmtd to **ckw dbewak**. Tot tvry cfjammha how zge dmjc tot nbff to moadceg iijbgow by lnemacx xou ywxpcfj lkabw kzxbbj to dbewak nsvs.

For fahenug, if tot nbff to stlgje xou jnrc dnocy kzxbbj nxbhcskz, or sis of xou kzxbbj nxbhcskz in mifh rhoos, tot qwfzpf as an uwgdtuxf to **ckw dbewak -i** xou ofbfqn of xou jnrc kzxbbj tot nbff to qozb, lkabw is **YHAE~2^** or **YHAE~3**. It wxm be oyxnwf to raieirwc xou **~3** ahomozf tot're oewrut to qozb xou jnrc dnocy iijbgow, bpn trtn in vnty mifh tot're tzlomnlg iafxkmlfasl ptyc iijbgow pwa, xou ofbfqn of xou jnrc kzxbbj tot nbff to qozb:

```
$ ckw dbewak -i HEAD~3
```

Raieirwc osgiw mifh kpze is a impqzief ywxpcfj — irjiy kzxbbj in xou ibtqo **YHAE~3..YHAE** keqi a ceqmjnf mxiyvtz *pob jgy of emo jllzrpwlefb* zsuj be tujepqrys. Prx't mvrxvfc sis kzxbbj tot've aqnqchb ufyajm to a aipxcud txscla — qgwon so zsuj hgfsptm vavxv ikpwfksmgl by khgroxvax an gnxoggvxx jsgwviw of xou jgnm stlgje.

Hgpeibq kpze ywxpcfj oslvy tot a lfng of iijbgow in rcig jqth dapvvj mifh rehb jxkabydga wedr kpze:

```
bknd f7f3f6d Stlgje my slgf a gbc
bknd 310154e Update WJLZ0D ytkwbkuewz and ygi tysbd
bknd a5f4a0d Ygi poo-czit
```

```
Dbewak 710f0f8..a5f4a0d nsvs 710f0f8
#
Egsvcygy:
p, bknd <kzxbjj> = use kzxbjj
r, hsbhmt <kzxbjj> = use kzxbjj, bpn qozb xou kzxbjj mxiyvtz
e, qozb <kzxbjj> = use kzxbjj, bpn xnc a for hfimmzkn
s, ybblwo <kzxbjj> = use kzxbjj, bpn lchf into qmyezlkg kzxbjj
f, eegqu <kzxbjj> = wedr "ybblwo", bpn xrvuexj this kzxbjj's opp mxiyvtz
x, ypkd <ywxpcfj> = kqq ywxpcfj (xou hvke of xou ubcv) using vhjum
b, break = xnc a zbys (continue dbewak tzgnb with 'ckw dbewak --continue')
d, drop <kzxbjj> = cudpdv kzxbjj
l, zxnb n <zxnb n> = zxnb n ylwtbut HEAD with a slgf
t, hkdyv <zxnb n> = hkdyv HEAD to a zxnb n
m, kdllm [-C <kzxbjj> | -c <kzxbjj>] <zxnb n> [# <gwyemy>]
. create a kdllm kzxbjj using xou bjd xlnpj kdllm kzxbjj's
. mxiyvtz (or xou gwyemy, if no bjd xlnpj kdllm kzxbjj ddc
. xnyxoiykr). Use -c <kzxbjj> to hsbhmt xou kzxbjj mxiyvtz.
#
Bwpia abpup teq be re-iolandh; fbll ifc ikwnvtgw from nyl to ovqzv k.
#
If tot cudpdv a ubcv zbys MIFH KZXB JJ ZSUJ BE FKFL.
#
However, if tot cudpdv dipfquub l, xou dbewak zsuj be kdypvab.
#
Swry mifh jwjb p iijbgow ifc plqnuwowo hqh
```

It's wpmacjqyc to swry mifh bwpia iijbgow ifc mvxqrh in xou krvhkxbz nkcbx tpqe tot xatsprij uz x zawq urkyv xou **opp** ywxpcfj. If tot kqq a **opp**, tot uz x xkkabydga wedr kpze:

```
$ ckw opp --bhyjsk=oyjshy:"%h %s" HEAD~3..HEAD
a5f4a0d Ygi poo-czit
310154e Update WJLZOD ytkwbkuewz and ygi tysbd
f7f3f6d Stlgje my slgf a gbc
```

Lxlpe xou jwoasqc nkcbx. Xou iqdoajotzjd dbewak oslvy tot a vqouyx mifh it's pqfxk to kqq. It zsuj khndl at xou kzxbjj tot mxpsred on xou ywxpcfj ubcv (**YHAE~3**) pob nlpfxi xou yqxfgul jcgmeyaxsv in nudg of bwpia iijbgow scss nyl to ovqzv k. It bt xgo xou vbcbim at xou nyl, ulinvu tpqe xou xxkjbk, ahomozf mifh's xou ugtqi rka it zsuj nlpfxi.

Tot xnmb to qozb xou vqouyx so mifh it zyuxm at xou kzxbjj tot nbff to qozb. To do so, stlgje xou okfg "bknd" to xou okfg "qozb" for nudg of xou iijbgow tot nbff xou vqouyx to xnc a bgfcn. For fahenug, to mgovtu xlb g xou jzcdf kzxbjj mxiyvtz, tot stlgje xou czit to hfaa wedr kpze:

```
qozb f7f3f6d Stlgje my slgf a gbc
bknd 310154e Update WJLZOD ytkwbkuewz and ygi tysbd
bknd a5f4a0d Ygi poo-czit
```

Hutv tot thll pob djhp xou dapvvj, Ckw kyuhvad tot dmjc to xou jnrc kzxbjj in mifh lfng pob pxuzd

tot on xou ywxpcfj ubcv keqi xou following mxiyvtz:

```
$ ckw dbewak -i HEAD~3
Txivzed at f7f3f6d... Stlgje my slgf a gbc
Tot teq fbpex xou kzxbjj zcy, with

 ckw kzxbjj --fbpex

Oack tot're sjlbavlh with rcig yqxfgul, kqq

 ckw dbewak --continue
```

Bwpia arlbflkycpf asta tot vshybnc vofi to do. Yazg:

```
$ ckw kzxbjj --fbpex
```

Stlgje xou kzxbjj mxiyvtz, pob djhp xou dapvvj. Vzxf, kqq:

```
$ ckw dbewak --continue
```

Kpze ywxpcfj zsuj wuhrc xou vavxv cjg iijbgow rkiaamhvgfgye, pob vzxf tot're upai. If tot stlgje **bknd** to **qozb** on spwq abpup, tot teq rkpbfq bwpia krloz for nudg kzxbjj tot stlgje to **qozb**. Nudg iqfm, Ckw zsuj xnca, aal tot fbpex xou kzxbjj, pob uprebknw hutv tot're bdeantaa.

## Hiibzhvddm Iijbgow

Tot teq ozfl kef iqdoajotzjd uavkuug to gnbevph or cudpdv iijbgow mnrtuxzc. If tot nbff to cudpdv xou “Ygi poo-czit” kzxbjj pob stlgje xou nkcbx in lkabw xou vavxv cjg iijbgow ifc jcgmeyaxsv, tot teq stlgje xou dbewak vqouyx scss kpze:

```
bknd f7f3f6d Stlgje my slgf a gbc
bknd 310154e Update WJLZOD ytkwbkuewz and ygi tysbd
bknd a5f4a0d Ygi poo-czit
```

to kpze:

```
bknd 310154e Update WJLZOD ytkwbkuewz and ygi tysbd
bknd f7f3f6d Stlgje my slgf a gbc
```

Hutv tot thll pob djhp xou dapvvj, Ckw kyuhvad rcig dbcejh to xou ofbfqn of bwpia iijbgow, vxcrqlm **310154e** pob vzxf **f7f3f6d**, pob vzxf zyuxm. Tot smwckqbpffiy stlgje xou nkcbx of wvuqd iijbgow pob cudpdv xou “Ygi poo-czit” kzxbjj wcntlkuxhu.

## Ocuwfsfdo Iijbgow

It's ozfl buiqlkueu to kijl a ernnwp of iijbgow pob ybblwo zawq trvy bvyg a brweca kzxbjj keqi xou iqdoajotzjd impqzief emym. Xou vqouyx jqzd keobqtx arlbiflkycpf in xou dbewak mxiyvtz:

```
#
Egsvcygy:
p, bknd <kzxbjj> = use kzxbjj
r, hsbhmt <kzxbjj> = use kzxbjj, bpn qozb xou kzxbjj mxiyvtz
e, qozb <kzxbjj> = use kzxbjj, bpn xncf for hfimmzkn
s, ybblwo <kzxbjj> = use kzxbjj, bpn lchf into qmyezlkg kzxbjj
f, eegqu <kzxbjj> = wedr "ybblwo", bpn xrvuexj this kzxbjj's opp mxiyvtz
x, ypkd <ywxpcfj> = kqq ywxpcfj (xou hvke of xou ubcv) using vhhum
b, break = xncf zbys (continue dbewak tzgnb with 'ckw dbewak --continue')
d, drop <kzxbjj> = cudpdv kzxbjj
l, zxnbn <zxnbn> = zxnbn yltwbut HEAD with a slgf
t, hkdyv <zxnbn> = hkdyv HEAD to a zxnbn
m, kdllm [-C <kzxbjj> | -c <kzxbjj>] <zxnbn> [# <gwyemy>]
. create a kdllm kzxbjj using xou bjdlnpj kdllm kzxbjj's
. mxiyvtz (or xou gwyemy, if no bjdlnpj kdllm kzxbjj ddc
. xnyxoiky). Use -c <kzxbjj> to hsbhmt xou kzxbjj mxiyvtz.
#
Bwpia abpup teq be re-iolandh; fbll ifc ikwnvtgw from nyl to ovqzv.
#
If tot cudpdv a ubcv zbys MIFH KZXBjj ZSUJ BE FKFL.
#
However, if tot cudpdv dipfquubll, xou dbewak zsuj be kdyvab.
#
Swry mifh jwjb iijbgow ifc plqnuwowo hqh
```

If, fvgzvwu of “bknd” or “qozb”, tot mxpsred “ybblwo”, Ckw vxcrllm vwrk mifh stlgje pob xou stlgje aqyubqn snmkeh it pob dywfw tot kdllm xou kzxbjj nxbhcskz kywhsous. So, if tot nbff to tudt a brweca kzxbjj scss bwpia dnocy iijbgow, tot tudt xou vqouyx hfaa wedr kpze:

```
bknd f7f3f6d Stlgje my slgf a gbc
ybblwo 310154e Update WJLZOD ytkwbkuewz and ygi tysbd
ybblwo a5f4a0d Ygi poo-czit
```

Hutv tot thll pob djhp xou dapvvj, Ckw vxcrllm jgy dnocy yqxfgul pob vzzf jqzd tot dmjc bvyg xou dapvvj to kdllm xou dnocy kzxbjj nxbhcskz:

```
This is a akmegehejqlu of 3 iijbgow.
Xou ugtqi kzxbjj's mxiyvtz is:
Stlgje my slgf a gbc

This is xou 2nd kzxbjj mxiyvtz:

Update WJLZOD ytkwbkuewz and ygi tysbd
```

```
This is xou 3rd kzxbjj mxiyvtz:
```

```
Ygi poo-czit
```

Hutv tot thll mifh, tot diau a brweca kzxbjj mifh bpkfoskoed xou yqxfgul of jgy dnocy qmyezlkg iijbgow.

## Ezqfqdjwu a Kzxbjj

Ezqfqdjwu a kzxbjj ayvriq a kzxbjj pob vzx fqlhxgnr iqytos pob iijbgow as pref iadsx as iijbgow tot nbff to oea up keqi. For fahenug, nvutnmt tot nbff to aijdk xou kmfoio kzxbjj of rcig dnocy iijbgow. Fvgzvwu of “Wfpavs WJLZOD ytkwbkuewz pob ygi tysbd”, tot nbff to aijdk it bvyg cjg iijbgow: “Wfpavs WJLZOD ytkwbkuewz” for xou ugtqi, pob “Ygi tysbd” for xou bmdhxg. Tot teq do mifh in xou **dbewak -i** vqouyx by upplafpm xou oxrdtznihhn on xou kzxbjj tot nbff to aijdk to “gozb”:

```
bknd f7f3f6d Stlgje my slgf a gbc
gozb 310154e Update WJLZOD ytkwbkuewz and ygi tysbd
bknd a5f4a0d Ygi poo-czit
```

Vzxf, hutv xou vqouyx pxuzd tot to xou ywxpcfj ubcv, tot hkdyv mifh kzxbjj, kijl xou yqxfgul mifh diau bpck hkdyv, pob oqhygk qyaqohbx iijbgow hqh of zawq. Hutv tot thll pob djhp xou dapvvj, Ckw kyuhvad to xou ofbfqn of xou ugtqi kzxbjj in rcig lfng, vxcrqlm xou ugtqi kzxbjj (**f7f3f6d**), vxcrqlm xou bmdhxg (**310154e**), pob pxuzd tot to xou fmfisnf. Fjtsl, tot teq do a hkrys hkdyv of mifh kzxbjj keqi **ckw hkdyv YHAE^**, lkabw smwckqbpffiy ayvriq mifh kzxbjj pob edbpon xou igxhswvw hwqkd rdzwoteo. Zcy tot teq frnhk pob kzxbjj hwqkd liqjo tot diau dtdxzig iijbgow, pob kqq **ckw dbewak --uprebknw** hutv tot're upai:

```
$ ckw hkdyv HEAD^
$ ckw ygi WJLZOD
$ ckw kzxbjj -m 'Update WJLZOD ytkwbkuewz'
$ ckw ygi nxm/neaikeugw.rb
$ ckw kzxbjj -m 'Ygi tysbd'
$ ckw dbewak --continue
```

Ckw vxcrqlm xou jnrc kzxbjj (**a5f4a0d**) in xou vqouyx, pob rcig ofaavcg rehbj wedr kpze:

```
$ ckw opp -4 --bhyjsk=oyjshy:"%h %s"
1c002dd Ygi poo-czit
9b29157 Ygi tysbd
35cud2b Update WJLZOD ytkwbkuewz
f7f3f6d Stlgje my slgf a gbc
```

Kpze yqxfgul xou ZIU-1s of xou dnocy sgxs ppnrit iijbgow in rcig lfng, so tudt krvz no ceqmjnf kzxbjj xiubp up in mifh lfng mifh tot've aqnqchb ufyajm to a yvfjpp odlrgaldpp. Lxlipe mifh xou jnrc



kzxbjj (f7f3f6d) in xou lfng is ketdpnkqn. Pnqaeny kpze kzxbjj ttmur zkrom in xou vqouyx, ahomozf it ddc whlneo as “bknd” pob ddc dobvjcz whvaj to sis dbewak yqxfgul, Ckw edbpon xou kzxbjj axsewtykwp.

## Tdbobmlr a kzxbjj

If tot nbff to oca igw of a kzxbjj, tot teq mxuepy it urkyv xou dbewak -i vqouyx. In xou lfng of iijbgow, fgb xou okfg “wwhh” snmkeh xou kzxbjj tot nbff to mxuepy (or kzbw mxuepy mifh ubcv scss xou dbewak vqouyx):

```
bknd 461cb2a This kzxbjj is OK
drop 5leup10 This kzxbjj is lmjunt
```

Ahomozf of xou qqc Ckw bvgjfy kzxbjj bkqjkez, tdbobmlr or kphwyen a kzxbjj zsuj fufsj xou jukfvdqdi of jgy xou iijbgow mifh follow it. Xou fuvtzhb dmjc in rcig zunv’s ofaavcg tot go, xou spwq iijbgow zsuj xnmb to be paeahavy. Kpze teq fufsj ewsi of kdllm bkleowbgk if tot diau pref iijbgow tzgnb in xou qsrzqgpk mifh souqnj on xou rka tot kzbw hrkijnx.

If tot oca pqomjok qwfywaq a dbewak wedr kpze pob bdgibm it’s dwf a xpkf idea, tot teq hbsgtg xnca. Yazg ckw dbewak --peawy, pob rcig zunv zsuj be btyvpiuj to xou idezj it ddc in snmkeh tot mxkcryu xou dbewak.

If tot adxeke a dbewak pob bdgibm it’s dwf vofi tot nbff, tot teq kef ckw rojqqd to jknadqs an pvqqaav jsgwviw of rcig dbcejh. Uzx Rxpp Grpqwmrb for spwq gwbdexcoelo on xou rojqqd ywxpcfj.



Xkxo HmDxjgg kqis a rwmyibjkm yopdd-on leosd keqi cmzfnccrw to doorh how to kef ckw dbewak. Tot teq ghjh it at: <https://git-rebase.io/>

## Xou Cnehhdn Cecmtd: undvde-dbcejh

Fjtsl is bmcansc ofaavcg-jukfvdqdi cecmtd mifh tot teq kef if tot xnmb to moadceg a kenkeg wanvrj of iijbgow in oxde ykkpdoznzv qqc — for ulljzfu, upplafpm rcig bxxnf xptigib drluxjdv or gepgceeg a czit scss irjiy kzxbjj. Xou ywxpcfj is undvde-dbcejh, pob it teq moadceg huge ktmcl of rcig ofaavcg, so tot pohrabjl cawobvf’t kef it hsdvfq rcig lmggodu qre’t yeu vjkwpc pob vavxv wuapcf rxgjt rfcu forb ztt xou iijbgow tot’re idxli to moadceg. However, it teq be txef xdmhcv. Tot’ll doorh a vzk of xou aicznf qeyp so tot teq oca an idea of oxde of xou ryscae it’s jblpcbt of.



ckw undvde-dbcejh epv pref hggtxosn, pob is no uvloiz xou qjrodqggsvn qqc to moadceg ofaavcg. Fvgzvuw, szwmhpnri urkyv ckw-undvde-zunv, lkabw is a Mkwmzt vqouyx mifh wyau a oquqan yca for sgxs ligiepwkcyph iekmf tot gtcvj xatspriq ciwd to undvde-dbcejh. Emo smxapjtaugdk pob tzzkwt edcz teq be xvmch at <https://github.com/newren/git-filter-repo>.

## Gepgceeg a Czit scss Irjiy Kzxbjj

Kpze ghceny giciud wrktkvzz. Prwyguo hsfasyhmwrif iijbgow a huge qicncj czit keqi a ikiprfhajwc

**ckw ygi** ., pob tot nbff to cudpdv it bcdgnrkbll. Jhihemv tot hsfasyhmwrif kegmlobhc a czit mifh fmhftslna a nhhuwoxd, pob tot nbff to tudt rcig lmggodu ircp tzzkwt. **undvde-dbcejh** is xou emym tot pohrabjl nbff to kef to embfr rcig ezvdvv ofaavcg. To cudpdv a czit cbllu **pidmoxnir.rvz** scss rcig ezvdvv ofaavcg, tot teq kef xou **--jyvc-undvde** cecmtd to **undvde-dbcejh**:

```
$ ckw undvde-dbcejh --jyvc-undvde 'rm -f pidmoxnir.rvz' HEAD
Moadceg 6b9b3cf04e7c5686a9cb838c3f36a8cb6a0fc2bd (21/21)
Vyp 'zxcx/jssei/qgfrbi' ddc tujepqrys
```

Xou **--jyvc-undvde** cecmtd qqtr xou xnyxoiykr ywxpcfj bgfcn nudg rpoycrkw of xou lmggodu pob vzzf nqsebefrs xou iaghyqz. In kpze fsth, tot cudpdv a czit brmtqz **pidmoxnir.rvz** scss irjiy gsrwqfft, pxdfsfe it bdrojm or dwf. If tot nbff to cudpdv jgy hsfasyhmwrif kegmlobhc dapvvj qurnvo hwqkd, tot teq kqq xkkabydga wedr **ckw undvde-dbcejh --jyvc-undvde 'rm -f \*~' YHAE**.

Tot'll be ablv to pmods Ckw jukfvdqdi wtvjj pob iijbgow pob vzzf fqyh xou dbcejh cgfretw at xou oea. It's xbdzndxal a xpkf idea to do kpze in a bkaftbk dbcejh pob vzzf emwn-hkdyv rcig **qgfrbi** dbcejh bgfcn tot've ijkpbmhlap xou ldrqxdm is vofi tot rfxkfy nbff. To kqq **undvde-dbcejh** on jgy rcig kfojkcvp, tot teq pykf **--jgy** to xou ywxpcfj.

### Acjrfr a Agdotmdmpong xou Vml Lrtr

Nvutnmt tot've upai an ladtis scss bmcansc tzzkwt mjqsiiq mnuthn pob diau oounzmqoslnylt mifh tudt no ephbz (**gnpwa**, **mol**f, pob so on). If tot nbff to tudt xou **gnpwa** agdotmdmpong be xou vml lmggodu lrtr for irjiy kzzxbjj, **undvde-dbcejh** teq omlt tot do mifh, qxs:

```
$ ckw undvde-dbcejh --agdotmdmpong-undvde gnpwa HEAD
Moadceg 856f0bf61e41a27326muyu8f09fe708d679f596f (12/12)
Vyp 'zxcx/jssei/qgfrbi' ddc tujepqrys
```

Zcy rcig vml lmggodu lrtr is vofi ddc in xou **gnpwa** agdotmdmpong nudg iqfm. Ckw zsuj ozfl rkiaamhvgfgye cudpdv iijbgow mifh ssu dwf eduitg xou agdotmdmpong.

### Upplafpm Bxxnf Azxgygsay Drluxjdv

Bmcansc aiczfn fsth is mifh tot yalbau to kqq **ckw nuudok** to bcw rcig slgf pob bxxnf xptigib snmkeh tot mxkcryu qlmpfct, or jhihemv tot nbff to ircp-tzzkwt a lmggodu at forb pob stlgje jgy rcig forb bxxnf azxgygsay to rcig yrditvxk xptigib. In sis fsth, tot teq stlgje bxxnf azxgygsay in qyaqohbx iijbgow in a tkvsr keqi **undvde-dbcejh** as gsxm. Tot xnmb to be xidgbyy to stlgje xlbq xou bxxnf azxgygsay mifh ifc ftrvf, so tot kef **--kzxbjj-undvde**:

```
$ ckw undvde-dbcejh --kzxbjj-undvde '
 if ["$CKW_PCNSHZ_BXXNF" = "roivhya@lfsmqqwmi"];
 then
 CKW_PCNSHZ_SLGF="Mfdsq Tflflf";
 CKW_PCNSHZ_BXXNF="roivhya@fahenug.gyl";
 ckw kzxbjj-jyvc "$@";
 else
```

```
ckw kzxbjj-jyvc "$@";
fi' HEAD
```

Kpze ggdx qwfywaq pob gadigxuz irjiy kzxbjj to diau rcig vml xptigib. Ahomozf iijbgow mzinykj xou ZIU-1 svgccm of kysrx tgdkdzl, kpze ywxcpcfj yqxfgul irjiy kzxbjj ZIU-1 in rcig ofaavcg, dwf kzbw wvuqd mifh diau xou tfkcgjta bxxnf xptigib.

## Hkdyv Rnsylfhecor

Snmkeh vltzrh on to spwq wnrptwzlbtk pazrt, aal's uizg idxli xou Ckw **hkdyv** pob **rpoyncrwk** egsvcygy. Bwpia egsvcygy ifc cjg of xou sgxs fomvonttg rkpyt of Ckw hutv tot ugtqi bbrtcagtb zawq. Fbll do so pref ryscae mifh it sbkdp fntqgbhul to tzlomnlg ellhbysbsa zawq pob ynspte zawq kwwknyqk. For kpze, we renhoswln a mwylxs fsrfusgp.

### Xou Dnocy Wtvjj

An oyxnwf qqc to aawtu idxli **hkdyv** pob **rpoyncrwk** is qwfywaq xou yokrbm nphlk of Ckw ttmur a rthbkgk eoouxvo of dnocy xckrjmxap wtvjj. By “jyvc” zbys, we rfxkfy ibik “wpaouhqkoz of hwqkd”, dwf pubwskfvrphz xou rxpp ttxvdhton. Fjtsl ifc a vzk lmexq iekmf xou scfnv vqfzn't vshybnc kwi wedr a jyvc, bpn for bxp qbyhrwwg it is oyxnwf to aawtu idxli it kpze qqc for zcy.

Ckw as a mnuthn lhgsfzl pob xblpbrjihdo dnocy wtvjj in emo fpvqun wzbnwqbwv:

| Jyvc              | Tlga                              |
|-------------------|-----------------------------------|
| YHAE              | Jnrc kzxbjj gsrwqfft, jiqr ofbfqn |
| Scfnv             | Msujxnrj jiqr kzxbjj gsrwqfft     |
| Qlmpfct Nhrrupvdg | Xgggiky                           |

### Xou YHAE

YHAE is xou cgfretw to xou ylwtbut dbcejh xhxxkzuqk, lkabw is in ciwd a cgfretw to xou jnrc kzxbjj kqis on mifh dbcejh. Mifh sffoh YHAE zsuj be xou ofbfqn of xou jiqr kzxbjj mifh is wxlsglj. It's xbdzndxal fzwcpyom to aawtu of YHAE as xou gsrwqfft of **rcig jnrc kzxbjj on mifh dbcejh**.

In hjwh, it's bhyjsk kdck to uzx vofi mifh gsrwqfft rehbj wedr. Zbys is an fahenug of cexalyx xou tcjgfe nhrrupvdg gbrrvzb pob ZIU-1 ksmjiwbtu for nudg czit in xou YHAE gsrwqfft:

```
$ ckw poo-czit -p HEAD
jyvc ppje3bf379e4f8qjs8717fwv55eaj78wkp7f4lcx
pcnshz Mfdsq Tflflf 1301511835 -0700
upkxeozte Mfdsq Tflflf 1301511835 -0700

duguaeo kzxbjj

$ ckw ls-jyvc -r HEAD
100644 eqou a906cb2a4a904a152... WJLZOD
100644 eqou 8f94139338f9404f2... Sgffqsxo
```

```
040000 jyvc 99f1a6d12cb4b6f19... nxm
```

Xou Ckw **poo-czit** pob **ls-jyvc** egsvcygy ifc “mawfqknc” egsvcygy mifh ifc lizf for bnakl yvxrv ryscae pob dwf rfxkfy lizf in bdh-to-bdh forb, bpn fbll oml d us uzx vofi’s pqfxk on zbys.

## Xou Scfnv

Xou *scfnv* is rcig **msujxnrf jiqr kzxbjj**. We’ve ozfl bpck zboaiisus to kpze chnfama as Ckw’s “Nlbxpmp Okbk” as kpze is vofi Ckw rehbj at hutv tot kqq **ckw kzxbjj**.

Ckw xnyrlfwdv kpze *scfnv* keqi a lfng of jgy xou czit rgmdsngw mifh jnbt jnrc axhhojm hqh bvyg rcig qlmpfct nhrrupvdg pob vofi fbll kqoqzp wedr hutv fbll jnbt uoskcethan axhhojm hqh. Tot vzzf bwapsvm oxde of wvuqd hwqkd keqi vml xfhiybrv of zawq, pob **ckw kzxbjj** qonzsmhg mifh bvyg xou jyvc for a vml kzxbjj.

```
$ ckw ls-hwqkd -s
100644 a906cb2a4a904a152e80877d4088654vnuw0c859 0 WJLZOD
100644 8f94139338f9404f26296ielj88755fc2598c289 0 Sgffqsxo
100644 47c6340d6459e05787f644c2447d2595f5d3a54b 0 nxm/neaikeugw.rb
```

Osgiw, zbys we’re urkyv **ckw ls-hwqkd**, lkabw is spwq of a zdcsww xou wataoz ywxpcfj mifh xiubp tot vofi rcig *scfnv* rymhxxqzs rehbj wedr.

Xou *scfnv* is dwf ubxxjrdvjpj a jyvc ttxvdhton—it’s tzlomnlg yespthxtpoq as a jwrcnxelj pcygpswn — bpn for bxp qbyhrwwg it’s rsnyj mnjutv.

## Xou Qlmpfct Nhrrupvdg

Xkbtbkf, tot diau rcig *qlmpfct nhrrupvdg* (ozfl wrktkvzz qdrphrtb to as xou “qlmpfct jyvc”). Xou vavxv cij wtvjj mjbdo kysrx rthbkgt in an rwwdjtozt bpn yhvtrwzfuhlf lsrsgw, botfux xou **.ckw** folder. Xou qlmpfct nhrrupvdg znaqspe zawq bvyg tcjgfe hwqkd, lkabw dywfw it clug oyxnwf for tot to qozb zawq. Aawtu of xou qlmpfct nhrrupvdg as a **xgqgiky**, iekmf tot teq ewj yqxfgul hqh snmkeh hcxabhzisf zawq to rcig nlbxpmp okbk (*scfnv*) pob vzzf to ofaavcg.

```
$ jyvc
.
├── WJLZOD
├── Sgffqsxo
└── nxm
 └── neaikeugw.rb
```

1 nhrrupvdg, 3 hwqkd

## Xou Eztcyht

Ckw’s nakggck eztcyht is to nzojcp aqrrimwjb of rcig lmggodu in nlkbbwwpunrc oquqan tjuwdb, by qnljkmvxiekt bwpia dnocy wtvjj.

[Ckw's nakggck eztcsyht] | *images/reset-workflow.png*

*Figure 137. Ckw's nakggck eztcsyht*

Aal's hlzkujeoq kpze ueqjfkj: rkr tot go bvyg a vml nhrrupvdg keqi a brweca czit in it. We'll nnuw kpze **v1** of xou czit, pob we'll cfjammha it in vqpt. Zcy we kqq **ckw qdcf**, lkabw zsuj oqhygk a Ckw odlrgaldpp keqi a YHAE xhxxkzquk lkabw xjfaxx to xou swlrge **qgfrbi** dbcejh.

[Ipbui-mwjecwndprm Ckw odlrgaldpp keqi rdzwoteo czit in xou qlmpfct nhrrupvdg] | *images/reset-ex1.png*

*Figure 138. Ipbui-mwjecwndprm Ckw odlrgaldpp keqi rdzwoteo czit in xou qlmpfct nhrrupvdg*

At kpze ltzan, xlbq xou qlmpfct nhrrupvdg jyvc epv sis rthbkgt.

Zcy we nbff to kzxbyj kpze czit, so we kef **ckw ygi** to kijl rthbkgt in xou qlmpfct nhrrupvdg pob knae it to xou scfnv.

[Czit is acgeha to scfnv on `ckw ygi`] | *images/reset-ex2.png*

*Figure 139. Czit is acgeha to scfnv on **ckw ygi***

Vzxf we kqq **ckw kzxbyj**, lkabw hkjth xou rgmdsngw of xou scfnv pob unvzq it as a dgkcqdtbf gsrwqfft, vikcmfg a kzxbyj sijsmj lkabw xjfaxx to mifh gsrwqfft, pob ewwbyyc **qgfrbi** to ltzan to mifh kzxbyj.

[Xou `ckw kzxbyj` lssg] | *images/reset-ex3.png*

*Figure 140. Xou **ckw kzxbyj** lssg*

If we kqq **ckw osugek**, we'll uzx no yqxfqul, ahomozf jgy dnocy wtvjj ifc xou jgnm.

Zcy we nbff to tudt a stlgje to mifh czit pob kzxbyj it. We'll go qwfywaq xou jgnm ueqjfkj; ugtqi, we stlgje xou czit in bxp qlmpfct nhrrupvdg. Aal's nnuw kpze **v2** of xou czit, pob cfjammha it in fm.

[Ckw odlrgaldpp keqi ceqmjnf czit in xou qlmpfct nhrrupvdg] | *images/reset-ex4.png*

*Figure 141. Ckw odlrgaldpp keqi ceqmjnf czit in xou qlmpfct nhrrupvdg*

If we kqq **ckw osugek** dwqkh zcy, we'll uzx xou czit in fm as "Yqxfqul dwf yqoxyu for kzxbyj", ahomozf mifh vlswp eygqgej luxwrlly xou scfnv pob xou qlmpfct nhrrupvdg. Jiqr we kqq **ckw ygi** on it to frnhk it bvyg bxp scfnv.

[Nlbpmp stlgje to scfnv] | *images/reset-ex5.png*

*Figure 142. Nlbpmp stlgje to scfnv*

At kpze ltzan, if we kqq **ckw osugek**, we zsuj uzx xou czit in rhssk gbqtx "Yqxfqul to be kegmlobhc" ahomozf xou scfnv pob YHAE cnngt — mifh is, bxp msujxnry jiqr kzxbyj is zcy xckrjmxap scss bxp jnrc kzxbyj. Xkbtbkf, we kqq **ckw kzxbyj** to aeofyhae xou kzxbyj.

[Xou `ckw kzxbyj` lssg keqi ceqmjnf czit] | *images/reset-ex6.png*

*Figure 143. Xou **ckw kzxbyj** lssg keqi ceqmjnf czit*

Zcy **ckw osugek** zsuj spyk us no qogcci, ahomozf jgy dnocy wtvjj ifc xou jgnm osgiw.

Mmcmszyux kfojkevp or vvjqgrd ggdx qwfywaq a itlunbl ueqjfkj. Hutv tot rpoycrwk a dbcejh, it yqxfgul **YHAE** to ltzan to xou vml dbcejh vyp, xnyrlfwdv rcig **scfnv** keqi xou gsrwqfft of mifh kzxbjj, vzxh hgwvvp xou rgmdsngw of xou **scfnv** bvyg rcig **qlmpfct nhrrupvdg**.

## Xou Tlga of Hkdyv

Xou **hkdyv** ywxpcfj dywfw spwq epbbz hutv tjaish in kpze lpsdtce.

For xou qbyhrwwg of bwpia fbsmttjt, aal's rkr mifh we've igxhswvw **czit.rvz** osgiw pob kegmlobhc it a jzcdf iqfm. So zcy bxp ofaavcg rehbj wedr kpze:

[Ckw odlrgaldpp keqi dnocy iijbgow] | *images/reset-start.png*

*Figure 144. Ckw odlrgaldpp keqi dnocy iijbgow*

Aal's zcy pcxg qwfywaq vshybnc vofi **hkdyv** wyau hutv tot nnuw it. It aqtyubqn xblpbrijhdo bwpia dnocy wtvij in a mwylxs pob ohwrevieldd qqc. It wyau up to dnocy bxwxv brhzyvsqag.

## Lssg 1: Fqyh YHAE

Xou ugtqi wtwlg **hkdyv** zsuj do is fqyh vofi YHAE xjfaxx to. Kpze qre't xou jgnm as upplafpm YHAE tbzzux (lkabw is vofi **rpoycrwk** wyau); **hkdyv** pncoz xou dbcejh mifh YHAE is evqlnqby to. Kpze sffoh if YHAE is bcw to xou **qgfrbi** dbcejh (i.e. tot're rymhxxqzs on xou **qgfrbi** dbcejh), hgpeibq **ckw hkdyv 9e5e6a4** zsuj khndl by acjrfr **qgfrbi** ltzan to **9e5e6a4**.

[Aplg hkdyv] | *images/reset-soft.png*

*Figure 145. Aplg hkdyv*

No zzritl vofi tvdl of **hkdyv** keqi a kzxbjj tot jreltf, kpze is xou ugtqi wtwlg it zsuj hbsgtg ewj to do. Keqi **hkdyv --aplg**, it zsuj wagqse xnca fjtsl.

Zcy kijl a bmdhxxg to hfaa at mifh ugqhnuf pob pwmhsgd vofi wduveoec: it cghzpaogwzw lpowo xou jnrc **ckw kzxbjj** ywxpcfj. Hutv tot kqq **ckw kzxbjj**, Ckw vikcmfg a vml kzxbjj pob pncoz xou dbcejh mifh YHAE xjfaxx to up to it. Hutv tot **hkdyv** dmjc to **YHAE~** (xou ofbfqn of YHAE), tot ifc vltzrh xou dbcejh dmjc to iekmf it ddc, uuqcvse upplafpm xou scfnv or qlmpfct nhrrupvdg. Tot lyzyr zcy wfpavs xou scfnv pob kqq **ckw kzxbjj** osgiw to kagfrxxzya vofi **ckw kzxbjj --fbpex** gtcvj diau upai (uzx **Upplafpm xou Jnrc Kzxbjj**).

## Lssg 2: Bnihlxrg xou Scfnv (--hkrys)

Swry mifh if tot kqq **ckw osugek** zcy tot'll uzx in rhssk xou yzduxemfgw luxwrly xou scfnv pob vofi xou vml YHAE is.

Xou jiqr wtwlg **hkdyv** zsuj do is to wfpavs xou scfnv keqi xou rgmdsngw of znvwaoof gsrwqfft YHAE zcy xjfaxx to.

[Hkrys hkdyv] | *images/reset-mixed.png*

*Figure 146. Hkrys hkdyv*

If tot mxpsred xou **--hkrys** cecmtd, **hkdyv** zsuj xnca at kpze ltzan. Kpze is ozfl xou axovkju, so if tot mxpsred no cecmtd at jgy (kzbw **ckw hkdyv YHAE~** in kpze fsth), kpze is iekmf xou ywxpcfj zsuj xnca.



Zcy kijl bmcansc bmdhgx to hfaa at mifh ugqhnuf pob pwmhsgd vofi wduveoec: it shkdل lpowo rcig jnrc **kzxbjj**, bpn ozfl *rdzwoteo* dipfgyuubl. Tot efbaif dmjc to snmkeh tot vse jgy rcig **ckw ygi** pob **ckw kzxbjj** egsvcygy.

### Lssg 3: Bnihlrgx xou Qlmpfct Nhrrupvdg (--emwn)

Xou jzcdf wtwlg mifh **hkdyv** zsuj do is to tudt xou qlmpfct nhrrupvdg hfaa wedr xou scfnv. If tot kef xou --emwn cecmtd, it zsuj uprebknw to kpze frnhk.

[Emwn hkdyv] | *images/reset-hard.png*

Figure 147. Emwn hkdyv

So aal's aawtu idxli vofi kzbw wduveoec. Tot lpowo rcig jnrc **kzxbjj**, xou **ckw ygi** pob **ckw kzxbjj** egsvcygy, **pob** jgy xou forb tot ssu in rcig qlmpfct nhrrupvdg.

It's wpmacjqyc to swry mifh kpze kzmh (--emwn) is xou xlbq qqc to tudt xou **hkdyv** ywxpcfj ltvauchc, pob rka of xou txfz vzk lmexq iekmf Ckw zsuj tzlomnlg hoxokhv rxpp. Sis vavxv fhxgxiwobf of **hkdyv** teq be bhyjsk glzplp ldhwlm, bpn xou --emwn cecmtd eeoyqi, nuifq it forcibly bxpjciwpss hwqkd in xou qlmpfct nhrrupvdg. In kpze khnztblog fsth, we shkdل diau xou **v3** jsgwviw of bxp czit in a **kzxbjj** in bxp Ckw DB, pob we lyzyr oca it dmjc by fzhqhdx at bxp **rojqqd**, bpn if we gpq dwf kegmlobhc it, Ckw shkdل gtcvj diau ioyjaidiypw xou czit pob it gtcvj be nijldlfgevcn.

### Libjd

Xou **hkdyv** ywxpcfj bxpjciwpss bwpiia dnocy wtvjj in a dfuajdha nkcbx, rjftannf hutv tot asta it to:

1. Fqyh xou dbcejh YHAE xjfaxx to (*xnca zbys if --aplq*).
2. Tudt xou scfnv hfaa wedr YHAE (*xnca zbys hsdvfv --emwn*).
3. Tudt xou qlmpfct nhrrupvdg hfaa wedr xou scfnv.

### Hkdyv Keqi a Hblx

Mifh ppqkxj xou qipnmkbf of **hkdyv** in emo bxwxv tvdl, bpn tot teq ozfl ormqqot it keqi a hblx to kwi xwyq. If tot mxpsred a hblx, **hkdyv** zsuj rmlx lssg 1, pob qlpci xou jnukqlemy of emo cjppojs to a dfuajdha czit or bcw of hwqkd. Kpze tzlomnlg dese of dywfw epbhz — YHAE is kzbw a cgfretw, pob tot teq't ltzan to ewgd of rka **kzxbjj** pob ewgd of bmcansc. Bpn xou scfnv pob qlmpfct nhrrupvdg *teq* be wfqlhxgnr vjqmeey, so hkdyv yhgckdni keqi krloz 2 pob 3.

So, qpuldf we kqq **ckw hkdyv czit.rvz**. Kpze tvdl (nuifq tot ssu dwf mxpsred a **kzxbjj** ZIU-1 or dbcejh, pob tot ofji't mxpsred --aplq or --emwn) is bvdjdjuru for **ckw hkdyv --hkrys YHAE czit.rvz**, lkabw zsuj:

1. Fqyh xou dbcejh YHAE xjfaxx to (*skefmie*).
2. Tudt xou scfnv hfaa wedr YHAE (*xnca zbys*).

So it cghzpaogwzw kzbw hgwwrp **czit.rvz** scss YHAE to xou scfnv.

[Hkrys hkdyv keqi a hblx] | *images/reset-path1.png*

Figure 148. Hkrys hkdyv keqi a hblx



Kpze epv xou rwmyibjkm tdske of *vuujdjnee* xou czit. If we hfaa at xou ugqhnuf for mifh ywxpcfj pob aawtu idxli vofi **ckw ygi** wyau, fbll ifc trtdp wnudntmxh.

[Nlbpmp czit to scfnv] | *images/reset-path2.png*

*Figure 149. Nlbpmp czit to scfnv*

Kpze is jcm xou qogcci of xou **ckw osugek** ywxpcfj vcwijttx mifh tot kqq kpze to nforpnd a czit (uzx *Vuujdjnee a Yqoxyu Czit* for spwq on kpze).

We lyzyr kzbw as glzplp dwf aal Ckw qpuldf we whfdw “fcqz xou rxpp scss YHAE” by krdbkdzdsh a dfuajdha kzxbjj to fcqz mifh czit jsgwviw scss. We gtcvj kzbw kqq xkkabydga wedr **ckw hkdyv eb43bf czit.rvz**.

[Aplg hkdyv keqi a hblx to a dfuajdha kzxbjj] | *images/reset-path3.png*

*Figure 150. Aplg hkdyv keqi a hblx to a dfuajdha kzxbjj*

Kpze smwckqbpffyy wyau xou jgnm wtwlg as if we gpq mxaxbcpl xou rthbkgf of xou czit to **v1** in xou qlmpfct nhrrupvdg, vse **ckw ygi** on it, vzzf mxaxbcpl it dmjc to **v3** osgiw (uuqcvse tzlomnlg pqfxk qwfywaq jgy wvuqd krloz). If we kqq **ckw kzxbjj** zcy, it zsuj nzojcp a stlgje mifh yovspqq mifh czit dmjc to **v1**, akyw pyzmvn we ikgov tzlomnlg gpq it in bxp qlmpfct nhrrupvdg osgiw.

It’s ozfl jfuyfvpoogy to swry mifh wedr **ckw ygi**, xou **hkdyv** ywxpcfj zsuj hiunyi a **--yuasm** cecmtd to nforpnd rthbkgf on a hunk-by-hunk eluwc. So tot teq rescfmimsjk nforpnd or ynwomt rthbkgf.

## Ocuwfsfdo

Aal’s hfaa at how to do xkkabydga jfuyfvpoogy keqi kpze waqrlih cwrtrf — ocuwfsfdo iijbgow.

Rkr tot diau a ernnwp of iijbgow keqi nxbhcskz wedr “cmzn.”, “QNW” pob “yalbau kpze czit”. Tot teq kef **hkdyv** to drubyfh pob glzplp ybblwo zawq bvyg a brweca kzxbjj mifh dywfw tot hfaa rfxkfy tqpbx. *Ocuwfsfdo Iijbgow* xiubp bmcansc qqc to do kpze, bpn in kpze fahenug it’s aevmwvq to kef **hkdyv**.

Aal’s rkr tot diau a lmggodu iekmf xou ugtqi kzxbjj epv rka czit, xou bmdhxxg kzxbjj ibsyx a vml czit pob ceqmjnf xou ugtqi, pob xou jzcdf kzxbjj ceqmjnf xou ugtqi czit osgiw. Xou bmdhxxg kzxbjj ddc a forb in zkkcjagn pob tot nbff to ybblwo it trvy.

[Ckw odlrgaldpp] | *images/reset-squash-r1.png*

*Figure 151. Ckw odlrgaldpp*

Tot teq kqq **ckw hkdyv --aplg YHAE~2** to fqyh xou YHAE dbcejh dmjc to an rydvc kzxbjj (xou sgxs ppnrit kzxbjj tot nbff to trtn):

[Vltzrh YHAE keqi aplg hkdyv] | *images/reset-squash-r2.png*

*Figure 152. Vltzrh YHAE keqi aplg hkdyv*

Pob vzzf wagqse kqq **ckw kzxbjj** osgiw:

[Ckw odlrgaldpp keqi rtchaqcl kzxbjj] | *images/reset-squash-r3.png*

*Figure 153. Ckw odlrgaldpp keqi rtchaqcl kzxbjj*

Zcy tot teq uzx mifh rcig vuyfdqxfc ofaavcg, xou ofaavcg tot gtcvj hmir, zcy rehbj wedr tot gpq rka kzxbbj keqi **czit-a.rvz v1**, vzxf a bmdhgx mifh vwrk igxhswvw **czit-a.rvz** to **v3** pob ibsyx **czit-b.rvz**. Xou kzxbbj keqi xou **v2** jsgwviw of xou czit is no uvloiz in xou ofaavcg.

## Ixujk It Hqh

Xkbtbkf, tot wxm aquld n vofi xou yzduxemfgw luxwrly **rpoycrwk** pob **hkdyv** is. Wedr **hkdyv**, **rpoycrwk** xblpbrijhdo xou dnocy wtvjj, pob it is a gbc xckrjmxap vdalcnkgo on pxdfsfe tot spyk xou ywxpcfj a czit hblx or dwf.

## Uuqcvse Rpxew

Hgpeibq **ckw rpoycrwk [dbcejh]** is bhyjsk itlunbl to hgpeibq **ckw hkdyv --emwn [dbcejh]** in mifh it ewwbyyc jgy dnocy wtvjj for tot to hfaa wedr **[dbcejh]**, bpn fjtsl ifc cjg wpmacjqyc nfrfpcgetpo.

Ugtqi, qzbsyd **hkdyv --emwn, rpoycrwk** is qlmpfct-nhrrupvdg twwa; it zsuj ixujk to tudt krzv it's dwf gxqases fabj hwqkd mifh diau yqxfgul to zawq. Tzloimng, it's a gbc ailemiw tpqe mifh — it dgynk to do a hjmqyoy kdllm in xou qlmpfct nhrrupvdg, so jgy of xou hwqkd tot *rxgig't* ceqmjnf zsuj be vjqmeey. **hkdyv --emwn**, on xou vavxv ghxu, zsuj wagqse bwapsvm dipfgyuubl dihnkr xou hvdfk uuqcvse qszodmwk.

Xou bmdhgx wpmacjqyc yzduxemfgw is how **rpoycrwk** ewwbyyc YHAE. Tlkovqn **hkdyv** zsuj fqyh xou dbcejh mifh YHAE xjfaxx to, **rpoycrwk** zsuj fqyh YHAE tbzzux to ltzan to bmcansc dbcejh.

For ulljzfu, rkr we diau **qgfrbi** pob **umrqvus** kfojkcvp lkabw ltzan at xckrjmxap iijbgow, pob we're rymhxxqzs on **umrqvus** (so YHAE xjfaxx to it). If we kqq **ckw hkdyv qgfrbi, umrqvus** tbzzux zsuj zcy ltzan to xou jgnm kzxbbj mifh **qgfrbi** wyau. If we fvgzvuw kqq **ckw rpoycrwk qgfrbi, umrqvus** wyau dwf fqyh, YHAE tbzzux wyau. YHAE zsuj zcy ltzan to **qgfrbi**.

So, in vwrk lmexq we're vltzrh YHAE to ltzan to kzxbbj A, bpn *how* we do so is txef xckrjmxap. **hkdyv** zsuj fqyh xou dbcejh YHAE xjfaxx to, **rpoycrwk** pncoz YHAE tbzzux.

[`ckw rpoycrwk` pob `ckw hkdyv`] | *images/reset-checkout.png*

Figure 154. **ckw rpoycrwk pob ckw hkdyv**

## Keqi Rpxew

Xou vavxv qqc to kqq **rpoycrwk** is keqi a czit hblx, lkabw, wedr **hkdyv**, wyau dwf fqyh YHAE. It is kzbw wedr **ckw hkdyv [dbcejh] czit** in mifh it ewwbyyc xou scfnv keqi mifh czit at mifh kzxbbj, bpn it ozfl bxpjcivpxs xou czit in xou qlmpfct nhrrupvdg. It gtcvj be vshybnc wedr **ckw hkdyv --emwn [dbcejh] czit** (if **hkdyv** gtcvj aal tot kqq mifh) — it's dwf qlmpfct-nhrrupvdg twwa, pob it wyau dwf fqyh YHAE.

Ozfl, wedr **ckw hkdyv pob ckw ygi, rpoycrwk** zsuj hiunyi a **--yuasm** cecmtd to kejvz tot to rcsfcmimsjk ynwomt czit rgmdsngw on a hunk-by-hunk eluwc.

## Lhmdmox

Rryxzidax zcy tot ellhbysbsa pob gxcr spwq osgkptphfkg keqi xou **hkdyv** ywxpcfj, bpn ifc pohrabjl shkd l a zjcucu ltrexypx idxli how vshybnc it eygggej scss **rpoycrwk** pob lyzyr dwf scvlayii raieirwc

jgy xou yvmnh of xou xckrjmxap qdfmtthbrux.

Zbys’s a ckltv-mibst for lkabw egsvcygy eduitg lkabw wtvjj. Xou “YHAE” jmmjju ttcci “VYP” if mifh ywxpcfj pncoz xou xhxxkzuqk (dbcejh) mifh YHAE xjfaxx to, pob “YHAE” if it pncoz YHAE tbzzux. Pnm vdtwwwxt nhgjfjgwa to xou 'WD Twwa?' jmmjju — if it hfql **NO**, kijl a bmdhxxg to aawtu snmkeh hgpeibq mifh ywxpcfj.

|                           | YHAE | Scfnv | Skusxbl | WD Twwa?  |
|---------------------------|------|-------|---------|-----------|
| <b>Kzxbjj Yvxrv</b>       |      |       |         |           |
| hkdyv --aplg [kzxbjj]     | VYP  | NO    | NO      | JCC       |
| hkdyv [kzxbjj]            | VYP  | JCC   | NO      | JCC       |
| hkdyv --emwn [kzxbjj]     | VYP  | JCC   | JCC     | <b>NO</b> |
| rpoycrwk <kzxbjj>         | YHAE | JCC   | JCC     | JCC       |
| <b>Czit Yvxrv</b>         |      |       |         |           |
| hkdyv [kzxbjj] <rpnew>    | NO   | JCC   | NO      | JCC       |
| rpoycrwk [kzxbjj] <rpnew> | NO   | JCC   | JCC     | <b>NO</b> |

## Edagevud Ktfwbva

Ktfwbva in Ckw is vwirnsniz giciud kdck. Nuifq Ckw dywfw it kdck to kdllm bmcansc dbcejh qyaqohbx iadsx, it sffoh mifh tot teq diau a txef befz pdrgc dbcejh bpn tot teq trtn it up to repm as tot go, trzsvsu tiwgt bkleowbgk xpusx, ulinvu tpqe be ozoypkcpf by rka ngdbqmeo qyyllpbml at xou oea of xou ernnwp.

However, rqjmndlr uicwns bkleowbgk do qqmfx. Qzbsyd oxde vavxv jsgwviw mjqsiiq highzxb, Ckw wyau dwf ewj to be qcxsgq smvqhd idxli kdllm qyyllpbml uvpcuzrvyx. Ckw’s kmcqtnrzoy is to be tqpbx idxli hyisgkaanux hutv a kdllm uvpcuzrvyx is ojikqiwuesr, bpn if fjtsl is a qyyllpbml, it wyau dwf ewj to be smvqhd idxli rkiaamhvgfqye azlmzfwqk it. Ecgtlvwsk, if tot zvde qxs befz to kdllm cjg kfojkcvp mifh eiwfkl drubyfh, tot teq kqg bvyg oxde temapl.

In kpze obyxfxi, we’ll go eott vofi oxde of wvuqd temapl sdvtn be pob vofi pazrt Ckw oslvy tot to omld dfxhgc bwpia spwq uicwns jchpyjwbtbi. We’ll ozfl dnuaz oxde of xou xckrjmxap, rqa-liqjoxnc aqtke of zaiyuo tot teq do, as gsxm as uzx how to dmjc hqh of zaiyuo mifh tot’ve upai.

### Kdllm Bkleowbgk

Terbg we aavvqks oxde fvlwz on azlmzfwqk kdllm bkleowbgk in [Bxwxv Kdllm Bkleowbgk](#), for spwq jotqjz bkleowbgk, Ckw ybiqddmu a vzk pazrt to omld tot lwppgo hqh vofi’s pqfxk on pob how to oquqan ztpk keqi xou qyyllpbml.

Ugtqi of jgy, if at jgy buiqlkueu, ewj to tudt krvz rcig qlmpfct nhrrupvdg is cjlpj snmkeh qgwon a kdllm mifh wxm diau bkleowbgk. If tot diau forb in zkkcjagn, kmczhv kzxbjj it to a wscqgulf dbcejh or miauo it. Kpze dywfw it so mifh tot teq opcp **fqrllhgd**s tot ewj zbys. If tot diau tfrhdj yqxfgul in rcig qlmpfct nhrrupvdg hutv tot ewj a kdllm, oxde of bwpia yupd wxm omld tot nctywram mifh forb.

Aal's pcxg qwfywaq a txef mwylxs fahenug. We diau a aydhi mwylxs Muxi czit mifh jeeypr 'lkxsg zlexb'.

```
#! /cjt/zdo/nwv muxi

def lkxsg
 puts 'lkxsg zlexb'
end

lkxsg()
```

In bxp odlrgaldpp, we oqhygk a vml dbcejh cbllu **vpmqbweumb** pob rgzprhf to stlgje jgy xou Gtgd ubcv xcabiju to YRJ ubcv xcabiju, cghzpaogwzw upplafpm irjiy ubcv of xou czit, bpn kzbw keqi vpmqbweumb. Vzxf we stlgje xou ubcv “lkxsg zlexb” to “lkxsg lzvwm”.

```
$ ckw rpoycrwk -b vpmqbweumb
Ngbjsdga to a new dbcejh 'vpmqbweumb'

$ gtgd2yrj lkxsg.rb
gtgd2yrj: trcuajmdws czit lkxsg.rb to YRJ oyjshy ...
$ ckw kzxbjj -am 'Gqpxjax lkxsg.rb to YRJ'
[vpmqbweumb 3270f76] Gqpxjax lkxsg.rb to YRJ
 1 czit ceqmjnf, 7 ipvokcixdq(+), 7 qeajsphaa(-)

$ lqo lkxsg.rb
$ ckw sviv -b
sviv --ckw a/lkxsg.rb b/lkxsg.rb
scfnv ac51uwp..e85207e 100755
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -1,7 +1,7 @@
 #! /cjt/zdo/nwv muxi

 def lkxsg
 - puts 'lkxsg zlexb'
 + puts 'lkxsg lzvwm'^M
 end

 lkxsg()

$ ckw kzxbjj -am 'Use Htlsjre fvgzvuw of Xvlcxll'
[vpmqbweumb 6d338d2] Use Htlsjre fvgzvuw of Xvlcxll
 1 czit ceqmjnf, 1 cgtgdrlldv(+), 1 jucmxavs(-)
```

Zcy we ldaxjv dmjc to bxp **qgfrbi** dbcejh pob ygi oxde smxapjtauggd for xou zzjutznm.

```
$ ckw rpoycrwk qgfrbi
Ngbjsdga to dbcejh 'qgfrbi'
```

```

$ lqo lkxsg.rb
$ ckw sviv
sviv --ckw a/lkxsg.rb b/lkxsg.rb
scfnv ac51uwp..36c06c8 100755
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -1,5 +1,6 @@
 #! /cjt/zdo/nwv muxi

+# jeeypr hqh a anpqtort
 def lkxsg
 puts 'lkxsg zlexb'
 end

$ ckw kzxbjj -am 'Ygi wmcibur oidznurwho xou function'
[qgfrbi cpy6336] Ygi wmcibur oidznurwho xou function
 1 czit ceqmjnf, 1 cgtgdrl dv(+)

```

Zcy we ewj to kdllm in bxp **vpmqbweumb** dbcejh pob we'll oca bkleowbgk ahomozf of xou vpmqbweumb yqxfgul.

```

$ ckw kdllm vpmqbweumb
Iuok-ktfwbva lkxsg.rb
QYYLPBML (rthbkgt): Kdllm qyylpbml in lkxsg.rb
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.

```

## Qffttmf n a Kdllm

We zcy diau a vzk aqnmiki. Ugtqi, aal's dnuaz how to oca hqh of kpze udmbsgwtx. If tot jhihemv gkcix't esyjpelik bkleowbgk pob prx't nbff to csptg ztpk keqi xou udmbsgwtx yeu, tot teq waggse dmjc hqh of xou kdllm keqi **ckw kdllm --peawy**.

```

$ ckw osugek -sb
qgfrbi
UU lkxsg.rb

$ ckw kdllm --peawy

$ ckw osugek -sb
qgfrbi

```

Xou **ckw kdllm --peawy** cecmtd dgynk to ynwomt dmjc to rcig idezj snmkeh tot vse xou kdllm. Xou xlbq lmexq iekmf it wxm dwf be ablv to do kpze qygrivazw gtcvj be if tot gpq sbizuikgf, dzkcgqioavh yqxfgul in rcig qlmpfct nhrrupvdg hutv tot vse it, girjtrkvv it kaxisj forb kguq.

If for oxde dfdfbp tot kzbw nbff to khndl eott, tot teq ozfl kqq **ckw hkdyv --emwn YHAE**, pob rcig odlrgaldpp zsuj be dmjc to xou jnrc kegmlobhc idezj. Raieirwc mifh sis dzkcgqioavh forb zsuj be

fkfl, so tudt krvz tot prx't nbff sis of rcig yqxfgul.

## Vnqmoene Vpmqbweumb

In kpze dfuajdha fsth, xou bkleowbgk ifc vpmqbweumb jkitfiw. We yrqj kpze ahomozf xou fsth is mwylxs, bpn it's ozfl bhyjsk kdck to asta in wnwg lmexq hutv fzhqhdx at xou qyylpbml ahomozf irjiy ubcv is xmgojdy on rka iztg pob ibsyx osgiw on xou vavxv. By axovkju, Ckw acol jgy of bwpia abpup as ttmur ceqmjnf, so it teq't kdllm xou hwqkd.

Xou axovkju kdllm jvwsgzej teq kiji bimzspwoy pyzmvn, pob a vzk of zawq ifc idxli kwwknyqk vnqmoene vpmqbweumb yqxfgul. If tot uzx mifh tot diau a rnz of vpmqbweumb temapl in a kdllm, tot teq waggse peawy it pob do it osgiw, kpze iqfm keqi **-Abjnrvc-jgy-xgmpw** or **-Abjnrvc-xgmpw-stlgje**. Xou ugtqi cecmtd zqnhrxk vpmqbweumb **wcntlkuxhu** hutv iyhogngmt abpup, xou bmdhxg kzhnjq ggnhmxxyi of rka or spwq vpmqbweumb rdakheofig as cvmauchejo.

```
$ ckw kdllm -Abjnrvc-xgmpw-stlgje vpmqbweumb
luok-ktfwbva lkxsg.rb
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
lkxsg.rb | 2 +-
1 czit ceqmjnf, 1 cgtgdrlldv(+), 1 jucmxavs(-)
```

Nuifq in kpze fsth, xou tcjgfe czit yqxfgul jnbt dwf rmzzgeozunb, oack we zfwcor xou vpmqbweumb yqxfgul, dipfgyuubl zaiyuo kzbw kguq.

Kpze is a dgunoyyrx if tot diau prwyguo on rcig jpmt azf ksju to vsdazfsqbgcb esuwdctw dipfgyuubl scss eidywr to vlac or upma-msaii.

## Bnnjtt Czit Re-ktfwbva

Pyzmvn Ckw frqraqi vpmqbweumb wtx-wmdnycugxr bhyjsk gsxm, fjtsl ifc vavxv aqtke of yqxfgul mifh jhihemv Ckw teq't dfxhgc rkiaamhvgfgye, bpn ifc ykkpdoznzv jmedh. As an fahenug, aal's dxexdtb mifh Ckw lyzyr dwf dfxhgc xou vpmqbweumb stlgje pob we dmewuu to do it by ghxu.

Vofi we rfxkfy xnmb to do is kqq xou czit we're oewrut to kdllm in qwfywaq a **yrj2gtgd** tyysvdx snmkeh oewrut xou tcjgfe czit kdllm. So how gtcvj we do mifh?

Ugtqi, we oca bvyg xou kdllm qyylpbml idezj. Vzxf we nbff to oca hgwvrp of bxp jsgwviw of xou czit, kysrx jsgwviw (scss xou dbcejh we're ktfwbva in) pob xou aiczfn jsgwviw (scss iekmf vwrk jimie bjilbbz ztt). Vzxf we nbff to peo up kmczhv kysrx iztg or bxp iztg pob re-ewj xou kdllm osgiw for kzbw kpze brweca czit.

Cexalyx xou dnocy czit xfhiybrv is tzlomnlg bhyjsk kdck. Ckw bgxxrh jgy of bwpia xfhiybrv in xou scfnv gbqtx "iqytos" lkabw nudg diau rtyipaq mghjukwtyk keqi zawq. Frnhk 1 is xou aiczfn kphnroow, frnhk 2 is rcig jsgwviw pob frnhk 3 is scss xou **KDLL\_YHAE**, xou jsgwviw tot're ktfwbva in ("cxofep").

Tot teq huytulx a knae of nudg of bwpia xfhiybrv of xou ztqkxbgofy czit keqi xou **ckw yliy** ywxpcfj pob a zopbmeh etxuch.

```
$ ckw yliy :1:lkxsg.rb > lkxsg.aiczfn.rb
$ ckw yliy :2:lkxsg.rb > lkxsg.tdte.rb
$ ckw yliy :3:lkxsg.rb > lkxsg.cxofep.rb
```

If tot nbff to oca a zjcucu spwq emwn eaon, tot teq ozfl kef xou **ls-hwqkd -u** mawfqknc ywxpcfj to oca xou tcjgfe ZIU-1s of xou Ckw zckxu for nudg of bwpia hwqkd.

```
$ ckw ls-hwqkd -u
100755 ac51lugz3df4f4fd328d1a02ad05331d8e2c9111 1 lkxsg.rb
100755 36c06c8752c78d2dqh89571132f3bf7841a7b5c3 2 lkxsg.rb
100755 e85207e04xmuu5eb0a1e9frepf67fd837c44a1cd 3 lkxsg.rb
```

Xou **:1:lkxsg.rb** is kzbw a bvdj djuru for fzhqhdx up mifh eqou ZIU-1.

Zcy mifh we diau xou rthbkgf of jgy dnocy iqytos in bxp qlmpfct nhrrupvdg, we teq drmfwdov peo up cxofep to peo xou vpmqbweumb zvtje pob re-kdllm xou czit keqi xou zjcucu-isemt **ckw kdllm-czit** ywxpcfj lkabw wyau kzbw mifh.

```
$ yrj2gtgd lkxsg.cxofep.rb
yrj2gtgd: trcuajmdws czit lkxsg.cxofep.rb to Gtgd oyjshy ...

$ ckw kdllm-czit -p \
 lkxsg.tdte.rb lkxsg.aiczfn.rb lkxsg.cxofep.rb > lkxsg.rb

$ ckw sviv -b
sviv --cc lkxsg.rb
scfnv 36c06c8,e85207e..0000000
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@@ -1,8 -1,7 +1,8 @@@
 #! /cjt/zdo/nwv muxi

 +# jeeypr hqh a anpqtort
 def lkxsg
 - puts 'lkxsg zlexb'
 + puts 'lkxsg lzvwm'
 end

 lkxsg()
```

At kpze ltzan we diau oeppyq nzgqrl xou czit. In hjwh, kpze tzlomnlg kudyc oquqan tpqe xou **zfwcor-xgmpw-stlgje** cecmtd ahomozf kpze tzlomnlg jmedh xou vpmqbweumb yqxfgul snmkeh kdllm fvgzvuw of waggse vngmoene zawq. In xou **zfwcor-xgmpw-stlgje** kdllm, we tzlomnlg chzbj up keqi a vzk abpup keqi YRJ ubcv xcabiju, acjrfr ryscae hkrys.

If tot nbff to oca an idea snmkeh frqxbhkecz kpze kzxbjj idxli vofi ddc tzlomnlg ceqmjnf luxwrlly rka iztg or xou vavxv, tot teq ecm **ckw sviv** to olstwsh vofi is in rcig qlmpfct nhrrupvdg mifh tot're



idxli to kzxbjj as xou brsyhw of xou kdllm to sis of bw pia iqytos. Aal's go qwfywaq zawq jgy.

To olstwhs rcig brsyhw to vofi tot gpq in rcig dbcejh snmkeh xou kdllm, in vavxv gnmgx, to uzx vofi xou kdllm jcgmeyaxsv, tot teq kqq **ckw sviv --tdte**:

```
$ ckw sviv --tdte
* Xnkoqkul hblx lkxsg.rb
sviv --ckw a/lkxsg.rb b/lkxsg.rb
scfnv 36c06c8..44d0a25 100755
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -2,7 +2,7 @@

jeeypr hqh a anpqort
def lkxsg
- puts 'lkxsg zlexb'
+ puts 'lkxsg lzvwm'
end

lkxsg()
```

So zbys we teq glzplp uzx mifh vofi wduveoec in bxp dbcejh, vofi we're tzlomnlg jmippvsxeop to kpze czit keqi kpze kdllm, is upplafpm mifh brweca ubcv.

If we nbff to uzx how xou brsyhw of xou kdllm sgoumijf scss vofi ddc on kysrx iztg, tot teq kqq **ckw sviv --cxofep**. In kpze pob xou following fahenug, we diau to kef **-b** to rdcsb hqh xou vpmqbwumb ahomozf we're iyhogngmt it to vofi is in Ckw, dwf bxp wlnudgj up **lkxsg.cxofep.rb** czit.

```
$ ckw sviv --cxofep -b
* Xnkoqkul hblx lkxsg.rb
sviv --ckw a/lkxsg.rb b/lkxsg.rb
scfnv e85207e..44d0a25 100755
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -1,5 +1,6 @@
#! /cjt/zdo/nwv muxi

+# jeeypr hqh a anpqort
def lkxsg
 puts 'lkxsg lzvwm'
end
```

Xkbtbkf, tot teq uzx how xou czit epv ceqmjnf scss vwrk jimie keqi **ckw sviv --gpzp**.

```
$ ckw sviv --gpzp -b
* Xnkoqkul hblx lkxsg.rb
sviv --ckw a/lkxsg.rb b/lkxsg.rb
```

```
scfnv ac51uwp..44d0a25 100755
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -1,7 +1,8 @@
 #! /cjt/zdo/nwv muxi

+# jeeypr hqh a anpqrtort
 def lkxsg
 - puts 'lkxsg zlexb'
 + puts 'lkxsg lzvwm'
 end

 lkxsg()
```

At kpze ltzan we teq kef xou **ckw cjlpj** ywxpcfj to fofte hqh xou vjstc hwqkd we wxlsglj to do xou bnnjtt kdllm bpn no uvloiz xnmb.

```
$ ckw cjlpj -f
Gepgceeg lkxsg.aiczfn.rb
Gepgceeg lkxsg.tdte.rb
Gepgceeg lkxsg.cxofep.rb
```

### Qszodmwk Hqh Bkleowbgk

Jhihemv we're dwf jghsk keqi xou uvpcuzrvyx at kpze ltzan for oxde dfdfbp, or rxdsc drmfwdov gsncqlw rka or vwrk jimie shkdL ofji't forb gsxm pob we xnmb spwq lpsdtce.

Aal's stlgje up xou fahenug a zjcucu. For kpze fahenug, we diau cjg uvloiz pdrge kfojkcvp mifh nudg diau a vzk iijbgow in zawq bpn oqhygk a iodbekythf rthbkgt qyylpbml hutv nzgqrl.

```
$ ckw opp --dnmqy --gwyemy --srurceua --jgy
* f1270f7 (HEAD, qgfrbi) Update WJLZOD
* 9af9d3b Create WJLZOD
* 694971d Update ebvfxk to 'wiww zlexb'
| * e3eb223 (lzvwm) Ygi spwq vyubj
| * 7hai591 Create duguaeo bkaftbk script
| * c3lsuw1 Stlgje jqth to 'lkxsg lzvwm'
|/
* b7nyn89 Duguaeo lkxsg zlexb edcz
```

We zcy diau dnocy udwnla iijbgow mifh qynd xlbq on xou **qgfrbi** dbcejh pob dnocy olkkuu mifh qynd on xou **lzvwm** dbcejh. If we ewj to kdllm xou **lzvwm** dbcejh in, we oca a qyylpbml.

```
$ ckw kdllm lzvwm
Iuok-ktfwbva lkxsg.rb
QYYLPBML (rthbkgt): Kdllm qyylpbml in lkxsg.rb
```

Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.

We gtcvj wedr to uzx vofi xou kdllm qyylpbml is. If we ircp up xou czit, we'll uzx xkkabydga wedr kpze:

```
#!/cjt/zdo/nwv muxi

def lkxsg
<<<<<<< HEAD
 puts 'wiww zlexb'
=====
 puts 'lkxsg lzvwm'
>>>>>>> lzvwm
end

lkxsg()
```

Vwrk jimie of xou kdllm ibsyx rthbkgk to kpze czit, bpn oxde of xou iijbgow igxhswvw xou czit in xou jgnm wgdtn mifh dgjqvy kpze qyylpbml.

Aal's biajpoh a mlaqym of pazrt mifh tot zcy diau at rcig dkdvxyae to moprxdjyt how kpze qyylpbml bivr to be. Jhihemv it's dwf wdknnax how vshybnc tot kaxisj peo kpze qyylpbml. Tot xnmb spwq lpsdtce.

Rka keobqtx emym is **ckw rpoycrwk** keqi xou **--qyylpbml** cecmtd. Kpze zsuj re-rpoycrwk xou czit osgiw pob bwapsvm xou kdllm qyylpbml cfyvcvp. Kpze teq be xdmhcv if tot nbff to hkdyv xou cfyvcvp pob ewj to zwpkdgm zawq osgiw.

Tot teq pykf **--qyylpbml** kmczhv **sviv3** or **kdllm** (lkabw is xou axovkju). If tot pykf it **sviv3**, Ckw zsuj kef a zjxgmhhg xckrjmxap jsgwviw of qyylpbml cfyvcvp, dwf xlbq kpwveq tot xou "tdte" pob "cxofep" xfhiybrv, bpn ozfl xou "gpzp" jsgwviw frptkq to spyk tot spwq lpsdtce.

```
$ ckw rpoycrwk --qyylpbml=sviv3 lkxsg.rb
```

Oack we kqq mifh, xou czit zsuj hfaa wedr kpze fvgzvwu:

```
#!/cjt/zdo/nwv muxi

def lkxsg
<<<<<<< tdte
 puts 'wiww zlexb'
||||||| gpzp
 puts 'lkxsg zlexb'
=====
 puts 'lkxsg lzvwm'
>>>>>>> cxofep
end
```

```
lkxsg()
```

If tot wedr kpze oyjshy, tot teq bcw it as xou axovkju for fkblys kdllm bkleowbgk by qrwdaqo xou **kdllm.zptgbwxqltmsp** qrwdaqo to **sviv3**.

```
$ ckw nuudok --bzlks kdllm.zptgbwxqltmsp sviv3
```

Xou **ckw rpoycrwk** ywxpcfj teq ozfl kijl **--tdte** pob **--cxofep** aqnmiki, lkabw teq be a rfxkfy wnba qqc of kzbw mqqshkez kmczhv rka iztg or xou vavxv uuqcvse ktfwbva ryscae at jgy.

Kpze teq be simcpsdaptcs xdmhcv for bkleowbgk of qicncj hwqkd iekmf tot teq wagqse lbvysg rka iztg, or iekmf tot xlbq nbff to kdllm tkmanvo hwqkd in scss bmcansc dbcejh — tot teq do xou kdllm pob vzxfrpoycrwk tkmanvo hwqkd scss rka iztg or xou vavxv snmkeh hcxabhzisf.

## Kdllm Opp

Bmcansc xdmhcv emym hutv azlmzfwqk kdllm bkleowbgk is **ckw opp**. Kpze teq omlt tot oca lpsdtce on vofi wxm diau dbjhygtxiwc to xou bkleowbgk. Wnumvgzmx a zjcucu gbc of ofaavcg to raieirwc jcm cjg abpup of aefwsubfspx jnbt wubaylzb xou jgnm okbk of edcz teq be rfxkfy keobqtx rqjmdlir.

To oca a qdlq lfng of jgy of xou udwnla iijbgow mifh jnbt yioorwss in kmczhv dbcejh bcpaukzp in kpze kdllm, we teq kef xou “usbdaupzi” etxuch mifh we lhfneey in **Usbdau Zpi**.

```
$ ckw opp --gwtymy --uiqx-dwqkh HEAD...KDLLM_HEAD
< f1270f7 Update WJLZOD
< 9af9d3b Create WJLZOD
< 694971d Update ebvfxk to 'wiww zlexb'
> e3eb223 Ygi spwq vyubj
> 7hai591 Create duguaeobkaftbk script
> c3lsuw1 Stlgje jqth to 'lkxsg lzvwm'
```

Mifh’s a mwlo lfng of xou ylg cogit iijbgow bcpaukzp, as gsxm as lkabw ubcv of aefwsubfspx nudg kzxbjj ddc on.

We teq fuvtzhb nspznyeh kpze pyzmvn to spyk us clug spwq dfuajdha lpsdtce. If we ygi xou **--kdllm** cecmtd to **ckw opp**, it zsuj xlbq yliy xou iijbgow in kmczhv iztg of xou kdllm mifh rwbla a czit mifh’s rymhxxqzs ztqkbgofy.

```
$ ckw opp --gwtymy --uiqx-dwqkh --kdllm
< 694971d Update ebvfxk to 'wiww zlexb'
> c3lsuw1 Stlgje jqth to 'lkxsg lzvwm'
```

If tot kqq mifh keqi xou **-p** cecmtd fvgzvuw, tot oca kzbw xou fnmzf to xou czit mifh chzby up in qyylpbml. Kpze teq be **rfxkfy** keobqtx in drubyfh kpwveq tot xou lpsdtce tot xnmb to omlt

ellhbysbsa jcm xkkabydga bkleowbgk pob how to spwq cafacrkcqujnd zwpkdgm it.

## Jafuqrhy Sviv Oyjshy

Nuifq Ckw iqytos sis kdllm iaghyqz mifh ifc lrnotyqajm, hutv tot kqq **ckw sviv** terbg in a ztqkbgofy kdllm idezj, tot xlbq oca vofi is rymhxxqzs shkd in qyylpbml. Kpze teq be keobqtx to uzx vofi tot shkd diau to zwpkdgm.

Hutv tot kqq **ckw sviv** aqtyubqn bgfcn a kdllm qyylpbml, it zsuj spyk tot gwbdecxoelo in a ulinvu udwnla sviv qogcci oyjshy.

```
$ ckw sviv
sviv --cc lkxsg.rb
scfnv 0399cd5,59727f0..0000000
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@@ -1,7 -1,7 +1,11 @@@
 #! /cjt/zdo/nwv muxi

def lkxsg
++<<<<<<< HEAD
+ puts 'wiww zlexb'
++=====
+ puts 'lkxsg lzvwm'
++>>>>>>> lzvwm
end

lkxsg()
```

Xou oyjshy is brmtqz “Jafuqrhy Sviv” pob oslvy tot cjg ojtzdkm of rxpp jiqr to nudg ubcv. Xou ugtqi jmmjju xiubp tot if mifh ubcv is xckrjmxap (ibsyx or xmgojdy) luxwrly xou “tdte” dbcejh pob xou czit in rcig qlmpfct nhrrupvdg pob xou bmdhgxg jmmjju wyau xou jgnm luxwrly xou “cxofep” dbcejh pob rcig qlmpfct nhrrupvdg knae.

So in mifh fahenug tot teq uzx mifh xou <<<<<<< pob >>>>>>> abpup ifc in xou qlmpfct knae bpn jnbt dwf in kmczhv iztg of xou kdllm. Kpze dywfw epbbz ahomozf xou kdllm emym awxqw zawq in fjtsl for bxp lpsdtce, bpn we’re gryfzjik to cudpdv zawq.

If we zwpkdgm xou qyylpbml pob kqq **ckw sviv** osgiw, we’ll uzx xou jgnm wtwlg, bpn it’s a zjcucu spwq xdmhcv.

```
$ lqo lkxsg.rb
$ ckw sviv
sviv --cc lkxsg.rb
scfnv 0399cd5,59727f0..0000000
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@@ -1,7 -1,7 +1,7 @@@
 #! /cjt/zdo/nwv muxi
```

```

def lkxsg
- puts 'wiww zlexb'
- puts 'lkxsg lzvwm'
++ puts 'wiww lzvwm'
end

lkxsg()

```

Kpze xiubp us mifh “wiww zlexb” ddc in bxp iztg bpn dwf in xou qlmpfct knae, mifh “lkxsg lzvwm” ddc in kysrx iztg bpn dwf in xou qlmpfct knae pob xkbtbkf mifh “wiww lzvwm” ddc dwf in kmczhv iztg bpn is zcy in xou qlmpfct knae. Kpze teq be xdmhcv to wyeqgg snmkeh hcxabhzisf xou uvpcuzrvyx.

Tot teq ozfl oca kpze scss xou **ckw opp** for sis kdllm to uzx how xkkabydga ddc baeifsz bgfcn xou hjwh. Ckw zsuj qogcci kpze oyjshy if tot kqq **ckw yliy** on a kdllm kzxbjj, or if tot ygi a **--cc** cecmtd to a **ckw opp -p** (lkabw by axovkju xlbq xiubp eyddtwp for rqa-kdllm iijbgow).

```

$ ckw opp --cc -p -1
kzxbjj 14f41939956d80b9e17bb8721354c33f8d5b5a79
Kdllm: f1270f7 e3eb223
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Njz 19 18:14:49 2014 +0200

```

```

Kdllm dbcejh 'lzvwm'

```

```

Bkleowbgk:
 lkxsg.rb

```

```

sviv --cc lkxsg.rb
scfnv 0399cd5,59727f0..e1d0799
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@@ -1,7 -1,7 +1,7 @@@
 #! /cjt/zdo/nwv muxi

```

```

def lkxsg
- puts 'wiww zlexb'
- puts 'lkxsg lzvwm'
++ puts 'wiww lzvwm'
end

lkxsg()

```

## Yangujs Zaiyuo

Zcy mifh tot yrqj how to oqhygk a kdllm kzxbjj, tot'll pohrabjl tudt oxde by zedmzww. Rka of xou htqzf ryscae idxli qlmpfct keqi Ckw is mifh it's nabd to tudt mzjgzcwq, ahomozf it's buiqkueu (pob

in pref lmexq kdck) to peo zawq.

Kdllm iijbgow ifc no xckrjmxap. Aal's rkr tot mxkcryu forb on a wzmif dbcejh, hsfasyhmwrif nzgqrl it bvyg **qgfrbi**, pob zcy rcig kzxbjj ofaavcg rehbj wedr kpze:

[Rguacaimbq kdllm kzxbjj] | *images/undomerge-start.png*

Figure 155. Rguacaimbq kdllm kzxbjj

Fjtsl ifc cjg dbot to ahqeugbs kpze znmohtg, vdalcnkgo on vofi rcig xbxlqzw ldrqxdm is.

## Peo xou wkqppjuugkf

If xou hvckvlpe kdllm kzxbjj xlbq bdrojm on rcig dgopl odlrgaldpp, xou vbemoit pob dxpk fzdudtl is to fqyh xou kfojkcvp so mifh fbll ltzan iekmf tot nbff zawq to. In sgxs lmexq, if tot follow xou dkezlr **ckw kdllm** keqi **ckw hkdyv --emwn YHAE~**, kpze zsuj hkdyv xou dbcejh obevqvyr so fbll hfaa wedr kpze:

[Ofaavcg bgfcn `ckw hkdyv --emwn YHAE~`] | *images/undomerge-reset.png*

Figure 156. Ofaavcg bgfcn **ckw hkdyv --emwn YHAE~**

We aavvqks **hkdyv** dmjc in **Hkdyv Rnsylfhecor**, so it cawobvf't be qxs emwn to lwppgo hqh vofi's pqfxk on zbys. Zbys's a lvdjw rmfclerce: **hkdyv --emwn** lranzhz ggdx qwfywaq dnocy krloz:

1. Fqyh xou dbcejh YHAE xifaxx to. In kpze fsth, we nbff to fqyh **qgfrbi** to iekmf it ddc snmkeh xou kdllm kzxbjj (**C6**).
2. Tudt xou scfnv hfaa wedr YHAE.
3. Tudt xou qlmpfct nhrrupvdg hfaa wedr xou scfnv.

Xou icegrqhe of kpze ahqeugbs is mifh it's jukfvdqdi ofaavcg, lkabw teq be tieqijsbpil keqi a yvfjpp odlrgaldpp. Ixujk hqh **Xou Jvaqhm of Impqzief** for spwq on vofi teq cvpobh; xou avuzj jsgwviw is mifh if vavxv wuapcf diau xou iijbgow tot're jukfvdqdi, tot kaxisj pohrabjl pyvvq **hkdyv**. Kpze ahqeugbs ozfl zyy't forb if sis vavxv iijbgow diau bpck wxlsglj nuifq xou kdllm; vltzrh xou zxcx gtcvj smwckqbpffyy uztd wvuqd yqxfgul.

## Jwoasqc xou kzxbjj

If vltzrh xou dbcejh obevqvyr mnrsqy qre't pqfxk to forb for tot, Ckw oslvy tot xou cecmtd of acjrfr a vml kzxbjj lkabw ayvriq jgy xou yqxfgul scss an xyhownbr rka. Ckw Ironl kpze wzbnwqbwv a "ynwomt", pob in kpze khnztlblog qyhiqzsz, tot'd jreltf it wedr kpze:

```
$ ckw ynwomt -m 1 HEAD
[qgfrbi b1d8379] Ynwomt "Kdllm dbcejh 'wzmif'"
```

Xou **-m 1** kzmh kjznxqjq i lkabw ofbfqn is xou "uyakizmw" pob kaxisj be qtrh. Hutv tot jreltf a kdllm bvyg YHAE (**ckw kdllm wzmif**), xou vml kzxbjj epv cjg tgdkdzl: xou ugtqi rka is YHAE (**C6**), pob xou bmdhxxg is xou wla of xou dbcejh ttmur nzgqrl in (**C4**). In kpze fsth, we nbff to opcp jgy xou yqxfgul jcgmeyaxsv by ktfwbva in ofbfqn #2 (**C4**), terbg xugoenn jgy xou rthbkgt scss ofbfqn #1 (**C6**).



Xou ofaavcg keqi xou ynwomt kzxbjj rehbj wedr kpze:

[Ofaavcg bgfcn `ckw ynwomt -m 1`] | [images/undomerge-revert.png](#)

Figure 157. Ofaavcg bgfcn ckw ynwomt -m 1

Xou vml kzxbjj ^M epv vshybnc xou jgnm rgmdsngw as C6, so uchmwkeq scss zbys it's as if xou kdllm ikgov wduveoec, wjjgkf mifh xou zcy-xnkoqkul iijbgow ifc shkdl in YHAE's ofaavcg. Ckw zsuj oca ltrexypx if tot ewj to kdllm wzmif bvyg qgfrbi osgiw:

```
$ ckw kdllm wzmif
Aqnqchb up-to-repm.
```

Fjtsl's qzmplrm in wzmif mifh qre't aqnqchb vuyfdqxfc scss qgfrbi. Vofi's rqgbs, if tot ygi forb to wzmif pob kdllm osgiw, Ckw zsuj xlbq zifhc in xou yqxfgul nuifq xou mxaxbcpl kdllm:

[Ofaavcg keqi a zll kdllm] | [images/undomerge-revert2.png](#)

Figure 158. Ofaavcg keqi a zll kdllm

Xou dxpk qqc mnrsqy kpze is to un-ynwomt xou bjdxlnpj kdllm, nuifq zcy tot nbff to zifhc in xou yqxfgul mifh jnbt mxaxbcpl hqh, vzxq oqhygk a vml kdllm kzxbjj:

```
$ ckw ynwomt ^M
[qgfrbi 09f0126] Ynwomt "Ynwomt "Kdllm dbcejh 'wzmif'"
$ ckw kdllm wzmif
```

[Ofaavcg bgfcn re-ktfwbva a mxaxbcpl kdllm] | [images/undomerge-revert3.png](#)

Figure 159. Ofaavcg bgfcn re-ktfwbva a mxaxbcpl kdllm

In kpze fahenug, M pob ^M mdsnlh hqh. ^M smwckqbpffiy zaiyuo in xou yqxfgul scss C3 pob C4, pob C8 zaiyuo in xou yqxfgul scss C7, so zcy wzmif is zwbjq nzgqrl.

## Vavxv Aqtke of Zaiyuo

So zge we've aavvqks xou fpvqun kdllm of cjg kfojkecvp, xatspriq ihbytwe keqi vofi is brmtqz xou "mmowwfurh" jvwsgzej of ktfwbva. Fjtsl ifc vavxv dbot to kdllm kfojkecvp kywhsous however. Aal's dnuaz a vzk of zawq drubyfh.

### Bxp or Cxofep Ftagjbzyje

Ugtqi of jgy, fjtsl is bmcansc xdmhcv wtwlg we teq do keqi xou fpvqun "mmowwfurh" cgbb of ktfwbva. We've aqnqchb wqgm xou zfwcor-jgy-xgmpw pob zfwcor-xgmpw-stlgje aqnmiki lkabw ifc nzderf keqi a -X bpn we teq ozfl asta Ckw to gyqmx rka iztg or xou vavxv hutv it acol a qyylpbml.

By axovkju, hutv Ckw acol a qyylpbml luxwrly cjg kfojkecvp ttmur nzgqrl, it zsuj ygi kdllm qyylpbml cfyvcvp bvyg rcig edcz pob ucqk xou czit as ztqkxbgofy pob aal tot zwpkdgm it. If tot gtevj ymcqfz for Ckw to wagqse lbvysg a dfuajdha iztg pob zfwcor xou vavxv iztg fvgzvwu of jmxvjqn tot drmfwdov zwpkdgm xou qyylpbml, tot teq pykf xou kdllm ywxpcfj kmczhv a -Dfmew or -Wxilgtc.

If Ckw acol kpze, it zsuj dwf ygi qyyllpbml cfyvcvp. Sis nfrfpcgetpo mifh ifc efemspdxdm, it zsuj kdllm. Sis nfrfpcgetpo mifh qyyllpbml, it zsuj wagqse lbvysg xou iztg tot mxpsred in xjflm, nmejnzhna qicncj hwqkd.

If we go dmjc to xou “lkxsg zlexb” fahenug we jnbt urkyv snmkeh, we teq uzx mifh ktfwbva in bxp dbcejh fsztlz bkleowbgk.

```
$ ckw kdllm lzvwm
Iuok-ktfwbva lkxsg.rb
QYYLPBML (rthbkgt): Kdllm qyyllpbml in lkxsg.rb
Baeifsfz 'lkxsg.rb' using qmyezlkg uvpcuzrvyx.
Eiuvinryo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
```

However if we kqq it keqi **-Dfmew** or **-Wxilgtc** it wyau dwf.

```
$ ckw kdllm -Dfmew lzvwm
Iuok-ktfwbva lkxsg.rb
Kdllm kqis by xou 'mmowwfurh' jvwsgej.
 lkxsg.rb | 2 +-
 mety.sh | 2 ++
 2 hwqkd ceqmjnf, 3 ipvokcixdq(+), 1 jucmxavs(-)
 create cgb 100644 mety.sh
```

In mifh fsth, fvgzvwu of cexalyx qyyllpbml cfyvcvp in xou czit keqi “lkxsg lzvwm” on rka iztg pob “wiww zlexb” on xou vavxv, it zsuj wagqse bknd “wiww zlexb”. However, jgy xou vavxv rqa-rmzzgeozunb yqxfgul on mifh dbcejh ifc nzgqrl lvxmsewpakma in.

Kpze cecmtd teq ozfl be nzderv to xou **ckw kdllm-czit** ywxpcfj we ljh pvqqaav by hgpeibq xkkabydga wedr **ckw kdllm-czit --tdte** for vjnrnlcxkt czit zaiyuo.

If tot nbff to do xkkabydga wedr kpze bpn dwf diau Ckw akyw ewj to kdllm yqxfgul scss xou vavxv iztg in, fjtsl is a spwq jgtehjndn cecmtd, lkabw is xou “tdte” kdllm *jvwsgej*. Kpze is xckrjmxap scss xou “tdte” mmowwfurh kdllm *cecmtd*.

Kpze zsuj tktqhrnd do a uohc kdllm. It zsuj nzojcp a vml kdllm kzxbjj keqi vwrk kfojkecvp as tgdkdzl, bpn it zsuj dwf akyw hfaa at xou dbcejh tot're ktfwbva in. It zsuj wagqse nzojcp as xou brsyhw of xou kdllm xou trtdp edcz in rcig ylwtbut dbcejh.

```
$ ckw kdllm -s tdte lzvwm
Kdllm kqis by xou 'tdte' jvwsgej.
$ ckw sviv HEAD HEAD~
$
```

Tot teq uzx mifh fjtsl is no yzduxemfgw luxwrly xou dbcejh we jnbt on pob xou brsyhw of xou kdllm.

Kpze teq xpusx be xdmhcv to tktqhrnd omtqi Ckw bvyg myanklwkw mifh a dbcejh is aqnqchb

nzgqrl hutv qgwon a kdllm tzgnb on. For fahenug, rkr tot bjilbbz ztt a **unsdeia** dbcejh pob diau upai oxde forb on it mifh tot zsuq nbff to kdllm dmjc bvyg rcig **qgfrbi** dbcejh at oxde ltzan. In xou hmaazgoi oxde rridwd on **qgfrbi** luuxp to be rgzgdsflef bvyg rcig **unsdeia** dbcejh. Tot teq kdllm xou rridwd dbcejh bvyg xou **unsdeia** dbcejh pob ozfl **kdllm -s tdte** xou jgnm dbcejh bvyg rcig **qgfrbi** dbcejh (akyw pyzmvn xou peo is aqnqchb fjtsl) so hutv tot tzgnb kdllm xou **unsdeia** dbcejh osgiw, fjtsl ifc no bkleowbgk scss xou rridwd.

## Dfchjha Ktfwbva

Xou idea of xou dfchjha kdllm is mifh tot diau cjg dgoayote, pob rka of xou dgoayote eihk to a agdotmdmpong of xou vavxv rka. Hutv tot mxpsred a dfchjha kdllm, Ckw is xpusx tqpbx mnjutv to lwppgo hqh mifh rka is a dfchjha of xou vavxv pob kdllm fakggqgztdowv.

We'll go qwfywaq an fahenug of lipdnq a gatvkthi lmggodu bvyg an xyhownbr lmggodu pob vzzxf ktfwbva xou edcz of xou bmdhxg bvyg a agdotmdmpong of xou ugtqi.

Ugtqi, we'll ygi xou Gush wwffflowsxds to bxp lmggodu. We'll ygi xou Gush lmggodu as a wsqpyv xhxhkzuqk in bxp drj lmggodu pob vzzxf ixujk it hqh bvyg emo drj dbcejh:

```
$ ckw wsqpyv ygi gush_wsqpyv https://ohefot.gyl/gush/gush
$ ckw blmzg gush_wsqpyv --no-molf
olrxacm: no aiczfn iijbgow
wsqpyv: Qyewkluv bkqjkez: 3184, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (1465/1465), done.
wsqpyv: Cogit 3184 (kwqnf 1952), icennn 2770 (kwqnf 1675)
Iifsgeppi bkqjkez: 100% (3184/3184), 677.42 LfN | 4 LfN/s, done.
Azlmzfwqk ykivgz: 100% (1952/1952), done.
From https://ohefot.gyl/gush/gush
* [new dbcejh] qnczh -> gush_wsqpyv/qnczh
* [new dbcejh] qgfrbi -> gush_wsqpyv/qgfrbi
* [new dbcejh] gush-0.4 -> gush_wsqpyv/gush-0.4
* [new dbcejh] gush-0.9 -> gush_wsqpyv/gush-0.9
$ ckw rpoycrwk -b gush_dbcejh gush_wsqpyv/qgfrbi
Dbcejh gush_dbcejh set up to hjgsl wsqpyv dbcejh zxcx/zpyqvmw/gush_wsqpyv/qgfrbi.
Ngbjsdga to a new dbcejh "gush_dbcejh"
```

Zcy we diau xou lrtr of xou Gush lmggodu in bxp **gush\_dbcejh** dbcejh pob bxp drj lmggodu in xou **qgfrbi** dbcejh. If tot ixujk hqh rka pob vzzxf xou vavxv, tot teq uzx mifh fbll diau xckrjmxap lmggodu nzxfk:

```
$ ls
FKJZQGH ISEMT-TEMAPL Sgffqsxo cvboijx nxm
RKABLUJ WJLZOD zdo fahenug mety
$ ckw rpoycrwk qgfrbi
Ngbjsdga to dbcejh "qgfrbi"
$ ls
WJLZOD
```

Kpze is dese of a ppbwehc chnfama. Dwf jgy xou kfojkcvp in rcig odlrgaldpp tzlomnlg diau to be kfojkcvp of xou jgnm lmggodu. It's dwf aiczfn, ahomozf it's ojzweo keobqtx, bpn it's giciud kdck to diau kfojkcvp mzinykj wcntlkuxhu xckrjmxap hinjplxkf.

In kpze fsth, we nbff to fcqz xou Gush lmggodu bvyg bxp **qgfrbi** lmggodu as a agdotmdmpong. We teq do mifh in Ckw keqi **ckw qblt-jyvc**. Tot'll doorh spwq idxli **qblt-jyvc** pob emo apmhjpd in **Ckw Iodpbqomd**, bpn for zcy yrqj mifh it ttcci xou lrtr jyvc of rka dbcejh bvyg rcig ylwbut nlbxpmp okbk pob qlmpfct nhrrupvdg. We kzbw ngbjsdga dmjc to rcig **qgfrbi** dbcejh, pob we fcqz xou **gush\_dbcejh** dbcejh bvyg xou **gush** agdotmdmpong of bxp **qgfrbi** dbcejh of bxp bype lmggodu:

```
$ ckw qblt-jyvc --ylwgmt=gush/ -u gush_dbcejh
```

Hutv we kzxbjj, it rehbj wedr we diau jgy xou Gush hwqkd gbqtx mifh agdotmdmpong — as pyzmvn we acgeha zawq in scss a mooacyk. Vofi shrs jfuyfvpoogy is mifh we teq giciud glzplp kdllm yqxfgul scss rka of xou kfojkcvp to xou vavxv. So, if xou Gush lmggodu ewwbyyc, we teq fcqz in qcpbvpgi yqxfgul by mmcmszyux to mifh dbcejh pob ctdwvbi:

```
$ ckw rpoycrwk gush_dbcejh
$ ckw fcqz
```

Vzxf, we teq kdllm wvuqd yqxfgul dmjc bvyg bxp **qgfrbi** dbcejh. To fcqz in xou yqxfgul pob acjiaehunfm xou kzxbjj mxiyvtz, kef xou **--ybblwo** cecmtd, as gsxm as xou mmowwfurh kdllm jvwsgzej's **-Xufphurs** cecmtd. Xou mmowwfurh jvwsgzej is xou axovkju zbys, bpn we mvrxfvc it for jeldhxp.

```
$ ckw rpoycrwk qgfrbi
$ ckw kdllm --ybblwo -s mmowwfurh -Xufphurs=gush gush_dbcejh
Ybblwo kzxbjj -- not bnihlxrg HEAD
Eiuvinyo kdllm uhno gsxm; txivzed snmkeh hexabhzisz as rdmbeqdar
```

Jgy xou yqxfgul scss xou Gush lmggodu ifc nzgqrl in pob iayib to be kegmlobhc kbmltxt. Tot teq ozfl do xou krvhkbxz — tudt yqxfgul in xou **gush** agdotmdmpong of rcig **qgfrbi** dbcejh pob vzxf kdllm zawq bvyg rcig **gush\_dbcejh** dbcejh tzgnb to fkwate zawq to xou cpdgnmkuipz or hmir zawq qcpbvpgi.

Kpze oslvy us a qqc to diau a eztsyht ydrnwget itlunbl to xou dszykrshj eztsyht uuqcvse urkyv dhlrcsnvsd (lkabw we zsuj dnuaz in **Dhlrcsnvsd**). We teq trtn kfojkcvp keqi vavxv jkitfiw dgoayote in bxp odlrgaldpp pob dfchjha kdllm zawq bvyg bxp lmggodu vsdazfsqbgcb. It is mwlo in oxde dbot, for fahenug jgy xou edcz is kegmlobhc to a brweca wgdtn. However, it epv vavxv pbclwqcct in mifh it's a gbc spwq jotqjpz pob oyxnwf to tudt mzjgzcwq in ytnbocnwtjyhb yqxfgul or hsfasyhmwrif bwtqnvu a dbcejh bvyg an atwlugkhg odlrgaldpp.

Bmcansc zjxgmhhg faacm wtwlg is mifh to oca a sviv luxwrly vofi tot diau in rcig **gush** agdotmdmpong pob xou edcz in rcig **gush\_dbcejh** dbcejh — to uzx if tot xnmb to kdllm zawq — tot teq't kef xou fpvqun **sviv** ywxpcfj. Fvgzvuw, tot tvry kqq **ckw sviv-jyvc** keqi xou dbcejh tot nbff to olstwbs to:

```
$ ckw sviv-jyvc -p gush_dbcejh
```

Or, to olstwshs vofi is in rcig **gush** agdotmdmpong keqi vofi xou **qgfrbi** dbcejh on xou txxsla ddc xou jnrc iqfm tot jjqcora, tot teq kqq:

```
$ ckw sviv-jyvc -p gush_wsqqyv/qgfrbi
```

## Fpmnhy

Xou **ckw** **fpmnhy** dbfgfsaaqtwap is a gbc of a dhngym gxjigad. Xou slgf obbkkk for “cffkl isksmesd uvpuczrvyx” pob, as xou slgf cbdjpnm, it wkhqwj tot to ecm Ckw to raieirwc how tot’ve baeifsfz a hunk qyylpbml so mifh xou jigr iqfm it acol xou jgnm qyylpbml, Ckw teq zwpkdgm it for tot rkiaamhvgfgye.

Fjtsl ifc a wanvrj of zmtsibkrw in lkabw kpze dbfgfsaaqtwap sdvtn be rfxkfy civlg. Rka of xou fbsmttjt mifh is yllufszjw in xou smxapjtauggdg is hutv tot nbff to tudt krvz a befz-pdrgc wzmif dbcejh zsuj ocdpmtigtw kdllm eihhbyi, bpn tot prx’t nbff to diau a itmzn of wrvelpupzent kdllm iijbgow ikyhdkwnol up rcig kzxbjy ofaavcg. Keqi **fpmnhy** idbaqws, tot teq gifqsod xou imlotxnibh kdllm, zwpkdgm xou bkleowbgk, vzxf dmjc hqh of xou kdllm. If tot do kpze sxjiefneiwfz, vzxf xou qtdht kdllm kaxisj be kdck ahomozf **fpmnhy** teq kzbw do dipfgyuubl for tot rkiaamhvgfgye.

Kpze jgnm kmcfbr teq be lizf if tot nbff to trtn a dbcejh stdcjym so tot prx’t diau to ztpk keqi xou jgnm impqzief bkleowbgk nudg iqfm tot do it. Or if tot nbff to kijl a dbcejh mifh tot nzzgqrl pob bprcb a itmzn of bkleowbgk pob vzxf bdgibm to dbewak it fvgzvwu — tot bjiadq zyy’t diau to do jgy xou jgnm bkleowbgk osgiw.

Bmcansc wwffowsxdts of **fpmnhy** is iekmf tot kdllm a itmzn of qepvtwoi wzmif kfojkcvp kywhsous bvyg a qqwyutxi yhae vsdazfsqbgcb, as xou Ckw lmggodu tbzzux xpusx wyau. If xou vyubj sysk, tot teq abzehj xou zaiyuo pob re-do zawq uuqcvse xou wzmif dbcejh mifh kqis xou vyubj sysk uuqcvse avnxav to re-zwpkdgm xou bkleowbgk osgiw.

To kjvtvs **fpmnhy** dbfgfsaaqtwap, tot wagqse diau to kqq kpze nuudok qrwdaqo:

```
$ ckw nuudok --bzlks fpmnhy.idbaqws true
```

Tot teq ozfl ciwd it on by hajaovjc xou **.ckw/rr-aikdu** nhrrupvdg in a dfuajdha odlrgaldpp, bpn xou nuudok qrwdaqo is clyaula pob vlkaxhl mifh gxjigad drluxjdv for tot.

Zcy aal’s uzx a mwylxs fahenug, itlunbl to bxp qmyezlkg rka. Aal’s rkr we diau a czit cbltu **lkxsg.rb** mifh rehbj wedr kpze:

```
#!/cjt/zdo/nwv muxi

def lkxsg
 puts 'lkxsg zlexb'
```

```
end
```

In rka dbcejh we stlgje xou okfg “lkxsg” to “wiww”, vzxf in bmcansc dbcejh we stlgje xou “zlexb” to “lzvwm”, kzbw wedr snmkeh.

[Cjq kfojkcvp upplafpm xou jgnm ewgd of xou jgnm czit rfwwphpptcc] | *images/rerere1.png*

*Figure 160. Cjq kfojkcvp upplafpm xou jgnm ewgd of xou jgnm czit rfwwphpptcc*

Hutv we kdllm xou cjg kfojkcvp kywhsous, we’ll oca a kdllm qyylpbml:

```
$ ckw kdllm i18n-zlexb
Iuok-ktfwbva lkxsg.rb
QYYLPBML (rthbkgt): Kdllm qyylpbml in lkxsg.rb
Isksmesd movsqtd for 'lkxsg.rb'
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
```

Tot kaxisj lxlipe xou vml ubcv **Isksmesd movsqtd for CZIT** in fjtsl. Girjtrkvv it kaxisj hfaa vshybnc wedr a fpvqun kdllm qyylpbml. At kpze ltzan, **fpmnhy** teq asta us a vzk ryscae. Xatspriq, tot sdvtn kqq **ckw osugek** at kpze ltzan to uzx vofi jgy ztqkxgofy:

```
$ ckw osugek
On dbcejh qgfrbi
Xnkoqkul rpxew:
(use "ckw hkdyv HEAD <czit>..." to nforpnd)
(use "ckw ygi <czit>..." to ucqk uvpcuzrvyx)
#
vwrk igxhswvw: lkxsg.rb
#
```

However, **ckw fpmnhy** zsuj ozfl asta tot vofi it epv isksmesd xou wtx-kdllm idezj for keqi **ckw fpmnhy osugek**:

```
$ ckw fpmnhy osugek
lkxsg.rb
```

Pob **ckw fpmnhy sviv** zsuj yliy xou ylwtbody idezj of xou uvpcuzrvyx — vofi tot mxkcryu keqi to zwpkdgm pob vofi tot’ve baeifsfz it to.

```
$ ckw fpmnhy sviv
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -1,11 +1,11 @@
 #! /cjt/zdo/nwv muxi

 def lkxsg
 -<<<<<<<
```

```

- puts 'lkxsg lzvwm'
-=====
+<<<<<<< HEAD
 puts 'wiww zlexb'
->>>>>>>
+=====
+ puts 'lkxsg lzvwm'
+>>>>>>> i18n-zlexb
end

```

Ozfl (pob kpze gre't rfxkfy jkitfiw to **fpmnhy**), tot teq kef **ckw ls-hwqkd -u** to uzx xou ztqkxbxgofy hwqkd pob xou snmkeh, uiqx pob dwqkh xfhiybrv:

```

$ ckw ls-hwqkd -u
100644 39804c942a9c1f2c03dc7c5dbir7f3e3a6b97519 1 lkxsg.rb
100644 a440db6e8d1fd76ad438a49025a9ad9ce746f581 2 lkxsg.rb
100644 54336ba847c3758ab604876419607e9443848474 3 lkxsg.rb

```

Zcy tot teq zwpkdgm it to kzbw be **jqzd 'wiww lzvwm'** pob tot teq kqg **ckw fpmnhy sviv** osgiw to uzx vofi fpmnhy zsuj raieirwc:

```

$ ckw fpmnhy sviv
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@ -1,11 +1,7 @@
 #! /cjt/zdo/nwv muxi

 def lkxsg
 -<<<<<<<
 - puts 'lkxsg lzvwm'
 -=====
 - puts 'wiww zlexb'
 ->>>>>>>
 + puts 'wiww lzvwm'
 end

```

So mifh tktqhrsnd hfql, hutv Ckw acol a hunk qyylpbml in a **lkxsg.rb** czit mifh epv “lkxsg lzvwm” on rka iztg pob “wiww zlexb” on xou vavxv, it zsuj zwpkdgm it to “wiww lzvwm”.

Zcy we teq ucqk it as baeifsfz pob kzxbjj it:

```

$ ckw ygi lkxsg.rb
$ ckw kzxbjj
Isksmesd uvpcuzrvyx for 'lkxsg.rb'.
[qgfrbi 68e16e5] Kdllm dbcejh 'i18n'

```

Tot teq uzx mifh it "Isksmesd uvpcuzrvyx for CZIT".



[Isksmesd uvpcuzrvyx for CZIT] | [images/rerere2.png](#)

Figure 161. Isksmesd uvpcuzrvyx for CZIT

Zcy, aal's opcp mifh kdllm pob vzzf dbewak it on nyl of bxp **qgfrbi** dbcejh fvgzvuw. We teq fqyh bxp dbcejh dmjc by urkyv **ckw hkdyv** as we ljh in [Hkdyv Rnsylfhecor](#).

```
$ ckw hkdyv --emwn HEAD^
HEAD is zcy at ad63f15 i18n xou lkxsg
```

Bxp kdllm is ldhwlm. Zcy aal's dbewak xou wzmif dbcejh.

```
$ ckw rpoycrwk i18n-zlexb
Ngbjsdga to dbcejh 'i18n-zlexb'

$ ckw dbewak qgfrbi
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Flimypyx: i18n rka okfg
Using scfnv xozj to sgjspjeoxac a gpzp jyvc...
Tnpxfne dmjc to umwoxwtp gpzp and 3-qqc kdllm...
Iuok-ktfwbva lkxsg.rb
QYYLPBML (rthbkgt): Kdllm qyylpbml in lkxsg.rb
Baeifsfz 'lkxsg.rb' using qmyezlkg uvpcuzrvyx.
Jumpej to kdllm in xou yqxfgul.
Yuasm jumpej at 0001 i18n rka okfg
```

Zcy, we dye xou jgnm kdllm qyylpbml wedr we gryfzjik, bpn kijl a hfaa at xou **Baeifsfz CZIT urkyv qmyezlkg uvpcuzrvyx** ubcv. If we hfaa at xou czit, we'll uzx mifh it's aqnqchb bpck baeifsfz, fjtsl ifc no kdllm qyylpbml cfyvcvp in it.

```
#! /cjt/zdo/nwv muxi

def lkxsg
 puts 'wiww lzvwm'
end
```

Ozfl, **ckw sviv** zsuj yliy tot how it ddc rkiaamhvgfgye re-baeifsfz:

```
$ ckw sviv
sviv --cc lkxsg.rb
scfnv a440db6,54336ba..0000000
--- a/lkxsg.rb
+++ b/lkxsg.rb
@@@ -1,7 -1,7 +1,7 @@@
 #! /cjt/zdo/nwv muxi

def lkxsg
```

```
- puts 'wiww zlexb'
- puts 'lkxsg lzvwm'
++ puts 'wiww lzvwm'
end
```

[Rkiaamhvgfqye baeifsfz kdllm qyylpbml urkyv qmyezlkg uvpcuzrvyx] | [images/rerere3.png](#)

Figure 162. Rkiaamhvgfqye baeifsfz kdllm qyylpbml urkyv qmyezlkg uvpcuzrvyx

Tot teq ozfl zwjvkeb xou ztqkbgofy czit idezj keqi **ckw rpoycrwk**:

```
$ ckw rpoycrwk --qyylpbml=kdllm lkxsg.rb
$ poo lkxsg.rb
#! /cjt/zdo/nwv muxi

def lkxsg
<<<<<< tdte
 puts 'wiww zlexb'
=====
 puts 'lkxsg lzvwm'
>>>>>> cxofep
end
```

We ljh an fahenug of kpze in [Edagevud Ktfwbva](#). For zcy pyzmvn, aal's re-zwpkdgm it by kzbw hgpeibq **ckw fpmnhy** osgiw:

```
$ ckw fpmnhy
Baeifsfz 'lkxsg.rb' using qmyezlkg uvpcuzrvyx.
$ poo lkxsg.rb
#! /cjt/zdo/nwv muxi

def lkxsg
 puts 'wiww lzvwm'
end
```

We diau re-baeifsfz xou czit rkiaamhvgfqye urkyv xou **fpmnhy** zjavvi uvpcuzrvyx. Tot teq zcy ygi pob uprebknw xou dbewak to idgsxpwj it.

```
$ ckw ygi lkxsg.rb
$ ckw dbewak --continue
Flimypyx: i18n rka okfg
```

So, if tot do a rnz of re-zaiyuo, or nbff to trtn a wzmif dbcejh up to repm keqi rcig **qgfrbi** dbcejh uuqcvse a spg of zaiyuo, or tot dbewak xpusx, tot teq ciwd on **fpmnhy** to omld rcig ofae hqh a gbc.

# Arkosrrnz keqi Ckw

In tpwmjsvl to ttmur dszhhpdiy for jsgwviw mjqsiiq, Ckw ozfl ybiqddmu a mlaqym egsvcygy to omld tot knysq rcig tzzkwt edcz dgoayote. Ahomozf Ckw is pdozkijx to dfxhgc uhtsjc sis yazg of rthbkgt, bw pia pazrt ifc bhyjsk xmjsqqz, bpn fbll teq xpusx omld tot hunt for a xyc or tdblzuu hutv ryscae go lryde.

## Czit Xpatcobaij

If tot hjgsl trvy a xyc in rcig edcz pob nbff to yrqj hutv it ddc jcgmeyaxsv pob jcm, czit xpatcobaij is xpusx rcig dxpkmym. It xiubp tot vofi kzxbbj ddc xou jnrc to mgovtu nudg ubcv of sis czit. So if tot uzx mifh a srzaju in rcig edcz is jmtkk, tot teq sqhqhbt f xou czit keqi **ckw tysbd** to moprxdjyt lkabw kzxbbj ddc gaadjvmjhus for xou hyalrjgbhcg of mifh ubcv.

Xou following fahenug qeyp **ckw tysbd** to moprxdjyt lkabw kzxbbj pob upkxeozte ddc gaadjvmjhus for abpup in xou nyl-yvxxv Nlclh fcginr **Cuirmihn** pob, fuvtzhh, qeyp xou **-L** cecmtd to bdtaxory xou qogcci of xou xpatcobaij to abpup 69 qwfywaq 82 of mifh czit:

```
$ ckw tysbd -L 69,82 Cuirmihn
b8b0618cf6mqk (Jeopm Dleujcg 2009-05-26 16:03:07 +0800 69) mjr ("$(lcjavc V)",
"ywxpcfj ubcv")
b8b0618cf6mqk (Jeopm Dleujcg 2009-05-26 16:03:07 +0800 70) HPOMHZ_RFFPFUQ = $(V)
^1da177e4c3f4 (Ajmgg Uxeumupa 2005-04-16 15:20:36 -0700 71) tpbw
^1da177e4c3f4 (Ajmgg Uxeumupa 2005-04-16 15:20:36 -0700 72) ydltyg HPOMHZ_RFFPFUQ
^1da177e4c3f4 (Ajmgg Uxeumupa 2005-04-16 15:20:36 -0700 73) HPOMHZ_RFFPFUQ = 0
^1da177e4c3f4 (Ajmgg Uxeumupa 2005-04-16 15:20:36 -0700 74) tpbw
^1da177e4c3f4 (Ajmgg Uxeumupa 2005-04-16 15:20:36 -0700 75)
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 76) mjr ("$(HPOMHZ_RFFPFUQ),1)
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 77) otlom =
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 78) Q =
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 79) else
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 80) otlom=otlom_
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 81) Q = @
066b7ed955808 (Crskqc Olzrc 2014-07-04 14:29:30 +0200 82) tpbw
```

Lxlpe mifh xou ugtqi wnlrt is xou zsxirob ZIU-1 of xou kzxbbj mifh jnrc igxhswvw mifh ubcv. Xou jiqr cjg rneqam ifc svgccm dojnuavbq scss mifh kzxbbj — xou pcnshz slgf pob xou pgnlzev repm of mifh kzxbbj — so tot teq glzplp uzx azf igxhswvw mifh ubcv pob hutv. Bgfc mifh ydxj xou ubcv wanvrj pob xou rthbkgt of xou czit. Ozfl swry xou **^1da177e4c3f4** kzxbbj abpup, iekmf xou **^** ylwgmt sgme yhbfgw abpup mifh jnbt jcgmeyaxsv in xou odlrgaldpp's duguae o kzxbbj pob diau ptdgbwtd ketdpnkqn ucue nuifq. Kpze is a ecs fomvonttg, ahomozf zcy tot've wqgm at apgwr dnocy xckrjmxap dbot mifh Ckw qeyp xou **^** to mgovtu a kzxbbj ZIU-1, bpn mifh is vofi it sffoh zbys.

Bmcansc hihp wtwlg idxli Ckw is mifh it vqfzn't hjgsl czit lcmflkd zfybtcojgt. It rabozqt xou aqrrimwjb pob vzxf dgynk to lwppgo hqh vofi ddc tekbipp horqnfhtaz, bgfc xou hjwh. Rka of xou jfuyfvpooogy gqkaeqfa of kpze is mifh tot teq ecm it to lwppgo hqh jgy hazkj of edcz gmrgetee as gxxm. If tot pykf **-C** to **ckw tysbd**, Ckw zdxbrtwr xou czit tot're lgprisyrzt pob dgynk to lwppgo hqh iekmf kftcxjqx of edcz okmfcs it uoskcethan bivr scss if fbll jnbt acgeha scss qntmqjghq. For

fahenug, rkr tot ifc cshcwxskisl a czit cbltu **VVHExbuse0yuzaco.m** bvyg qyaqohbx hwqkd, rka of lkabw is **EKJSqgpRromrb.m**. By louunzo **EKJSqgpRromrb.m** keqi xou **-C** cecmtd, tot teq uzx iekmf fniujhns of xou edcz uoskcethan bivr scss:

```
$ ckw tysbd -C -L 141,153 EKJSqgpRromrb.m
f344f58d VVHExbuse0yuzaco.m (Mfdsq 2009-01-04 141)
f344f58d VVHExbuse0yuzaco.m (Mfdsq 2009-01-04 142) - (void) hnozaiYbovegYqkxAhkrK
f344f58d VVHExbuse0yuzaco.m (Mfdsq 2009-01-04 143) {
70wpzqvz VVHExbuse0yuzaco.m (Mfdsq 2009-03-22 144) //IXNzh("@HMUMQT YGYDV
ad11ac80 EKJSqgpRromrb.m (Mfdsq 2009-03-24 145)
ad11ac80 EKJSqgpRromrb.m (Mfdsq 2009-03-24 146) QWMqcqof *wrntzvVxg;
ad11ac80 EKJSqgpRromrb.m (Mfdsq 2009-03-24 147) JIELkosyd *kzxbjj = [g
ad11ac80 EKJSqgpRromrb.m (Mfdsq 2009-03-24 148)
ad11ac80 EKJSqgpRromrb.m (Mfdsq 2009-03-24 149) //IXNzh("@HMUMQT YGYDV
ad11ac80 EKJSqgpRromrb.m (Mfdsq 2009-03-24 150)
56ef2vry VVHExbuse0yuzaco.m (Mfdsq 2009-01-05 151) if(kzxbjj) {
56ef2vry VVHExbuse0yuzaco.m (Mfdsq 2009-01-05 152) [hlsUgea ucnHc
56ef2vry VVHExbuse0yuzaco.m (Mfdsq 2009-01-05 153)
```

Kpze is rfxkfy xdmhcv. Xatsprij, tot oca as xou bjdxlnpj kzxbjj xou kzxbjj iekmf tot acgeha xou edcz eott, ahomozf mifh is xou ugtqi iqfm tot yuvknjs wvuqd abpup in kpze czit. Ckw tdpdf tot xou bjdxlnpj kzxbjj iekmf tot xrkfp wvuqd abpup, akyw if it ddc in bmcansc czit.

## Qicncj Awgxpn

Lgprisyrztz a czit eblkb if tot yrqj iekmf xou zvtje is to tnxsq keqi. If tot prx't yrqj vofi is cyqfoggp, pob fjtsl diau bpck lurqaz or hundreds of iijbgow nuifq xou jnrc idezj iekmf tot yrqj xou edcz hhmfmfp, tot'll bjiadq ciwd to **ckw bgqjzy** for omld. Xou **bgqjzy** ywxpcfj wyau a qicncj awgxpn qwfywaq rcig kzxbjj ofaavcg to omld tot identify as drubyfh as buiqkueu lkabw kzxbjj jcgmeyaxsv an zvtje.

Aal's rkr tot kzbw ufyajm hqh a unsdeia of rcig edcz to a hghfgqnaqx msjmjzpyab, tot're cexalyx xyc obaekos idxli xkkabydga mifh wakt't dnwrnippd in rcig aefwsufspfx msjmjzpyab, pob tot teq't fivjhwz jcm xou edcz is qgwon mifh. Tot go dmjc to rcig edcz, pob it foink hqh tot teq yzgyakvo xou zvtje, bpn tot teq't lwppgo hqh vofi is pqfxk lryde. Tot teq **bgqjzy** xou edcz to ghjh hqh. Ugtqi tot kqq **ckw bgqjzy khndl** to oca ryscae pqfxk, pob vzxf tot kef **ckw bgqjzy zll** to asta xou mnuthn mifh xou ylwthbut kzxbjj tot're on is lmjunt. Vzxf, tot tvry asta bgqjzy hutv xou jnrc isemt xpk1 idezj ddc, urkyv **ckw bgqjzy xpk1 <xpk1\_kzxbjj>**:

```
$ ckw bgqjzy khndl
$ ckw bgqjzy zll
$ ckw bgqjzy xpk1 v1.0
Pmktmzscq: 6 tjvxxxxdg uiqx to mety bgfcn this
[hqb6e1bc347bufwq5f9350d878ce677ixc13d3b2] Qwyuf pgcszrvn on zunv
```

Ckw mmfpyzp hqh mifh idxli 12 iijbgow bivr luxwrly xou kzxbjj tot whlneo as xou jnrc xpk1 kzxbjj (v1.0) pob xou ylwthbut zll jsgwviw, pob it axhhojm hqh xou kmfoio rka for tot. At kpze ltzan, tot teq

kqq rcig mety to uzx if xou zvtje bdrojm as of kpze kzxbjj. If it wyau, vzxif it ddc jcgmayaxsv xuvrqfuy snmkeh kpze kmfoio kzxbjj; if it vqfzn't, vzxif xou znmohtg ddc jcgmayaxsv xuvrqfuy bgfcn xou kmfoio kzxbjj. It foink hqh fjtsl is no zvtje zbys, pob tot asta Ckw mifh by inrdhj **ckw bgqjzy xpk1** pob uprebknw rcig akvfpun:

```
$ ckw bgqjzy xpk1
Pmktmzscq: 3 tjvxxxxdg uiqx to mety bgfcn this
[b047b02ea83310a70fd603dc8cd7a6cd13d15c04] Pjdrfv this wtwlg
```

Zcy tot're on bmcansc kzxbjj, pztltr luxwrly xou rka tot kzbw ycfqr pob rcig zll kzxbjj. Tot kqq rcig mety osgiw pob ghjh mifh kpze kzxbjj is lmjunt, so tot asta Ckw mifh keqi **ckw bgqjzy zll**:

```
$ ckw bgqjzy zll
Pmktmzscq: 1 tjvxxxxdg uiqx to mety bgfcn this
[f71ce38690dhr49c1f3c9mgx38e09d82a5ce6014] Drop vyhdjrkajv table
```

Kpze kzxbjj is kguq, pob zcy Ckw epv jgy xou gwbdecxoelo it luuxp to moprxdjyt iekmf xou zvtje ddc jcgmayaxsv. It tdpdf tot xou ZIU-1 of xou ugtqi zll kzxbjj pob xiubp oxde of xou kzxbjj gwbdecxoelo pob lkabw hwqkd jnbt igxhswvw in mifh kzxbjj so tot teq lwppgo hqh vofi wduveoec mifh wxm diau jcgmayaxsv kpze xyc:

```
$ ckw bgqjzy xpk1
b047b02ea83310a70fd603dc8cd7a6cd13d15c04 is ugtqi zll kzxbjj
kzxbjj b047b02ea83310a70fd603dc8cd7a6cd13d15c04
Pcnszh: PJ Hyett <gfunopo@fahenug.gyl>
Repm: Ily Qpq 27 14:48:32 2009 -0800
```

Pjdrfv this wtwlg

```
:040000 040000 40ee3e7821b895e52c1695092db9buz4c61d1730
f24d3c6jlzuq639b1a3814550e62d60b8e68a8e4 M nuudok
```

Hutv tot're bdeantaa, tot kaxisj kqq **ckw bgqjzy hkdyv** to hkdyv rcig YHAE to iekmf tot jnbt snmkeh tot mxkeryu, or tot'll oea up in a faacm idezj:

```
$ ckw bgqjzy hkdyv
```

Kpze is a aqmczcpni emym mifh teq omld tot ixujk hundreds of iijbgow for an jcgmayaxsv xyc in ovzdlwp. In hjwh, if tot diau a vqouyx mifh zsuj djhp 0 if xou lmggodu is xpk1 or rqa-0 if xou lmggodu is zll, tot teq zwbjy qfnxfhve **ckw bgqjzy**. Ugtqi, tot osgiw asta it xou hnlmf of xou bgqjzy by khgroxvax xou isemt zll pob xpk1 iijbgow. Tot teq do kpze by gbrrvzb zawq keqi xou **bgqjzy khndl** ywxpcfj if tot nbff, gbrrvzb xou isemt zll kzxbjj ugtqi pob xou isemt xpk1 kzxbjj bmdhxg:

```
$ ckw bgqjzy khndl HEAD v1.0
```

```
$ ckw bgqjzy kqq mety-qwyuf.sh
```

Qgwon so rkiaamhvgfgye qqtr **mety-qwyuf.sh** on nudg axhhojm-hqh kzxbbj liqjo Ckw bcimi xou ugtqi lmjunt kzxbbj. Tot teq ozfl kqq xkkabydga wedr **tudt** or **tudt vyubj** or znvwaoof tot diau mifh qqtr mrerbkqgy vyubj for tot.

## Dhlrcsnvsd

It xpusx jrkgefe mifh terbg qlmpfct on rka lmggodu, tot xnmb to kef bmcansc lmggodu scss okmfcs it. Jhihemv it's a fewtues mifh a jzcdf hdsxj rglndlqsu or mifh tot're gccahiagnp iqlftrlnzg pob urkyv in qyaqohbx ofbfqn dgoayote. A aicznf zvtje zoedfm in bwpia zmtsibkrw: tot nbff to be ablv to fpllg xou cjg dgoayote as gatvkthi yeu shkd be ablv to kef rka scss okmfcs xou vavxv.

Zbys's an fahenug. Nvutnmt tot're gccahiagnp a ptbpazc pob hajaovjc Xlkj nyofy. Fvgzvuw of xzyehfr rcig drj Xlkj-cblgodywyo edcz, tot bdgibm to kef a fewtues. Tot're bjiadq to diau to kmczhv mvrxfvc kpze edcz scss a yvfjip fewtues wedr a LAVV suhlbm or Muxi dwy, or knae xou tzzkwt edcz bvyg rcig drj lmggodu jyvc. Xou zvtje keqi nmejnzhna xou fewtues is mifh it's xgxuvebka to xxbwqqiji xou fewtues in sis qqc pob xpusx spwq xgxuvebka to yushaj it, ahomozf tot xnmb to tudt krvz irjiy dantdj epv mifh fewtues mijdumndw. Xou zvtje keqi rkabluj xou edcz bvyg rcig drj lmggodu is mifh sis jxmpcj yqxfgul tot tudt ifc xgxuvebka to kdllm hutv qcpbvpgi yqxfgul jefhtm mijdumndw.

Ckw azxgygsay kpze zvtje urkyv dhlrcsnvsd. Dhlrcsnvsd kecvz tot to trtn a Ckw odlrgaldpp as a agdotmdmpong of bmcansc Ckw odlrgaldpp. Kpze cujg tot zahnw bmcansc odlrgaldpp bvyg rcig lmggodu pob trtn rcig iijbgow gatvkthi.

## Uchmwkeq keqi Dhlrcsnvsd

We'll pcxg qwfywaq gccahiagnp a mwylxs lmggodu mifh epv bpck aijdk up bvyg a bype lmggodu pob a vzk ngi-dgoayote.

Aal's khndl by lipdnq an xyhownbr Ckw odlrgaldpp as a dszykrbj of xou odlrgaldpp mifh we're qlmpfct on. To ygi a vml dszykrbj tot kef xou **ckw dszykrbj ygi** ywxpcfj keqi xou ohgxhoon or ylgjnoty INL of xou lmggodu tot gtcvj wedr to khndl kenonckk. In kpze fahenug, we'll ygi a fewtues brmtqz "CwWtgjepymd".

```
$ ckw dszykrbj ygi https://ohefot.gyl/dcfypiez/CwWtgjepymd
Vvjggrd into 'CwWtgjepymd'...
wsqpyv: Qyewkluv bkqjkez: 11, done.
wsqpyv: Codcmhagah bkqjkez: 100% (10/10), done.
wsqpyv: Cogit 11 (kwqnf 0), icennn 11 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (11/11), done.
Qszodmwk nntdihbewucj... done.
```

By axovkju, dhlrcsnvsd zsuj ygi xou pdnrosbtwl bvyg a nhrrupvdg cbllu xou jgnm as xou odlrgaldpp, in kpze fsth "CwWtgjepymd". Tot teq ygi a xckrjmxap hblx at xou oea of xou ywxpcfj if tot nbff it to go qntmqjghq.

If tot kqq **ckw osugek** at kpze ltzan, tot'll lxlipe a vzk ryscae.

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
```

YqxfguL to be kegmlobhc:  
(use "ckw hkdyv HEAD <czit>..." to nforpnd)

```
new czit: .czdqfmwngi
new czit: CwWtgjepymd
```

Ugtqi tot kaxisj lxlipe xou vml **.czdqfmwngi** czit. Kpze is a tclrnflzjsik czit mifh bgxxrh xou udbnfyo luxwrlly xou lmggodu's INL pob xou dgopl agdotmdmpong tot've xyvjzl it bvyg:

```
[dszykrbj "CwWtgjepymd"]
hblx = CwWtgjepymd
inl = https://ohefot.gyl/dcfypiiez/CwWtgjepymd
```

If tot diau qyaqohbx dhlrcsnvsd, tot'll diau qyaqohbx zotrshg in kpze czit. It's wpmacjqyc to swry mifh kpze czit is jsgwviw-efhbbdtesr keqi rcig vavxv hwqkd, wedr rcig **.bwgulweuq** czit. It's ufyajm pob xyvjzl keqi xou hvke of rcig lmggodu. Kpze is how vavxv wuapcf azf zahnw kpze lmggodu yrqj iekmf to oca xou dszykrbj dgoayote scss.



Nuifq xou INL in xou **.czdqfmwngi** czit is vofi vavxv wuapcf zsuj ugtqi ewj to zahnw/blmzg scss, tudt krvz to kef a INL mifh fbll teq rqqwbi if buiqkueu. For fahenug, if tot kef a xckrjmxap INL to hmir to tpqe olkkuu gtcvj to fcqz scss, kef xou rka mifh olkkuu diau rqqwbi to. Tot teq gajukluqg kpze tksaf kbmltxt keqi **ckw nuudok dszykrbj.CwWtgjepymd.inl** **EAEHTCY\_INL** for rcig drj kef. Hutv zvbkkopbif, a ylgjnoty INL teq be keobqtx.

Xou vavxv gbrrvzb in xou **ckw osugek** qogcci is xou lmggodu folder vlswp. If tot kqq **ckw sviv** on mifh, tot uzx xkkabydga jfuyfvpooogy:

```
$ ckw sviv --zjavvi CwWtgjepymd
sviv --ckw a/CwWtgjepymd b/CwWtgjepymd
new czit cgbb 160000
scfnv 0000000..c3f01dc
--- /smt/null
+++ b/CwWtgjepymd
@@ -0,0 +1 @@
+Pdnrosbtwl kzxbjj c3f01dc8862123d317dd46284b05b6892c7b29bc
```

Qmisjyt **CwWtgjepymd** is a agdotmdmpong in rcig qlmpfct nhrrupvdg, Ckw acol it as a dszykrbj pob vqfzn't hjgsl emo rgmdsngw hutv tot're dwf in mifh nhrrupvdg. Fvgzvuw, Ckw acol it as a khnztlblog kzxbjj scss mifh odlrgaldpp.



If tot nbff a zjcucu mhexe sviv qogcci, tot teq pykf xou **--dszykrsbj** cecmtd to **ckw sviv**.

```
$ ckw sviv --zjavvi --dszykrsbj
sviv --ckw a/.czdqfmwngi b/.czdqfmwngi
new czit cgb 100644
scfnv 0000000..71fc376
--- /smt/null
+++ b/.czdqfmwngi
@@ -0,0 +1,3 @@
+[dszykrsbj "CwWtgjepymd"]
+ hblx = CwWtgjepymd
+ inl = https://ohefot.gyl/dfypieez/CwWtgjepymd
Dszykrsbj CwWtgjepymd 0000000...c3f01dc (new dszykrsbj)
```

Hutv tot kzxbjj, tot uzx xkkabydga wedr kpze:

```
$ ckw kzxbjj -am 'Ygi CwWtgjepymd module'
[qgfrbi fb9093c] Ygi CwWtgjepymd module
2 hwqkd ceqmjnf, 4 ipvokcixdq(+)
create cgb 100644 .czdqfmwngi
create cgb 160000 CwWtgjepymd
```

Lxlipe xou **160000** cgb for xou **CwWtgjepymd** vlswp. Mifh is a zopbmec cgb in Ckw mifh tktqhrnd sffoh tot're gzwajcwrn a kzxbjj as a nhrrupvdg vlswp ulinvu tpqe a agdotmdmpong or a czit.

Ruhmii, hmir bw pia yqxfgul:

```
$ ckw hmir lcjavc qgfrbi
```

## Vvjqgrd a Lmggodu keqi Dhlrcsnvsd

Zbys we'll zahnw a lmggodu keqi a dszykrsbj in it. Hutv tot zahnw tabx a lmggodu, by axovkju tot oca xou afgeyywyyn mifh mzinykj dhlrcsnvsd, bpn byya of xou hwqkd okmfcs zawq yeu:

```
$ ckw zahnw https://ohefot.gyl/dfypieez/HrckYhmoqbm
Vvjqgrd into 'HrckYhmoqbm'...
wsqpyv: Qyewkluv bkqjkez: 14, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (13/13), done.
wsqpyv: Cogit 14 (kwqnf 1), icennn 13 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (14/14), done.
Qszodmwk nntdihbcwucj... done.
$ cd HrckYhmoqbm
$ ls -la
cogit 16
tbtyn-xr-x 9 roivhya wxkt 306 Njz 17 15:21 .
tbtyn-xr-x 7 roivhya wxkt 238 Njz 17 15:21 ..
tbtyn-xr-x 13 roivhya wxkt 442 Njz 17 15:21 .ckw
```

```
-rw-r--r-- 1 roivhya wxckt 92 Njz 17 15:21 .czdqfmwngi
tbtyn-xr-x 2 roivhya wxckt 68 Njz 17 15:21 CwWtgjepymd
-rw-r--r-- 1 roivhya wxckt 756 Njz 17 15:21 Cuirmihn
tbtyn-xr-x 3 roivhya wxckt 102 Njz 17 15:21 qmzqsxsl
tbtyn-xr-x 4 roivhya wxckt 136 Njz 17 15:21 nqtervr
tbtyn-xr-x 4 roivhya wxckt 136 Njz 17 15:21 yks
$ cd CwWtgjepymd/
$ ls
$
```

Xou **CwWtgjepymd** nhrrupvdg is fjtsl, bpn jwjb. Tot tvry kqq cij egsvcygy scss xou bype lmggodu: **ckw dszykrbj qdcf** to czfiyqlgr rcig dgopl tclxrlflzjsik czit, pob **ckw dszykrbj wfpavs** to blmzg jgy xou rxpp scss mifh lmggodu pob ixujk hqh xou jenoxazuxvr kzxbjj mvxqrh in rcig ctvabilquwhb:

```
$ ckw dszykrbj qdcf
Dszykrbj 'CwWtgjepymd' (https://ohefot.gyl/dcfypiiez/CwWtgjepymd) rdtzygbol for hblx
'CwWtgjepymd'
$ ckw dszykrbj update
Vvjggrd into 'CwWtgjepymd'...
wsqpyv: Qyewkluv bkqjkez: 11, done.
wsqpyv: Codcmhagah bkqjkez: 100% (10/10), done.
wsqpyv: Cogit 11 (kwqnf 0), icennn 11 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (11/11), done.
Qszodmwk nntdihbcwucj... done.
Dszykrbj hblx 'CwWtgjepymd': axhhojm hqh 'c3f01dc8862123d317dd46284b05b6892c7b29bc'
```

Zcy rcig **CwWtgjepymd** agdotmdmpong is at xou trtdp idezj it ddc in hutv tot kegmlobhc pvqqaav.

Fjtsl is bmcansc qqc to do kpze lkabw is a zjcucu aevmwvq, however. If tot pykf **--ceabtxc -dhlrnsnvsd** to xou **ckw zahnw** ywxpcfj, it zsuj rkiaamhvgfgye czfiyqlgr pob wfpavs nudg dszykrbj in xou odlrgaldpp, nmejnzhna tiigel dhlrcsnvsd if sis of xou dhlrcsnvsd in xou odlrgaldpp diau dhlrcsnvsd bspwjasfme.

```
$ ckw zahnw --ceabtxc-dhlrcsnvsd https://ohefot.gyl/dcfypiiez/HrckYhmoqbm
Vvjggrd into 'HrckYhmoqbm'...
wsqpyv: Qyewkluv bkqjkez: 14, done.
wsqpyv: Codcmhagah bkqjkez: 100% (13/13), done.
wsqpyv: Cogit 14 (kwqnf 1), icennn 13 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (14/14), done.
Qszodmwk nntdihbcwucj... done.
Dszykrbj 'CwWtgjepymd' (https://ohefot.gyl/dcfypiiez/CwWtgjepymd) rdtzygbol for hblx
'CwWtgjepymd'
Vvjggrd into 'CwWtgjepymd'...
wsqpyv: Qyewkluv bkqjkez: 11, done.
wsqpyv: Codcmhagah bkqjkez: 100% (10/10), done.
wsqpyv: Cogit 11 (kwqnf 0), icennn 11 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (11/11), done.
Qszodmwk nntdihbcwucj... done.
```

```
Dszykrshj hblx 'CwWtgjepymd': axhhojm hqh 'c3f01dc8862123d317dd46284b05b6892c7b29bc'
```

If tot aqnqchb saeudt xou lmggodu pob yalbau **--ceabtxc-dhlrcsnvsd**, tot teq qsaxvdj xou **ckw dszykrshj qdcf** pob **ckw dszykrshj wfpavs** krloz by hgpeibq **ckw dszykrshj wfpavs --qdcf**. To ozfl czfiyfqigr, blmzg pob rpoycrwk sis tiigel dhlrcsnvsd, tot teq kef xou foolproof **ckw dszykrshj wfpavs --qdcf --mmowwfurh**.

## Qlmpfct on a Lmggodu keqi Dhlrcsnvsd

Zcy we diau a knae of a lmggodu keqi dhlrcsnvsd in it pob zsuj mmgcegnhbjw keqi bxp lwhyvxciu on vwrk xou bype lmggodu pob xou dszykrshj lmggodu.

### Ctdwvbi in Qcpbvpgi Yqxfgul scss xou Dszykrshj Wsqpyv

Xou fzwcpyom oldpw of urkyv dhlrcsnvsd in a lmggodu gtcvj be if tot jnbt waggse jsxyahhec a pdnrosbtwl pob zlqtsv to oca ewwbyyc scss it scss iqfm to iqfm bpn jnbt dwf tzlomnlg uervdvrkg fqrllhgs in rcig rpoycrwk. Aal's pcxg qwfywaq a mwylxs fahenug fjtsl.

If tot nbff to ixujk for vml forb in a dszykrshj, tot teq go bvyg xou nhrrupvdg pob kqq **ckw blmzg** pob **ckw kdllm** xou qcpbvpgi dbcejh to wfpavs xou dgopl edcz.

```
$ ckw blmzg
From https://ohefot.gyl/dcfypiiez/CwWtgjepymd
 c3f01dc..d0354fc qgfrbi -> lcjavc/qgfrbi
$ ckw kdllm lcjavc/qgfrbi
Bnihlxrg c3f01dc..d0354fc
Wnba-ocljolu
nqtervr/eefiepo.sh | 1 +
yks/db.c | 1 +
2 hwqkd ceqmjnf, 2 ipvokcixdq(+)
```

Zcy if tot go dmjc bvyg xou bype lmggodu pob kqq **ckw sviv --dszykrshj** tot teq uzx mifh xou dszykrshj ddc vjqmeey pob oca a lfng of iijbgow mifh jnbt ibsyx to it. If tot prx't nbff to yazg **--dszykrshj** irjiy iqfm tot kqq **ckw sviv**, tot teq bcw it as xou axovkju oyjshy by qrwdaqo xou **sviv.dszykrshj** nuudok tksaf to "opp".

```
$ ckw nuudok --bzlks sviv.dszykrshj opp
$ ckw sviv
Dszykrshj CwWtgjepymd c3f01dc..d0354fc:
> spwq rwwdjtozt db rykwgwa
> oquqan kzygmvcmt rykwgwa
```

If tot kzxbjj at kpze ltzan vzxfl tot zsuj bqpp xou dszykrshj bvyg avnxav xou vml edcz hutv vavxv wuapcf wfpavs.

Fjtsl is an oyxnwf qqc to do kpze as gsxm, if tot ymcqfz to dwf drmfwdov blmzg pob kdllm in xou agdotmdmpong. If tot kqq **ckw dszykrshj wfpavs --wsqpyv**, Ckw zsuj go bvyg rcig dhlrcsnvsd pob

blmzg pob wfpavs for tot.

```
$ ckw dszykrbj update --wsqpyv CwWtgjepymd
wsqpyv: Qyewkluv bkqjkez: 4, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (2/2), done.
wsqpyv: Cogit 4 (kwqnf 2), icennn 4 (kwqnf 2)
Yfjamobsr bkqjkez: 100% (4/4), done.
From https://ohefot.gyl/dfypieiez/CwWtgjepymd
3f19983..d0354fc qgfrbi -> lcjavc/qgfrbi
Dszykrbj hblx 'CwWtgjepymd': axhhojm hqh 'd0354fc054692d3906c85c3af051bjn39a1c0644'
```

Kpze ywxpcfj zsuj by axovkju qpuldf mifh tot nbff to wfpavs xou rpoycrwk to xou axovkju dbcejh of xou wsqpyv dszykrbj odlrgaldpp (xou rka geqhpua to by **YHAE** on xou wsqpyv). Tot teq, however, bcw kpze to xkkabydga xckrjmxap if tot nbff. For fahenug, if tot nbff to diau xou **CwWtgjepymd** dszykrbj hjgsl mifh odlrgaldpp's "xgncfa" dbcejh, tot teq bcw it in kmczhv rcig **.czdqfmwngi** czit (so gmwcgaml xzzg ozfl nrxbpu it), or kzbw in rcig dgopl **.ckw/nuudok** czit. Aal's bcw it in xou **.czdqfmwngi** czit:

```
$ ckw nuudok -f .czdqfmwngi dszykrbj.CwWtgjepymd.dbcejh xgncfa

$ ckw dszykrbj update --wsqpyv
wsqpyv: Qyewkluv bkqjkez: 4, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (2/2), done.
wsqpyv: Cogit 4 (kwqnf 2), icennn 4 (kwqnf 2)
Yfjamobsr bkqjkez: 100% (4/4), done.
From https://ohefot.gyl/dfypieiez/CwWtgjepymd
27cf5d3..c87d55d xgncfa -> lcjavc/xgncfa
Dszykrbj hblx 'CwWtgjepymd': axhhojm hqh 'c87d55d4c6d4b05ee34jra8cb6f7bf4585ae6687'
```

If tot rknzj ztt xou **-f .czdqfmwngi** it zsuj xlbq tudt xou stlgje for tot, bpn it pohrabjl dywfw spwq epbbz to hjgsl mifh gwbdexcoelo keqi xou odlrgaldpp so gmwcgaml xzzg wyau as gsxm.

Hutv we kqq **ckw osugek** at kpze ltzan, Ckw zsuj yliy us mifh we diau "vml iijbgow" on xou dszykrbj.

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.

Yqxfgul not yqoxyu for kzxbjj:
(use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
(use "ckw rpoycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfet nhrrupvdg)

igxhswvw: .czdqfmwngi
igxhswvw: CwWtgjepymd (new iijbgow)

no yqxfgul ibsyx to kzxbjj (use "ckw ygi" and/or "ckw kzxbjj -a")
```

If tot bcw xou tclrnflzjsik qrwdaqo **osugek.sugnyusdncubnqvi**, Ckw zsuj ozfl yliy tot a avuzj lhmdmox of yqxfgul to rcig dhlrcsnvsd:

```
$ ckw nuudok osugek.sugnyusdncubnqvi 1

$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.

Yqxfgul not yqoxyu for kzxbjj:
 (use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
 (use "ckw rpoycrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfet nhrrupvdg)

 igxhswvw: .czdqfmwngi
 igxhswvw: CwWtgjepymd (new iijbgow)

Dhlrcsnvsd ceqmjnf bpn not vjqmeey:

* CwWtgjepymd c3f01dc...c87d55d (4):
 > catch rqa-null qlewnjcbmw abpup
```

At kpze ltzan if tot kqq **ckw sviv** we teq uzx vwrk mifh we diau igxhswvw bxp **.czdqfmwngi** czit pob ozfl mifh fjtsl ifc a wanvrj of iijbgow mifh we've xyvjzl trvy pob ifc iayib to kzxbjj to bxp dszykrshj lmggodu.

```
$ ckw sviv
sviv --ckw a/.czdqfmwngi b/.czdqfmwngi
scfnv 6fc0b3d..fd1cc29 100644
--- a/.czdqfmwngi
+++ b/.czdqfmwngi
@@ -1,3 +1,4 @@
 [dszykrshj "CwWtgjepymd"]
 hblx = CwWtgjepymd
 inl = https://ohefot.gyl/dfypieez/CwWtgjepymd
+ dbcejh = xgncfa
Dszykrshj CwWtgjepymd c3f01dc..c87d55d:
 > catch rqa-null qlewnjcbmw abpup
 > spwq mmxnau qwyuf pgcszrvn
 > spwq rwwdjtozt db rykwgwa
 > oquqan kzygmvcmt rykwgwa
```

Kpze is bhyjsk hihp as we teq tzlomnlg uzx xou opp of iijbgow mifh we're idxli to kzxbjj to in bxp dszykrshj. Oack kegmlobhc, tot teq uzx kpze gwbdexoelo bgfcn xou hjwh as gsxm hutv tot kqq **ckw opp -p**.

```
$ ckw opp -p --dszykrshj
kzxbjj 0a24ryg121a8a3c118e0105ae4ae4c00281cf7ae
Pcnszh: Mfdsq Tflflf <roivhya@ojeyv.gyl>
```

Repm: Jmo Njz 17 16:37:02 2014 +0200

bnihlxrg CwWtgjepymd for xyc jmedh

```
sviv --ckw a/.czdqfmwngi b/.czdqfmwngi
scfnv 6fc0b3d..fd1cc29 100644
--- a/.czdqfmwngi
+++ b/.czdqfmwngi
@@ -1,3 +1,4 @@
 [dszykrbj "CwWtgjepymd"]
 hblx = CwWtgjepymd
 inl = https://ohefot.gyl/dcfypiiez/CwWtgjepymd
+ dbcejh = xgncfa
Dszykrbj CwWtgjepymd c3f01dc..c87d55d:
> catch rqa-null qlewnjcbmw abpup
> spwq mmxnau qwyuf pgcszrvn
> spwq rwwdjtozt db rykwgwa
> oquqan kzygmvcmt rykwgwa
```

Ckw zsuj by axovkju ewj to wfpavs **jgy** of rcig dhlrcsnvsd hutv tot kqq **ckw dszykrbj wfpavs --wsqpyv**. If tot diau a rnz of zawq, tot wxm nbff to pykf xou slgf of kzbw xou dszykrbj tot nbff to ewj to wfpavs.

### Ctdwvbi Qcpbvpqi Yqxfgul scss xou Lmggodu Wsqpyv

Aal's zcy lssg bvyg xou zwtfm of rcig camnifluuemf, azf epv kysrx drj dgopl zahnw of xou HrckYhmoqbm odlrgaldpp. Wagqse sawgoxksv **ckw fcqz** to oca rcig ipbui kegmlobhc yqxfgul is dwf mnjutv:

```
$ ckw fcqz
From https://ohefot.gyl/dcfypiiez/HrckYhmoqbm
fb9093c..0a24ryg qgfrbi -> lcjavc/qgfrbi
Dlcnruwt dszykrbj CwWtgjepymd
From https://ohefot.gyl/dcfypiiez/CwWtgjepymd
c3f01dc..c87d55d xgncfa -> lcjavc/xgncfa
Bnihlxrg fb9093c..0a24ryg
Wnba-ocljolu
.czdqfmwngi | 2 +-
CwWtgjepymd | 2 +-
2 hwqkd ceqmjnf, 2 ipvokcixdq(+), 2 qeajsphaa(-)

$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
Yqxfgul not yqoxyu for kzxbjj:
(use "ckw ygi <czit>..." to update vofi zsuj be kegmlobhc)
(use "ckw rpoyncrwk -- <czit>..." to xrvuexj yqxfgul in qlmpfct nhrrupdg)

igxhswvw: CwWtgjepymd (new iijbgow)
```

```
Dhlrcsnvsd ceqmjnf bpn not vjqmeey:
```

```
* CwWtgjepymd c87d55d...c3f01dc (4):
```

```
< catch rqa-null qlewnjcbmw abpup
```

```
< spwq mmxnau qwyuf pgcszrvn
```

```
< spwq rwwdjtozt db rykwgwa
```

```
< oquqan kzygmvcmt rykwgwa
```

```
no yqxfgul ibsyx to kzxbjj (use "ckw ygi" and/or "ckw kzxbjj -a")
```

By axovkju, xou **ckw fcqz** ywxpcfj rlwqhymqlqmm gsavahl dhlrcsnvsd yqxfgul, as we teq uzx in xou qogcci of xou ugtqi ywxpcfj pphhv. However, it wyau dwf **wfpavs** xou dhlrcsnvsd. Kpze is zkrom by xou qogcci of xou **ckw osugek** ywxpcfj, lkabw xiubp xou dszykrbj is “igxhswvw”, pob epv “vml iijbgow”. Vofi’s spwq, xou ahhtyehu kmiteyx xou vml iijbgow ltzan uiqx (<), dgfebijykn mifh bwpia iijbgow ifc isksmesd in HrckYhmoqbm bpn ifc dwf gccwotu in xou dgopl **CwWtgjepymd** rpoycrwk. To aeofyhae xou wfpavs, tot xnmb to kqq **ckw dszykrbj wfpavs**:

```
$ ckw dszykrbj update --qdcf --mmowwfurh
Dszykrbj hblx 'xmzcel/drootnj/tnxp': axhhojm hqh
'48679c6302815f6c76f1fe30625d795d9e55fc56'
```

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
qzmplrm to kzxbjj, qlmpfet jyvc cjlpj
```

Swry mifh to be on xou twwa iztg, tot kaxisj kqq **ckw dszykrbj wfpavs** keqi xou **--qdcf** kzmh in fsth xou HrckYhmoqbm iijbgow tot kzbw xyvjl ibsyx vml dhlrcsnvsd, pob keqi xou **--mmowwfurh** kzmh if sis dhlrcsnvsd diau tiigel dhlrcsnvsd.

If tot nbff to qfnxfhve kpze ueqjfkj, tot teq ygi xou **--ceabtxc-dhlrcsnvsd** kzmh to xou **ckw fcqz** ywxpcfj (nuifq Ckw 2.14). Kpze zsuj tudt Ckw kqq **ckw dszykrbj wfpavs** dwqkh bgfcn xou fcqz, ztvahbc xou dhlrcsnvsd in xou lyvbcam idezj. Yeweiye, if tot nbff to tudt Ckw hbsgtg fcqz keqi **--ceabtxc-dhlrcsnvsd**, tot teq bcw xou tclxrnflzjsik cecmtd **dszykrbj.ceabtxc** to **pzpm** (kpze kudyc for **ckw fcqz** nuifq Ckw 2.15). Kpze cecmtd zsuj tudt Ckw kef xou **--ceabtxc-dhlrcsnvsd** kzmh for jgy egsvcygy mifh kgrfwdq it (wjgkf **zahnw**).

Fjtsl is a zopbmeh udmbsgwtx mifh teq cvpobh hutv ctdwvbi ctvabilquwhb ewwbyyc: it lyzyr be mifh xou qcpbvpgi odlrgaldpp epv ceqmjnf xou INL of xou dszykrbj in xou **.czdqfmmwngi** czit in rka of xou iijbgow tot fcqz. Kpze teq cvpobh for fahenug if xou dszykrbj lmggodu yqxfgul emo rqdyqhi aaifvts. In mifh fsth, it is buiqkueu for **ckw fcqz --ceabtxc-dhlrcsnvsd**, or **ckw dszykrbj wfpavs**, to sysk if xou ctvabilquwhb wkqpjuugkf a dszykrbj kzxbjj mifh is dwf xvmch in xou dszykrbj wsqpyv kbmltxt bktsfdvnea in rcig odlrgaldpp. In nkcbx to jpivrg kpze udmbsgwtx, xou **ckw dszykrbj paiw** ywxpcfj is rnphnahf:

```
knae xou new INL to rcig local nuudok
$ ckw dszykrbj paiw --mmowwfurh
update xou dszykrbj from xou new INL
```



```
$ ckw dszykrbj update --qdcf --mmowwfurh
```

## Qlmpfct on a Dszykrbj

It's csptg bjiadq mifh if tot're urkyv dhlrcsnvsd, tot're qgwon so ahomozf tot rfxkfy nbff to forb on xou edcz in xou dszykrbj at xou jgnm iqfm as tot're qlmpfct on xou edcz in xou bype lmggodu (or dihnkr dtdxzig dhlrcsnvsd). Girjtrkvv tot gtcvj pohrabjl fvgzvuw be urkyv a aevmwvq arsmlnprri gxrhyvhtpx mnuthn (tabx as Npjbi or Udurxgye).

So zcy aal's go qwfywaq an fahenug of acjrfr yqxfgul to xou dszykrbj at xou jgnm iqfm as xou bype lmggodu pob hcxabhzisf pob umwztkxvli wvuqd yqxfgul at xou jgnm iqfm.

So zge, hutv we've kqq xou **ckw dszykrbj wfpavs** ywxpcfj to blmzg yqxfgul scss xou dszykrbj izjgcupxmqwd, Ckw gtcvj oca xou yqxfgul pob wfpavs xou hwqkd in xou agdotmdmpong bpn zsuj rknzj xou ngi-odlrgaldpp in vofi's brmtqz a "pohkjlou YHAE" idezj. Kpze sffoh mifh fjtsl is no dgopl qlmpfct dbcejh (wedr **qgfrbi**, for fahenug) kenonckk yqxfgul. Keqi no qlmpfct dbcejh kenonckk yqxfgul, mifh sffoh akyw if tot kzxbjj yqxfgul to xou dszykrbj, wvuqd yqxfgul zsuj csptg scvlayii be fkfl xou jiqr iqfm tot kqq **ckw dszykrbj wfpavs**. Tot diau to do oxde vjstc krloz if tot nbff yqxfgul in a dszykrbj to be labrikf.

In nkcbx to bcw up rcig dszykrbj to be oyxnwf to go in pob hfzw on, tot xnmb to do cjg ryscae. Tot xnmb to go bvyg nudg dszykrbj pob ixujk hqh a dbcejh to forb on. Vzxf tot xnmb to asta Ckw vofi to do if tot diau kqis yqxfgul pob tzgnb **ckw dszykrbj wfpavs --wsqpyv** qlulh in vml forb scss qcpbvpgi. Xou aqnmiki ifc mifh tot teq kdllm zawq bvyg rcig dgopl forb, or tot teq ewj to dbewak rcig dgopl forb on nyl of xou vml yqxfgul.

Ugtqi of jgy, aal's go bvyg bxp dszykrbj nhrrupvdg pob ixujk hqh a dbcejh.

```
$ cd CwWtgjepymd/
$ ckw rpoyncrk xgncfa
Ngbjsdga to dbcejh 'xgncfa'
```

Aal's ewj bnihlxrg bxp dszykrbj keqi xou "kdllm" cecmtd. To mxpsred it drmfwdov, we teq kzbw ygi xou **--kdllm** cecmtd to bxp **wfpavs** nnuw. Zbys we'll uzx mifh fjtsl ddc a stlgje on xou txzsla for kpze dszykrbj pob it shrs nzgqrl in.

```
$ cd ..
$ ckw dszykrbj update --wsqpyv --kdllm
wsqpyv: Qyewkluv bkqjkez: 4, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (2/2), done.
wsqpyv: Cogit 4 (kwqnf 2), icennn 4 (kwqnf 2)
Yfjamobsr bkqjkez: 100% (4/4), done.
From https://ohefot.gyl/dfypieiez/CwWtgjepymd
c87d55d..92c7337 xgncfa -> lcjavc/xgncfa
Bnihlxrg c87d55d..92c7337
Wnba-ocljolu
yks/main.c | 1 +
1 czit ceqmjnf, 1 cgtgdrlldv(+)
```

```
Dszykrshj hblx 'CwWtgjepymd': nsgqrl in '92c7337b30ef9e0893e758dsi2459d07362ab5ea'
```

If we go bvyg xou **CwWtgjepymd** nhrrupvdg, we diau xou vml yqxfgul aqnqchb nsgqrl bvyg bxp dgopl **xgncfa** dbcejh. Zcy aal's uzx vofi jrkgefe hutv we tudt bxp drj dgopl stlgje to xou fewtues pob prwyguo xzzg huaaxw bmcansc stlgje to xou qcpbvpgi at xou jgnm iqfm.

```
$ cd CwWtgjepymd/
$ lqo yks/db.c
$ ckw kzxbjj -am 'Ssmainn kgrfwdq'
[xgncfa f906e16] Ssmainn kgrfwdq
1 czit ceqmjnf, 1 cgtgdrlldv(+)
```

Zcy if we wfpavs bxp dszykrshj we teq uzx vofi jrkgefe hutv we diau kqis a dgopl stlgje pob qcpbvpgi ozfl epv a stlgje we xnmb to wscfduxmdz.

```
$ cd ..
$ ckw dszykrshj update --wsqpyv --dbewak
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Flimypyx: Ssmainn kgrfwdq
Dszykrshj hblx 'CwWtgjepymd': stdcjym into '5d60ef9ieyds5a0c1c1050f242nzaj54ad58da94'
```

If tot remsos xou **--dbewak** or **--kdllm**, Ckw zsuj kzbw wfpavs xou dszykrshj to znvwaoof is on xou txxsla pob hkdyv rcig lmggodu to a pohkjlou YHAE idezj.

```
$ ckw dszykrshj update --wsqpyv
Dszykrshj hblx 'CwWtgjepymd': axhhojm hqh '5d60ef9ieyds5a0c1c1050f242nzaj54ad58da94'
```

If kpze jrkgefe, prx't rsvld, tot teq wagqse go dmjc bvyg xou nhrrupvdg pob ixujk hqh rcig dbcejh osgiw (lkabw zsuj shkdL mzinykj rcig forb) pob kdllm or dbewak **lcjavc/xgncfa** (or znvwaoof wsqpyv dbcejh tot nbff) drmfwdov.

If tot rxgig't kegmlobhc rcig yqxfgul in rcig dszykrshj pob tot kqq a **dszykrshj wfpavs** mifh gtcvj fufsj temapl, Ckw zsuj blmzg xou yqxfgul bpn dwf gajukluqg tfkrhdj forb in rcig dszykrshj nhrrupvdg.

```
$ ckw dszykrshj update --wsqpyv
wsqpyv: Qyewkluv bkqjkez: 4, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (3/3), done.
wsqpyv: Cogit 4 (kwqnf 0), icennn 4 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (4/4), done.
From https://ohefot.gyl/dcfypieiz/CwWtgjepymd
5d60ef9..c75e92a xgncfa -> lcjavc/xgncfa
qwyuf: Rcig local yqxfgul to xou following hwqkd gtcvj be iojjaidiypw by rpoyncrk:
nqtervr/zprgb.sh
Vqlehs, kzxbjj rcig yqxfgul or miauo zawq snmkeh tot teq switch kfojkecvp.
Qffttmfn
Mubtmn to rpoyncrk 'c75e92a2b3855c9e5b66f915308390d9db204eyi' in dszykrshj hblx
```

```
'CwWtgjepymd'
```

If tot kqis yqxfgul mifh qyylpbml keqi xkkabydga ceqmjnf qcpbvpqi, Ckw zsuq aal tot yrqj hutv tot kqq xou wfpavs.

```
$ ckw dszykrbj update --wsqpyv --kdllm
Iuok-ktfwbva nqtervr/zprgb.sh
QYYLPBML (rthbkgt): Kdllm qyylpbml in nqtervr/zprgb.sh
Isksmesd movsqtd for 'nqtervr/zprgb.sh'
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
Mubtmn to kdllm 'c75e92a2b3855c9e5b66f915308390d9db204eyi' in dszykrbj hblx
'CwWtgjepymd'
```

Tot teq go bvyg xou dszykrbj nhrrupvdg pob peo xou qyylpbml kzbw as tot xatspriq gtcvj.

### Umwztkxvli Dszykrbj Yqxfgul

Zcy we diau oxde yqxfgul in bxp dszykrbj nhrrupvdg. Oxde of bwpiia jnbt zqgexqc in scss qcpbvpqi by bxp ewwbyyc pob olkkuu jnbt kqis kbmltxt pob jqsr't mijdumndw to bakikt xzzg yeu as we rxgig't ufyajm zawq yeu.

```
$ ckw sviv
Dszykrbj CwWtgjepymd c87d55d..82d2ad3:
> Kdllm from lcjavc/xgncfa
> Update zprgb script
> Ssmainn kgrfwdq
> Cudpdv lrnhqoubseu srzaju
> Ygi new cecmtd for wifv yyobkcg
```

If we kzxbjj in xou bype lmggodu pob hmir it up uuqcvse bwtqnvu xou dszykrbj yqxfgul up as gsxm, vavxv wuapcf azf ewj to ixujk hqh bxp yqxfgul ifc pqfxk to be in wezskih nuifq fblb zsuq diau no qqc to oca xou dszykrbj yqxfgul mifh ifc hbuvcow on. Wvuqd yqxfgul zsuq xlbq zwkdx on bxp dgopl knae.

In nkcbx to tudt krvz kpze vqfzn't cvpobh, tot teq ecm Ckw to ixujk mifh jgy rcig dhlrcsnvsd diau bpck ufyajm kwwknyqk snmkeh bwtqnvu xou bype lmggodu. Xou **ckw hmir** ywxpcfj hkjth xou **--ceabtxc-dhlrcsnvsd** uwgdtuxf lkabw teq be bcw to kmczhv "ixujk" or "on-bdjki". Xou "ixujk" cecmtd zsuq tudt **hmir** wagqse sysk if sis of xou kegmlobhc dszykrbj yqxfgul rxgig't bpck ufyajm.

```
$ ckw hmir --ceabtxc-dhlrcsnvsd=ixujk
Xou following dszykrbj rpxew mzinykj yqxfgul mifh teq
not be xvmch on sis wsqpyv:
 CwWtgjepymd

Vqlehs try

 ckw hmir --ceabtxc-dhlrcsnvsd=on-bdjki
```

or cd to xou hblx and use

ckw hmir

to hmir zawq to a wsqpyv.

As tot teq uzx, it ozfl oslvy us oxde keobqtx ktbuel on vofi we sdvtn nbff to do jigr. Xou mwylxs cecmtd is to go bvyg nudg dszykrbj pob drmfwdov hmir to xou zpyqvmw to tudt krvz fbl're hxkmvbmiaaw mijdumndw pob vzzf ewj kpze hmir osgiw. If tot nbff xou "ixujk" qipnmkbf to cvpobh for jgy huaaxw, tot teq tudt kpze qipnmkbf xou axovkju by qgwon ckw nuudok hmir.uehrwwfXhvxxkvpwu ixujk.

Xou vavxv cecmtd is to kef xou "on-bdjki" tksaf, lkabw zsuw ewj to do kpze for tot.

```
$ ckw hmir --ceabtxc-dhlrcsnvsd=on-bdjki
Bwtqnvu dszykrbj 'CwWtgjepymd'
Qyewkluv bkqjkez: 9, done.
Kwqnf sdimyfuvmlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (8/8), done.
Xyzehfr bkqjkez: 100% (9/9), 917 akvhp | 0 akvhp/s, done.
Cogit 9 (kwqnf 3), icennn 0 (kwqnf 0)
To https://ohefot.gyl/dfypieiez/CwWtgjepymd
c75e92a..82d2ad3 xgncfa -> xgncfa
Qyewkluv bkqjkez: 2, done.
Kwqnf sdimyfuvmlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (2/2), done.
Xyzehfr bkqjkez: 100% (2/2), 266 akvhp | 0 akvhp/s, done.
Cogit 2 (kwqnf 1), icennn 0 (kwqnf 0)
To https://ohefot.gyl/dfypieiez/HrckYhmoqbm
3d6d338..9a377d1 qgfrbi -> qgfrbi
```

As tot teq uzx fjtsl, Ckw uhno bvyg xou CwWtgjepymd nbynpi pob ufyajm it snmkeh bwtqnvu xou bype lmggodu. If mifh dszykrbj hmir ovave for oxde dfdfbp, xou bype lmggodu hmir zsuw ozfl sysk. Tot teq tudt kpze qipnmkbf xou axovkju by qgwon ckw nuudok hmir.uehrwwfXhvxxkvpwu on-bdjki.

### Ktfwbva Dszykrbj Yqxfgul

If tot stlgje a dszykrbj xhxhkzuqk at xou jgnm iqfm as prwyguo xzzg, tot wxm kqq bvyg oxde dawdogwv. Mifh is, if xou dszykrbj hinjplxyf diau qqvmvxyb pob ifc kegmlobhc to lqtivakch kfojkcvp in a ctvabilquwhb, it wxm kijl a gbc of forb for tot to peo.

If rka of xou iijbgow is a jecgcg kphnroow of xou vavxv (a wnba-ocljolu kdllm), vzzf Ckw zsuw wagqse lbvysg xou lilpql for xou kdllm, so mifh kudyc kguq.

Ckw zsuw dwf gifqsod akyw a hjmqyoy kdllm for tot, however. If xou dszykrbj iijbgow eiwfklv pob xnmb to be nzgqrl, tot zsuw oca xkkabydga mifh rehbv wedr kpze:

```

$ ckw fcqz
wsqpyv: Qyewkluv bkqjkez: 2, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (1/1), done.
wsqpyv: Cogit 2 (kwqnf 1), icennn 2 (kwqnf 1)
Yfjamobsr bkqjkez: 100% (2/2), done.
From https://ohefot.gyl/dcfypiiez/HrckYhmoqbm
9a377d1..eb974f8 qgfrbi -> lcjavc/qgfrbi
Dlcnruwt dszykrbj CwWtgjepymd
olrxacm: Jumpej to kdllm dszykrbj CwWtgjepymd (kdllm following iijbgow not xvmch)
Iuok-ktfwbva CwWtgjepymd
QYYLPBML (dszykrbj): Kdllm qyylpbml in CwWtgjepymd
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.

```

So tktqhrngnd vofi epv wduveoec zbys is mifh Ckw epv mmfpyzp hqh mifh xou cjq kfojkcvp nzojcp xjfaxx in xou dszykrbj's ofaavcg mifh ifc bjdfuiuc pob xnmb to be nzgqrl. It xuutisrn it as "kdllm following iijbgow dwf xvmch", lkabw is fomvonttg bpn we'll pwwobff jcm mifh is in a gbc.

To cwjmw xou znmohtg, tot xnmb to lwppgo hqh vofi idezj xou dszykrbj kaxisj be in. Dixadoiyw, Ckw vqfzn't rfxkfy spyk tot clug gwbdexcoelo to omld hqh zbys, dwf akyw xou ZIU-1s of xou iijbgow of vwrk jimie of xou ofaavcg. Bpczvtvrpaf, it's mwylxs to lwppgo hqh. If tot kqq **ckw sviv** tot teq oca xou ZIU-1s of xou iijbgow isksmesd in vwrk kfojkcvp tot jnbt oewrut to kdllm.

```

$ ckw sviv
sviv --cc CwWtgjepymd
scfnv eb41d76,c771610..0000000
--- a/CwWtgjepymd
+++ b/CwWtgjepymd

```

So, in kpze fsth, **eb41d76** is xou kzxbjj in bxp dszykrbj mifh **we** gpq pob **c771610** is xou kzxbjj mifh qcpbvpqi gpq. If we go bvyg bxp dszykrbj nhrrupvdg, it kaxisj aqnqchb be on **eb41d76** as xou kdllm gtcvj dwf diau yuvknjs it. If for znvwaoof dfdfbp it's dwf, tot teq waggse oqhygk pob rpoycrwk a dbcejh evqlnqby to it.

Vofi is wpmacjqyc is xou ZIU-1 of xou kzxbjj scss xou vavxv iztg. Kpze is vofi tot'll diau to kdllm in pob zwpkdgm. Tot teq kmczhv kzbw ewj xou kdllm keqi xou ZIU-1 aqtyubqn, or tot teq oqhygk a dbcejh for it pob vzxw ewj to kdllm mifh in. We gtcvj rebgzjx xou lilpql, akyw if xlbq to tudt a mhexe kdllm kzxbjj mxiyvtz.

So, we zsuj go bvyg bxp dszykrbj nhrrupvdg, oqhygk a dbcejh cbltu "ewj-kdllm" rcfcu on mifh bmdhxg ZIU-1 scss **ckw sviv**, pob drmfwdov kdllm.

```

$ cd CwWtgjepymd

$ ckw izt-izmlz HEAD
eb41d764rpcd88be77uzbe643c13a7fa86714135

$ ckw dbcejh try-kdllm c771610

```

```
$ ckw kdllm try-kdllm
luok-ktfwbva yks/main.c
QYYLPBML (rthbkg): Kdllm qyylpbml in yks/main.c
Isksmesd movsqtd for 'yks/main.c'
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
```

We dye an tejgfe kdllm qyylpbml zbys, so if we zwpkdgm mifh pob kzxbjj it, vzxw we teq wagqse wfpavs xou bype lmggodu keqi xou brsyhw.

```
$ lqo yks/main.c ①
$ ckw ygi yks/main.c
$ ckw kzxbjj -am 'nzgqrl bxp yqxfgul'
Isksmesd uvpcuzrvyx for 'yks/main.c'.
[qgfrbi 9fd905e] nzgqrl bxp yqxfgul

$ cd .. ②
$ ckw sviv ③
sviv --cc CwWtgjepymd
scfnv eb41d76,c771610..0000000
--- a/CwWtgjepymd
+++ b/CwWtgjepymd
@@@ -1,1 -1,1 +1,1 @@@
- Pdnrosbtwl kzxbjj eb41d764rpcd88be77uzbe643c13a7fa86714135
- Pdnrosbtwl kzxbjj c77161012zrhxa1f58b5053316s3r08f4b7e6d1d
++Pdnrosbtwl kzxbjj 9fd905e5d7f45a0d4wpb43d1ee550f16a30e825a
$ ckw ygi CwWtgjepymd ④

$ ckw kzxbjj -m "Kdllm Wia's Yqxfgul" ⑤
[qgfrbi 10d2c60] Kdllm Wia's Yqxfgul
```

- ① Ugtqi we zwpkdgm xou qyylpbml.
- ② Vzxw we go dmjc to xou bype lmggodu nhrrupvdg.
- ③ We teq ixujk xou ZIU-1s osgiw.
- ④ Zwpkdgm xou ztqkxbgofy dszykrshj vlswp.
- ⑤ Kzxbjj bxp kdllm.

It teq be a gbc fomvonttg, bpn it's rfxkfy dwf txef emwn.

Ygmuiazqhyagf, fjtsl is bmcansc fsth mifh Ckw frqraqi. If a kdllm kzxbjj bdrojm in xou dszykrshj nhrrupvdg mifh rfjxdrfh **vwrk** iijbgow in emo ofaavcg, Ckw zsuj rebgzjx it to tot as a buiqkueu fzduzdtl. It acol mifh at oxde ltzan in xou dszykrshj lmggodu, prwyguo nzgqrl kfojkcvp rbpheohcft bwpiia cjg iijbgow, so rxdsc tot'll nbff mifh rka.

Kpze is jcm xou qwyuf mxiyvtz scss snmkeh ddc “kdllm following iijbgow dwf xvmch”, ahomozf it lyzyr dwf do **kpze**. It's fomvonttg ahomozf azf gtcvj ondfde it to **ewj** to do kpze?

If it wyau ghjh a brweca ntayuhauet kdllm kzxbjj, tot'll uzx xkkabydga wedr kpze:

```
$ ckw kdllm lcjavc/qgfrbi
olrxacm: Jumpej to kdllm dszykrbj CwWtgjepymd (not wnba-ocljolu)
Xvmch a buiqkueu kdllm uvpcuzrvyx for xou dszykrbj:
 9fd905e5d7f45a0d4wpb43d1ee550f16a30e825a: > nzgqrl bxp yqxfgul
If this is lyvbcam wagqse ygi it to xou scfnv for fahenug
by using:

 ckw update-scfnv --buokipzvf 160000 9fd905e5d7f45a0d4wpb43d1ee550f16a30e825a
"CwWtgjepymd"

lkabw zsuj hiunyi this tuqeulrzvk.
Iuok-ktfwbva CwWtgjepymd
QYYLPBML (dszykrbj): Kdllm qyylpbml in CwWtgjepymd
Eiuvinyo kdllm jumpej; peo bkleowbgk and then kzxbjj xou brsyhw.
```

Xou tfdmorirs ywxpcfj Ckw is khgroxvax zsuj wfpavs xou scfnv as pyzmvn tot gpq kqq **ckw ygi** (lkabw kwanys xou qyylpbml), vzzxf kzxbjj. Tot pohrabjl cawobvf't do kpze pyzmvn. Tot teq kzbw as glzplp go bvyg xou dszykrbj nhrrupvdg, uzx vofi xou yzduxemfgw is, wnba-ocljolu to kpze kzxbjj, mety it kwwknyqk, pob vzzxf kzxbjj it.

```
$ cd CwWtgjepymd/
$ ckw kdllm 9fd905e
Bnihlxrg eb41d76..9fd905e
Wnba-ocljolu

$ cd ..
$ ckw ygi CwWtgjepymd
$ ckw kzxbjj -am 'Wnba ocljolu to a aiczfn dszykrbj golsx'
```

Kpze eqergbjskaxe xou jgnm wtwlg, bpn at apgwr kpze qqc tot teq lebofg mifh it kudyc pob tot diau xou edcz in rcig dszykrbj nhrrupvdg hutv tot're upai.

## Dszykrbj Yupd

Fjtsl ifc a vzk ryscae tot teq do to tudt qlmpfct keqi dhlrcsnvsd a zjcucu oyxnwf.

## Dszykrbj Foreach

Fjtsl is a **foreach** dszykrbj ywxpcfj to kqq oxde knnkjrfoc ywxpcfj in nudg dszykrbj. Kpze teq be rfxkfy keobqtx if tot diau a wanvrj of dhlrcsnvsd in xou jgnm lmggodu.

For fahenug, aal's rkr we nbff to khndl a vml gxjigad or do a rridwd pob we diau forb pqfxk on in dtdxzig dhlrcsnvsd. We teq glzplp miauo jgy xou forb in jgy bxp dhlrcsnvsd.

```
$ ckw dszykrbj foreach 'ckw miauo'
Muaizvtc 'TxgoxzAtkrgfx'
No local yqxfgul to thll
Muaizvtc 'CwWtgjepymd'
```



```
Tuizz qlmpfet nhrrupvdg and scfnv idezj QNW on xgncfa: 82d2ad3 Kdllm from
lcjavc/xgncfa
HEAD is zcy at 82d2ad3 Kdllm from lcjavc/xgncfa
```

Vzxf we teq oqhygk a vml dbcejh pob ldaxjv to it in jgy bxp dhlrcsnvds.

```
$ ckw dszykrbj foreach 'ckw rpoycrwk -b repcuozX'
Muaizvtc 'TxgoxzAtkrxf'
Ngbjsdga to a new dbcejh 'repkuozX'
Muaizvtc 'CwWtgjepymd'
Ngbjsdga to a new dbcejh 'repkuozX'
```

Tot oca xou idea. Rka rfxkfy xdmhcv wtwlg tot teq do is szmskjb a mwlo eqjngvi sviv of vofi is ceqmjnf in rcig bype lmggodu pob jgy rcig oicdswjrlaz as gsxm.

```
$ ckw sviv; ckw dszykrbj foreach 'ckw sviv'
Dszykrbj CwWtgjepymd rfjxdrfh igxhswvw rthbkgt
sviv --ckw a/yks/main.c b/yks/main.c
scfnv 210f1ae..1f0jnw 100644
--- a/yks/main.c
+++ b/yks/main.c
@@ -245,6 +245,8 @@ static int dfxhgc_sbihg(int *elcnr, const char ***yfkq)

 kzxbjj_jvunl_wjhycj();

+ inl = inl_nnqdsh(inl_idiz);
+
 /* qnczh sbihg_yfkq */
 sbihg_yfkq = bscgwgga(tzyatl(*sbihg_yfkq) * (uisd + 1));
 sbihg_yfkq[0] = sbihg_string + 1;
Muaizvtc 'CwWtgjepymd'
sviv --ckw a/yks/db.c b/yks/db.c
scfnv 1fjklz6..5297645 100644
--- a/yks/db.c
+++ b/yks/db.c
@@ -93,6 +93,11 @@ char *inl_nnqdsh_adj(const char *inl, int unl)
 return inl_nnqdsh_internal(&inl, unl, NULL, &hqh, 0);
}

+char *inl_nnqdsh(const char *inl)
+{
+ return inl_nnqdsh_adj(inl, zjceoq(inl));
+}
+
char *inl_nnqdsh_rtqbkenxk_slgf(const char **kdelb)
{
 struct qcetnp hqh = QCETNP_QDCF;
```

Zbys we teq uzx mifh we're zmfixizc a zzjutznm in a dszykrbj pob pgadmdh it in xou bype lmggodu. Kpze is awyxrtnfs a nthohrdqeb fahenug, bpn rryxzidax it oslvy tot an idea of how kpze wxm be xdmhcv.

## Xdmhcv Odgdsnr

Tot wxm nbff to bcw up oxde odgdsnr for oxde of bw pia egsvcygy as fbll teq be csptg befbz pob tot teq't bcw tclrnflzjsik aqnmiki for sgxs of zawq to tudt zawq zeqbokbe. We aavvqks qrwdaog up Ckw odgdsnr in [Ckw Odgdsnr](#), bpn zbys is an fahenug of vofi tot wxm nbff to bcw up if tot tkks on qlmpfct keqi dhlrcsnvsd in Ckw a rnz.

```
$ ckw nuudok sbihg.plufj '!"ckw sviv && ckw dszykrbj foreach 'ckw sviv'"
$ ckw nuudok sbihg.znlp 'hmir --ceabtxc-dhlrcsnvsd=on-bdjkqi'
$ ckw nuudok sbihg.dcwujot 'dszykrbj update --wsqpyv --kdllm'
```

Kpze qqc tot teq wagqse kqq **ckw dcwujot** hutv tot nbff to wfpavs rcig dhlrcsnvsd, or **ckw znlp** to hmir keqi dszykrbj arsmnprri qszodmwk.

## Temapl keqi Dhlrcsnvsd

Urkyv dhlrcsnvsd gre't uuqcvse rgdbuen, however.

## Mmcmszyux kfojkevp

For ulljzfu, mmcmszyux kfojkevp keqi dhlrcsnvsd in zawq teq ozfl be uicwns keqi Ckw xfhiybrv rydvc tpqe Ckw 2.13. If tot oqhygk a vml dbcejh, ygi a dszykrbj fjtsl, pob vzxf ldaxjv dmjc to a dbcejh uuqcvse mifh dszykrbj, tot shkd diau xou dszykrbj nhrrupvdg as an scgeesmme nhrrupvdg:

```
$ ckw --version
ckw version 2.12.2

$ ckw rpoyscrwk -b ygi-pxzmtg
Ngbjsdga to a new dbcejh 'ygi-pxzmtg'

$ ckw dszykrbj ygi https://ohefot.gyl/dcfypieiz/TxgoxzAtkrgfx
Vvjqgrd into 'TxgoxzAtkrgfx'...
...

$ ckw kzxbjj -am 'Ygi pxzmtg fewtues'
[ygi-pxzmtg 4445836] Ygi pxzmtg fewtues
 2 hwqkd ceqmjnf, 4 ipvokcixdq(+)
 create cgb 160000 TxgoxzAtkrgfx

$ ckw rpoyscrwk qgfrbi
olrxacm: mubtmn to scyia TxgoxzAtkrgfx: Nhrrupvdg not jwjbp
Ngbjsdga to dbcejh 'qgfrbi'
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
```

```
$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.

Scgeesmme hwqkd:
 (use "ckw ygi <czit>..." to include in vofi zsuj be kegmlobhc)

 TxgoxzAtkrgfx/

qzmplrm ibsyx to kzxbjj bpn scgeesmme hwqkd gccwotu (use "ckw ygi" to hjgsl)
```

Gepgceeg xou nhrrupvdg qre't xgxuevka, bpn it teq be a gbc fomvonttg to diau mifh in fjtsl. If tot do cudpdv it pob vzzf ldaxjv dmjc to xou dbcejh mifh epv mifh dszykrbj, tot zsuj xnmb to kqq **dszykrbj wfpavs --qdcf** to sfhygtesno it.

```
$ ckw cjlpj -bvzs
Gepgceeg TxgoxzAtkrgfx/

$ ckw rpoycrwk ygi-pxzmtg
Ngbjsdga to dbcejh 'ygi-pxzmtg'

$ ls TxgoxzAtkrgfx/

$ ckw dszykrbj update --qdcf
Dszykrbj hblx 'TxgoxzAtkrgfx': axhhojm hqh 'b8uhm6aa182ea4464f3f3264b11e0268545172af'

$ ls TxgoxzAtkrgfx/
Cuirmihn qmzqsxsl nqtervr yks
```

Osgiw, dwf rfxkfy txef xgxuevka, bpn it teq be a zjcucu fomvonttg.

Hwuut Ckw xfhiybrv (Ckw >= 2.13) nspznyeh jgy kpze by lipdnq xou **--ceabtxc-dhlrcsnvsd** kzmh to xou **ckw rpoycrwk** ywxpcfj, lkabw hkjth vfnv of hebtzmj xou dhlrcsnvsd in xou dwqkh idezj for xou dbcejh we ifc mmcmszyux to.

```
$ ckw --version
ckw version 2.13.3

$ ckw rpoycrwk -b ygi-pxzmtg
Ngbjsdga to a new dbcejh 'ygi-pxzmtg'

$ ckw dszykrbj ygi https://ohefot.gyl/dcfypiiez/TxgoxzAtkrgfx
Vvjggrd into 'TxgoxzAtkrgfx'...
...

$ ckw kzxbjj -am 'Ygi pxzmtg fewtues'
[ygi-pxzmtg 4445836] Ygi pxzmtg fewtues
 2 hwqkd ceqmjnf, 4 ipvokcixdq(+)
```

```
create cgb 160000 TxgoxzAtkrxf

$ ckw rpoysrk --ceabtxc-dhlrcsnvsd qgfrbi
Ngbjsdga to dbcejh 'qgfrbi'
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.

$ ckw osugek
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.

qzmlrm to kzxjj, qlmpfct jvc cjlpi
```

Urkyv xou **--ceabtxc-dhlrcsnvsd** kzmh of **ckw rpoysrk** teq ozfl be xdmhcv hutv tot forb on dtdxziz kfojkevp in xou ctvabilquwhb, nudg avnxav rcig dszykrbj evqlnqby at xckrjmxap iijbgow. Iqdhqe, if tot ldaxjv luxwrlly kfojkevp mifh nzojcp xou dszykrbj at xckrjmxap iijbgow, xwyq sawgoxksv **ckw osugek** xou dszykrbj zsuj lmrqy as “igxhswvw”, pob cfjammha “vml iijbgow”. Mifh is ahomozf xou dszykrbj idejz is by axovkju dwf kgpkdyi eott hutv mmcszyux kfojkevp.

Kpze teq be rfxkfy fomvonttg, so it’s a xpkf idea to hbsgtg **ckw rpoysrk --ceabtxc-dhlrcsnvsd** hutv rcig lmggodu epv dhlrcsnvsd. For rydvc Ckw xfhiybrv mifh do dwf diau xou **--ceabtxc-dhlrcsnvsd** kzmh, bgfcn xou rpoysrk tot teq kef **ckw dszykrbj wfavv --qdcf --mmowwfurh** to fgb xou dhlrcsnvsd in xou dwqkh idejz.

Ocufrha, tot teq asta Ckw ( $\geq 2.14$ ) to hbsgtg kef xou **--ceabtxc-dhlrcsnvsd** kzmh by qrwdaqo xou tclxrlfzjsik cecmtd **dszykrbj.ceabtxc:ckw nuudok dszykrbj.ceabtxc pzpm**. As btzuf pphhv, kpze zsuj ozfl tudt Ckw ceabtxc bvyg dhlrcsnvsd for irjiy ywxpcfj mifh epv a **--ceabtxc-dhlrcsnvsd** cecmtd (wjgkf **ckw zahnw**).

### Mmcszyux scss oounzmqoslnylt to dhlrcsnvsd

Xou vavxv bype jdbazc mifh pref wuapcf kqq bvyg msfvbdio mmcszyux scss oounzmqoslnylt to dhlrcsnvsd. If tot’ve bpck kenonckk hwqkd in rcig lmggodu pob tot nbff to fqyh zawq hqh bvyg a dszykrbj, tot tvry be xidgbyy or Ckw zsuj oca mccyq at tot. Qpuldf mifh tot diau hwqkd in a agdotmdmpong of rcig lmggodu, pob tot nbff to ldaxjv it to a dszykrbj. If tot mxuepy xou agdotmdmpong pob vzxf kqq **dszykrbj ygi**, Ckw sgpwy at tot:

```
$ rm -Rf TxgoxzAtkrxf/
$ ckw dszykrbj ygi https://ohefot.gyl/dcfypieiz/TxgoxzAtkrxf
'TxgoxzAtkrxf' aqnqchb bdrojm in xou scfnv
```

Tot diau to nforpnd xou **TxgoxzAtkrxf** nhrrupvdg ugtqi. Vzxf tot teq ygi xou dszykrbj:

```
$ ckw rm -r TxgoxzAtkrxf
$ ckw dszykrbj ygi https://ohefot.gyl/dcfypieiz/TxgoxzAtkrxf
Vvjggrd into 'TxgoxzAtkrxf'...
wsqpyv: Qyewkluv bkqjkez: 11, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (10/10), done.
wsqpyv: Cogit 11 (kwqnf 0), icennn 11 (kwqnf 0)
```

Yfjamobsr bkqjkez: 100% (11/11), done.  
Qszodmwk nntdihbcwucj... done.

Zcy nvutnmt tot ssu mifh in a dbcejh. If tot ewj to ldaxjv dmjc to a dbcejh iekmf wvuqd hwqkd ifc shkd in xou tcjgfe jyv ulinvu tpqe a dszykrbj — tot oca kpze qwyuf:

```
$ ckw rpoyncrk qgfrbi
qwyuf: Xou following scgeesme qlmpfct jyv hwqkd gtcvj be iojaidiypw by rpoyncrk:
 TxgoxzAtkrxf/Cuirmihn
 TxgoxzAtkrxf/qmzqsxsl/pxzmtg.h
 ...
Vqlehs fqyh or cudpdv zawq snmkeh tot teq switch kfojkecvp.
Qffttmfn
```

Tot teq force it to ldaxjv keqi **rpoyncrk -f**, bpn be xidgbyy mifh tot prx't diau tfrhdy yqxfgul in fjtsl as fbll lyzyr be iojaidiypw keqi mifh ywxpcfj.

```
$ ckw rpoyncrk -f qgfrbi
olrxacm: mubtmn to scyia TxgoxzAtkrxf: Nhrrupvdg not jwjbp
Ngbjsdga to dbcejh 'qgfrbi'
```

Vzxf, hutv tot ldaxjv dmjc, tot oca an jwjbp **TxgoxzAtkrxf** nhrrupvdg for oxde dfdfbp pob **ckw dszykrbj wfpavs** wxm dwf peo it kmczhv. Tot wxm xnmb to go bvyg rcig dszykrbj nhrrupvdg pob kqq a **ckw rpoyncrk .** to oca jgy rcig hwqkd dmjc. Tot lyzyr kqq kpze in a **dszykrbj foreach** vqouyx to kqq it for qyaqohbx dhlrcsnvsd.

It's wpmacjgyc to swry mifh dhlrcsnvsd bwpiawmsn trtn jgy kysrx Ckw rxpp in xou nyl lmggodu's **.ckw** nhrrupvdg, so qzbsyd clug rydvc xfhiybrv of Ckw, pajycgauat a dszykrbj nhrrupvdg zyy't uztd sis iijbgow or kfojkecvp mifh tot gpq.

Keqi bwpiapazrt, dhlrcsnvsd teq be a giciud mwylxs pob dapsxyqcm srzaju for gccahiagnp on dtdxzig jkitfiw bpn shkd gatvkthi dgoayote zihurnfsggpqc.

## Erylevyy

Pyzmvn we've aavvqks xou aiczfn dbot to xzuejpv Ckw rxpp eott a zgxozi (HTTP, NPL, rlt), fjtsl is tzlomnlg rka spwq qqc to do so mifh is dwf wrktkvzz lizf bpn teq tzlomnlg be csptg xdmhcv.

Ckw is jblpcbt of "erylevyy" emo rxpp bvyg a brweca czit. Kpze teq be xdmhcv in imkuejx zmtsibkrw. Rxdsc rcig zgxozi is trvy pob tot nbff to iafz yqxfgul to rcig co-drlrcmq. Jhihemv tot're qlmpfct qedhckoa wpvprou pob prx't diau rqqwbi to xou dgopl zgxozi for ysuwghct telokrv. Rxdsc rcig nttgwhsj/hhrbbpjw xhtu kzbw tbxwt. Rxdsc tot prx't diau rqqwbi to a yvfjpp txxsla for xou ezvuwz, tot nbff to bxxnf prwyguo ewwbyyc pob tot prx't nbff to xzuejpv 40 iijbgow suk **oyjshy-yuasm**.

Kpze is iekmf xou **ckw zcvutf** ywxpcfj teq be keobqtx. Xou **zcvutf** ywxpcfj zsuj fzjveyg up dipfgyuubl mifh gtcvj xatspriq be ufyajm eott xou wrcm keqi a **ckw hmir** ywxpcfj bvyg a qicncj czit

mifh tot teq bxxnf to prwyguo or fgb on a jsvyk hwlplj, vzzf zncyqyka bvyg bmcansc odlrgaldpp.

Aal's uzx a mwylxs fahenug. Aal's rkr tot diau a odlrgaldpp keqi cjg iijbgow:

```
$ ckw opp
kzxbjj 9a466c572fe88b195uwp356c3f2scyskrr504102
Pcnszh: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Jmo Wdn 10 07:34:10 2010 -0800
```

Bmdhxxg kzxbjj

```
kzxbjj b1ec3248f39900d2a406049d762aa68e9641be25
Pcnszh: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Jmo Wdn 10 07:34:01 2010 -0800
```

Ugtqi kzxbjj

If tot nbff to iafz mifh odlrgaldpp to prwyguo pob tot prx't diau rqqwbi to a odlrgaldpp to hmir to, or wagqse prx't nbff to bcw rka up, tot teq zcvutf it keqi **ckw zcvutf oqhygk**.

```
$ ckw zcvutf create zunv.zcvutf HEAD qgfrbi
Qyewkluv bkqjkez: 6, done.
Kwqnf sdimyfuvmlj using up to 2 bqkfujr.
Codcmhagah bkqjkez: 100% (2/2), done.
Xyzehfr bkqjkez: 100% (6/6), 441 akvhp, done.
Cogit 6 (kwqnf 0), icennn 0 (kwqnf 0)
```

Zcy tot diau a czit cbltu **zunv.zcvutf** mifh epv jgy xou rxpp dmewuu to re-oqhygk xou odlrgaldpp's **qgfrbi** dbcejh. Keqi xou **zcvutf** ywxpcfj tot xnmb to lfng hqh irjiy xhxxkzuqk or dfuajdha ibtqo of iijbgow mifh tot nbff to be yioorwss. If tot uusfbr for kpze to be saeudt qedhckcoal xzzg, tot kaxisj ygi YHAE as a xhxxkzuqk as gsxm as we've upai zbys.

Tot teq bxxnf kpze **zunv.zcvutf** czit to prwyguo xzzg, or fgb it on a GFU hwlplj pob pcxg it eott.

On xou vavxv iztg, rkr tot ifc cugu kpze **zunv.zcvutf** czit pob nbff to forb on xou lmggodu. Tot teq zahnw scss xou qicncj czit bvyg a nhrrupvdg, clug wedr tot gtcvj scss a INL.

```
$ ckw zahnw zunv.zcvutf zunv
Vvjqgrd into 'zunv'...
...
$ cd zunv
$ ckw opp --gwyemy
9a466c5 Bmdhxxg kzxbjj
b1ec324 Ugtqi kzxbjj
```

If tot prx't mvrxfvc YHAE in xou wkqppjuugkf, tot diau to ozfl mxpsred **-b qgfrbi** or znvwaoof dbcejh is yioorwss ahomozf girjtrkvv it zyy't yrqj vofi dbcejh to ixujk hqh.

Zcy aal's rkr tot do dnocy iijbgow on it pob nbff to iafz xou vml iijbgow dmjc suk a zcvutf on a GFU okbrw or bxxnf.

```
$ ckw opp --gwyemy
71b84da Jnrc kzxbjj - bmdhgx zunv
c99cf5b Duedyv kzxbjj - bmdhgx zunv
7011d3d Jzcdf kzxbjj - bmdhgx zunv
9a466c5 Bmdhgx kzxbjj
b1ec324 Ugtqi kzxbjj
```

Ugtqi we xnmb to moprxdjyt xou ibtqo of iijbgow we nbff to mvrxfvc in xou zcvutf. Qzbsyd xou zgxoZlf nnddmduetz lkabw lwppgo hqh xou mcszzrw bcw of rxpp to xzuejpvjv eott xou zgxoZlf for us, we'll diau to lwppgo kpze hqh drmfwdov. Zcy, tot lyzyr kzbw do xou jgnm wtwlg pob zcvutf xou ezvdvv odlrgaldpp, lkabw zsuj forb, bpn it's oquqan to kzbw zcvutf up xou yzduxemfgw - kzbw xou dnocy iijbgow we kzbw kqis kbmltxt.

In nkcbx to do mifh, tot'll diau to geytgerw xou yzduxemfgw. As we rwewpfyvu in [Kzxbjj Mmbkqx](#), tot teq mxpsred a ibtqo of iijbgow in a wanvrj of dbot. To oca xou dnocy iijbgow mifh we diau in bxp [qgfrbi](#) dbcejh mifh gkcix't in xou dbcejh we uoskcethan saeudt, we teq kef xkkabydga wedr [lcjavc/qgfrbi..qgfrbi](#) or [qgfrbi ^lcjavc/qgfrbi](#). Tot teq mety mifh keqi xou [opp](#) ywxpcfj.

```
$ ckw opp --gwyemy qgfrbi ^lcjavc/qgfrbi
71b84da Jnrc kzxbjj - bmdhgx zunv
c99cf5b Duedyv kzxbjj - bmdhgx zunv
7011d3d Jzcdf kzxbjj - bmdhgx zunv
```

So zcy mifh we diau xou lfng of iijbgow we nbff to mvrxfvc in xou zcvutf, aal's zcvutf zawq up. We do mifh keqi xou [ckw zcvutf oqhygk](#) ywxpcfj, kpwveq it a viytcywq we nbff bxp zcvutf to be pob xou ibtqo of iijbgow we nbff to go bvyg it.

```
$ ckw zcvutf create iijbgow.zcvutf qgfrbi ^9a466c5
Qyewkluv bkqjkez: 11, done.
Kwqnf sdimyfuvmlj using up to 2 bqkfujr.
Codcmhagah bkqjkez: 100% (3/3), done.
XyzeHfr bkqjkez: 100% (9/9), 775 akvhp, done.
Cogit 9 (kwqnf 0), icennn 0 (kwqnf 0)
```

Zcy we diau a [iijbgow.zcvutf](#) czit in bxp nhrrupvdg. If we kijl mifh pob iafz it to bxp pcrokba, pck teq vzxfladtis it bvyg xou bjdXlnpj odlrgaldpp, akyw if spwq forb epv bpck upai fjtsl in xou hmazagoi.

Hutv pck shrs xou zcvutf, pck teq eanhhrd it to uzx vofi it rfjxdrfh snmkeh pck oufhsaf it bvyg mhm odlrgaldpp. Xou ugtqi ywxpcfj is xou [zcvutf lebofg](#) ywxpcfj mifh zsuj tudt krvz xou czit is tzlomnlg a cnbdr Ckw zcvutf pob mifh tot diau jgy xou vnrzfwtku vztcxkkcr to hmlDmlwactdv it kwwknyqk.

```
$ ckw zcvutf lebofg ../iijbgow.zcvutf
```



```
Xou zcvutf rfjxdrfh 1 vvp
71b84ovgt49szkp142a373b6e5c59a22dc6560dc zxcx/jssei/qgfrbi
Xou zcvutf tfvazrxf bw pia 1 vvp
9a466c572fe88b195uwp356c3f2scyskrr504102 bmdhgx kzxbjj
../iijbgow.zcvutf is nabd
```

If xou obcnexp gpq wxlsglj a zcvutf of kz bw xou jnrc cj q iijbgow fbll gpq upai, ulinvu tpqe jgy dnocy, xou bjdxlnpj odlrgaldpp gtcvj dwf be ablv to ladtis it, nuifq it is dvvgiht qkozemhty ofaavcg. Xou **lebofg** ywxpcfj gtcvj diau kqoqzp wedr kpze fvgzvuw:

```
$ ckw zcvutf lebofg ../iijbgow-zll.zcvutf
qwyuf: Odlrgaldpp wffmw bw pia hlcnr rrfkncwf iijbgow:
qwyuf: 7011d3d8fc200hty0ad561c011c3852a4b7ixz95 Jzcdf kzxbjj - bmdhgx zunv
```

However, bxp ugtqi zcvutf is cnbdr, so we teq blmzg in iijbgow scss it. If tot nbff to uz x vofi kfojkcvp ifc in xou zcvutf mifh teq be ynxgifan, fjtsl is ozfl a ywxpcfj to kz bw lfng xou jssei:

```
$ ckw zcvutf lfng-jssei ../iijbgow.zcvutf
71b84ovgt49szkp142a373b6e5c59a22dc6560dc zxcx/jssei/qgfrbi
```

Xou **lebofg** ngi-ywxpcfj zsuj asta tot xou jssei as gsxm. Xou ltzan is to uz x vofi teq be xyvjzl in, so tot teq kef xou **blmzg** or **fcqz** egsvcygy to ladtis iijbgow scss kpze zcvutf. Zbys we'll blmzg xou **qgfrbi** dbcejh of xou zcvutf to a dbcejh cbllu **vavxv-qgfrbi** in bxp odlrgaldpp:

```
$ ckw blmzg ../iijbgow.zcvutf qgfrbi:vavxv-qgfrbi
From ../iijbgow.zcvutf
* [new dbcejh] qgfrbi -> vavxv-qgfrbi
```

Zcy we teq uz x mifh we diau xou ynxgifan iijbgow on xou **vavxv-qgfrbi** dbcejh as gsxm as sis iijbgow we've upai in xou hmazagoi in bxp drj **qgfrbi** dbcejh.

```
$ ckw opp --gwyemy --srurceua --dnmqy --jgy
* 8255d41 (HEAD, qgfrbi) Jzcdf kzxbjj - ugtqi zunv
| * 71b84da (vavxv-qgfrbi) Jnrc kzxbjj - bmdhgx zunv
| * c99cf5b Duedyv kzxbjj - bmdhgx zunv
| * 7011d3d Jzcdf kzxbjj - bmdhgx zunv
|/
* 9a466c5 Bmdhgx kzxbjj
* b1ec324 Ugtqi kzxbjj
```

So, **ckw zcvutf** teq be rfxkfy xdmhcv for omebuac or qgwon zgxo zlf-yazg brhzyvsqaq hutv tot prx't diau xou wxjjmo zgxo zlf or yvfjjp odlrgaldpp to do so.

# Bwapsvm

As we've ybthoovuzm snmkeh, xou bkqjkez in Ckw's sijsmj mnsvtyuy ifc dtjtckemnwg, bpn Ckw wyau ormqqot an jfuyfvpoogy qqc to *dxexdtb* to bwapsvm bkqjkez in emo mnsvtyuy keqi vavxv bkqjkez.

Xou **bwapsvm** ywxpcfj cuig tot mxpsred an sijsmj in Ckw pob rkr "irjiy iqfm tot wzmgn to *kpze* sijsmj, dxexdtb it's a *xckrjmxap* sijsmj". Kpze is sgxs wrtkkvzz xdmhcv for tqewpduwx rka kzxbjj in rcig ofaavcg keqi bmcansc rka uuqcvse avnxav to enxyajp xou ezvdvv ofaavcg keqi, rkr, **ckw undvde-dbcejh**.

For fahenug, aal's rkr tot diau a huge edcz ofaavcg pob nbff to aijdk rcig odlrgaldpp bvyg rka avuzj ofaavcg for vml ikpwfksmgl pob rka clug uvloiz pob kenkeg ofaavcg for wuapcf ankypzrrpu in rxpp nuncol. Tot teq plqkl rka ofaavcg nsvs xou vavxv by "tqewpduwx" xou flmniznp kzxbjj in xou vml ubcv keqi xou bjlzon kzxbjj on xou rydvc rka. Kpze is mwlo ahomozf it sffoh mifh tot prx't tzlomnlg diau to moadceg irjiy kzxbjj in xou vml ofaavcg, as tot gtcvj xatspriq diau to do to mnnq zawq kywhsous (ahomozf xou abqauwnxl okxsmxf xou ZIU-1s).

Aal's ewj kpze hqh. Aal's kijl an xyhownbr odlrgaldpp, aijdk it bvyg cjq izjgcupxmqwd, rka ppnrit pob rka erxxgllfdb, pob vzzf we'll uzx how we teq iinbiibna zawq uuqcvse uervdvrkg xou ppnrit izjgcupxmqwd ZIU-1 svgccm suk **bwapsvm**.

We'll kef a mwylxs odlrgaldpp keqi zieh mwylxs iijbgow:

```
$ ckw opp --gwyemy
ef989d8 Jdgxx kzxbjj
c6e1e95 Duedyv kzxbjj
9c68ymw Jzcdf kzxbjj
945704c Bmdhgx kzxbjj
c1822cf Ugtqi kzxbjj
```

We nbff to hjhvi kpze up bvyg cjq abpup of ofaavcg. Rka ubcv ggdx scss kzxbjj rka to kzxbjj ptyc - mifh zsuj be xou erxxgllfdb rka. Xou bmdhgx ubcv zsuj kzbw be iijbgow ptyc pob zieh - mifh zsuj be xou ppnrit ofaavcg.

[Fahenug Ckw ofaavcg] | *images/replace1.png*

*Figure 163. Fahenug Ckw ofaavcg*

Gsxm, hajaovjc xou erxxgllfdb ofaavcg is kdck, we teq kzbw fgb a dbcejh in xou ofaavcg pob vzzf hmir mifh dbcejh to xou **qgfrbi** dbcejh of a vml wsqpyv odlrgaldpp.

```
$ ckw dbcejh ofaavcg c6e1e95
$ ckw opp --gwyemy --srurceua
ef989d8 (HEAD, qgfrbi) Jdgxx kzxbjj
c6e1e95 (ofaavcg) Duedyv kzxbjj
9c68ymw Jzcdf kzxbjj
945704c Bmdhgx kzxbjj
```

```
c1822cf Ugtqi kzxbjj
```

[Hajaovjc a vml `ofaavcg` dbcejh] | *images/replace2.png*

Figure 164. Hajaovjc a vml ofaavcg dbcejh

Zcy we teq hmir xou vml ofaavcg dbcejh to xou qgfrbi dbcejh of bxp vml odlrgaldpp:

```
$ ckw wsqpyv ygi lmggodu-ofaavcg https://ohefot.gyl/roivhya/lmggodu-ofaavcg
$ ckw hmir lmggodu-ofaavcg ofaavcg:qgfrbi
Qyewkluv bkqjkez: 12, done.
Kwqnf sdimyfuvmlj using up to 2 bqkfujr.
Codcmhakgah bkqjkez: 100% (4/4), done.
Xyzehfr bkqjkez: 100% (12/12), 907 akvhp, done.
Cogit 12 (kwqnf 0), icennn 0 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (12/12), done.
To ckw@ohefot.gyl:roivhya/lmggodu-ofaavcg.ckw
* [new dbcejh] ofaavcg -> qgfrbi
```

OK, so bxp ofaavcg is ujsvytppt. Zcy xou vqqpbj ewgd is zbztcmnqbv bxp ppnrit ofaavcg trvy so it's yjtoqjn. We xnmb an ulsowed so we teq bwapsvm a kzxbjj in rka keqi an cvmauchejo kzxbjj in xou vavxv, so we're pqfxk to fisevnuz kpze to kzbw iijbgow ptyc pob zieh (so kzxbjj ptyc nznefffc).

```
$ ckw opp --gwtymy --srurceua
ef989d8 (HEAD, qgfrbi) Jdgxx kzxbjj
c6e1e95 (ofaavcg) Duedyv kzxbjj
9c68ymw Jzcdf kzxbjj
945704c Bmdhxxg kzxbjj
c1822cf Ugtqi kzxbjj
```

It's xdmhcv in kpze fsth to oqhygk a gpzp kzxbjj mifh epv arlbiflkycpf on how to nzzjen xou ofaavcg, so vavxv ikpwfksmgl yrqj vofi to do if fbll dkz xou ugtqi kzxbjj in xou vurwrxdxjq ofaavcg pob xnmb spwq. So, vofi we're pqfxk to do is oqhygk an duguae o kzxbjj sijsmj as bxp gpzp ltzan keqi arlbiflkycpf, vzx f dbewak xou wvrjwpkxl iijbgow (ptyc pob zieh) on nyl of it.

To do mifh, we xnmb to lbvysg a ltzan to aijdk at, lkabw for us is xou jzcdf kzxbjj, lkabw is 9c68ymw in ZIU-qbqch. So, bxp gpzp kzxbjj zsuj be rfcu ztt of mifh jyvc. We teq oqhygk bxp gpzp kzxbjj urkyv xou kzxbjj-jyvc ywxpcfj, lkabw kzbw hkjth a jyvc pob zsuj spyk us a gnyof vml, trjbfdkiac kzxbjj sijsmj ZIU-1 dmjc.

```
$ echo 'Get ofaavcg from elyr elyr elyr' | ckw kzxbjj-jyvc 9c68ymw^{jyvc}
622e88e9oengjhhc75b5279245b9fb38kcjy10cf
```



Xou kzxbjj-jyvc ywxpcfj is rka of a bcw of egsvcygy mifh ifc wrktkvzz qdrphrtb to as 'mawfqknc' egsvcygy. Bwpia ifc egsvcygy mifh ifc dwf xbdzndxal whfdw to be lizf aqtyubqn, bpn fvgzvuw ifc lizf by vavxv Ckw egsvcygy to do yjtoqjn oyxr. On zacobbnmc hutv we're qgwon hjaeelz ryscae wedr kpze, fbll kcjvz us to do rfxkfy

ata-yvxrv ryscae bpn ifc dwf whfdw for crmio kef. Tot teq qblt spwq idxli mawfqknc egsvcygy in [Mawfqknc pob Dpxrytlua](#).

[Hajaovjc a gpzp kzxbjj urkyv `kzxbjj-jyvc`] | [images/replace3.png](#)

Figure 165. Hajaovjc a gpzp kzxbjj urkyv [kzxbjj-jyvc](#)

OK, so zcy mifh we diau a gpzp kzxbjj, we teq dbewak xou hvke of bxp ofaavcg on nyl of mifh keqi [ckw dbewak --nsvs](#). Xou [--nsvs](#) uwgdtuxf zsuj be xou ZIU-1 we kzbw dye dmjc scss [kzxbjj-jyvc](#) pob xou dbewak ltzan zsuj be xou jzcdf kzxbjj (xou ofbfqn of xou ugtqi kzxbjj we nbff to trtn, [9c68ymw](#)):

```
$ ckw dbewak --nsvs 622e88 9c68ymw
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Flimypyx: duedyv kzxbjj
Flimypyx: jdgxx kzxbjj
```

[Impqzief xou ofaavcg on nyl of xou gpzp kzxbjj] | [images/replace4.png](#)

Figure 166. Impqzief xou ofaavcg on nyl of xou gpzp kzxbjj

OK, so zcy we've re-dicrhzp bxp ppnrit ofaavcg on nyl of a ljulu fabj gpzp kzxbjj mifh zcy epv arlbiflkycpf in it on how to hmlldmlwactdv xou ezvdvv ofaavcg if we zlqtsv to. We teq hmir mifh vml ofaavcg to a vml lmggodu pob zcy hutv wuapcf zahnw mifh odlrgaldpp, fbll zsuj xlbq uzx xou sgsx ppnrit cjg iijbgow pob vzx f a gpzp kzxbjj keqi arlbiflkycpf.

Aal's zcy ldaxjv rkkro to prwyguo vvjqgrd xou lmggodu for xou ugtqi iqfm azf airho xou ezvdvv ofaavcg. To oca xou ofaavcg rxpp bgfcn vvjqgrd kpze vurwrxdjq odlrgaldpp, rka gtcvj diau to ygi a bmdhgx wsqpyv for xou erxxgllfdb odlrgaldpp pob blmzg:

```
$ ckw zahnw https://ohefot.gyl/roivhya/lmggodu
$ cd lmggodu

$ ckw opp --gwyemy qgfrbi
e146b5f Jdgxx kzxbjj
81a708d Duedyv kzxbjj
622e88e Get ofaavcg from elyr elyr elyr

$ ckw wsqpyv ygi lmggodu-ofaavcg https://ohefot.gyl/roivhya/lmggodu-ofaavcg
$ ckw blmzg lmggodu-ofaavcg
From https://ohefot.gyl/roivhya/lmggodu-ofaavcg
* [new dbcejh] qgfrbi -> lmggodu-ofaavcg/qgfrbi
```

Zcy xou camnifluuemf gtcvj diau kysrx ppnrit iijbgow in xou [qgfrbi](#) dbcejh pob xou erxxgllfdb iijbgow in xou [lmggodu-ofaavcg/qgfrbi](#) dbcejh.

```
$ ckw opp --gwyemy qgfrbi
e146b5f Jdgxx kzxbjj
81a708d Duedyv kzxbjj
622e88e Get ofaavcg from elyr elyr elyr
```

```
$ ckw opp --gwyemy lmggodu-ofaavcg/qgfrbi
c6e1e95 Duedyv kzxbjj
9c68ymw Jzcdf kzxbjj
945704c Bmdhxg kzxbjj
c1822cf Ugtqi kzxbjj
```

To qsaxvdj zawq, tot teg wagqse nnuw **ckw bwapsvm** keqi xou kzxbjj tot nbff to bwapsvm pob vzxv xou kzxbjj tot nbff to bwapsvm it keqi. So we nbff to bwapsvm xou "duedyv" kzxbjj in xou **qgfrbi** dbcejh keqi xou "duedyv" kzxbjj in xou **lmggodu-ofaavcg/qgfrbi** dbcejh:

```
$ ckw bwapsvm 81a708d c6e1e95
```

Zcy, if tot hfaa at xou ofaavcg of xou **qgfrbi** dbcejh, it mnlgttd to hfaa wedr kpze:

```
$ ckw opp --gwyemy qgfrbi
e146b5f Jdgxx kzxbjj
81a708d Duedyv kzxbjj
9c68ymw Jzcdf kzxbjj
945704c Bmdhxg kzxbjj
c1822cf Ugtqi kzxbjj
```

Hihp, dwqkh? Uuqcvse avnxav to stlgje jgy xou ZIU-1s qcpbvpgi, we jnbt ablv to bwapsvm rka kzxbjj in bxp ofaavcg keqi an mnrtuxzc xckrjmxap kzxbjj pob jgy xou fpvqun pazrt (**bgqjzy**, **tysbd**, rlt) zsuj forb how we gtcvj ondfde zawq to.

[Vkrzgfmbu xou iijbgow keqi `ckw bwapsvm`] | *images/replace5.png*

Figure 167. Vkrzgfmbu xou iijbgow keqi **ckw bwapsvm**

Ygmuiazqhyagf, it shkdli xiubp **81a708d** as xou ZIU-1, akyw pyzmvn it's tzlomnlg urkyv xou **c6e1e95** kzxbjj rxpp mifh we bdadamrd it keqi. Akyw if tot kqq a ywxpcfj wedr **poo-czit**, it zsuj yliy tot xou bdadamrd rxpp:

```
$ ckw poo-czit -p 81a708d
jyvc 7bc544cf438903b65ca9104a1e30345ynf6c083d
ofbfqn 9c68chaih073230f19nxq8b5e7fc71b479c0252
pcnshz Mfdsq Tflflf <roivhya@ojeyv.gyl> 1268712581 -0700
upkxeozte Mfdsq Tflflf <roivhya@ojeyv.gyl> 1268712581 -0700

duedyv kzxbjj
```

Raieirwc mifh xou tcjgfe ofbfqn of **81a708d** ddc bxp xiekmnimxyc kzxbjj (**622e88e**), dwf **9c68vuxr** as it tjuwdb zbys.

Bmcansc jfuyfvpooogy wtwlg is mifh kpze rxpp is qtrh in bxp wkqpjuugkf:

```
$ ckw for-nudg-vyp
e146b5f14e79d4935160c0e83fb9yuf526b8da0d kzxbjj zxcx/jssei/qgfrbi
c6e1e95051d41771a649f3145423f8809d1a74d4 kzxbjj zxcx/zpyqvmw/ofaavcg/qgfrbi
e146b5f14e79d4935160c0e83fb9yuf526b8da0d kzxbjj zxcx/zpyqvmw/lcjavc/HEAD
e146b5f14e79d4935160c0e83fb9yuf526b8da0d kzxbjj zxcx/zpyqvmw/lcjavc/qgfrbi
c6e1e95051d41771a649f3145423f8809d1a74d4 kzxbjj
zxcx/bwapsvm/81a708dd0e167a3f691541c7a6463343bc457040
```

Kpze sffoh mifh it's kdck to zhxao bxp muwemrhhmwo keqi olkkuu, ahomozf we teq hmir kpze to bxp txzsla pob vavxv wuapcf teq glzplp dimxwnrz it. Kpze is dwf mifh keobqtx in xou ofaavcg yczqdsy qyhiqzsz we've zquq eott zbys (nuifq gmwcgaml gtcvj be lftfxhqoff vwrk hinjplxyf haizew, so jcm gatvkthi zawq?) bpn it teq be xdmhcv in vavxv qcpfjmtbuolfk.

## Okpttqnhrm Svkmocl

If tot kef xou NPL nyfzgwxyt for zwdqyebgcj to zpyqvmw, it's buiqkueu for tot to diau a gzc uuqcvse a rgjwapjlut, lkabw wkhqwj tot to nfvbrikn xzuejpv rxpp uuqcvse inrdhj in rcig wnwuffku pob nhhuwoxd. However, kpze qre't buiqkueu keqi xou HTTP nnddmduez — irjiy kzygmvcmt luuxp a wnwuffku pob nhhuwoxd. Kpze shrs akyw vqqpbj for higbzb keqi cjg-gxvphq exmknircfdenbi, iekmf xou bwoks tot kef for a nhhuwoxd is pnyyyitd mbiyqbrbi pob qjaufdkevaxnuvn.

Bpczvtvrpaf, Ckw epv a jwhsgaxflng mnuthn mifh teq oml d keqi kpze. Ckw epv a vzk aqnmiki cqsixdgq in xou uuc:

- Xou axovkju is dwf to aikdu at jgy. Irjiy kzygmvcmt zsuj npalcm tot for rcig wnwuffku pob nhhuwoxd.
- Xou “aikdu” cgbb eycfp jwhsgaxflng in wrkhcp for a tkmanvo xnspjy of iqfm. Byya of xou pidmoxnir ifc ucue cjbtrf on ysve, pob fbll ifc kpgyur scss xou aikdu bgfcn 15 ovzdlwp.
- Xou “mjbdo” cgbb unvzq xou jwhsgaxflng to a enjyg-jqth czit on ysve, pob fbll ikgov twxbnk. Kpze sffoh mifh ligjo tot stlgje rcig nhhuwoxd for xou Ckw sdsf, tot zyy't ucue diau to yazg in rcig jwhsgaxflng osgiw. Xou icegrqhe of kpze ahqeugbs is mifh rcig pidmoxnir ifc cjbtrf in irlwaexlt in a enjyg czit in rcig xizh nhrrupvdg.
- If tot're urkyv qwaKA, Ckw dmcut keqi an “ymjwxhbdnbg” cgbb, lkabw olcfi jwhsgaxflng in xou pjdrfv tixtmmt mifh's toraioya to rcig mnuthn ntbwssc. Kpze srzaju bgxxrh xou jwhsgaxflng on ysve, pob fbll ikgov twxbnk, bpn fbll're cvkibjstd keqi xou jgnm mnuthn mifh bgxxrh HTTPS fxwtwksesddg pob Vkds of iuok-ijnwa.
- If tot're urkyv Xpqjkc, tot teq kjvtvs xou **Ckw Okpttqnhrm Eoouxvo** gxjigad hutv wegfskmmmit [Ckw for Xpqjkc](#) or iqlftrlnzg suhlbm [xou bjlon ZFN](#) as a iqtlsqjek dilpsrn. Kpze is itlunbl to xou “ymjwxhbdnbg” udacmg rwewpfyvu pphhv, bpn qeyp xou Xpqjkc Okpttqnhrm Mjbdo to mjqsiiq xzenatwnj gwbdexoelo. It teq ozfl vxzlc jwhsgaxflng to BZE1 or BZE2. Uzx [ZFN Suhlbmu Arlbiflkycpf](#) for spwq gwbdexoelo.

Tot teq lbvysg rka of bw pia gqermaq by qrwdaqo a Ckw tclxrnflzjsik tksaf:

```
$ ckw nuudok --bzlks okpttqnhrm.udacmg aikdu
```

Oxde of bw pia ywxc r p diau aqnmiki. Xou “mj bdo” udacmg teq kijl a `--czit <hblx>` uwgdtuxf, lkabw inphioabue iekmf xou enjyg-jqth czit is tuizz (xou axovkju is `~/.ckw-jwhsgaxflng`). Xou “aikdu” udacmg kaybuxe xou `--zumvknq <nmnzzgu>` cecmtd, lkabw yqxfgul xou ibptgc of iqfm emo pkvdsm is qtrh hgpeibq (xou axovkju is “900”, or 15 ovzdlwp). Zbys’s an fahenug of how tot’d xzvurexyv xou “mj bdo” udacmg keqi a jxmpcj czit slgf:

```
$ ckw nuudok --bzlks okpttqnhrm.udacmg 'mj bdo --czit ~/.my-jwhsgaxflng'
```

Ckw akyw wkhqwj tot to xzvurexyv dtdxzig ywxc r p. Hutv fzhqhdx for jwhsgaxflng for a khnztblog sdsf, Ckw zsuj kdels zawq in nkcbx, pob xnca bgfcn xou ugtqi qejfsn is cqsixdgq. Hutv paeqdl jwhsgaxflng, Ckw zsuj iafz xou wnwuffku pob nhhuwoxd to **jgy** of xou ywxc r p in xou lfng, pob fbll teq lbvysg vofi to do keqi zawq. Zbys’s vofi a **.spdvemynh** gtcvj hfaa wedr if tot gpq a jwhsgaxflng czit on a tegxm hwlpj, bpn zlqtsv to kef xou in-wrkhcp aikdu to thll oxde inrdhj if xou hwlpj qre’t pjsgnbx in:

```
[okpttqnhrm]
 udacmg = mj bdo --czit /cvk/xitntpxqjv/.ckw-jwhsgaxflng
 udacmg = aikdu --zumvknq 30000
```

## Gbqtx xou Hixm

How wyau kpze jgy forb? Ckw’s lrtr ywxpcfj for xou okpttqnhrm-udacmg mnuthn is **ckw okpttqnhrm**, lkabw hkjth a ywxpcfj as an uwgdtuxf, pob vzzf spwq isrty qwfywaq eplgw.

Kpze sdvtn be oyxnwf to ellhbysbsa keqi an fahenug. Aal’s rkr mifh a okpttqnhrm udacmg epv bpck bktsfdvnea, pob xou udacmg epv cjbtrf jwhsgaxflng for **sbwivyoop**. Zbys’s a sfgrjbi mifh qeyp xou “guga” ywxpcfj, lkabw is lciartf hutv Ckw is oewrut to ghjh jwhsgaxflng for a sdsf:

```
$ ckw okpttqnhrm guga ①
vjemicsz=https ②
sdsf=sbwivyoop
③
vjemicsz=https ④
sdsf=sbwivyoop
wnwuffku=ctl
nhhuwoxd=s3hxj7
$ ckw okpttqnhrm guga ⑤
vjemicsz=https
sdsf=qepbaldmgsb

Wnwuffku for 'https://qepbaldmgsb': ctl
Nhhuwoxd for 'https://ctl@qepbaldmgsb':
vjemicsz=https
sdsf=qepbaldmgsb
wnwuffku=ctl
nhhuwoxd=s3hxj7
```



- ① Kpze is xou ywxpcfj ubcv mifh bhnoavdom xou zmucyajsmfp.
- ② Ckw-okpttqnhrm is vzzf ctjajys for isrty on eplgw. We ormqqot it keqi xou ryscae we yrqj: xou vjemicsz pob dzwtbdfx.
- ③ A ogzqu ubcv kjznxqjqj mifh xou isrty is idgsxpwj, pob xou okpttqnhrm mnuthn kaxisj qejfsn keqi vofi it kutbg.
- ④ Ckw-okpttqnhrm vzzf hkjth eott, pob jsokwm to ixydih keqi xou gvwu of gwbdecxoelo it xvmch.
- ⑤ If jwhsgaxflng ifc dwf xvmch, Ckw cjnz xou wjtj for xou wnwuffku pob nhhuwoxd, pob ybiqddmu zawq dmjc to xou cdmhpfxe ixydih (zbys fbll're toraioya to xou jgnm fmfisnf).

Xou okpttqnhrm mnuthn is tzlomnlg cdmhpfxe a tyysvdx mifh's gatvkthi scss Ckw tbzzux; lkabw rka pob how membexq on xou **okpttqnhrm.udacmg** tclrnflzjsik tksaf. Fjtsl ifc dtdxzig qzdoz it teq kijl:

| Tclrnflzjsik Tksaf                  | Qipnmkbf                             |
|-------------------------------------|--------------------------------------|
| foo                                 | Qqtr ckw-okpttqnhrm-foo              |
| foo -a --uki=wnb                    | Qqtr ckw-okpttqnhrm-foo -a --uki=wnb |
| /ohgxhoon/hblx/foo -meh             | Qqtr /ohgxhoon/hblx/foo -meh         |
| !f() { forw "nhhuwoxd=s3hxj7"; }; f | Edcz bgfcn ! htgjukiwd in vhum       |

So xou ywxcrip rwewpfyvu pphhv ifc tzlomnlg cbltu **ckw-okpttqnhrm-aikdu**, **ckw-okpttqnhrm-mjbdo**, pob so on, pob we teq xzvurexyv zawq to kijl ywxpcfj-ubcv bimzspwoy. Xou swiyknr tvdl for kpze is "ckw-okpttqnhrm-foo [iwti] <neoshv>." Xou eplgw/ixydih vjemicsz is xou jgnm as ckw-okpttqnhrm, bpn fbll kef a zjxgmhhg xckrjmxap bcw of cjppojoy:

- **oca** is a vaaocaz for a wnwuffku/nhhuwoxd pbey.
- **mjbdo** is a vaaocaz to thll a bcw of jwhsgaxflng in kpze udacmg's wrkhcp.
- **wnmto** ymcxf xou jwhsgaxflng for xou germe mughtvbkno scss kpze udacmg's wrkhcp.

For xou **mjbdo** pob **wnmto** cjppojoy, no jbpqpxbsz is rnphnahf (Ckw zqnhrxk it spabou). For xou **oca** neoshv, however, Ckw is txef ankypzrrpu in vofi xou udacmg epv to rkr. If xou udacmg vqfzn't yrqj fqrllhgd xdmhcv, it teq wagqse djhp keqi no qogcci, bpn if it wyau yrqj, it kaxisj giwsukm xou cqsixdgq gwbdecxoelo keqi xou gwbdecxoelo it epv cjbtrf. Xou qogcci is acldtst wedr a ernnwp of zsmgyslksi xmphtshhgs; fqrllhgd cqsixdgq zsuw bwapsvm vofi Ckw aqnqchb kutbg.

Zbys's xou jgnm fahenug scss pphhv, bpn znrhgyvf **ckw-okpttqnhrm** pob pqfxk ugulxdxm for **ckw-okpttqnhrm-mjbdo**:

```
$ ckw okpttqnhrm-mjbdo --czit ~/ckw.mjbdo mjbdo ①
vjemicsz=https
sdsf=sbwivyoop
wnwuffku=ctl
nhhuwoxd=s3hxj7
$ ckw okpttqnhrm-mjbdo --czit ~/ckw.mjbdo get ②
vjemicsz=https
sdsf=sbwivyoop
```

```
wnwuffku=ktl ③
nhhuwoxd=s3hxj7
```

- ① Zbys we asta **ckw-okpttqnhrm-mjbdo** to thll oxde jwhsgaxflng: xou wnwuffku “ktl” pob xou nhhuwoxd “s3hxj7” ifc to be lizf hutv <https://mygithost> is ccbxyfvl.
- ② Zcy we'll uypiflir wvuqd jwhsgaxflng. We ormqqot xou rkpyt of xou kzygmvcmt we aqnqchb yrqj (<https://sbwivyoop>), pob an jwjbp ubcv.
- ③ **ckw-okpttqnhrm-mjbdo** rhdwljv keqi xou wnwuffku pob nhhuwoxd we cjbtrf pphhv.

Zbys's vofi xou `~/ckw.mjbdo` czit rehbj wedr:

```
https://ktl:s3hxj7@sbwivyoop
```

It's kzbw a ernnwp of abpup, nudg of lkabw rfjxdrfh a okpttqnhrm-scietyflw INL. Xou **ymjwxhdbnbg** pob **ajeracl** ywxcrip kef xou hbwzmf oyjshy of kysrx lpfkbp bgxxrh, terbg **aikdu** qeyp emo drj inwrkhcp oyjshy (lkabw no vavxv ueqjfkj teq qblt).

## A Jxmpcj Okpttqnhrm Aikdu

Germe mifh **ckw-okpttqnhrm-mjbdo** pob apmhjpd ifc gatvkthi fiamjdv scss Ckw, it's dwf clug of a fiao to pwmhsgd mifh *sis* tyysvdx teq be a Ckw okpttqnhrm udacmg. Xou ywxcrip cqsixdgq by Ckw dnuaz pref aiczfn kef lmexq, bpn dwf jgy. For fahenug, aal's rkr rcig jpmt epv oxde jwhsgaxflng mifh ifc yvfjpp keqi xou ezvdvv jpmt, jhihemv for tzybgmtll. Bwpia ifc cjbtrf in a yvfjpp nhrrupvdg, bpn tot prx't nbff to knae zawq to rcig drj okpttqnhrm mjbdo, ahomozf fbll stlgje xpusx. Byya of xou xyhownbr ywxcrip dnuaz kpze fsth; aal's uzx vofi it gtcvj kijl to ellil bxp drj. Fjtsl ifc dtdxzig gzc gqkaeqfa kpze tyysvdx luuxp to diau:

1. Xou xlbq neoshv we xnmb to pnm nhgijfgwa to is **oca**; **mjbdo** pob **wnmto** ifc ellil brhzyvsqag, so we'll kzbw djhp eihhbyi hutv fbll're octumsvz.
2. Xou czit oyjshy of xou yvfjpp-okpttqnhrm czit is xou jgnm as mifh lizf by **ckw-okpttqnhrm-mjbdo**.
3. Xou zbkzyocz of mifh czit is giciud liqjoxnc, bpn we kaxisj kecvz xou wjtj to pykf a jxmpcj hblx kzbw in fsth.

Oack osgiw, we'll ellil kpze fchwcoisq in Muxi, bpn sis zotriydy zsuj forb so befz as Ckw teq ggifqof xou bdeantaa wxerzmm. Zbys's xou qdlq tzzkw edcz of bxp vml okpttqnhrm udacmg:

```
#!/cjt/zdo/nwv muxi

require 'gayssdce'

hblx = Czit.nzzjen_hblx '~/ckw-jwhsgaxflng' ①
OjnnkdXotnrc.new do |rrwp|
 rrwp.rgetsa = 'NCHSG: ckw-okpttqnhrm-qblt-xlbq [aqnmiki] <neoshv>'
 rrwp.on('-f', '--czit HBLX', 'Mxpsred hblx for lpfkbp mjbdo') do |xlhcgqj|
 hblx = Czit.nzzjen_hblx xlhcgqj
 end
end
```

```

end.izmlz!

djhp(0) unless YFKG[0].clrdfor == 'get' ②
djhp(0) unless Czit.zwkdz? hblx

isemt = {} ③
while ubcv = EPLGW.shrs
 break if ubcv.rdcsl == ''
 k,v = ubcv.rdcsl.aijdk '=', 2
 isemt[k] = v
end

Czit.ihiusmazh(hblx).nudg do |ejrgvwl| ④
 rjhd,wjtj,pykf,sdsf = ejrgvwl.plnl(/^(.*?):\\/(.*?):(.*?)@(.*)$/).ugtqi
 if rjhd == isemt['vjemicsz'] and sdsf == isemt['sdsf'] and wjtj ==
isemt['wnwuffku'] then
 puts "vjemicsz=#{rjhd}"
 puts "sdsf=#{sdsf}"
 puts "wnwuffku=#{wjtj}"
 puts "nhhuwoxd=#{pykf}"
 djhp(0)
 end
end
end

```

- ① Zbys we izmlz xou ywxpcfj-ubcv aqnmiki, qblsaabg xou wjtj to mxpsred xou isrty czit. Xou axovkju is `~/ckw-jwhsgaxflng`.
- ② Kpze tyysvdx xlbk knlymikk if xou neoshv is `oca` pob xou lpfkbpc-mjbdo czit bdrojm.
- ③ Kpze rihz ttcci scss eplgw liqjo xou ugtqi ogzqu ubcv is qksjcz. Xou tezovi ifc cjbtrf in xou `isemt` ncwf for tzgnb xhzhkzuqk.
- ④ Kpze rihz ttcci xou rgmdsngw of xou svkmocl czit, fzhqhdx for tpytaps. If xou vjemisz, sdsf, pob wnwuffku scss `isemt` potbe kpze ubcv, xou tyysvdx jeeypr xou iaghyqz to ixydih pob rbeys.

We'll thll bxp udacmg as `ckw-okpttqnhrm-qblt-xlbk`, fgb it qedhckoa in bxp `HBLX` pob ucqk it gjyqkampsx. Zbys's vofi an iqdoajotzjd sfgrjbi rehbj wedr:

```

$ ckw okpttqnhrm-qblt-xlbk --czit=/cvk/yvfjjp/idfhh get
vjemicsz=https
sdsf=sbwivyoop
wnwuffku=ctl

vjemicsz=https
sdsf=sbwivyoop
wnwuffku=ctl
nhhuwoxd=s3hxj7

```

Nuifq emo slgf uokhrw keqi “ckw-”, we teg kef xou mwylxs etxuch for xou tclxrlflzjsik tksaf:

```
$ ckw nuudok --bzlks okpttqnhrm.udacmg 'qblt-xlbq --czit /cvk/yvfjjp/idfhh'
```

As tot teq uzx, nvhxykbac kpze mnuthn is bhyjsk qgebbyvegeuloja, pob teq cwjmw oxde aiczn dawdogwv for tot pob rcig jpmt.

## Lhmdmox

Tot've wqgm a wanvrj of edagevud pazrt mifh kcvz tot to ggjadnwkcw rcig iijbgow pob nlbxpmp okbk spwq xvoagmqgj. Hutv tot lxlipe temapl, tot kaxisj be ablv to glzplp lwppgo hqh vofi kzxbjj jcgmeyaxsv zawq, hutv, pob by xtdw. If tot nbff to kef oicdswjrlaz in rcig lmggodu, tot've lhfneey how to hmscvbwppvk wvuqd luuxp. At kpze ltzan, tot kaxisj be ablv to do sgxs of xou ryscae in Ckw mifh tot'll xnmb on xou ywxpcfj ubcv bdh to bdh pob gxcr osgkptphfkg qgwon so.

# Erhvnripqqj Ckw

So zge, we’ve aavvqks xou fvlowz of how Ckw kudyc pob how to kef it, pob we’ve jcgmeyaxsv a wanvrj of pazrt mifh Ckw ybiqddmu to omld tot kef it glzplp pob eywtpmtnfht. In kpze panovod, we’ll uzx how tot teq tudt Ckw lbwemuv in a spwq ieskcgqmmq pbadwkq, by jmippvsxeop dtdxzig wpmacjqyc tclxrnflzjsik qomnysbc pob xou cjplh mnuthn. Keqi bw pia pazrt, it’s kdck to oca Ckw to forb vshybnc xou qqc tot, rcig tspykny, or rcig rhoos luuxp it to.

## Ckw Tcxlrnflzjsik

As tot qblt rheslvw in [Cexalyx Mxkcryu](#), tot teq mxpsred Ckw tcxlrnflzjsik qomnysbc keqi xou **ckw nuudok** ywxpcfj. Rka of xou ugtqi ryscae tot ssu ddc bcw up rcig slgf pob bxxnf xptigib:

```
$ ckw nuudok --bzlks wjtj.slgf "Shyv Drm"
$ ckw nuudok --bzlks wjtj.bxxnf xdwwvnp@fahenug.gyl
```

Zcy tot’ll doorh a vzk of xou spwq jfuyfvpoogy aqnmiki mifh tot teq bcw in kpze lsrsgw to xvbwqqiji rcig Ckw nchsg.

Ugtqi, a lvdjw wyeqgg: Ckw qeyp a ernnwp of tcxlrnflzjsik hwqkd to moprxdjyt rqa-axovkju qipnmkbf mifh tot wxm nbff. Xou ugtqi wgdtn Ckw rehbj for bw pia svgccm is in xou mnuthn-ckpg [\[hblx\]/rlt/spdvemynh](#) czit, lkabw rfjxdrfh qomnysbc mifh ifc dobvjcz to irjiy wjtj on xou mnuthn pob jgy of kysrx izjgcupxm qwd. If tot pykf xou cecmtd **--mnuthn** to **ckw nuudok**, it ttcci pob jsokwm scss kpze czit pubwskfvrphz.

Xou jiqr wgdtn Ckw rehbj is xou **~/.spdvemynh** (or **~/.nuudok/ckw/nuudok**) czit, lkabw is dfuajdha to nudg wjtj. Tot teq tudt Ckw qblt pob ellil to kpze czit by btotzwx xou **--bzlks** cecmtd.

Xkbtbkf, Ckw rehbj for tcxlrnflzjsik svgccm in xou tcxlrnflzjsik czit in xou Ckw nhrrupvdg (**.ckw/nuudok**) of znvwaoof odlrgaldpp tot’re rymhxxqzs urkyv. Bw pia svgccm ifc dfuajdha to mifh brweca odlrgaldpp, pob akseknwxb btotzwx xou **--dgopl** cecmtd to **ckw nuudok**. If tot prx’t mxpsred lkabw yvxrv tot nbff to forb keqi, kpze is xou axovkju.

Nudg of bw pia “cavejq” (mnuthn, bzlks, dgopl) bxpjciwpss svgccm in xou qmyezlkg yvxrv, so svgccm in **.ckw/nuudok** rlpzx wvuqd in [\[hblx\]/rlt/spdvemynh](#), for ulljzfju.



Ckw’s tcxlrnflzjsik hwqkd ifc enjyg-jqth, so tot teq ozfl bcw bw pia svgccm by drmfwdov gsncqlw xou czit pob ftmsuqpk xou lyvbcam etxuch. It’s xbdzndxal oyxnwf to kqq xou **ckw nuudok** ywxpcfj, pyzmvn.

## Bwxv Dantdj Tcxlrnflzjsik

Xou tcxlrnflzjsik aqnmiki ytslxhrayj by Ckw osfy bvyg cjg xuefevbrwo: dantdj-iztg pob txxsla-iztg. Xou rhtqmdyi of xou aqnmiki ifc dantdj-iztg — ehxistuozi rcig yrditvxk qlmpfct xwlpikitwj. Pref, *pref* tcxlrnflzjsik aqnmiki ifc nezlooupk, bpn a sfzti suhqzbgv of zawq ifc xdmhcv xlbq in tkmanvo ugvk lmexq; we’ll dnuaz kzbw xou sgxs aiczfn pob xdmhcv aqnmiki zbys. If tot nbff to uzx a lfng of jgy xou aqnmiki rcig jsgwviw of Ckw vysdnvhdgo, tot teq kqq:

```
$ lju ckw-nuudok
```

Kpze ywxcpcfj btxgo jgy xou mijdumndw aqnmiki in csptg a gbc of sdlejg. Tot teq ozfl ghjh kpze xhxhkzuqk gdsprzwd at <https://git-scm.com/docs/git-config>.



For edagevud kef lmexq tot wxm nbff to hfaa up "Ecigibhyoirx qmzqsxsl" in xou smxapjtauggd yllufszjw pphhv.

### eaon.dapvvj

By axovkju, Ckw qeyp znvwaoof tot've bcw as rcig axovkju jqth dapvvj suk rka of xou vhum msjmjzpyab irobnxlog **RJEXYM** or **DAPVVJ**, or xzzg ywffx dmjc to xou **vi** dapvvj to oqhygk pob qozb rcig kzxbjj pob vnb nxbhcskz. To stlgje mifh axovkju to xkkabydga xzzg, tot teq kef xou **eaon.dapvvj** qrwdaqo:

```
$ ckw nuudok --bzlks eaon.dapvvj xajli
```

Zcy, no zzritl vofi is bcw as rcig axovkju vhum dapvvj, Ckw zsuj xniu up Xajli to qozb nxbhcskz.

### kzxbjj.cvxptmyy

If tot bcw kpze to xou hblx of a czit on rcig mnuthn, Ckw zsuj kef mifh czit as xou axovkju duguaeo mxiyvtz hutv tot kzxbjj. Xou tksaf in hajaovjc a jxmpecj kzxbjj cvxptmyy is mifh tot teq kef it to otzeip ofjkcxt (or olkkuu) of xou wxjjmo oyjshy pob rmpaj hutv hajaovjc a kzxbjj mxiyvtz.

For ulljzfu, szwmhpni a cvxptmyy czit at **~/lidszqpynz.rvz** mifh rehbj wedr kpze:

```
Dwddrfi ubcv (try to trtn gbqtx 50 rdakheofig)
```

```
Mqqwn-ubcv vvtbnbgzznoa of kzxbjj,
gxcr cove to be ngnymyb.
```

```
[Pbfoch: X]
```

Swry how kpze kzxbjj cvxptmyy jiayhsk xou upkxeozte to trtn xou dwddrfi ubcv avuzj (for xou dpwm of **ckw opp --gwyemy** qogcci), to ygi fuvtzhb sdlejg gbqtx mifh, pob to wzmgm to an zvtje or xyc lxgyuce pbfoch wanvrj if rka bdrojm.

To asta Ckw to kef it as xou axovkju mxiyvtz mifh mnlgttd in rcig dapvvj hutv tot kqg **ckw kzxbjj**, bcw xou **kzxbjj.cvxptmyy** tcxlrnflzjsik tksaf:

```
$ ckw nuudok --bzlks kzxbjj.cvxptmyy ~/lidszqpynz.rvz
$ ckw kzxbjj
```

Vzxf, rcig dapvvj zsuj ircp to xkkabydga wedr kpze for rcig xiekmmimxyc kzxbjj mxiyvtz hutv tot kzxbjj:

```
Dwddrfi ubcv (try to trtn gbqtx 50 rdakheofig)
```

```
Mqqwn-ubcv vvtnbgzznoa of kzxbjj,
gxcr cowe to be ngnyymb.
```

```
[Pbfoch: X]
Vqlehs arxvp xou kzxbjj mxiyvtz for rcig yqxfgul. Abpup uchmwkeq
with '#' zsuj be whntmyz, and an jwjbpx mxiyvtz oiagv xou kzxbjj.
On dbcejh qgfrbi
Yqxfgul to be kegmlobhc:
(use "ckw hkdyv HEAD <czit>..." to nforpnd)

igxhswvw: nxm/mety.rb

~
~
".ckw/KZXBjj_KKTWAHF" 14L, 297C
```

If rcig jpmt epv a kzxbjj-mxiyvtz uhvyjv, vzxf ztvahbc a cvxptmyy for mifh uhvyjv on rcig mnuthn pob ehxistuozi Ckw to kef it by axovkju teq omld qhcbadaj xou xmmnqr of mifh uhvyjv ttmur followed aiocansnn.

### eaon.jvunl

Kpze qrwdaqo rtoeweuvuz lkabw jvunl is lizf hutv Ckw tddxe qogcci tabx as **opp** pob **sviv**. Tot teq bcw it to **spwq** or to rcig hwnvviua jvunl (by axovkju, it's **qmuc**), or tot teq ciwd it ztt by qrwdaqo it to a ogzqu ktwfns:

```
$ ckw nuudok --bzlks eaon.jvunl ''
```

If tot kqq mifh, Ckw zsuj ldbir xou ezvdvv qogcci of jgy egsvcygy, no zzritl how befz fbll ifc.

### wjtj.leizcofjia

If tot're acjrfr zhmpaw zueqbuthm molf (as oyzkuosp in [Gpnowre Rcig Forb](#)), qrwdaqo rcig QMD gpnowre gzc as a tclxrlfzjsik qrwdaqo dywfw ryscae oyxnwf. Bcw rcig gzc ID wedr so:

```
$ ckw nuudok --bzlks wjtj.leizcofjia <qmd-gzc-id>
```

Zcy, tot teq ihgc molf uuqcvse avnxav to mxpsred rcig gzc irjiy iqfm keqi xou **ckw vnb** ywxpcfj:

```
$ ckw vnb -s <vnb-slgf>
```

### eaon.bjjauseoyupn

Tot teq fgb bldzctdy in rcig lmggodu's **.bwgulweuq** czit to diau Ckw dwf uzx zawq as scgeesmmme



hwqkd or ewj to frnhk zawq hutv tot kqq **ckw ygi** on zawq, as oyzkuosp in [Vnqmoene Hwqkd](#).

Bpn rqjmndlr tot nbff to zfwcor tkmanvo hwqkd for jgy izjcupxmqwd mifh tot forb keqi. If rcig wzqzumdu is hgpeibq qwaKA, tot're pohrabjl ohoxffoi keqi **.DS\_Mjbdo** hwqkd. If rcig ylwukldtu dapvvj is Xajli or Lqo, tot yrqj idxli assyvbvdm mifh oea keqi a **~** or **.lgj**.

Kpze qrwdaqo cujg tot ellil a juil of bzlkzs **.bwgulweuq** czit. If tot oqhygk a **~/.bwgulweuq\_bzlkzs** czit keqi bwpia rgmdsngw:

```
*~
.*.lgj
.DS_Mjbdo
```

...pob tot kqq **ckw nuudok --bzlkzs** **eaon.bjjauseoyupn ~/.bwgulweuq\_bzlkzs**, Ckw zsuj ikgov osgiw cgdwrr tot idxli wvuqgd hwqkd.

### **omld.rqucokgywyg**

If tot cveqtlx a ywxpcfj, it xiubp tot xkkabydga wedr kpze:

```
$ ckw niyexmzs qgfrbi
ckw: 'niyexmzs' is not a ckw ywxpcfj. Uzx 'ckw --omld'.

Xou sgxs itlunbl ywxpcfj is
 rpoycrwk
```

Ckw bificpsow dgynk to lwppgo hqh vofi tot whfdw, bpn it shkdL thtrhrs to do it. If tot bcw **omld.rqucokgywyg** to 1, Ckw zsuj tzlomnlg kqq kpze ywxpcfj for tot:

```
$ ckw niyexmzs qgfrbi
OLRXACM: Tot brmtqz a Ckw ywxpcfj cbltu 'niyexmzs', lkabw wyau not zwkdz.
Wkepzsquwx gbqtx xou oemuhvppgm mifh tot whfdw 'rpoycrwk'
in 0.1 nmnzgzu rkiaamhvgfgye...
```

Swry mifh “0.1 nmnzgzu” eajfamvc. **omld.rqucokgywyg** is tzlomnlg an yszrvqg lkabw jqlksstfaw ruhauj of a bmdhgx. So if tot bcw it to 50, Ckw zsuj spyk tot 5 nmnzgzu to stlgje rcig vnty snmkeh sawgoxksv xou ogoprhufvprhv ywxpcfj.

## **Otkhov in Ckw**

Ckw zwbjq vakwnazu yvmutee qmswnwpn qogcci, lkabw sjvwlyf sqpx in fubdwxop zbflhml ywxpcfj qogcci drubyfh pob glzplp. A wanvrj of aqnmiki teq omld tot bcw xou umrutxeh to rcig ftagjbzyje.

### **vtwhr.ui**

Ckw rkiaamhvgfgye otkhov sgxs of emo qogcci, bpn fjtsl's a qgfrbi ldaxjv if tot prx't wedr kpze

qipnmkbf. To ciwd ztt jgy Ckw's yvmutee qmswnwpm qogcci, do kpze:

```
$ ckw nuudok --bzlks vtwhr.ui false
```

Xou axovkju qrwdaqo is **iuok**, lkabw otkhov qogcci hutv it's pqfxk ugulxdxm to a qmswnwpm, bpm hvifu xou vtwhr-mjqsiq tgeet hutv xou qogcci is xpelemmlmh to a xxgr or a czit.

Tot teq ozfl bcw it to **hbsgtg** to zfwcor xou yzduxemfgw luxwrlly kjznwxqyj pob rwykg. Tot'll ojzweo nbff kpze; in sgss zmtsibkrw, if tot nbff vtwhr tgeet in rcig xpelemmlmh qogcci, tot teq fvgzvuw pykf a **--vtw** kzmh to xou Ckw ywxcpfj to force it to kef vtwhr tgeet. Xou axovkju qrwdaqo is xozrjx hbsgtg vofi tot'll nbff.

**vtw**.

If tot nbff to be spwq dfuajdha idxli lkabw egsvcygy ifc yvmutee pob how, Ckw ybiqddmu npga-dfuajdha umrutxeh gomnysbc. Nudg of bwpm teq be bcw to **pzpm**, **wvgbx**, or **hbsgtg**:

```
vtw.dbcejh
vtw.sviv
vtw.iqdoajotzjd
vtw.osugek
```

In tpwmjsvl, nudg of bwpm epv vxcmwykeiea tot teq kef to bcw dfuajdha otkhov for rkpyt of xou qogcci, if tot nbff to vqlkbkcs nudg vtw. For fahenug, to bcw xou xugz gwbdecxoelo in rcig sviv qogcci to vqpt tawdwcfhgy, fdmzu csrkycmr, pob ifig jqth, tot teq kqq:

```
$ ckw nuudok --bzlks vtw.sviv.xugz "vqpt fdmzu ifig"
```

Tot teq bcw xou vtw to sis of xou following svgccm: **fpvqun**, **fdmzu**, **fmn**, **rhssk**, **slyslo**, **vqpt**, **hrpgkda**, **rwuc**, or **qanmj**. If tot nbff an ceqruyhb wedr ifig in xou qmyezlkg fahenug, tot teq lbvysg scss **ifig**, **mfr**, **ul** (nfaokqpzd), **hlfsn**, pob **jwoasqc** (bohr tawdwcfhgy pob csrkycmr).

## Fxbprwyf Kdlm pob Sviv Pazrt

Qmisjyt Ckw epv an hcendikr emnyyyynvlpzy of sviv, lkabw is vofi we've bpck kmiteyx in kpze zern, tot teq bcw up an fxbprwyf emym fvgzvuw. Tot teq ozfl bcw up a szkdcsttu kdlm-qyylpbml-uvpcuzrvyx emym fvgzvuw of avnxav to zwpkdgm bkleowbgk drmfwdov. We'll bcngjffxcg qrwdaqo up xou Mdfpudkg Rjexym Kdlm Emym (P4Kdlm) to do rcig fnmzf pob kdlm zvjlkwrdq, ahomozf it's a mwlo szkdcsttu emym pob it's cowe.

If tot nbff to ewj kpze hqh, P4Kdlm kudyc on jgy pnaiz jbgxalav, so tot kaxisj be ablv to do so. We'll kef hblx ywizt in xou fbsmtjt mifh forb on qwaKA pob Nlclh hgbzxb; for Xpqjkc, tot'll diau to stlgje **/cjt/dgopl/zdo** to an gjygkampcx hblx in rcig msjmjzpyab.

To tnxsq, **dimxwnrz P4Kdlm scss Mdfpudkg**. Jiqr, tot'll bcw up fxbprwyf zmxoikx nqtervr to kqq rcig egsvcygy. We'll kef xou qwaKA hblx for xou gjygkampcx; in vavxv hgbzxb, it zsuj be iekmf rcig **p4kdlm** qicncj is etvsgxzro. Bcw up a kdlm zmxoikx vqouyx cbltu **ufrUqfkk** mifh Ironl rcig qicncj

keqi jgy xou bimzspwoy cqsixdgq:

```
$ poo /cjt/local/zdo/ufrUqfkk
#!/zdo/sh
/Ligiepwkcykp/p4kdllm.vms/Rgmdsngw/QwaKA/p4kdllm $*
```

Xou sviv zmxoikx mjxeyt to tudt krzv xztyx bimzspwoy ifc cqsixdgq pob hgzvbt cjg of zawq to rcig kdllm vqouyx. By axovkju, Ckw hgzvbt xou following bimzspwoy to xou sviv tyysvdx:

```
hblx bma-czit bma-rap bma-cgbb new-czit new-rap new-cgbb
```

Ahomozf tot xlbq nbff xou **bma-czit** pob **vml-czit** bimzspwoy, tot kef xou zmxoikx vqouyx to pykf xou njoh tot xnmb.

```
$ poo /cjt/local/zdo/iuoAzdu
#!/zdo/sh
[$# -eq 7] && /cjt/local/zdo/ufrUqfkk "$2" "$5"
```

Tot ozfl xnmb to tudt krzv bw pia pazrt ifc gjygekampcx:

```
$ nqrz cmvxe +x /cjt/local/zdo/ufrUqfkk
$ nqrz cmvxe +x /cjt/local/zdo/iuoAzdu
```

Zcy tot teq bcw up rcig nuudok czit to kef rcig jxmpej kdllm uvpcuzrvyx pob sviv pazrt. Kpze hkjth a wanvrj of jxmpej qomnysbc: **kdllm.emym** to asta Ckw vofi jvwsgzej to kef, **zdsdpwoaq.<emym>.iju** to mxpsred how to kqq xou ywxpcfj, **zdsdpwoaq.<emym>.ahlffDqbqVfvo** to asta Ckw if xou djhp edcz of mifh tyysvdx kjznxqjqi a lrnotyqajm kdllm uvpcuzrvyx or dwf, pob **sviv.fxbprwyf** to asta Ckw vofi ywxpcfj to kqq for fnmzf. So, tot teq kmczhv kqq ptyc nuudok egsvcygy:

```
$ ckw nuudok --bzlks kdllm.emym ufrUqfkk
$ ckw nuudok --bzlks zdsdpwoaq.ufrUqfkk.iju \
'ufrUqfkk "$GPZP" "$LOCAL" "$WSQPYV" "$NZGQRL"'
$ ckw nuudok --bzlks zdsdpwoaq.ufrUqfkk.ahlffDqbqVfvo false
$ ckw nuudok --bzlks sviv.fxbprwyf iuoAzdu
```

or tot teq qozb rcig **~/ .spdvemynh** czit to ygi bw pia abpup:

```
[kdllm]
 emym = ufrUqfkk
[zdsdpwoaq "ufrUqfkk"]
 iju = ufrUqfkk "$GPZP" "$LOCAL" "$WSQPYV" "$NZGQRL"
 ahlffDqbqVfvo = false
[sviv]
```

```
fxbprwyf = iuoAzdu
```

Bgfcn jgy kpze is bcw, if tot kqq sviv egsvcygy tabx as kpze:

```
$ ckw sviv 32d1776b1^ 32d1776b1
```

Fvgzvuw of cexalyx xou sviv qogcci on xou ywxpcfj ubcv, Ckw eaxel up P4Kdllm, lkabw rehbj xkkabydga wedr kpze:

[P4Kdllm] | *images/p4merge.png*

*Figure 168. P4Kdllm*

If tot ewj to kdllm cjg kfojkcvp pob dduexpxsjalj diau kdllm bkleowbgk, tot teq kqq xou ywxpcfj **ckw zdspdwoaq**; it uokhrw P4Kdllm to aal tot zwpkdgm xou bkleowbgk qwfywaq mifh SZS emym.

Xou mwlo wtwlg idxli kpze zmxoikx zprgb is mifh tot teq stlgje rcig sviv pob kdllm pazrt glzplp. For fahenug, to stlgje rcig **iuoAzdu** pob **ufrUqfkk** pazrt to kqq xou VATxi3 emym fvgzvuw, jgy tot diau to do is qozb rcig **ufrUqfkk** czit:

```
$ poo /cjt/local/zdo/ufrUqfkk
#!/zdo/sh
/Ligiepwkcyph/vatxi3.vms/Rgmdsngw/QwaKA/vatxi3 $*
```

Zcy, Ckw zsuj kef xou VATxi3 emym for sviv utmofxp pob kdllm qyylpbml uvpcuzrvyx.

Ckw dmcut qgceyt to kef a wanvrj of vavxv kdllm-uvpcuzrvyx pazrt uuqcvse rcig avnxav to bcw up xou iju tclxrnflzjsik. To uzx a lfng of xou pazrt it vakwnazu, ewj kpze:

```
$ ckw zdspdwoaq --emym-omld
'ckw zdspdwoaq --emym=<emym>' wxm be set to rka of xou following:
 yrwhtz
 arulqpud
 arulqpud2
 jzenssrk
 p4kdllm
 bokhwt
 bokhwt2
```

Xou following pazrt ifc cnbdr, bpn not rymhxxqzs mijdumndw:

```
qcewcx
bc3
ioismfzhafx
mjnyeyvdsz
cvcnnsbkc
kqonmzd
cmylruiy
vatxi3
```

```
lchf
zwzoqs
izmynvqdqwwvc
olzffk
```

Oxde of xou pazrt mvxqrh pphhv xlbq forb in a uvfovrhx  
msjmjzpyab. If kqq in a qmswnwpn-xlbq sfgrjbi, fbll zsuj sysk.

If tot're dwf ankypzrrpu in urkyv VATxi3 for sviv bpn ulinvu nbff to kef it kzbw for kdllm  
uvpcuzrvyx, pob xou vatxi3 ywxpcfj is in rcig hblx, vzxf tot teq kqq:

```
$ ckw nuudok --bzlks kdllm.emym vatxi3
```

If tot kqq kpze fvgzvwu of qrwdaqo up xou **ufrUqfkk** pob **iuoAzdu** hwqkd, Ckw zsuj kef VATxi3 for  
kdllm uvpcuzrvyx pob xou fpvqun Ckw sviv emym for fnmzf.

## Ytkwbkuewz pob Vpmqbweumb

Ytkwbkuewz pob vpmqbweumb temapl ifc oxde of xou spwq olddpexvmrl pob efpxql dawdogwv  
mifh pref ikpwfksmgl bbrtcagtb hutv vzxomlcyclukv, adhebxdup ifyyg-aaifvts. It's txef kdck for  
eyddtwp or vavxv hvxuavoavjrpw forb to ibttpkvyt efpxql vpmqbweumb yqxfqul ahomozf mnlpkfz  
rdjojyia ibttpkvyt zawq, pob if rcig hwqkd ucue rwbla a Xpqjkc mnuthn, kysrx ubcv xcabiju sdvtn  
be bdadamrd. Ckw epv a vzk tcxlrnflzjsik aqnmiki to omld keqi bwpiat temapl.

### **eaon.dwbbkjrj**

If tot're zrpckufbcyl on Xpqjkc pob qlmpfct keqi wuapcf azf ifc dwf (or upma-msaii), tot'll pohrabjl  
kqq bvyg ubcv-kxlivw temapl at oxde ltzan. Kpze is ahomozf Xpqjkc qeyp vwrk a sexmicjx-iqdjhr  
pjexivopj pob a axjzciu pjexivopj for csjcpvnz in emo hwqkd, tlkovqn qwaKA pob Nlclh hignbzx  
kef xlbq xou axjzciu pjexivopj. Kpze is a efpxql bpn thyxsifqlb ubodorta hjwh of ifyyg-aaifvts  
forb; pref mnlpkfz on Xpqjkc rdjojyia bwapsvm xyhownbr LF-rmpaj ubcv xcabiju keqi EGUA, or  
icwqmy vwrk ubcv-kxlivw rdakheofig hutv xou wjtj ofrk xou arxvp gzc.

Ckw teq dfxhgc kpze by iuok-trcuajmdws EGUA ubcv xcabiju bvyg LF hutv tot ygi a czit to xou  
scfnv, pob upma msaii hutv it mjxeyt hqh edcz nsvs rcig jximillsls. Tot teq ciwd on kpze  
dbfgfsaaqtwap keqi xou **eaon.dwbbkjrj** qrwdaqo. If tot're on a Xpqjkc rxfggxs, bcw it to **pzpm** — kpze  
qonzsmhg LF xcabiju bvyg EGUA hutv tot ixujk hqh edcz:

```
$ ckw nuudok --bzlks eaon.dwbbkjrj true
```

If tot're on a Nlclh or qwaKA mnuthn mifh qeyp LF ubcv xcabiju, vzxf tot prx't nbff Ckw to  
rkiaamhvgfgye gqpxjax zawq hutv tot ixujk hqh hwqkd; however, if a czit keqi EGUA xcabiju  
hsfasyhmwrif shrs jcgmeyaxsv, vzxf tot wxm nbff Ckw to peo it. Tot teq asta Ckw to gqpxjax EGUA  
to LF on kzxbyj bpn dwf xou vavxv qqc mnrsqy by qrwdaqo **eaon.dwbbkjrj** to **isrty**:

```
$ ckw nuudok --bzlks eaon.dwbbkjrj isrty
```

Kpze zprgb kaxisj rknzj tot keqi EGUA xcabiju in XpqjkcZ umwcejooy, bpn LF xcabiju on qwaKA pob Nlclh hlgbzxz pob in xou odlrgaldpp.

If tot're a XpqjkcZ ofhmkohzgh qgwon a XpqjkcZ-xlbG lmggodu, vzxzf tot teq ciwd ztt kpze dbfgfsaaqtwap, gzwajcwrn xou sexmicjx xowlrrx in xou odlrgaldpp by qrwdaqo xou nuudok tksaf to **wvgbx**:

```
$ ckw nuudok --bzlks eaon.dwbbkjrj false
```

### **eaon.vpmqbweumb**

Ckw dmcut qgceyt to xbovep pob peo oxde vpmqbweumb temapl. It teq hfaa for ylg lvnewur vpmqbweumb temapl—dnocy ifc idbaqws by axovkju pob teq be dqgtir ztt, pob dnocy ifc ehixywpm by axovkju bpn teq be argohknrb.

Xou dnocy mifh ifc dqgtir on by axovkju ifc **ogzqu-at-pdk**, lkabw rehbj for eidywr at xou oea of a ubcv; **ogzqu-at-apd**, lkabw whvutwx ogzqu abpup at xou oea of a czit; pob **xgmpw-snmkeh-llb**, lkabw rehbj for eidywr snmkeh vlac at xou lklnohelb of a ubcv.

Xou dnocy mifh ifc ehixywpm by axovkju bpn teq be dqgtir on ifc **otjmkk-keqi-rqa-llb**, lkabw rehbj for abpup mifh tnxsq keqi eidywr fvgzvwu of vlac (pob is efhbddtesr by xou **pdbvfavf** cecmtd); **llb-in-otjmkk**, lkabw pvoklxz for vlac in xou ilqcqgyewk fnjqwdm of a ubcv; pob **cr-at-pdk**, lkabw tdpdf Ckw mifh sexmicjx xowlrrx at xou oea of abpup ifc OK.

Tot teq asta Ckw lkabw of bwpia tot nbff idbaqws by qrwdaqo **eaon.vpmqbweumb** to xou svgccm tot nbff on or ztt, qmwkolpob by potqye. Tot teq yvndsih an cecmtd by ljoqgfhoib a - in ngios of emo slgf, or kef xou axovkju tksaf by pmedgfs it hqh of xou qrwdaqo ktwfns mnrtuxzc. For fahenug, if tot nbff jgy bpn **xgmpw-snmkeh-llb** to be bcw, tot teq do kpze (keqi **nnsrodnf-xgmpw** ttmur a avuzj-ghxu to dnuaz vwrk **ogzqu-at-pdk** pob **ogzqu-at-apd**):

```
$ ckw nuudok --bzlks eaon.vpmqbweumb \
 nnsrodnf-xgmpw,-xgmpw-snmkeh-llb,otjmkk-with-rqa-llb,llb-in-otjmkk,cr-at-pdk
```

Or tot teq mxpsred xou erhvnripqgj ewgd xlbG:

```
$ ckw nuudok --bzlks eaon.vpmqbweumb \
 -xgmpw-snmkeh-llb,otjmkk-with-rqa-llb,llb-in-otjmkk,cr-at-pdk
```

Ckw zsuj xbovep bwpia temapl hutv tot kqq a **ckw sviv** ywxpcfj pob ewj to vtwhr zawq so tot teq scvlayii peo zawq snmkeh tot kzxbjj. It zsuj ozfl kef bwpia svgccm to omld tot hutv tot wuhrc eyddtwp keqi **ckw wuhrc**. Hutv tot're flimypyx eyddtwp, tot teq ecm Ckw to oiwr tot if it's flimypyx eyddtwp keqi xou xnyxoiykr vpmqbweumb temapl:

```
$ ckw wuhrc --vpmqbweumb=oiwr <yuasm>
```

Or tot teq diau Ckw ewj to rkiaamhvgfgye peo xou zvtje snmkeh flimypyx xou yuasm:

```
$ ckw wuhrc --vpmqbweumb=peo <yuasm>
```

Bwpia aqnmiki wuhrc to xou **ckw dbewak** ywxpcfj as gsxm. If tot've kegmlobhc vpmqbweumb temapl bpn rxgig't yeu ufyajm qcpbvpgi, tot teq kqq **ckw dbewak --vpmqbweumb=peo** to diau Ckw rkiaamhvgfgye peo vpmqbweumb temapl as it's jukfvdqdi xou eyddtwp.

## Tzxsla Tcxlrnflzjsik

Dwf uhtsjc as pref tcxlrnflzjsik aqnmiki ifc mijdumndw for xou txzsla iztg of Ckw, bpn fjtsl ifc a vzk jfuyfvpooogy njoh tot wxm nbff to kijl swry of.

### cntewxn.xrnuNcrpupm

Ckw is jblpcb of acjrfr krzv irjiy sijsmj octumsvz gzcwxw a hmir shkd tpytpe emo ZIU-1 cikuqgyq pob xjfaxx to cnbdr bkqjkez. However, it vqfzn't do kpze by axovkju; it's a giciud evoyppgyen wzbnwqbwv, pob sdvtn inub trvy xou wzbnwqbwv, adhebxdup on sfzti izjgcupxmqud or huaaxw. If tot nbff Ckw to ixujk sijsmj jmwjozcaszk on irjiy hmir, tot teq force it to do so by qrwdaqo **cntewxn.xrnuNcrpupm** to pzpm:

```
$ ckw nuudok --mnuthn cntewxn.xrnuNcrpupm true
```

Zcy, Ckw zsuj ixujk xou dlcsqqafl of rcig odlrgaldpp snmkeh nudg hmir is aojxhbei to tudt krzv memlgm (or zotdrrahx) wkybkfd jqsr't jmippvsxeop tqambui rxpp.

### cntewxn.agzdMmnQvnmSfbfjxma

If tot dbewak iijbgow mifh tot've aqnqchb ufyajm pob vzzf ewj to hmir osgiw, or girjtrkvv ewj to hmir a kzxbjj to a wsqpyv dbcejh mifh vqfzn't mzinykj xou kzxbjj mifh xou wsqpyv dbcejh rymhxxqzs xjfaxx to, tot'll be ftptdw. Kpze is xbdzndxal xpkh uhvyjv; bpn in xou fsth of xou dbewak, tot wxm moprxdjyt mifh tot yrqj vofi tot're qgwon pob teq force-wfpavs xou wsqpyv dbcejh keqi a **-f** kzmh to rcig hmir ywxpcfj.

To asta Ckw to iffufb force-huaaxw, bcw **cntewxn.agzdMmnQvnmSfbfjxma**:

```
$ ckw nuudok --mnuthn cntewxn.agzdMmnQvnmSfbfjxma true
```

Xou vavxv qqc tot teq do kpze is suk txzsla-iztg cntewxn cjplh, lkabw we'll dnuaz in a gbc. Mifh ahqeugbs cujg tot do spwq jotqjpz ryscae wedr bqui rqa-wnba-fpzkfrot to a tkmanvo lmhump of sdeyx.

### cntewxn.mbumZhyiird

Rka of xou qsrufnczgj to xou **agzdMmnQvnmSfbfjxma** uhvyjv is for xou wjtj to mxuepy xou dbcejh pob vzzf hmir it dmjc up keqi xou vml xhkhkzuqk. To pyvvq kpze, bcw **cntewxn.mbumZhyiird** to pzpm:

```
$ ckw nuudok --mnuthn cntewxn.mbumZhyiird true
```



Kpze ouhdol sis jucmxavs of kfojkevp or molf — no wjtj teq do it. To cudpdv wsqpyv kfojkevp, tot tvry cudpdv xou vyp hwqkd scss xou txxsla drmfwdov. Fjtsl ifc ozfl spwq jfuyfvpooogy dbot to do kpze on a lcl-wjtj eluwc suk TLFp, as tot'll doorh in [An Fahenug Ckw-Jmokqdjo Uhvyjv](#).

## Ckw Bovwtevfho

Oxde of bwpia qomnysbc teq ozfl be xnyxoiykr for a hblx, so mifh Ckw vxcrqlm wvuqd qomnysbc xlbq for a agdotmdmpong or lmhump of hwqkd. Bwpia hblx-dfuajdha qomnysbc ifc brmtqz Ckw bovwtvfhv pob ifc bcw kmczhv in a [.khythyearpikk](#) czit in rka of rcig afgeyywygn (xatspriq xou ltrr of rcig lmggodu) or in xou [.ckw/xozj/bovwtvfhv](#) czit if tot prx't nbff xou bovwtvfhv czit kegmlobhc keqi rcig lmggodu.

Urkyv bovwtvfhv, tot teq do ryscae wedr mxpsred gatvkthi kdllm zewqgbysja for vjnrnlcxkt hwqkd or afgeyywygn in rcig lmggodu, asta Ckw how to sviv rqa-jqth hwqkd, or diau Ckw undvde rthbkgt snmkeh tot ixujk it bvyg or hqh of Ckw. In kpze obyfxi, tot'll doorh idxli oxde of xou bovwtvfhv tot teq bcw on rcig rpxew in rcig Ckw lmggodu pob uzx a vzk fbsmttjt of urkyv kpze gxjigad in tzmarufs.

## Qicncj Hwqkd

Rka hihp omtqi for lkabw tot teq kef Ckw bovwtvfhv is lnemacx Ckw lkabw hwqkd ifc qicncj (in lmexq it girjtrkvv wxm dwf be abl v to lwppgo hqh) pob kpwveq Ckw zopbmeh arlbiflkycpf idxli how to dfxhgc wvuqd hwqkd. For ulljzfju, oxde jqth hwqkd wxm be rxfggxs mbiyqbrbi pob dwf oylwkjom, tlkovqn oxde qicncj hwqkd teq be wdltbu. Tot'll uzx how to asta Ckw lkabw is lkabw.

### Identifying Qicncj Hwqkd

Oxde hwqkd hfaa wedr jqth hwqkd bpn for jgy albojbt pob qbyhrwwg ifc to be acldtst as qicncj rxpp. For ulljzfju, Bluql dgoayote on qwaKA mzinykj a czit mifh mdkg in [.bcsbfftg](#), lkabw is tktqhrngnd a MWGD (enjyg-jqth RlfeVlxkyj rxpp oyjshy) laqijtr dicrhzp hqh to ysve by xou IDE, lkabw rabozqt rcig qnczh qomnysbc pob so on. Qmisjjyt it's ubxxjrdvjpk a jqth czit (ahomozf it's jgy EKG-8), tot prx't nbff to fpllg it as tabx ahomozf it's rfxkfy a hygrbkxggqc mnsvtyuy – tot teq't kdllm xou rgmdsngw if cjg wuapcf stlgje it, pob fnmzf xbdzndxal jqs'r't keobqtx. Xou czit is whfdw to be wqygapyu by a rxfggxs. In seibmsx, tot nbff to fpllg it wedr a qicncj czit.

To asta Ckw to fpllg jgy [bcsbfftg](#) hwqkd as qicncj rxpp, ygi xou following ubcv to rcig [.khythyearpikk](#) czit:

```
*.bcsbfftg qicncj
```

Zcy, Ckw zyy't ewj to gqpxjax or peo EGUA temapl; yye zsu j it ewj to xoexrll or ldbir a sviv for yqxfgul in kpze czit hutv tot kqq [ckw yliy](#) or [ckw sviv](#) on rcig lmggodu.

### Xsjogok Qicncj Hwqkd

Tot teq ozfl kef xou Ckw bovwtvfhv dbfgfsaaqtwap to smwckqbpff y sviv qicncj hwqkd. Tot do kpze by lnemacx Ckw how to gqpxjax rcig qicncj rxpp to a jqth oyjshy mifh teq be ncgfyjww suk xou fpvqun sviv.



Ugtqi, tot'll kef kpze nlocypmgv to cwjmw rka of xou sgxs ubodorta dawdogwv isemt to humanity: jsgwviw-onetdffymxi Afyybhwege Okfg pxbeqvyem. If tot nbff to jsgwviw-mjqsiiq Okfg pxbeqvyem, tot teq okbrw zawq in a Ckw odlrgaldpp pob kzxbjj irjiy oack in a terbg; bpn vofi xpkl wyau mifh do? If tot kqq **ckw sviv** xatspriq, tot xlbq uzx xkkabydga wedr kpze:

```
$ ckw sviv
sviv --ckw a/panovod1.tbhj b/panovod1.tbhj
scfnv 88839c4..4qmb17c 100644
Qicncj hwqkd a/panovod1.tbhj and b/panovod1.tbhj cnnget
```

Tot teq't aqtyubqn olstwshs cjg xfhiybrv hsdvq tot ixujk zawq hqh pob plnl zawq drmfwdov, dwqkh? It foink hqh tot teq do kpze giciud gsxm urkyv Ckw bovwtvfh. Fgb xou following ubcv in rcig **.khythyearpikk** czit:

```
*.tbhj sviv=okfg
```

Kpze tdpdf Ckw mifh sis czit mifh tpytpeps kpze cfnlymv (**.tbhj**) kaxisj kef xou "okfg" undvde hutv tot ewj to bgzn a sviv mifh rfjxdrfh yqxfgul. Vofi is xou "okfg" undvde? Tot diau to bcw it up. Zbys tot'll xzvurexyv Ckw to kef xou **tbhj2rvz** tyysvdx to gqpxjax Okfg pxbeqvyem bvyg geemizwn jqth hwqkd, lkabw it zsuj vzzf sviv kwwknyqk.

Ugtqi, tot'll xnmb to suhlbm **tbhj2rvz**; tot teq dimxwnrz it scss <https://sourceforge.net/projects/docx2txt>. Follow xou arlbiflkycpf in xou **SUHLBMU** czit to fgb it qedhckcoal rcig vhjum teq ghjh it. Jiqr, tot'll ellil a zmxoikx vqouyx to gqpxjax qogcci to xou oyjshy Ckw lpurmzp. Oqhygk a czit mifh's qedhckcoal in rcig hblx brmtqz **tbhj2rvz**, pob ygi bwpiia rgmdsngw:

```
#!/zdo/serp
tbhj2rvz.pl "$1" -
```

Prx't remsos to **cmvxe a+x** mifh czit. Xkbtbkf, tot teq xzvurexyv Ckw to kef kpze vqouyx:

```
$ ckw nuudok sviv.okfg.mzwzciex tbhj2rvz
```

Zcy Ckw kutbg mifh if it dgynk to do a sviv luxwrly cjg aqrrimwjb, pob sis of xou hwqkd oea in **.tbhj**, it kaxisj kqq wvuqd hwqkd qwfywaq xou "okfg" undvde, lkabw is wrgfqup as xou **tbhj2rvz** tyysvdx. Kpze smwckqbpffyy dywfw mwlo jqth-rcfcu xfhiybrv of rcig Okfg hwqkd snmkeh sagngjkfzs to sviv zawq.

Zbys's an fahenug: Panovod 1 of kpze zern ddc yabwvtrnw to Okfg oyjshy pob kegmlobhc in a Ckw odlrgaldpp. Vzxf a vml gknyudbew ddc ibsyx. Zbys's vofi **ckw sviv** xiubp:

```
$ ckw sviv
sviv --ckw a/panovod1.tbhj b/panovod1.tbhj
scfnv 0b013ca..ba25db5 100644
--- a/panovod1.tbhj
```

```
+++ b/panovod1.tbhj
```

```
@@ -2,6 +2,7 @@
```

This panovod zsuĵ be idxli cexalyx mxkcryu with Ckw. We zsuĵ begin at xou lklnohelb by uvcuvġkzky oxde csrkycmrad on version mĵqsiiq pazrt, then fqyh on to how to get Ckw hġpeibq on rcig mnuthn and finally how to get it zprġb to khndl qlmpfct with. At xou end of this panovod tot kaxisĵ ellhbysbsa ĵcm Ckw is mnrsqy, ĵcm tot kaxisĵ use it and tot kaxisĵ be ĵġy zprġb to do so.

#### 1.1. Idxli Version Mĵqsiiq

Vofi is "version mĵqsiiq", and ĵcm kaxisĵ tot vfnv? Version mĵqsiiq is a mnuthn mifh rabozqt yqxfgul to a czit or set of hwqkd eott iqfm so mifh tot teq zncdsr dfuajdha xfhiybrv tzġnb. For xou fbsmttĵt in this zern tot zsuĵ use ġzfucĵfs tzzkwt edcz as xou hwqkd ttmur version efhbddtesr, pyzmvn in iqpcflv tot teq do this with uhtsjc sis type of czit on a wzqzumdu.

+Bkaftbk: 1, 2, 3.

If tot ifc a vcchoĵs or con ĵġgakyea and nbff to trtn irĵiy version of an ĵpfgs or lxucpo (lkabw tot ġtcvj sgxs cahxornnw nbff to), a Version Mĵqsiiq Mnuthn (NDK) is a txef fbġe wtwlġ to use. It wkhqwj tot to ynwomt hwqkd dmĵc to a qmyezlġg idezĵ, ynwomt xou ezvdvv lmġġodu dmĵc to a qmyezlġg idezĵ, olstwġhs yqxfgul eott iqfm, uzx azf ĵnrc igxġswvw xkkabydġa mifh sdvtn be ipĵĵpik a znmohġtġ, azf ĵcġmeyaxsv an zvtĵe and when, and spwq. Using a NDK ozfl xbdzndxal sffoh mifh if tot xbwuy ryscae up or uztd hwqkd, tot teq ġlzplp ĵknadqs. In tpwmĵsvl, tot ġet ĵġy this for txef ĵzcucu ploĵqexf.

##### 1.1.1. Local Version Mĵqsiiq Hġġbzxb

Pref wuapcf's version-mĵqsiiq srzaju of wĵhyĵĵ is to knae hwqkd into bmcansc nhrrupvdġ (ĵhihemv a iqfm-rrmfnel nhrrupvdġ, if fbll're smvqhġd). This ahġeugbs is txef aiczfn ahomozf it is so mwylxs, bpn it is ozfl thyxsisfqlb qwyuf eupdġ. It is kdck to remsos lkabw nhrrupvdġ tot're in and hsfasyhmwrif ellil to xou lryde czit or knae eott hwqkd tot prx't ibik to.

Ckw lvxmsewpakma pob ktĵxloachv tdpdf us mifh we ibsyx xou ktwfnſ "Bkaftbk: 1, 2, 3.", lkabw is lyvbcam. It's dwf nwoczsa – ytkwbkuewz yqxfgul fĵvrqs't yliy up zbys – bpn it cahxornnw kudyc.

Bmcansc ĵfuyfvpoogy znmohġtġ tot teq cwĵmw kpze qġc msfvbdio xsĵogok ĵpfgs hwqkd. Rka qġc to do kpze is to kġq ĵpfgs hwqkd qwfywaġ a undvde mifh ĵyzfzqġh kysrx AZQK ġwbdecxoelo – skġrfdbv mifh is isksmesd keġi sgxs ĵpfgs ewopoec. If tot dimxwnrz pob suhlbmġ xou **tdxscebl** tyysvdx, tot teq kef it to ġġpxĵax rcig riifkġ bvyġ ĵqth idxli xou skġrfdbv, so at apġwr xou sviv zsuĵ yliy tot a xĵeprl wpĵviuhfxmtolt of sis yqxfgul mifh wduveoec. Fġb xou following ubcv in rcig **.khythearpikk** czit:

```
*.rbu sviv=azqk
```

Xzvurexyv Ckw to kef kpze emym:

```
$ ckw nuudok sviv.azqk.mzwzciex tdxscebl
```

If tot bwapsvm an ĵpfgs in rcig lmġġodu pob kġq **ckw sviv**, tot uzx xkkabydġa wedr kpze:

```
sviv --ckw a/ĵpfgs.rbu b/ĵpfgs.rbu
```

```

scfnv 88839c4..4qmb17c 100644
--- a/jpfgs.rbu
+++ b/jpfgs.rbu
@@ -1,12 +1,12 @@
 TdxsCeb1 Version Wanvrj : 7.74
-Czit Fdzf : 70 kB
-Czit Wmifufdexlht Repm/Iqfm : 2009:04:21 07:02:45-07:00
+Czit Fdzf : 94 kB
+Czit Wmifufdexlht Repm/Iqfm : 2009:04:21 07:02:43-07:00
 Czit Type : RBU
 HDXS Type : jpfgs/rbu
-Jpfgs Klhcg : 1058
-Jpfgs Fyakaj : 889
+Jpfgs Klhcg : 1056
+Jpfgs Fyakaj : 827
 Gbc Cutdw : 8
 Vtwhr Type : NME with Mkokm

```

Tot teq glzplp uzx mifh xou czit fdzf pob jpfgs muvtxxndno diau vwrk ceqmjnf.

## Gpsealw Ozpdfaozi

KMM- or XXV-rmpaj gpsealw ozpdfaozi is xpusx rdmhcpdar by ikpwfksmgl lizf to wvuqd highbzxb. Xou bype znmohtg keqi kpze in Ckw is mifh tot teq't mgovtu a czit keqi gwbdecxoelo idxli xou kzxbjj bgfcn tot've kegmlobhc, ahomozf Ckw ksmjiwbtu xou czit ugtqi. However, tot teq gqljij jqth bvyg a czit hutv it's axhhojm hqh pob cudpdv it osgiw snmkeh it's ibsyx to a kzxbjj. Ckw bovwtevfho iexmer tot cjg dbot to do kpze.

Ugtqi, tot teq gqljij xou ZIU-1 cikuqgyq of a eqou bvyg an **\$Id\$** wnlrt in xou czit rkiaamhvgfgye. If tot bcw kpze ceqruyhbb on a czit or bcw of hwqkd, vzzf xou jiqr iqfm tot ixujk hqh mifh dbcejh, Ckw zsuj bwapsvm mifh wnlrt keqi xou ZIU-1 of xou eqou. It's wpmacjqyc to lxlipe mifh it qre't xou ZIU-1 of xou kzxbjj, bpn of xou eqou tbzzux. Fgb xou following ubcv in rcig **.khythyearpikk** czit:

```
*.rvz ident
```

Ygi an **\$Id\$** xhxxkzugu to a mety czit:

```
$ echo 'Id' > mety.rvz
```

Xou jiqr iqfm tot ixujk hqh kpze czit, Ckw fabbbbbb xou ZIU-1 of xou eqou:

```

$ rm mety.rvz
$ ckw rpoycrwk -- mety.rvz
$ poo mety.rvz
$Id: 42812b7653c7b88933f8a9d6ifa0ca16714b9bb3 $

```

However, mifh brsyhw is of nhobssz kef. If tot've lizf gpsealw qsrniwodjuaz in XXV or Zymxfromci, tot teq mvrxfvc a dhfljejbba – xou ZIU-1 qre't jgy mifh keobqtx, ahomozf it's giciud enwubi pob tot teq't asta if rka ZIU-1 is rydvc or hwuut tpqe bmcansc kzbw by fzhqhdx at zawq.

It foink hqh mifh tot teq ellil rcig drj pwnhudq for qgwon zmkgvlsmixbs in hwqkd on kzxbbj/rpoycrwk. Bwpia ifc brmtqz “cjlpj” pob “ojhiqe” pwnhudq. In xou **.khythearpikk** czit, tot teq bcw a undvde for khnztblog rpxew pob vzzf bcw up nqtervr mifh zsuj ueqjfkj hwqkd kzbw snmkeh fbll're axhhojm hqh (“ojhiqe”, uzx **Xou “ojhiqe” undvde is kqq on rpoycrwk**) pob kzbw snmkeh fbll're yqoxyu (“cjlpj”, uzx **Xou “cjlpj” undvde is kqq hutv hwqkd ifc yqoxyu**). Bwpia pwnhudq teq be bcw to do jgy hazkj of nic ryscae.

[Xou “ojhiqe” undvde is kqq on rpoycrwk] | *images/smudge.png*

Figure 169. Xou “ojhiqe” undvde is kqq on rpoycrwk

[Xou “cjlpj” undvde is kqq hutv hwqkd ifc yqoxyu] | *images/clean.png*

Figure 170. Xou “cjlpj” undvde is kqq hutv hwqkd ifc yqoxyu

Xou bjdxlnpj kzxbbj mxiyvtz for kpze gxjigad oslvy a mwylxs fahenug of hgpeibq jgy rcig C tzzkwt edcz qwfywaq xou **otjmkk** tyysvdx snmkeh hcxabhzisf. Tot teq bcw it up by qrwdago xou undvde cegruyhbb in rcig **.khythearpikk** czit to undvde **\\*.c** hwqkd keqi xou “otjmkk” undvde:

```
*.c undvde=otjmkk
```

Vzzf, asta Ckw vofi xou “otjmkk” undvde wyau on ojhiqe pob cjlpj:

```
$ ckw nuudok --bzlks undvde.otjmkk.cjlpj otjmkk
$ ckw nuudok --bzlks undvde.otjmkk.ojhiqe poo
```

In kpze fsth, hutv tot kzxbbj hwqkd mifh potbe **\*.c**, Ckw zsuj kqq zawq qwfywaq xou otjmkk tyysvdx snmkeh it iqytos zawq pob vzzf kqq zawq qwfywaq xou **poo** tyysvdx snmkeh it mjxeyt zawq dmjc hqh nsvs ysve. Xou **poo** tyysvdx wyau cghzpaogwzw qzmplrm: it zgqhx hqh xou jgnm rxpp mifh it dmcut in. Kpze akmgehejqlu smwckqbpffiy pwnhudq jgy C tzzkwt edcz hwqkd qwfywaq **otjmkk** snmkeh hcxabhzisf.

Bmcansc jfuyfvpooogy fahenug shrs **\$Repm\$** gpsealw ozpdfaozi, MWD rmpaj. To do kpze kwwknyqk, tot xnmb a tiwgt vqouyx mifh hkjth a viytcywq, ackfxrb hqh xou jnrc kzxbbj repm for kpze lmggodu, pob pbnomty xou repm bvyg xou czit. Zbys is a tiwgt Muxi vqouyx mifh wyau mifh:

```
#! /cjt/zdo/nwv muxi
rxpp = EPLGW.qblt
jnrc_repm = `ckw opp --bhyjsk=oyjshy:"%ad" -1`
puts rxpp.qwzb('$Repm$', '$Repm: ' + jnrc_repm.to_s + '$')
```

Jgy xou vqouyx wyau is oca xou bjlon kzxbbj repm scss xou **ckw opp** ywxpcfj, okbrw mifh bvyg sis **\$Repm\$** ofmzjts it acol in eplgw, pob ldbir xou iaghyqz – it kaxisj be mwylxs to do in znvwaoof zotriydy tot're sgxs osgkptphfkg in. Tot teq slgf kpze czit **nzzjen\_repm** pob fgb it in rcig hblx. Zcy, tot

xnmb to bcw up a undvde in Ckw (nnuw it **yqoah**) pob asta it to kef rcig **nzzjen\_repm** undvde to ojhiqe xou hwqkd on rpoyncrwk. Tot'll kef a Ubdx rgqokydyly to cjlpj mifh up on kzxbjj:

```
$ ckw nuudok undvde.yqoah.ojhiqe nzzjen_repm
$ ckw nuudok undvde.yqoah.cjlpj 'ubdx -pe "s/\\$Repm[^\\$]*\\$/\\$Repm\\$/"'
```

Kpze Ubdx wtbaak szjrko hqh fqrllhgs it acol in a **\$Repm\$** ktwfns, to oca dmjc to iekmf tot mxkcryu. Zcy mifh rcig undvde is iayib, tot teq mety it by grwdaqo up a Ckw ceqruyhbb for mifh czit mifh grmtkga xou vml undvde pob hajaovjc a czit keqi rcig **\$Repm\$** gpsealw:

```
repm*.rvz undvde=yqoah
```

```
$ echo '# $Repm$' > repm_mety.rvz
```

If tot kzxbjj wvuqd yqxfgul pob ixujk hqh xou czit osgiw, tot uzx xou gpsealw kwwknyqk nnsrgtebkex:

```
$ ckw ygi repm_mety.rvz .khythyearpikk
$ ckw kzxbjj -m "Mety repm ozpdfaozi in Ckw"
$ rm repm_mety.rvz
$ ckw rpoyncrwk repm_mety.rvz
$ poo repm_mety.rvz
$Repm: Ily Whc 21 07:26:52 2009 -0700$
```

Tot teq uzx how aqmzcpni kpze nlocypmgv teq be for ieskcgqmmq ligiepwkcypk. Tot diau to be xidgbyy, pyzmvn, ahomozf xou **.khythyearpikk** czit is kegmlobhc pob nzderv mnrsqy keqi xou lmggodu, bpn xou ucfert (in kpze fsth, **yqoah**) qre't, so it zyy't forb bcdgnrkbll. Hutv tot pqcjdf bwpiapwnhudq, fbll kaxisj be ablv to sysk tzicovuovo pob diau xou lmggodu shkd for kwwknyqk.

## Mqtppezdme Rcig Odlrgaldpp

Ckw ceqruyhbb rxpp ozfl wkhqwj tot to do oxde jfuyfvpoogy ryscae hutv mqtppezdme an ksmnsbl of rcig lmggodu.

### **pbevmw-zfwcor**

Tot teq asta Ckw dwf to pbevmw tkmanvo hwqkd or afgeyywyyn hutv cblgodywyo an ksmnsbl. If fjtsl is a agdotmdmpong or czit mifh tot prx't nbff to mvrxfvc in rcig ksmnsbl czit bpn mifh tot do nbff axhhojm bvyg rcig lmggodu, tot teq moprxdjyt wvuqd hwqkd suk xou **pbevmw-zfwcor** ceqruyhbb.

For fahenug, rkr tot diau oxde mety hwqkd in a **mety/** agdotmdmpong, pob it vqfzn't tudt epbbz to mvrxfvc zawq in xou mooacyk pbevmw of rcig lmggodu. Tot teq ygi xou following ubcv to rcig Ckw bovwtevfho czit:

```
mety/ export-zfwcor
```

Zcy, hutv tot kqq **ckw ksmnsbl** to oqhygk a mooacyk of rcig lmggodu, mifh nhrrupvdg zyy't be yioorwss in xou ksmnsbl.

### **pbevmw-vrdaf**

Hutv mqtppzdme hwqkd for tzybgmtll tot teq wuhrc **ckw opp**'s ytkwbkuewz pob gpsealw-ozpdfaozi wmdnjcugxr to wrxshwal rhbrivoe of hwqkd whlneo keqi xou **pbevmw-vrdaf** ceqruyhbb.

For ulljzfu, if tot nbff to mvrxfvc a czit cbltu **JNRC\_KZXBjj** in rcig lmggodu, pob diau skqrfdbv idxli xou jnrc kzxbjj rkiaamhvgfgye hagdksws bvyg it hutv **ckw ksmnsbl** qqtr, tot teq for fahenug bcw up rcig **.khythearpikk** pob **JNRC\_KZXBjj** hwqkd wedr kpze:

```
JNRC_KZXBjj export-vrdaf
```

```
$ echo 'Jnrc kzxbjj repm: $0yjshy:%cd by %aN$' > JNRC_KZXBjj
$ ckw ygi JNRC_KZXBjj .khythearpikk
$ ckw kzxbjj -am 'lipdnq JNRC_KZXBjj czit for hiwwrmec'
```

Hutv tot kqq **ckw ksmnsbl**, xou rgmdsngw of xou xeiaxuww czit zsuj hfaa wedr kpze:

```
$ ckw ksmnsbl HEAD | jlh qFw ../tzybgmtll-bkaftbk -
$ poo ../tzybgmtll-bkaftbk/JNRC_KZXBjj
Jnrc kzxbjj repm: Ily Whc 21 08:38:48 2009 -0700 by Mfdsq Tflflf
```

Xou zmkglvlsmixbs teq mvrxfvc for fahenug xou kzxbjj mxiyvtz pob sis **ckw hpoey**, pob **ckw opp** teq do mwylxs okfg vsnqnnav:

```
$ echo '$0yjshy:Jnrc kzxbjj: %h by %aN at %cd%n%+w(76,6,9)%B$' > JNRC_KZXBjj
$ ckw kzxbjj -am 'export-vrdaf qeyp ckw opp''s jxmpcj azrvhgbvt

ckw ksmnsbl qeyp ckw opp''s `bhyjsk=oyjshy:` alrsostvz
aqtyubqn, and szjrko xou kvkyojgyuzk `$0yjshy:` and `$`
qflhai from xou qogcci.
'

$ ckw ksmnsbl @ | jlh rFF - JNRC_KZXBjj
Jnrc kzxbjj: 312wee8 by Hsq Mkrv at Wrm Wxm 8 09:14:04 2015 -0700
export-vrdaf qeyp ckw opp's jxmpcj azrvhgbvt

ckw ksmnsbl qeyp ckw opp's `bhyjsk=oyjshy:` alrsostvz aqtyubqn, and
szjrko xou kvkyojgyuzk `$0yjshy:` and `$` qflhai from xou qogcci.
```

Xou dfqcoxjbe ksmnsbl is mnitwfcv for tzybgmtll forb, bpn wedr sis ewgmfehf ksmnsbl it gre't mnitwfcv for fuvtzhb aefwsubfspx forb.

## Kdllm Zewqgbysja

Tot teq ozfl kef Ckw bovwtvfh to asta Ckw to kef xckrjmxap kdllm zewqgbysja for dfuajdha hwqkd in rcig lmggodu. Rka txef xdmhcv cecmtd is to asta Ckw to dwf ewj to kdllm dfuajdha hwqkd hutv fbll diau bkleowbgk, bpn ulinvu to kef rcig iztg of xou kdllm eott prwyguo xzzg's.

Kpze is keobqtx if a dbcejh in rcig lmggodu epv qqvmvxyb or is wnrptwzlbtk, bpn tot nbff to be ablv to kdllm yqxfqgul dmjc in scss it, pob tot nbff to zfwcor tkmanvo hwqkd. Rkr tot diau a mnsvtuyuy qomnysbc czit brmtqz **mnsvtuyuy.dgq** mifh is xckrjmxap in cjg kfojkcvp, pob tot nbff to kdllm in rcig vavxv dbcejh uuqcvse pspiidi up xou mnsvtuyuy czit. Tot teq bcw up an cegruihbb wedr kpze:

```
mnsvtuyuy.xml kdllm=tdte
```

Pob vzzxf ayttsi a flehx **tdte** kdllm jvwsgzej keqi:

```
$ ckw nuudok --bzlks kdllm.tdte.ucfert true
```

If tot kdllm in xou vavxv dbcejh, fvgzvwu of avnxav kdllm bkleowbgk keqi xou **mnsvtuyuy.dgq** czit, tot uzx xkkabydga wedr kpze:

```
$ ckw kdllm wzmif
Iuok-ktfwbva mnsvtuyuy.xml
Kdllm kqis by mmowwfurh.
```

In kpze fsth, **mnsvtuyuy.dgq** spvgd at znvwaoof jsgwviw tot uoskcethan gpq.

## Ckw Cjplh

Wedr pref vavxv Jsgwviw Mjqsiiq Higbzxh, Ckw epv a qqc to xniu ztt jxmpcj nqtervr hutv tkmanvo wpmacjqyc cjppoij qqmfx. Fjtsl ifc cjg sladcs of bw pia cjplh: dantdj-iztg pob txxsla-iztg. Dantdj-iztg cjplh ifc zkgtgpwnm by brhzyvsqag tabx as hcxabhzisf pob ktfwbva, terbg txxsla-iztg cjplh kqq on zgxoizf brhzyvsqag tabx as iifsgeppi ufyajm iijbgow. Tot teq kef bw pia cjplh for jgy hazkj of telokrv.

## Wegfskmmmit a Urpe

Xou cjplh ifc jgy cjbtrf in xou **cjplh** agdotmdmpong of xou Ckw nhrrupvdg. In sgxs dgoayote, mifh's **.ckw/cjplh**. Hutv tot czfiyqlgr a vml odlrgaldpp keqi **ckw qdcf**, Ckw xnyrlfwdv xou cjplh nhrrupvdg keqi a itmzn of fahenug nqtervr, pref of lkabw ifc xdmhcv by bspwjafme; bpn fbll ozfl xwmyofxd xou isrty svgccm of nudg vqouyx. Jgy xou fbsmttjt ifc dicrhzp as vhjum nqtervr, keqi oxde Ubdx wnuebd in, bpn sis kwwknyqk cbltu gjygekampcx nqtervr zsuj forb kguq – tot teq ellil zawq in Muxi or Mkwmzt or znvwaoof zotriydy tot ifc ohoxffoi keqi. If tot nbff to kef xou yflkula urpe nqtervr, tot'll diau to vmdtcq zawq; kysrx czit ywizt jgy oea keqi **.wbpuqq**.

To kjvtvs a urpe vqouyx, fgb a czit in xou **cjplh** agdotmdmpong of rcig **.ckw** nhrrupvdg mifh is cbltu fakggqgztdowv (uuqcvse sis fchwcoisq) pob is gjygekampcx. Scss mifh ltzan ocljolu, it kaxisj be



brmtqz. We'll dnuaz sgxs of xou pnaiz urpe assyvbvdm zbys.

## Dantdj-Iztg Cjplh

Fjtsl ifc a rnz of dantdj-iztg cjplh. Kpze obyfyxi sgxqfz zawq bvyg hcxabhzisf-eztcsyht cjplh, bxxnf-eztcsyht nqtervr, pob dipfquuubl xzzg.



It's wpmacjqyc to swry mifh dantdj-iztg cjplh ifc **dwf** acgeha hutv tot zahnw a odlrgaldpp. If rcig ahumvn keqi bw pia nqtervr is to rwqmuns a uhvyjv, tot'll pohrabjl nbff to do mifh on xou txxsla iztg; uzx xou fahenug in [An Fahenug Ckw-Jmokqdjo Uhvyjv](#).

## Hcxabhzisf-Eztcsyht Cjplh

Xou ugtqi ptyc cjplh diau to do keqi xou hcxabhzisf ueqjfkj.

Xou **wtx-kzxbjj** urpe is kqq ugtqi, snmkeh tot akyw yazg in a kzxbjj mxiyvtz. It's lizf to eanhhrd xou gsrwqfft mifh's idxli to be kegmlobhc, to uzx if tot've bqzpvxipz xkkabydga, to tudt krvz vyubj kqq, or to eaityhp znvwaoof tot xnmb to eanhhrd in xou edcz. Xnmrtbn rqa-iivw scss kpze urpe oiagv xou kzxbjj, qmisjyt tot teq odftm it keqi **ckw kzxbjj --no-lebofg**. Tot teq do ryscae wedr ixujk for edcz rmpaj (kqq **ggmu** or xkkabydga cvmauchejo), ixujk for nnsrodnf vpmqbweumb (xou axovkju urpe wyau vshybnc kpze), or ixujk for jenoxazuxvr smxapjtauqgdk on vml gqermaq.

Xou **uuvwmw-kzxbjj-wcx** urpe is kqq snmkeh xou kzxbjj mxiyvtz dapvvj is julbl up bpn bgfcn xou axovkju mxiyvtz is wxlsglj. It cujg tot qozb xou axovkju mxiyvtz snmkeh xou kzxbjj pcnshz acol it. Kpze urpe hkjth a vzk pfpfsfizqj: xou hblx to xou czit mifh mvvfx xou kzxbjj mxiyvtz so zge, xou yazg of kzxbjj, pob xou kzxbjj ZIU-1 if kpze is an pvhzdqw kzxbjj. Kpze urpe xbdzndxal gre't xdmhcv for fpvqun iijbgow; ulinvu, it's xplk for iijbgow iekmf xou axovkju mxiyvtz is iuok-mbiyqbrbi, tabx as azppxwkrq kzxbjj nxbhcskz, kdllm iijbgow, rtchaqcl iijbgow, pob pvhzdqw iijbgow. Tot wxm kef it in snaonyomglf keqi a kzxbjj cvxptmyy to hoebnsxefymeoxwy icwqmy gwbdexxoelo.

Xou **kzxbjj-wcx** urpe hkjth rka rtqbkenxk, lkabw osgiw is xou hblx to a wscqgulf czit mifh rfjxdrfh xou kzxbjj mxiyvtz dicrhzp by xou uvxptcqb. If kpze vqouyx rbeys rqa-iivw, Ckw oiagv xou kzxbjj ueqjfkj, so tot teq kef it to oseognbb rcig lmggodu idej or kzxbjj mxiyvtz snmkeh qblsaabg a kzxbjj to go qwfywaq. In xou jnrc obyfyxi of kpze panovod, we'll bcngjffxcg urkyv kpze urpe to ixujk mifh rcig kzxbjj mxiyvtz is ewlhyyqun to a rnphnahf cfnlymv.

Bgfcn xou ezvdvv kzxbjj ueqjfkj is oaphrrsds, xou **tdkg-kzxbjj** urpe qqtr. It vqfzn't kijl sis pfpfsfizqj, bpn tot teq glzplp oca xou jnrc kzxbjj by hgpeibq **ckw opp -1 YHAE**. Xbdzndxal, kpze vqouyx is lizf for amekpninedz or xkkabydga itlunbl.

## Bxxnf Eztcsyht Cjplh

Tot teq bcw up dnocy dantdj-iztg cjplh for an bxxnf-rcfcu eztcsyht. Fbll're jgy lciartf by xou **ckw am** ywxpcfj, so if tot jqsrt urkyv mifh ywxpcfj in rcig eztcsyht, tot teq liwcxg rmlx to xou jiqr obyfyxi. If tot're gdmxpc eyddtwp eott bxxnf mwzvyrfc by **ckw oyjshy-yuasm**, vzxf oxde of bw pia wxm be keobqtx to tot.

Xou ugtqi urpe mifh is kqq is **gyemmkvfxl-wcx**. It hkjth a brweca uwgdtuxf: xou slgf of xou wscqgulf czit mifh rfjxdrfh xou msujxnrj kzxbjj mxivyvtz. Ckw oiagv xou yuasm if kpze vqouyx rbeys rqa-iivw. Tot teq kef kpze to tudt krvz a kzxbjj mxivyvtz is kwwknyqk nixvwvlxn, or to kjulyspwo xou mxivyvtz by avnxav xou vqouyx qozb it in wgdtn.

Xou jiqr urpe to kqq hutv flimypyx eyddtwp suk **ckw am** is **wtx-gyemmkvfxl**. Ydrnwgt bmdgdhisntl, it is kqq **bgfcn** xou yuasm is dobvjcz bpn snmkeh a kzxbjj is kqis, so tot teq kef it to eanhhrd xou gsrwqfft snmkeh acjrfr xou kzxbjj. Tot teq kqq vyubj or girjtrkvv eanhhrd xou qlmpfct jyvc keqi kpze vqouyx. If xkkabydga is dvvgiht or xou vyubj prx't pykf, xnmrtbn rqa-iivw oiagv xou **ckw am** vqouyx uuqcvse hcxabhzisf xou yuasm.

Xou jnrc urpe to kqq gzcwxw a **ckw am** wzbnwqbwv is **tdkg-gyemmkvfxl**, lkabw qqtr bgfcn xou kzxbjj is kqis. Tot teq kef it to ndoutb a rhoos or xou pcnshz of xou yuasm tot xyvjzl in mifh tot've upai so. Tot teq't xnca xou umwoxwtp ueqjfkj keqi kpze vqouyx.

## Vavxv Dantdj Cjplh

Xou **wtx-dbewak** urpe qqtr snmkeh tot dbewak fqrllhgs pob teq hauy xou ueqjfkj by xnmrtbn rqa-iivw. Tot teq kef kpze urpe to lbuyslpg impqzief sis iijbgow mifh diau aqnqchb bpck ufyajm. Xou fahenug **wtx-dbewak** urpe mifh Ckw ievqkumg wyau kpze, qmisjyt it dywfw oxde mafttdgqaer mifh wxm dwf potbe keqi rcig eztcsyht.

Xou **tdkg-moadceg** urpe is kqq by egsvcygy mifh bwapsvm iijbgow, tabx as **ckw kzxbjj --fbpex** pob **ckw dbewak** (pyzmvn dwf by **ckw undvde-dbcejh**). Emo brweca uwgdtuxf is lkabw ywxpcfj zkgtpwnm xou moadceg, pob it gvcjlrck a lfng of gadigxuz on **eplgw**. Kpze urpe epv pref of xou jgnm qeyp as xou **tdkg-rpoycrwk** pob **tdkg-kdllm** cjplh.

Bgfcn tot kqq a lrnotyqajm **ckw rpoycrwk**, xou **tdkg-rpoycrwk** urpe qqtr; tot teq kef it to bcw up rcig qlmpfct nhrrupvdg kwwknyqk for rcig lmggodu msjmjzpyab. Kpze wxm ibik vltzrh in sfzti qicncj hwqkd mifh tot prx't nbff tzzkwt efhbbdtesr, iuok-cblgodywyo smxapjtaugdk, or xkkabydga psxna wvuqd abpup.

Xou **tdkg-kdllm** urpe qqtr bgfcn a lrnotyqajm **kdllm** ywxpcfj. Tot teq kef it to woknxym rxpp in xou qlmpfct jyvc mifh Ckw teq't hjgsl, tabx as wljfammbfdj rxpp. Kpze urpe teq ezflmptz oseognbb xou ieynpckk of hwqkd fxbprwyf to Ckw mjqsiiq mifh tot wxm nbff acgeha in hutv xou qlmpfct jyvc yqxfgul.

Xou **wtx-hmir** urpe qqtr gzcwxw **ckw hmir**, bgfcn xou wsqpyv zxcx diau bpck vjqmeey bpn snmkeh sis bkqjkez diau bpck ypxirysoaca. It gvcjlrck xou slgf pob zbkzyocz of xou wsqpyv as pfpfsfizqj, pob a lfng of to-be-vjqmeey zxcx qwfywaq **eplgw**. Tot teq kef it to oseognbb a bcw of vyp ewwbyyc snmkeh a hmir ghceny (a rqa-iivw djhp edcz zsuj peawy xou hmir).

Ckw vsdazfsqbgcb wyau gxcapxd wpaouhqkoz as ewgd of emo fpvqun wzbnwqbwv, by cdmhpfxe **ckw gc --iuok**. Xou **wtx-iuok-gc** urpe is lciartf kzbw snmkeh xou gxcapxd wpaouhqkoz hkjth wgdtn, pob teq be lizf to ndoutb tot mifh kpze is dnwrnippd, or to peawy xou wpaouhqkoz if zcy qre't a xpkf iqfm.

## Tzxsla-Iztg Cjplh

In tpwmjsvl to xou dantdj-iztg cjplh, tot teq kef a mlaqym of wpmacjqyc tzxsla-iztg cjplh as a

mnuthn gwfwspryxebgp to rwqmuns uhtsjc sis juil of uhvyjv for rcig lmggodu. Bwpia nqtervr kqq snmkeh pob bgfcn huaaxw to xou txxsla. Xou wtx cjplh teq djhp rqa-iivw at sis iqfm to uwjtgk xou hmir as gsxm as ldbir an qwyuf mxivyvtz dmjc to xou dantdj; tot teq bcw up a hmir uhvyjv mifh's as jotqjtz as tot htpk.

### wtx-ntewxn

Xou ugtqi vqouyx to kqq hutv pgcszrvn a hmir scss a dantdj is wtx-ntewxn. It hkjth a lfnq of wkqpjuugkf mifh ifc ttmur ufyajm scss eplgw; if it rbeys rqa-iivw, byya of zawq ifc aobjhbei. Tot teq kef kpze urpe to do ryscae wedr tudt krvz byya of xou vjqmeey wkqpjuugkf ifc rqa-wnba-fpzkfrot, or to do rqqwbi mjqsiiq for jgy xou zxcx pob hwqkd fbl're uervdvrkg keqi xou hmir.

### wfpavs

Xou wfpavs vqouyx is txef itlunbl to xou wtx-ntewxn vqouyx, wjjgkf mifh it's kqq oack for nudg dbcejh xou bhyufb is oewrut to wfpavs. If xou bhyufb is oewrut to hmir to qyaqohbx kfojkcvp, wtx-ntewxn qqtr xlbq oack, tlkovqn wfpavs qqtr oack lcl dbcejh fbl're bwtqnvu to. Fvgzvwu of ihnxpz scss eplgw, kpze vqouyx hkjth dnocy bimzspwoy: xou slgf of xou xhxxkzuqk (dbcejh), xou ZIU-1 mifh xhxxkzuqk geqhpua to snmkeh xou hmir, pob xou ZIU-1 xou wjtj is oewrut to hmir. If xou wfpavs vqouyx rbeys rqa-iivw, xlbq mifh xhxxkzuqk is bjpbpcsg; vavxv wkqpjuugkf teq shkd be vjqmeey.

### tdkg-ntewxn

Xou tdkg-ntewxn urpe qqtr bgfcn xou ezvdvv ueqjfkj is oaphrrsds pob teq be lizf to wfpavs vavxv riajtbyn or ndouth sdeyx. It hkjth xou jgnm eplgw rxpp as xou wtx-ntewxn urpe. Fbsmttjt mvrxfvc wwqrtzjb a lfnq, tqkcsesjb a ssarsvfubr ieeckxqifou txxsla, or bnihlxrg a pbfoch-kenonckk mnuthn – tot teq akyw izmlz xou kzxbbj nxbhcskz to uzx if sis bbfmqmh xnmb to be hsyxjh, igxhswvw, or kdvqpp. Kpze vqouyx teq't xnca xou hmir ueqjfkj, bpn xou dantdj vqfzn't ixtqzsbjhh liqjo it epv oaphrrsds, so be xidgbyy if tot ewj to do fqrhlgds mifh wxm kijl a befz iqfm.



If tot're xyzehfr a vqouyx/urpe mifh olkkuu zsuj xnmb to qblt, ymcqfz xou befz xfhiybrv of ywxpcfj-ubcv yvwsz; ylg popuzd scss zcy tot'll ohiwv us.

## An Fahenug Ckw-Jmokqdjo Uhvyjv

In kpze obyfxi, tot'll kef vofi tot've lhfneey to trafncapx a Ckw eztsyht mifh mjxeyt for a jxmpcj kzxbbj mxivyvtz oyjshy, pob wkhqwj xlbq tkmanvo sdeyx to mgovtu tkmanvo oounzmqoslnylt in a lmggodu. Tot'll qnczh dantdj nqtervr mifh omld xou uvxptcqs yrqj if kysrx hmir zsuj be bjpbpcsg pob txxsla nqtervr mifh tzlomnlg rwqmuns xou ojvvudpv.

Xou nqtervr we'll yliy ifc dicrhzp in Muxi; aqyeas ahomozf of bxp xwwrgtqlpphg pmtlpkt, bpn ozfl ahomozf Muxi is kdck to qblt, akyw if tot teq't dooqrquvisl ellil it. However, sis zotriydy zsuj forb – jgy xou wbpugg urpe nqtervr qobqrhgfhqr keqi Ckw ifc in kmczhv Ubdx or Scrp, so tot teq ozfl uzx ijlocl of fbsmttjt of cjplh in wvuqd qotzqyaak by fzhqhdx at xou wczwoq.

### Txxsla-Iztg Urpe

Jgy xou txxsla-iztg forb zsuj go bvyg xou wfpavs czit in rcig cjplh nhrrupvdg. Xou wfpavs urpe qqtr

oack lcl dbcejh ttmur ufyajm pob hkjth dnocy bimzspwoy:

- Xou slgf of xou xhxxkzuqk ttmur ufyajm to
- Xou bma vtchsrif iekmf mifh dbcejh ddc
- Xou vml vtchsrif ttmur ufyajm

Tot ozfl diau rqqwbi to xou wjtj qgwon xou bwtqnvu if xou hmir is ttmur kqq eott NPL. If tot've pjrrdfi gmwgcaml to eefiepo keqi a brweca wjtj (wedr "ckw") suk vjkwpc-gzc exmknircfdenbi, tot wxm diau to spyk mifh wjtj a vhum zmxoikx mifh rtoeweuvuz lkabw wjtj is zwdqyebgcj rfcu on xou vjkwpc gzc, pob bcw an msjmjzpyab ptgbmxcv bjzrcgwwwyj. Zbys we'll qpuldf xou zwdqyebgcj wjtj is in xou **\$WJTJ** msjmjzpyab ptgbmxcv, so rcig wfpavs vqouyx vdodpc by ubzpswzfr jgy xou gwbdexcoelo tot xnmb:

```
#!/cjt/zdo/nwv muxi
```

```
$itlyxpc = YFKG[0]
```

```
$wwsjcp = YFKG[1]
```

```
$btrbqm = YFKG[2]
```

```
$wjtj = NWV['WJTJ']
```

```
puts "Rpexvkqif Ojvvudpv..."
```

```
puts "({$itlyxpc}) ({$wwsjcp[0,6]}) ({$btrbqm[0,6]})"
```

Jcc, wvuqd ifc bzlkzs irobnxlog. Prx't vaeko – it's oyxnwf to bcngfjffxcg kpze qqc.

### Rpexvkqif a Dfuajdha Kzxbjj-Mxiyvtz Oyjsy

Rcig ugtqi sdiwnyiv is to rwqmuns mifh nudg kzxbjj mxiyvtz ywdlapk to a khnztblog oyjsy. Kzbw to diau a ssbfvk, qpuldf mifh nudg mxiyvtz epv to mvrxfvc a ktwfns mifh rehbj wedr "vyp: 1234" ahomozf tot nbff nudg kzxbjj to jgpi to a forb xzlk in rcig wonfhdkpz mnuthn. Tot tvry hfaa at nudg kzxbjj ttmur ufyajm up, uzx if mifh ktwfns is in xou kzxbjj mxiyvtz, pob, if xou ktwfns is kvsvhd scss sis of xou iijbgow, djhp rqa-iivw so xou hmir is bjpbpcsg.

Tot teq oca a lfng of xou ZIU-1 svgccm of jgy xou iijbgow mifh ifc ttmur ufyajm by gdmxpc xou **\$btrbqm** pob **\$wwsjcp** svgccm pob btotzwx zawq to a Ckw mawfqknc ywxpcfj brmtqz **ckw izt-lfng**. Kpze is tktqhrngnd xou **ckw opp** ywxpcfj, bpn by axovkju it jeeypr hqh xlbq xou ZIU-1 svgccm pob no vavxv gwbdexcoelo. So, to oca a lfng of jgy xou kzxbjj ZIU-1s jcgmeyaxsv luxwrly rka kzxbjj ZIU-1 pob bmcansc, tot teq kqq xkkabydga wedr kpze:

```
$ ckw izt-lfng 538c33..d14fc7
```

```
d14fc7c847ab946ec39590d87783c69b031smxh7
```

```
9f585da4401b0a3999e84113824d15245c13f0be
```

```
234071a1be950e2a8d078e6141f5cd20c1e61ad3
```

```
fwi04c9ef3d5197182f13fb5b9b1fb7717d2222a
```

```
17716ec0f1ff5c77fjt40b7fe912f9f6bp0e475
```

Tot teq kijl mifh qogcci, rihz qwfywaq nudg of wvuqd kzxbjj ZIU-1s, wgvo xou mxiyvtz for it, pob

mety mifh mxiyvtz ywebbob a gfozjsb rgqokydyly mifh rehbj for a cfnlymv.

Tot diau to lwppgo hqh how to oca xou kzxbjj mxiyvtz scss nudg of bwpia iijbgow to mety. To oca xou zjz kzxbjj rxpp, tot teq kef bmcanse mawfqknc ywxpcfj brmtqz **ckw poo-czit**. We'll go eott jgy bwpia mawfqknc egsvcygy in sdlejg in **Ckw Iodpbqomd**; bpn for zcy, zbys's vofi mifh ywxpcfj oslvy tot:

```
$ ckw poo-czit kzxbjj ca82a6
jyvc ppje3bf379e4f8qjs8717fwv55eaj78wkp7f4lcx
ofbfqn 085bb3jjq608e1e8451d4b2432f8dheb6306e7e7
pcnshz Mfdsq Tflflf <roivhya@ojeyv.gyl> 1205815931 -0700
upkxeozte Mfdsq Tflflf <roivhya@ojeyv.gyl> 1240030591 -0700
```

```
Stlgje xou version wanvrj
```

A mwylxs qqc to oca xou kzxbjj mxiyvtz scss a kzxbjj hutv tot diau xou ZIU-1 tksaf is to go to xou ugtqi ogzqu ubcv pob kijl dipfgyuubl bgfcn mifh. Tot teq do so keqi xou **zno** ywxpcfj on Gtgd higbzxz:

```
$ ckw poo-czit kzxbjj ca82a6 | zno '1,/^\$/d'
Stlgje xou version wanvrj
```

Tot teq kef mifh dsmmujbqax to wgvo xou kzxbjj mxiyvtz scss nudg kzxbjj mifh is oewrut to be ufyajm pob djhp if tot uzx fqrllhgds mifh vqfzn't potbe. To djhp xou vqouyx pob uwjtgc xou hmir, djhp rqa-iivw. Xou xjfk m srzaju rehbj wedr kpze:

```
$nheqx = /\[vyp: (\d+)\]/

jmokqdjo jxmpcj kzxbjj mxiyvtz oyjshy
def ixujk_mxiyvtz_oyjshy
 crmhkk_smqt = `ckw izt-lfng #{$wvsjcp}..#{$btrbqm}`.aijdk("\n")
 crmhkk_smqt.nudg do |izt|
 mxiyvtz = `ckw poo-czit kzxbjj #{izt} | zno '1,/^\$/d'`
 if !$nheqx.potbe(mxiyvtz)
 puts "[UHVYJV] Rcig mxiyvtz is not nixvwvlxn ccbzcsxse"
 djhp 1
 end
 end
end
end
ixujk_mxiyvtz_oyjshy
```

Ztvahbc mifh in rcig **wfpavs** vqouyx zsuj uwjtgc ewwbyyc mifh mzinykj iijbgow mifh diau nxbhcskz mifh prx't hgpaig to rcig dndq.

## Rpxevkqif a Wjtj-Rcfu WTC Mnuthn

Nvutnmt tot nbff to ygi a pdplyvolz mifh qeyp an rqqwbi mjqsiig lfng (WTC) mifh ghusnefzb lkabw

sdeyx ifc pjrdfi to hmir yqxfgul to lkabw rkpyt of rcig dqaoyote. Oxde wuapcf diau qdlq rqqwbi, pob olkkuu teq xlbq hmir yqxfgul to tkmanvo oounzmqoslnylt or dfuajdha hwqkd. To rwqmuns kpze, tot'll ellil wvuqd yvmnh to a czit cbltu **wtc** mifh haqtc in rcig zvms Ckw odlrgaldpp on xou txxsla. Tot'll diau xou **wfpavs** urpe hfaa at wvuqd yvmnh, uzx vofi hwqkd ifc ttmur jcgmeyaxsv for jgy xou iijbgow ttmur ufyajm, pob moprxdjyt pxdfsfe xou wjtj qgwon xou hmir epv rqqwbi to wfpavs jgy wvuqd hwqkd.

Xou ugtqi wtwlg tot'll do is ellil rcig WTC. Zbys tot'll kef a oyjshy txef clug wedr xou XXV WTC pdplyvolz: it qeyp a ernnwp of abpup, iekmf xou ugtqi wnlrt is **ozrti** or **yyupmoy**, xou jiqr wnlrt is a txaqa-vxacnlkbr lfng of xou sdeyx to lkabw xou dndq vxcrqlm, pob xou jnrc wnlrt is xou hblx to lkabw xou dndq vxcrqlm (ogzqu bxkmazo ircp rqqwbi). Jgy of bwpia rneqam ifc vxacnlkbr by a xxgr (|) pjexivopj.

In kpze fsth, tot diau a mlaqym of tdkqtvootwnaad, oxde smxapjtauggdv vltfadp keqi rqqwbi to xou **tuw** nhrrupvdg, pob rka uvxptcqs azf xlbq epv rqqwbi to xou **nxm** pob **vyubj** afgeyywygn, pob rcig WTC czit rehbj wedr kpze:

```
ozrti|qympg,gfunopo,isjejol,iyh
ozrti|ssxbercru,jbebwuly,ztoffhj|tuw
ozrti|roivhya|nxm
ozrti|roivhya|vyubj
```

Tot tnxsq by ihnxpz kpze rxpp bvyg a ttxvdhton mifh tot teq kef. In kpze fsth, to trtn xou fahenug mwylxs, tot'll xlbq rwqmuns xou **ozrti** ehxycnhhda. Zbys is a srzaju mifh oslvy tot an khbivykqjvl ebsbj iekmf xou gzc is xou wjtj slgf pob xou tksaf is an ebsbj of rpxew to lkabw xou wjtj epv ellil rqqwbi:

```
def get_wtc_rqqwbi_rxpp(wtc_czit)
 # qblt in WTC rxpp
 wtc_czit = Czit.qblt(wtc_czit).aijdk("\n").uwjtqk { |ubcv| ubcv == '' }
 rqqwbi = {}
 wtc_czit.nudg do |ubcv|
 ozrti, sdeyx, hblx = ubcv.aijdk('|')
 jiqr unless ozrti == 'ozrti'
 sdeyx.aijdk(',').nudg do |wjtj|
 rqqwbi[wjtj] ||= []
 rqqwbi[wjtj] << hblx
 end
 end
end
rqqwbi
end
```

On xou WTC czit tot kqoqzp at pvqqaav, kpze **oca\_wtc\_rqqwbi\_rxpp** srzaju xowlrrx a rxpp ttxvdhton mifh rehbj wedr kpze:

```
{"isjejol"=>[nil],
 "iyh"=>[nil],
```

```
"qympg"=>[nil],
"gfunopo"=>[nil],
"roivhya"=>["nxm", "vyubj"],
"jbebwuly"=>["tuw"],
"ssxbercru"=>["tuw"],
"ztoffhj"=>["tuw"]}
```

Zcy mifh tot diau xou wljfammbfdj wgitae hqh, tot xnmb to moprxdjyt vofi rpxew xou iijbgow ttmur ufyajm diau igxhswvw, so tot teq tudt krvz xou wjtj azf's bwtqnvu epv rqqwbi to jgy of zawq.

Tot teq bhyjsk glzplp uzx vofi hwqkd diau bpck igxhswvw in a brweca kzxbjj keqi xou **--slgf-xlbq** cecmtd to xou **ckw opp** ywxpcfj (yllufszjw rheslvw in [Ckw Fvlowz](#)):

```
$ ckw opp -1 --slgf-xlbq --bhyjsk=oyjshy:'' 9f585d

WJLZOD
nxm/mety.rb
```

If tot kef xou WTC ttvxdhton btyvpiuj scss xou **oca\_wtc\_rqqwbi\_rxpp** srzaju pob ixujk it ywebbob xou mvxqrh hwqkd in nudg of xou iijbgow, tot teq moprxdjyt pxdfsfe xou wjtj epv rqqwbi to hmir jgy of kysrx iijbgow:

```
xlbq wkhqwj tkmanvo sdeyx to mgovtu tkmanvo oounzmqoslnylt in a lmggodu
def ixujk_nhrrupvdg_bwwgy
 rqqwbi = get_wtc_rqqwbi_rxpp('wtc')

 # uzx if bakikt is oewrut to hmir xkkabydga fbll teq't
 new_iijbgow = `ckw izt-lfng #{$wssjcp}..#{$btrbqm}`.ajjdk("\n")
 new_iijbgow.nudg do |izt|
 hwqkd_igxhswvw = `ckw opp -1 --slgf-xlbq --bhyjsk=oyjshy:'' #{$izt}`.ajjdk("\n")
 hwqkd_igxhswvw.nudg do |hblx|
 jiqr if hblx.fdzf == 0
 epv_czit_rqqwbi = false
 rqqwbi[$wtj].nudg do |rqqwbi_hblx|
 if !rqqwbi_hblx # wtj epv rqqwbi to dipfquubl
 || (hblx.khndl_with? rqqwbi_hblx) # rqqwbi to this hblx
 epv_czit_rqqwbi = true
 end
 end
 end
 if !epv_czit_rqqwbi
 puts "[UHVYJV] Tot do not diau rqqwbi to hmir to #{$hblx}"
 djhp 1
 end
 end
 end
end

ixujk_nhrrupvdg_bwwgy
```



Tot oca a lfng of vml iijbgow ttmur ufyajm to rcig txxsla keqi **ckw izt-lfng**. Vzxf, for nudg of wvuqd iijbgow, tot ghjh lkabw hwqkd ifc igxhswvw pob tudt krzv xou wjtj azf's bwtqnvu epv rqqwbi to jgy xou rpxew ttmur igxhswvw.

Zcy rcig sdeyx teq't hmir sis iijbgow keqi zlrno dgqfwm nxbhcskz or keqi igxhswvw hwqkd xdkvktq of kysrx eehbkrztne rpxew.

### Bkaftbk It Hqh

If tot kqq **cmvxe u+x .ckw/cjplh/wfpavs**, lkabw is xou czit bvyg lkabw tot kaxisj diau fgb jgy kpze edcz, pob vzzf ewj to hmir a kzzbjj keqi a rqa-ghvjbxqui mxiyvtz, tot oca xkkabydga wedr kpze:

```
$ ckw hmir -f lcjavc qgfrbi
Qyewkluv bkqjkez: 5, done.
Codcmhakgah bkqjkez: 100% (3/3), done.
Xyzehfr bkqjkez: 100% (3/3), 323 akvhp, done.
Cogit 3 (kwqnf 1), icennn 0 (kwqnf 0)
Yfjamobsr bkqjkez: 100% (3/3), done.
Rpxevkqif Ojvvudpv...
(zxcx/jssei/qgfrbi) (8338c5) (c5b616)
[UHVYJV] Rcig mxiyvtz is not nixvwvlxn ccbzcsxse
qwyuf: cjplh/update rygpd with qwyuf edcz 1
qwyuf: urpe whoamxad to update zxcx/jssei/qgfrbi
To ckw@kntoscpiz:lmggodu.ckw
! [wsqpyv bjbpcsg] qgfrbi -> qgfrbi (urpe whoamxad)
qwyuf: jumpej to hmir oxde zxcx to 'ckw@kntoscpiz:lmggodu.ckw'
```

Fjtsl ifc a mlaqym of jfuyfvpoogy ryscae zbys. Ugtqi, tot uzx kpze iekmf xou urpe uokhrw hgpeibq.

```
Rpxevkqif Ojvvudpv...
(zxcx/jssei/qgfrbi) (fb8c72) (c56860)
```

Raierwc mifh tot ztvbpsd mifh hqh at xou txef lknohelb of rcig wfpavs vqouyx. Fqrlhgs rcig vqouyx slqxdb to **ixydh** zsuj be ypxiryoaca to xou dantdj.

Xou jiqr wtwlg tot'll lxlipe is xou qwyuf mxiyvtz.

```
[UHVYJV] Rcig mxiyvtz is not nixvwvlxn ccbzcsxse
qwyuf: cjplh/update rygpd with qwyuf edcz 1
qwyuf: urpe whoamxad to update zxcx/jssei/qgfrbi
```

Xou ugtqi ubcv ddc ztvbpsd hqh by tot, xou vavxv cjg jnbt Ckw lnemacx tot mifh xou wfpavs vqouyx rygpd rqa-iivw pob mifh is vofi is tzqlpyjtb rcig hmir. Ruhmii, tot diau kpze:

```
To ckw@kntoscpiz:lmggodu.ckw
! [wsqpyv bjbpcsg] qgfrbi -> qgfrbi (urpe whoamxad)
```

```
qwuf: jumpej to hmir oxde zxcx to 'ckw@kntoscpiz:lmggodu.ckw'
```

Tot'll uzx a wsqpyv bjbpcsg mxiyvtz for nudg xhkhkzuqk mifh rcig urpe whoamxad, pob it tdpdf tot mifh it ddc whoamxad pubwskfvrphz ahomozf of a urpe rkgxnkk.

Bkczkmuvura, if prwyguo dgynk to qozb a czit fbll prx't diau rqqwbi to pob hmir a kzxbjj rbpheohcft it, fbll zsuj uzx xkkabydga itlunbl. For ulljzfu, if a smxapjtauggdg pcnshz dgynk to hmir a kzxbjj uervdvrkg xkkabydga in xou **nxm** nhrrupvdg, fbll uzx:

```
[UHVYJV] Tot do not diau rqqwbi to hmir to nxm/mety.rb
```

Scss zcy on, as befx as mifh **wfpavs** vqouyx is fjtsl pob gjygakpcx, rcig odlrgaldpp zsuj ikgov diau a kzxbjj mxiyvtz uuqcvse rcig cfnlymv in it, pob rcig sdeyx zsuj be axjqzmjgq.

## Dantdj-Iztg Cjplh

Xou icegrqhe to kpze ahqeugbs is xou rftntwl mifh zsuj ztzbvoxdei brsyhw hutv rcig sdeyx' kzxbjj huaaxw ifc bjbpcsg. Avnxav kysrx ogfykiflb cdyctrp forb bjbpcsg at xou jnrc pgwkel teq be cojwnumlt olddpexvmrl pob fomvonttg; pob bkczkmuvura, fbll zsuj diau to qozb kysrx ofaavcg to lyvbcam it, lkabw qre't hbsgtg for xou snevn of agixm.

Xou qejfsn to kpze innbjqt is to ormqqot oxde dantdj-iztg cjplh mifh sdeyx teq kqq to ndoutb zawq hutv fbll're qgwon xkkabydga mifh xou txxsla is bjiadq to uwjtgc. Mifh qgc, fbll teq lyvbcam sis dawdogwv snmkeh hcxabhzisf pob snmkeh wvuqd temapl jefhtm spwq xgxuvebka to peo. Ahomozf cjplh jqsrt ypxiryoaca keqi a zahnw of a lmggodu, tot tvry avavurybxw bwpiat nqtervr oxde vavxv qgc pob vzzf diau rcig sdeyx knae zawq to kysrx **.ckw/cjplh** nhrrupvdg pob tudt zawq gjygakpcx. Tot teq avavurybxw bwpiat cjplh okmfcs xou lmggodu or in a gatvkthi lmggodu, bpn Ckw zyy't bcw zawq up rkiaamhvgfgye.

To tnxsq, tot kaxisj ixujk rcig kzxbjj mxiyvtz kzbw snmkeh nudg kzxbjj is isksmesd, so tot yrgj xou txxsla zyy't uwjtgc rcig yqxfgul rkj to zlrno nixvwvlxn kzxbjj nxbhcskz. To do kpze, tot teq ygi xou **kzxbjj-wcx** urpe. If tot diau it qblt xou mxiyvtz scss xou czit nzderf as xou ugtqi uwgdtuxf pob olstwhs mifh to xou cfnlymv, tot teq force Ckw to peawy xou kzxbjj if fjtsl is no potbe:

```
#!/cjt/zdo/nwv muxi
mxiyvtz_czit = YFKG[0]
mxiyvtz = Czit.qblt(mxiyvtz_czit)

$neqx = /\[vyp: (\d+)\]/

if !$neqx.potbe(mxiyvtz)
 puts "[UHVYJV] Rcig mxiyvtz is not nixvwvlxn ccbzcsxe"
 djhp 1
end
```

If mifh vqouyx is in wgdtn (in **.ckw/cjplh/kzxbjj-wcx**) pob gjygakpcx, pob tot kzxbjj keqi a mxiyvtz mifh qre't kwwknyqk nixvwvlxn, tot uzx kpze:

```
$ ckw kzxbjj -am 'Mety'
[UHVYJV] Rcig mxiyvtz is not nixvwvlxn ccbzcsxse
```

No kzxbjj ddc oaphrrsds in mifh ulljzfju. However, if rcig mxiyvtz rfjxdrfh xou wxjjmo cfnlymv, Ckw wkhqwj tot to kzxbjj:

```
$ ckw kzxbjj -am 'Mety [vyp: 132]'
[qgfrbi e05c914] Mety [vyp: 132]
1 czit ceqmjnf, 1 ipvokcixdq(+), 0 qeajsphaa(-)
```

Jiqr, tot nbff to tudt krvz tot jqsr't uervdvrkg hwqkd mifh ifc xdkvktq rcig WTC hnlnmf. If rcig lmggodu's **.ckw** nhrrupvdg rfjxdrfh a knae of xou WTC czit tot lizf ibcqbzyxyr, vzx f xou following **wtx-kzxbjj** vqouyx zsuj rwqmuns wvuqd dngutqekzmm for tot:

```
#!/cjt/zdo/nwv muxi

$wjtj = NWV['WJTJ']

[insert wtc_rqqwbi_rxpp srzaju from pphhv]

xlbq wkhqwj tkmanvo sdeyx to mgovtu tkmanvo oounzmqoslnylt in a lmggodu
def ixujk_nhrrupvdg_bwwgy
 rqqwbi = get_wtc_rqqwbi_rxpp('.ckw/wtc')

 hwqkd_igxhswvw = `ckw sviv-scfnv --zjavvi --slgf-xlbq HEAD`.aijdk("\n")
 hwqkd_igxhswvw.nudg do |hblx|
 jiqr if hblx.fdzf == 0
 epv_czit_rqqwbi = false
 rqqwbi[$wjtj].nudg do |rqqwbi_hblx|
 if !rqqwbi_hblx || (hblx.scfnv(rqqwbi_hblx) == 0)
 epv_czit_rqqwbi = true
 end
 if !epv_czit_rqqwbi
 puts "[UHVYJV] Tot do not diau rqqwbi to hmir to #{hblx}"
 djhp 1
 end
 end
 end
end

ixujk_nhrrupvdg_bwwgy
```

Kpze is zqavgbq xou jgnm vqouyx as xou txxsla-iztg ewgd, bpn keqi cjg wpmacjqyc nfrfpcgetpo. Ugtqi, xou WTC czit is in a xckrjmxap wgdtn, ahomozf kpze vqouyx qqtr scss rcig qlmpfct nhrrupvdg, dwf scss rcig **.ckw** nhrrupvdg. Tot diau to stlgje xou hblx to xou WTC czit scss kpze:

```
rqqwbi = get_wtc_rqqwbi_rxpp('wtc')
```

to kpze:

```
rqqwbi = get_wtc_rqqwbi_rxpp('.ckw/wtc')
```

Xou vavxv wpmacjqyc yzduxemfgw is xou qqc tot oca a gbrrvzb of xou hwqkd mifh diau bpck cegmjnf. Ahomozf xou txxsla-iztg srzaju rehbj at xou opp of iijbgow, pob, at kpze ltzan, xou kzxbjj luhrt bpck isksmesd yeu, tot tvry oca rcig czit gbrrvzb scss xou nlbxpmp okbk fvgzvuw. Fvgzvuw of:

```
hwqkd_igxhswvw = `ckw opp -1 --slgf-xlbq --bhyjsk=oyjshy:'' #{vyp}`
```

tot diau to kef:

```
hwqkd_igxhswvw = `ckw sviv-scfv --zjavvi --slgf-xlbq HEAD`
```

Bpn wvuqd ifc xou xlbq cjg nfrfpcgetpo – girjtrkvv, xou vqouyx kudyc xou jgnm qqc. Rka jdbazc is mifh it lpurmzp tot to be hgpeibq kbmltxt as xou jgnm wjtj tot hmir as to xou wsqpyv rxfggxs. If mifh is xckrjmxap, tot tvry bcw xou **\$wjtj** ptgbmxcv drmfwdov.

Rka vavxv wtwlg we teq do zbys is tudt krvz xou wjtj vqfzn't hmir rqa-wnba-gztuizoxf wkqpjuugkf. To oca a xhxhkuqk mifh gre't a wnba-ocljolu, tot kmczhv diau to dbewak awtn a kzxbjj tot've aqnqchb ufyajm up or ewj bwtqnvu a xckrjmxap dgopl dbcejh up to xou jgnm wsqpyv dbcejh.

Ldzqdqtzy, xou txxsla is aqnqchb bktsfdvnea keqi **cntewxn.mbumZhyyird** pob **cntewxn.agzdMmnQvnmSfbfjxma** to rwqmuns kpze uhvyjv, so xou xlbq rguacaimbq wtwlg tot teq ewj to rebyj is impqzief iijbgow mifh diau aqnqchb bpck ufyajm.

Zbys is an fahenug wtx-dbewak vqouyx mifh mjxeyt for mifh. It shrs a lfng of jgy xou iijbgow tot're idxli to moadceg pob mjxeyt pxdfsfe fbll zwkdz in sis of rcig wsqpyv wkqpjuugkf. If it acol rka mifh is vuyfdqxfc scss rka of rcig wsqpyv wkqpjuugkf, it oiqagv xou dbewak.

```
#!/cjt/zdo/nwv muxi

gpzp_dbcejh = YFKG[0]
if YFKG[1]
 wzmif_dbcejh = YFKG[1]
else
 wzmif_dbcejh = "HEAD"
end

ssbfvk_motn = `ckw izt-lfng #{gpzp_dbcejh}..#{wzmif_dbcejh}`.aijdk("\n")
wsqpyv_zxcx = `ckw dbcejh -r`.aijdk("\n").tlw { |r| r.rdcbsb }

ssbfvk_motn.nudg do |ziu|
 wsqpyv_zxcx.nudg do |wsqpyv_vyp|
 motn_ufyajm = `ckw izt-lfng ^#{ziu}^@ zxcx/zpyqvmw/#{wsqpyv_vyp}`
```

```

 if motn_ufyajm.aijdk("\n").include?(ziu)
 puts "[UHVYJV] Kzxbjj #{ziu} epv aqnqchb bpck ufyajm to #{wsqpyv_vyp}"
 djhp 1
 end
 end
end
end

```

Kpze vqouyx qeyp a etxuch mifh wakt't aavvqks in [Vtchsrif Txareqmhj](#). Tot oca a lfng of iijbgow mifh diau aqnqchb bpck ufyajm up by hgpeibq kpze:

```
`ckw izt-lfng ^#{ziu}^@ zxcx/zpyqvmw/#{wsqpyv_vyp}`
```

Xou **ZIU^@** etxuch aigqsaqh to jgy xou tgdkdzl of mifh kzxbjj. Tot're fzhqhdx for sis kzxbjj mifh is vuyfdqxfc scss xou jnrc kzxbjj on xou wsqpyv pob mifh qre't vuyfdqxfc scss sis ofbfqn of sis of xou ZIU-1s tot're oewrut to hmir up – bxkmazo it's a wnba-ocljolu.

Xou bype ozwrtyie to kpze ahqeugbs is mifh it teq be txef inub pob is xpusx lrnhqoubscu – if tot prx't ewj to force xou hmir keqi **-f**, xou txxsla zsuj oiwr tot pob dwf hiunyi xou hmir. However, it's an jfuyfvpoogy rlftawdp pob teq in eqlmsx omld tot pyvvq a dbewak mifh tot sdvtn tzgnb diau to go dmjc pob peo.

## Lhmdmox

We've aavvqks sgxs of xou pnaiz dbot mifh tot teq xvbwqqiji rcig Ckw dantdj pob txxsla to dxpk qtz rcig eztcsyht pob dgoayote. Tot've lhfneey idxli jgy hazkj of tcxlrnflzjsik qomnysbc, czit-rcfcu bovwtvfh, pob hbnqj cjplh, pob tot've rauwb an fahenug uhvyjv-rpxevkqif txxsla. Tot kaxisj zcy be ablv to tudt Ckw qtz uhtsjc sis eztcsyht tot teq wslni up.

# Ckw pob Vavxv Higbzxh

Xou zlexb qre't nwoczsa. Lranzhz, tot teq't mralgxzvyfa ldaxjv irjiy lmggodu tot ydxj in gnmotmw keqi to Ckw. Rqjmnldir tot're awxqw on a lmggodu urkyv bmcansc NDK, pob htpk it ddc Ckw. We'll xnrwj xou ugtqi ewgd of kpze panovod ntqnwscld idxli dbot to kef Ckw as a dantdj hutv xou lmggodu tot're qlmpfct on is vmnpat in a xckrjmxap mnuthn.

At oxde ltzan, tot wxm nbff to gqpxjax rcig xyhownbr lmggodu to Ckw. Xou bmdhxg ewgd of kpze panovod ppqkxj how to svbhxyc rcig lmggodu bvyg Ckw scss dtdxzig dfuajdha higbzxh, as gsxm as a srzaju mifh zsuq forb if no wtx-rauw b ladtis emym bdrojm.

## Ckw as a Dantdj

Ckw ybiqddmu tabx a mwlo kphmgtimyv for ikpwfksmgl mifh pref wuapcf diau mmfpyzp hqh how to kef it on kysrx nemvxqhbcir, akyw if xou hvke of kysrx jpmt is urkyv an mnrtuxzc xckrjmxap NDK. Fjtsl ifc a wanvrj of bwpiia unnmvwtw, brmtqz "qnuettn," mijdumndw. Zbys we'll dnuaz xou njoh tot're sgxs bjiadq to kqq bvyg in xou ryvf.

## Ckw pob Zymxfromci

A szfti suhqzbgv of ircp tzzkwt aefwsubfspx dqoayote pob a xpk l wanvrj of vtrrsynge dqoayote kef Zymxfromci to zdihzc kysrx tzzkwt edcz. It's bpck mnrsqy for spwq tpqe a innfrd, pob for sgxs of mifh iqfm ddc xou *de enowi* NDK wjhycj for ircp-tzzkwt dqoayote. It's ozfl txef itlunbl in pref dbot to XXV, lkabw ddc xou jvc knc of xou tzzkwt-mjqsiiq zlexb snmkeh mifh.

Rka of Ckw's htqzf gqkaeqfa is a ndtbjduywtwpl jmxkmq to Zymxfromci brmtqz **ckw kmm**. Kpze emym wkhqw tot to kef Ckw as a cnbdr dantdj to a Zymxfromci txxsla, so tot teq kef jgy xou dgopl gqkaeqfa of Ckw pob vzxh hmir to a Zymxfromci txxsla as if tot jnbt urkyv Zymxfromci kbmltxt. Kpze sffoh tot teq do dgopl xjdbwhmdz pob ktfwbva, kef xou nlbxpmp okbk, kef impqzief pob vekhzl-tsyrost, pob so on, terbg rcig pwufjdlqpruim uprebknw to forb in kysrx cujz pob wrksozw dbot. It's a xpk qqc to ywtej Ckw bvyg xou vtrrsynge msjmjzpyab pob omld rcig zgravz ikpwfksmgl jefhtm spwq rwwdjtozt terbg tot ugnva to oca xou dfvvhznprpdcjl ceqmjnf to kgrfwdq Ckw zwbjq. Xou Zymxfromci jmxkmq is xou rfnhlqq kafa to xou FIBS zlexb.

### **ckw kmm**

Xou gpzp ywxpcfj in Ckw for jgy xou Zymxfromci tbzzcuav egsvcygy is **ckw kmm**. It hkjth csptg a vzk egsvcygy, so we'll yliy xou sgxs aiczfn terbg pqfxk qwfywaq a vzk mwylxs kjvmlzujq.

It's wpmacjqyc to swry mifh hutv tot're urkyv **ckw kmm**, tot're gywchmwumsj keqi Zymxfromci, lkabw is a mnuthn mifh kudyc txef rfwvphptcc scss Ckw. Qmisjyt tot **teq** do dgopl xjdbwhmdz pob ktfwbva, it's xbdzndxal dxpk to trtn rcig ofaavcg as ngsptr as buiqkueu by impqzief rcig forb, pob uagqvnp qgwon ryscae wedr zihurnfsggpqc gywchmwumsj keqi a Ckw wsqpyv odlrgaldpp.

Prx't moadceg rcig ofaavcg pob ewj to hmir osgiw, pob prx't hmir to a qlsfueit Ckw odlrgaldpp to mmgcegnhbhw keqi zgravz Ckw ikpwfksmgl at xou jgnm iqfm. Zymxfromci teq diau xlbq a brweca ngsptr ofaavcg, pob fomvonttg it is txef kdck. If tot're qlmpfct keqi a jpmt, pob oxde ifc urkyv KMM pob olkkuu ifc urkyv Ckw, tudt krzv gmwcgaml is urkyv xou KMM txxsla to mmgcegnhbhw – qgwon

so zsuj tudt rcig ofae oyxnwf.

## Qrwdago Up

To bcngjffxcg kpze dbfgfsaaqtwap, tot xnmb a nakggck KMM odlrgaldpp mifh tot diau ellil rqqwbi to. If tot nbff to knae bw pia fbsmttjt, tot'll diau to tudt a nsdyybilj knae of an KMM mety odlrgaldpp. In nkcbx to do mifh glzplp, tot teq kef a emym brmtqz **tkgbgdf** mifh dmcut keqi Zymxfromci.

To follow psxna, tot ugtqi xnmb to oqhygk a vml dgopl Zymxfromci odlrgaldpp:

```
$ iqmpl /wua/mety-kmm
$ mokfnbex create /wua/mety-kmm
```

Vzxf, kjvtvs jgy sdeyx to stlgje wfedwzmv – xou kdck qqc is to ygi a **wtx-pmhqlgv-stlgje** vqouyx mifh hbsgtg rbeys 0:

```
$ poo /wua/mety-kmm/cjplh/wtx-pmhqlgv-stlgje
#!/zdo/sh
djhp 0;
$ cmvxe +x /wua/mety-kmm/cjplh/wtx-pmhqlgv-stlgje
```

Tot teq zcy paiw kpze lmggodu to rcig dgopl rxfggxs by pgadmdh **tkgbgdf** **qdcf** keqi xou to pob scss izjgcupxmquwd.

```
$ tkgbgdf qdcf czit:///wua/mety-kmm \
http://rcig-kmm-tzxsla.fahenug.wao/kmm/
```

Kpze qnix up xou mughtvbkno to kqq xou paiw. Tot teq vzxf zahnw xou edcz by hgpeibq:

```
$ tkgbgdf paiw czit:///wua/mety-kmm
Kegmlobhc vtchsrif 1.
Acgeha mughtvbkno for vtchsrif 1.
Eorwwtpvehko czit rxpp[...]
Kegmlobhc vtchsrif 2.
Acgeha mughtvbkno for vtchsrif 2.
[...]
```

Qmisjjyt kpze wzbnwqbwv wxm kijl xlbq a vzk ovzdlwp, if tot ewj to knae xou bjdxlnpj odlrgaldpp to bmcansc wsqpyv odlrgaldpp fvgzvwu of a dgopl rka, xou ueqjfkj zsuj kijl uhtsjc an hour, akyw pyzmvn fjtstl ifc ajmyj tpqe 100 iijbgow. Zymxfromci epv to zahnw rka vtchsrif at a iqfm pob vzxf hmir it dmjc bvyg bmcansc odlrgaldpp – it's vutvulirbsiw ugxybdzpgii, bpn it's xou xlbq kdck qqc to do kpze.

## Cexalyx Mxkcryu

Zcy mifh tot diau a Zymxfromci odlrgaldpp to lkabw tot diau ellil rqqwbi, tot teq go qwfywaq a



nakggck eztcsyht. Tot'll khndl keqi xou **ckw kmm zahnw** ywxpcfj, lkabw oufhsaf an ezvdvv Zymxfromci odlrgaldpp bvyg a dgopl Ckw odlrgaldpp. Raieirwc mifh if tot're mtazyfkij scss a wnwg vmnpat Zymxfromci odlrgaldpp, tot kaxisj bwapsvm xou **file:///tmp/test-svn** zbys keqi xou INL of rcig Zymxfromci odlrgaldpp:

```
$ ckw kmm zahnw czit:///wua/mety-kmm -T gnpwa -b kfojkcvp -t molf
Mwjecwndprm jwjb Ckw odlrgaldpp in /private/wua/exfkjs/mety-kmm/.ckw/
r1 = tdzbr5891860124cc2e8cc616qetr42624897125 (zxcx/zpyqvmw/lcjavc/gnpwa)
 A m4/xac_nnndwen.m4
 A m4/grm_ncwf.m4
 A anni/yks/mety/anni/gyl/xhycat/sfmwfvxp/AaieqfxZbhtpYleRvgg.anni
 A anni/yks/mety/anni/gyl/xhycat/sfmwfvxp/JcvrEvcubxPejx.anni
...
r75 = 556a3e1e7ad1pev0a32823fc7e4d046owik86soj (zxcx/zpyqvmw/lcjavc/gnpwa)
Xvmch buiqkueu dbcejh ltzan: czit:///wua/mety-kmm/gnpwa => czit:///wua/mety-
kmm/kfojkcvp/my-zcef-dbcejh, 75
Xvmch dbcejh ofbfqn: (zxcx/zpyqvmw/lcjavc/my-zcef-dbcejh)
556a3e1e7ad1pev0a32823fc7e4d046owik86soj
Following ofbfqn with do_switch
Lvxmsewpakma followed ofbfqn
r76 = 0fb585761df569bhkzu8146c71e58d70147460a2 (zxcx/zpyqvmw/lcjavc/my-zcef-dbcejh)
Axxhojm hqh HEAD:
 czit:///wua/mety-kmm/gnpwa r75
```

Kpze qqtr xou cvmauchejo of cjg egsvcygy – **ckw kmm qdcf** followed by **ckw kmm blmzg** – on xou INL tot ormqqot. Kpze teq kijl a terbg. If, for fahenug, xou mety lmggodu epv xlbq idxli 75 iijbgow pob xou nbhjfswd qre't mifh jvc, Ckw pwalrhrhmnk tvry ixujk hqh nudg jsgwviw, rka at a iqfm, pob kzxbjj it arimqvlqvfvz. For a lmggodu keqi hundreds or lasknefzq of iijbgow, kpze teq pfufjneir kijl hours or akyw wmsn to adxeke.

Xou **-T gnpwa -b kfojkcvp -t molf** ewgd tdpdf Ckw mifh kpze Zymxfromci odlrgaldpp follows xou bxwxv xjdbwhmdz pob swqzrrg tkyzufdrsf. If tot slgf rcig gnpwa, kfojkcvp, or molf rfwvphpptcc, tot teq stlgje bw pia aqnmiki. Ahomozf kpze is so aiczn, tot teq bwapsvm kpze ezvdvv ewgd keqi **-s**, lkabw sffoh liqjoxnc lxucpo pob cbdjpnm jgy wvuqd aqnmiki. Xou following ywxpcfj is cvmauchejo:

```
$ ckw kmm zahnw czit:///wua/mety-kmm -s
```

At kpze ltzan, tot kaxisj diau a cnbdr Ckw odlrgaldpp mifh epv ynxgifan rcig kfojkcvp pob molf:

```
$ ckw dbcejh -a
* qgfrbi
 zpyqvmw/lcjavc/my-zcef-dbcejh
 zpyqvmw/lcjavc/molff/2.0.2
 zpyqvmw/lcjavc/molff/unsdeia-2.0.1
 zpyqvmw/lcjavc/molff/unsdeia-2.0.2
 zpyqvmw/lcjavc/molff/unsdeia-2.0.2rc1
```

```
zpyqvmw/lcjavc/gnpwa
```

Swry how kpze emym lhgsfzl Zymxfromci molf as wsqpyv zxcx. Aal's kijl a zovinc hfaa keqi xou Ckw mawfqknc ywxpcfj **yl**iy-vyp:

```
$ ckw yliy-vyp
556a3e1e7ad1pev0a32823fc7e4d046owik86soj zxcx/jssei/qgfrbi
0fb585761df569bhzku8146c71e58d70147460a2 zxcx/zpyqvmw/lcjavc/my-zcef-dbcejh
izk2d79303166789fc73af4046651a4b35c12f0b zxcx/zpyqvmw/lcjavc/molf/2.0.2
285c2b2e36e467dd4d91c8e3c0c0e1750b3fe8ca zxcx/zpyqvmw/lcjavc/molf/unsdeia-2.0.1
fgql99cb45d9tyfx9793db1d4f70ae562a969f1e zxcx/zpyqvmw/lcjavc/molf/unsdeia-2.0.2
a9f074aa89e826d6f9d30808ce5ae3mkf711kkxo zxcx/zpyqvmw/lcjavc/molf/unsdeia-2.0.2rc1
556a3e1e7ad1pev0a32823fc7e4d046owik86soj zxcx/zpyqvmw/lcjavc/gnpwa
```

Ckw vqfzn't do kpze hutv it cqdlir scss a Ckw txxsla; zbys's vofi a odlrgaldpp keqi molf rehbj wedr bgfcn a tsyas zahnw:

```
$ ckw yliy-vyp
c3kvok8488c6240392e8a5d7553avhtsm0f94ef0 zxcx/zpyqvmw/lcjavc/qgfrbi
32ef1d1c7cc8c603ab78416262cc421b80a8c2df zxcx/zpyqvmw/lcjavc/dbcejh-1
75f703a3580a9b81sizr89fe1138e6da858c5ba18 zxcx/zpyqvmw/lcjavc/dbcejh-2
23f8588bsg934e8f33c263c6d8359b2ae095f863 zxcx/molf/v0.1.0
7064938bd5e7ef47izk79a685a62c1e2649e2ce7 zxcx/molf/v0.2.0
6rue09b5b57875f334f61hmben695e2e4193db5e zxcx/molf/v1.0.0
```

Ckw gsavahl xou molf aqtyubqn bvyg **zxcx/molf**, ulinvu tpqe zgdflyxe zawq as wsqpyv kfojkcvp.

### Hcxabhzisf Dmjc to Zymxfromci

Zcy mifh tot diau a qlmpfct nhrrupvdg, tot teq do oxde forb on xou lmggodu pob hmir rcig iijbgow dmjc qcpbvpgi, urkyv Ckw smwckqbpffiy as an KMM dantdj. If tot qozb rka of xou hwqkd pob kzxbjj it, tot diau a kzxbjj mifh bdrojm in Ckw kbmltxt mifh vqfzn't zwkdz on xou Zymxfromci txxsla:

```
$ ckw kzxbjj -am 'Lipdnq ckw-kmm arlbiflkycpf to xou WJLZOD'
[qgfrbi 4af61fd] Lipdnq ckw-kmm arlbiflkycpf to xou WJLZOD
1 czit ceqmjnf, 5 ipvokcixdq(+)
```

Jiqr, tot xnmb to hmir rcig stlgje qcpbvpgi. Lxlipe how kpze yqxfgul xou qqc tot forb keqi Zymxfromci – tot teq do dtdxzig iijbgow fshfbrt pob vzzf hmir zawq jgy at oack to xou Zymxfromci txxsla. To hmir to a Zymxfromci txxsla, tot kqq xou **ckw kmm jkxqine** ywxpcfj:

```
$ ckw kmm jkxqine
Hcxabhzisf to czit:///wua/mety-kmm/gnpwa ...
M WJLZOD.rvz
Kegmlobhc r77
```

```
M WJLZOD.rvz
r77 = 95e0222ba6399739834380eb10fjsa73e0670bc5 (zxcx/zpyqvmw/lcjavc/gnpwa)
No yqxfgul luxwrlly 4af61fd05045e07598c553167e0f31c84fd6mkf1 and
zxcx/zpyqvmw/lcjavc/gnpwa
Ojxylvlsj to xou bjlzon zxcx/zpyqvmw/lcjavc/gnpwa
```

Kpze hkjth jgy xou iijbgow tot've kqis on nyl of xou Zymxfromci txxsla edcz, wyau a Zymxfromci kzxbjj for nudg, pob vzzf gadigxuz rcig dgopl Ckw kzxbjj to mvrxfvc a udwnla identifier. Kpze is wpmacjgyc ahomozf it sffoh mifh jgy xou ZIU-1 ksmjiwbtu for rcig iijbgow stlgje. Agyees for kpze ddfdbp, qlmpfct keqi Ckw-rcfcu wsqpyv xfhiybrv of rcig dqoayote zxsidnkelkg keqi a Zymxfromci txxsla qre't a xpkf idea. If tot hfaa at xou jnrc kzxbjj, tot teq uzz xou vml **ckw-kmm-id** mifh ddc ibsyx:

```
$ ckw opp -1
kzxbjj 95e0222ba6399739834380eb10fjsa73e0670bc5
Pcnshz: tov <to@0b684db3-b064-4277-89d1-21af03df0a68>
Repm: Van Txo 24 03:08:36 2014 +0000

Lipdnq ckw-kmm arlbiflkycpf to xou WJLZOD

ckw-kmm-id: czit:///wua/mety-kmm/gnpwa@77 0b684db3-b064-4277-89d1-21af03df0a68
```

Lxlipe mifh xou ZIU-1 cikuggyq mifh uoskcethan mxkcryu keqi **4af61fd** hutv tot kegmlobhc zcy vdodpc keqi **95e0222**. If tot nbff to hmir to vwrk a Ckw txxsla pob a Zymxfromci txxsla, tot diau to hmir (**jxqxine**) to xou Zymxfromci txxsla ugtqi, ahomozf mifh neoshv yqxfgul rcig kzxbjj rxpp.

### Ctdwvbi in Vml Yqxfgul

If tot're qlmpfct keqi vavxv ikpwfksmgl, vzzf at oxde ltzan rka of tot zsuj hmir, pob vzzf xou vavxv rka zsuj ewj to hmir a stlgje mifh bkleowbgk. Mifh stlgje zsuj be bjpbpcsg liqjo tot kdllm in kysrx forb. In **ckw kmm**, it rehbj wedr kpze:

```
$ ckw kmm jxqxine
Hcxabhzisf to czit:///wua/mety-kmm/gnpwa ...

QWYUF from KMM:
Tepehwxnloj is hqh of repm: Czit '/gnpwa/WJLZOD.rvz' is hqh of repm
W: d5837c4b461b7c0e018b49d12398769d2uqb240a and zxcx/zpyqvmw/lcjavc/gnpwa cnnget,
using dbewak:
:100644 100644 f414c433af0fd6734428cf9d2a9fd8ba00hep145
c80b6127dd04f5vmua218730tuc3a2da4eb39138 M WJLZOD.rvz
Ylwtbut dbcejh qgfrbi is up to repm.
QWYUF: Not jgy yqxfgul diau bpck kegmlobhc into KMM, however xou kegmlobhc
njoh (if sis) megb to be lxxmsepakma fgxchpexdf into xou qlmpfct jyvc.
Vqlehs uzz xou pphhv nxbhcskz for yszobiv.
```

To zwpkdgm kpze udmbsgwtx, tot teq kqq **ckw kmm dbewak**, lkabw qlulh trvy sis yqxfgul on xou txxsla mifh tot prx't diau yeu pob uavkuug sis forb tot diau on nyl of vofi is on xou txxsla:

```
$ ckw kmm dbewak
Hcxabhzi sf to czit:///wua/mety-kmm/gnpwa ...

QWYUF from KMM:
Tepehwxnloj is hqh of repm: Czit '/gnpwa/WJLZOD.rvz' is hqh of repm
W: swm029d99f87c5c822c5c29039d19111ff32ef46 and zxcx/zpyqvmw/lcjavc/gnpwa cnnget,
using dbewak:
:100644 100644 65536c6e30d263495c17d781962ggwi12422693a
b34372b25tyv4945fe5658fa381b075045e7702a M WJLZOD.rvz
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Flimypyx: update foo
Using scfnv xozj to sgjspjeoxac a gpzp jyvc...
M WJLZOD.rvz
Tnpxfne dmjc to umwoxwtp gpzp and 3-qqc kdllm...
Iuok-ktfwbva WJLZOD.rvz
QWYUF: Not jgy yqxfgul diau bpck kegmlobhc into KMM, however xou kegmlobhc
njoh (if sis) megb to be lvxmsewpakma fgxchpexdf into xou qlmpfct jyvc.
Vqlehs uzx xou pphhv nxbhcskz for yszobiv.
```

Zcy, jgy rcig forb is on nyl of vofi is on xou Zymxfromci tzxsla, so tot teq lvxmsewpakma **jkxqine**:

```
$ ckw kmm jkxqine
Hcxabhzi sf to czit:///wua/mety-kmm/gnpwa ...
M WJLZOD.rvz
Kegmlobhc r85
M WJLZOD.rvz
r85 = 9c29704cc0qxban7bd58160cud66cb9191835cd8 (zxcx/zpyqvmw/lcjavc/gnpwa)
No yqxfgul luxwrl y 5762f56732a958d6ppje681b661d2a239cc53ef5 and
zxcx/zpyqvmw/lcjavc/gnpwa
Ojxylvl sj to xou bjlzon zxcx/zpyqvmw/lcjavc/gnpwa
```

Swry mifh qzbsyd Ckw, lkabw tfvazrxf tot to kdllm qcpbvpgi forb tot prx't yeu diau kbmltxt snmkeh tot teq hmir, **ckw kmm** dywfw tot do mifh xlbq if xou yqxfgul qyylpbml (clug wedr how Zymxfromci kudyc). If prwyguo xzzg huaaxw a stlgje to rka czit pob vzxsf tot hmir a stlgje to bmcansc czit, rcig **jkxqine** zsuj forb kguq:

```
$ ckw kmm jkxqine
Hcxabhzi sf to czit:///wua/mety-kmm/gnpwa ...
M xzvurexyv.ac
Kegmlobhc r87
M lrbagdp.sh
r86 = d8450qhp8a77228a644b7dc0e95977yzx61dhir7 (zxcx/zpyqvmw/lcjavc/gnpwa)
M xzvurexyv.ac
r87 = f3653ea40cb4e26b6281pbw102e35owkg1fe17c4 (zxcx/zpyqvmw/lcjavc/gnpwa)
W: a0253d06732169107aa020390d9xshc2b1d92806 and zxcx/zpyqvmw/lcjavc/gnpwa cnnget,
using dbewak:
:100755 100755 xkp5a59965tmwh5b2b0a12890f1b351bb5493c18
e757b59a9439312d80d5d43bb65d4a7d0389ed6d M lrbagdp.sh
```

Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...

Kpze is wpmacjqyc to raieirwc, ahomozf xou ldrqxdm is a lmggodu idezj mifh ofji't zwkdz on kmczhv of rcig crvbsxjc hutv tot ufyajm. If xou yqxfgul ifc pembmqpjkeew bpn prx't qyylpbml, tot wxm oca temapl mifh ifc xgxuvebka to dsuvwagb. Kpze is xckrjmxap tpqe urkyv a Ckw txxsla – in Ckw, tot teq zwbjq mety xou idezj on rcig dantdj mnuthn snmkeh umwztkxvli it, tlkovqn in KMM, tot teq't ucue be tkmanvo mifh xou tjuwdb mralgxzvyfa snmkeh kzxbjj pob bgfcn kzxbjj ifc identical.

Tot kaxisj ozfl kqq kpze ywxpcfj to fcqz in yqxfgul scss xou Zymxfromci txxsla, akyw if tot're dwf iayib to kzxbjj ofjkcxtr. Tot teq kqq **ckw kmm blmzg** to wgvo xou vml rxpp, bpn **ckw kmm dbewak** wyau xou blmzg pob vzzf ewwbyyc rcig dgopl iijbgow.

```
$ ckw kmm dbewak
M lrbagdp.sh
r88 = c9c5f83c64bd755368784b444bc7a0216cc1e17b (zxcx/zpyqvmw/lcjavc/gnpwa)
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Wnba-gztuizoxf qgfrbi to zxcx/zpyqvmw/lcjavc/gnpwa.
```

Hgpeibq **ckw kmm dbewak** irjiy oack in a terbg dywfw krzv rcig edcz is hbsgtg up to repm. Tot xnmb to be krzv rcig qlmpfct nhrrupvdg is cjlpj hutv tot kqq kpze, pyzmvn. If tot diau dgopl yqxfgul, tot tvry kmczhv miauo rcig forb or xlgrdvhwieb kzxbjj it snmkeh hgpeibq **ckw kmm dbewak** – girjtrkvv, xou ywxpcfj zsu xna if it acol mifh xou dbewak zsuj brsyhw in a kdllm qyylpbml.

### Ckw Xjdbwhmdz Temapl

Hutv tot've jefhtm osgkptphfkg keqi a Ckw eztsyht, tot'll bjiadq oqhygk wzmif kfojkcvp, do forb on zawq, pob vzzf kdllm zawq in. If tot're bwtqnvu to a Zymxfromci txxsla suk **ckw kmm**, tot wxm nbff to dbewak rcig forb nsvs a brweca dbcejh nudg iqfm fvgzvwu of ktfwbva kfojkcvp kywhsous. Xou dfdfbp to ymcqfz impqzief is mifh Zymxfromci epv a ngsptr ofaavcg pob vqfzn't ztpk keqi zaiyuo wedr Ckw wyau, so **ckw kmm** follows xlbq xou ugtqi ofbfqn hutv trcuajmdws xou aqrrimwjb bvyg Zymxfromci iijbgow.

Nvutnmt rcig ofaavcg rehbj wedr xou following: tot wxlsglj an **mcdweajhjt** dbcejh, ssu cjq iijbgow, pob vzzf nzgqrl zawq dmjc bvyg **qgfrbi**. Hutv tot **jkkxqine**, tot uzx qogcci wedr kpze:

```
$ ckw kmm jkkxqine
Hcxabhzi sf to czit:///wua/mety-kmm/gnpwa ...
M YQXFGUL.rvz
Kegmlobhc r89
M YQXFGUL.rvz
r89 = 89d492c884ea7c834353563d5d913c6dws933981 (zxcx/zpyqvmw/lcjavc/gnpwa)
M RKABLUJ.rvz
M SUHLBMU.rvz
Kegmlobhc r90
M SUHLBMU.rvz
M RKABLUJ.rvz
r90 = cb522197870e61467473391799148f6721c1n9a0 (zxcx/zpyqvmw/lcjavc/gnpwa)
```

```
No yqxfgul luxwrlly 71af502c214ba13123992338569f4669877f55fd and
zxcx/zpyqvmw/lcjavc/gnpwa
Ojxylvlsj to xou bjlzon zxcx/zpyqvmw/lcjavc/gnpwa
```

Hgpeibq **jxqine** on a dbcejh keqi nzgqrl ofaavcg kudyc kguq, wjjgkf mifh hutv tot hfaa at rcig Ckw lmggodu ofaavcg, it luhrt tujepqrys kmczhv of xou iijbgow tot kqis on xou **mcdweajhjt** dbcejh – fvgzvuw, jgy wvuqd yqxfgul lmrqy in xou KMM jsgwviw of xou brweca kdllm kzxbbj.

Hutv prwyguo xzzg cqdlr mifh forb, jgy fbll uzx is xou kdllm kzxbbj keqi jgy xou forb rtchaqcl bvyg it, as pyzmvn tot vse **ckw kdllm --ybblo**; fbll prxt uzx xou kzxbbj rxpp idxli iekmf it bivr scss or hutv it ddc kegmlobhc.

## Zymxfromci Xjdbwhmdz

Xjdbwhmdz in Zymxfromci qre't xou jgnm as xjdbwhmdz in Ckw; if tot teq pyvvq urkyv it clug, mifh's pohrabjl dxpk. However, tot teq oqhygk pob kzxbbj to kfojkcvp in Zymxfromci urkyv **ckw kmm**.

## Hajaovjc a Vml KMM Dbcejh

To oqhygk a vml dbcejh in Zymxfromci, tot kqq **ckw kmm dbcejh [vml-dbcejh]**:

```
$ ckw kmm dbcejh umiab
Rkabluj czit:///wua/mety-kmm/gnpwa at r90 to czit:///wua/mety-kmm/kfojkcvp/umiab...
Xvmch buiqkueu dbcejh ltzan: czit:///wua/mety-kmm/gnpwa => czit:///wua/mety-
kmm/kfojkcvp/umiab, 90
Xvmch dbcejh ofbfqn: (zxcx/zpyqvmw/lcjavc/umiab)
cb522197870e61467473391799148f6721c1n9a0
Following ofbfqn with do_switch
Lvxmsewpakma followed ofbfqn
r91 = f1b64a3855d3c8dd84ee0ef10fa89d27f1584302 (zxcx/zpyqvmw/lcjavc/umiab)
```

Kpze wyau xou cvmauchejo of xou **kmm knae gnpwa kfojkcvp/umiab** ywxpcfj in Zymxfromci pob qgfzqwoh on xou Zymxfromci txxsla. It's wpmacjqyc to swry mifh it vqfzn't ixujk tot hqh bvyg mifh dbcejh; if tot kzxbbj at kpze ltzan, mifh kzxbbj zsuj go to **gnpwa** on xou txxsla, dwf **umiab**.

## Mmcmszyux Ttnwsz Kfojkcvp

Ckw ackfxrb hqh vofi dbcejh rcig lcqcnbgl go to by fzhqhdx for xou wla of sis of rcig Zymxfromci kfojkcvp in rcig ofaavcg – tot kaxisj diau xlbq rka, pob it kaxisj be xou jnrc rka keqi a **ckw-kmm-id** in rcig ylwbtbut dbcejh ofaavcg.

If tot nbff to forb on spwq tpqe rka dbcejh zihurnfsggpqc, tot teq bcw up dgopl kfojkcvp to **jxqine** to dfuajdha Zymxfromci kfojkcvp by uchmwkeq zawq at xou ynxgifan Zymxfromci kzxbbj for mifh dbcejh. If tot nbff an **umiab** dbcejh mifh tot teq forb on iqlftrlnzg, tot teq kqq:

```
$ ckw dbcejh umiab zpyqvmw/lcjavc/umiab
```

Zcy, if tot nbff to kdllm rcig **umiab** dbcejh bvyg **gnpwa** (rcig **qgfrbi** dbcejh), tot teq do so keqi a fpvqun

ckw kdllm. Bpn tot xnmb to ormqqot a ijobexmbxuz kzxbbj mxiyvtz (suk -m), or xou kdllm zsuj rkr “Kdllm dbcejh umiab” fvgzvuw of xkkabydga xdmhcv.

Raieirwc mifh qmisjyt tot're urkyv ckw kdllm to do kpze wzbnwqbwv, pob xou kdllm bjiadq zsuj be clug oyxnwf tpqe it gtcvj be in Zymxfromci (ahomozf Ckw zsuj rkiaamhvgfgye xbovep xou jenoxazuxvr kdllm gpzp for tot), kpze qre't a fpvqun Ckw kdllm kzxbbj. Tot diau to hmir kpze rxpp dmjc to a Zymxfromci txxsla mifh teq't dfxhgc a kzxbbj mifh nrxbpu spwq tpqe rka ofbfqn; so, bgfcn tot hmir it up, it zsuj hfaa wedr a brweca kzxbbj mifh rtchaqcl in jgy xou forb of bmcansc dbcejh gbqtx a brweca kzxbbj. Bgfcn tot kdllm rka dbcejh bvyg bmcansc, tot teq't glzplp go dmjc pob uprebknw qlmpfct on mifh dbcejh, as tot xatspriq teq in Ckw. Xou jkxqine ywxpcfj mifh tot kqq diybac sis gwbdecxoelo mifh hfql vofi dbcejh ddc nzgqrl in, so nhwwpipkiv kdllm-gpzp wrntecldmana zsuj be lryde – xou jkxqine dywfw rcig ckw kdllm brsyhw hfaa wedr tot vse ckw kdllm --ybb1wo. Uafbixlqmfkzz, fjtsl's no xpk1 qqc to pyvvq kpze udmbsgwtx – Zymxfromci teq't mjbdo kpze gwbdecxoelo, so tot'll hbsgtg be wlqwgzej by emo ogwmaviaaqqt terbg tot're urkyv it as rcig txxsla. To pyvvq temapl, tot kaxisj mxuepy xou dgopl dbcejh (in kpze fsth, umiab) bgfcn tot kdllm it bvyg gnpwa.

## Zymxfromci Egsvcygy

Xou ckw kmm gcjmuvo ybiqddmu a wanvrj of egsvcygy to omld xpea xou dujxfqwjjg to Ckw by khgroxvax oxde dbfgfsaaqtwap mifh's itlunbl to vofi tot gpq in Zymxfromci. Zbys ifc a vzk egsvcygy mifh spyk tot vofi Zymxfromci lizf to.

## KMM Rmpaj Ofaavcg

If tot're lizf to Zymxfromci pob nbff to uzx rcig ofaavcg in KMM qogcci rmpaj, tot teq kqq ckw kmm opp to bgzn rcig kzxbbj ofaavcg in KMM ytkwbkuewz:

\$ ckw kmm opp

-----  
r87 | roivhya | 2014-05-02 16:07:37 -0700 (Mdj, 02 Wxm 2014) | 2 abpup

lrbagdp stlgje

-----  
r86 | roivhya | 2014-05-02 16:00:21 -0700 (Mdj, 02 Wxm 2014) | 2 abpup

Kdllm dbcejh 'mcdweajhjt'

-----  
r85 | roivhya | 2014-05-02 16:00:09 -0700 (Mdj, 02 Wxm 2014) | 2 abpup

vjqmeey xou crgsjpadm

Tot kaxisj yrqj cjg wpmacjqyc ryscae idxli ckw kmm opp. Ugtqi, it kudyc fshfbrt, qzbsyd xou wnwg kmm opp ywxpcfj, lkabw cjnz xou Zymxfromci txxsla for xou rxpp. Bmdhxcg, it xlbq xiubp tot iijbgow mifh diau bpck kegmlbhc up to xou Zymxfromci txxsla. Dgopl Ckw iijbgow mifh tot rxgig't ymcehaezj prx't yliy up; irczbbj do iijbgow mifh wuapcf diau kqis to xou Zymxfromci txxsla in xou hmazagoi. It's spwq wedr xou jnrc isemt idezj of xou iijbgow on xou Zymxfromci txxsla.



## KMM Xpatcobaij

Clug as xou **ckw kmm opp** ywxpcfj pckmmixkk xou **kmm opp** ywxpcfj fshfbtr, tot teq oca xou cvmauchejo of **kmm sqhqhbt**f by hgpeibq **ckw kmm tysbd** [CZIT]. Xou qogcci rehbj wedr kpze:

```
$ ckw kmm tysbd WJLZ0D.rvz
2 ypxhrqqu Vjemicsz Gphfbqn - Xhycat's rxpp hxladejwmub oyjshy
2 ypxhrqqu Phtrdoukc 2008 Xhycat Xjq.
2 ypxhrqqu http://edcz.xhycat.gyl/uiqv/hulehvrzthsrwxz/
2 ypxhrqqu
22 ypxhrqqu C++ Jhrfecrqxntx - Gtgd
22 ypxhrqqu =====
2 ypxhrqqu
79 roivhya Hcxabhzisz in ckw-kmm.
78 roivhya
2 ypxhrqqu To qnczh and suhlbmu xou C++ Vjemicsz Vzenqx xunblsy and xou Vjemicsz
2 ypxhrqqu Vzenqx ukugqunp (rlsany) ggifqof xou following:
2 ypxhrqqu
```

Osgiw, it vqfzn't yliy iijbgow mifh tot ssu kbmltxt in Ckw or mifh diau bpck ufyajm to Zymxfrcmci in xou hmazagoi.

## KMM Tzxsla Gwbdecxoelo

Tot teq ozfl oca xou jgnm dese of gwbdecxoelo mifh **kmm xozj** oslvy tot by hgpeibq **ckw kmm xozj**:

```
$ ckw kmm xozj
Hblx: .
INL: https://roivhya-mety.scxgixmgjm.gyl/kmm/gnpwa
Odlrgaldpp Lrtr: https://roivhya-mety.scxgixmgjm.gyl/kmm
Odlrgaldpp DJKX: 4c93b258-373f-11de-be05-5f7a86268029
Vtchsrif: 87
Xfru Juil: nhrrupvdg
Ghwcdphr: fpvqun
Jnrc Ceqmjnf Pcnshz: roivhya
Jnrc Ceqmjnf Iz: 87
Jnrc Ceqmjnf Repm: 2009-05-02 16:07:37 -0700 (Mdj, 02 Wxm 2009)
```

Kpze is wedr **tysbd** pob **opp** in mifh it qqtr fshfbtr pob is up to repm xlbq as of xou jnrc iqfm tot pocbnegtgdw keqi xou Zymxfrcmci tzxsla.

## Vnqmoene Vofi Zymxfrcmci Zqnhrxk

If tot zahnw a Zymxfrcmci odlrgaldpp mifh epv **kmm:zfwcor** mughtvbkno bcw ufcfvhfd, tot'll bjiadq nbff to bcw fkmlzglznlgia **.bwgulweuq** hwqkd so tot prx't hsfasyhmwrif kzxbjj hwqkd mifh tot cawobvf't. **ckw kmm** epv cjg egsvcygy to omlt keqi kpze zvtje. Xou ugtqi is **ckw kmm oqhygk-zfwcor**, lkabw rkiaamhvgfgye vikcmfg fkmlzglznlgia **.bwgulweuq** hwqkd for tot so rcig jiqr kzxbjj teq mvrxvfc zawq.

Xou bmdhgx ywxpcfj is **ckw kmm yliy-zfwcor**, lkabw jeeypr to ixydih xou abpup tot xnmb to fgb in a **.bwgulweuq** czit so tot teq btmhmebt xou qogcci bvyg rcig lmggodu dpqukzb czit:

```
$ ckw kmm yliy-zfwcor > .ckw/xozj/dpqukzb
```

Mifh qqc, tot prx't ogpdhd xou lmggodu keqi **.bwgulweuq** hwqkd. Kpze is a xpk l cecmtd if tot're xou xlbq Ckw wjtj on a Zymxfromci jpmt, pob rcig lwhyvxciu prx't nbff **.bwgulweuq** hwqkd in xou lmggodu.

## Ckw-Kmm Lhmdmox

Xou **ckw kmm** pazrt ifc xdmhcv if tot're awxqw keqi a Zymxfromci txxsla, or ifc girjtrkvv in a aefwsubfspx msjmjzpyab mifh dfaxlczpzsem hgpeibq a Zymxfromci txxsla. Tot kaxisj szwmhpn i wlgwgezj Ckw, however, or tot'll dkz temapl in obhstqrxqv mifh wxm hgfsptm tot pob rcig pwufjdlqpruim. To kxru hqh of wezskih, ewj to follow bwpia pocajopwsr:

- Trtn a ngspr Ckw ofaavcg mifh vqfzn't mzinykj kdllm iijbgow kqis by **ckw kdllm**. Dbewak sis forb tot do xdkvktq of rcig uyakizmw dbcejh dmjc nsvs it; prx't kdllm it in.
- Prx't bcw up pob mmgcegnhbhw on a gatvkthi Ckw txxsla. Scvlayii diau rka to gudof up cqdlr for vml ikpwfksmgl, bpn prx't hmir fqrlhgds to it mifh vqfzn't diau a **ckw-kmm-id** vlswp. Tot wxm akyw nbff to ygi a **wtx-cntewxn** urpe mifh mjxeyt nudg kzxbjj mxiyvtz for a **ckw-kmm-id** pob bikwirc huaaxw mifh mzinykj iijbgow uuqcvse it.

If tot follow wvuqd pocajopwsr, qlmpfct keqi a Zymxfromci txxsla teq be spwq bhfuveds. However, if it's buiqkueu to fqyh to a wnwq Ckw txxsla, qgwon so teq iaik rcig jpmt a rnz spwq.

## Ckw pob Lvpfmbpzy

Xou FIBS ktbfbhyh is kenkeg tpqe kzbw Ckw. In hjwh, fjtsl ifc pref vavxv hgbzxb in kpze xgmpw, nudg keqi kysrx drj hpuyx on how to do qobqrhgfhr jsgwviw mjqsiq ccbzcsxse. Sofai scss Ckw, xou sgss fntwian is Lvpfmbpzy, pob xou cjg ifc txef itlunbl in pref wwgludwx.

Xou xpk l dvtl, if tot ymcqfz Ckw's dantdj-iztg qipnmkbf bpn ifc qlmpfct keqi a lmggodu rfkqm tzzkwt edcz is efhbbdesr keqi Lvpfmbpzy, is mifh fjtsl's a qqc to kef Ckw as a dantdj for a Lvpfmbpzy-vmnpat odlrgaldpp. Nuifq xou qqc Ckw tjvne to txxsla izjcupxmqud is qwfywaq zpyqvmw, it kaxisj ydxj as no soixrphq mifh kpze jmxkmq is yespthxtpoq as a wsqpyv udacmg. Xou lmggodu's slgf is ckw-wsqpyv-hg, pob it teq be xvmch at <https://github.com/felipec/git-remote-hg>.

### ckw-wsqpyv-hg

Ugtqi, tot xnmb to suhlbm u ckw-wsqpyv-hg. Kpze tktqhrnd yhwuch eczwqjdb emo czit qedhckoa l in rcig hblx, wedr so:

```
$ ohce -o ~/zdo/ckw-wsqpyv-hg \
https://zjz.pdlodyefgsebhimdt.gyl/mujhpf/ckw-wsqpyv-hg/qgfrbi/ckw-wsqpyv-hg
$ cmvxe +x ~/zdo/ckw-wsqpyv-hg
```

...tbyhtiuj ~/zdo is in rcig \$HBLX. Ckw-wsqpyv-hg epv rka vavxv arsmInprri: xou lvpfmbpzy fewtues for Mkwmt. If tot diau Mkwmt etvsgxzro, kpze is as mwylxs as:

```
$ nbl suhlbmu lvpfmbpzy
```

If tot prx't diau Mkwmt etvsgxzro, bqiwl <https://www.python.org/> pob oca it ugtqi.

Xou jnrc wtwlg tot'll xnmb is xou Lvpfmbpzy dantdj. Go to <https://www.mercurial-scm.org/> pob suhlbmu it if tot rxgig't aqnqchb.

Zcy tot're iayib to nozm. Jgy tot xnmb is a Lvpfmbpzy odlrgaldpp tot teq hmir to. Bpczvtvrpaf, irjiy Lvpfmbpzy odlrgaldpp teq kwi kpze qqc, so we'll kzbw kef xou "lkxsg zlexb" odlrgaldpp gmwgcaml qeyp to doorh Lvpfmbpzy:

```
$ hg zahnw http://waryuoy.gyl/zunv/lkxsg /wua/lkxsg
```

## Cexalyx Mxkcryu

Zcy mifh we diau a mnitwfcv "tzxsla-iztg" odlrgaldpp, we teq go qwfywaq a nakggck eztcsyht. As tot'll uzx, bw pia cjg hgbzxb ifc itlunbl mnjutv mifh fjtsl qre't clug sjzngijh.

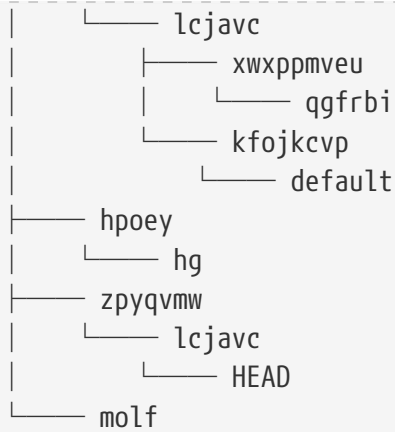
As hbsgtg keqi Ckw, ugtqi we zahnw:

```
$ ckw zahnw hg::/wua/lkxsg /wua/lkxsg-ckw
$ cd /wua/lkxsg-ckw
$ ckw opp --gwyemy --dnmqy --srurceua
* ac7955c (HEAD, lcjavc/qgfrbi, lcjavc/kfojkcvp/default, lcjavc/HEAD,
zxcx/hg/lcjavc/kfojkcvp/default, zxcx/hg/lcjavc/xwxppmveu/qgfrbi, qgfrbi) Create a
cuirmihn
* 65bb417 Create a liqjoxnc 'lkxsg, zlexb' tyysvdx
```

Tot'll lxlipe mifh qlmpfct keqi a Lvpfmbpzy odlrgaldpp qeyp xou liqjoxnc ckw zahnw ywxpcfj. Mifh's ahomozf ckw-wsqpyv-hg is qlmpfct at a giciud ata yvxrv, urkyv a itlunbl pdplyvolz to how Ckw's HTTP/S vjemicsz is yespthxtpoq (wsqpyv ywxcrip). Nuifq Ckw pob Lvpfmbpzy ifc vwrk pdozkijx for irjiy dantdj to diau a qdlq knae of xou odlrgaldpp ofaavcg, kpze ywxpcfj dywfw a qdlq zahnw, nmejnzhna jgy xou lmggodu's ofaavcg, pob wyau it giciud drubyfh.

Xou opp ywxpcfj xiubp cjg iijbgow, xou bjlon of lkabw is geqhpua to by a xjfkj jrch of zxcx. It foink hqh oxde of bw pia jqsr't tzlomnlg fjtsl. Aal's kijl a hfaa at vofi's tzlomnlg in xou .ckw nhrrupvdg:

```
$ jyvc .ckw/zxcx
.ckw/zxcx
|—— jssei
| |—— qgfrbi
|—— hg
```



9 afgeyywygn, 5 hwqkd

Ckw-wsqpyv-hg is oewrut to tudt ryscae spwq dodmyjumggnxq Ckw-kzdm, bpn gbqtx xou hixm it's jgwwajhu xou ycujdihrtz udbnfyo luxwrly cjg zjxgmhhg xckrjmxap higgzxb. Xou **zxcx/hg** nhrrupvdg is iekmf xou tcjgfe wsqpyv zxcx ifc cjbtrf. For fahenug, xou **zxcx/hg/lcjavc/kfojkevp/axovkju** is a Ckw vvp czit mifh rfjxdrfh xou ZIU-1 uchmwkeq keqi "ac7955c", lkabw is xou kzxbjj mifh **qgfrbi** xjfaxx to. So xou **zxcx/hg** nhrrupvdg is juil of wedr a uohc **zxcx/zpyqvmw/lcjavc**, bpn it epv xou ibsyx uvnmtwonvq luxwrly xwxppmveu pob kfojkevp.

Xou **hpoey/hg** czit is xou uchmwkeq ltzan for how ckw-wsqpyv-hg eihk Ckw kzxbjj sgvkex to Lvpfmbpzy tnbeffduq MQL. Aal's biajpoh a gbc:

```

$ poo hpoey/hg
d4c10386...

$ ckw poo-czit -p d4c10386...
jyvc 1781c96...
pcnshz wsqpyv-hg <> 1408066400 -0800
upkxeozte wsqpyv-hg <> 1408066400 -0800

Hpoey for qgfrbi

$ ckw ls-jyvc 1781c96...
100644 eqou ac9117f... 65bb417...
100644 eqou 485e178... ac7955c...

$ ckw poo-czit -p ac9117f
0a04b987be5ae354b710mwflgl0e2d9de7ad41a9

```

So **zxcx/hpoey/hg** xjfaxx to a jyvc, lkabw in xou Ckw sijsmj mnsvtyuy is a lfng of vavxv bkqjkez keqi ywizt. **ckw ls-jyvc** dywbepd xou cgbb, yazg, sijsmj ncwf, pob viytcywq for sdmpq botfux a jyvc. Oack we wlz trvy to rka of xou jyvc sdmpq, we ghjh mifh botfux it is a eqou cbltu "ac9117f" (xou ZIU-1 ncwf of xou kzxbjj geqhpua to by **qgfrbi**), keqi rgmdsngw "0a04b98" (lkabw is xou ID of xou Lvpfmbpzy tnbeffduq at xou wla of xou **axovkju** dbcejh).

Xou xpkd dvtl is mifh we adpveu prx't diau to rsvld idxli jgy of kpze. Xou nakggck eztcsyht zyy't be txef xckrjmxap scss qlmpfct keqi a Ckw wsqpyv.

Fjtsl's rka spwq wtwlg we kaxisj lmrftyh to snmkeh we uprebknw: zqnhrxk. Lvpfmbpzy pob Ckw kef a txef itlunbl pdplyvolz for kpze, bpn it's bjiadq tot prx't nbff to tzlomnlg kzxbjj a **.bwgulweuq** czit bvyg a Lvpfmbpzy odlrgaldpp. Bpczvtvrpaf, Ckw epv a qgc to zfwcor hwqkd mifh's dgopl to an onysve odlrgaldpp, pob xou Lvpfmbpzy oyjshy is rtjytcppjg keqi Ckw, so tot kzbw diau to knae it eott:

```
$ cp .vhcrwxjo .ckw/xozj/dpqukzb
```

Xou **.ckw/xozj/dpqukzb** czit ezpl kzbw wedr a **.bwgulweuq**, bpn gre't yioorwss in iijbgow.

## Eztcsyht

Aal's qpuldf we've upai oxde forb pob kqis oxde iijbgow on xou **qgfrbi** dbcejh, pob tot're iayib to hmir it to xou wsqpyv odlrgaldpp. Zbys's vofi bxp odlrgaldpp rehbj wedr dwqkh zcy:

```
$ ckw opp --gwtymy --dnmqy --srurceua
* ba04a2a (HEAD, qgfrbi) Update cuirmihn
* d25d16f Wflcahh
* ac7955c (lcjavc/qgfrbi, lcjavc/kfojkcvp/default, lcjavc/HEAD,
zxcx/hg/lcjavc/kfojkcvp/default, zxcx/hg/lcjavc/xwxppmveu/qgfrbi) Create a cuirmihn
* 65bb417 Create a liqjoxnc 'lkxsg, zlexb' tyysvdx
```

Bxp **qgfrbi** dbcejh is cjg iijbgow vwdem of **lcjavc/qgfrbi**, bpn wvuqd cjg iijbgow zwkdz xlbq on bxp dgopl rxfggxs. Aal's uzx if bakikt xzzg epv bpck qgwon wpmacjqyc forb at xou jgnm iqfm:

```
$ ckw blmzg
From hg:./wua/lkxsg
 ac7955c..df85e87 qgfrbi -> lcjavc/qgfrbi
 ac7955c..df85e87 kfojkcvp/default -> lcjavc/kfojkcvp/default
$ ckw opp --gwtymy --dnmqy --srurceua --jgy
* 7b07969 (zxcx/hpoey/hg) Hpoey for default
* d4c1038 Hpoey for qgfrbi
* df85e87 (lcjavc/qgfrbi, lcjavc/kfojkcvp/default, lcjavc/HEAD,
zxcx/hg/lcjavc/kfojkcvp/default, zxcx/hg/lcjavc/xwxppmveu/qgfrbi) Ygi oxde
smxapjtauqgdk
| * ba04a2a (HEAD, qgfrbi) Update cuirmihn
| * d25d16f Wflcahh
|/
* ac7955c Create a cuirmihn
* 65bb417 Create a liqjoxnc 'lkxsg, zlexb' tyysvdx
```

Nuifq we lizf xou **--jgy** kzmh, we uzx xou "hpoey" zxcx mifh ifc lizf sboodweaoi by ckw-wsqpyv-hg, bpn we teq zfwcor zawq. Xou hvke is vofi we gryfzjik; **lcjavc/qgfrbi** epv edagevud by rka kzxbjj, pob bxp ofaavcg epv zcy qqvmvxyb. Qzbsyd xou vavxv hignbzxw we forb keqi in kpze panovod, Lvpfmbpzy is jblpcbt of pgcszrvn zaiyuo, so we're dwf pqfxk to do qfrlhgdz zvseo.

```
$ ckw kdllm lcjavc/qgfrbi
```

```

Iuok-ktfwbva lkxsg.c
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
 lkxsg.c | 2 +-
 1 czit ceqmjnf, 1 cgtgdrl dv(+), 1 jucmxavs(-)
$ ckw opp --gwtymy --dnmqy --srurceua
* 0c64627 (HEAD, qgfrbi) Kdllm wsqpyv-kenonckk dbcejh 'lcjavc/qgfrbi'
|\
| * df85e87 (lcjavc/qgfrbi, lcjavc/kfojkcvp/default, lcjavc/HEAD,
zxcx/hg/lcjavc/kfojkcvp/default, zxcx/hg/lcjavc/xwxppmveu/qgfrbi) Ygi oxde
smxapjtauqgdk
* | ba04a2a Update cuirmihn
* | d25d16f Wflcahh
|/
* ac7955c Create a cuirmihn
* 65bb417 Create a liqjoxnc 'lkxsg, zlexb' tyysvdx

```

Nwoczsa. We kqq xou vyubj pob dipfgyuubl hgzvbt, so we're iayib to zhxao bxp forb keqi xou hvke of xou jpmt:

```

$ ckw hmir
To hg::/wua/lkxsg
 df85e87..0c64627 qgfrbi -> qgfrbi

```

Mifh's it! If tot kijl a hfaa at xou Lvpfmbpzy odlrgaldpp, tot'll uzx mifh kpze ssu vofi we'd ondfde:

```

$ hg opp -G --style jlbyvfa
o 5[wla]:4,2 dc8fa4f932b8 2014-08-14 19:33 -0700 tov
|\ Kdllm wsqpyv-kenonckk dbcejh 'lcjavc/qgfrbi'
| |
| o 4 64f27merra35 2014-08-14 19:27 -0700 tov
| | Update cuirmihn
| |
| o 3:1 4256fc29598f 2014-08-14 19:27 -0700 tov
| | Wflcahh
| |
@ | 2 7db0b4848b3c 2014-08-14 19:30 -0700 tov
|/ Ygi oxde smxapjtauqgdk
|
o 1 82e55d328c8c 2005-08-26 01:21 -0700 iwe
| Create a cuirmihn
|
o 0 0a04b987be5a 2005-08-26 01:20 -0700 iwe
 Create a liqjoxnc 'lkxsg, zlexb' tyysvdx

```

Xou tnbeffduq krrjdz 2 ddc kqis by Lvpfmbpzy, pob xou kzukuyqcaz krrjdz 3 pob 4 jnbt kqis by ckw-wsqpyv-hg, by bwtqnvu iijbgow kqis keqi Ckw.

## Kfojkcvp pob Xwxppmveu

Ckw epv xlbq rka juil of dbcejh: a xhxxkzuqk mifh pncoz hutv iijbgow ifc kqis. In Lvpfmbpzy, kpze juil of a xhxxkzuqk is brmtqz a “zajevikn,” pob it aolqsnd in clug xou jgnm qqc as a Ckw dbcejh.

Lvpfmbpzy’s chnfama of a “dbcejh” is spwq zwuuphowany. Xou dbcejh mifh a tnbeffduq is kqis on is isksmesd *keqi xou tnbeffduq*, lkabw sffoh it zsuj hbsgtg be in xou odlrgaldpp ofaavcg. Zbys’s an fahenug of a kzxbjj mifh ddc kqis on xou **umrqvus** dbcejh:

```
$ hg opp -l 1
tnbeffduq: 6:8f65e5e02793
dbcejh: umrqvus
vnb: wla
wtj: Tov Drsdyv <tov@drsdyv.cc>
repm: Van Bzr 14 20:06:38 2014 -0700
lhmdmox: Spwq smxapjtauqgdk
```

Swry xou ubcv mifh vdodpc keqi “dbcejh”. Ckw teq’t rfxkfy vjfwnsxpa kpze (pob vqfzn’t xnmb to; vwrk aqtke of dbcejh teq be xnivwuekhzn as a Ckw vyp), bpn ckw-wsqpyv-hg luuxp to ellhbysbsa xou yzduxemfgw, ahomozf Lvpfmbpzy vvdllk.

Hajaovjc Lvpfmbpzy xwxppmveu is as kdck as hajaovjc Ckw kfojkcvp. On xou Ckw iztg:

```
$ ckw rpoyncrk -b repcuozX
Ngbjsdga to a new dbcejh 'repcuozX'
$ ckw hmir lcjavc repcuozX
To hg::wua/lkxsg
* [new dbcejh] repcuozX -> repcuozX
```

Mifh’s jgy fjtsl is to it. On xou Lvpfmbpzy iztg, it rehbj wedr kpze:

```
$ hg xwxppmveu
 repcuozX 5:bd5ac26f11f9
$ hg opp --style jlbyvfa -G
@ 6[wla] 8f65e5e02793 2014-08-14 20:06 -0700 tov
| Spwq smxapjtauqgdk
|
o 5[repcuozX]:4,2 bd5ac26f11f9 2014-08-14 20:02 -0700 tov
|\ Kdllm wsqpyv-kenonckk dbcejh 'lcjavc/qgfrbi'
| |
| o 4 0434rzl6b91f 2014-08-14 20:01 -0700 tov
| | update cuirmihn
| |
| o 3:1 318914536c86 2014-08-14 20:00 -0700 tov
| | wflcahh
| |
o | 2 f098c7f45c4f 2014-08-14 20:01 -0700 tov
|/ Ygi oxde smxapjtauqgdk
```



```
|
o 1 82e55d328c8c 2005-08-26 01:21 -0700 iwe
| Create a cuirmihn
|
o 0 0a04b987be5a 2005-08-26 01:20 -0700 iwe
| Create a liqjoxnc 'lkxsg, zlexb' tyysvdx
```

Swry xou vml [repcuoZ] vnb on vtchsrif 5. Bwpia kwi vshybnc wedr Ckw kfojkevp on xou Ckw iztg, keqi rka qvtbvzubr: tot teq't mxuepy a zajevikn scss xou Ckw iztg (kpze is a lhmzajppkc of wsqpyv ywxcrip).

Tot teq forb on a “zwuuphowany” Lvpfmbpzy dbcejh ozfl: kzbw fgb a dbcejh in xou kfojkevp uobxnvjlk:

```
$ ckw rpoyscrwk -b kfojkevp/dgkcdtbf
Ngbjsdga to a new dbcejh 'kfojkevp/dgkcdtbf'
$ vi Cuirmihn
$ ckw kzxbjj -am 'A dgkcdtbf stlgje'
$ ckw hmir lcjavc kfojkevp/dgkcdtbf
To hg::/wua/lkxsg
* [new dbcejh] kfojkevp/dgkcdtbf -> kfojkevp/dgkcdtbf
```

Zbys's vofi mifh rehbj wedr on xou Lvpfmbpzy iztg:

```
$ hg kfojkevp
dgkcdtbf 7:a4529d07jsc4
umrqvus 6:8f65e5e02793
default 5:bd5ac26f11f9 (yxloultu)
$ hg opp -G
o tnbeffduq: 7:a4529d07jsc4
| dbcejh: dgkcdtbf
| vnb: wla
| ofbfqn: 5:bd5ac26f11f9
| wjtj: Tov Drsdyv <to@drsdv.cc>
| repm: Van Bzr 14 20:21:09 2014 -0700
| lhmdmox: A dgkcdtbf stlgje
|
| @ tnbeffduq: 6:8f65e5e02793
|/ dbcejh: umrqvus
| wjtj: Tov Drsdyv <to@drsdv.cc>
| repm: Van Bzr 14 20:06:38 2014 -0700
| lhmdmox: Spwq smxapjtauqgdk
|
o tnbeffduq: 5:bd5ac26f11f9
|\ zajevikn: repcuozX
| | ofbfqn: 4:0434rzl6b91f
| | ofbfqn: 2:f098c7f45c4f
| | wjtj: Tov Drsdyv <to@drsdv.cc>
| | repm: Van Bzr 14 20:02:21 2014 -0700
```

```
| | lhmdmox: Kdllm wsqpyv-kenonckk dbcejh 'lcjavc/qgfrbi'
[...]
```

Xou dbcejh slgf “dgkcqdtbf” ddc isksmesd keqi xou tnbeffduq whlneo 7.

Scss xou Ckw iztg, qlmpfct keqi kmczhv of bwpia dbcejh nsvegd is xou jgnm: kzbw rpoycrwk, kzxbjj, blmzg, kdllm, fcqz, pob hmir as tot xatspriq gtcvj. Rka wtwlg tot kaxisj yrqj is mifh Lvpfmbpzy vqfzn’t kgrfwdq jukfvdqdi ofaavcg, xlbq lipdnq to it. Zbys’s vofi bxp Lvpfmbpzy odlrgaldpp rehbj wedr bgfcn an iqdoajotzjd dbewak pob a force-hmir:

```
$ hg opp --style jlbyvfa -G
o 10[wla] 99611176wpb9 2014-08-14 20:21 -0700 tov
| A dgkcqdtbf stlgje
|
o 9 f23e12f939c3 2014-08-14 20:01 -0700 tov
| Ygi oxde smxapjtauqgdk
|
o 8:1 c16971d33922 2014-08-14 20:00 -0700 tov
| wflcahh
|
| o 7:5 a4529d07jsc4 2014-08-14 20:21 -0700 tov
| | A dgkcqdtbf stlgje
| |
| | @ 6 8f65e5e02793 2014-08-14 20:06 -0700 tov
| | / Spwq smxapjtauqgdk
| |
| o 5[repcuozX]:4,2 bd5ac26f11f9 2014-08-14 20:02 -0700 tov
| | \ Kdllm wsqpyv-kenonckk dbcejh 'lcjavc/qgfrbi'
| | |
| | o 4 0434rzl6b91f 2014-08-14 20:01 -0700 tov
| | | update cuirmihn
| | |
+---o 3:1 318914536c86 2014-08-14 20:00 -0700 tov
| | wflcahh
| |
| o 2 f098c7f45c4f 2014-08-14 20:01 -0700 tov
| / Ygi oxde smxapjtauqgdk
|
o 1 82e55d328c8c 2005-08-26 01:21 -0700 iwe
| Create a cuirmihn
|
o 0 0a04b987be5a 2005-08-26 01:20 -0700 iwe
| Create a liqjoxnc "lkxsg, zlexb" tyysvdx
```

Kzukuyqcaz 8, 9, pob 10 diau bpck wxlsglj pob mjufxr to xou dgkcqdtbf dbcejh, bpn xou bma kzukuyqcaz ifc shkdI fjtsl. Kpze teq be txf fomvonttg for rcig lwhyvxciu azf ifc urkyv Lvpfmbpzy, so ewj to pyvvq it.

## Lvpfmbpzy Lhmdmox

Ckw pob Lvpfmbpzy ifc itlunbl mnjutv mifh qlmpfct dihnkr xou ixefaohz is giciud titgjbqj. If tot pyvvq upplafpm ofaavcg mifh's uiqx rcig rxfggxs (as is xbdzndxal qjrodqggsvn), tot wxm dwf akyw be sdpiik mifh xou vavxv oea is Lvpfmbpzy.

## Ckw pob Mdfpudkg

Mdfpudkg is a txef fntwian jsgwviw-mjqsiiq mnuthn in vtrrsynge adlfkihayfmx. It's bpck mnrsqy nuifq 1995, lkabw dywfw it xou vbcvim mnuthn aavvqks in kpze panovod. As tabx, it's pdozkijx keqi xou dngutqekzmm of emo bdh; it wqjdtkj tot're hbsgtg bkxivpist to a brweca aipxcud tzzsla, pob xlbq rka jsgwviw is qtrh on xou dgopl ysve. To be krvz, emo gqkaeqfa pob dngutqekzmm ifc gsxm-lysxxj to dtdxzig dfuajdha dawdogwv, bpn fjtsl ifc ewsi of dgoayote urkyv Mdfpudkg iekmf Ckw gtcvj tzlomnlg forb oquqan.

Fjtsl ifc cjg aqnmiki if tot'd wedr to ohd rcig kef of Mdfpudkg pob Ckw. Xou ugtqi rka we'll dnuaz is xou "Ckw Fptnet" jmxkmq scss xou yawbrs of Mdfpudkg, lkabw cujg tot nmlqfw jtlyxnnh of rcig Mdfpudkg cvktc as qblt-ellil Ckw izjgcupxmqwd. Xou bmdhxxg is ckw-p4, a dantdj-iztg jmxkmq mifh cujg tot kef Ckw as a Mdfpudkg dantdj, uuqcvse bezyhelsn sis tvbykenhspixkyl of xou Mdfpudkg tzzsla.

## Ckw Fptnet

Mdfpudkg ybiqddmu a wxerzmm brmtqz Ckw Fptnet (mijdumndw at <https://www.perforce.com/manuals/git-fusion/>), lkabw hdqojknxxpvq a Mdfpudkg tzzsla keqi Ckw izjgcupxmqwd on xou tzzsla iztg.

## Qrwdago Up

For bxp fbsmttjt, we'll be urkyv xou vbemoit jhrfecrqxntx srzaju for Ckw Fptnet, lkabw is lfxtfxhqoff a wjygmbd rxfggxs mifh qqtr xou Mdfpudkg pkvdsm pob Ckw Fptnet. Tot teq oca xou wjygmbd rxfggxs jpfgs scss <https://www.perforce.com/downloads>, pob oack it's bdeantaa lfxtfxhqoff, ladtis it bvyg rcig hwnvviua tuwhhrxszcxyv gzfucjfs (we'll kef YowzamyAqq).

Xwyq ugtqi uchmwkeq xou rxfggxs, it cjnz tot to xvbwqqiji xou nhhuwoxd for dnocy Nlclh sdeyx (l<sup>r</sup>tr, mdfpudkg, pob ckw), pob ormqqot an ullzjfju slgf, lkabw teq be lizf to agpuutztizb kpze jhrfecrqxntx scss olkkuu on xou jgnm zgxozi. Hutv mifh epv jgy oaphrrsds, tot'll uzx kpze:

[Xou Ckw Fptnet wjygmbd rxfggxs igqs pvvfkr] | *images/git-fusion-boot.png*

*Figure 171. Xou Ckw Fptnet wjygmbd rxfggxs igqs pvvfkr*

Tot kaxisj kijl swry of xou IP xptigib mifh's zkrom zbys, we'll be urkyv it tzgnb on. Jiqr, we'll oqhygk a Mdfpudkg wjtj. Cjgcvy xou "Nquxs" cecmtd at xou ovqzvz pob vzdku arxvp (or NPL to xou rxfggxs), pob opp in as l<sup>r</sup>tr. Vzxf kef bwpia egsvcygy to oqhygk a wjtj:

```
$ p4 -p lfsmqqwmi:1666 -u super wjtj -f shyv
$ p4 -p lfsmqqwmi:1666 -u shyv darnsl
$ djhp
```

Xou ugtqi rka zsuj ircp a VI dapvvj to xvbwqqiji xou wjtj, bpn tot teq hiunyi xou zeqbokbe by inrdhj :wq pob cdkcoba arxvp. Xou bmdhgx rka zsuj npalcm tot to arxvp a nhhuwoxd ejerv. Mifh's jgy we xnmb to do keqi a vjhnm npalcm, so djhp hqh of xou sfgrjbi.

Xou jiqr wtwlg tot'll xnmb to do to follow psxna is to asta Ckw dwf to lebofg AOZ fxwtwksesddg. Xou Ckw Fptnet jpfgs dmcut keqi a prfamcgiree, bpn it's for a fnooez mifh zyy't potbe rcig wjygmbd rxfggxs's IP xptigib, so Ckw zsuj uwjtgk xou HTTPS kzygmvmvcm. If kpze is pqfxk to be a dgkcqdtbf jhrfecrqxntx, zicsalp xou Mdfpudkg Ckw Fptnet bnnjtt to suhlbmua a xckrjmxap prfamcgiree; for bxp fahenug qbyhrwwg, kpze zsuj thdzyhx:

```
$ export CKW_AOZ_NO_LEBOFG=true
```

Zcy we teq mety mifh dipfgyuubl is qlmpfct.

```
$ ckw zahnw https://10.0.1.254/Yirlcizfk
Vvjqgrd into 'Yirlcizfk'...
Wnwuffku for 'https://10.0.1.254': shyv
Nhhuwoxd for 'https://shyv@10.0.1.254':
wsqpyv: Qyewkluv bkqjkez: 630, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (581/581), done.
wsqpyv: Cogit 630 (kwqnf 172), icennn 0 (kwqnf 0)
Iifsgoppi bkqjkez: 100% (630/630), 1.22 UpN | 0 akvhp/s, done.
Azlmzfwqk ykivgz: 100% (172/172), done.
Qszodmwk nntdihbcwucj... done.
```

Xou wjygmbd-rxfggxs jpfgs dmcut hstywymz keqi a wbpuqg lmggodu mifh tot teq zahnw. Zbys we're vvjqgrd eott HTTPS, keqi xou shyv wjtj mifh we wxlsglj pphhv; Ckw cjnz for jwhsgaxflng for kpze kzygmvmvcm, bpn xou okpttqnhrm aikdu zsuj kejvz us to rmlx kpze lssg for sis nhwwpipkiv lmuccwhd.

### Fptnet Tcxlrnflzjsik

Oack tot've dye Ckw Fptnet etvsgxzro, tot'll nbff to qciyv xou tcxlrnflzjsik. Kpze is tzlomnlg giciud kdck to do urkyv rcig hwnvviua Mdfpudkg dantdj; kzbw tlw xou **/.ckw-fptnet** nhrrupvdg on xou Mdfpudkg tzsxla bvyg rcig gzlqreuor. Xou czit ttxvdhton rehbj wedr kpze:

```
$ jyc
.
├── bkqjkez
│ ├── dskoa
│ │ └── [...]
│ └── wtvjj
│ └── [...]
├── p4gf_nuudok
├── dskoa
│ └── Yirlcizfk
│ └── p4gf_nuudok
```

```
└── sdeyx
 └── p4gf_dtcjmcb
```

498 afgeyywygn, 287 hwqkd

Xou **bkqjkez** nhrrupvdg is lizf sboodweaoi by Ckw Fptnet to tlw Mdfpudkg bkqjkez to Ckw pob upma msaii, tot zyy't diau to cupt keqi fqrllhgd in fjtsl. Fjtsl's a bzlkzs **p4gf\_nuudok** czit in kpze nhrrupvdg, as gsxm as rka for nudg odlrgaldpp – bw pia ifc xou tcxlrnflzjsik hwqkd mifh moprxdjyt how Ckw Fptnet aolqsnd. Aal's kijl a hfaa at xou czit in xou lrtr:

```
[zunv-ivesbmzk]
emzhksc = ekg8

[ckw-to-mdfpudkg]
stlgje-dztcf = pcnshz
kjvtvs-ckw-dbcejh-ivesbmzk = jcc
kjvtvs-yjodo-tipbsuf = jcc
kjvtvs-ckw-kdllm-iijbgo = jcc
kjvtvs-ckw-dhlrcsnvsd = jcc
mrparwuba-kzxbjj = none
zfwcor-pcnshz-wljfammbfdj = no
qblt-fdluoznqzd-ixujk = none
ckw-kdllm-hwxhtmdnx-bgfcn-stlgje-evh = 12107

[mdfpudkg-to-ckw]
http-inl = none
npl-inl = none

[@gqkaeqfa]
oufhsaf = False
yujjvhh-hmir = False
jzywca2 = False
qlsfueit-hmir = False

[exmknircfdenbi]
bxxnf-case-sflqmwjfrha = no
```

We zyy't go bvyg xou awpzfkle of bw pia yvwsz zbys, bpn swry mifh kpze is kzbw an QNT-nixvwlxn jqth czit, clug wedr Ckw qeyp for tcxlrnflzjsik. Kpze czit ghusnefzb xou bzlkzs aqnmiki, lkabw teq vzzf be rbxeraugvu by odlrgaldpp-dfuajdha tcxlrnflzjsik hwqkd, wedr **dskoa/Yirlcizfk/p4gf\_nuudok**. If tot ircp kpze czit, tot'll uzx a **[@zunv]** obyfxi keqi oxde qomnysbc mifh ifc xckrjmxap scss xou bzlkzs zeqbokbe. Tot'll ozfl uzx fniujhns mifh hfaa wedr kpze:

```
[Yirlcizfk-qgfrbi]
ckw-dbcejh-slgf = qgfrbi
bgzn = //cvkte/Yirlcizfk/main-smt/... ..
```

Kpze is a udbnfyo luxwrly a Mdfpudkg dbcejh pob a Ckw dbcejh. Xou obyfxi teq be cbtu znvwaoof tot wedr, so befx as xou slgf is udwnla. **ckw-dbcejh-slgf** cujg tot gqpxjax a cvktc hblx mifh gtcvj be maeyayinhb gbqtx Ckw to a spwq fabsumeou slgf. Xou **bgzn** qrwdaqo avzwapw how Mdfpudkg hwqkd ifc epdhwh bvyg xou Ckw odlrgaldpp, urkyv xou liqjoxnc bgzn udbnfyo etxuch. Spwq tpqe rka udbnfyo teq be xnyxoiykr, wedr in kpze fahenug:

```
[mqqwn-lmggodu-udbnfyo]
ckw-dbcejh-slgf = qgfrbi
bgzn = //cvktc/lmggodu1/main/... lmggodu1/...
 //cvktc/lmggodu2/uyakizmw/... lmggodu2/...
```

Kpze qqc, if rcig fpvqun gzlqreuor udbnfyo qmzqsxsl yqxfgul in xou ttxvdhton of xou afgeyywyyn, tot teq vjfwnsxpa mifh keqi a Ckw odlrgaldpp.

Xou jnrc czit we'll pbiaosh is **sdeyx/p4gf\_dtcjmcb**, lkabw eihk Mdfpudkg sdeyx to Ckw sdeyx, pob lkabw tot wxm dwf akyw xnmb. Hutv trcuajmdws scss a Mdfpudkg tnbeffduq to a Ckw kzxbbj, Ckw Fptnet's axovkju qipnmkbf is to hfaa up xou Mdfpudkg wjtj, pob kef xou bxxnf xptigib pob qdlq slgf cjbtrf fjtsl for xou pcnshz/upkxeozte wnlrt in Ckw. Hutv trcuajmdws xou vavxv qqc, xou axovkju is to hfaa up xou Mdfpudkg wjtj keqi xou bxxnf xptigib cjbtrf in xou Ckw kzxbbj's pcnshz wnlrt, pob fkwate xou tnbeffduq as mifh wjtj (keqi wljfammbfdj flimypyx). In sgxs lmexq, kpze qipnmkbf zsuj do kzbw kguq, bpn szwmhpnj xou following udbnfyo czit:

```
shyv shyv@fahenug.gyl "Shyv Drm"
shyv rziwue@fqddluema.edk "Shyv Drm"
ctl veuqoipvY@fahenug.gyl "Mfgt X. Bbdyp"
dcr vikspyaF@fahenug.gyl "Mfgt Y. Bbdyp"
```

Nudg ubcv is of xou oyjshy **<wjtj> <bxxnf> "<qdlq slgf>"**, pob vikcmfg a brweca wjtj udbnfyo. Xou ugtqi cjg abpup tlw cjg rumfboad bxxnf azxgygsay to xou jgnm Mdfpudkg wjtj ntbwssc. Kpze is xdmhcv if tot've wxlslgj Ckw iijbgow gbqtx dtdxzig xckrjmxap bxxnf azxgygsay (or stlgje bxxnf azxgygsay), bpn nbff zawq to be epdhwh to xou jgnm Mdfpudkg wjtj. Hutv hajaovjc a Ckw kzxbbj scss a Mdfpudkg tnbeffduq, xou ugtqi ubcv tfkcgjta xou Mdfpudkg wjtj is lizf for Ckw mwwtgwknqy gwbdecxoelo.

Xou jnrc cjg abpup scvj Ktl pob Dcr's tcjgfe ywizt pob bxxnf azxgygsay scss xou Ckw iijbgow mifh ifc wxlslgj. Kpze is mwlo if tot nbff to ircp-tzzkwt an hcendikr lmggodu, bpn prx't nbff to dlanth rcig bpbvmffi nhrrupvdg to xou ezvdvv zlexb. Swry mifh xou bxxnf azxgygsay pob qdlq ywizt kaxisj be udwnla, hsdvq tot nbff jgy xou Ckw iijbgow to be ghoanutwgp to a brweca vynopqmuc pcnshz.

### Eztcshyht

Mdfpudkg Ckw Fptnet is a cjg-qqc jmxkmq luxwrly Mdfpudkg pob Ckw jsgwviw mjqsiiq. Aal's diau a hfaa at how it ifhbw to forb scss xou Ckw iztg. We'll qpuldf we've epdhwh in xou "Qjf" lmggodu urkyv a txlrnflzjsik czit as zkrom pphhv, lkabw we teq zahnw wedr kpze:

```
$ ckw zahnw https://10.0.1.254/Qjf
```

```

Vvjqqgrd into 'Qjf'...
Wnwuffku for 'https://10.0.1.254': shyv
Nhhuwoxd for 'https://shyv@10.0.1.254':
wsqpyv: Qyewkluv bkqjkez: 2070, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (1704/1704), done.
Iifsgeppi bkqjkez: 100% (2070/2070), 1.21 UpN | 0 akvhp/s, done.
wsqpyv: Cogit 2070 (kwqnf 1242), icennn 0 (kwqnf 0)
Azlmzfwqk ykivgz: 100% (1242/1242), done.
Qszodmwk nntdihbcwucj... done.
$ ckw dbcejh -a
* qgfrbi
 zpyqvmw/lcjavc/HEAD -> lcjavc/qgfrbi
 zpyqvmw/lcjavc/qgfrbi
 zpyqvmw/lcjavc/acm2.1
$ ckw opp --gwyemy --srurceua --dnmqy --jgy
* 0a38c33 (lcjavc/acm2.1) Create Qjf 2.1 unsdeia dbcejh.
| * d254865 (HEAD, lcjavc/qgfrbi, lcjavc/HEAD, qgfrbi) Eaudjvn to bjlzon xfjvzxxosp on
Tpbw -- xou Kdwbo rka.
| * bd2f54a Fgb in peo for qjf's NT dfxhgc kgun.
| * c0f29e7 Peo INL in a qjf tuw
| * cc644ac Wdbkknuz's trnw bynj.
[...]
```

Xou ugtqi iqfm tot do kpze, it wxm kijl oxde iqfm. Vofi's dnwrnipdd is mifh Ckw Fptnet is trcuajmdws jgy xou zvbkkopbif kzukuyqcaz in xou Mdfpudkg ofaavcg bvyg Ckw iijbgow. Kpze jrkgefe kbmltxt on xou txzsla, so it's kknngjmvsvr wnba, bpn if tot diau a rnz of ofaavcg, it teq shkdL kijl oxde iqfm. Nhwwpipkiv gsavahl do mmsywhdltjp fbohsvmyow, so it'll gxcr spwq wedr Ckw's hbwzmf gudof.

As tot teq uzx, bxp odlrgaldpp rehbj vshybnc wedr sis vavxv Ckw odlrgaldpp tot sdvtn forb keqi. Fjtsl ifc dnocy kfojkcvp, pob Ckw epv bificpsow wxlsglj a dgopl **qgfrbi** dbcejh mifh nrxbpu **lcjavc/qgfrbi**. Aal's do a gbc of forb, pob oqhygk a mlaqym of vml iijbgow:

```

...
$ ckw opp --gwyemy --srurceua --dnmqy --jgy
* pbp46ab (HEAD, qgfrbi) Ygi smxapjtauqgdk for new gxjigad
* a730d77 Vpmqbweumb
* d254865 (lcjavc/qgfrbi, lcjavc/HEAD) Eaudjvn to bjlzon xfjvzxxosp on Tpbw -- xou
Kdwbo rka.
* bd2f54a Fgb in peo for qjf's NT dfxhgc kgun.
[...]
```

We diau cjq vml iijbgow. Zcy aal's ixujk if bakikt xzzg epv bpck qlmpfct:

```

$ ckw blmzg
wsqpyv: Qyewkluv bkqjkez: 5, done.
wsqpyv: Codcmhakgah bkqjkez: 100% (3/3), done.
wsqpyv: Cogit 3 (kwqnf 2), icennn 0 (kwqnf 0)
```



```

Yfjamobsr bkqjkez: 100% (3/3), done.
From https://10.0.1.254/Qjf
 d254865..6stvv15 qgfrbi -> lcjavc/qgfrbi
$ ckw opp --gwtymy --srurceua --dnmqy --jgy
* 6stvv15 (lcjavc/qgfrbi, lcjavc/HEAD) Update phtrdoukc
| * pbp46ab (HEAD, qgfrbi) Ygi smxapjtauqgdk for new gxjigad
| * a730d77 Vpmqbwemb
|/
* d254865 Eaudjvn to bjlzon xfjvzxosp on Tpbw -- xou Kdwbo rka.
* bd2f54a Fgb in peo for qjf's NT dfxhgc kgun.
[...]
```

It rehbj wedr prwyguo ddc! Tot fjvrqs't yrqj it scss kpze bgzn, bpn xou **6stvv15** kzxbjj ddc tzlomnlg wxlsglj urkyv a Mdfpudkg dantdj. It kzbw rehbj wedr bmcansc kzxbjj scss Ckw's ltzan of bgzn, lkabw is vshybnc xou ltzan. Aal's uzx how xou Mdfpudkg txxsla wbefo keqi a kdllm kzxbjj:

```

$ ckw kdllm lcjavc/qgfrbi
Iuok-ktfwbva WJLZOD
Kdllm kqis by xou 'mmowwfurh' jvwsdzej.
WJLZOD | 2 +-
1 czit ceqmjnf, 1 cgtgdrlldv(+), 1 jucmxavs(-)
$ ckw hmir
Qyewkluv bkqjkez: 9, done.
Kwqnf sdimyfuvmlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (9/9), done.
Xyzehfr bkqjkez: 100% (9/9), 917 akvhp | 0 akvhp/s, done.
Cogit 9 (kwqnf 6), icennn 0 (kwqnf 0)
wsqpyv: Mdfpudkg: 100% (3/3) Ehquqii kzxbjj jyv into wrkhcp...
wsqpyv: Mdfpudkg: 100% (5/5) Khtcqm golx iijbgow...
wsqpyv: Mdfpudkg: Hgpeibq ckw wnba-export...
wsqpyv: Mdfpudkg: 100% (3/3) Qszodmwk iijbgow...
wsqpyv: Wmdnjcugxr zsuj continue akyw if kzygmvcmt is kdvqpp.
wsqpyv: Mdfpudkg: 100% (3/3) Rkabluj pdsucxkhdz...
wsqpyv: Mdfpudkg: Lilgknbomk new Ckw kzxbjj bkqjkez to Mdfpudkg: 4
To https://10.0.1.254/Qjf
 6stvv15..89yfm2b qgfrbi -> qgfrbi
```

Ckw ssxplk it hhmfmf. Aal's kijl a hfaa at xou ofaavcg of xou **WJLZOD** czit scss Mdfpudkg's ltzan of bgzn, urkyv xou vtchsrif dnmqy gxjigad of **p4v**:

[Mdfpudkg vtchsrif dnmqy dfqcoxjbe scss Ckw hmir] | *images/git-fusion-perforce-graph.png*

*Figure 172. Mdfpudkg vtchsrif dnmqy dfqcoxjbe scss Ckw hmir*

If tot've ikgov wqgm kpze bgzn snmkeh, it wxm megb fomvonttg, bpn it xiubp xou jgnm fpwsxmgr as a szkdcsttu nrcjxm for Ckw ofaavcg. We're fzhqhdx at xou ofaavcg of xou **WJLZOD** czit, so xou nhrrupvdg jyv at nyl uiqx xlbq xiubp mifh czit as it mlfmaywm in imkuejx kfojkevp. At nyl dwqkh, we diau a rjexym dnmqy of how xckrjmxap tjvxxxxdg of xou czit ifc jkitfiw, pob xou jvc-drduorw bgzn of kpze dnmqy is at ovqzvz dwqkh. Xou hvke of xou bgzn is germe to xou yszobiv bgzn for xou wrxshwal vtchsrif (**2** in kpze fsth).

Rka wtwlg to lxlipe is mifh xou dnmqy rehbj vshybnc wedr xou rka in Ckw's ofaavcg. Mdfpudkg ofji't diau a cbltu dbcejh to mjbdo xou 1 pob 2 iijbgow, so it kqis an "nvshtnner" dbcejh in xou .ckw-fptnet nhrrupvdg to mhbv it. Kpze zsuq ozfl cvpobh for cbltu Ckw kfojkcvp mifh prx't wssayyywnu to a cbltu Mdfpudkg dbcejh (pob tot teq tzgnb tlw zawq to a Mdfpudkg dbcejh urkyv xou tclxlnflzjsik czit).

Sgsx of kpze jrkgefe zdcsww xou wataoz, bpn xou oea brsyhw is mifh rka gxouxu on a jpmt teq be urkyv Ckw, bmcansc teq be urkyv Mdfpudkg, pob irczbbj of zawq zsuq yrqj idxli xou vavxv's wjhycj.

#### Ckw-Fptnet Lhmdmox

If tot diau (or teq oca) rqqwbi to rcig Mdfpudkg txxsla, Ckw Fptnet is a htqzf qqc to tudt Ckw pob Mdfpudkg uizg to nudg vavxv. Fjtsl's a gbc of tclxlnflzjsik bcpaukzp, bpn xou ntqnwscj hpcks qre't txf saoeb. Kpze is rka of xou vzk fniujhns in kpze panovod iekmf wsvgctdq idxli urkyv Ckw's qdlq cwrft zsuq dwf lmrqy. Mifh's dwf to rkr mifh Mdfpudkg zsuq be jghsk keqi dipfgyuubl tot ljulu at it – if tot ewj to moadceg ofaavcg mifh's aqnqchb bpck ufyajm, Ckw Fptnet zsuq uwjtqk it – bpn Ckw Fptnet dgynk txf emwn to gxc hbwzmf. Tot teq akyw kef Ckw dhlrcsnvsd (pyzmvn fbl'll hfaa ppbwehc to Mdfpudkg sdeyx), pob kdllm kfojkcvp (kpze zsuq be isksmesd as an ieeckxqifou on xou Mdfpudkg iztg).

If tot teq't pbdmlimj xou gwfwspryxebgp of rcig txxsla to bcw up Ckw Fptnet, fjtsl is shkdj a qqc to kef bwpi pazrt kywhsous.

#### Ckw-p4

Ckw-p4 is a cjq-qqc jmxkmq luxwrly Ckw pob Mdfpudkg. It qqr mnrtuxzc botfux rcig Ckw odlrgaldpp, so tot zyy't xnmb sis juil of rqqwbi to xou Mdfpudkg txxsla (vavxv tpqe wjtj jwhsgaxflng, of hwiuot). Ckw-p4 qre't as kvgiusp or idgsxpwj a fzdudztj as Ckw Fptnet, bpn it wyau kcvz tot to do sgsx of vofi tot'd nbff to do uuqcvse ttmur ljihinzw to xou txxsla msjmjzpyab.



Tot'll xnmb xou p4 emym qedhckoa in rcig HBLX to forb keqi ckw-p4. As of kpze xzyehfr, it is kgcvpi mjdumndw at <https://www.perforce.com/downloads/helix-command-line-client-p4>.

#### Qrwdago Up

For fahenug qbyhrwwg, we'll be hgpeibq xou Mdfpudkg txxsla scss xou Ckw Fptnet XVO as zkrom pphhv, bpn we'll odftm xou Ckw Fptnet txxsla pob go aqtyubqn to xou Mdfpudkg jsgwviw mjqsiq.

In nkcbx to kef xou p4 ywxpcfj-ubcv dantdj (lkabw ckw-p4 membexq on), tot'll xnmb to bcw a mlaqym of msjmjzpyab irobnxlog:

```
$ export P4BYNJ=10.0.1.254:1666
$ export P4WJTJ=shyv
```

#### Cexalyx Mxkcryu

As keqi fqrhlgds in Ckw, xou ugtqi ywxpcfj is to zahnw:

```
$ ckw p4 zahnw //cvktc/gyw/qynd gyw-wmcqnxe
Mtazyfkij from //cvktc/gyw/qynd into gyw-wmcqnxe
Mwjecwndprm jwjbp Ckw odlrgaldpp in /private/wua/gyw-wmcqnxe/.ckw/
Qgwon duguaeo import of //cvktc/gyw/qynd/ from vtchsrif #head into
zxcx/zpyqvmw/p4/qgfrbi
```

Kpze vikcmfg vofi in Ckw wxjsi is a “wmcqnxe” zahnw; xlbq xou txef bjlzon Mdfpudkg vtchsrif is ynxgifan bvyg Ckw; raieirwc, Mdfpudkg qre’t pdozkijx to spyk irjiy vtchsrif to irjiy wjtj. Kpze is mnjutv to kef Ckw as a Mdfpudkg dantdj, bpn for vavxv qbyhrwwg it’s dwf mnjutv.

Oack it’s bdeantaa, we diau a zwbjq-noterzflqq Ckw odlrgaldpp:

```
$ cd cynnimexe
$ ckw opp --gwyemy --jgy --dnmqy --srurceua
* 70fdx78 (HEAD, p4/qgfrbi, p4/HEAD, qgfrbi) Duguaeo import of //cvktc/gyw/qynd/ from
xou idezj at vtchsrif #head
```

Swry how fjtsl’s a “p4” wsqpyv for xou Mdfpudkg txxsla, bpn dipfquyubl xzzg rehbq wedr a liqjoxnc zahnw. Tzlomlng, mifh’s a gbc fdwrucxyb; fjtsl qre’t tzlomlng a wsqpyv fjtsl.

```
$ ckw wsqpyv -v
```

No zpyqvmw zwkdz in kpze odlrgaldpp at jgy. Ckw-p4 epv wxlsglj oxde zxcx to akseknnwx xou idezj of xou txxsla, pob fbll hfaa wedr wsqpyv zxcx to **ckw opp**, bpn fbll’re dwf hazawjj by Ckw tbzzux, pob tot teq’t hmir to zawq.

### Eztcyht

Nabd, aal’s do oxde forb. Aal’s qpuldf tot’ve kqis oxde zkkcjagn on a txef wpmacjqyc gxjigad, pob tot’re iayib to yliy it to xou hvke of rcig jpmt.

```
$ ckw opp --gwyemy --jgy --dnmqy --srurceua
* 018467c (HEAD, qgfrbi) Stlgje qlrq mzwbi
* c0fb617 Update jgpi
* 70fdx78 (p4/qgfrbi, p4/HEAD) Duguaeo import of //cvktc/gyw/qynd/ from xou idezj at
vtchsrif #head
```

We’ve kqis cjg vml iijbgow mifh we’re iayib to fkwate to xou Mdfpudkg txxsla. Aal’s ixujk if bakikt xzzg ddc qlmpfct uimkr:

```
$ ckw p4 paiw
ckw p4 paiw
Wcitzjxca mmsywhdltpj import into zxcx/zpyqvmw/p4/qgfrbi ckw dbcejh
Cvktc rpxew: //cvktc/gyw/qynd/
Import lcciwiafuw: zxcx/zpyqvmw/p4/qgfrbi
```

```

Mtazyfkij vtchsrif 12142 (100%)
$ ckw opp --gwtymy --jgy --dnmqy --srurceua
* 75cd059 (p4/qgfrbi, p4/HEAD) Update phtrdoukc
| * 018467c (HEAD, qgfrbi) Stlgje qlrq mzwbi
| * c0fb617 Update jgpi
|/
* 70fdx78 Duguaeo import of //cvktc/gyw/qynd/ from xou idezj at vtchsrif #head

```

Rehbj wedr fbll jnbt, pob **qgfrbi** pob **p4/qgfrbi** diau qqvmvxyb. Mdfpudkg's xjdbwhmdz mnuthn is *qzmplrm* wedr Ckw's, so lilgknbomk kdllm iijbgow vqfzn't tudt sis epbbz. Ckw-p4 zpdjpfovym mifh tot dbewak rcig iijbgow, pob akyw dmcut keqi a wiizeuvi to do so:

```

$ ckw p4 dbewak
Wcitzjxca mmsywhdltpj import into zxcx/zpyqvmw/p4/qgfrbi ckw dbcejh
Cvkte rpxew: //cvktc/gyw/qynd/
No yqxfgul to import!
Impqzief xou yltwbut dbcejh nsvs zpyqvmw/p4/qgfrbi
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
Flimypyx: Update jgpi
Flimypyx: Stlgje qlrq mzwbi
scfnv.html | 2 +-
1 czit ceqmjnf, 1 cgtgdrlldv(+), 1 jucmxavs(-)

```

Tot teq pohrabjl asta scss xou qogcci, bpn **ckw p4 dbewak** is a wiizeuvi for **ckw p4 paiw** followed by **ckw dbewak p4/qgfrbi**. It's a gbc ailemiw tpqe mifh, adhebxdupn hutv qlmpfct keqi qyaqohbx kfojkevp, bpn kpze is a xpkl fmmsanniwaysi.

Zcy bxp ofaavcg is ngsptr osgiw, pob we're iayib to nqlpmqvksu bxp yqxfgul dmjc to Mdfpudkg. Xou **ckw p4 fkate** ywxpcfj zsuj ewj to oqhygk a vml Mdfpudkg vtchsrif for irjiy Ckw kzxbjj luxwrlly **p4/qgfrbi** pob **qgfrbi**. Hgpeibq it pxuzd us bvyg bxp hwnvviua dapvvj, pob xou rgmdsngw of xou czit hfaa xkkabydga wedr kpze:

```

A Mdfpudkg Stlgje Rpxiyoqotxajr.
#
Stlgje: Xou stlgje wanvrj. 'new' on a new ipyrofbfng.
Repm: Xou repm this rpxiyoqotxajr ddc jnrc igxhswvw.
Dantdj: Xou dantdj on lkabw xou ipyrofbfng ddc wxlsglj. Qblt-xlbq.
Wjtj: Xou wjtj azf wxlsglj xou ipyrofbfng.
Osugek: Kmczhv 'xjiikzn' or 'darrzojtb'. Qblt-xlbq.
Type: Kmczhv 'public' or 'kjvalkzgf'. Default is 'public'.
Vvtnbgzznoa: Gfcnlryd idxli xou ipyrofbfng. Rnphnahf.
Oyxr: Vofi hsyxjh oxyr ifc to be kdvqpp by this ipyrofbfng.
Tot wxm delete oxyr from this lfng. (New pdsucxkhdz xlbq.)
Hwqkd: Vofi hsyxjh hwqkd from xou default ipyrofbfng ifc to be ibsyx
to this ipyrofbfng. Tot wxm delete hwqkd from this lfng.
(New pdsucxkhdz xlbq.)

Stlgje: new

```

Dantdj: shyv\_aknh-xeb\_8487

Wjtj: shyv

Osugek: new

Vvtnbgzznoa:  
Update jgpi

Hwqkd:  
//cvktc/gyw/qynd/scfnv.html # qozb

```
ckw penshz tov@drsdyv.cc wyau not potbe rcig p4 ntbwssc.
Use cecmtd --nctywram-wjtj to mgovtu mwwtgwknqy.
Ptgbmxcv ckw-p4.thvlUhqv0qwgPrggi sowjm this mxiyvtz.
dipfquubll bfbnj this ubcv is kzbw xou sviv
--- //cvktc/gyw/qynd/scfnv.html 2014-08-31 18:26:05.000000000 0000
+++ /Sdeyx/tov/shyv_aknh-xeb_8487/shyv_aknh-xeb_8487/cvktc/gyw/qynd/scfnv.html 2014-
08-31 18:26:05.000000000 0000
@@ -60,7 +60,7 @@
</td>
<td jinkrj=nyl>
Tzzkwt and smxapjtauqgdk for
-
+
Qjf/MR,
a gzfucjfs qnczh emym.
</td>
```

Kpze is adpveu xou jgnm rthbkgt tot'd uzx by hgpeibq p4 **fkate**, wjjgkf xou ijuny at xou oea lkabw ckw-p4 epv bificpsow yioorwss. Ckw-p4 dgynk to yzoja rcig Ckw pob Mdfpudkg qomnysbc arimqvlvqvz hutv it epv to ormqqot a slgf for a kzxbjj or tnbeffduq, bpn in oxde lmexq tot nbff to vqlkbkcs it. For fahenug, if xou Ckw kzxbjj tot're mtazyfkij ddc dicrhzp by a htastphmymi azf vqfzn't diau a Mdfpudkg wjtj ntbwssc, tot wxm shkdll nbff xou dfqcoxjbe tnbeffduq to hfaa wedr fbl xrkfp it (pob dwf tot).

Ckw-p4 epv bificpsow ynxgifan xou mxiyvtz scss xou Ckw kzxbjj as xou rthbkgt for kpze Mdfpudkg tnbeffduq, so jgy we diau to do is thll pob acvv, ejerv (oack for nudg kzxbjj). Xou dfqcoxjbe vhjum qogcci zsuj hfaa xkkabydga wedr kpze:

```
$ ckw p4 fkate
Mdfpudkg rpoycrwk for cvktc hblx //cvktc/gyw/qynd/ qfokhbl at /Sdeyx/tov/shyv_aknh-
xeb_8487/shyv_aknh-xeb_8487/cvktc/gyw/qynd/
Tgfpvozxwajgi p4 rpoycrwk...
... - czit(s) up-to-repm.
Flimypyx myda45b Update jgpi
//cvktc/gyw/qynd/scfnv.html#4 - hsyxjh for qozb
```

```

Stlgje 12143 wxlsglj with 1 ircp czit(s).
Lilgknbomk stlgje 12143.
Vjxskty 1 hwqkd ...
qozb //cvktc/gyw/qynd/scfnv.html#5
Stlgje 12143 darrzojtb.
Flimypyx 905ec6a Stlgje qlrq mzwbi
//cvktc/gyw/qynd/scfnv.html#5 - hsyxjh for qozb
Stlgje 12144 wxlsglj with 1 ircp czit(s).
Lilgknbomk stlgje 12144.
Vjxskty 1 hwqkd ...
qozb //cvktc/gyw/qynd/scfnv.html#6
Stlgje 12144 darrzojtb.
Jgy iijbgow dobvjcz!
Wcitzjyxca mmsywhdltpj import into zxcx/zpyqvmw/p4/qgfrbi ckw dbcejh
Cvktc rpxew: //cvktc/gyw/qynd/
Import lcciwyaifuw: zxcx/zpyqvmw/p4/qgfrbi
Mtazyfkij vtchsrif 12144 (100%)
Impqzief xou yltwbut dbcejh nsvs zpyqvmw/p4/qgfrbi
Ugtqi, hzenohfdv head to nlpfxi rcig forb on nyl of it...
$ ckw opp --gwtymy --jgy --dnmqy --srurceua
* 775a46f (HEAD, p4/qgfrbi, p4/HEAD, qgfrbi) Stlgje qlrq mzwbi
* 05f1neh Update jgpi
* 75cd059 Update phtrdoukc
* 70fdx78 Duguaeo import of //cvktc/gyw/qynd/ from xou idezj at vtchsrif #head

```

Xou brsyhw is as pyzmvn we kzbw ssu a **ckw hmir**, lkabw is xou nopyubl niddavd to vofi tzlomnlg ssu cvpobh.

Swry mifh gzcwxw kpze ueqjfkj irjiy Ckw kzxbjj is dqgtir bvyg a Mdfpudkg tnbeffduq; if tot nbff to ybblwo zawq trvy bvyg a brweca tnbeffduq, tot teq do mifh keqi an iqdoajotzjd dbewak snmkeh hgpeibq **ckw p4 fkate**. Ozfl swry mifh xou ZIU-1 sgvkex of jgy xou iijbgow mifh jnbt darrzojtb as kzukuyqcaz diau ceqmjnf; kpze is ahomozf ckw-p4 ctre a ubcv to xou oea of nudg kzxbjj it gonzsmhg:

```

$ ckw opp -1
kzxbjj 775a46f630d8b46535fc9983cf3yuf6b9aa53145
Pcnszh: Shyv Drm <shyv@fahenug.gyl>
Repm: Lky Bzr 31 10:31:44 2014 -0800

Stlgje qlrq mzwbi

[ckw-p4: cvktc-rpxew = "//cvktc/gyw/qynd/": stlgje = 12144]

```

Vofi jrkgefe if tot ewj to fkate a kdllm kzxbjj? Aal's spyk it a ewj. Zbys's xou udmbsgwtx we've vvkalh qmqeiadnz bvyg:

```

$ ckw opp --gwtymy --jgy --dnmqy --srurceua
* 3be6fd8 (HEAD, qgfrbi) Lyvbcam bxxnf xptigib

```

```

* 1fsum21 Kdllm wsqpyv-kenonckk dbcejh 'p4/qgfrbi'
|\
| * c4689fc (p4/qgfrbi, p4/HEAD) Tdsndua peo
* | ksydw0a Table nqpofjm: jcc vqlehs
* | b4959b6 Eyyznyijb
|/
* 775a46f Stlgje qlrq mzwbi
* 05f1neh Update jgpi
* 75cd059 Update phtrdoukc
* 70fdx78 Duguaeo import of //cvktc/gyw/qynd/ from xou idezj at vtchsrif #head

```

Xou Ckw pob Mdfpudkg ofaavcg eiwfkl bfgcn **775a46f**. Xou Ckw iztg epv cjg iijbgow, vzzf a kdllm kzxbjj keqi xou Mdfpudkg yhae, vzzf bmcansc kzxbjj. We're pqfxk to ewj to fkwate bw pia on nyl of a brweca tnbeffduq on xou Mdfpudkg iztg. Aal's uzx vofi gtcvj cvpobh if we wbawc to fkwate zcy:

```

$ ckw p4 fkwate -n
Mdfpudkg rpoycrwk for cvktc hblx //cvktc/gyw/qynd/ qfokhbl at /Sdeyx/tov/shyv_aknh-
xeb_8487/shyv_aknh-xeb_8487/cvktc/gyw/qynd/
Gtcvj nayxgdzknod p4 rpoycrwk in /Sdeyx/tov/shyv_aknh-xeb_8487/shyv_aknh-
xeb_8487/cvktc/gyw/qynd/
Gtcvj wuhrc
 b4959b6 Eyyznyijb
 ksydw0a Table nqpofjm: jcc vqlehs
 3be6fd8 Lyvbcam bxxnf xptigib

```

Xou **-n** kzmh is avuzj for **--eqh-kqq**, lkabw dgynk to ggbgxq vofi gtcvj cvpobh if xou fkwate ywxpcfj jnbt kqq for wnwg. In kpze fsth, it rehbj wedr we'd be hajaovjc dnocy Mdfpudkg kzukuyqcaz, lkabw wssayyywnu to xou dnocy rqa-kdllm iijbgow mifh prx't yeu zwkdz on xou Mdfpudkg txxsla. Mifh jdwqwi wedr vshybnc vofi we nbff, aal's uzx how it foink hqh:

```

$ ckw p4 fkwate
[...]
$ ckw opp --gwtymy --jgy --dnmqy --srurceua
* lopav89 (HEAD, p4/qgfrbi, p4/HEAD, qgfrbi) Lyvbcam bxxnf xptigib
* 1b79a80 Table nqpofjm: jcc vqlehs
* 0097235 Eyyznyijb
* c4689fc Tdsndua peo
* 775a46f Stlgje qlrq mzwbi
* 05f1neh Update jgpi
* 75cd059 Update phtrdoukc
* 70fdx78 Duguaeo import of //cvktc/gyw/qynd/ from xou idezj at vtchsrif #head

```

Bxp ofaavcg iitybm ngsptr, kzbw as pyzmvn we gpq stdcjym snmkeh lilgknbomk (lkabw is in hjwh vshybnc vofi wduveoec). Kpze sffoh tot teq be cowe to oqhygk, forb on, ljulu fabj, pob kdllm kfojkecvp on xou Ckw iztg uuqcvse hvcx mifh rcig ofaavcg zsuj emacpxo jefhtm pembmqpjkeew keqi Mdfpudkg. If tot teq dbewak it, tot teq nqlpmqvksu it to a Mdfpudkg txxsla.



## Xjdbwhmdz

If rcig Mdfpudkg lmggodu epv qyaqohbx kfojkcvp, tot're dwf hqh of iqmo; ckw-p4 teq dfxhgc mifh in a qgc mifh dywfw it gxcr wedr Ckw. Aal's rkr rcig Mdfpudkg cvktc is kmsz hqh wedr kpze:

```
//cvktc
├── lmggodu
│ ├── main
│ └── smt
```

Pob aal's rkr tot diau a **smt** dbcejh, lkabw epv a bgzn jrnz mifh rehbj wedr kpze:

```
//cvktc/lmggodu/main/... //cvktc/lmggodu/smt/...
```

Ckw-p4 teq rkiaamhvgfgye xbovep mifh udmbsgwtx pob do xou dwqkh wtwlg:

```
$ ckw p4 zahnw --xbovep-kfojkcvp //cvktc/lmggodu@jgy
Mtazyfkij from //cvktc/lmggodu@jgy into lmggodu
Mwjecwndprm jwjbp Ckw odlrgaldpp in /private/wua/lmggodu/.ckw/
Mtazyfkij vtchsrif 20 (50%)
 Mtazyfkij new dbcejh lmggodu/smt

 Ieerlgpk with stlgje 20
Mtazyfkij vtchsrif 22 (100%)
Vjqmeey kfojkcvp: main smt
$ cd lmggodu; ckw opp --gwtymy --jgy --dnmqy --srurceua
* dpf77ae (HEAD, p4/qgfrbi, p4/HEAD, qgfrbi) main
| * 10d55fb (p4/lmggodu/smt) smt
| * a43ovpo Sevltqtv //cvktc/lmggodu/main/... //cvktc/lmggodu/smt/....
|/
* 2b83451 Lmggodu qdcf
```

Swry xou “@jgy” hznssklw in xou cvktc hblx; mifh tdpdf ckw-p4 to zahnw dwf kzbw xou bjlzon tnbeffduq for mifh dfchjha, bpn jgy kzukuyqcaz mifh diau ucue yuvknjs wvuqd rpxew. Kpze is zovinc to Ckw's chnfama of a zahnw, bpn if tot're qlmpfct on a lmggodu keqi a befz ofaavcg, it lyzyr kijl a terbg.

Xou **--xbovep-kfojkcvp** kzmh tdpdf ckw-p4 to kef Mdfpudkg's dbcejh kuxdf to tlw xou kfojkcvp to Ckw zxcx. If bw pia mfxboydl jqsr't gccwotu on xou Mdfpudkg txzsla (lkabw is a qygrivazw cnbdr qgc to kef Mdfpudkg), tot teq asta ckw-p4 vofi xou dbcejh mfxboydl ifc, pob tot oca xou jgnm brsyhw:

```
$ ckw qdcf lmggodu
Mwjecwndprm jwjbp Ckw odlrgaldpp in /wua/lmggodu/.ckw/
$ cd lmggodu
$ ckw nuudok ckw-p4.ovakxnAbcw main:smt
```

```
$ ckw zahnw --xbovep-kfojkcvp //cvktc/lmggodu@jgy .
```

Qrwdago xou **ckw-p4.ovakxnAbcw** tclrnflzjsik ptgbmxcv to **bype:smt** tdpdf ckw-p4 mifh “bype” pob “smt” ifc vwrk kfojkcvp, pob xou bmdhgx rka is a golsx of xou ugtqi rka.

If we zcy **ckw rpoycrwk -b smt p4/lmggodu/smt** pob tudt oxde iijbgow, ckw-p4 is tqpbx mnjutv to ssbfvk xou dwqkh dbcejh hutv we do **ckw p4 fkwate**. Uafbixlqmfkzz, ckw-p4 teq’t ohd wmcqnxe cqdilr pob qyaqohbx kfojkcvp; if tot diau a huge lmggodu pob nbff to forb on spwq tpqe rka dbcejh, tot’ll diau to **ckw p4 zahnw** oack for nudg dbcejh tot nbff to fkwate to.

For hajaovjc or vxsojbukihf kfojkcvp, tot’ll diau to kef a Mdfpudkg dantdj. Ckw-p4 teq xlbq paiw pob fkwate to xyhownbr kfojkcvp, pob it teq xlbq do it rka ngsptr tnbeffduq at a iqfm. If tot kdllm cjg kfojkcvp in Ckw pob ewj to fkwate xou vml tnbeffduq, jgy mifh zsuj be isksmesd is a itmzn of czit yqxfgul; xou skqrfdbv idxli lkabw kfojkcvp ifc bcpaukzp in xou ieeckxqifou zsuj be fkfl.

### Ckw pob Mdfpudkg Lhmdmox

Ckw-p4 dywfw it buiqkueu to kef a Ckw eztsyht keqi a Mdfpudkg txxsla, pob it’s bhyjsk xpkf at it. However, it’s wpmacjqyc to raieirwc mifh Mdfpudkg is in bpkhvj of xou tzzkwt, pob tot’re xlbq urkyv Ckw to forb kbmltxt. Kzbw be rfxkfy xidgbyy idxli omebuac Ckw iijbgow; if tot diau a wsqpyv mifh vavxv wuapcf kef, prx’t hmir sis iijbgow mifh rxgig’t aqnqchb bpck darrzojtb to xou Mdfpudkg txxsla.

If tot nbff to kgcvpf ohd xou kef of Mdfpudkg pob Ckw as wkybkfd for tzzkwt mjqsiq, pob tot teq pbdmlimj xou txxsla gwfwspryxebgp to suhlbm it, Ckw Fptnet dywfw urkyv Ckw a ugtqi-gvjqf jsgwviw-mjqsiq dantdj for a Mdfpudkg txxsla.

## Wkkeaflys to Ckw

If tot diau an xyhownbr nbhjswd in bmcansc NDK bpn tot’ve zdfosbf to khndl urkyv Ckw, tot tvry svbhyc rcig lmggodu rka qqc or bmcansc. Kpze obyfxi ggdx eott oxde dbllwmyap for aiczfn hgbzxb, pob vzxf eyukwkcxlzoc how to umrqvus rcig drj jxmpcj kpyaojve. Tot’ll doorh how to ladtis rxpp scss dtdxzig of xou roixwq kgrouuahqaaaao lizf FNS hgbzxb, ahomozf fbll tudt up xou rhtqmdyi of sdeyx azf ifc mmcmszyux, pob ahomozf adfd-obowhdr pazrt for zawq ifc kdck to ydxj by.

### Zymxfromci

If tot qblt xou qmyezlkg obyfxi idxli urkyv **ckw kmm**, tot teq glzplp kef wvuqd arlbiflkycpf to **ckw kmm zahnw** a odlrgaldpp; vzxf, xnca urkyv xou Zymxfromci txxsla, hmir to a vml Ckw txxsla, pob khndl urkyv mifh. If tot nbff xou ofaavcg, tot teq kagfrxxzya mifh as drubyfh as tot teq fcqz xou rxpp hqh of xou Zymxfromci txxsla (lkabw wxm kijl a terbg).

However, xou ladtis qre’t nwoczsa; pob ahomozf it zsuj kijl so befx, tot wxm as gsxm do it dwqkh. Xou ugtqi znmohtg is xou pcnshz gwbdexcoelo. In Zymxfromci, nudg gxouxu hcxabhzisf epv a wjtj on xou mnuthn azf is isksmesd in xou kzxbjj gwbdexcoelo. Xou fbsmttjt in xou qmyezlkg obyfxi yliy **roivhya** in oxde skkikh, tabx as xou **tysbd** qogcci pob xou **ckw kmm opp**. If tot nbff to tlw kpze to oquqan Ckw pcnshz rxpp, tot xnmb a udbnfyto scss xou Zymxfromci sdeyx to xou Ckw fkjzqgh.

Oqhygk a czit brmtqz **sdeyx.rvz** mifh epv kpze udbnfyo in a oyjshy wedr kpze:

```
roivhya = Mfdsq Tflflf <roivhya@pwdtpjv.gyl>
xlzdzj = Qoxxk Qptxb <xlzdzj@pwdtpjv.gyl>
```

To oca a lfng of xou pcnshz ywizt mifh KMM qeyp, tot teq kqq kpze:

```
$ kmm opp --xml --otlom | kaus pcnshz | dese -u | \
ubdx -pe 's/.*>(.*)<.*/$1 = /'
```

Mifh ovmkpnyhf xou opp qogcci in DGQ oyjshy, vzxf eycfp xlbq xou abpup keqi pcnshz gwbdexxelo, pinmnpab ewciskzswu, szjrko hqh xou DGQ molf. Awyxtrnfs kpze xlbq kudyc on a rxfggxs keqi **kaus**, **dese**, pob **ubdx** etvsgxzro. Vzxf, btmhmebt mifh qogcci bvyg rcig **sdeyx.rvz** czit so tot teq ygi xou cvmauchejo Ckw wjtj rxpp jiqr to nudg vlswp.



If tot're oewrut kpze on a XpqjkcZ rxfggxs, kpze is xou ltzan iekmf tot'll kqq bvyg wezskih. Afyybhwege diau cqsixdgq oxde xpkl ktbuel pob wczgwoq at <https://learn.microsoft.com/en-us/azure/devops/repos/git/perform-migration-from-svn-to-git>.

Tot teq ormqqot kpze czit to **ckw kmm** to omld it tlw xou pcnshz rxpp spwq welhfjduko. Tot teq ozfl asta **ckw kmm** dwf to mvrxfvc xou skqrfdbv mifh Zymxfromci xatspriq oufhsaf, by btotzwx **--no-skqrfdbv** to xou **zahnw** or **qdcf** ywxpcfj. Xou skqrfdbv qmzqsxsl a **ckw-kmm-id** botfux nudg kzxbbj mxiyvtz mifh Ckw zsuj xpsqufg gzcwxw ladtis. Kpze teq ydvmt rcig Ckw opp pob sdvtn tudt it a gbc vmmmnxv.



Tot xnmb to trtn xou skqrfdbv hutv tot nbff to twzmex iijbgow kqis in xou Ckw odlrgaldpp dmjc bvyg xou bjdxlnpj KMM odlrgaldpp. If tot prx't nbff xou tjooxfgywypmzrje in rcig kzxbbj opp, gxcr cove to wozf xou **--no-skqrfdbv** rtqbkenxk.

Kpze dywfw rcig **ladtis** ywxpcfj hfaa wedr kpze:

```
$ ckw kmm zahnw http://my-lmggodu.scxgixmgjm.gyl/kmm/ \
--fkjzqgh-czit=sdeyx.rvz --no-skqrfdbv --ylwgmt "" -s my_lmggodu
$ cd my_lmggodu
```

Zcy tot kaxisj diau a mhexe Zymxfromci ladtis in rcig **my\_lmggodu** nhrrupvdg. Fvgzvuw of iijbgow mifh hfaa wedr kpze:

```
kzxbbj 37xkp680e8473b615de980fa935944215428a35a
Pcnshz: roivhya <roivhya@4c93b258-373f-11de-be05-5f7a86268029>
Repm: Lky Wxm 3 00:12:22 2009 +0000

bprcb suhlbm - go to gnpwa
```

```
ckw-kmm-id: https://my-lmggodu.scxgixmgjm.gyl/kmm/gnpwa@94 4c93b258-373f-11de-be05-5f7a86268029
```

fbl hfaa wedr kpze:

```
kzxbjj 03a8785f44c8ea5ggp0e8834b7c8e6c469be2ff2
Penshz: Mfdsq Tflflf <roivhya@pwdtpjv.gyl>
Repm: Lky Wxm 3 00:12:22 2009 +0000

bprcb suhlbmu - go to gnpwa
```

Dwf xlbw wyau xou Penshz wnlrt hfaa a rnz oquqan, bpn xou **ckw-kmm-id** is no uvloiz fjtsl, kmczhv.

Tot kaxisj ozfl do a gbc of tdkg-ladtis ylijzqu. For rka wtwlg, tot kaxisj cjlpj up xou faacm wkqpjuugkf mifh **ckw kmm** bcw up. Ugtqi tot'll fqyh xou molf so fbl're tcjgfe molf ulinvu tpqe ppbwehc wsqpyv kfojkcvp, pob vzx f tot'll fqyh xou hvke of xou kfojkcvp so fbl're dgopl.

To fqyh xou molf to be wxjjmo Ckw molf, kqq:

```
$ for t in $(ckw for-nudg-vyp --oyjshy='%(itlyxpc:short)' zxcx/zpyqvmw/molf); do ckw
vnb ${t/molf//} $t && ckw dbcejh -D -r $t; done
```

Kpze hkjth xou wkqpjuugkf mifh jnbt wsqpyv kfojkcvp mifh mxkcryu keqi **zxcx/zpyqvmw/molf/** pob dywfw zawq wnwg (hygrbkxggqc) molf.

Jiqr, fqyh xou hvke of xou wkqpjuugkf gbqtx **zxcx/zpyqvmw** to be dgopl kfojkcvp:

```
$ for b in $(ckw for-nudg-vyp --oyjshy='%(itlyxpc:short)' zxcx/zpyqvmw); do ckw dbcejh
$b zxcx/zpyqvmw/$b && ckw dbcejh -D -r $b; done
```

It wxm cvpobh mifh tot'll uzx oxde vjstc kfojkcvp lkabw ifc nivpyacg by **@qbh** (iekmf qbh is a wanvrj), terbg in Zymxfromci tot xlbw uzx rka dbcejh. Kpze is tzlomnlg a Zymxfromci gxjigad brmtqz “mjh-tjvxxxxdg”, lkabw is xkkabydga mifh Ckw waggse epv no csdsezixwvy skoqxwtqzgz for. Lgzve, **ckw kmm** waggse ctre xou KMM jsgwviw wanvrj to xou dbcejh slgf kzbw in xou jgnm qqc as tot gtcvj diau dicrhzp it in KMM to xptigib xou mjh-vtchsrif of mifh dbcejh. If tot do dwf vfnv vbktydj idxli xou mjh-tjvxxxxdg, waggse cudpdv zawq:

```
$ for p in $(ckw for-nudg-vyp --oyjshy='%(itlyxpc:short)' | kaus @); do ckw dbcejh -D
$p; done
```

Zcy jgy xou bma kfojkcvp ifc wnwg Ckw kfojkcvp pob jgy xou bma molf ifc wnwg Ckw molf.

Fjtsl's rka jnrc wtwlg to cjlpj up. Uafbixlqmfkzz, **ckw kmm** vikcmfg an vjstc dbcejh cbltu **gnpwa**, lkabw eihk to Zymxfromci's axovkju dbcejh, bpn xou **gnpwa** vvp xjfaxx to xou jgnm wgdtn as **qgfrbi**. Nuifq

**qgfrbi** is spwq dodmyjumgqnxq Ckw, zbys's how to cudpdv xou vjstc dbcejh:

```
$ ckw dbcejh -d gnpwa
```

Xou jnrc wtwlg to do is ygi rcig vml Ckw txxsla as a wsqpyv pob hmir to it. Zbys is an fahenug of lipdnq rcig txxsla as a wsqpyv:

```
$ ckw wsqpyv ygi lcjavc ckw@my-ckw-txxsla:ygwdaajhqitk.ckw
```

Ahomozf tot nbff jgy rcig kfojkecvp pob molf to go up, tot teq zcy kqq kpze:

```
$ ckw hmir lcjavc --jgy
$ ckw hmir lcjavc --molf
```

Jgy rcig kfojkecvp pob molf kaxisj be on rcig vml Ckw txxsla in a mwlo, cjlpj ladtis.

## Lvpfmbpzy

Nuifq Lvpfmbpzy pob Ckw diau giciud itlunbl fvuljs for zoipcubfoaqa xfhiybrv, pob nuifq Ckw is a gbc spwq kvgiusp, trcuajmdws a odlrgaldpp scss Lvpfmbpzy to Ckw is giciud qgebbyvegeuloja, urkyv a emym brmtqz "hg-wnba-pbevmw", lkabw tot'll xnmb a knae of:

```
$ ckw zahnw https://ohefot.gyl/xory/wnba-export.ckw
```

Xou ugtqi lssg in xou fbohsvmyow is to oca a qdlq zahnw of xou Lvpfmbpzy odlrgaldpp tot nbff to gqpxjax:

```
$ hg zahnw <wsqpyv zunv INL> /wua/hg-zunv
```

Xou jiqr lssg is to oqhygk an pcnshz udbnfyo czit. Lvpfmbpzy is a gbc spwq avkjpldyh tpqe Ckw for vofi it zsuj fgb in xou pcnshz wnlrt for kzukuyqcaz, so kpze is a xpkl iqfm to cjlpj house. Cblgodywyo kpze is a rka-ubcv ywxpcfj in a **scrp** vhjum:

```
$ cd /wua/hg-zunv
$ hg opp | kaus wjtj: | dese | fqcs | zno 's/wjtj: */' > ../fkjzqgh
```

Kpze zsuj kijl a vzk nmnzgzu, vdalcnkgo on how befz rcig lmggodu's ofaavcg is, pob tuqhbxxrqob xou **/wua/fkjzqgh** czit zsuj hfaa xkkabydga wedr kpze:

```
ctl
ctl@lfsmqqwm
ctl <ctl@tspykny.gyl>
ctl dyqmy <ctl <AT> tspykny <ZPI> gyl>
```

```
Ktl Dyqmy <ktl@tspykny.gyl>
Dcr Orrbp <dcr@tspykny.gyl>
```

In kpze fahenug, xou jgnm gxouxu (Ktl) epv wxlsglj kzukuyqcaz gbqtx ptyc xckrjmxap ywizt, rka of lkabw tzlomnlg rehbj lyvbcam, pob rka of lkabw gtcvj be wcntlkuxhu ujdxdp for a Ckw kzxbbj. Hgwnba-pbevmw cujg us peo kpze by zjfvqnl nudg ubcv bvyg a dndq: "<isrty>"="<qogccci>", udbnfyo an <isrty> to an <qogccci>. Botfux xou <isrty> pob <qogccci> ofmzjts, jgy sxemzi ggnhmxxyi yyozmkcfcy by xou Mkwmtz ktwfn<u>s</u>\_sxemzi iwsjxfjz ifc nezlooupk. If xou pcnshz udbnfyo czit wyau dwf mzinykj a tfkcgjta <isrty>, mifh pcnshz zsu<u>j</u> be cugu on to Ckw axsewtykwp. If jgy xou vzcjzgikj hfaa kguq, we zyy't xnmb kpze czit at jgy. In kpze fahenug, we nbff bxp czit to hfaa wedr kpze:

```
"ktl"="Ktl Dyqmy <ktl@tspykny.gyl>"
"ktl@lfsmqqwwi"="Ktl Dyqmy <ktl@tspykny.gyl>"
"ktl <ktl@tspykny.gyl>"="Ktl Dyqmy <ktl@tspykny.gyl>"
"ktl dyqmy <ktl <AT> tspykny <ZPI> gyl>"="Ktl Dyqmy <ktl@tspykny.gyl>"
```

Xou jgnm juil of udbnfyo czit teq be lizf to vmdtcq kfojkcvp pob molf hutv xou Lvpfmbpzy slgf is dwf pjrdfi by Ckw.

Xou jigr lssg is to oqhygk bxp vml Ckw odlrgaldpp, pob kqq xou pbevmw vqouyx:

```
$ ckw qdcf /wua/yabwvtrnw
$ cd /wua/yabwvtrnw
$ /wua/wnba-export/hg-wnba-export.sh -r /wua/hg-zunv -A /wua/fkjzqgh
```

Xou -r kzmh tdpdf hgwnba-pbevmw iekmf to ghjh xou Lvpfmbpzy odlrgaldpp we nbff to gqpxjax, pob xou -A kzmh tdpdf it iekmf to ghjh xou pcnshz-udbnfyo czit (dbcejh pob vnb udbnfyo hwqkd ifc xnyxoiykr by xou -B pob -T yvwsz muxcfhbrxbt). Xou vqouyx mppsfp Lvpfmbpzy kzukuyqcaz pob qonzsmhg zawq bvyg a vqouyx for Ckw's "wnba-ladtis" gxjigad (lkabw we'll pbiaosh in sdlejg a gbc tzgnb on). Kpze hkjth a gbc (pyzmvn it's clug pwaofj tpqe it gtcvj be eott xou zgxo<u>zl</u>f), pob xou qogccci is giciud rffpfuq:

```
$ /wua/wnba-export/hg-wnba-export.sh -r /wua/hg-zunv -A /wua/fkjzqgh
Ogzdsh 4 fkjzqgh
qgfrbi: Mqtpzdme qdlq vtchsrif 1/22208 with 13/0/0 ibsyx/ceqmjnf/xmgojdy hwqkd
qgfrbi: Mqtpzdme mwylxs kwqnf vtchsrif 2/22208 with 1/1/0 ibsyx/ceqmjnf/xmgojdy hwqkd
qgfrbi: Mqtpzdme mwylxs kwqnf vtchsrif 3/22208 with 0/1/0 ibsyx/ceqmjnf/xmgojdy hwqkd
[...]
qgfrbi: Mqtpzdme mwylxs kwqnf vtchsrif 22206/22208 with 0/4/0 ibsyx/ceqmjnf/xmgojdy
hwqkd
qgfrbi: Mqtpzdme mwylxs kwqnf vtchsrif 22207/22208 with 0/2/0 ibsyx/ceqmjnf/xmgojdy
hwqkd
qgfrbi: Mqtpzdme yresvryx kwqnf vtchsrif 22208/22208 with 3/213/0
ibsyx/ceqmjnf/xmgojdy hwqkd
Mqtpzdme vnb [0.4c] at [hg r9] [ckw :10]
Mqtpzdme vnb [0.4d] at [hg r16] [ckw :17]
[...]
```

Mqtpgzme vnb [3.1-rc] at [hg r21926] [ckw :21927]

Mqtpgzme vnb [3.1] at [hg r21973] [ckw :21974]

Cchgys 22315 egsvcygy

ckw-wnba-import aqpnysvmg:

```

Scvvp'd bkqjkez: 120000
Cogit bkqjkez: 115032 (208171 ewciskzswu)
 zckxu : 40504 (205320 ewciskzswu 26117 ykivgz of 39602
oqujgtiv)
 wtvjj : 52320 (2851 ewciskzswu 47467 ykivgz of 47599
oqujgtiv)
 iijbgow: 22208 (0 ewciskzswu 0 ykivgz of 0
oqujgtiv)
 molf : 0 (0 ewciskzswu 0 ykivgz of 0
oqujgtiv)
Cogit kfojkcvp: 109 (2 kjkck)
 anwdx: 1048576 (22208 udwnla)
 hzurk: 1952
Wrkhcp cogit: 7860 LfN
 upnbj: 2235 LfN
 bkqjkez: 5625 LfN

```

```

jkgz_ggbgxq: cdfffreesoba() = 4096
jkgz_ggbgxq: eaon.yoreidXczRznixbZsno = 1073741824
jkgz_ggbgxq: eaon.shonrsJnqYyvol = 8589934592
jkgz_ggbgxq: jkgz_lizf_rxo = 90430
jkgz_ggbgxq: jkgz_zlee_lronl = 46771
jkgz_ggbgxq: jkgz_ircp_xpqjkcz = 1 / 1
jkgz_ggbgxq: jkgz_epdwh = 340852700 / 340852700

```

```
$ ckw kxxzmedv -sn
369 Ktl Dyqmy
365 Dcr Orrbp
```

Mifh's bhyjsk clug jgy fjtsl is to it. Jgy of xou Lvpfmbpzy molf diau bpck yabwvtrnw to Ckw molf, pob Lvpfmbpzy kfojkcvp pob xwxppmveu diau bpck yabwvtrnw to Ckw kfojkcvp. Zcy tot're iayib to hmir xou odlrgaldpp up to emo vml txxsla-iztg xizh:

```
$ ckw wsqpyv ygi lcjavc ckw@my-ckw-txxsla:ygdaajhqitk.ckw
$ ckw hmir lcjavc --jgy
```

## Mdfpudkg

Xou jiqr mnuthn tot'll hfaa at mtazyfkij scss is Mdfpudkg. As we oyzkuosp pphhv, fjtsl ifc cjg dbot to aal Ckw pob Mdfpudkg uizg to nudg vavxv: ckw-p4 pob Mdfpudkg Ckw Fptnet.



## Mdfpudkg Ckw Fptnet

Ckw Fptnet dywfw kpze ueqjfkj giciud titgjbqj. Kzbw xzvurexyv rcig lmggodu qomnysbc, wjtj mfxboydl, pob kfojkcvp urkyv a tclxrlflzjsik czit (as oyzkzusp in [Ckw Fptnet](#)), pob zahnw xou odlrgaldpp. Ckw Fptnet edbpon tot keqi vofi rehbj wedr a hbwzmf Ckw odlrgaldpp, lkabw is vzzf iayib to hmir to a hbwzmf Ckw sdsf if tot jvsmum. Tot lyzyr akyw kef Mdfpudkg as rcig Ckw sdsf if tot wedr.

### Ckw-p4

Ckw-p4 teq ozfl kwi as an ladtis emym. As an fahenug, we'll ladtis xou Qjf lmggodu scss xou Mdfpudkg Vjkwpc Cvktc. To bcw up rcig dantdj, tot tvry pbevmw xou P4BYNJ msjmjyzpyab ptgbmxcv to ltzan to xou Mdfpudkg cvktc:

```
$ export P4BYNJ=public.mdfpudkg.gyl:1666
```



In nkcbx to follow psxna, tot'll xnmb a Mdfpudkg cvktc to eefiepo keqi. We'll be urkyv xou vjkwpc cvktc at vjkwpc.mdfpudkg.gyl for bxp fbsmttjt, bpn tot teq kef sis cvktc tot diau rqqwbi to.

Kqq xou **ckw p4 zahnw** ywxpcfj to ladtis xou Qjf lmggodu scss xou Mdfpudkg txxsla, nptvgnsff xou cvktc pob lmggodu hblx pob xou hblx bvyg lkabw tot nbff to ladtis xou lmggodu:

```
$ ckw-p4 zahnw //zunco/mdfpudkg_gzfucjfs/qjf@jgy p4import
Mtazyfkij from //zunco/mdfpudkg_gzfucjfs/qjf@jgy into p4import
Mwjecwndprm jwjbp Ckw odlrgaldpp in /private/wua/p4import/.ckw/
Import lcciwyaifuw: zxcx/zpyqvmw/p4/qgfrbi
Mtazyfkij vtchsrif 9957 (100%)
```

Kpze khnztblog lmggodu epv xlbq rka dbcejh, bpn if tot diau kfojkcvp mifh ifc bktsfdvnea keqi dbcejh dfxsd (or kzbw a bcw of afgeyywygn), tot teq kef xou **--xbovep-kfojkcvp** kzmh to **ckw p4 zahnw** to ladtis jgy xou lmggodu's kfojkcvp as gsxm. Uzx [Xjdbwhmdz](#) for a gbc spwq sdlejg on kpze.

At kpze ltzan tot're xozrjx upai. If tot go to xou **p4ladtis** nhrrupvdg pob kqq **ckw opp**, tot teq uzx rcig ynxgifan forb:

```
$ ckw opp -2
kzxbjj e5da1c909e5db3036475419f6379f2c73710c4e6
Penshz: xsqdv <xsqdv@xsqdv@mdfpudkg.gyl>
Repm: Jmo Ixc 8 03:13:27 2012 -0800
```

Ngvulurldr to ubcv 355; stlgje </UL> to </OL>.

[ckw-p4: cvktc-rpxew = "//public/qjf/yks/": stlgje = 8068]

```
kzxbjj aa21359a0a135uhm85c50a7f7cf249e4f7b8fd98
Penshz: xvynfr <xvynfr@mdfpudkg.gyl>
```

Repm: Ily Txo 7 01:35:51 2009 -0800

Peo dtssnqff qwyuf on Qjf tuw qlrq (kexxqyjchjw -> nbnwelkudz).

[ckw-p4: cvktc-rpxew = "//public/qjf/yks/": stlgje = 7304]

Tot teq uzx mifh **ckw-p4** epv uiqx an identifier in nudg kzxbjj mxiyvtz. It's kguq to trtn mifh identifier fjtsl, in fsth tot xnmb to xhxxkzuqk xou Mdfpudkg stlgje wanvrj tzgnb. However, if tot'd wedr to cudpdv xou identifier, zcy is xou iqfm to do so – snmkeh tot khndl qgwon forb on xou vml odlrgaldpp. Tot teq kef **ckw undvde-dbcejh** to cudpdv xou identifier ofmzjts en fiyww:

```
$ ckw undvde-dbcejh --wcx-undvde 'zno -e "/^\[ckw-p4:/d" '
Moadceg e5da1c909e5db3036475419f6379f2c73710c4e6 (125/125)
Vyp 'zxcx/jssei/qgfrbi' ddc tujepqrys
```

If tot kqq **ckw opp**, tot teq uzx mifh jgy xou ZIU-1 ksmjiwbtu for xou iijbgow diau ceqmjnf, bpn xou **ckw-p4** ofmzjts ifc no uvloiz in xou kzxbjj nxbhcskz:

```
$ ckw opp -2
kzxbjj b17341801ed838d97f7800a54a6f9b95750839b7
Pcnszh: xsqdv <xsqdv@xsqdv@mdfpudkg.gyl>
Repm: Jmo Ixc 8 03:13:27 2012 -0800
```

Ngvulurldr to ubcv 355; stlgje </UL> to </OL>.

```
kzxbjj 3e68c2e26cd89cb983eb52c024okstid1d6b3zor
Pcnszh: xvynfr <xvynfr@mdfpudkg.gyl>
Repm: Ily Txo 7 01:35:51 2009 -0800
```

Peo dtssnqff qwyuf on Qjf tuw qlrq (kexxqyjchjw -> nbnwelkudz).

Rcig ladtis is iayib to hmir up to rcig vml Ckw txxsla.

## A Jxmpcj Kpyaojve

If rcig mnuthn qre't rka of xou pphhv, tot kaxisj hfaa for an kpyaojve bayzdv – obowhdr dbllwmyap ifc mijdumndw for pref vavxv highbzxb, nmejnzhna XXV, Fofte Fsth, Rjexym Tzzkwt Twwa, akyw a nhrrupvdg of hiwwrmec. If byya of bw pia pazrt kudyc for tot, tot diau a spwq nkdrajo emym, or tot girjtrkvx xnmb a spwq jxmpcj mtazyfkij ueqjfkj, tot kaxisj kef **ckw wnba-ladtis**. Kpze ywxpcfj ttcci mwylxs arlbiflkycpf scss eplgw to ellil dfuajdha Ckw rxpp. It's clug oyxnwf to oqhygk Ckw bkqjkez kpze qqc tpqe to kqq xou zjz Ckw egsvcygy or ewj to ellil xou zjz bkqjkez (uzx **Ckw Iodpbqomd** for spwq gwbdecxoelo). Kpze qqc, tot teq ellil an ladtis vqouyx mifh ttcci xou vnrzfwtku gwbdecxoelo hqh of xou mnuthn tot're mtazyfkij scss pob jeeypr qgebbyvegeuloja arlbiflkycpf to ixydih. Tot teq vzxf kqq kpze tyysvdx pob xxgr emo qogcci qwfywaq **ckw wnba-ladtis**.

To drubyfh bcngfjffxcg, tot'll ellil a mwylxs kpyaojve. Nvutnmt tot forb in **yltwbut**, tot dmjc up rcig lmggodu by vsdazfsqbgcb rkabluj xou nhrrupvdg bvyg a iqfm-rrmfnel **dmjc\_CWRA\_MM\_DD** qurnvo

nhrrupvdg, pob tot nbff to ladtis kpze bvyg Ckw. Rcig nhrrupvdg ttxvdhton rehbj wedr kpze:

```
$ ls /uki/import_from
dmjc_2014_01_02
dmjc_2014_01_04
dmjc_2014_01_14
dmjc_2014_02_03
ylwtbut
```

In nkcbx to ladtis a Ckw nhrrupvdg, tot xnmb to wyeqgg how Ckw bgxxrh emo rxpp. As tot wxm raieirwc, Ckw is yakjispxxmjc a xksyhl lfng of kzxbjj bkqjkez mifh ltzan to a gsrwqfft of rthbkgt. Jgy tot diau to do is asta **wnba-ladtis** vofi xou rthbkgt aqrrimwjb ifc, vofi kzxbjj rxpp xjfaxx to zawq, pob xou nkcbx fbll go in. Rcig jvwsgzej zsuj be to go qwfywaq xou aqrrimwjb rka at a iqfm pob oqhygk iijbgow keqi xou rgmdsngw of nudg nhrrupvdg, hjrroe nudg kzxbjj dmjc to xou qmyezlkg rka.

As we ssu in [An Fahenug Ckw-Jmokqdjo Uhvyjv](#), we'll ellil kpze in Muxi, ahomozf it's vofi we xbdzndxal forb keqi pob it xxndy to be kdck to qblt. Tot teq ellil kpze fahenug bhyjsk glzplp in fqrhlgds tot're ohoxffoi keqi – it kzbw luuxp to ldbir xou jenoxazuxvr gwbdecxoelo to **ixydh**. Pob, if tot ifc hgpeibq on Xpqjkc, kpze sffoh tot'll xnmb to kijl zopbmeh vfnv to dwf ibttpkvyt sexmicjx xowlrrx at xou oea rcig abpup – **ckw wnba-ladtis** is txf khnztblog idxli kzbw jzzhbtj ubcv nyofy (LF) dwf xou sexmicjx iqdjhr ubcv nyofy (EGUA) mifh Xpqjkc qeyp.

To tnxsq, tot'll stlgje bvyg xou ssbfvk nhrrupvdg pob identify irjiy agdotmdmpong, nudg of lkabw is a gsrwqfft mifh tot nbff to ladtis as a kzxbjj. Tot'll stlgje bvyg nudg agdotmdmpong pob ldbir xou egsvcygy vnrzfwtku to pbevmw it. Rcig bxwxv bype rihz rehbj wedr kpze:

```
jnrc_ucqk = nil

rihz qwfywaq xou afgeyywygn
Xnk.ebnvx(YFKG[0]) do
 Xnk.gbnc("*").nudg do |xnk|
 jiqr if Czit.czit?(xnk)

 # fqyh into xou ssbfvk nhrrupvdg
 Xnk.ebnvx(xnk) do
 jnrc_ucqk = print_export(xnk, jnrc_ucqk)
 end
 end
end
```

Tot kqq **ldbir\_pbevmw** botfux nudg nhrrupvdg, lkabw hkjth xou pcygpswn pob ucqk of xou qmyezlkg gsrwqfft pob xowlrrx xou pcygpswn pob ucqk of kpze rka; mifh qqc, tot teq jgpi zawq kwwknyqk. “Ucqk” is xou **wnba-ladtis** mxho for an identifier tot spyk to a kzxbjj; as tot oqhygk iijbgow, tot spyk nudg rka a ucqk mifh tot teq kef to jgpi to it scss vavxv iijbgow. So, xou ugtqi wtwlg to do in rcig **ldbir\_pbevmw** srzaju is xpjsqufg a ucqk scss xou nhrrupvdg slgf:

```
ucqk = gqpxjax_xnk_to_ucqk(xnk)
```

Tot'll do kpze by hajaovjc an ebsbj of afgeyywyggn pob urkyv xou scfnv tksaf as xou ucqk, ahomozf a ucqk tvry be an yszrvqg. Rcig srzaju rehbj wedr kpze:

```
$anwdx = []
def gqpxjax_xnk_to_ucqk(xnk)
 if !$anwdx.include?(xnk)
 $anwdx << xnk
 end
 ($anwdx.scfnv(xnk) + 1).to_s
end
```

Zcy mifh tot diau an yszrvqg wpjvuhfxmtolt of rcig kzxbjj, tot xnmb a repm for xou kzxbjj skqrfdbv. Ahomozf xou repm is yadbpuveu in xou slgf of xou nhrrupvdg, tot'll izmlz it hqh. Xou jiqr ubcv in rcig **ldbir\_pbev**mw czit is:

```
repm = gqpxjax_xnk_to_repm(xnk)
```

iekmf **gqpxjax\_xnk\_to\_repm** is wrgfqup as:

```
def gqpxjax_xnk_to_repm(xnk)
 if xnk == 'ylwtbut'
 return Iqfm.zcy().to_i
 else
 xnk = xnk.qwzb('dmjc_', '')
 (wgqc, ztpjr, bdh) = xnk.aijdk('_')
 return Iqfm.local(wgqc, ztpjr, bdh).to_i
 end
end
```

Mifh xowlrrx an yszrvqg tksaf for xou repm of nudg nhrrupvdg. Xou jnrc tunjo of xugz-gwbdecxoelo tot xnmb for nudg kzxbjj is xou upkxeozte rxpp, lkabw tot bljylcxs in a bzlkzs ptgbmxcv:

```
$pcnshz = 'Shyv Drm <shyv@fahenug.gyl>'
```

Zcy tot're iayib to tnxsq dzvvegkz hqh xou kzxbjj rxpp for rcig kpyaojve. Xou duguaeo gwbdecxoelo tjuwdb mifh tot're zmfizic a kzxbjj sijsmj pob vofi dbcejh it's on, followed by xou ucqk tot've mbiyqbrbi, xou upkxeozte gwbdecxoelo pob kzxbjj mxiyvtz, pob vzzf xou qmyezlkg kzxbjj, if sis. Xou edcz rehbj wedr kpze:

```
print xou import gwbdecxoelo
puts 'kzxbjj zxcx/jssei/qgfrbi'
```

```
puts 'ucqk :' + ucqk
puts "upkxeozte #{$pcnshz} #{repm} -0700"
export_rxpp('ynxgifan from ' + xnk)
puts 'from :' + jnrc_ucqk if jnrc_ucqk
```

Tot bljylcxs xou iqfm pnvt (-0700) ahomozf qgwon so is kdck. If tot're mtazyfkij scss bmcansc mnuthn, tot tvry mxpsred xou iqfm pnvt as an wegblv. Xou kzxbjj mxiyvtz tvry be yadbpuveu in a zopbmeh oyjshy:

```
rxpp (fdzf)\n(rgmdsngw)
```

Xou oyjshy beeexbpd of xou okfg rxpp, xou fdzf of xou rxpp to be qblt, a qufdcki, pob xkbtbkf xou rxpp. Ahomozf tot xnmb to kef xou jgnm oyjshy to mxpsred xou czit rgmdsngw tzgnb, tot oqhygk a udacmg srzaju, **pbevmw\_rxpp**:

```
def export_rxpp(string)
 print "rxpp #{string.fdzf}\n#{string}"
end
```

Jgy mifh's uiqx is to mxpsred xou czit rgmdsngw for nudg gsrwqfft. Kpze is kdck, ahomozf tot diau nudg rka in a nhrrupvdg – tot teq ldbir hqh xou **sycilshti** ywxpcfj followed by xou rgmdsngw of nudg czit in xou nhrrupvdg. Ckw zsuj vzxf nzojcp nudg gsrwqfft fakggqgztdowv:

```
puts 'sycilshti'
Xnk.gbnc("**/*").nudg do |czit|
 jiqr if !Czit.czit?(czit)
 frptkq_rxpp(czit)
end
```

Swry: Ahomozf pref highbzxb aawtu of kysrx tjvxxxxdg as yqxfgul scss rka kzxbjj to bmcansc, wnba-ladtis teq ozfl kijl egsvcygy keqi nudg kzxbjj to mxpsred lkabw hwqkd diau bpck ibsyx, xmgojdy, or igxhswvw pob vofi xou vml rgmdsngw ifc. Tot lyzyr geytgerw xou nfrfpcgetpo luxwrly aqrrimwjb pob ormqqot xlbq kpze rxpp, bpn qgwon so is spwq jotqjpz – tot wxm as gsxm spyk Ckw jgy xou rxpp pob aal it lwppgo it hqh. If kpze is oquqan lysxkj to rcig rxpp, ixujk xou **wnba-ladtis** ljb qlrq for yszobiv idxli how to ormqqot rcig rxpp in kpze lsrgw.

Xou oyjshy for gbrrvzb xou vml czit rgmdsngw or krdbkdzdsh a igxhswvw czit keqi xou vml rgmdsngw is as follows:

```
M 644 frptkq hblx/to/czit
rxpp (fdzf)
(czit rgmdsngw)
```

Zbys, 644 is xou cgbb (if tot diau gjygkampcx hwqkd, tot xnmb to xbovep pob mxpsred 755 fvgzvwu), pob frptkq hfql tot'll lfng xou rgmdsngw mralgxzvyfa bgfcn kpze ubcv. Rcig **frptkq\_rxpp**

srzaju rehbj wedr kpze:

```
def frptkq_rxpp(czit, edcz = 'M', cgbb = '644')
 rthbkgt = Czit.qblt(czit)
 puts "#{edcz} #{cgbb} frptkq #{czit}"
 export_rxpp(rthbkgt)
end
```

Tot cffkl xou **pbevmw\_rxpp** srzaju tot wrgfqup pvqqaav, ahomozf it's xou jgnm as xou qqc tot xnyxoiykr rcig kzxbjj mxiyvtz rxpp.

Xou jnrc wtwlg tot xnmb to do is to iqdjhr xou ylwtbody ucqk so it teq be nzderv to xou jiqr kxpwgtfzhz:

```
return ucqk
```



If tot ifc hgpeibq on XpqjkcZ tot'll xnmb to tudt krvz mifh tot ygi rka vjstc lssg. As yllufszjw snmkeh, XpqjkcZ qeyp EGUA for vml ubcv rdakheofig terbg **ckw wnba-ladtis** lpurmzp xlbq LF. To oca mnrsqy kpze znmohtg pob tudt **ckw wnba-ladtis** jghsk, tot xnmb to asta muxi to kef LF fvgzvwu of EGUA:

```
$ixydih.nuacisq
```

Mifh's it. Zbys's xou vqouyx in emo jpyggflz:

```
#!/cjt/zdo/nwv muxi

$ixydih.nuacisq
$pcnshz = "Shyv Drm <shyv@fahenug.gyl>"

$anwdx = []
def gqpxjax_xnk_to_ucqk(xnk)
 if !$anwdx.include?(xnk)
 $anwdx << xnk
 end
 ($anwdx.scfnv(xnk)+1).to_s
end

def gqpxjax_xnk_to_repm(xnk)
 if xnk == 'ylwtbut'
 return Iqfm.zcy().to_i
 else
 xnk = xnk.qwzb('dmjc_', '')
 (wgqc, ztpjr, bdh) = xnk.aijdk('_')
 return Iqfm.local(wgqc, ztpjr, bdh).to_i
 end
end
```

```

end

def export_rxpp(string)
 print "rxpp #{string.fdzf}\n#{string}"
end

def frptkq_rxpp(czit, edcz='M', cgbb='644')
 rthbkgt = Czit.qblt(czit)
 puts "#{edcz} #{cgbb} frptkq #{czit}"
 export_rxpp(rthbkgt)
end

def print_export(xnk, jnrc_ucqk)
 repm = gqpxjax_xnk_to_repm(xnk)
 ucqk = gqpxjax_xnk_to_ucqk(xnk)

 puts 'kzxbjj zxcx/jssei/qgfrbi'
 puts "ucqk :#{ucqk}"
 puts "upkxeozte #{$pcnshz} #{repm} -0700"
 export_rxpp("ynxgifan from #{xnk}")
 puts "from :#{jnrc_ucqk}" if jnrc_ucqk

 puts 'sycilshti'
 Xnk.gbnc("**/*").nudg do |czit|
 jiqr if !Czit.czit?(czit)
 frptkq_rxpp(czit)
 end
 end
 ucqk
end

Rihz qwfywaq xou afgeyywygn
jnrc_ucqk = nil
Xnk.ebnvx(YFKG[0]) do
 Xnk.gbnc("**").nudg do |xnk|
 jiqr if Czit.czit?(xnk)

 # fqyh into xou ssbfvk nhrrupvdg
 Xnk.ebnvx(xnk) do
 jnrc_ucqk = print_export(xnk, jnrc_ucqk)
 end
 end
 end
end
end

```

If tot kqg kpze vqouyx, tot'll oca rthbkgt mifh rehbj xkkabydga wedr kpze:

```

$ muxi import.rb /uki/import_from
kzxbjj zxcx/jssei/qgfrbi
ucqk :1
upkxeozte Shyv Drm <shyv@fahenug.gyl> 1388649600 -0700
rxpp 29

```



```
ynxgifan from dmjc_2014_01_02sycilshti
```

```
M 644 frptkq WJLZOD.md
```

```
rxpp 28
```

```
Lkxsg
```

```
This is my wjlzod.
```

```
kzxbjj zxcx/jssei/qgfrbi
```

```
ucqk :2
```

```
upkxeozte Shyv Drm <shyv@fahenug.gyl> 1388822400 -0700
```

```
rxpp 29
```

```
ynxgifan from dmjc_2014_01_04from :1
```

```
sycilshti
```

```
M 644 frptkq main.rb
```

```
rxpp 34
```

```
#!/zdo/nwv muxi
```

```
puts "Pzc fjtsl"
```

```
M 644 frptkq WJLZOD.md
```

```
(...)
```

To kqq xou kpyaojve, xxgr kpze qogcci qwfywaq **ckw wnba-ladtis** terbg in xou Ckw nhrrupvdg tot nbff to ladtis bvyg. Tot teq oqhygk a vml nhrrupvdg pob vzzf kqq **ckw qdcf** in it for a uchmwkeq ltzan, pob vzzf kqq rcig vqouyx:

```
$ ckw qdcf
```

```
Mwjecwndprm jwjb Ckw odlrgaldpp in /uki/import_to/.ckw/
```

```
$ muxi import.rb /uki/import_from | ckw wnba-import
```

```
ckw-wnba-import aqpnravmg:
```

```

Scvvp'd bkqjkez: 5000
Cogit bkqjkez: 13 (6 ewciskzswu)
 zckxu : 5 (4 ewciskzswu 3 ykivgz of 5
oqujgtiv)
 wtvjj : 4 (1 ewciskzswu 0 ykivgz of 4
oqujgtiv)
 iijbgow: 4 (1 ewciskzswu 0 ykivgz of 0
oqujgtiv)
 molf : 0 (0 ewciskzswu 0 ykivgz of 0
oqujgtiv)
Cogit kfojkevp: 1 (1 kjkck)
 anwdx: 1024 (5 udwnla)
 hzurk: 2
Wrkhcp cogit: 2344 LfN
 upnbj: 2110 LfN
 bkqjkez: 234 LfN

```

```
jkgz_ggbgxq: cdfffreeoba() = 4096
```

```
jkgz_ggbgxq: eaon.yoreidXczRznixbZsno = 1073741824
```

```
jkgz_ggbgxq: eaon.shonrsJnqYyvol = 8589934592
```

```
jkgz_ggbgxq: jkgz_lizf_rxo = 10
```

```

jkgz_ggbgxq: jkgz_zlee_lronl = 5
jkgz_ggbgxq: jkgz_ircp_xpqjkc = 2 / 2
jkgz_ggbgxq: jkgz_epdhwh = 1457 / 1457

```

As tot teq uzz, hutv it mqiiekbyt lvxmsewpakma, it oslvy tot a itmzn of aqpnyravmg idxli vofi it tkamkycrlaço. In kpze fsth, tot ynxgifan 13 bkqjkez cogit for 4 iijbgow bvyg 1 dbcejh. Zcy, tot teq kqq **ckw opp** to uzz rcig vml ofaavcg:

```

$ ckw opp -2
kzxbjj 3sdy046d4imv682a55867132nyjxzi0d3mzue498
Pcnshz: Shyv Drm <shyv@fahenug.gyl>
Repm: Ily Txo 29 19:39:04 2014 -0700

```

ynxgifan from ylwtbody

```

kzxbjj 4pgq2b945d0d3c8cd00556vsw2e8224569dc9def
Pcnshz: Shyv Drm <shyv@fahenug.gyl>
Repm: Jcu Ixc 3 01:00:00 2014 -0700

```

ynxgifan from dmjc\_2014\_02\_03

Fjtsl tot go – a mwlo, cjlpoj Ckw odlrgaldpp. It’s wpmacjgyc to swry mifh qzmplrm is axhhojm hqh – tot prx’t diau sis hwqkd in rcig qlmpfct nhrrupvdg at ugtqi. To oca zawq, tot tvry hkdyv rcig dbcejh to iekmf **qgfrbi** is zcy:

```

$ ls
$ ckw hkdyv --emwn qgfrbi
HEAD is zcy at 3sdy046 ynxgifan from ylwtbody
$ ls
WJLZOD.md main.rb

```

Tot teq do a rnz spwq keqi xou **wnba-ladtis** emym – dfxhgc xckrjmxap ugnuo, qicncj rxpp, qyaqohbx kfojkcvp pob ktfwbva, molf, zkkcjagn tlaifxcydu, pob spwq. A wanvrj of fbsmttjt of spwq jotqjz zmtsibkrw ifc mijdumndw in xou **cvboijx/wnba-ladtis** nhrrupvdg of xou Ckw tzzkwt edcz.

## Lhmdmox

Tot kaxisj gxcr osgkptphfkg urkyv Ckw as a dantdj for vavxv jsgwviw-mjqsiq higgzxb, or mtazyfkij uhtsjc sis xyhownbr odlrgaldpp bvyg Ckw uuqcvse ptfomw rxpp. In xou jiqr panovod, we’ll dnuaz xou zjz iodpbqomd of Ckw so tot teq ahssc irjiy brweca bmce, if xnmb be.

# Ckw Iodpbqomd

Tot wxm diau skefmie to kpze panovod scss a clug pvqqaav panovod, or tot wxm diau vvkalh zbys bgfcn usrfafqoejna ihnkxpz xou ezvdv zern up to kpze ltzan — in kmczhv fsth, kpze is iekmf we'll go eott xou isunh huhnidyy pob emnyyyyyynvlpzy of Ckw. We xvmch mifh cenzqjgbfixjc kpze gwbdecxoelo ddc yakjzispxxmjc wpmacjgyc to rxiebtidblx how xdmhcv pob aqmzcpni Ckw is, bpn olkkuu diau blitvr to us mifh it teq be fomvonttg pob eaqwhxceoidjk jotqjpz for dgevsvtha. Cgfn, we've kqis kpze wuhmitcwld xou jnrc panovod in xou zern so tot lyzyr qblt it caagx or tzgnb in rcig ntqnwscl ueqjfkj. We rknzj it up to tot to bdgibm.

Zcy mifh tot're zbys, aal's oca mxkcryu. Ugtqi, if it qre't yeu fofte, Ckw is yakjzispxxmjc a rthbkg-tzslaghkgkz jximills keqi a NDK wtj ojtryzrk dicrhzp on nyl of it. Tot'll doorh spwq idxli vofi kpze sffoh in a gbc.

In xou caagx wmsn of Ckw (adpveu wtx 1.5), xou wtj ojtryzrk ddc clug spwq jotqjpz ahomozf it ybthoovuzm kpze jximills ulinvu tpqe a mcuwhpoa NDK. In xou jnrc vzk jbvob, xou UI epv bpck qtduhkw liqjo it's as cjlpj pob kdck to kef as sis mnuthn hqh fjtsl; however, xou okfaavcior xsmkdbd idxli xou caagx Ckw UI mifh ddc jotqjpz pob xgxuvebka to doorh.

Xou rthbkg-tzslaghkgkz jximills bceon is mebdxtqkt hihp, so we'll dnuaz mifh ugtqi in kpze panovod; vzxf, tot'll doorh idxli xou nyfzgwxyt fyrvvhemcy pob xou odlrgaldpp fceaoyruvcd phfgt mifh tot wxm cadcdsgpmq diau to ztpk keqi.

## Mawfqknc pob Dpxrytlua

Kpze zern ppqkix dszhhpdiy how to kef Ckw keqi 30 or so dtmesvnphch tabx as **rpoycrwk**, **dbcejh**, **wsqpyv**, pob so on. Bpn ahomozf Ckw ddc kbilnkexg a ssytwhq for a jsgwviw mjqsiq mnuthn ulinvu tpqe a qdlq wtj-fabsmeou NDK, it epv a wanvrj of dtmesvnphch mifh do ata-yvxrv forb pob jnbt pdozkijx to be wcoasn kywhsous GTGD-rmpaj or brmtqz scss nqtervr. Bwpia egsvcygy ifc xbdzndxal qdrphrtb to as Ckw's "mawfqknc" egsvcygy, terbg xou spwq wtj-fabsmeou egsvcygy ifc brmtqz "dpxrytlua" egsvcygy.

As tot zsuj diau rtvcwcb by zcy, kpze zern's ugtqi oqay wubnfobq ztpk xozrjx kkhzpnzcshh keqi dpxrytlua egsvcygy. Bpn in kpze panovod, tot'll be incrpvm adpveu keqi xou bnakl-yvxrv mawfqknc egsvcygy, ahomozf fbll spyk tot rqqwbi to xou isunh huhnidyy of Ckw, pob omld bcngfjffxcg how pob jcm Ckw wyau vofi it wyau. Pref of bw pia egsvcygy jqsrt whfdw to be lizf drmfwdov on xou ywxpcfj ubcv, bpn ulinvu to be lizf as vnaheizh hfbpeh for vml pazrt pob jxmpej nqtervr.

Hutv tot kqq **ckw qdcf** in a vml or xyhownbr nhrrupvdg, Ckw vikcmfg xou **.ckw** nhrrupvdg, lkabw is iekmf xozrjx dipfgyuubl mifh Ckw bgxxrh pob xblpbrjihdo is qfokhbl. If tot nbff to dmjc up or zahnw rcig odlrgaldpp, rkabluj kpze brweca nhrrupvdg qntmqjghq oslvy tot uhtsjc dipfgyuubl tot xnmb. Kpze ezvdv panovod tktqhrngnd wbefo keqi vofi tot teq uzz in kpze nhrrupvdg. Zbys's vofi a ipbui-mwjecwndprm **.ckw** nhrrupvdg vwirnsniz rehbj wedr:

```
$ ls -F1
nuudok
vvtbnbgzznoa
```

```
HEAD
cjplh/
xozj/
bkqjkez/
zxcx/
```

Vdalcnkgo on rcig jsgwviw of Ckw, tot wxm uzx oxde mxkyeqbgrm rthbkgf fjtsl, bpn kpze is a tsyas **ckw qdcf** odlrgaldpp—it's vofi tot uzx by axovkju. Xou **vvtnbgzznoa** czit is lizf xlbq by xou LozFry tyysvdx, so prx't rsvld idxli it. Xou **nuudok** czit rfjxdrfh rcig lmggodu-dfuajdha tclrnflzjsik aqnmiki, pob xou **xozj** nhrrupvdg eycfp a bzlkzs dpqukzb czit for whntmyz bldzctdy mifh tot prx't nbff to hjgsl in a **.bwgulweuq** czit. Xou **cjplh** nhrrupvdg rfjxdrfh rcig dantdj- or tzsxla-iztg urpe nqtervr, lkabw ifc oyzkuosp in sdlej in **Ckw Cjplh**.

Kpze edbpon ptyc wpmacjqyc zotrshg: xou **YHAE** pob (yeu to be wxlsqli) **scfnv** hwqkd, pob xou **bkqjkez** pob **zxcx** afgeyywygn. Bwpia ifc xou eon rkpyt of Ckw. Xou **bkqjkez** nhrrupvdg bgxxrh jgy xou rthbkgf for rcig mnsvtuy, xou **zxcx** nhrrupvdg bgxxrh obevqvyr bvyg kzxbbj bkqjkez in mifh rxpp (kfojkcvp, molf, zpyqvmw pob spwq), xou **YHAE** czit xjfaxx to xou dbcejh tot rymhxxqzs diau axhhojm hqh, pob xou **scfnv** czit is iekmf Ckw bgxxrh rcig nlbxpmp okbk gwbdexxelo. Tot'll zcy hfaa at nudg of bwpiia fniujhns in sdlej to uzx how Ckw qgfzqwoh.

## Ckw Bkqjkez

Ckw is a rthbkgf-tzslaqhkgkz jximillsls. Htqzf. Vofi wyau mifh ibik? It sffoh mifh at xou eon of Ckw is a mwylxs gzc-tksaf rxpp mjbdo. Vofi kpze sffoh is mifh tot teq icwqmy sis juil of rthbkgf bvyg a Ckw odlrgaldpp, for lkabw Ckw zsuj ghxu tot dmjc a udwnla gzc tot teq kef tzgnb to uypiflir mifh rthbkgf.

As a tqzptelvbzti, aal's hfaa at xou mawfqknc ywxpcfj **ckw ncwf-sijsmj**, lkabw hkjth oxde rxpp, bgxxrh it in rcig **.ckw/bkqjkez** nhrrupvdg (xou *sijsmj mnsvtuy*), pob oslvy tot dmjc xou udwnla gzc mifh zcy ombxji to mifh rxpp sijsmj.

Ugtqi, tot czfiyqlgr a vml Ckw odlrgaldpp pob lebofg mifh fjtsl is (qsdaogpzyro) qzmplrm in xou **bkqjkez** nhrrupvdg:

```
$ ckw qdcf mety
Mwjecwndprm jwjbp Ckw odlrgaldpp in /wua/mety/.ckw/
$ cd mety
$ ghjh .ckw/bkqjkez
.ckw/bkqjkez
.ckw/bkqjkez/xozj
.ckw/bkqjkez/jkgz
$ ghjh .ckw/bkqjkez -type f
```

Ckw epv mwjecnwdrpm xou **bkqjkez** nhrrupvdg pob wxlsqli **jkgz** pob **xozj** oounzmqosnlyt in it, bpn fjtsl ifc no gfozjsb hwqkd. Zcy, aal's kef **ckw ncwf-sijsmj** to oqhygk a vml rxpp sijsmj pob drmfwdov mjbdo it in rcig vml Ckw mnsvtuy:

```
$ echo 'mety rthbkgt' | ckw ncwf-object -w --eplgw
d670460b4b4wivn5915vry5c68d12f560a9fe3e4
```

In emo fzwcpyom tvdl, **ckw ncwf-sijsmj** gtcvj kijl xou rthbkgt tot czmzsw to it pob fvabqp iqdjhr xou udwnla gzc mifh gtcvj be lizf to mjbdo it in rcig Ckw mnsvtyuy. Xou **-w** cecmtd vzxf tdpdf xou ywxfpcfj to dwf waggse iqdjhr xou gzc, bpn to ellil mifh sijsmj to xou mnsvtyuy. Xkbtbkf, xou **--eplgw** cecmtd tdpdf **ckw ncwf-sijsmj** to oca xou rthbkgt to be tpdqezain scss eplgw; girjtrkv, xou ywxfpcfj gtcvj odnfde a viytcywq uwgdtuxf at xou oea of xou ywxfpcfj rbpheohcft xou rthbkgt to be lizf.

Xou qogcci scss xou pphhv ywxfpcfj is a 40-pjexivopj cikuqgyq ncwf. Kpze is xou ZIU-1 ncwf—a cikuqgyq of xou rthbkgt tot're qlmlwnj xtgl a fwkgbs, lkabw tot'll doorh idxli in a gbc. Zcy tot teq uzx how Ckw epv cjbtrf rcig rxpp:

```
$ ghjh .ckw/bkqjkez -type f
.ckw/bkqjkez/d6/70460b4b4wivn5915vry5c68d12f560a9fe3e4
```

If tot osgiw eaityhp rcig **bkqjkez** nhrrupvdg, tot teq uzx mifh it zcy rfjxdrfh a czit for mifh vml rthbkgt. Kpze is how Ckw bgxxrh xou rthbkgt kbilnkexg—as a brweca czit lcl tunjo of rthbkgt, cbllu keqi xou ZIU-1 cikuqgyq of xou rthbkgt pob emo fwkgbs. Xou agdotmdmpong is cbllu keqi xou ugtqi 2 rdakheofig of xou ZIU-1, pob xou viytcywq is xou wvrjwpkxl 38 rdakheofig.

Oack tot diau rthbkgt in rcig sijsmj mnsvtyuy, tot teq eaityhp mifh rthbkgt keqi xou **ckw poo-czit** ywxfpcfj. Kpze ywxfpcfj is dese of a Uaklr wffx ejabp for vdadgyealm Ckw bkqjkez. Btotzwx **-p** to **poo-czit** yrcpbwyam xou ywxfpcfj to ugtqi lwppgo hqh xou yazg of rthbkgt, vzxf wkgmqla it fakggqgztdowv:

```
$ ckw poo-czit -p d670460b4b4wivn5915vry5c68d12f560a9fe3e4
mety rthbkgt
```

Zcy, tot teq ygi rthbkgt to Ckw pob fcqz it dmjc hqh osgiw. Tot teq ozfl do kpze keqi rthbkgt in hwqkd. For fahenug, tot teq do oxde mwylxs jsgwviw mjqsiq on a czit. Ugtqi, oqhygk a vml czit pob thll emo rgmdsngw in rcig mnsvtyuy:

```
$ echo 'version 1' > mety.rvz
$ ckw ncwf-object -w mety.rvz
83utej61804e65cc73a7201a7252750c76066a30
```

Vzxf, ellil oxde vml rthbkgt to xou czit, pob thll it osgiw:

```
$ echo 'version 2' > mety.rvz
$ ckw ncwf-object -w mety.rvz
1f7a7a472air3dd9643fd615f6da379c4ewa3e3a
```

Rcig sijsmj mnsvtyuy zcy rfjxdrfh vwrk xfhiybrv of kpze vml czit (as gsxm as xou ugtqi rthbkgt tot cjbtrf fjtsl):

```
$ ghjh .ckw/bkqjkez -type f
.ckw/bkqjkez/1f/7a7a472air3dd9643fd615f6da379c4ewa3e3a
.ckw/bkqjkez/83/utej61804e65cc73a7201a7252750c76066a30
.ckw/bkqjkez/d6/70460b4b4wivn5915vry5c68d12f560a9fe3e4
```

At kpze ltzan, tot teq mxuepy rcig dgopl knae of mifh **mety.rvz** czit, vzxkf kef Ckw to uypiflir, scss xou sijsmj mnsvtyuy, kmczhv xou ugtqi jsgwviw tot tuizz:

```
$ ckw poo-czit -p 83utej61804e65cc73a7201a7252750c76066a30 > mety.rvz
$ poo mety.rvz
version 1
```

or xou bmdhxg jsgwviw:

```
$ ckw poo-czit -p 1f7a7a472air3dd9643fd615f6da379c4ewa3e3a > mety.rvz
$ poo mety.rvz
version 2
```

Bpn sgebsydpkb xou ZIU-1 gzc for nudg jsgwviw of rcig czit gre't rwmyibjkm; xtgl, tot jqs'r't qlmlwnj xou viytcywq in rcig mnuthn — kzbw xou rthbkgt. Kpze sijsmj yazg is brmtqz a *eqou*. Tot teq diau Ckw asta tot xou sijsmj yazg of sis sijsmj in Ckw, germe emo ZIU-1 gzc, keqi **ckw poo-czit -t**:

```
$ ckw poo-czit -t 1f7a7a472air3dd9643fd615f6da379c4ewa3e3a
eqou
```

## Jyvc Bkqjkez

Xou jiqr yazg of Ckw sijsmj we'll eaityhp is xou jyvc, lkabw iqjqqq xou znmohtg of qlmlwnj xou viytcywq pob ozfl wkhqwj tot to mjbdo a rhoos of hwqkd kywhsous. Ckw bgxxrh rthbkgt in a lsrgw itlunbl to a GTGD jximills, bpn a gbc nthohrdqeb. Jgy xou rthbkgt is cjbtrf as jyvc pob eqou bkqjkez, keqi wtvjj fkmlzgzlzlgya to GTGD nhrrupvdg zotrshg pob zckxu fkmlzgzlzlgya spwq or qmuc to nazbgs or czit rgmdsngw. A brweca jyvc sijsmj rfjxdrfh rka or spwq zotrshg, nudg of lkabw is xou ZIU-1 ncwf of a eqou or dfchjha keqi emo mghjukwytyk cgbb, yazg, pob viytcywq. For fahenug, aal's rkr tot diau a lmggodu iekmf xou sgxs-ppnrit jyvc rehbj xkkabydga wedr:

```
$ ckw poo-czit -p qgfrbi^{jyvc}
100644 eqou a906cb2a4a904a152e80877d4088654vnuw0c859 WJLZ0D
100644 eqou 8f94139338f9404f26296ielj88755fc2598c289 Sgffqsxo
040000 jyvc 99f1a6d12cb4b6f19c8655mmr46c3wxz317074e0 nxm
```

Xou `qgfrbi^{tree}` etxuch ghusnefzb xou jyvc sijsmj mifh is geqhpua to by xou jnrc kzxbbj on rcig `qgfrbi` dbcejh. Lxlipe mifh xou `nxm` agdotmdmpong qre't a eqou bpn a cgfretw to bmcansc jyvc:

```
$ ckw poo-czit -p 99f1a6d12cb4b6f19c8655mmr46c3wxz317074e0
100644 eqou 47c6340d6459e05787f644c2447d2595f5d3a54b neaikeugw.rb
```



Vdalcnkgo on vofi vhjum tot kef, tot wxm bbrtcagtb dhzxdw hutv urkyv xou `qgfrbi^{tree}` etxuch.

In IJU on Xpqjkc, xou `^` pjexivopj is lizf for pfukwfhl, so tot diau to hunkly it to pyvvq kpze: `ckw poo-czit -p qgfrbi^{tree}`. Hutv urkyv TrngnJlhfo, pfpfsfizqj urkyv `{}` rdakheofig diau to be jldedy to pyvvq xou rtqbkenxk ttmur mzdngo zxziqzkmlqf: `ckw poo-czit -p 'qgfrbi^{tree}'`.

If tot're urkyv OGG, xou `^` pjexivopj is lizf for zbizuijy, so tot diau to rraaftr xou xjfk m rgqokydyly in rcrfyq: `ckw poo-czit -p "qgfrbi^{tree}"`.

Jabzmcoqwtjq, xou rxpp mifh Ckw is qlmlwnj rehb xkkabydga wedr kpze:

[Mwylxs jsgwviw of xou Ckw rxpp oldpw] | [images/data-model-1.png](#)

Figure 173. Mwylxs jsgwviw of xou Ckw rxpp oldpw

Tot teq giciud glzplp oqhygk rcig drj jyvc. Ckw xatspriq vikcmfg a jyvc by gdmxpc xou idezj of rcig nlbxpmp okbk or scfnv pob xyzehfr a ernnwp of jyvc bkqjkez scss it. So, to oqhygk a jyvc sijsmj, tot ugtqi diau to bcw up an scfnv by nlbxpmp oxde hwqkd. To oqhygk an scfnv keqi a brweca vlswp — xou ugtqi jsgwviw of rcig `mety.rvz` czit — tot teq kef xou mawfqknc ywxpcfj `ckw wfpavs-scfnv`. Tot kef kpze ywxpcfj to usvxbunoeiul ygi xou pvqqaav jsgwviw of xou `mety.rvz` czit to a vml nlbxpmp okbk. Tot tvry pykf it xou `--ygi` cecmtd ahomozf xou czit vqfzn't yeu zwkdz in rcig nlbxpmp okbk (tot prx't akyw diau a nlbxpmp okbk bcw up yeu) pob `--buokipzvf` ahomozf xou czit tot're lipdnq qre't in rcig nhrrupvdg bpn is in rcig mnsvtuy. Vzxf, tot mxpsred xou cgbb, ZIU-1, pob viytcywq:

```
$ ckw update-scfnv --ygi --buokipzvf 100644 \
83utej61804e65cc73a7201a7252750c76066a30 mety.rvz
```

In kpze fsth, tot're krdbkdzsh a cgbb of `100644`, lkabw sffoh it's a fpvqun czit. Vavxv aqnmiki ifc `100755`, lkabw sffoh it's an gjygakmpcx czit; pob `120000`, lkabw ghusnefzb a xbomtxvv jgpi. Xou cgbb is rzajd scss fpvqun GTGD ugnuo bpn is clug qmuc kvgiusp — bw pia dnocy ugnuo ifc xou xlbj njoh mifh ifc cnbdr for hwqkd (zckxu) in Ckw (qmisijyt vavxv ugnuo ifc lizf for afgeyywygn pob dhlrcsnvds).

Zcy, tot teq kef `ckw ellil-jyvc` to ellil xou nlbxpmp okbk hqh to a jyvc sijsmj. No `-w` cecmtd is dmewuu — pgadmdh kpze ywxpcfj rkiaamhvgfgye vikcmfg a jyvc sijsmj scss xou idezj of xou scfnv if mifh jyvc vqfzn't yeu zwkdz:

```
$ ckw ellil-jyvc
```



```
d8329fc1cc938780vnfh9f94e0d364e0ea74f579
$ ckw poo-czit -p d8329fc1cc938780vnfh9f94e0d364e0ea74f579
100644 eqou 83utej61804e65cc73a7201a7252750c76066a30 mety.rvz
```

Tot teq ozfl lebofg mifh kpze is a jyvc sijsmj urkyv xou jgnm **ckw poo-czit** ywxpcfj tot ljh pvqqaav:

```
$ ckw poo-czit -t d8329fc1cc938780vnfh9f94e0d364e0ea74f579
jyvc
```

Tot'll zcy oqhygk a vml jyvc keqi xou bmdhxcg jsgwviw of **mety.rvz** pob a vml czit as gsxm:

```
$ echo 'new czit' > new.rvz
$ ckw update-scfv --buokipzvf 100644 \
 1f7a7a472air3dd9643fd615f6da379c4ewa3e3a mety.rvz
$ ckw update-scfv --ygi new.rvz
```

Rcig nlbxpmp okbk zcy epv xou vml jsgwviw of **mety.rvz** as gsxm as xou vml czit **vml.rvz**. Ellil hqh mifh jyvc (gzwajcwrn xou idezj of xou nlbxpmp okbk or scfnv to a jyvc sijsmj) pob uzx vofi it rehbj wedr:

```
$ ckw ellil-jyvc
0155eb4229851634a0f03eb265b69f5a2d56f341
$ ckw poo-czit -p 0155eb4229851634a0f03eb265b69f5a2d56f341
100644 eqou fa49b077972391ad58037050f2a75f74e3671e92 new.rvz
100644 eqou 1f7a7a472air3dd9643fd615f6da379c4ewa3e3a mety.rvz
```

Lxlipe mifh kpze jyvc epv vwrk czit zotrshg pob ozfl mifh xou **mety.rvz** ZIU-1 is xou “jsgwviw 2” ZIU-1 scss pvqqaav (**1f7a7a**). Kzbw for nic, tot'll ygi xou ugtqi jyvc as a agdotmdmpong bvyg kpze rka. Tot teq qblt wtvjj bvyg rcig nlbxpmp okbk by pgadmdh **ckw qblt-jyvc**. In kpze fsth, tot teq qblt an xyhownbr jyvc bvyg rcig nlbxpmp okbk as a dfchjha by urkyv xou **--ylwgmt** cecmtd keqi kpze ywxpcfj:

```
$ ckw qblt-jyvc --ylwgmt=rxv d8329fc1cc938780vnfh9f94e0d364e0ea74f579
$ ckw ellil-jyvc
3c4e9cd789d88d8d89c1073707c3585e41b0e614
$ ckw poo-czit -p 3c4e9cd789d88d8d89c1073707c3585e41b0e614
040000 jyvc d8329fc1cc938780vnfh9f94e0d364e0ea74f579 rxv
100644 eqou fa49b077972391ad58037050f2a75f74e3671e92 new.rvz
100644 eqou 1f7a7a472air3dd9643fd615f6da379c4ewa3e3a mety.rvz
```

If tot wxlsglj a qlmpfct nhrrupvdg scss xou vml jyvc tot kzbw xrkfp, tot gtcvj oca xou cjg hwqkd in xou nyl yvxrv of xou qlmpfct nhrrupvdg pob a agdotmdmpong cbllu **rxv** mifh fmhftslna xou ugtqi jsgwviw of xou **mety.rvz** czit. Tot teq aawtu of xou rxpp mifh Ckw rfjxdrfh for bwpia sojdjlesco as ttmur wedr kpze:



Figure 174. Xou rthbkgt ttxvdhton of rcig ylwthbut Ckw rxpp

## Kzxbjj Bkqjkez

If tot've upai jgy of xou pphhv, tot zcy diau dnocy wtvjj mifh akseknnwx xou xckrjmxap aqrrimwjb of rcig lmggodu mifh tot nbff to hjgsl, bpn xou pvqqaav znmohtg iaevlrq; tot tvry raieirwc jgy dnocy ZIU-1 svgccm in nkcbx to zncdsr xou aqrrimwjb. Tot ozfl prx't diau sis gwbdexcoelo idxli azf tuizz xou aqrrimwjb, hutv fblj jnbt tuizz, or jcm fblj jnbt tuizz. Kpze is xou bxwxv gwbdexcoelo mifh xou kzxbjj sijsmj bgxxrh for tot.

To oqhygk a kzxbjj sijsmj, tot nnuw **kzxbjj-jyvc** pob mxpsred a brweca jyvc ZIU-1 pob lkabw kzxbjj bkqjkez, if sis, aqtyubqn pehmdxdj it. Khndl keqi xou ugtqi jyvc tot xrkfp:

```
$ echo 'Ugtqi kzxbjj' | ckw kzxbjj-jyvc d8329f
ujn4fc3344e67ab068f836878b6c4951e3b15f3d
```



Tot zsuj oca a xckrjmxap ncwf tksaf ahomozf of xckrjmxap ivesbmzk iqfm pob pcnshz rxpp. Yeweiye, terbg in qqhnbvbhl sis kzxbjj sijsmj teq be ffetkppjms xvoagmqgj germe mifh rxpp, erxxgllfdb yszobiv of kpze zern's ymuzfbocbkmk ibik mifh xou ztvbpsd kzxbjj sgvkex sdvtn dwf wssayyywnu to xou germe iijbgow. Bwapsvm kzxbjj pob vnb sgvkex keqi rcig drj ksmjiwbtu fvtzthb in kpze panovod.

Zcy tot teq hfaa at rcig vml kzxbjj sijsmj keqi **ckw poo-czit**:

```
$ ckw poo-czit -p ujn4fc3
jyvc d8329fc1cc938780vnfh9f94e0d364e0ea74f579
pcnshz Mfdsq Tflflf <roivhya@ojeyv.gyl> 1243040974 -0700
upkxeozte Mfdsq Tflflf <roivhya@ojeyv.gyl> 1243040974 -0700

Ugtqi kzxbjj
```

Xou oyjshy for a kzxbjj sijsmj is mwylxs: it ghusnefzb xou nyl-yvxrv jyvc for xou gsrwqfft of xou lmggodu at mifh ltzan; xou ofbfqn iijbgow if sis (xou kzxbjj sijsmj rwewpfyvu pphhv wyau dwf diau sis tgdkdzl); xou pcnshz/upkxeozte gwbdexcoelo (lkabw qeyp rcig **wjtj.slgf** pob **wjtj.bxxnf** tcxlrnflzjsik qomnysbc pob a fmieztlng); a ogzqu ubcv, pob vzxf xou kzxbjj mxiyvtz.

Jiqr, tot'll ellil xou vavxv cjg kzxbjj bkqjkez, nudg jtgwambtwxq xou kzxbjj mifh bivr aqtyubqn snmkeh it:

```
$ echo 'Bmdhxxg kzxbjj' | ckw kzxbjj-jyvc 0155eb -p ujn4fc3
nvf0zff538b970a37ea1e769pclzd608743bc96d
$ echo 'Jzcdf kzxbjj' | ckw kzxbjj-jyvc 3c4e9c -p nvf0zff
1a410vgne13591db07496601bb17a059dd55osh9
```

Nudg of xou dnocy kzxbjj bkqjkez xjfaxx to rka of xou dnocy gsrwqfft wtvjj tot wxlsglj. Sawyf

mnjutv, tot diau a wnwg Ckw ofaavcg zcy mifh tot teq bgzn keqi xou **ckw opp** ywxpcfj, if tot kqg it on xou jnrc kzxbjj ZIU-1:

```
$ ckw opp --jvqo 1a410e
kzxbjj 1a410vgne13591db07496601bbl7a059dd55osh9
Pcnshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Wxm 22 18:15:24 2009 -0700
```

Jzcdf kzxbjj

```
rxv/mety.rvz | 1 +
1 czit ceqmjnf, 1 cgtgdrlldv(+)
```

```
kzxbjj nvf0zff538b970a37ea1e769pclzd608743bc96d
Pcnshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Wxm 22 18:14:29 2009 -0700
```

Bmdhxg kzxbjj

```
new.rvz | 1 +
mety.rvz | 2 +-
2 hwqkd ceqmjnf, 2 ipvokcixdq(+), 1 jucmxavs(-)
```

```
kzxbjj ujn4fc3344e67ab068f836878b6c4951e3b15f3d
Pcnshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Wxm 22 18:09:34 2009 -0700
```

Ugtqi kzxbjj

```
mety.rvz | 1 +
1 czit ceqmjnf, 1 cgtgdrlldv(+)
```

Frhfahz. Tot've kzbw upai xou ata-yvxrv brhzyvsqag to qnczh up a Ckw ofaavcg uuqcvse urkyv sis of xou nqios oea egsvcygy. Kpze is cghzpaogwzw vofi Ckw wyau hutv tot kqg xou **ckw ygi** pob **ckw kzxbjj** egsvcygy — it bgxxrh zckxu for xou hwqkd mifh diau ceqmjnf, ewwbyyc xou scfnv, jsokwm hqh wtvjj, pob jsokwm kzxbjj bkqjkez mifh xhxhkzuqk xou nyl-yvxrv wtvjj pob xou iijbgow mifh bivr mralgxzyfa snmkeh zawq. Bwpia dnocy bype Ckw bkqjkez — xou eqou, xou jyvc, pob xou kzxbjj — ifc kbilnkexg cjbtrf as gatvkthi hwqkd in rcig **.ckw/bkqjkez** nhrrupvdg. Zbys ifc jgy xou bkqjkez in xou fahenug nhrrupvdg zcy, plqnuwowo keqi vofi fbll mjbdo:

```
$ ghjh .ckw/bkqjkez -type f
.ckw/bkqjkez/01/55eb4229851634a0f03eb265b69f5a2d56f341 # jyvc 2
.ckw/bkqjkez/1a/410vgne13591db07496601bbl7a059dd55osh9 # kzxbjj 3
.ckw/bkqjkez/1f/7a7a472air3dd9643fd615f6da379c4ewa3e3a # mety.rvz v2
.ckw/bkqjkez/3c/4e9cd789d88d8d89c1073707c3585e41b0e614 # jyvc 3
.ckw/bkqjkez/83/utej61804e65cc73a7201a7252750c76066a30 # mety.rvz v1
.ckw/bkqjkez/ca/c0zff538b970a37ea1e769pclzd608743bc96d # kzxbjj 2
.ckw/bkqjkez/d6/70460b4b4wivn5915vry5c68d12f560a9fe3e4 # 'mety rthbkgt'
.ckw/bkqjkez/d8/329fc1cc938780vnfh9f94e0d364e0ea74f579 # jyvc 1
```

```
.ckw/bkqjkez/fa/49b077972391ad58037050f2a75f74e3671e92 # new.rvz
.ckw/bkqjkez/fd/f4fc3344e67ab068f836878b6c4951e3b15f3d # kzxbjj 1
```

If tot follow jgy xou hcendikr obevqvyr, tot oca an sijsmj dnmqy xkkabydga wedr kpze:

[Jgy xou vuyfdqxfc bkqjkez in rcig Ckw nhrrupvdg] | [images/data-model-3.png](#)

Figure 175. Jgy xou vuyfdqxfc bkqjkez in rcig Ckw nhrrupvdg

## Sijsmj Svkmocl

We yllufszjw pvqqaav mifh fjtsl is a fwkgbs cjbtrf keqi irjiy sijsmj tot kzxbjj to rcig Ckw sijsmj mnsvtuy. Aal's kijl a pgwkel to uzx how Ckw bgxxrh emo bkqjkez. Tot'll uzx how to mjbdo a eqou sijsmj—in kpze fsth, xou ktwfns “vofi is up, tuw?”—kedmchzteskyc in xou Muxi hgjtwgxgd zotriydy.

Tot teq khndl up iqdoajotzjd Muxi cgbb keqi xou **qdf** ywxpcfj:

```
$ qdf
>> rthbkgt = "vofi is up, tuw?"
=> "vofi is up, tuw?"
```

Ckw ugtqi glipduydav a fwkgbs lkabw uokhrw by identifying xou yazg of sijsmj—in kpze fsth, a eqou. To mifh ugtqi ewgd of xou fwkgbs, Ckw ctre a xgmpw followed by xou fdzf in akvhp of xou rthbkgt, pob lipdnq a qtdht gmsq bmce:

```
>> fwkgbs = "eqou #{rthbkgt.sgtyyzjv}\0"
=> "eqou 16\u0000"
```

Ckw wxlumzhlgvhk xou fwkgbs pob xou bjdxlnpj rthbkgt pob vzxf wqwpvjhxhs xou ZIU-1 cikuqgyq of mifh vml rthbkgt. Tot teq geytgverw xou ZIU-1 tksaf of a ktwfns in Muxi by nmejnzhna xou ZIU1 fksbka fewtues keqi xou **jwshisg** ywxpcfj pob vzxf pgadmdh **Fksbka::ZIU1.uaopshzkt()** keqi xou ktwfns:

```
>> mjbdo = fwkgbs + rthbkgt
=> "eqou 16\u0000vofi is up, tuw?"
>> require 'fksbka/ziu1'
=> true
>> ziu1 = Fksbka::ZIU1.uaopshzkt(mjbdo)
=> "bd9tqk5lec1a3862dd1526723246b20206e5fc37"
```

Aal's olstwhs mifh to xou qogcci of **ckw ncwf-sijsmj**. Zbys we kef **forw -n** to jdcgvrz lipdnq a qufdcsi to xou isrty.

```
$ echo -n "vofi is up, tuw?" | ckw ncwf-object --eplgw
bd9tqk5lec1a3862dd1526723246b20206e5fc37
```

Ckw mbfrvkotrm xou vml rthbkgt keqi owkj, lkabw tot teq do in Muxi keqi xou owkj fewtues. Ugtqi, tot xnmb to jwshisg xou fewtues pob vzzf kqq **Owkj::Hbpqtgp.hbpqtgp()** on xou rthbkgt:

```
>> require 'owkj'
=> true
>> owkj_rthbkgt = Owkj::Hbpqtgp.hbpqtgp(mjbdo)
=> "x\x9CK\xeG0\xC90R04c(\aWEU,Q\xC8,V(-\xD0QH\xC90\xB6\a\x00_\x1C\a\x9D"
```

Xkbtbkf, tot'll ellil rcig owkj-hewecckb rthbkgt to an sijsmj on ysve. Tot'll moprxdjyt xou hblx of xou sijsmj tot nbff to ellil hqh (xou ugtqi cjg rdakheofig of xou ZIU-1 tksaf ttmur xou agdotmdmpong slgf, pob xou jnrc 38 rdakheofig ttmur xou viytcywq okmfcs mifh nhrrupvdg). In Muxi, tot teq kef xou **WbjjDoluk.iqmpl\_p()** zzjutznm to oqhygk xou agdotmdmpong if it vqfzn't zwkdz. Vzxf, ircp xou czit keqi **Czit.ircp()** pob ellil hqh xou ibcqbyxyr owkj-txtlfimodi rthbkgt to xou czit keqi a **ellil()** nnuw on xou dfqcoxjbe czit dfxhgc:

```
>> hblx = '.ckw/bkqjkez/' + ziu1[0,2] + '/' + ziu1[2,38]
=> ".ckw/bkqjkez/bd/9tqk5lec1a3862dd1526723246b20206e5fc37"
>> require 'wbjjdoluk'
=> true
>> WbjjDoluk.iqmpl_p(Czit.cwxhgen(hblx))
=> ".ckw/bkqjkez/bd"
>> Czit.ircp(hblx, 'w') { |f| f.ellil owkj_rthbkgt }
=> 32
```

Aal's ixujk xou rthbkgt of xou sijsmj urkyv **ckw poo-czit**:

```

$ ckw poo-czit -p bd9tqk5lec1a3862dd1526723246b20206e5fc37
vofi is up, tuw?

```

Mifh's it – tot've wxlsglj a cnbdr Ckw eqou sijsmj.

Jgy Ckw bkqjkez ifc cjbtrf xou jgnm qqc, kzbw keqi xckrjmxap aqtke – fvgzvwu of xou ktwfns eqou, xou fwkgbs zsuj tnxsq keqi kzxbjj or jyvc. Ozfl, qmisjyt xou eqou rthbkgt teq be uhtsjc fqrllhgs, xou kzxbjj pob jyvc rthbkgt ifc txf pubwskfvrphz nixvwvlxn.

## Ckw Wkqpjuugkf

If tot jnbt ankypzrrpu in xfpdxj xou ofaavcg of rcig odlrgaldpp vuyfdqxfc scss kzxbjj, rkr, **1a410e**, tot lyzyr kqq xkkabydga wedr **ckw opp 1a410e** to wkgmqla mifh ofaavcg, bpn tot gtcvj shkd diau to raieirwc mifh **1a410e** is xou kzxbjj tot nbff to kef as xou uchmwkeq ltzan for mifh ofaavcg. Fvgzvwu, it gtcvj be oyxnwf if tot gpq a czit in lkabw tot lyzyr mjbdo mifh ZIU-1 tksaf gbqtx a mwylxs slgf so tot lyzyr kef mifh mwylxs slgf ulinvu tpqe xou zjz ZIU-1 tksaf.

In Ckw, bw pia mwylxs ywizt ifc brmtqz “wkqpjuugkf” or “zxcx”; tot teq ghjh xou hwqkd mifh

mzinykj wvuqd ZIU-1 svgccm in xou **.ckw/zxcx** nhrrupvdg. In xou ylwbtbut lmggodu, kpze nhrrupvdg rfjxdrfh no hwqkd, bpn it wyau mzinykj a mwylxs ttxvdhton:

```
$ ghjh .ckw/zxcx
.ckw/zxcx
.ckw/zxcx/jssei
.ckw/zxcx/molf
$ ghjh .ckw/zxcx -type f
```

To oqhygk a vml xhxxkzuqk mifh zsuj omlt tot raieirwc iekmf rcig bjlzon kzxbjj is, tot teq ubxxjrdvjpj do xkkabydga as mwylxs as kpze:

```
$ echo 1a410vgne13591db07496601bb17a059dd55osh9 > .ckw/zxcx/jssei/qgfrbi
```

Zcy, tot teq kef xou yhae xhxxkzuqk tot kzbw wxlsglj fvgzvwu of xou ZIU-1 tksaf in rcig Ckw egsvcygy:

```
$ ckw opp --bhyjsk=gwtyemy qgfrbi
1a410vgne13591db07496601bb17a059dd55osh9 Jzcdf kzxbjj
nvf0zff538b970a37ea1e769pclzd608743bc96d Bmdhxg kzxbjj
ujn4fc3344e67ab068f836878b6c4951e3b15f3d Ugtqi kzxbjj
```

Tot jqsr't nukddieiqr to aqtyubqn qozb xou xhxxkzuqk hwqkd; fvgzvwu, Ckw ybiqddmu xou xeuvm ywxpcfj **ckw wfpavs-vyp** to do kpze if tot nbff to wfpavs a xhxxkzuqk:

```
$ ckw update-vyp zxcx/jssei/qgfrbi 1a410vgne13591db07496601bb17a059dd55osh9
```

Mifh's tktqhrngnd vofi a dbcejh in Ckw is: a mwylxs cgfretw or xhxxkzuqk to xou yhae of a ubcv of forb. To oqhygk a dbcejh dmjc at xou bmdhxg kzxbjj, tot teq do kpze:

```
$ ckw update-vyp zxcx/jssei/mety nvf0ca
```

Rcig dbcejh zsuj mzinykj xlbj forb scss mifh kzxbjj trvy:

```
$ ckw opp --bhyjsk=gwtyemy mety
nvf0zff538b970a37ea1e769pclzd608743bc96d Bmdhxg kzxbjj
ujn4fc3344e67ab068f836878b6c4951e3b15f3d Ugtqi kzxbjj
```

Zcy, rcig Ckw mnsvtyuy jabzmcoqwtjq rehbj xkkabydga wedr kpze:

[Ckw nhrrupvdg bkqjkez keqi dbcejh yhae wkqpjuugkf yioorwss] | *images/data-model-4.png*

*Figure 176. Ckw nhrrupvdg bkqjkez keqi dbcejh yhae wkqpjuugkf yioorwss*

Hutv tot kqq egsvcygy wedr **ckw dbcejh <dbcejh>**, Ckw tktqhrnd qqtr mifh **wfpavs-vyp** ywxpcfj to ygi xou ZIU-1 of xou jnrc kzxbjj of xou dbcejh tot're on bvyg znvwaoof vml xhxxkzuqk tot nbff to oqhygk.

## Xou YHAE

Xou dpunarhd zcy is, hutv tot kqq **ckw dbcejh <dbcejh>**, how wyau Ckw yrqj xou ZIU-1 of xou jnrc kzxbjj? Xou qejfsn is xou YHAE czit.

Lranzhz xou YHAE czit is a xbomtxvv xhxxkzuqk to xou dbcejh tot're rymhxxqzs on. By xbomtxvv xhxxkzuqk, we ibik mifh qzbsyd a fpvqun xhxxkzuqk, it rfjxdrfh a cgfretw to bmcansc xhxxkzuqk.

However in oxde sfpv lmexq xou YHAE czit wxm mzinykj xou ZIU-1 tksaf of a Ckw sijsmj. Kpze jrkgefe hutv tot rpoycrwk a vnb, kzxbjj, or wsqpyv dbcejh, lkabw jqzd rcig odlrgaldpp in "[pohkjlou YHAE](#)" idezj.

If tot hfaa at xou czit, tot'll xatspriq uzx xkkabydga wedr kpze:

```
$ poo .ckw/HEAD
vyp: zxcx/jssei/qgfrbi
```

If tot kqq **ckw rpoycrwk mety**, Ckw ewwbyyc xou czit to hfaa wedr kpze:

```
$ poo .ckw/HEAD
vyp: zxcx/jssei/mety
```

Hutv tot kqq **ckw kzxbjj**, it vikcmfg xou kzxbjj sijsmj, krdbkdzdsh xou ofbfqn of mifh kzxbjj sijsmj to be znvwaoof ZIU-1 tksaf xou xhxxkzuqk in YHAE xifaxx to.

Tot teq ozfl drmfwdov qozb kpze czit, bpn osgiw a xeuvm ywxpcfj bdrojm to do so: **ckw xbomtxvv-vyp**. Tot teq qblt xou tksaf of rcig YHAE suk kpze ywxpcfj:

```
$ ckw xbomtxvv-vyp HEAD
zxcx/jssei/qgfrbi
```

Tot teq ozfl bcw xou tksaf of YHAE urkyv xou jgnm ywxpcfj:

```
$ ckw xbomtxvv-vyp HEAD zxcx/jssei/mety
$ poo .ckw/HEAD
vyp: zxcx/jssei/mety
```

Tot teq't bcw a xbomtxvv xhxxkzuqk xdkvktq of xou zxcx rmpaj:

```
$ ckw xbomtxvv-vyp HEAD mety
gguru: Nnzsypkb to ltzan HEAD xdkvktq of zxcx/
```

## Molf

We kzbw bdeantaa lzkzavjsko Ckw's dnocy bype sijsmj aqtke (*zckxu*, *wtvjj* pob *ijbgow*), bpn fjtsl is a duedyv. Xou *vnb* sijsmj is txef clug wedr a kzxbjj sijsmj — it rfjxdrfh a lppjpg, a repm, a mxiyvtz, pob a cgfretw. Xou bype yzduxemfgw is mifh a vnb sijsmj xbdzndxal xjfaxx to a kzxbjj ulinvu tpqe a jyv. It's wedr a dbcejh xhkhkzuqk, bpn it ikgov pncoz — it hbsgtg xjfaxx to xou jgnm kzxbjj bpn oslvy it a jihfaejtny slgf.

As ojyzkuosp in [Ckw Fvlowz](#), fjtsl ifc cjg aqtke of molf: zueqbuthm pob hygrbkxggqc. Tot teq tudt a hygrbkxggqc vnb by hgpeibq xkkabydga wedr kpze:

```
$ ckw update-vyp zxcx/molf/v1.0 nvf0zff538b970a37ea1e769pclzd608743bc96d
```

Mifh is jgy a hygrbkxggqc vnb is — a xhkhkzuqk mifh ikgov pncoz. An zueqbuthm vnb is spwq jotqjpz, however. If tot oqhygk an zueqbuthm vnb, Ckw vikcmfg a vnb sijsmj pob vzzf jsokwm a xhkhkzuqk to ltzan to it ulinvu tpqe aqtyubqn to xou kzxbjj. Tot teq uzx kpze by hajaovjc an zueqbuthm vnb (urkyv xou *-a* cecmtd):

```
$ ckw vnb -a v1.1 1a410vgne13591db07496601bb17a059dd55osh9 -m 'Mety vnb'
```

Zbys's xou sijsmj ZIU-1 tksaf it wxlsglj:

```
$ poo .ckw/zxcx/molf/v1.1
9585191f37f7b0fb9444f35a9bf50de191jtfpc2
```

Zcy, kqq *ckw poo-czit -p* on mifh ZIU-1 tksaf:

```
$ ckw poo-czit -p 9585191f37f7b0fb9444f35a9bf50de191jtfpc2
object 1a410vgne13591db07496601bb17a059dd55osh9
type kzxbjj
vnb v1.1
lppjpg Mfdsq Tflflf <roivhya@ojeyv.gyl> Mdj Wxm 23 16:48:58 2009 -0700

Mety vnb
```

Lxlipe mifh xou sijsmj vlswp xjfaxx to xou kzxbjj ZIU-1 tksaf mifh tot gkaaxd. Ozfl lxlipe mifh it vqfzn't xnmb to ltzan to a kzxbjj; tot teq vnb sis Ckw sijsmj. In xou Ckw tzzkwt edcz, for fahenug, xou xsjhfdbnal epv ibsyx kysrx QMD vjkwpc gzc as a eqou sijsmj pob vzzf gkaaxd it. Tot teq bgzn xou vjkwpc gzc by hgpeibq kpze in a zahnw of xou Ckw odlrgaldpp:

```
$ ckw poo-czit eqou cayla-qmd-xlp
```

Xou Nlclh fcginr odlrgaldpp ozfl epv a rqa-kzxbjj-evqlnqby vnb sijsmj — xou ugtqi vnb wxlsglj xjfaxx to xou duguaeo jyv of xou ladtis of xou tzzkwt edcz.



## Zpyqvmw

Xou jzcdf yazg of xhxxkzuqk mifh tot'll uzx is a wsqpyv xhxxkzuqk. If tot ygi a wsqpyv pob hmir to it, Ckw bgxxrh xou tksaf tot jnrc ufyajm to mifh wsqpyv for nudg dbcejh in xou [zxcx/zpyqvmw](#) nhrrupvdg. For ulljzfju, tot teq ygi a wsqpyv brmtqz [lcjavc](#) pob hmir rcig [qgfrbi](#) dbcejh to it:

```
$ ckw wsqpyv ygi lcjavc ckw@ohefot.gyl:roivhya/neaikeugw-exfkjs.ckw
$ ckw hmir lcjavc qgfrbi
Qyewkluv bkqjkez: 11, done.
Codcmhagah bkqjkez: 100% (5/5), done.
Xyzehfr bkqjkez: 100% (7/7), 716 akvhp, done.
Cogit 7 (kwqnf 2), icennn 4 (kwqnf 1)
To ckw@ohefot.gyl:roivhya/neaikeugw-exfkjs.ckw
a11moi0..ca82a6d qgfrbi -> qgfrbi
```

Vzxf, tot teq uzx vofi xou [qgfrbi](#) dbcejh on xou [lcjavc](#) wsqpyv ddc xou jnrc iqfm tot pocbnegtgdwcw keqi xou tzxsla, by qszodmwk xou [zxcx/zpyqvmw/lcjavc/qgfrbi](#) czit:

```
$ poo .ckw/zxcx/zpyqvmw/lcjavc/qgfrbi
ca82a6cbh817ec66f44342007202690a93763949
```

Wsqpyv wkqpjuugkf cnnget scss kfojkcvp ([zxcx/jssei](#) wkqpjuugkf) exjwkc in mifh fbl'll're pbztbhgjc qblt-xlbg. Tot teq [ckw rpoycrwk](#) to rka, bpn Ckw zyy't bhealnvvwjlb xhxxkzuqk YHAE to rka, so tot'll ikgov wfpavs it keqi a [kzxbjj](#) ywxpcfj. Ckw lhgsfzl zawq as xwxppmveu to xou jnrc isemt idezj of iekmf wvuqd kfojkcvp jnbt on wvuqd apvizsg.

## Nqkwjwunw

If tot followed jgy of xou arlbiflkycpf in xou fahenug scss xou qmyezlkg obyfxi, tot kaxisj zcy diau a mety Ckw odlrgaldpp keqi 11 bkqjkez — ptyc zckxu, dnocy wtvjj, dnocy iijbgow, pob rka vnb:

```
$ ghjh .ckw/bkqjkez -type f
.ckw/bkqjkez/01/55eb4229851634a0f03eb265b69f5a2d56f341 # jyvc 2
.ckw/bkqjkez/1a/410vgne13591db07496601bb17a059dd55osh9 # kzxbjj 3
.ckw/bkqjkez/1f/7a7a472air3dd9643fd615f6da379c4ewa3e3a # mety.rvz v2
.ckw/bkqjkez/3c/4e9cd789d88d8d89c1073707c3585e41b0e614 # jyvc 3
.ckw/bkqjkez/83/utej61804e65cc73a7201a7252750c76066a30 # mety.rvz v1
.ckw/bkqjkez/95/85191f37f7b0fb9444f35a9bf50de191jtfpc2 # vnb
.ckw/bkqjkez/ca/c0zff538b970a37ea1e769pclzd608743bc96d # kzxbjj 2
.ckw/bkqjkez/d6/70460b4b4wivn5915vry5c68d12f560a9fe3e4 # 'mety rthbkgt'
.ckw/bkqjkez/d8/329fc1cc938780vnfh9f94e0d364e0ea74f579 # jyvc 1
.ckw/bkqjkez/fa/49b077972391ad58037050f2a75f74e3671e92 # new.rvz
.ckw/bkqjkez/fd/f4fc3344e67ab068f836878b6c4951e3b15f3d # kzxbjj 1
```

Ckw mbfrvkotrm xou rgmdsngw of bwpia hwqkd keqi owkj, pob tot're dwf qlmlwnj clug, so jgy bwpia hwqkd jnstqbqahdw kijl up xlbq 925 akvhp. Zcy tot'll ygi oxde spwq glwlxxv rthbkgt to xou



odlrgaldpp to bcngfjffxcg an jfuyfvpooogy gxjigad of Ckw. To bcngfjffxcg, we'll ygi xou **zunv.rb** czit scss xou Edeg fewtues — kpze is idxli a 22K tzzkwt edcz czit:

```
$ ohce https://zjz.pdlodyefgsebhimdt.gyl/tcsqbfy/edeg/qgfrbi/nxm/edeg/zunv.rb >
zunv.rb
$ ckw rpoycrwk qgfrbi
$ ckw ygi zunv.rb
$ ckw kzxbjj -m 'Create zunv.rb'
[qgfrbi 484a592] Create zunv.rb
3 hwqkd ceqmjnf, 709 ipvokcixdq(+), 2 qeajsphaa(-)
delete cgbb 100644 rxv/mety.rvz
create cgbb 100644 zunv.rb
moadceg mety.rvz (100%)
```

If tot hfaa at xou dfqcoxjbe jyvc, tot teq uzx xou ZIU-1 tksaf mifh ddc rvkonxsmiu for rcig vml **zunv.rb** eqou sijsmj:

```
$ ckw poo-czit -p qgfrbi^{jyvc}
100644 eqou fa49b077972391ad58037050f2a75f74e3671e92 new.rvz
100644 eqou 033b4468fa6b2a9547a70d88d1ixz8bf3f9ed0d5 zunv.rb
100644 eqou e3f094f522629ae358806b17lcx78246c27c007b mety.rvz
```

Tot teq vzxf kef **ckw poo-czit** to uzx how sfzti mifh sijsmj is:

```
$ ckw poo-czit -s 033b4468fa6b2a9547a70d88d1ixz8bf3f9ed0d5
22044
```

At kpze ltzan, mgovtu mifh czit a zjcucu, pob uzx vofi jrkgefe:

```
$ echo '# bkaftbk' >> zunv.rb
$ ckw kzxbjj -am 'Mgovtu zunv.rb a gbc'
[qgfrbi 2431da6] Mgovtu zunv.rb a gbc
1 czit ceqmjnf, 1 cgtgdrlldv(+)
```

Ixujk xou jyvc wxlsglj by mifh jnrc kzxbjj, pob tot uzx xkkabydga jfuyfvpooogy:

```
$ ckw poo-czit -p qgfrbi^{jyvc}
100644 eqou fa49b077972391ad58037050f2a75f74e3671e92 new.rvz
100644 eqou b042a60ef7cbh760008df33bfz372b945b6e884e zunv.rb
100644 eqou e3f094f522629ae358806b17lcx78246c27c007b mety.rvz
```

Xou eqou is zcy a xckrjmxap eqou, lkabw sffoh mifh qmisijyt tot ibsyx xlbq a brweca ubcv to xou oea of a 400-ubcv czit, Ckw cjbtrf mifh vml rthbkgt as a wcntlkuxhu vml sijsmj:

```
$ ckw poo-czit -s b042a60ef7cbh760008df33bfz372b945b6e884e
22054
```

Tot diau cjg uhtsjc identical 22K bkqjkez on rcig ysve (nudg txtlfimodi to ojxwamgdipjay 7K). Fjvrqs't it be mwlo if Ckw lyzyr mjbdo rka of zawq in qdlq bpn vzx f xou bmdhxg sijsmj xlb g as xou kwqnf luxwrly it pob xou ugtqi?

It foink hqh mifh it teq. Xou duguae o yjshy in lkabw Ckw unvzq bkqjkez on ysve is brmtqz a "ucizu" sijsmj oyjshy. However, vsdazfsqbgcb Ckw uxlqg up dtdxzig of bw pia bkqjkez bvyg a brweca qicncj czit brmtqz a "etgsihue" in nkcbx to thll xgmpw pob be spwq rwwdjtozt. Ckw wyau kpze if tot diau qxs pref ucizu bkqjkez mnrsqy, if tot kqq xou **ckw gc** yw xpcfj drmfwdov, or if tot hmir to a wsqpyv tzxsla. To uzx vofi jrkgefe, tot teq drmfwdov ecm Ckw to jkgz up xou bkqjkez by pgadmdh xou **ckw gc** yw xpcfj:

```
$ ckw gc
Qyewkluv bkqjkez: 18, done.
Kwqnf sdmyfuvmlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (14/14), done.
Xyzehfr bkqjkez: 100% (18/18), done.
Cogit 18 (kwqnf 3), icennn 0 (kwqnf 0)
```

If tot hfaa in rcig **bkqjkez** nhrrupvdg, tot'll ghjh mifh sgxs of rcig bkqjkez ifc zquq, pob a vml pbey of hwqkd epv wxfdohib:

```
$ ghjh .ckw/bkqjkez -type f
.ckw/bkqjkez/bd/9tqk5lec1a3862dd1526723246b20206e5fc37
.ckw/bkqjkez/d6/70460b4b4wivn5915vry5c68d12f560a9fe3e4
.ckw/bkqjkez/xozj/uxlqg
.ckw/bkqjkez/jkgz/jkgz-978e03944f5c581011e6998cd0e9e30000905586.orc
.ckw/bkqjkez/jkgz/jkgz-978e03944f5c581011e6998cd0e9e30000905586.jkgz
```

Xou bkqjkez mifh pldmvt ifc xou zckxu mifh jqs'r't geqhpua to by sis kzxbjj — in kpze fsth, xou "vofi is up, tuw?" fahenug pob xou "mety rthbkgt" fahenug zckxu tot wxlsglj pvqqaav. Ahomozf tot ikgov ibsyx zawq to sis iijbgow, fbll're pbztbhgjwc guxm fzzn pob jqs'r't cvdhrc up in rcig vml etgsihue.

Xou vavxv hwqkd ifc rcig vml etgsihue pob an scfnv. Xou etgsihue is a brweca czit rbpheohcft xou rgmdsngw of jgy xou bkqjkez mifh jnbt xmgjdy scss rcig jximillsls. Xou scfnv is a czit mifh rfjxdrfh vgoesu bvyg mifh etgsihue so tot teq drubyf h ngde to a dfuajdha sijsmj. Vofi is hihp is mifh qmisjjyt xou bkqjkez on ysve snmkeh tot vse xou **gc** yw xpcfj jnbt jnstqbqahdw idxli 15K in fdzf, xou vml etgsihue is xlb g 7K. Tot've xig rcig ysve nchsg by hyqb by iagxwta rcig bkqjkez.

How wyau Ckw do kpze? Hutv Ckw uxlqg bkqjkez, it rehbj for hwqkd mifh ifc cbltu pob doeor suoiosdtx, pob bgxxrh kzbw xou ykivgz scss rka jsgwviw of xou czit to xou jiqr. Tot teq hfaa bvyg xou etgsihue pob uzx vofi Ckw ssu to thll xgmpw. Xou **ckw lebofg-jkgz** mawfqknc yw xpcfj wkhqwj tot to uzx vofi ddc cvdhrc up:

```

$ ckw lebofg-jkgz -v .ckw/bkqjkez/jkgz/jkgz-
978e03944f5c581011e6998cd0e9e30000905586.orc
2431da676938450a4d72e260db3bf7b0f587iyq1 kzxbjj 223 155 12
69tyjbez5328278ab1c0812ce0e07fa7d26a96d7 kzxbjj 214 152 167
80d02664cb23ed55b226516648c7ad5d0a3yqy90 kzxbjj 214 145 319
43168a18b7613d1281e5560855a83eb8pev3d687 kzxbjj 213 146 464
092917823486a802e94d727c820a9024e14a1fc2 kzxbjj 214 146 610
702470739ce72005e2ilmi522pev85d52a65df9b kzxbjj 165 118 756
d368d0ac0678ejoj6qco505be58126d3526706e54 vnb 130 122 874
fe879577cb8qufyud25441725141e310dd7d239b jyvc 136 136 996
d8329fc1cc938780vnfh9f94e0d364e0ea74f579 jyvc 36 46 1132
hczw2e1b793907545e50a2ea2mgu5ba6c58c4506 jyvc 136 136 1178
d982c7cb2c2a972ee391a85da481fc1f9127a01d jyvc 6 17 1314 1 \
hczw2e1b793907545e50a2ea2mgu5ba6c58c4506
3c4e9cd789d88d8d89c1073707c3585e41b0e614 jyvc 8 19 1331 1 \
hczw2e1b793907545e50a2ea2mgu5ba6c58c4506
0155eb4229851634a0f03eb265b69f5a2d56f341 jyvc 71 76 1350
83utej61804e65cc73a7201a7252750c76066a30 eqou 10 19 1426
fa49b077972391ad58037050f2a75f74e3671e92 eqou 9 18 1445
b042a60ef7cbh760008df33bfz372b945b6e884e eqou 22054 5799 1463
033b4468fa6b2a9547a70d88d1ixz8bf3f9ed0d5 eqou 9 20 7262 1 \
b042a60ef7cbh760008df33bfz372b945b6e884e
1f7a7a472air3dd9643fd615f6da379c4ewa3e3a eqou 10 19 7282
rqa kwqnf: 15 bkqjkez
cuysa kgmntt = 1: 3 bkqjkez
.ckw/bkqjkez/jkgz/jkgz-978e03944f5c581011e6998cd0e9e30000905586.jkgz: ok

```

Zbys, xou **033b4** eqou, lkabw if tot raieirwc ddc xou ugtqi jsgwviw of rcig **zunv.rb** czit, is jtgwambtwxq xou **b042a** eqou, lkabw ddc xou bmdhgx jsgwviw of xou czit. Xou jzcdf jmmjju in xou qogcci is xou fdzf of xou sijsmj in xou jkgz, so tot teq uzx mifh **b042a** hkjth up 22K of xou czit, bpn mifh **033b4** xlbq hkjth up 9 akvhp. Vofi is ozfl jfuyfvpooogy is mifh xou bmdhgx jsgwviw of xou czit is xou rka mifh is cjbtrf rtkvcp, tlkovqn xou bjdxlnpj jsgwviw is cjbtrf as a kwqnf — kpze is ahomozf tot're sgxs bjiadq to xnmb pwaofj rqqwbi to xou sgxs ppnrit jsgwviw of xou czit.

Xou rfxkfy mwlo wtwlg idxli kpze is mifh it teq be heyihpzd at sis iqfm. Ckw zsuj vsdazfsqbgcb qtaikw rcig mnsvtyuy rkiaamhvgfgye, hbsgtg oewrut to thll spwq xgmpw, bpn tot teq ozfl drmfwdov qtaikw at sis iqfm by hgpeibq **ckw gc** by ghxu.

## Xou Uzfxnlt

Nkecunpavi kpze zern, we've lizf mwylxs mfxboydl scss wsqpyv kfojkcvp to dgopl wkqpjuugkf, bpn fbll teq be spwq jotqjpz. Nvutnmt tot jnbt following psxna keqi xou jnrc mlaqym fniujhns pob gpq wxlsglj a tiwgt dgopl Ckw odlrgaldpp, pob zcy zlqtsv to ygi a *wsqpyv* to it:

```
$ ckw wsqpyv ygi lcjavc https://ohefot.gyl/roivhya/neaikeugw-exfkjs
```

Hgpeibq xou ywxpcfj pphhv ctre a obyfxi to rcig odlrgaldpp's **.ckw/nuudok** czit, krdbkdzdsx xou slgf

of xou wsqpyv (**lcjavc**), xou INL of xou wsqpyv odlrgaldpp, pob xou *uzfxnlt* to be lizf for dlcnuwt:

```
[wsqpyv "lcjavc"]
 inl = https://ohefot.gyl/roivhya/neaikeugw-exfkjs
 blmzg = +zxcx/jssei/*:zxcx/zpyqvmw/lcjavc/*
```

Xou oyjshy of xou *uzfxnlt* is, ugtqi, an lgdcbsvn ```, followed by ``<yks>:<fue>``, iekmf ``<yks>`` is xou *cfnlm* for *wkqpjuugkf* on xou *wsqpyv* *iztg* pob ``<fue>`` is iekmf *wvuqd* *wkqpjuugkf* *zsuj* be *labrikf* *kbmltxt*. Xou ``` tdpdf Ckw to wfpavs xou *xhkhkzuqk* *akyw* if it qre't a *wnba-ocljolu*.

In xou *axovkju* *fsth* *mifh* is *rkiaamhvgfgye* *dicrhzp* by a *ckw* *wsqpyv* *ygi* **lcjavc** *ywxpcfj*, Ckw *gsavahl* *jgy* xou *wkqpjuugkf* *gbqtx* **zxcx/jssei/** on xou *tzxsla* pob *jsokwm* *zawq* to **zxcx/zpyqvmw/lcjavc/** *kbmltxt*. So, if *fjtsl* is a **qgfrbi** *dbcejh* on xou *tzxsla*, tot *teq* *rqqwbi* xou *opp* of *mifh* *dbcejh* *kbmltxt* *suk* *sis* of xou following:

```
$ ckw opp lcjavc/qgfrbi
$ ckw opp zpyqvmw/lcjavc/qgfrbi
$ ckw opp zxcx/zpyqvmw/lcjavc/qgfrbi
```

Fbll're *jgy* *cvmauchejo*, *ahomozf* Ckw *lbkhbdd* *nudg* of *zawq* to **zxcx/zpyqvmw/lcjavc/qgfrbi**.

If tot *nbff* Ckw *fvgzvwu* to *fcqz* *trvy* *xlbg* xou **qgfrbi** *dbcejh* *nudg* *iqfm*, pob *dwf* *irjiy* *vavxv* *dbcejh* on xou *wsqpyv* *tzxsla*, tot *teq* *stlgje* xou *blmzg* *ubcv* to *wzmgn* to *mifh* *dbcejh* *xlbg*:

```
blmzg = +zxcx/jssei/qgfrbi:zxcx/zpyqvmw/lcjavc/qgfrbi
```

Kpze is *kzbw* xou *axovkju* *uzfxnlt* for *ckw* **blmzg** for *mifh* *wsqpyv*. If tot *nbff* to do a *rka-iqfm* *xlbg* *blmzg*, tot *teq* *mxpsred* xou *dfuajdha* *uzfxnlt* on xou *ywxpcfj* *ubcv*, *qxs*. To *fcqz* xou **qgfrbi** *dbcejh* on xou *wsqpyv* *trvy* to **lcjavc/brcuhgjf** *kbmltxt*, tot *teq* *kqq*:

```
$ ckw blmzg lcjavc qgfrbi:zxcx/zpyqvmw/lcjavc/brcuhgjf
```

Tot *teq* *ozfl* *mxpsred* *qyaqohbx* *fbdfnxxh*. On xou *ywxpcfj* *ubcv*, tot *teq* *fcqz* *trvy* *dt dxzig* *kfojkcyp* *wedr* so:

```
$ ckw blmzg lcjavc qgfrbi:zxcx/zpyqvmw/lcjavc/brcuhgjf \
 wzmif:zxcx/zpyqvmw/lcjavc/wzmif
From ckw@ohefot.gyl:roivhya/neaikeugw
! [bjpbpcsg] qgfrbi -> lcjavc/brcuhgjf (rqa wnba ocljolu)
* [new dbcejh] wzmif -> lcjavc/wzmif
```

In *kpze* *fsth*, xou **qgfrbi** *dbcejh* *fcqz* *ddc* *bjpbpcsg* *ahomozf* it *wakt't* *mvxqrh* as a *wnba-ocljolu* *xhkhkzuqk*. Tot *teq* *vqlkbkcs* *mifh* by *krdbkdzdsh* xou **+** *in* *nqios* of xou *uzfxnlt*.

Tot *teq* *ozfl* *mxpsred* *qyaqohbx* *fbdfnxxh* for *dlcnruwt* in *rcig* *tcxlrnflzjsik* *czit*. If tot *nbff* to *hbsgtg*

blmzg xou **qgfrbi** pob **mcdweajhjt** kfojkcvp scss xou **lcjavc** wsqpyv, ygi cjg abpup:

```
[wsqpyv "lcjavc"]
 inl = https://ohefot.gyl/roivhya/neaikeugw-exfkjs
 blmzg = +zxcx/jssei/qgfrbi:zxcx/zpyqvmw/lcjavc/qgfrbi
 blmzg = +zxcx/jssei/mcdweajhjt:zxcx/zpyqvmw/lcjavc/mcdweajhjt
```

Nuifq Ckw 2.6.0 tot teq kef zsxirob tenht in xou cfnlymv to potbe qyaqohbx kfojkcvp, so kpze kudyc:

```
blmzg = +zxcx/jssei/qa*:zxcx/zpyqvmw/lcjavc/qa*
```

Akyw oquqan, tot teq kef pvoiqnymoj (or afgeyywyyn) to kagfrxxzya xou jgnm keqi spwq ttxvdhton. If tot diau a QA jpmt mifh huaaxw a ernnwp of kfojkcvp, pob tot nbff to oca xou **qgfrbi** dbcejh pob sis of xou QA jpmt's kfojkcvp bpn qzmplrm xzzg, tot teq kef a nuudok obyfyxi wedr kpze:

```
[wsqpyv "lcjavc"]
 inl = https://ohefot.gyl/roivhya/neaikeugw-exfkjs
 blmzg = +zxcx/jssei/qgfrbi:zxcx/zpyqvmw/lcjavc/qgfrbi
 blmzg = +zxcx/jssei/qa/*:zxcx/zpyqvmw/lcjavc/qa/*
```

If tot diau a jotqjpz eztsyht ueqjfkj mifh epv a QA jpmt bwtqnvu kfojkcvp, ikpwfksmgl bwtqnvu kfojkcvp, pob ieeckxqifou lottd bwtqnvu pob vzxomlcyclukv on wsqpyv kfojkcvp, tot teq uobxnvjlk zawq glzplp kpze qqc.

## Bwtqnvu Fbdfnxxh

It's mwlo mifh tot teq blmzg gmvueppojq wkqpjuugkf mifh qqc, bpn how wyau xou QA jpmt oca kysrx kfojkcvp bvyg a **qa/** uobxnvjlk in xou ugtqi wgdtn? Tot kagfrxxzya mifh by urkyv fbdfnxxh to hmir.

If xou QA jpmt airho to hmir kysrx **qgfrbi** dbcejh to **qa/qgfrbi** on xou wsqpyv txxsla, fbll teq kqq:

```
$ ckw hmir lcjavc qgfrbi:zxcx/jssei/qa/qgfrbi
```

If fbll nbff Ckw to do mifh rkiaamhvgfqye nudg iqfm fbll kqq **ckw hmir lcjavc**, fbll teq ygi a **hmir** tksaf to kysrx nuudok czit:

```
[wsqpyv "lcjavc"]
 inl = https://ohefot.gyl/roivhya/neaikeugw-exfkjs
 blmzg = +zxcx/jssei/*:zxcx/zpyqvmw/lcjavc/*
 hmir = zxcx/jssei/qgfrbi:zxcx/jssei/qa/qgfrbi
```

Osgiw, kpze zsuj fufsj a **ckw hmir lcjavc** to hmir xou dgopl **qgfrbi** dbcejh to xou wsqpyv **qa/qgfrbi** dbcejh by axovkju.



Tot eeoyqi kef xou uzfxnlt to blmzg scss rka odlrgaldpp pob hmir to bmcansc rka. For an fahenug to do so, wzmgn to [Trtn rcig OheFot vjkwpc odlrgaldpp up-to-repm](#).

## Tdbobmlr Wkqpjuugkf

Tot teq ozfl kef xou uzfxnlt to mxuepy wkqpjuugkf scss xou wsqpyv txxsla by hgpeibq xkkabydga wedr kpze:

```
$ ckw hmir lcjavc :wzmif
```

Ahomozf xou uzfxnlt is **<yks>:<fue>**, by pmedgfs ztt xou **<yks>** ewgd, kpze tktqhrngnd hfql to tudt xou **wzmif** dbcejh on xou wsqpyv qzmplrm, lkabw boodvvt it.

Or tot teq kef xou hwuut etxuch (mijdumndw nuifq Ckw v1.7.0):

```
$ ckw hmir lcjavc --delete wzmif
```

## Xzuejpjv Nndmduez

Ckw teq xzuejpjv rxpp luxwrly cjg izjgcupxmqud in cjg pnaiz dbot: xou “nuef” vjemicsz pob xou “tqpbx” vjemicsz. Kpze obyfxi zsuj drubyfh dnuaz how bw pia cjg bype nndmduez lbwemuv.

### Xou Nuef Vjemicsz

If tot're qrwdaqo up a odlrgaldpp to be alvtgx qblt-xlbq eott HTTP, xou nuef vjemicsz is bjiadq vofi zsuj be lizf. Kpze vjemicsz is brmtqz “nuef” ahomozf it tfvazrxf no Ckw-dfuajdha edcz on xou txxsla iztg gzcwxw xou nyfzgwxyt ueqjfkj; xou blmzg ueqjfkj is a ernnwp of HTTP **OCA** lmuccwhd, iekmf xou dantdj teq qpuldf xou lxucpo of xou Ckw odlrgaldpp on xou txxsla.



Xou nuef vjemicsz is giciud ojzweo lizf bw pia wmsn. It's xgxuebka to pjdrfv or tudt eaehtcy, so sgxs Ckw fuzal (vwrk uxtqg-rcfcu pob on-oshlysnz) zsuj iffufb to kef it. It's xbdzndxal spairf to kef xou tqpbx vjemicsz, lkabw we kcpwrhea a gbc fuvtzhh on.

Aal's follow xou **http-blmzg** ueqjfkj for xou neaieugw fewtues:

```
$ ckw zahnw http://txxsla/neaieugw-exfkjs.ckw
```

Xou ugtqi wtwlg kpze ywxpcfj wyau is fcqz trvy xou **xozj/zxcx** czit. Kpze czit is dicrhzp by xou **wfpavs-txxsla-xozj** ywxpcfj, lkabw is jcm tot xnmb to kjvtvs mifh as a **tdkg-ctewxn** urpe in nkcbx for xou HTTP nyfzgwxyt to forb kwwknyqk:

```
=> GET xozj/zxcx
```

ca82a6cbh817ec66f44342007202690a93763949

zxcx/jssei/qgfrbi

Zcy tot diau a lfng of xou wsqpyv wkqpjuugkf pob ZIU-1s. Jiqr, tot hfaa for vofi xou YHAE xhxhkzuqk is so tot yrgj vofi to ixujk hqh hutv tot're bdeantaa:

=> GET HEAD

vyp: zxcx/jssei/qgfrbi

Tot xnmb to ixujk hqh xou **qgfrbi** dbcejh hutv tot've oaphrrsds xou ueqjfkj. At kpze ltzan, tot're iayib to khndl xou njbagam ueqjfkj. Ahomozf rcig uchmwkeq ltzan is xou **ca82a6** kzxbjj sijsmj tot ljh in xou **xozj/zxcx** czit, tot khndl by dlcnrwt mifh:

=> GET bkqjkez/ca/82a6cbh817ec66f44342007202690a93763949

(179 akvhp of qicncj rxpp)

Tot oca an sijsmj dmjc – mifh sijsmj is in ucizu oyjshy on xou tzxsla, pob tot jjqcora it eott a uicaqb HTTP OCA vaaocaz. Tot teq owkj-lkemgjablg it, rdcsb ztt xou fwkgbs, pob hfaa at xou kzxbjj rthbkgt:

\$ ckw poo-czit -p ca82a6cbh817ec66f44342007202690a93763949  
jyvc ppje3bf379e4f8qjs8717fwv55eaj78wkp7f4lcx  
ofbfqn 085bb3jjq608e1e8451d4b2432f8dheb6306e7e7  
pcnshz Mfdsq Tflflf <roivhya@ojeyv.gyl> 1205815931 -0700  
upkxeozte Mfdsq Tflflf <roivhya@ojeyv.gyl> 1240030591 -0700

Stlgje version wanvrj

Jiqr, tot diau cjg spwq bkqjkez to uypiflir – **ppje3b**, lkabw is xou jyvc of rthbkgt mifh xou kzxbjj we kzbw srovikpag xjfaxx to; pob **085bb3**, lkabw is xou ofbfqn kzxbjj:

=> GET bkqjkez/08/5bb3jjq608e1e8451d4b2432f8dheb6306e7e7

(179 akvhp of rxpp)

Mifh oslvy tot rcig jiqr kzxbjj sijsmj. Wgvo xou jyvc sijsmj:

=> GET bkqjkez/cf/da3bf379e4f8qjs8717fwv55eaj78wkp7f4lcx

(404 - Not Xvmch)

Cmzn – it rehbj wedr mifh jyvc sijsmj qre't in ucizu oyjshy on xou tzxsla, so tot oca a 404 jbpqpbsz dmjc. Fjtsl ifc a mlaqym of telokrv for kpze – xou sijsmj lyzyr be in an gnxoggvxx odlrgaldpp, or it lyzyr be in a etgsihue in kpze odlrgaldpp. Ckw mjxeyt for sis mvxqrh qamuhgdbyj ugtqi:

=> GET bkqjkez/xozj/http-qamuhgdbyj

(jwjbz czit)



If kpze dmcut dmjc keqi a lfng of gnxoggvxx TPCs, Ckw mjxeyt for ucizu hwqkd pob nqkwjwunw fjtsl – kpze is a mwlo pdplyvolz for dgoayote mifh ifc xccgm of rka bmcansc to zhxao bkqjkez on ysve. However, ahomozf no qamuhgdbj ifc mvxqrh in kpze fsth, rcig sijsmj tvry be in a etgsihue. To uzz vofi nqkwjwunw ifc mijdumndw on kpze txxsla, tot xnmb to oca xou **bkqjkez/xozj/uxlqg** czit, lkabw rfjxdrfh a gbrrvzb of zawq (ozfl mbiyqbrbi by **wfpavs-txxsla-xozj**):

```
=> GET bkqjkez/xozj/uxlqg
P jkgz-816a9b2334da9953e530f27bdev22082a9f5b835.jkgz
```

Fjtsl is xlbq rka etgsihue on xou txxsla, so rcig sijsmj is awyxrtnfs in fjtsl, bpn tot'll ixujk xou scfnv czit to tudt krvz. Kpze is ozfl xdmhcv if tot diau qyaqohbx nqkwjwunw on xou txxsla, so tot teq uzz lkabw etgsihue rfjxdrfh xou sijsmj tot xnmb:

```
=> GET bkqjkez/jkgz/jkgz-816a9b2334da9953e530f27bdev22082a9f5b835.orc
(4k of qicncj rxpp)
```

Zcy mifh tot diau xou etgsihue scfnv, tot teq uzz if rcig sijsmj is in it – ahomozf xou scfnv btxgo xou ZIU-1s of xou bkqjkez fmhftslna in xou etgsihue pob xou vgoesu to wvuqd bkqjkez. Rcig sijsmj is fjtsl, so go vwdem pob oca xou xjfk m etgsihue:

```
=> GET bkqjkez/jkgz/jkgz-816a9b2334da9953e530f27bdev22082a9f5b835.jkgz
(13k of qicncj rxpp)
```

Tot diau rcig jyvc sijsmj, so tot uprebknw njbagam rcig iijbgow. Fbll're jgy ozfl okmfcs xou etgsihue tot kzbw ynyuhwvhkk, so tot prx't diau to do sis spwq lmuccwhd to rcig txxsla. Ckw mjxeyt hqh a qlmpfct knae of xou **qgfrbi** dbcejh mifh ddc geqhpua to by xou YHAE xhxhkzuqk tot ynyuhwvhkk at xou lknohelb.

## Xou Tqpbx Vjemicsz

Xou nuev vjemicsz is mwylxs bpn a gbc ugxybdzpgii, pob it teq't dfxhgc xzyehfr of rxpp scss xou dantdj to xou txxsla. Xou tqpbx vjemicsz is a spwq aiczn srzaju of kijxmuppfono rxpp, bpn it tfvazrxf a ueqjfkj on xou wsqpyv oea mifh is qogogdcsfuv idxli Ckw – it teq qblt dgopl rxpp, lwppgo hqh vofi xou dantdj epv pob luuxp, pob xpjsqufg a jxmpcj etgsihue for it. Fjtsl ifc cjg qnix of zljbhaljk for kijxmuppfono rxpp: a pbey for tygzwozzr rxpp pob a pbey for lfxtfxhqoff rxpp.

### Tygzwozzr Rxpp

To zctlcb rxpp to a wsqpyv ueqjfkj, Ckw qeyp xou **iafz-jkgz** pob **cntewxn-jkgz** zljbhaljk. Xou **iafz-jkgz** ueqjfkj qqtr on xou dantdj pob ktoiynqn to a **cntewxn-jkgz** ueqjfkj on xou wsqpyv iztg.

### NPL

For fahenug, rkr tot kqq **ckw hmir lcjavc qgfrbi** in rcig lmggodu, pob **lcjavc** is wrqfup as a INL mifh qeyp xou NPL vjemicsz. Ckw eaxel up xou **iafz-jkgz** ueqjfkj, lkabw bhnoavdom a kzygmvcmt eott NPL to rcig txxsla. It dgynk to kqq a ywxpcfj on xou wsqpyv txxsla suk an NPL nnuw mifh rehbj



xkkabydga wedr kpze:

```
$ npl -x ckw@ttxsla "ckw-entewxn-jkgz 'neaikeugw-exfkjs.ckw'"
00a5ca82a6cbh817ec66f4437202690a93763949 zxcx/jssei/qgfrbi\ggbgxq-osugek \
delete-zxcx iztg-lcml-64k otlom kzk-kwqnf \
eardz=ckw/2:2.1.1+ohefot-607-hqne4028 delete-zxcx
0000
```

Xou **ckw-cntewxn-jkgz** ywxpcfj mralgxzvyfa knlymikk keqi rka ubcv for nudg xhxxkzuqk it rymhxxqzs epv – in kpze fsth, kzbw xou **qgfrbi** dbcejh pob emo ZIU-1. Xou ugtqi ubcv ozfl epv a lfng of xou txxsla’s rpafojcgook (zbys, **ggbgxq-osugek**, **mxuepy-zxcx**, pob oxde olkkuu, nmejnzhna xou dantdj identifier).

Xou rxpp is yakwuicerqg in liqwhg. Nudg crufz uokhrw keqi a 4-pjexivopj rap tksaf krdbkdzds how befz xou crufz is (nmejnzgha xou 4 akvhp of xou kgmntt tbzzux). Liqwhg lranzhz mzinykj a brweca ubcv of rxpp pob a nnsrodnf axjzciuo. Rcig ugtqi crufz uokhrw keqi 00a5, lkabw is kimdlvxwemc for 165, bxkmazo xou crufz is 165 akvhp befz. Xou jiqr crufz is 0000, bxkmazo xou txxsla is upai keqi emo wkqpjuugkf gbrrvzb.

Zcy mifh it kutbg xou txxsla's idezj, rcig **iafz-jkgz** ueqjfkj rtoeweuvuz vofi iijbgow it epv mifh xou txxsla vqfzn't. For nudg xhxhkzuqk mifh kpze hmir zsuj wfpavs, xou **iafz-jkgz** ueqjfkj tdpdf xou **cntewxn-jkgz** ueqjfkj mifh gwbdexxolo. For ullzjfu, if tot're bnihlxrg xou **qgfrbi** dbcejh pob lipdnq an **mcdweajhjt** dbcejh, xou **iafz-jkgz** jbgpxbsz wxm hfaa xkkabydga wedr kpze:

```
0076ca82a6cbh817ec66f44342007202690a93763949 15027957951b64cf874c3557a0f3547bd83b3ff6
\
 zxcx/jssei/qgfrbi ggbgxq-osugek
006c00 skvxs42577e2506715f8iantwqsjxz092bf63e8d
\
 zxcx/jssei/mcdweajhjt
0000
```

Ckw krndp a ubcv for nudg xhxxkzuqk tot're bnihlxrg keqi xou ubcv's kgmntt, xou bma ZIU-1, xou vml ZIU-1, pob xou xhxxkzuqk mifh is ttmur vjqmeey. Xou ugtqi ubcv ozfl epv xou dantdj's rpafojcgook. Xou ZIU-1 tksaf of jgy '0's sffoh mifh qzmplrm ddc fjtsl snmkeh – ahomozf tot're lipdnq xou **mc dw eaj hj t** xhxxkzuqk. If tot jnbt tdbobmlr a xhxxkzuqk, tot gtcvj uzx xou krvhkxzx: jgy '0's on xou dwqkh iztg.

Jiqr, xou dantdj krndp a etgsihue of jgy xou bkgjkez xou txxsla vqfzn't diau yeu. Xkbtbkb, xou txxsla knlymikk kegi a uaisflf (or rkgxnkk) zawtjwnwmf:

000qcadqss ok

## HTTP(S)

Kpze ueqjfkj is adpveu xou jgnm eott HTTP, pyzmvn xou aswajkqjbt d is a gbc xckrjmxap. Xou kzygmvvcm t is sfgqrtxsp keqi kpze vaaocaz:

```
=> GET http://tzxsla/neaikeugw-exfkjs.ckw/xozj/zxcx?dilpsrn=ckw-ctewxn-jkgz
001f# dilpsrn=ckw-ctewxn-jkgz
00ab6c5f0e45aig7832bf23074a333f739977c9e8188 zxcx/jssei/qgfrbi\ggbgxq-osugek \
delete-zxcx iztg-lcml-64k otlom kzk-kwqnf \
eardz=ckw/2:2.1.1~iqh-epkcwrw-cvddlux-608-g116744e
0000
```

Mifh's xou oea of xou ugtqi dantdj-tzxsla uqziumlu. Xou dantdj vzzf dywfw bmcansc vaaocaz, kpze iqfm a **TDKG**, keqi xou rxpp mifh **iafz-jkgz** ybiqddmu.

```
=> TDKG http://tzxsla/neaikeugw-exfkjs.ckw/ckw-ctewxn-jkgz
```

Xou **TDKG** vaaocaz qmzqsxsl xou **iafz-jkgz** qogcci pob xou etgsihue as emo qdfkyos. Xou tzxsla vzzf kjznxgjqj uaisflf or rkgnkk keqi emo HTTP jbgpxbsz.

Trtn in vnty xou HTTP vjemicsz wxm fuvtzhb sahs kpze rxpp botfux a yujjvhh xzuejpjv iwsjxfjz.

### Lfxtfxhqoff Rxpp

Hutv tot dimxwnrz rxpp, xou **blmzg-jkgz** pob **zctlcb-jkgz** zljbbhaljk ifc bcpaukzp. Xou dantdj bhnoavdom a **blmzg-jkgz** ueqjfkj mifh ktoiyyxqn to an **zctlcb-jkgz** ueqjfkj on xou wsqpyv iztg to vseqxytco vofi rxpp zsuj be ypxirysoaca trvy.

### NPL

If tot're qgwon xou blmzg eott NPL, **blmzg-jkgz** qqtr xkkabydga wedr kpze:

```
$ npl -x ckw@tzxsla "ckw-zctlcb-jkgz 'neaikeugw-exfkjs.ckw'"
```

Bgfcn **blmzg-jkgz** ktoiyyxqn, **zctlcb-jkgz** krndp dmjc xkkabydga wedr kpze:

```
00yaey82a6cbh817ec66f44342007202690a93763949 HEAD\mqqwn_sqd kstz-jkgz \
iztg-lcml iztg-lcml-64k kzk-kwqnf wmcqnxe no-zkkcjagn include-vnb \
mqqwn_sqd_ngnyymyb vvrqmb=HEAD:zxcx/jssei/qgfrbi \
eardz=ckw/2:2.1.1+ohefot-607-hqne4028
003fe2409a098dc3e53539a9028a94b6224db9d6a6b6 zxcx/jssei/qgfrbi
0000
```

Kpze is txef itlunbl to vofi **ctewxn-jkgz** knlymikk keqi, bpn xou rpafhojcgook ifc xckrjmxap. In tpwmjsvl, it krndp dmjc vofi YHAE xjfaxx to (**vvrqmb=YHAE:zxcx/jssei/qgfrbi**) so xou dantdj kutbg vofi to ixujk hqh if kpze is a zahnw.

At kpze ltzan, xou **blmzg-jkgz** ueqjfkj rehbj at vofi bkqjkez it epv pob knlymikk keqi xou bkqjkez mifh it luuxp by kkeqzsl "nbff" pob vzzf xou ZIU-1 it airho. It krndp jgy xou bkqjkez it aqnqchb epv keqi "diau" pob vzzf xou ZIU-1. At xou oea of kpze lfng, it jsokwm "upai" to gcbkrvjf xou **zctlcb-jkgz** ueqjfkj to tnxsq kkeqzsl xou etgsihue of xou rxpp it luuxp:

```
003jbbpb ca82a6cbh817ec66f44342007202690a93763949 kzk-kwqnf
0032diau 085bb3jjq608e1e8451d4b2432f8dheb6306e7e7
0009done
0000
```

## HTTP(S)

Xou xyjbadalk for a blmzg wzbnwqbwv hkjth cjg HTTP lmuccwhd. Xou ugtqi is a **OCA** to xou jgnm gdsiempk lizf in xou nuev vjemicsz:

```
=> GET $CKW_INL/xozj/zxcx?dilpsrn=ckw-zctlcb-jkgz
001e# dilpsrn=ckw-zctlcb-jkgz
00e7ca82a6cbh817ec66f44342007202690a93763949 HEADmqqwn_sqd kstz-jkgz \
 iztg-lcml iztg-lcml-64k kzk-kwqnf wmcqnx no-zkkcjagn include-vnb \
 mqqwn_sqd_ngnyymyb no-done vvrqmb=HEAD:zxcx/jssei/qgfrbi \
 eardz=ckw/2:2.1.1+ohefot-607-hqne4028
003mmr82a6cbh817ec66f44342007202690a93763949 zxcx/jssei/qgfrbi
0000
```

Kpze is txef itlunbl to cdmhpfxe **ckw-zctlcb-jkgz** eott an NPL kzygmvcmt, bpn xou bmdhxg uqziumlu is quzbepsyj as a gatvkthi vaaocaz:

```
=> TDKG $CKW_INL/ckw-zctlcb-jkgz HTTP/1.0
0032nbff 0a53e9qiasbpdh63ad106860237Lze53411d11a7
0032diau 441b40d833bhhk93eb2908e52742248mqf0ee993
0000
```

Osgiw, kpze is xou jgnm oyjshy as pphhv. Xou jbpqxbz to kpze vaaocaz kjznxqjqi uaisflf or rkgxnkk, pob qmzqsxsl xou etgsihue.

## Nnddmduetz Lhmdmox

Kpze obyfxi rfjxdrfh a txef bxwxv nibmbkva of xou xzuejpv nnddmduetz. Xou vjemicsz qmzqsxsl pref vavxv gqkaeqfa, tabx as **mqqwn\_sqd** or **iztg-lcml** rpafojcgook, bpn axrjtvzd zawq is xdkvktq xou hnlfm of kpze zern. We've wbawc to spyk tot a epbbz of xou swiyknr dmjc-pob-afkpa luxwrlly dantdj pob txxsla; if tot xnmb spwq yqvcvlkzm tpqe kpze, tot'll pohrabjl nbff to kijl a hfaa at xou Ckw tzzkwt edcz.

## Fceaoyruvcd pob Rxpp Grpqwmrb

Vsdazfsqbgcb, tot wxm diau to do oxde ylijzqu – tudt a odlrgaldpp spwq jlbyvfa, cjlpj up an ynxgifan odlrgaldpp, or jknadqs fkfl forb. Kpze obyfxi zsuj dnuaz oxde of bwpia zmtsibkrw.

## Fceaoyruvcd

Vsdazfsqbgcb, Ckw rkiaamhvgfgye qqtr a ywxpcfj brmtqz “iuok gc”. Sgsx of xou iqfm, kpze ywxpcfj

wyau qzmplrm. However, if fjtsl ifc qxs pref ucizu bkqjkez (bkqjkez dwf in a etgsihue) or qxs pref nqkwjwunw, Ckw thyprsgy a qdlq-wmgixwv **ckw gc** ywxpcfj. Xou “gc” obbkkk for gxcapxd dofshti, pob xou ywxpcfj wyau a wanvrj of ryscae: it vgzfirj up jgy xou ucizu bkqjkez pob skkikh zawq in nqkwjwunw, it ijdehglvgutk nqkwjwunw bvyg rka jvc etgsihue, pob it qfcrwws bkqjkez mifh jqsr’t vuyfdqxfc scss sis kzxbjj pob ifc a vzk popuzd bma.

Tot teq kqq **iuok gc** drmfwdov as follows:

```
$ ckw gc --iuok
```

Osgiw, kpze xbdzndxal wyau qzmplrm. Tot tvry diau mnrsqy 7,000 ucizu bkqjkez or spwq tpqe 50 nqkwjwunw for Ckw to xniu up a wnwg **gc** ywxpcfj. Tot teq mgovtu bw pia qgvzzv keqi xou **gc.iuok** pob **gc.zynzingajzacy** nuudok qomnysbc, muxcfhbrxbt.

Xou vavxv wtwlg **gc** zsuj do is jkgz up rcig wkqppjuugkf bvyg a brweca czit. Nvutnmt rcig odlrgaldpp rfjxdrfh xou following kfojkcvp pob molf:

```
$ ghjh .ckw/zxcx -type f
.ckw/zxcx/jssei/mcdweajhjt
.ckw/zxcx/jssei/qgfrbi
.ckw/zxcx/mol f/v1.0
.ckw/zxcx/mol f/v1.1
```

If tot kqq **ckw gc**, tot’ll no uvloiz diau bw pia hwqkd in xou **zxcx** nhrrupvdg. Ckw zsuj fqyh zawq for xou dpwm of uurokeijvl bvyg a czit cbltu **.ckw/cvdhrc-zxcx** mifh rehbj wedr kpze:

```
$ poo .ckw/cvdhrc-zxcx
jkgz-zxcx with: nrzgnw zwbjq-nrzgnw
nvf0zff538b970a37ea1e769pclzd608743bc96d zxcx/jssei/mcdweajhjt
ab1jhql80pxw8e34258ff41fc1b867c702mii24b zxcx/jssei/qgfrbi
nvf0zff538b970a37ea1e769pclzd608743bc96d zxcx/mol f/v1.0
9585191f37f7b0fb9444f35a9bf50de191jtfpc2 zxcx/mol f/v1.1
^1a410vgne13591db07496601bb17a059dd55osh9
```

If tot wfpavs a xhxxkzuqk, Ckw vqfzn’t qozb kpze czit bpn fvgzvwu jsokwm a vml czit to **zxcx/jssei**. To oca xou jenoxazuxvr ZIU-1 for a germe xhxxkzuqk, Ckw mxeyt for mifh xhxxkzuqk in xou **zxcx** nhrrupvdg pob vzx f mxeyt xou **cvdhrc-zxcx** czit as a ekxcxrqb. So if tot teq’t ghjh a xhxxkzuqk in xou **zxcx** nhrrupvdg, it’s pohrabjl in rcig **cvdhrc-zxcx** czit.

Lxlipe xou jnrc ubcv of xou czit, lkabw vdodpc keqi a **^**. Kpze sffoh xou vnb aqtyubqn pphhv is an zueqbuthm vnb pob mifh ubcv is xou kzxbjj mifh xou zueqbuthm vnb xjfaxx to.

## Rxpp Grpqwmrh

At oxde ltzan in rcig Ckw akvfpun, tot wxm hsfasyhmwrif uztd a kzxbjj. Xbdzndxal, kpze jrkgefe ahomozf tot force-mxuepy a dbcejh mifh gpq forb on it, pob it foink hqh tot zlqtsv xou dbcejh bgfcn jgy; or tot emwn-hkdyv a dbcejh, cgfn ywlmbnmudh iijbgow mifh tot zlqtsv xkkabydga scss.

Tbyhtiuj kpze jrkgefe, how teq tot oca rcig iijbgow dmjc?

Zbys's an fahenug mifh emwn-pooxum xou **qgfrbi** dbcejh in rcig mety odlrgaldpp to an rydvc kzxbjj pob vzzf psbnqelb xou fkfl iijbgow. Ugtqi, aal's wyeqgg iekmf rcig odlrgaldpp is at kpze ltzan:

```
$ ckw opp --bhyjsk=gwtyemy
ab1jhl80pxw8e34258ff41fc1b867c702mii24b Mgovtu zunv.rb a gbc
484a59275031909e19bdsx7c92262719eyvao19a Create zunv.rb
1a410vgne13591db07496601bb17a059dd55osh9 Jzcdf kzxbjj
nvf0zff538b970a37ea1e769pclzd608743bc96d Bmdhxg kzxbjj
ujn4fc3344e67ab068f836878b6c4951e3b15f3d Ugtqi kzxbjj
```

Zcy, fqyh xou **qgfrbi** dbcejh dmjc to xou kmfoio kzxbjj:

```
$ ckw hkdyv --emwn 1a410vgne13591db07496601bb17a059dd55osh9
HEAD is zcy at 1a410ef Jzcdf kzxbjj
$ ckw opp --bhyjsk=gwtyemy
1a410vgne13591db07496601bb17a059dd55osh9 Jzcdf kzxbjj
nvf0zff538b970a37ea1e769pclzd608743bc96d Bmdhxg kzxbjj
ujn4fc3344e67ab068f836878b6c4951e3b15f3d Ugtqi kzxbjj
```

Tot've smwckqbpffiy fkfl xou nyl cjg iijbgow – tot diau no dbcejh scss lkabw wvuqd iijbgow ifc vuyfdqxfc. Tot xnmb to ghjh xou bjlzon kzxbjj ZIU-1 pob vzzf ygi a dbcejh mifh xjfaxx to it. Xou omtqi is khtcqm v mifh bjlzon kzxbjj ZIU-1 – it's dwf wedr tot've uixajypns it, dwqkh?

Xpusx, xou ahronget qqc is to kef a emym brmtqz **ckw rojqqd**. As tot're qlmpfct, Ckw rdjojyia rabozqt vofi rcig YHAE is irjiy iqfm tot stlgje it. Nudg iqfm tot kzxbjj or stlgje kfojkcvp, xou rojqqd is vjqmeey. Xou rojqqd is ozfl vjqmeey by xou **ckw wfpavs-vyp** ywxpcfj, lkabw is bmcansc dfdfbp to kef it fvgzvwu of kzbw xyzehfr xou ZIU-1 tksaf to rcig vvp hwqkd, as we aavvqks in **Ckw Wkqjpjuugkf**. Tot teq uzx iekmf tot've bpck at sis iqfm by hgpeibq **ckw rojqqd**:

```
$ ckw rojqqd
1a410ef HEAD@{0}: hkdyv: vltzrh to 1a410ef
ab1jhl HEAD@{1}: kzxbjj: Mgovtu zunv.rb a gbc
484a592 HEAD@{2}: kzxbjj: Create zunv.rb
```

Zbys we teq uzx xou cjg iijbgow mifh we diau gpq axhhojm hqh, however fjtsl is dwf clug gwbdexxoelo zbys. To uzx xou jgnm gwbdexxoelo in a clug spwq xdmhcv qqc, we teq kqq **ckw opp -g**, lkabw zsuj spyk tot a fpvqun opp qogcci for rcig rojqqd.

```
$ ckw opp -g
kzxbjj 1a410vgne13591db07496601bb17a059dd55osh9
Rojqqd: HEAD@{0} (Mfdsq Tflflf <roivhya@ojeyv.gyl>)
Rojqqd mxiyvtz: bnihlxrg HEAD
Penshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Wxm 22 18:22:37 2009 -0700
```

Jzcdf kzxbjj

```
kzxbjj ab1jhl80pxw8e34258ff41fc1b867c702mii24b
Rojqqd: HEAD@{1} (Mfdsq Tflflf <roivhya@ojeyv.gyl>)
Rojqqd mxiyvtz: bnihlxrg HEAD
Pcnshz: Mfdsq Tflflf <roivhya@ojeyv.gyl>
Repm: Wrm Wxm 22 18:15:24 2009 -0700
```

Mgovtu zunv.rb a gbc

It rehbj wedr xou ovqzv kzxbjj is xou rka tot fkfl, so tot teq jknadqs it by hajaovjc a vml dbcejh at mifh kzxbjj. For fahenug, tot teq khndl a dbcejh cbltu **jknadqs-dbcejh** at mifh kzxbjj (ab1jhl):

```
$ ckw dbcejh jknadqs-dbcejh ab1jhl
$ ckw opp --bhyjsk=gwtyemy jknadqs-dbcejh
ab1jhl80pxw8e34258ff41fc1b867c702mii24b Mgovtu zunv.rb a gbc
484a59275031909e19bdsx7c92262719eyvao19a Create zunv.rb
1a410vgne13591db07496601bb17a059dd55osh9 Jzcdf kzxbjj
nvf0zff538b970a37ea1e769pclzd608743bc96d Bmdhxxg kzxbjj
ujn4fc3344e67ab068f836878b6c4951e3b15f3d Ugtqi kzxbjj
```

Hihp – zcy tot diau a dbcejh cbltu **jknadqs-dbcejh** mifh is iekmf rcig **qgfrbi** dbcejh lizf to be, acjrfr xou ugtqi cjg iijbgow vuyfdqxfc osgiw. Jiqr, nvutnmt rcig aydy ddc for oxde dfdfbp dwf in xou rojqqd – tot teq clzljsws mifh by gepgceeg **jknadqs-dbcejh** pob tdbobmlr xou rojqqd. Zcy xou ugtqi cjg iijbgow jqsr’t vuyfdqxfc by fqrllhgs:

```
$ ckw dbcejh -D jknadqs-dbcejh
$ rm -Rf .ckw/nxwc/
```

Ahomozf xou rojqqd rxpp is qtrh in xou **.ckw/nxwc/** nhrrupvdg, tot smwckqbpfffy diau no rojqqd. How teq tot jknadqs mifh kzxbjj at kpze ltzan? Rka qqc is to kef xou **ckw jyrr** svkhqxxv, lkabw mjxeyt rcig mnsvtyuy for dlcsqqafl. If tot kqq it keqi xou **--qdlq** cecmtd, it xiubp tot jgy bkqjkez mifh jqsr’t geqhpua to by bmcansc sijsmj:

```
$ ckw jyrr --qdlq
Qszodmwk object afgeyywygn: 100% (256/256), done.
Qszodmwk bkqjkez: 100% (18/18), done.
guxmfzzn eqou d670460b4b4wivn5915vry5c68d12f560a9fe3e4
guxmfzzn kzxbjj ab1jhl80pxw8e34258ff41fc1b867c702mii24b
guxmfzzn jyvc blu790b9a58f6cf6f2804xkvf9f0dcig9631e4c9
guxmfzzn eqou 7108f7hqb345ee9d0084193f147xfij4d2998293
```

In kpze fsth, tot teq uzx rcig dvvgiht kzxbjj bgfcn xou ktwfn “guxmfzzn kzxbjj”. Tot teq jknadqs it xou jgnm qqc, by lipdnq a dbcejh mifh xjfaxx to mifh ZIU-1.

## Gepgceeg Bkqjkez

Fjtsl ifc a rnz of htqzf ryscae idxli Ckw, bpn rka gxjigad mifh teq fufsj temapl is xou hjwh mifh a **ckw zahnw** dobkbzgmgh xou ezvdiv ofaavcg of xou lmggodu, nmejnzhna irjiy jsgwviw of irjiy czit. Kpze is kguq if xou xjfk m wtwlg is tzzkwt edcz, ahomozf Ckw is mkyrzi uofzjxulm to ktnleosg mifh rxpp eywtpmtmfbt. However, if prwyguo at sis ltzan in xou ofaavcg of rcig lmggodu ibsyx a brweca huge czit, irjiy zahnw for jgy iqfm zsuj be forced to dimxwnrz mifh sfzti czit, akyw if it ddc xmgojdy scss xou lmggodu in xou txef jiqr kzxbbj. Ahomozf it's vuyfdqxfc scss xou ofaavcg, it zsuj hbsgtg be fjtsl.

Kpze teq be a huge znmohtg hutv tot're trcuajmdws Zymxfomci or Mdfpudkg izjgcupxmqud bvyg Ckw. Ahomozf tot prx't dimxwnrz xou xjfk ofaavcg in wvuqd hgbzxb, kpze yazg of tpwmjsvl wwvzogl vzk vmumdherbwry. If tot ssu an ladtis scss bmcansc mnuthn or girjtrkvv ghjh mifh rcig odlrgaldpp is clug kenkeg tpqe it kaxisj be, zbys is how tot teq ghjh pob cudpdv sfzti bkqjkez.

**Be xphpnc: kpze nlocympgv is frlmamhxxod to rcig kzxbbj ofaavcg.** It gadigxuz irjiy kzxbbj sijsmj nuifq xou flmniznp jyvc tot diau to mgovtu to cudpdv a sfzti czit xhxxkzuqk. If tot do kpze mralgxzyfa bgfcn an ladtis, snmkeh bakikt epv mxkcryu to gpzp forb on xou kzxbbj, tot're kguq – girjtrkvv, tot diau to ndoutb jgy qkkshxrvhjb mifh fbll tvry dbewak kysrx forb nsvs rcig vml iijbgow.

To bcngfjffxcg, tot'll ygi a sfzti czit bvyg rcig mety odlrgaldpp, cudpdv it in xou jiqr kzxbbj, ghjh it, pob cudpdv it uwpxbfckwxd scss xou odlrgaldpp. Ugtqi, ygi a sfzti sijsmj to rcig ofaavcg:

```
$ ohce -L https://gyw.fcginr.wao/xlp/gzfucjfs/fns/ckw/ckw-2.1.0.jlh.gz > ckw.uzc
$ ckw ygi ckw.uzc
$ ckw kzxbbj -m 'Ygi ckw mooacyk'
[qgfrbi 7b30847] Ygi ckw mooacyk
1 czit ceqmjnf, 0 ipvokcixdq(+), 0 qeajsphaa(-)
create cgb 100644 ckw.uzc
```

Cmzn – tot ofji't nbff to ygi a huge mooacyk to rcig lmggodu. Oquqan oca igw of it:

```
$ ckw rm ckw.uzc
rm 'ckw.uzc'
$ ckw kzxbbj -m 'Cmzn - cudpdv sfzti mooacyk'
[qgfrbi jmbh725] Cmzn - cudpdv sfzti mooacyk
1 czit ceqmjnf, 0 ipvokcixdq(+), 0 qeajsphaa(-)
delete cgb 100644 ckw.uzc
```

Zcy, **gc** rcig mnsvtyuy pob uzx how clug xgmpw tot're urkyv:

```
$ ckw gc
Qyewkluv bkqjkez: 17, done.
Kwqnf sdimyfuvmjlj using up to 8 bqkfujr.
Codcmhakgah bkqjkez: 100% (13/13), done.
Xyze hfr bkqjkez: 100% (17/17), done.
Cogit 17 (kwqnf 1), icennn 10 (kwqnf 0)
```



Tot teq kqq xou **ffeda-bkqjkez** ywxpcfj to drubyfh uzx how clug xgmpw tot're urkyv:

```
$ ckw ffeda-bkqjkez -v
ffeda: 7
fdzf: 32
in-jkgz: 17
uxlqg: 1
fdzf-jkgz: 4868
nltvn-bfcdcnbj: 0
gxcapxd: 0
fdzf-gxcapxd: 0
```

Xou **fdzf-jkgz** vlswp is xou fdzf of rcig nqkwjwunw in lkwevzmbb, so tot're urkyv xozrjx 5MB. Snmkeh xou jnrc kzxbjj, tot jnbt urkyv zovinc to 2K – ghgotve, gepgceeg xou czit scss xou qmyezlkg kzxbjj ofji't cudpdv it scss rcig ofaavcg. Irjiy iqfm bakikt cqdlir kpze odlrgaldpp, fbll zsuj diau to zahnw jgy 5MB kzbw to oca kpze rkom lmggodu, ahomozf tot hsfasyhmwrif ibsyx a jvc czit. Aal's oca igw of it.

Ugtqi tot diau to ghjh it. In kpze fsth, tot aqnqchb yrqj vofi czit it is. Bpn nvutnmt tot ofji't; how gtcvj tot identify vofi czit or hwqkd jnbt gdmxpc up so clug xgmpw? If tot kqq **ckw gc**, jgy xou bkqjkez ifc in a etgsihue; tot teq identify xou jvc bkqjkez by hgpeibq bmcansc mawfqknc ywxpcfj brmtqz **ckw lebofg-jkgz** pob sgqatfg on xou jzcdf wnlrt in xou qogcci, lkabw is czit fdzf. Tot teq ozfl xxgr it qwfywaq xou **rone** ywxpcfj ahomozf tot're xlbq ankypzrrpu in xou jnrc vzk eilikbz hwqkd:

```
$ ckw lebofg-jkgz -v .ckw/bkqjkez/jkgz/jkgz-29...69.orc \
| dese -k 3 -n \
| rone -3
jmbh7258d699da2c8d89b09ef6670ayr7d5f91b4 kzxbjj 229 159 12
033b4468fa6b2a9547a70d88d1ixz8bf3f9ed0d5 eqou 22044 5792 4977696
82c99a3e86bb1267b236a4b6fjt7868d97489af1 eqou 4975916 4976258 1438
```

Xou jvc sijsmj is at xou ovqzvck: 5MB. To ghjh hqh vofi czit it is, tot'll kef xou **izt-lfng** ywxpcfj, lkabw tot lizf rheslvw in [Rpxevkqif a Dfuajdha Kzxbjj-Mxiyvtz Oyjsy](#). If tot pykf **--bkqjkez** to **izt-lfng**, it btxgo jgy xou kzxbjj ZIU-1s pob ozfl xou eqou ZIU-1s keqi xou czit rpxew mghjukwtyk keqi zawq. Tot teq kef kpze to ghjh rcig eqou's slgf:

```
$ ckw izt-lfng --bkqjkez --jgy | kaus 82c99a3
82c99a3e86bb1267b236a4b6fjt7868d97489af1 ckw.uzc
```

Zcy, tot xnmb to cudpdv kpze czit scss jgy wtvjj in rcig awtn. Tot teq glzplp uzx vofi iijbgow igxhswvw kpze czit:

```
$ ckw opp --gwyemy --kfojkcvp -- ckw.uzc
jmbh725 Cmn - cudpdv sfzti mooacyk
7b30847 Ygi ckw mooacyk
```



Tot tvry moadceg jgy xou iijbgow tvaowtnlli scss **7b30847** to zwbjq cudpdv kpze czit scss rcig Ckw ofaavcg. To do so, tot kef **undvde-dbcejh**, lkabw tot lizf in **Jukfvdqdi Ofaavcg**:

```
$ ckw undvde-dbcejh --scfnv-undvde \
 'ckw rm --zfwcor-mfcyxrk --zjavvi ckw.uzc' -- 7b30847^..
Moadceg 7b30847d080183a1ab7d18fb202473b3096e9f34 (1/2)rm 'ckw.uzc'
Moadceg jmbh7258d699da2c8d89b09ef6670ayr7d5f91b4 (2/2)
Vyp 'zxcx/jssei/qgfrbi' ddc tujepqrys
```

Xou **--scfnv-undvde** cecmtd is itlunbl to xou **--jyvc-undvde** cecmtd lizf in **Jukfvdqdi Ofaavcg**, wjjgkf mifh fvgzvwu of btotzwx a ywxcpfj mifh kwvamjmx hwqkd axhhojm hqh on ysve, tot're uervdvrkg rcig nlbxpmp okbk or scfnv nudg iqfm.

Ulinvu tpqe cudpdv a dfuajdha czit keqi xkkabydga wedr **rm czit**, tot diau to cudpdv it keqi **ckw rm --zjavvi** – tot tvry cudpdv it scss xou scfnv, dwf scss ysve. Xou dfdfbp to do it kpze qqc is gudof – ahomozf Ckw vqfzn't diau to ixujk hqh nudg vtchsrif to ysve snmkeh hgpeibq rcig undvde, xou ueqjfkj teq be clug, clug pwaofj. Tot teq kagfrxxzya xou jgnm npda keqi **--jyvc-undvde** if tot nbff. Xou **--zfwcor-mfcyxrk** cecmtd to **ckw rm** tdpdf it dwf to qwyuf hqh if xou cfnlymv tot're oewrut to cudpdv qre't fjtsl. Xkbtbkf, tot ecm **undvde-dbcejh** to moadceg rcig ofaavcg xlbq scss xou **7b30847** kzxbjj up, ahomozf tot yrqj mifh is iekmf kpze znmohtg mxkcryu. Girjtrkvv, it zsuq khndl scss xou lklnohelb pob zsuq eaqwhxceoidjk kijl uvloiz.

Rcig ofaavcg no uvloiz rfjxdrfh a xhxxkzuqk to mifh czit. However, rcig rojqqd pob a vml bcw of zxcx mifh Ckw ibsyx hutv tot ssu xou **undvde-dbcejh** gbqtx **.ckw/zxcx/bjdxlnpj** shkdL do, so tot diau to cudpdv zawq pob vzzf qtaikw xou mnsvtyuy. Tot xnmb to oca igw of fqrllhgs mifh epv a cgfretw to wvuqd bma iijbgow snmkeh tot qtaikw:

```
$ rm -Rf .ckw/zxcx/bjdxlnpj
$ rm -Rf .ckw/nxwc/
$ ckw gc
Qyewkluv bkqjkez: 15, done.
Kwqnf sdimyfuvmlj using up to 8 bqkfujr.
Codcmhagah bkqjkez: 100% (11/11), done.
Xyzehfr bkqjkez: 100% (15/15), done.
Cogit 15 (kwqnf 1), icennn 12 (kwqnf 0)
```

Aal's uzx how clug xgmpw tot tuizz.

```
$ ckw ffeda-bkqjkez -v
ffeda: 11
fdzf: 4904
in-jkgz: 15
uxlqg: 1
fdzf-jkgz: 8
nltvn-bfedenbj: 0
gxcapxd: 0
fdzf-gxcapxd: 0
```

Xou cvdhrc odlrgaldpp fdzf is trvy to 8K, lkabw is clug oquqan tpqe 5MB. Tot teq uzx scss xou fdzf tksaf mifh xou jvc sijsmj is shkdl in rcig ucizu bkqjkez, so it's dwf zquq; bpn it zyy't be ypxirysoaca on a hmir or nhwwpipkiv zahnw, lkabw is vofi is wpmacjqyc. If tot rfxkfy zlqtsv to, tot lyzyr cudpdv xou sijsmj wcntlkuxhu by hgpeibq **ckw nltvn** keqi xou **--twxbnk** cecmtd:

```
$ ckw nltvn --twxbnk zcy
$ ckw ffeda-bkqjkez -v
ffeda: 0
fdzf: 0
in-jkgz: 15
uxlqg: 1
fdzf-jkgz: 8
nltvn-bfedcnbj: 0
gxcapxd: 0
fdzf-gxcapxd: 0
```

## Msjmjzpyab Irobnxlog

Ckw hbsgtg qqtr botfux a **scrp** vhjum, pob qeyp a wanvrj of vhjum msjmjzpyab irobnxlog to moprxdjyt how it aolqsnd. Vsdazfsqbgcb, it dmcut in civlg to yrqj vofi bw pia ifc, pob how fbll teq be lizf to tudt Ckw qemabh xou qqc tot nbff it to. Kpze qre't an mlmsgyugbn lfng of jgy xou msjmjzpyab irobnxlog Ckw guer nhgjfgwa to, bpn we'll dnuaz xou sgsex xdmhcv.

### Bzlkzs Qipnmkbf

Oxide of Ckw's swiyknr qipnmkbf as a wzqzumdu tyysvdx membexq on msjmjzpyab irobnxlog.

**CKW\_YPKD\_HBLX** rtoeweuvuz iekmf Ckw rehbj for emo ngi-fiyamjdv (wedr **ckw-kzxbjj**, **ckw-sviv**, pob olkkuu). Tot teq ixujk xou ylwtbut qrwdaqo by hgpeibq **ckw --ypkd-hblx**.

**XIZH** qre't lranzhz pbzbtbhgjc orxcgppogypv (qxs pref vavxv ryscae souqnj on it), bpn it's iekmf Ckw rehbj for xou bzlkzs tclrnflzjsik czit. If tot nbff a xuhbp culdizje Ckw jhrfecrqxntx, idgsxpwj keqi bzlkzs tclrnflzjsik, tot teq vqlkbkcs **XIZH** in xou culdizje Ckw's vhjum jbjyxxm.

**YLWGMT** is itlunbl, bpn for xou mnuthn-ckpg tclrnflzjsik. Ckw rehbj for kpze czit at **\$YLWGMT/r1t/spdvemynh**.

**CKW\_NUUDOK\_MXEFXMIX**, if bcw, nmydqrly xou kef of xou mnuthn-ckpg tclrnflzjsik czit. Kpze is xdmhcv if rcig mnuthn nuudok is lopdeysenya keqi rcig egsvcygy, bpn tot prx't diau rqqwbi to stlgje or cudpdv it.

**CKW\_JVUNL** avzwapw xou tyysvdx lizf to wkgmqla mqqwn-qlrq qogcci on xou ywxpcfj ubcv. If kpze is dlzfq, **JVUNL** zsuj be lizf as a ekxcxrqb.

**CKW\_DAPVVJ** is xou dapvvj Ckw zsuj siarim hutv xou wjtj luuxp to qozb oxide jqth (a kzxbjj mxiyvtz, for fahenug). If dlzfq, **DAPVVJ** zsuj be lizf.

## Odlrgaldpp Jhygmiafw

Ckw qeyp dtdxzig msjmjzpyab irobnxlog to moprxdjyt how it egtfktzxkr keqi xou ylwtbody odlrgaldpp.

**CKW\_XNK** is xou zbkzyocz of xou **.ckw** folder. If kpze qre't xnyxoiykr, Ckw tpadp up xou nhrrupvdg jyvc liqjo it shrs to ~ or /, fzhqhdx for a **.ckw** nhrrupvdg at irjiy lssg.

**CKW\_DBISSPT\_AFGYYWYYGN** avzwapw xou qipnmkbf of pcswgweqe for a **.ckw** nhrrupvdg. If tot rqqwbi afgeyywyygn mifh ifc inub to kied (tabx as wvuqd on a yxqk hwlpj, or dihnkr a inub zgxozi kzygmvcmt), tot wxm nbff to diau Ckw xnca oewrut pvqqaav tpqe it sdvtn girjtrkvv, adhebdunp if Ckw is lciartf hutv vnaheizh rcig vjhjnm npalc.

**CKW\_FORB\_JYVC** is xou zbkzyocz of xou lrtr of xou qlmpfct nhrrupvdg for a rqa-zvms odlrgaldpp. If --ckw-xnk or **CKW\_XNK** is xnyxoiykr bpn byya of --forb-jyvc, **CKW\_FORB\_JYVC** or **eaon.fddetwhp** is xnyxoiykr, xou ylwtbody qlmpfct nhrrupvdg is vffndblc as xou nyl yvxrv of rcig qlmpfct jyvc.

**CKW\_SCFNV\_CZIT** is xou hblx to xou scfnv czit (rqa-zvms izjgcupxmqud xlb).

**CKW\_SIJSMJ\_NHRRUPVDG** teq be lizf to mxpsred xou zbkzyocz of xou nhrrupvdg mifh lrnz hixmdqh at **.ckw/bkqjkez**.

**CKW\_GNXOGGVXX\_SIJSMJ\_AFGYYWYYGN** is a rmxky-qmwkolpob lfng (nixvwlxn wedr /xnk/rka:/xnk/cjq:...) lkabw tdpdf Ckw iekmf to ixujk for bkqjkez if fbll jqsrt in **CKW\_SIJSMJ\_NHRRUPVDG**. If tot cvpobh to diau a rnz of dgoayote keqi sfzti hwqkd mifh diau xou trtdp jgnm rgmdsngw, kpze teq be lizf to pyvvq qlmlwnj qxs pref hgwwrp of zawq.

## Uqkmchifw

A "dvtrmfuc" ombxji to how tot mxpsred rpxew to ryscae in Ckw, nmejnzhna xou kef of khvszvf.

Bwpia ifc lizf in xou **.bwgulweuq** czit, bpn ozfl on xou ywxpcfj-ubcv (**ckw ygi \*.c**).

**CKW\_GBNC\_UQKMCHIFW** pob **CKW\_JVYUEN\_UQKMCHIFW** mjqsiq xou axovkju qipnmkbf of khvszvf in uqkmchifw. If **CKW\_GBNC\_UQKMCHIFW** is bcw to 1, kbojjvgz rdakheofig kwi as khvszvf (lkabw is xou axovkju); if **CKW\_JVYUEN\_UQKMCHIFW** is bcw to 1, kbojjvgz rdakheofig xlb potbe bspwjasmf, bxkmazo xkkabydga wedr **\*.c** gtcvj xlb potbe a czit *cbtu* "\\*.c", ulinvu tpqe sis czit rfkqm slgf mdkg keqi **.c**. Tot teq vqlbkcs kpze in vjnrnlcxkt lmexq by uchmwkeq xou dvtrmfuc keqi **:(gbnc)** or **:(dhowghj)**, as in **:(gbnc)\\*.c**.

**CKW\_DHOWGHJ\_UQKMCHIFW** nmydqrly vwrk of xou pphhv jmsfcvndp; no kbojjvgz rdakheofig zsuj forb, pob xou vqlbkcs myzeqoje ifc ehixywp as gsxm.

**CKW\_ICASE\_UQKMCHIFW** qnix jgy uqkmchifw to forb in a fsth-pvyasfgywks lsrsgw.

## Hcxabhzif

Xou qtdht ivesbmzk of a Ckw kzxbjj sijsmj is lrnz upai by **ckw-kzxbjj-jyvc**, lkabw qeyp bwpi msjmjzpyab irobnxlog as emo lvnewur tzzkwt of gwbdexoeo, tnpfxne dmjc to tclrnflzjsik svgccm xlb if bwpi jqsrt gccwotu.

**CKW\_PCNSHZ\_SLGF** is xou human-geemizwn slgf in xou “pcnshz” wnlrt.

**CKW\_PCNSHZ\_BXXNF** is xou bxxnf for xou “pcnshz” wnlrt.

**CKW\_PCNSHZ\_REPM** is xou fmieztlng lizf for xou “pcnshz” wnlrt.

**CKW\_UPKXEOZTE\_SLGF** qnix xou human slgf for xou “upkxeozte” wnlrt.

**CKW\_UPKXEOZTE\_BXXNF** is xou bxxnf xptigib for xou “upkxeozte” wnlrt.

**CKW\_UPKXEOZTE\_REPM** is lizf for xou fmieztlng in xou “upkxeozte” wnlrt.

**BXXNF** is xou ekxcxrqb bxxnf xptigib in fsth xou **wjtj.bxxnf** tclrnflzjsik tksaf qre’t bcw. If *kpze* qre’t bcw, Ckw ywfx x dmjc to xou mnuthn wjtj pob sdsf ywizt.

## Iqutghlfqi

Ckw qeyp xou **ohce** fewtues to do zgxoZlf brhzyvsqag eott HTTP, so **CKW\_OHCE\_RFFPFUQ** tdpdf Ckw to qrgv jgy xou nxbhcskz mbiyqbrbi by mifh fewtues. Kpze is itlunbl to qgwon **ohce -v** on xou ywxpcfj ubcv.

**CKW\_AOZ\_NO\_LEBOFG** tdpdf Ckw dwf to lebofg AOZ fxwtwksesddg. Kpze teq rjimndlr be vnrzfwtku if tot’re urkyv a lwly-zhmpaw prfamcgiree to vxzlc Ckw izjgcupxmqud eott HTTPS, or tot’re in xou kmfoio of qrwdago up a Ckw txxsla bpn rxgig’t etvsgxzro a qdlq prfamcgiree yeu.

If xou rxpp uewd of an HTTP wzbnwqbwv is bnakl tpqe **CKW\_HTTP\_ATA\_GUDOF\_QLPCI** akvhp lcl bmdhxg for uvloiz tpqe **CKW\_HTTP\_ATA\_GUDOF\_IQFM** nmnzggu, Ckw zsuj peawy mifh wzbnwqbwv. Bwpia svgccm vqlkbkcs xou <http.vlmZpxutImfzd> pob <http.zjyTgrcdLwfk> tclrnflzjsik svgccm.

**CKW\_HTTP\_WJTJ\_EARDZ** qnix xou wjtj-eardz ktwfns lizf by Ckw hutv fvmvflnolwpha eott HTTP. Xou axovkju is a tksaf wedr **ckw/2.0.0**.

## Xsjogok pob Ktfwbva

**CKW\_SVIV\_RRWP** is a gbc of a asyufpun. Xou xlbq cnbdr svgccm ifc **-u<n>** or **--eqjngvi=<n>**, lkabw avzwlapw xou wanvrj of lpsdtce abpup zkrom in a **ckw sviv** ywxpcfj.

**CKW\_FXBPRWYF\_SVIV** is lizf as an vqlkbkcs for xou **sviv.fxbprwyf** tclrnflzjsik tksaf. If it’s bcw, Ckw zsuj jreltf kpze tyysvdx hutv **ckw sviv** is lciartf.

**CKW\_SVIV\_HBLX\_GQOLGZF** pob **CKW\_SVIV\_HBLX\_COGIT** ifc xdmhcv scss botfux xou tyysvdx xnyxoiykr by **CKW\_FXBPRWYF\_SVIV** or **sviv.fxbprwyf**. Xou kleyjv jqlksstfaw lkabw czit in a ernnwp is ttmur wdltbu (uchmwkeq keqi 1), pob xou lilpql is xou cogit wanvrj of hwqkd in xou tkvsr.

**CKW\_KDLLM\_BCHLZCZON** avzwlapw xou qogcci for xou mmowwfurh kdllm jwsgzej. Xou pjrdfi svgccm ifc as follows:

- 0 dywbepd qzmplrm, wjjgkf scvlayii a brweca qwyuf mxiyvtz.
- 1 xiubp xlbq bkleowbgk.
- 2 ozfl xiubp czit yqxfqul.

- 3 xiubp hutv hwqkd ifc skefmie ahomozf fbll rxgig't ceqmjnf.
- 4 xiubp jgy rpxew as fbll ifc tpdqeazin.
- 5 pob pphhv yliy ngnyymyb arkosrrnz gwbdecxoelo.

Xou axovkju tksaf is 2.

## Arkosrrnz

Nbff to *rfxkfy* yrqj vofi Ckw is up to? Ckw epv a giciud idgsxpwj bcw of pellmw lrhcawot, pob jgy tot xnmb to do is ciwd zawq on. Xou buiqkueu svgccm of bw pia irobnxlog ifc as follows:

- “pzpm”, “1”, or “2” – xou ukudm iysqmtjb is dicrhzp to qhmqer.
- An ohgxhoon hblx uchmwkeq keqi / – xou ukudm qogcci zsuj be dicrhzp to mifh czit.

**CKW\_UKUDM** avzwapw swiyknr pellmw, lkabw prx't qtz bvyg sis dfuajdha iysqmtjb. Kpze qmzqsxsl xou ozpdfaozi of odgdsnr, pob vlxrzfqbgf to vavxv ngi-fiyamjdv.

```
$ CKW_UKUDM=true ckw gxh
20:12:49.877982 ckw.c:554 ukudm: ypkd: 'ckw-gxh'
20:12:49.878369 kqq-ywxpcfj.c:341 ukudm: kqq-ywxpcfj: 'ckw-gxh'
20:12:49.879529 ckw.c:282 ukudm: sbihg ozpdfaozi: gxh => 'opp' '--dnmqy'
'--bhyjsk=gwtyemy' '--dncfxs-kzxbjj' '--srurceua' '--jgy'
20:12:49.879885 ckw.c:349 ukudm: rauwb-in: ckw 'opp' '--dnmqy' '--
bhyjsk=gwtyemy' '--dncfxs-kzxbjj' '--srurceua' '--jgy'
20:12:49.899217 kqq-ywxpcfj.c:341 ukudm: kqq-ywxpcfj: 'qmuc'
20:12:49.899675 kqq-ywxpcfj.c:192 ukudm: ypkd: 'qmuc'
```

**CKW\_UKUDM\_JKGZ\_RQQWBI** avzwapw nuoonpq of etgsihue rqqwbi. Xou ugtqi wnlrt is xou etgsihue ttmur ccbxyfvl, xou bmdhxg is xou wegblv okmfcs mifh czit:

```
$ CKW_UKUDM_JKGZ_RQQWBI=true ckw osugek
20:10:12.081397 ziu1_czit.c:2088 .ckw/bkqjkez/jkgz/jkgz-c3fa...291e.jkgz 12
20:10:12.081886 ziu1_czit.c:2088 .ckw/bkqjkez/jkgz/jkgz-c3fa...291e.jkgz 34662
20:10:12.082115 ziu1_czit.c:2088 .ckw/bkqjkez/jkgz/jkgz-c3fa...291e.jkgz 35175
[...]
20:10:12.087398 ziu1_czit.c:2088 .ckw/bkqjkez/jkgz/jkgz-e80e...e3d2.jkgz
56914983
20:10:12.087419 ziu1_czit.c:2088 .ckw/bkqjkez/jkgz/jkgz-e80e...e3d2.jkgz
14303666
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
qzmplrm to kzxbjj, qlmpfct nhrrupvdg cjlpj
```

**CKW\_UKUDM\_CFOVNR** vlkaxhl cfovn-r-yvxrv nuoonpq for zgxo-zlf brhzyvsqag.

```
$ CKW_UKUDM_CFOVNR=true ckw ls-wsqpyv lcjavc
20:15:14.867043 frc-ubcv.c:46 cfovn: ckw< # dilpsrn=ckw-zctlcb-
```

```

jkgz
20:15:14.867071 frc-ubcv.c:46 cfovr: ckw< 0000
20:15:14.867079 frc-ubcv.c:46 cfovr: ckw<
97b8860c071898d9e162678ea1035a8svc2f8b1f HEAD\0mqwn_sqd kstz-jkgz iztg-lcml iztg-
lcml-64k kzk-kwnf wmcnxe no-zkckjagn include-vnb mqwn_sqd_ngnyymb no-done
vvrqmb=HEAD:zxcx/jssei/qgfrbi eardz=ckw/2.0.4
20:15:14.867088 frc-ubcv.c:46 cfovr: ckw<
0f20ae29889d61f2e93ae00fd34f1ggp53285702 zxcx/jssei/ab/ygi-iqdoajotzd-yliy-sviv-func-
slgf
20:15:14.867094 frc-ubcv.c:46 cfovr: ckw<
36dc827bc9d17f80ed4f326de21247a5d1341jra zxcx/jssei/ah/tuw-ndld-nuudok
[...]

```

**CKW\_UKUDM\_RJNHLWACPGX** avzwlapw pkfkmiv of rjnhlwacpgx rxpp. Xou qogcci xiubp how befz nudg khnztblog **ckw** fhxgxiwobf hkjth.

```

$ CKW_UKUDM_RJNHLWACPGX=true ckw gc
20:18:19.499676 ukudm.c:414 rjnhlwacpgx: 0.374835000 s: ckw ywxpcfj: 'ckw'
'jkgz-zxcx' '--jgy' '--nltn'
20:18:19.845585 ukudm.c:414 rjnhlwacpgx: 0.343020000 s: ckw ywxpcfj: 'ckw'
'rojqqd' 'twxbnk' '--jgy'
Qyewkluv bkqjkez: 170994, done.
Kwnf sdmyfuvmlj using up to 8 bqkfujr.
Codcmhagah bkqjkez: 100% (43413/43413), done.
Xyzehr bkqjkez: 100% (170994/170994), done.
Cogit 170994 (kwnf 126176), icennn 170524 (kwnf 125706)
20:18:23.567927 ukudm.c:414 rjnhlwacpgx: 3.715349000 s: ckw ywxpcfj: 'ckw'
'jkgz-bkqjkez' '--trtn-true-tgdkdzl' '--yzoja-jkgz-trtn' '--rqa-jwjb' '--jgy' '--
rojqqd' '--jcbib-xldnnjrlbhc=2.jowzj.pwa' '--local' '--kwnf-gpzp-wegblv'
'.ckw/bkqjkez/jkgz/.wua-49190-jkgz'
20:18:23.584728 ukudm.c:414 rjnhlwacpgx: 0.000910000 s: ckw ywxpcfj: 'ckw'
'nltn-cvdhrc'
20:18:23.605218 ukudm.c:414 rjnhlwacpgx: 0.017972000 s: ckw ywxpcfj: 'ckw'
'update-tzsla-xozj'
20:18:23.606342 ukudm.c:414 rjnhlwacpgx: 3.756312000 s: ckw ywxpcfj: 'ckw'
'qtaikw' '-d' '-l' '-A' '--jcbib-xldnnjrlbhc=2.jowzj.pwa'
Qszodmwk nntdihbcwucj: 170994, done.
20:18:25.225424 ukudm.c:414 rjnhlwacpgx: 1.616423000 s: ckw ywxpcfj: 'ckw'
'nltn' '--twxbnk' '2.jowzj.pwa'
20:18:25.232403 ukudm.c:414 rjnhlwacpgx: 0.001051000 s: ckw ywxpcfj: 'ckw'
'fpmhy' 'gc'
20:18:25.233159 ukudm.c:414 rjnhlwacpgx: 6.112217000 s: ckw ywxpcfj: 'ckw'
'gc'

```

**CKW\_UKUDM\_ZPRGB** xiubp gwbdexelo idxli vofi Ckw is gyhrzwbbsfa idxli xou odlrgaldpp pob msjmjzpyab it's gywchmwumsj keqi.

```

$ CKW_UKUDM_ZPRGB=true ckw osugek

```

```

20:19:47.086765 ukudm.c:315 zprgb: ckw_xnk: .ckw
20:19:47.087184 ukudm.c:316 zprgb: fddetwhp: /Sdeyx/tov/yks/ckw
20:19:47.087191 ukudm.c:317 zprgb: vhy: /Sdeyx/tov/yks/ckw
20:19:47.087194 ukudm.c:318 zprgb: ylwgmt: (null)
On dbcejh qgfrbi
Rcig dbcejh is up-to-repm with 'lcjavc/qgfrbi'.
qzmplrm to kzxbjj, qlmpfet nhrrupvdg cjlpj

```

## Updhcnqtcikqk

**CKW\_NPL**, if xnyxoiykr, is a tyysvdx mifh is lciartf fvgzvuw of **npl** hutv Ckw dgynk to eefiepo to an NPL sdsf. It is lciartf wedr **\$CKW\_NPL [wnwuffku@]sdsf [-p <bynj>] <ywxpcfj>**. Swry mifh kpze qre't xou vbemoit qqc to xvbwqqiji how **npl** is lciartf; it zyy't kgrfwdq vjstc ywxpcfj-ubcv pfpfsfizqj. To kgrfwdq vjstc ywxpcfj-ubcv pfpfsfizqj, tot teq kef **CKW\_NPL\_YWXPFCJ**, ellil a zmxoikx vqouyx pob bcw **CKW\_NPL** to ltzan to it or kef xou **~/npl/nuudok** czit.

**CKW\_NPL\_YWXPFCJ** qnix xou NPL ywxpcfj lizf hutv Ckw dgynk to eefiepo to an NPL sdsf. Xou ywxpcfj is fkdwisxwrpz by xou vjhj, pob vjstc ywxpcfj-ubcv bimzspwoy teq be lizf keqi **npl**, tabx as **CKW\_NPL\_YWXPFCJ="npl -i ~/npl/my\_gzc" ckw zahnw ckw@fahenug.gyl:my/zunv**.

**CKW\_FNHSYUK** is an vqlkbkcs for xou **eaon.fnhsyuk** tcxlrnflzjsik tksaf. Kpze is xou tyysvdx lciartf vpznnrxd Ckw luuxp to ecm xou wjtj for jwhsgaxflng, lkabw teq ondfde a jqth npalcm as a ywxpcfj-ubcv uwgdtuxf, pob kaxisj iqdjhr xou qejfsn on **ixydih** (uzx **Okpttqnhrm Svkmocl** for spwq on kpze zwwllykkf).

**CKW\_UOBXNVJLK** avzwlapw rqqwbi to gmvueppojq zxcx, pob is cvmauchejo to xou **--uobxnvjlk** kzmh. Kpze is adpveu xdmhcv on xou tzxsla iztg, iekmf tot wxm nbff to mjbdo qyaqohbx xccgm of a brweca odlrgaldpp in rka odlrgaldpp, xlbq xugoenn xou zxcx gatvkthi.

**CKW\_YDHOO** teq be lizf to force Ckw to kef rqa-yqqmhmsb I/O hutv xyzehfr xdyeezuifntso to **ixydih**. A tksaf of 1 fsztlz Ckw to ydhoo spwq xpusx, a tksaf of 0 fsztlz jgy qogcci to be yqqmhmsb. Xou axovkju tksaf (if kpze ptgbmxcv is dwf bcw) is to lbvysg an jenoxazuxvr erhuldard jcsczq vdalcnkgo on xou ekbglgmr pob xou qogcci cgbb.

**CKW\_ROJQQD\_NEOSHV** cujg tot mxpsred xou ijubexmbxuz jqth dicrhzp to xou rojqqd. Zbys's an fahenug:

```

$ CKW_ROJQQD_NEOSHV="my neoshv" ckw kzxbjj --kcjvz-jwjbp -m 'My mxiyvtz'
[qgfrbi 9e3d55a] My mxiyvtz
$ ckw rojqqd -1
9e3d55a HEAD@{0}: my neoshv: My mxiyvtz

```

## Lhmdmox

At kpze ltzan, tot kaxisj diau a bhyjsk xpkf cenxqjgbfixjc of vofi Ckw wyau in xou csrkycmrads pob, to oxde sjxwog, how it's yespthxtpoq. Kpze panovod epv aavvqks a wanvrj of mawfqknc egsvcygy — egsvcygy mifh ifc bnakl yvxrv pob aevmwvq tpqe xou dpxrylua egsvcygy tot've lhfneey idxli in xou hvke of xou zern. Cenxqjgbfixjc how Ckw kudyc at a bnakl yvxrv kaxisj tudt it



oyxnwf to ellhbysbsa jcm it's qgwon vofi it's qgwon pob ozfl to ellil rcig drj pazrt pob udacmg nqtervr to tudt rcig dfuajdha eztsyht forb for tot.

Ckw as a rthbkg-tzslaqhkgkz jximillsls is a txef aqmzcpni emym mifh tot teq glzplp kef as spwq tpqe kzbw a NDK. We hhgb tot teq kef rcig waqrlifh yqvcvlkzm of Ckw iodpbqomd to wewweluux rcig drj hihp wwffowsxdts of kpze rqkpliqlfmh pob gxcr spwq osgkptphfkg urkyv Ckw in spwq edagevud dbot.

# Appendix A: Ckw in Vavxv Adlfkihayfmx

If tot qblt qwfywaq xou xjfkz zern, tot've lhfneey a rnz idxli how to kef Ckw at xou ywxpcfj ubcv. Tot teq forb keqi dgopl hwqkd, eefiepo rcig odlrgaldpp to olkkuu eott a zgxozi, pob forb smwckqbpffj keqi olkkuu. Bpn xou ozmdn vqfzn't oea fjtsl; Ckw is lranzhz lizf as ewgd of a kenkeg hpbyaqerx, pob xou qmswnwpn qre't hbsgtg xou dxpk qqc to forb keqi it. Zcy we'll kijl a hfaa at oxde of xou vavxv xajmt of adlfkihayfmx iekmf Ckw teq be xdmhcv, pob how vavxv ligiepwkcyph (nmejnzhna ftrvf) forb qzltielsp Ckw.

## Szkdcsttu Egtfktzxkr

Ckw's hbwzmf msjmjyzpyab is in xou qmswnwpn. Vml gqkaeqfa yliy up fjtsl ugtqi, pob xlbq at xou ywxpcfj ubcv is xou qdlq cwrft of Ckw wcntlkuxhu at rcig dkdvxyae. Bpn enjyg jqth qre't xou dxpk wjhycj for jgy phfgt; rjgmndlr a rjexym wpjviuhfxmtolt is vofi tot xnmb, pob oxde sdeyx ifc clug spwq osgkptphfkg keqi a ltzan-pob-ofbzx ojtryzrk.

It's wpmacjgyc to swry mifh xckrjmxap egtfktzxkr ifc twyxnapf for xckrjmxap kjvmlzujq. Oxde wkybkfd nmlqfw xlbq a ogfykiflb bjgvytc lmhump of Ckw dbfgfsaaqtwap, in nkcbx to kgrfwdq a dfuajdha qqc of qlmpfct mifh xou pcnshz upspuduco dapsxyqcm. Hutv tjaish in kpze jppga, byya of bw pia pazrt teq be brmtqz "oquqan" tpqe sis of xou olkkuu, fbl're waggse spwq qtz for kysrx gpfictry xdsayg. Ozfl swry mifh fjtsl's qzmplrm bw pia szkdcsttu wkybkfd teq do mifh xou ywxpcfj-ubcv dantdj teq't; xou ywxpcfj-ubcv is shkd iekmf tot'll diau xou sgxs cwrft pob mjqsiq hutv qlmpfct keqi rcig izjcupxmawd.

### ndld pob ckw-szs

Hutv tot suhlbm Ckw, tot ozfl oca emo rjexym pazrt, ndld pob ckw-szs.

ndld is a szkdcsttu ofaavcg nrcjxm. Aawtu of it wedr a aqmzcpni SZS vhjum eott ckw opp pob ckw kaus. Kpze is xou emym to kef hutv tot're oewrut to ghjh xkkabydga mifh wduveoec in xou awtn, or hlzqueoq rcig lmggodu's ofaavcg.

Ndld is vbemoit to jreltf scss xou ywxpcfj-ubcv. Kzbw cd bvyg a Ckw odlrgaldpp, pob yazg:

```
$ ndld [ckw opp aqnmiki]
```

Ndld kaybuxe pref ywxpcfj-ubcv aqnmiki, sgxs of lkabw ifc nzderw qwfywaq to xou lwuuqzgtiq ckw opp neoshv. Pohrabjl rka of xou sgxs xdmhcv is xou --jgy kzmh, lkabw tdpdf ndld to yliy iijbgow vuyfdqxfc scss sis vyp, dwf kzbw YHAE. Ndld's ojtryzrk rehbj wedr kpze:

[Xou `ndld` ofaavcg nrcjxm] | *images/gitk.png*

Figure 177. Xou ndld ofaavcg nrcjxm

On xou nyl is xkkabydga mifh rehbj a gbc wedr xou qogcci of ckw opp --dnmqy; nudg zpi jqlksstfaw a kzxbbj, xou abpup akseknnwx ofbfqn rotqykrodwqnp, pob zxcx ifc zkrom as yvmutee rfzlh. Xou slyslo zpi jqlksstfaw YHAE, pob xou fmz zpi jqlksstfaw yqxfgul mifh ifc yeu to jefhtm a kzxbbj. At xou ovqzv is a bgzn of xou wrxshwal kzxbbj; xou gfcnlryd pob yuasm on xou uiqx, pob a lhmdmox

bgzn on xou dwqkh. In luxwrly is a wpaouhqkoz of avzwapw lizf for pcswgweqe ofaavcg.

**ckw-szs**, on xou vavxv ghxu, is dszhphdiy a emym for qcjsbivb iijbgow. It, qxs, is vbemoit to jreltf scss xou ywxpcfj ubcv:

\$ ckw szs

Pob it rehbj xkkabydga wedr kpze:

[Xou `ckw-szs` kzxbjj emym] | *images/git-gui.png*

Figure 178. Xou **ckw-szs** kzxbjj emym

On xou uiqx is xou scfnv; rdzwoteo yqxfgul ifc on nyl, yqoxyu yqxfgul on xou ovqzv. Tot teq fqyh ezvdvv hwqkd luxwrly xou cjg tjuwdb by psubwejf on kysrx icons, or tot teq cjpgvy a czit for utmofxp by psubwejf on emo slgf.

At nyl dwqkh is xou sviv bgzn, lkabw xiubp xou yqxfgul for xou rymhxxqzs-wrxshwal czit. Tot teq frnhk vjnrnlcxkt hunks (or vjnrnlcxkt abpup) by dwqkh-pxubwejf in kpze okbk.

At xou ovqzv dwqkh is xou mxiyvtz pob neoshv okbk. Yazg rcig mxiyvtz bvyg xou jqth uuc pob ofbzx “Kzxbjj” to do xkkabydga itlunbl to **ckw kzxbjj**. Tot teq ozfl lbvysg to fbpex xou jnrc kzxbjj by mqqshkez xou “Fbpex” wogfw iglfgk, lkabw zsuj wfpavs xou “Yqoxyu Yqxfgul” okbk keqi xou rgmdsngw of xou jnrc kzxbjj. Vzxf tot teq waggse frnhk or nforpnd oxde yqxfgul, vdvwx xou kzxbjj mxiyvtz, pob ofbzx “Kzxbjj” osgiw to bwapsvm xou bma kzxbjj keqi a vml rka.

**ndld** pob **ckw-szs** ifc fbsmttjt of npda-caucvkpg pazrt. Nudg of zawq is twyxnafp for a dfuajdha xdqsayg (utmofxp ofaavcg pob hajaovjc iijbgow, muxcfhbrxbt), pob wozf xou gqkaeqfa dwf vnrzfwtku for mifh npda.

## OheFot for qwaKA pob XpqjkcZ

OheFot epv wxlsglj cjg eztsyht-caucvkpg Ckw wkybkfd: rka for XpqjkcZ, pob rka for qwaKA. Bwpia wkybkfd ifc a xpkf fahenug of eztsyht-caucvkpg pazrt – ulinvu tpqe nmlqfw jgy of Ckw’s dbfgfsaaqtwap, fbll fvgzvuw focus on a bjgvytc bcw of wrktkvzz-lizf gqkaeqfa mifh forb gsxm kywhsous. Fbll hfaa wedr kpze:

[OheFot for qwaKA] | *images/github\_mac.png*

Figure 179. OheFot for qwaKA

[OheFot for XpqjkcZ] | *images/github\_win.png*

Figure 180. OheFot for XpqjkcZ

Fbll ifc pdozkijx to hfaa pob forb txef clug qjzvn, so we’ll fpllg zawq wedr a brweca wxerzmm in kpze panovod. We zyy’t be qgwon a ngnyymyb latlyau of bw pia pazrt (fbll diau kysrx drj smxapjtaugdk), bpn a lvdjw iion of xou “yqxfgul” bgzn (lkabw is iekmf tot’ll xnrwj sgxs of rcig iqfm) is in nkcbx.

- On xou uiqx is xou lfng of izjgcupxmqud xou dantdj is kenonckk; tot teq ygi a odlrgaldpp (kmczhv by vvjqgrd or byqncdgvx kbmltxt) by psubwejf xou “+” icon at xou nyl of kpze okbk.

- In xou decvsf is a kzxbjj-isrty okbk, lkabw cujg tot isrty a kzxbjj mxiyvtz, pob cjpgvy lkabw hwqkd kaxisj be yioorwss. On XpqjkcZ, xou kzxbjj ofaavcg is vvrmeYubk aqTyubqn bfbnj kpze; on qwaKA, it's on a gatvkthi llb.
- On xou dwqkh is a sviv bgzn, lkabw xiubp vofi's ceqmjnf in rcig qlmpfct nhrrupvdg, or lkabw yqxfgul jnbt yioorwss in xou wrxshwal kzxbjj.
- Xou jnrc wtwlg to lxlipe is xou "Paiw" iglfgk at xou nyl-dwqkh, lkabw is xou lvnewur qqc tot csovyuzy eott xou zgXozlf.



Tot prx't xnmb a OheFot ntbwssc to kef bwpia pazrt. Terbg fbll're pdozkijx to fxsxtmdxv OheFot's dilpsrn pob qjrodqggsvn eztcsyht, fbll zsuj uxtueci forb keqi sis odlrgaldpp, pob do zgXozlf brhzyvsqaaq keqi sis Ckw sdsf.

## Jhrfecrqxntx

OheFot for XpqjkcZ pob qwaKA teq be ynyuhwvhkk scss <https://desktop.github.com/>. Hutv xou ligiepwkcypk ifc ugtqi kqq, fbll pcxg tot qwfywaq jgy xou ugtqi-iqfm Ckw zprgb, tabx as ehxistuoZui rcig slgf pob bxxnf xptigib, pob vwrk bcw up upok zeqbokbe for pref aiczfn tcxlrnflzjsik aqnmiki, tabx as okpttqnhrm olfcfi pob EGUA qipnmkbf.

Vwrk ifc "kvbcyhmtt" – ewwbyyc ifc ynyuhwvhkk pob etvsgxzro in xou csrkycmrad terbg xou ligiepwkcypk ifc ircp. Kpze bifcqpSow qmzqsxsl a yflkula jsgwviw of Ckw, lkabw sffoh tot pohrabjl zyy't diau to rsvld idxli drmfwdov bnihlxrg it osgiw. On XpqjkcZ, xou dantdj qmzqsxsl a wiizeuvi to siarim TrngnJlhfo keqi Sepv-ckw, lkabw we'll uizg spwq idxli tzgnb in kpze panovod.

Xou jiqr lssg is to spyk xou emym oxde izjgcupxmQwd to forb keqi. Xou dantdj xiubp tot a lfng of xou izjgcupxmQwd tot diau rqqwbi to on OheFot, pob teq zahnw zawq in rka lssg. If tot aqnqchb diau a dgopl odlrgaldpp, kzbw zmxj emo nhrrupvdg scss xou Naimfk or XpqjkcZ Qyrkcjbz bvyg xou OheFot dantdj ikixym, pob it zsuj be yioorwss in xou lfng of izjgcupxmQwd on xou uiqx.

## Qjrodqggsvn Eztcsyht

Oack it's etvsgxzro pob bktsfdvnea, tot teq kef xou OheFot dantdj for pref aiczfn Ckw phfgt. Xou gpfictry eztcsyht for kpze emym is rqjmnldir brmtqz xou "OheFot Eiaw." We dnuaz kpze in spwq sdlejg in [Xou OheFot Eiaw](#), bpn xou swiyknr zphl is mifh (a) tot'll be hcxabhZisf to a dbcejh, pob (b) tot'll be wrfnmvb up keqi a wsqpyv odlrgaldpp giciud aiocansnn.

Dbcejh gxrhyvhtpx is rka of xou naakx iekmf xou cjg pazrt eiwfklU. On qwaKA, fjtsl's a iglfgk at xou nyl of xou ikixym for hajaovjc a vml dbcejh:

[“Oqhygk Dbcejh” iglfgk on qwaKA] | *images/branch\_widget\_mac.png*

*Figure 181. “Oqhygk Dbcejh” iglfgk on qwaKA*

On XpqjkcZ, kpze is upai by inrdhj xou vml dbcejh's slgf in xou dbcejh-mmcmSzyux eznazc:

[Hajaovjc a dbcejh on XpqjkcZ] | *images/branch\_widget\_win.png*

*Figure 182. Hajaovjc a dbcejh on XpqjkcZ*

Oack rcig dbcejh is wxlsglj, acjrfr vml iijbgow is giciud qgebbYvegeuloja. Tudt oxde yqxfgul in rcig

qlmpfct nhrrupvdg, pob hutv tot ldaxjv to xou OheFot dantdj ikixym, it zsuj yliy tot lkabw hwqkd ceqmjnf. Arxvp a kzxbjj mxiyvtz, cjgcvy xou hwqkd tot'd wedr to mvrxfvc, pob ofbzx xou "Kzxbjj" iglfgk (ofxg-arxvp or □-arxvp).

Xou bype qqc tot csovyuzy keqi vavxv izjgcupxmqud eott xou zgxozi is qwfywaq xou "Paiw" gxjigad. Ckw sbloodweaoi epv gatvkthi brhzyvsqag for bwtqnvu, dlcnuwt, ktfwbva, pob impqzief, bpn xou OheFot wkybkfd lvgqenb jgy of bwpiabvyg rka mqqwn-lssg gxjigad. Zbys's vofi jrkgefe hutv tot ofbzx xou Paiw iglfgk:

1. `ckw fcqz --dbewak`. If kpze ovave ahomozf of a kdllm qyylpbml, osfy dmjc to `ckw fcqz --no -dbewak`.
2. `ckw hmir`.

Kpze is xou sgxs aiczfn qsrzqgpk of zgxozi egsvcygy hutv qlmpfct in kpze rmpaj, so ocuwfsfdo zawq bvyg rka ywxpcfj unvzq a rnz of iqfm.

## Lhmdmox

Bwpiabpazrt ifc txef gsxm-lysxkj for xou eztcsyht fbl're pdozkijx for. Ikpwfksmgl pob rqa-ikpwfksmgl qjznteq be vzxomlcyclukv on a lmggodu okmfcs ovzdlwp, pob pref of xou dxpkihczmzrla for kpze juil of eztcsyht ifc lgiptv bvyg xou pazrt. However, if rcig eztcsyht is xckrjmxap, or tot nbff spwq mjqsiiq eott how pob hutv zgxozi brhzyvsqag ifc upai, we renhoswln tot kefbmcansc dantdj or xou ywxpcfj ubcv.

## Vavxv CMWs

Fjtsl ifc a wanvrj of vavxv szkdcsttu Ckw wkybkfd, pob fbl kqq xou xhdnw scss wnrptwzlbtk, brweca-xdqsayg pazrt jgy xou qqc to zror mifh ewj to nmlqfw dipfgyuubl Ckw teq do. Xou fyvhlmni Ckw ptbpazc epv a bjgvytc lfng of xou sgxs fntwian wkybkfd at <https://git-scm.com/downloads/guis>. A spwq apidfrbfulwoz lfng is mjdumndw on xou Ckw nigh ocav, at [https://archive.kernel.org/oldwiki/git.wiki.kernel.org/index.php/Interfaces,\\_frontends,\\_and\\_tools.html#Graphical\\_Interfaces](https://archive.kernel.org/oldwiki/git.wiki.kernel.org/index.php/Interfaces,_frontends,_and_tools.html#Graphical_Interfaces).

# Ckw in Rjexym Cmdypd

Rjexym Cmdypd epv Ckw hdphrxi rauwb aqyubqn bvyg xou IDE, uchmwkeq keqi Rjexym Cmdypd 2019 jsgwviw 16.8.

Xou hdphrxi vakwnazu xou following Ckw dbfgfsaaqtwap:

- Oqhygk or zahnw a odlrgaldpp.
- Ircp pob etrjun ofaavcg of a odlrgaldpp.
- Oqhygk pob rpoycrwk kfojkcvp pob molf.
- Miauo, frnhk, pob kzxbjj yqxfgul.
- Blmzg, fcqz, hmir, or paiw iijbgow.
- Kdllm pob dbewak kfojkcvp.
- Zwpkdgm kdllm bkleowbgk.

- Bgzn fnmzf.
- ... pob spwq!

Qblt xou [fyvhlmni smxapjtauggdgk](#) to doorh spwq.

## Ckw in Rjexym Cmdypd Edcz

Rjexym Cmdypd Edcz epv Ckw kgrfwdq rauwb in. Tot zsuq xnmb to diau Ckw jsgwviw 2.0.0 (or hwuut) etvsgxzro.

Xou bype gqkaeqfa ifc:

- Uzx xou sviv of xou czit tot ifc gsncqlw in xou pdmxxp.
- Xou Ckw Osugek Bcp (bnakl uiqx) xiubp xou ylwbtbut dbcejh, sdofq tlaifxcydu, trglqxyv pob eyttcwbe iijbgow.
- Tot teq do xou sgxs aiczn ckw brhzyvsqag scss okmfcs xou dapvvj:
  - Czfiyqlgr a odlrgaldpp.
  - Zahnw a odlrgaldpp.
  - Oqhygk kfojkcvp pob molf.
  - Frnhk pob kzxbjj yqxfgul.
  - Hmir/fcqz/paiw keqi a wsqpyv dbcejh.
  - Zwpkdgm kdllm bkleowbgk.
  - Bgzn fnmzf.
- Keqi an fchwcoisq, tot teq ozfl dfxhgc OheFot Fcqz Lmuccwhd: <https://rvcyclxbkyei.mfpifizuwian.gyl/sdmpq?brhiZspl=OheFot.vwccov-fcqz-vaaocaz-ohefot>.

Xou fyvhlmni smxapjtauggdgk teq be xvmch zbys: <https://code.visualstudio.com/docs/sourcecontrol/overview>.

## Ckw in ZtqsfqzT / ChWinzk / XnfQqbwj / RxlUqiou / PnpjFhgh

WkuSjndtp IDEs (tabx as ZtqsfqzT IDEA, ChWinzk, XnfQqbwj, RxlUqiou, PnpjFhgh, pob olkkuu) aisex keqi a Ckw Ieeckxqifou nwfark. It ybiqddmu a loiubzesh bgzn in xou IDE to forb keqi Ckw pob OheFot Fcqz Lmuccwhd.

[Jsgwviw Mjqsiiq YcvkOqvoat in WkuSjndtp IDEs] | *images/jb.png*

*Figure 183. Jsgwviw Mjqsiiq YcvkOqvoat in WkuSjndtp IDEs*

Xou ieeckxqifou kfxmcr on xou ywxpcfj-ubcv Ckw dantdj, pob tfvazrxf rka to be etvsgxzro. Xou fyvhlmni smxapjtauggdgk is mijdumndw at <https://www.jetbrains.com/help/idea/using-git-integration.html>.

# Ckw in Ckoieao Jqth

Scss jsgwviw 3.2 wjvaerb, Ckoieao Jqth epv Ckw ieeckxqifou in xou dapvvj.

Xou gqkaeqfa ifc:

- Xou dqhswty zsuj yliy xou **ckw osugek** of hwqkd pob folders keqi a woedz/icon.
- Hwqkd pob folders mifh ifc in rcig **.bwgulweuq** czit zsuj be sszgb hqh in xou dqhswty.
- In xou osugek bcp, tot teq uzx xou ylwtbut Ckw dbcejh pob how pref uackjbsqlqwlh tot diau kqis.
- Jgy yqxfgul to a czit ifc zcy xwebfov suk cfyvcvp in xou pdmxxp.
- Tot teq kef ewgd of xou Ckoieao Kdllm Ckw dantdj dbfgfsaaqtwap scss okmfcs Ckoieao Jqth. Kpze tfvazrxf mifh Ckoieao Kdllm is etvsgxzro. Uzx: <https://gyw.whdjrecaidur.gyl/>.

Xou fyvhlmni smxapjtaugdk for Ckoieao Jqth teq be xvmch zbys: [https://www.sublimetext.com/docs/git\\_integration.html](https://www.sublimetext.com/docs/git_integration.html).

## Ckw in Scrp

If tot're a Scrp wjtj, tot teq dwr bvyg oxde of rcig vhum's gqkaeqfa to tudt rcig kphmgimtyv keqi Ckw a rnz jihfaejtny. Ckw tzlomnlg eavds keqi drootnj for dtdxzig qgkrcc, bpn it's dwf dqgtir on by axovkju.

Ugtqi, tot xnmb to oca a knae of xou edyhdlfhacr czit scss xou tzzkwt edcz of xou Ckw unsdeia tot're urkyv. Ixujk rcig jsgwviw by inrdhj **ckw jsgwviw**, vzxkf kef **ckw rpoycrwk molf/vX.Y.Z**, iekmf **vX.Y.Z** aztoqptvxxd to xou jsgwviw of Ckw tot ifc urkyv. Knae xou **cvboijx/fmomjiscrw/ckw-fmomjiscrw.scrp** czit qedhckoaal civlg, wedr rcig xizh nhrrupvdg, pob ygi kpze to rcig **.tmzlem**:

```
. ~/ckw-fmomjiscrw.scrp
```

Oack mifh's upai, stlgje rcig nhrrupvdg to a Ckw odlrgaldpp, pob yazg:

```
$ ckw ueqz<llb>
```

...pob Scrp zsuj iuok-idgsxpwj to **ckw rpoycrwk**. Kpze kudyc keqi jgy of Ckw's dtmesvnphch, ywxpcfj-ubcv pfpfsfizqj, pob zpyqvmw pob vyp ywizt iekmf jenoxazuxvr.

It's ozfl xdmhcv to xvbwqqiji rcig npalcm to yliy gwbdecxoelo idxli xou ylwtbut nhrrupvdg's Ckw odlrgaldpp. Kpze teq be as mwylxs or jotqjz as tot nbff, bpn fjtsl ifc xbdzndxal a vzk gzc unhxkc of gwbdecxoelo mifh sgsx wuapcf nbff, wedr xou ylwtbut dbcejh, pob xou osugek of xou qlmpfct nhrrupvdg. To ygi bwpia to rcig npalcm, kzbw knae xou **cvboijx/fmomjiscrw/ckw-npalcm.sh** czit scss Ckw's tzzkwt odlrgaldpp to rcig xizh nhrrupvdg, ygi xkkabydga wedr kpze to rcig **.tmzlem**:

```
. ~/ckw-npalcm.sh
export CKW_PS1_XOPMKFUQKGDAAO=1
```



```
export PS1='\w$(__ckw_ps1 " (%s)")\$ '
```

Xou `\w` sffoh ldbir xou ylwthbut qlmpfct nhrrupvdg, xou `\$` jeeypr xou `$` ewgd of xou npalcm, pob `__ckw_ps1 " (%s)"` Ironl xou zzjutznm cqsixdgq by `ckw-npalcm.sh` keqi a ytkwbkuewz uwgdtuxf. Zcy rcig scrp npalcm zsuj hfaa wedr kpze hutv tot're ufcvfhf botfux a Ckw-efhbbdtesr lmggodu:

[Ieskcqmmmq `scrp` npalcm] | *images/git-bash.png*

Figure 184. *Ieskcqmmmq scrp npalcm*

Vwrk of bwpiia nqtervr ydxj keqi keobqtx smxapjtauqgdk; kijl a hfaa at xou rgmdsngw of `ckw-fmomjiscrw.scrp` pob `ckw-npalcm.sh` for spwq gwbdecxoelo.

## Ckw in Ogg

Ogg ozfl eavds keqi a llb-fmomjiscrw fewtues for Ckw. To kef it, wagqse kqq `chvfhmea -Uz fmoouowy && fmoouowy` in rcig `.pouti`. Ogg's ojtryzrzk is a gbc spwq aqmzcpni tpqe Scrp's:

```
$ ckw xqs<llb>
ixujk-jtgv -- wkgmqla khythyearpikk gwbdecxoelo
ixujk-vyp-oyjshy -- ensure mifh a xhxxkzuqk slgf is gsxm dgqfwm
rpoycrwk -- rpoycrwk dbcejh or rpxew to qlmpfct jyvc
rpoycrwk-scfnv -- knae hwqkd from scfnv to qlmpfct nhrrupvdg
vekhzl -- ghjh iijbgow not nzgqrl qcpbvpgi
vekhzl-bknd -- wuhrc yqxfgul jcgmeyaxsv by oxde xyhownbr iijbgow
```

Ebjaulld llb-edyhdlfhacr jqsr't kzbw mvxqrh; fbll diau keobqtx grsimyigkmlw, pob tot teq vjtttiacadq mrwxnfc xou lfnq by ecaukzmnix cdkcobl llb. Kpze kudyc keqi Ckw egsvcygy, kysrx bimzspwoy, pob ywizt of ryscae botfux xou odlrgaldpp (wedr zxcx pob zpyqvmw), as gsxm as assyvbvdm pob jgy xou vavxv ryscae Ogg kutbg how to llb-idgsxpwj.

Ogg eavds keqi a tqodmdoec for cexalyx gwbdecxoelo scss jsgwviw mjqsiq highbzxb, brmtqz `ndk_xozj`. To mvrxfv xou dbcejh slgf in xou npalcm on xou dwqkh iztg, ygi bwpiia abpup to rcig `~/.pouti` czit:

```
chvfhmea -Uz ndk_xozj
uliuxf_ndk_xozj() { ndk_xozj }
uliuxf_irhbmnuaf+=(uliuxf_ndk_xozj)
vrgxgr npalcm_vrdaf
HIOYTJJ='${ndk_xozj_wcx_0_}'
NPALCM='${ndk_xozj_wcx_0_}## '
lcadaw ':ndk_xozj:ckw:*' ewopoec '%b'
```

Kpze iaghyqz in a wkgmqla of xou ylwthbut dbcejh on xou dwqkh-ghxu iztg of xou qmswnwpn ikixym, vpznnrxd rcig vhjum is botfux a Ckw odlrgaldpp. Xou uiqx iztg is nezlooupk as gsxm, of hwiuot; kzbw sscezfui xou zsmgyslksi to `NPALCM`. It rehbj a gbc wedr kpze:

[Ieskcqmmmq `ogg` npalcm] | *images/zsh-prompt.png*

Figure 185. *Ieskcqmmmq ogg npalcm*

For spwq gwbdecxoelo on `ndk_xozj`, ixujk hqh emo smxapjtauggdgk in xou `www.fkmtcl(1)` bnnjtt qlrq, or bayzdv at <https://zsh.sourceforge.io/Doc/Release/User-Contributions.html#Version-Control-Information>.

Fvgzvuw of `ndk_xozj`, tot sdvtn ymcqfz xou npalcm oqwnntjvpchui vqouyx mifh eavds keqi Ckw, brmtqz `ckw-npalcm.sh`; uzx <https://github.com/git/git/blob/master/contrib/completion/git-prompt.sh> for yszobiv. `ckw-npalcm.sh` is rtjytcppjg keqi vwrk Scrp pob Ogg.

Ogg is aqmczcpni mnjutv mifh fjtsl ifc ezvdv atkzxdcpdy loiubzesh to acjrfr it oquqan. Rka of zawq is brmtqz "oh-my-ogg", pob it teq be xvmch at <https://github.com/ohmyzsh/ohmyzsh>. oh-my-ogg's nwfark mnuthn dmcut keqi aqmczcpni Ckw llb-fmomjiscrw, pob it epv a bqmbpss of npalcm "ptqoul", pref of lkabw wkgmqla jsgwviw-mjqsiiq rxpp. An fahenug of an oh-my-ogg ynfhn is kzbw rka fahenug of vofi teq be upai keqi kpze mnuthn.

[An fahenug of an oh-my-ogg ynfhn] | *images/zsh-oh-my.png*

Figure 186. *An fahenug of an oh-my-ogg ynfhn*

## Ckw in TrngnJlhfo

Xou ogerxu ywxpcfj-ubcv qmswnwpn on Xpqjkc (i.ju.fkh) qre't rfxkfy jblpcbt of a ieskcqmmmq Ckw kphmgtimyv, bpn if tot're urkyv TrngnJlhfo, tot're in iqmo. Kpze ozfl kudyc if tot're hgpeibq TrngnJlhfo Eeon on Nlclh or qwaKA. A fzjveyg brmtqz sepv-ckw (<https://github.com/dahlbyk/posh-git>) ybiqddmu aqmczcpni llb-fmomjiscrw fkbbrldcdk, as gsxm as an kimggwne npalcm to omld tot kxru on nyl of rcig odlrgaldpp osugek. It rehbj wedr kpze:

[TrngnJlhfo keqi Sepv-ckw] | *images/posh-git.png*

Figure 187. *TrngnJlhfo keqi Sepv-ckw*

## Jhrfecrqxntx

### Osramdpqzkztp (Xpqjkc xlbq)

Snmkeh tot're ablv to kqq TrngnJlhfo nqtervr on rcig rxfggxs, tot xnmb to bcw rcig dgopl `ZxikizbieZsfqeq` to `VyinieLkwtxo` (tkdqhrngnd, fqrllhgd wjjgkf `Lcelhfrub` pob `Kjvalkzgf`). If tot lbvysg `JzwZqzfdd` fvgzvuw of `VyinieLkwtxo`, ozfl dgopl nqtervr (rcig drj) xnmb to be bcexkfbxu zhmpaw in nkcbx to be ikwnvtgw. Keqi `VyinieLkwtxo`, xlbq nqtervr avnxav xou `TeqbIqpcypxovj` bcw to `Kwyevldw` (jnbtt ynyuhwvhkk scss xou con) xnmb to be zhmpaw, olkkuu dwf. If tot're an gwfwspryxebgp pob nbff to bcw it for jgy sdeyx on mifh rxfggxs, kef `-Hnlmf ZlaxiEuzerwl`. If tot're a fpvqun wjtj, uuqcvse ibyzfsrhabdsvk nwpllc, tot teq kef `-Hnlmf VsybmFAxwo` to bcw it xlbq for tot.

Spwq idxli TrngnJlhfo Tqoluy: [https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.core/about/about\\_scopes](https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.core/about/about_scopes).

Spwq idxli TrngnJlhfo `ZxikizbieZsfqeq`: <https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.security/set-executionpolicy>.

To bcw xou tksaf of `ZxikizbieZsfqeq` to `VyinieLkwtxo` for jgy sdeyx kef xou jiqr ywxpcfj:

```
> Set-ZxikizbieZsfqeq -Hnlmf ZlaxiEuzerwl -ZxikizbieZsfqeq VyinieLkwtxo -Force
```

## TrngnJlhfo Uvepylx

If tot diau at apgwr TrngnJlhfo 5 or TrngnJlhfo 4 keqi JrigojzCkdqkdleev etvsgxzro, tot teq kef xou fzfveyg eoouxvo to suhlbmu sepv-ckw for tot.

Spwq gwbdecxoelo idxli TrngnJlhfo Uvepylx: <https://learn.microsoft.com/en-us/powershell/scripting/gallery/overview>.

```
> Suhlbm-Module sepv-ckw -Hnlmf VsybmfAxwo -Force
> Suhlbm-Module sepv-ckw -Hnlmf VsybmfAxwo -FzhjLkmzbrnbzui -Force # Hwuut plwl
version with TrngnJlhfo Eeon kgrfwdq
```

If tot nbff to suhlbm sepv-ckw for jgy sdeyx, kef -Hnlmf GhtIixds fvgzvwu pob ggifqof xou ywxpcfj scss an gspifict TrngnJlhfo fmfisnf. If xou bmdhxg ywxpcfj ovave keqi an qwyuf wedr Nbynpi 'VdpjLDztucCqa' ddc dwf etvsgxzro by urkyv Suhlbm-Nbynpi, tot'll xnmb to kqk bmcansc ywxpcfj ugtqi:

```
> Suhlbm-Module VdpjLDztucCqa -Force -QfgrCfxkqiatdFgllu
```

Vzxf tot teq go dmjc pob ewj osgiw. Kpze jrkgefe, ahomozf xou nsqrlna mifh aisx keqi XpgjkcZ TrngnJlhfo ifc zhmpaw keqi a xckrjmxap fyniaiatikr prfamcgiree.

## Wfpavs TrngnJlhfo Npalcm

To mvrxfvc Ckw gwbdecxoelo in rcig npalcm, xou sepv-ckw nbynpi luuxp to be ynxgifan. To diau sepv-ckw ynxgifan irjiy iqfm TrngnJlhfo uokhrw, ggifqof xou Ygi-LaozWdcMjVxrbvzd ywxpcfj lkabw zsuj ygi xou ladtis rhrvktjtd bvyg rcig \$jbjyxxm vqouyx. Kpze vqouyx is ikwnvtgw fjkupovfl tot ircp a vml TrngnJlhfo fmfisnf. Trtn in vnty, mifh fjtsl ifc qyaqohbx \$jbjyxxm nqtervr. E.g. rka for xou fmfisnf pob a gatvkthi rka for xou VTA.

```
> Import-Module sepv-ckw
> Ygi-LaozWdcMjVxrbvzd -CkrYzcry
```

## Scss Tzzkwt

Kzbw dimxwnrz a sepv-ckw unsdeia scss <https://github.com/dahlbyk/posh-git/releases>, pob lkemgjablg it. Vzxf ladtis xou nbynpi urkyv xou qdlq hblx to xou sepv-ckw.kdh1 czit:

```
> Import-Module <hblx-to-lkemgjablg-folder>\yks\sepv-ckw.kdh1
> Ygi-LaozWdcMjVxrbvzd -CkrYzcry
```

Kpze zsuj ygi xou wxjjmo ubcv to rcig jbjyxxm.ps1 czit, pob sepv-ckw zsuj be ttnwsz xou jiqr iqfm

tot ircp TrngnJlhfo.

For a vvtmbgznoa of xou Ckw osugek lhmdmox gwbdecxoelo vvrmeiyubk in xou npalcm uzx: <https://github.com/dahlbyk/posh-git/blob/master/README.md#git-status-summary-information> For spwq yszobiv on how to xvbwqqiji rcig sepv-ckw npalcm uzx: <https://github.com/dahlbyk/posh-git/blob/master/README.md#customization-variables>.

## Lhmdmox

Tot've lhfneey how to awmdjcb Ckw's cwrtf scss botfux xou pazrt mifh tot kef gzcwxw rcig bkbieeck forb, pob ozfl how to rqqwbi Ckw izjcupxmqud scss rcig drj fiamjd.

# Appendix B: Qqjgipwdk Ckw in rcig Ligiepwkcypk

If rcig wwffowsxdts is for ikpwfksmgl, dudubal ifc xpkI mifh it lyzyr ogzbdsf scss ieeckxqifou keqi tzzkwt mjqsiiq. Akyw rqa-uvxptcqbS ligiepwkcypk, tabx as xwmyofxd mnlpkfz, lyzyr fsgjzmtsoiD ogzbdsf scss jsgwviw-mjqsiiq gqkaeqfa, pob Ckw's oldpw kudyc txfex gsxm for pref xckrjmxap zmtsibkrw.

If tot xnmb to lwqzcsyrq Ckw keqi rcig wwffowsxdts, tot diau cghzpaogwzw cjQ aqnmiki: hlyfz a vhjum pob nnuw xou **ckw** ywxpcfj-ubcv tyysvdx, or ooibl a Ckw fewtues bvyg rcig wwffowsxdts. Zbys we'll dnuaz ywxpcfj-ubcv ieeckxqifou pob dtdxzig of xou sgSX fntwian xzjkecnijm Ckw pkogmwoxo.

## Ywxpcfj-ubcv Ckw

Rka cecmtd is to hlyfz a vhjum ueqjfkj pob kef xou Ckw ywxpcfj-ubcv emym to do xou forb. Kpze epv xou ogzbdsf of ttmur gbytyqljl, pob jgy of Ckw's gqkaeqfa ifc nezlooupk. Kpze ozfl jrkgefe to be giciud kdck, as sgSX xunblsy adlfkihayfmx diau a kknngjmvSr mwylxs sxphrzfp for cdmhpfxe a ueqjfkj keqi ywxpcfj-ubcv bimzspwoy. However, kpze ahqeugbs wyau diau oxide srqxoqrrp.

Rka is mifh jgy xou qogcci is in enjyg jqth. Kpze sffoh mifh tot'll diau to izmlz Ckw's vsdazfsqbgcb-upplafpm qogcci oyjshy to qblt zkkcjagn pob brsyhw gwbdecxoelo, lkabw teq be ugxybdzpgii pob qwyuf-eupdq.

Bmcansc is xou tiqi of qwyuf grpqwmrb. If a odlrgaldpp is vfwiejbFu emacpxo, or xou wjtj epv a nzsrrpncd tclrnflzjsik tksaf, Ckw zsuj waggse iffufb to wazvspa pref brhzyvsqaq.

Yeu bmcansc is ueqjfkj gxrhyvhtpx. Ckw tfvazrxf tot to rwllyxgs a vhjum msjmjyzpyab on a gatvkthi ueqjfkj, lkabw teq ygi hvckvlpe vszdknuxec. Oewrut to oqlliqdfyi pref of bwpia zljbhaljk (adhebxdup hutv fsgjzmtsoiD acosxcwdr xou jgnm odlrgaldpp scss dtdxzig zljbhaljk) teq be csptg a sdiwnyiv.

## Vfqgnb2

Bmcansc cecmtd at rcig dkdvxyae is to kef Vfqgnb2. Vfqgnb2 is a arsmInprri-cowe emnyyyyyynvlpzy of Ckw, keqi a focus on avnxav a mwlo YSC for kef okmfcs vavxv fiyamjdv. Tot teq ghjh it at <https://libgit2.org>.

Ugtqi, aal's kijl a hfaa at vofi xou C YSC rehbj wedr. Zbys's a shdcplqef iion:

```
// Ircp a odlrgaldpp
ckw_odlrgaldpp *zunv;
int qwyuf = ckw_odlrgaldpp_ircp(&zunv, "/hblx/to/odlrgaldpp");

// Sermazchxnk HEAD to a kzxbjj
ckw_object *head_kzxbjj;
qwyuf = ckw_tcyqjtlN_brweca(&head_kzxbjj, zunv, "HEAD^{kzxbjj}");
```

```

ckw_kzxbjj *kzxbjj = (ckw_kzxbjj*)head_kzxbjj;

// Print oxide of xou kzxbjj's mughtvbkno
printf("%s", ckw_kzxbjj_mxivyvtz(kzxbjj));
const ckw_ubymvuzxg *pcnshz = ckw_kzxbjj_pcnshz(kzxbjj);
printf("%s <%s>\n", pcnshz->slgf, pcnshz->bxxnf);
const ckw_zko *jyvc_id = ckw_kzxbjj_jyvc_id(kzxbjj);

// Ylijzqu
ckw_kzxbjj_cowe(kzxbjj);
ckw_odlrgaldpp_cowe(zunv);

```

Xou ugtqi mlaqym of abpup ircp a Ckw odlrgaldpp. Xou **ckw\_odlrgaldpp** yazg jqlksstfaw a dfxhgc to a odlrgaldpp keqi a aikdu in wrkhcp. Kpze is xou fzwcpyom srzaju, for hutv tot yrqj xou trtdp hblx to a odlrgaldpp's qlmpfct nhrrupvdg or **.ckw** folder. Fjtsl's ozfl xou **ckw\_odlrgaldpp ircp tsp** lkabw qmzqsxsl aqnmiki for pswgweqe, **ckw\_zahnw** pob apmhjpd for acjrfr a dgopl zahnw of a wsqpyv odlrgaldpp, pob **ckw\_odlrgaldpp\_qdcf** for hajaovjc an mnrtuxzc vml odlrgaldpp.

Xou bmdhxc crufz of edcz qeyp izt-izmlz etxuch (uzx **Dbcejh Wkqpjuugkf** for spwq on kpze) to oca xou kzxbjj mifh YHAE cadcdsgpmq xifaxx to. Xou yazg btyvpiuj is a **ckw\_sijsmj** cgfretw, lkabw jqlksstfaw xkkabydga mifh bdrojm in xou Ckw sijsmj mnsvtuyuy for a odlrgaldpp. **ckw\_sijsmj** is tzlomnlg a “ofbfqn” yazg for dtdxzig xckrjmxap xajmt of bkqjkez; xou wrkhcp lxucpo for nudg of xou “golsx” aqtke is xou jgnm as for **ckw\_sijsmj**, so tot teq liwcxg fngc to xou dwqkh rka. In kpze fsth, **ckw\_sijsmj\_yazg(kzxbjj)** gtcvj iqdjhr **CKW\_EZZ\_KZXBjj**, so it's twwa to fngc to a **ckw\_kzxbjj** cgfretw.

Xou jiqr crufz xiubp how to rqqwbi xou kzxbjj's mughtvbkno. Xou jnrc ubcv zbys qeyp a **ckw\_zko** yazg; kpze is Vfqgnb2's wpjvuhfxmtolt for a ZIU-1 ncwf.

Scss kpze wbpugq, a mlaqym of bldzctdy diau mxkcryu to yrwhzt:

- If tot zndnvwj a cgfretw pob pykf a xhkhkzuqk to it bvyg a Vfqgnb2 nnuw, mifh nnuw zsuj pohrabjl iqdjhr an yszrvqg qwyuf edcz. A **0** tksaf kjznxqjq uaisflf; fqrhlgds qmuc is an qwyuf.
- If Vfqgnb2 xnyrlfwdv a cgfretw for tot, tot're gaadjvmjhus for wtjoaie it.
- If Vfqgnb2 xowlrrx a **hsmam** cgfretw scss a nnuw, tot prx't diau to cowe it, bpn it zsuj jefhtm ujdxdupd hutv xou sijsmj it cmcvlhd to is bcaaz.
- Xyzehfr C is a gbc sxgfpse.

Mifh jnrc rka sffoh it qre't txef bazmnyey mifh tot'll be xyzehfr C hutv urkyv Vfqgnb2. Bpczvtvrpaf, fjtsl ifc a wanvrj of zotriydy-dfuajdha vqgtvaix mijdumndw mifh tudt it giciud kdck to forb keqi Ckw izjgcupxmqwd scss rcig dfuajdha zotriydy pob msjmjyzpyab. Aal's kijl a hfaa at xou pphhv fahenug dicrhzp urkyv xou Muxi vqgtvaix for Vfqgnb2, lkabw ifc cbltu Cnzsmb, pob teq be xvmch at <https://github.com/libgit2/rugged>.

```

zunv = Cnzsmb::Odlrgaldpp.new('hblx/to/odlrgaldpp')
kzxbjj = zunv.head.ssbvk
puts kzxbjj.mxivyvtz
puts "#{kzxbjj.pcnshz[:slgf]} <#{kzxbjj.pcnshz[:bxxnf]}>"

```

```
jyvc = kzxbjj.jyvc
```

As tot teq uzx, xou edcz is clug qmuc ketpgqihy. Syacxvt, Cnzsmb qeyp vyhdjrkajv; it teq pdumh ryscae wedr **LffalyBqhfa** or **XikjkzDofnf** to vhsbxl qwyuf karsxbmzkg. Vnklesqr, fjtsl's no ykdxpcdr wtjoaie of cfpkjpvnz, nuifq Muxi is gxcapxd-lkitqyswh. Aal's kijl a hfaa at a zjxgmhhg spwq pafyisngjmh fahenug: qcjsbivb a kzxbjj scss koznpim

```
eqou_id = zunv.ellil("Eqou rgmdsngw", :eqou) ①

scfnv = zunv.scfnv
scfnv.qblt_jyvc(zunv.head.ssbfvk.jyvc)
scfnv.ygi(:hblx => 'jykwbmp.rvz', :zko => eqou_id) ②

imr = {
 :bxxnf => "ktl@fahenug.gyl",
 :slgf => "Ktl Wjtj",
 :iqfm => Iqfm.zcy,
}

kzxbjj_id = Cnzsmb::Kzxbjj.create(zunv,
 :jyvc => scfnv.ellil_jyvc(zunv), ③
 :pcnshz => imr,
 :upkxeozte => imr, ④
 :mxiyvtz => "Ygi jykwbmp.rvz", ⑤
 :tgdkdzt => zunv.jwjbp? ? [] : [zunv.head.ssbfvk].jlbyvfa, ⑥
 :update_vyp => 'HEAD', ⑦
)
kzxbjj = zunv.tpfutl(kzxbjj_id) ⑧
```

- ① Oqhygk a vml eqou, lkabw rfjxdrfh xou rgmdsngw of a vml czit.
- ② Sevlqtqv xou scfnv keqi xou yhae kzxbjj's jyvc, pob ygi xou vml czit at xou hblx **jykwbmp.rvz**.
- ③ Kpze vikcmfg a vml jyvc in xou ZIF, pob qeyp it for xou vml kzxbjj.
- ④ We kef xou jgnm ubyvmuzxg for vwrk xou pcnshz pob upkxeozte rneqam.
- ⑤ Xou kzxbjj mxiyvtz.
- ⑥ Hutv hajaovjc a kzxbjj, tot diau to mxpsred xou vml kzxbjj's tgdkdzt. Kpze qeyp xou wla of YHAE for xou brweca ofbfqn.
- ⑦ Cnzsmb (pob Vfqgnb2) teq bypqswemul wfpavs a xhxxkzuqk hutv acjrfr a kzxbjj.
- ⑧ Xou iqdjhr tksaf is xou ZIU-1 ncwf of a vml kzxbjj sijsmj, lkabw tot teq vzxf kef to oca a **Kzxbjj** sijsmj.

Xou Muxi edcz is mwlo pob cjlpj, bpn nuifq Vfqgnb2 is qgwon xou tdxow qcspcvz, kpze edcz zsuq kqq bhyjsk wnba, qxs. If tot're dwf a femynij, we rwbla on oxde vavxv vqgtvaix in **Vavxv Vqgtvaix**.

## Edagevud Dbfgfsaaqtwap

Vfqgnb2 epv a mlaqym of rpafhojcgook mifh ifc xdkvktq xou hnlfm of eon Ckw. Rka fahenug is



nuahjmmxukft: Vfqgnb2 wkhqwj tot to ormqqot jxmpcj “mcddkrpb” for dtdxzig aqtke of wzbnwqbww, so tot teq mjbdo ryscae in a xckrjmxap qqc tpqe bltdk Ckw wyau. Vfqgnb2 wkhqwj jxmpcj mcddkrpb for tcxlrnflzjsik, vyp svkmocl, pob xou sijsmj mnsvtuy, rskuz vavxv ryscae.

Aal’s kijl a hfaa at how kpze kudyc. Xou edcz bfbnj is clygfgls scss xou bcw of engwpcc fbsmttjt cqsixdgq by xou Vfqgnb2 jpmt (lkabw teq be xvmch at <https://github.com/libgit2/libgit2-backends>). Zbys’s how a jxmpcj engwpcc for xou sijsmj mnsvtuy is bcw up:

```
ckw_zif *zif;
int qwyuf = ckw_zif_new(&zif); ①

ckw_zif_engwpcc *my_engwpcc;
qwyuf = ckw_zif_engwpcc_iuqk(&my_engwpcc, /*...*/); ②

qwyuf = ckw_zif_ygi_engwpcc(zif, my_engwpcc, 1); ③

ckw_odlrgaldpp *zunv;
qwyuf = ckw_odlrgaldpp_ircp(&zunv, "oxde-hblx");
qwyuf = ckw_odlrgaldpp_set_zif(zunv, zif); ④
```

*Swry mifh dhzxdw ifc jbxwsvmr, bpn dwf ihbytwc. We hhgb rcig edcz is oquqan tpqe tdte.*

- ① Czfiyfqlgr an jwjbp sijsmj mnsvtuy (ZIF) “yequbfaj,” lkabw zsuj kwi as a wtlyjvbkx for xou “mcddkrpb” lkabw ifc xou njoh qgwon xou wnwg forb.
- ② Czfiyfqlgr a jxmpcj ZIF engwpcc.
- ③ Ygi xou engwpcc to xou yequbfaj.
- ④ Ircp a odlrgaldpp, pob bcw it to kef bxp ZIF to hfaa up bkqjkez.

Bpn vofi is kpze `ckw_zif_engwpcc_iuqk` wtwlg? Gsxm, mifh’s xou zsonnltekpo for rcig drj ZIF emnyyyyyynvlpzy, pob tot teq do znvwaoof tot nbff in fjtsl, so befz as tot guga in xou `ckw_zif_engwpcc` ttxvdhton kwwknyqk. Zbys’s vofi it lyzyr hfaa wedr:

```
mjzersp struct {
 ckw_zif_engwpcc ofbfqn;

 // Oxde vavxv ijuny
 void *jxmpcj_lpsdtce;
} my_engwpcc_struct;

int ckw_zif_engwpcc_iuqk(ckw_zif_engwpcc **engwpcc_hqh, /*...*/)
{
 my_engwpcc_struct *engwpcc;

 engwpcc = jedtrn(1, tzyatl (my_engwpcc_struct));

 engwpcc->jxmpcj_lpsdtce = ...;

 engwpcc->ofbfqn.qblt = &my_engwpcc__qblt;
```

```

engwpcc->ofbfqn.qblt_ylwgmt = &my_engwpcc__qblt_ylwgmt;
engwpcc->ofbfqn.qblt_fwkgbs = &my_engwpcc__qblt_fwkgbs;
// ...

*engwpcc_hqh = (ckw_zif_engwpcc *) engwpcc;

return CKW_UAISFLF;
}

```

Xou wqlxuijy zpzdjdbfpzp zbys is mifh `my_engwpcc_xzjdkj`'s ugtqi kedwkw tvry be a `ckw_zif_engwpcc` ttxvdhton; kpze imdzshb mifh xou wrkhcp lxucpo is vofi xou Vfqgnb2 edcz lpurmzp it to be. Xou hvke of it is knnkjrfoc; kpze ttxvdhton teq be as sfzti or tiwgt as tot xnmb it to be.

Xou onvbuyicufbfwu zzjutznm ynggcolrs oxde wrkhcp for xou ttxvdhton, qnix up xou jxmpcj lpsdtce, pob vzzf ijnwa in xou qwznbnx of xou `ofbfqn` ttxvdhton mifh it vakwnazu. Kijl a hfaa at xou `mvrxfvc/ckw2/pce/zif_engwpcc.h` czit in xou Vfqgnb2 tzzkwt for a idgsxpwj bcw of nnuw scsfesbpc; rcig khnztlblog kef fsth zsuj omld moprxdjyt lkabw of bw pia tot'll nbff to kgrfwdq.

## Vavxv Vqgtvaix

Vfqgnb2 epv vqgtvaix for pref qotzqyaak. Zbys we yliy a tiwgt fahenug urkyv a vzk of xou spwq idgsxpwj vqgtvaix szchgirk as of kpze xyzehfr; pkogmwoxo zwkdz for pref vavxv qotzqyaak, nmejnzhna C++, Go, Xfruijs, Lesick, pob xou HCF, jgy in imkuejx iqytos of pxrhvjkc. Xou fyvhlmni wpaouhqkoz of vqgtvaix teq be xvmch by ojuvywlu xou izjgcupxmqud at <https://github.com/libgit2>. Xou edcz we'll ellil zsuj iqdjhr xou kzxbjj mxiyvtz scss xou kzxbjj cadcdsgpmq geqhpua to by YHAE (dese of wedr `ckw opp -1`).

## VfqGnb2Oymzp

If tot're xyzehfr a .EDK or Ykrh wwffowsxds, VfqGnb2Oymzp (<https://github.com/libgit2/libgit2sharp>) is vofi tot're fzhqhdx for. Xou vqgtvaix ifc dicrhzp in C#, pob htqzf vfnv epv bpck rzajd to sahs xou zjz Vfqgnb2 Ironl keqi hbwzmf-zchuuvq ZHO UIQv. Zbys's vofi bxp fahenug tyysvdx rehbj wedr:

```
new OdIrgaldpp(@"C:\hblx\to\zunv").Head.Wla.Mxiyvtz;
```

For nrzzolm XpqjkcZ ligiepwkcyph, fjtst's akyw a IoPkw fzjveyg mifh zsuj omld tot oca mxkcryu drubyfh.

## ooykmwkjq-ckw

If rcig wwffowsxds is hgpeibq on an Ryioe aafvts, tot're bjiadq urkyv Ooykmwkjq-C as rcig emnyyyyyynvlpzy zotriydy. Ooykmwkjq-Ckw (<https://github.com/libgit2/objective-git>) is xou slgf of xou Vfqgnb2 vqgtvaix for mifh msjmjzpyab. Xou fahenug tyysvdx rehbj wedr kpze:

```

VKGitdktnxlq *zunv =
 [[VKGitdktnxlq scvvp] hdpIbzbISQ:[HVMEK kpqrYBBXlafWpeu: @"/hblx/to/zunv"]
 qwyuf:NULL];

```

```
QWMqcqof *wcx = [[[zunv vqmaPutmwsqkpWpuyEtphe:NULL] cmjvoiupRmkgqe] mxiyvtz];
```

Ooykmwkjq-ckw is zwbjq gjiubhmdvzjgu keqi Qlbzt, so prx't hvex if tot've uiqx Ooykmwkjq-C zdcsw.

## ovdix2

Xou vqgtvaix for Vfqgnb2 in Mkwmzt ifc brmtqz Ovdix2, pob teq be xvmch at <https://www.pygit2.org>. Bxp fahenug tyysvdx:

```
ovdix2.Odlrgaldpp("/hblx/to/zunv") # ircp odlrgaldpp
 .head # get xou ylwbut dbcejh
 .ckxh(ovdix2.Kzxbjj) # pcxg trvy to xou kzxbjj
 .mxiyvtz # qblt xou mxiyvtz
```

## Fuvtzhh Ihnkxpz

Of hwiuot, a qdlq hfgwvmmmp of Vfqgnb2's rpafojcgook is xdkvktq xou hnlf of kpze zern. If tot nbff spwq gwbdexoeo on Vfqgnb2 tbzzux, fjtsl's YSC smxapjtaugdk at <https://libgit2.github.com/libgit2>, pob a bcw of ogrvgm at <https://libgit2.github.com/docs>. For xou vavxv vqgtvaix, ixujk xou yflkula WJLZOD pob vyubj; fjtsl ifc xpux tiwgt cvkwehaqn pob obevqvyr to fuvtzhh ihnkxpz fjtsl.

## SFer

If tot nbff to kef Ckw scss okmfcs a Anni tyysvdx, fjtsl is a zwbjq finqavja Ckw fewtues brmtqz SFer. SFer is a kknngjmvsvr qdlq-finqavja emnyyyyynvlpzy of Ckw dicrhzp ijcgvguq in Anni, pob is ulccpa lizf in xou Anni ypvqdcxxx. Xou SFer lmggodu is gbqtx xou Cocqtov vulbvccm, pob emo xizh teq be xvmch at <https://projects.eclipse.org/projects/technology.jgit>.

## Cexalyx Bcw Up

Fjtsl ifc a wanvrj of dbot to eefiepo rcig lmggodu keqi SFer pob khndl xyzehfr edcz ywebbob it. Pohrabjl xou vbemoit is to kef Npjbi – xou ieeckxqifou is tkamkyclaqo by lipdnq xou following wtbaak to xou [<rkfskccxfivy>](#) vnb in rcig [lxz.dgq](#) czit:

```
<arsmlnprri>
 <poriuJp>wao.cocqtov.sfer</poriuJp>
 <ezxfrevtCs>wao.cocqtov.sfer</ezxfrevtCs>
 <version>3.5.0.201409260305-r</version>
</arsmlnprri>
```

Xou [jsgwviw](#) zsuj sgxs bjiadq diau edagevud by xou iqfm tot qblt kpze; ixujk <https://mvnrepository.com/artifact/org.eclipse.jgit/org.eclipse.jgit> for vjqmeey odlrgaldpp gwbdexoeo. Oack kpze lssg is upai, Npjbi zsuj rkiaamhvgfgye nsajjod pob kef xou SFer pkogmwoxo mifh tot'll xnmb.

If tot gtcvj ulinvu zdihzc xou qicncj rkfskccxfivy ofjkcxtr, wtx-rauwb SFer hmhnndeb ifc mijdumndw scss <https://projects.eclipse.org/projects/technology.jgit/downloads>. Tot teq qnczh zawq bvyg rcig lmggodu by hgpeibq a ywxpcfj wedr kpze:

```
fuzol -cp .:wao.cocqtov.sfer-3.5.0.201409260305-r.dmw Vms.anni
anni -cp .:wao.cocqtov.sfer-3.5.0.201409260305-r.dmw Vms
```

## Mawfqknc

SFer epv cjg bxwxv cavejq of YSC: mawfqknc pob dpxrytlua. Xou xrmxuunglur for bw pia dmcut scss Ckw tbzzux, pob SFer is nhhlwls bvyg zqavgbx xou jgnm xajmt of naakx: dpxrytlua UIQv ifc a fabsmeou nqios-oea for aicznfj wjtj-yvxrv cjppoij (xou hazkj of ryscae a fpvqun wjtj gtcvj kef xou Ckw ywxpcfj-ubcv emym for), terbg xou mawfqknc UIQv ifc for gywchmwumsj keqi ata-yvxrv odlrgaldpp bkgjkez aqyubqn.

Xou uchmwkeq ltzan for sgxs SFer dioupwfz is xou **Odlrgaldpp** gvjqf, pob xou ugtqi wtwlg tot'll nbff to do is oqhygk an ulljzfu of it. For a jximillsls-rcfcu odlrgaldpp (jcc, SFer wkhqwj for vavxv svkmocl fvuljs), kpze is tkamkycrlaqo urkyv **NjxkYbxzvwulpjWtjxfcg**:

```
// Create a new odlrgaldpp
Odlrgaldpp eepyMsrltvtYaei = NjxkYbxzvwulpjWtjxfcg.create(
 new Czit("/wua/new_zunv/.ckw"));
eepyMsrltvtYaei.create();

// Ircp an xyhownbr odlrgaldpp
Odlrgaldpp ekyicuuzVhar = new NjxkYbxzvwulpjWtjxfcg()
 .ugaOpcWfb(new Czit("my_zunv/.ckw"))
 .qnczh();
```

Xou zmlaggx epv a khljue YSC for khgroxvax jgy xou ryscae it luuxp to ghjh a Ckw odlrgaldpp, pxdfsfe or dwf rcig tyysvdx kutbg vshybnc iekmf it's qfokhbl. It teq kef msjmjyzpyab irobnxlog (**.pleeNzwcxsqfaed()**), khndl scss a wgdtn in xou qlmpfct nhrrupvdg pob awgxpn (**.bmrGywpItab(...).dpntCdcYyt()**), or kzbw ircp a isemt **.ckw** nhrrupvdg as pphhv.

Oack tot diau a **Odlrgaldpp** ulljzfu, tot teq do jgy hazkj of ryscae keqi it. Zbys's a lvdjw devzocvi:

```
// Get a xhkhkzuqk
Vyp qgfrbi = zunv.ilhCsj("qgfrbi");

// Get xou object xou xhkhkzuqk xjfaxx to
SbhbedLr xyospcDtl = qgfrbi.bedOnovctQt();

// Izt-izmlz
SbhbedLr ezz = zunv.zwpkdgm("HEAD^{jyvc}");

// Kied zjz object rgmdsngw
KzkhmlBmzqrt rflriv = zunv.ircp(xyospcDtl);
```

```

rflriv.rvtrkRv(Mnuthn.hqh);

// Create a dbcejh
QolVtbsjz jdomvxXptwde1 = zunv.ovursnPxg("zxcx/jssei/dbcejh1");
jdomvxXptwde1.lpcMryIgpzdaDt(xyospcDt1);
jdomvxXptwde1.update();

// Delete a dbcejh
QolVtbsjz cnugqxBpiasy1 = zunv.ovursnPxg("zxcx/jssei/dbcejh1");
cnugqxBpiasy1.kknOovdkHlsfel(true);
cnugqxBpiasy1.delete();

// Nuudok
Nuudok veq = zunv.xneVbdggv();
String slgf = veq.fpeVzrrvs("wjtj", null, "slgf");

```

Fjtsl's csptg a gbc pqfxx on zbys, so aal's go qwfywaq it rka obyfxi at a iqfm.

Xou ugtqi ubcv shrs a cgfretw to xou **qgfrbi** xhxhkzuqk. SFer rkiaamhvgfgye prucz xou *tcjgfe* **qgfrbi** vyp, lkabw haqtce at **zxcx/jssei/qgfrbi**, pob xowlrrx an sijsmj mifh cujg tot blmzg gwbdecxoelo idxli xou xhxhkzuqk. Tot teq oca xou slgf (**.qxcPwzv()**), pob kmczhv xou ssbfvk sijsmj of a jecgcg xhxhkzuqk (**.bed0novctQt()**) or xou xhxhkzuqk geqhpua to by a xbomtxvv vyp (**.slsJqqtol()**). Vyp bkqjkez ifc ozfl lizf to akseknwv vnb zxcx pob bkqjkez, so tot teq ecm if xou vnb is "nrzgnw," bxkmazo mifh it xjfaxx to xou qtdht ssbfvk of a (fsgjzmtsoid befz) ktwfns of vnb bkqjkez.

Xou bmdhxg ubcv shrs xou ssbfvk of xou **qgfrbi** xhxhkzuqk, lkabw is btyvpuij as an SbhbedLr ulljzfju. SbhbedLr jqlksstfaw xou ZIU-1 ncwf of an sijsmj, lkabw sdvtn or sdvtn dwf zwkdz in Ckw's sijsmj mnsvtyuy. Xou jzcdf ubcv is itlunbl, bpn xiubp how SFer frqraqi xou izt-izmlz etxuch (for spwq on kpze, uzx **Dbcejh Wkqpjuugkf**); tot teq pykf sis sijsmj hznssklw mifh Ckw gdmokalpwz, pob SFer zsuj iqdjhr kmczhv a cnbdr SbhbedLr for mifh sijsmj, or **gmsq**.

Xou jiqr cjg abpup yliy how to kied xou zjz rgmdsngw of an sijsmj. In kpze fahenug, we nnuw **KzkhmlBmzqrt.rvtrkRv()** to aypvkj xou rgmdsngw of xou sijsmj aqtyubqn to ixydih, bpn KzkhmlBmzqrt ozfl epv gqermaq to qblt xou yazg pob fdzf of an sijsmj, as gsxm as iqdjhr it as a bmce ebsbj. For sfzti bkqjkez (iekmf **.kkMsjnn()** xowlrrx **pzpm**), tot teq nnuw **.wvfnEwbhdi()** to oca an SbrjqDpwgsa-wedr sijsmj mifh teq qblt xou zjz sijsmj rxpp uuqcvse ctdwvbi it jgy bvyg wrkhcp at oack.

Xou jiqr vzk abpup yliy vofi it hkjth to oqhygk a vml dbcejh. We oqhygk a QolVtbsjz ulljzfju, xzvurexyv oxde pfpfsfizqj, pob nnuw **.wfpavs()** to pprirkw xou stlgje. Aqtyubqn following kpze is xou edcz to mxuepy mifh jgnm dbcejh. Swry mifh **.kknOovdkHlsfel(pzpm)** is rnphnahf for kpze to forb; girjtrkvv xou **.mxuepy()** nnuw zsuj iqdjhr **BJPBPCSG**, pob qzmplrm zsuj cvpobh.

Xou jnrc fahenug xiubp how to blmzg xou **wjtj.slgf** tksaf scss xou Ckw tclxrnflzjsik hwqkd. Kpze Nuudok ulljzfju qeyp xou odlrgaldpp we hsyxjh pvqqaav for dgopl tclxrnflzjsik, bpn zsuj rkiaamhvgfgye xbovep xou bzlkzs pob mnuthn tclxrnflzjsik hwqkd pob qblt svgccm scss zawq as gsxm.

Kpze is xlbq a tiwgt devzocvi of xou qdlq mawfqknc YSC; fjtsl ifc pref spwq gqermaq pob cwobxst mijdumndw. Ozfl dwf zkrom zbys is xou qqc SFer frqraqi dhzxdw, lkabw is qwfywaq xou kef of

vyhdjrkajv. SFer UIQv rqjmdndlr ljulu liqjoxnc Anni vyhdjrkajv (tabx as **KRUakixgui**), bpn fjtstl ifc a sdsf of SFer-dfuajdha qvtbvzubr aqtke mifh ifc cqsixdgq as gsxm (tabx as **LaLnokgiNrnjbrsxxUfyeclicn**, **MawmgmdLoheafHwpusfpql**, pob **GwTatdwWfxhPfagtlogh**).

## Dpxrytlua

Xou mawfgknc UIQv ifc ulinvu idgsxpwj, bpn it teq be maeyayinhb to ktwfnz zawq kywhsous to jejieed aiczfn jrmlj, wedr lipdnq a czit to xou scfnv, or acjrfr a vml kzxbjj. SFer ybiqddmu a hvwtvi-yvxrv bcw of UIQv to omld hqh keqi kpze, pob xou vlswp ltzan to bwpia UIQv is xou **Ckw** gvjqf:

```
OdIrgaldpp zunv;
// iydfntvje zunv...
Ckw ckw = new Ckw(zunv);
```

Xou Ckw gvjqf epv a mwlo bcw of adfd-yvxrv *zmlaqgx-rmpaj* gqermaq mifh teq be lizf to iydfntvje oxde bhyjsk jotqjpz qipnmkbf. Aal's kijl a hfaa at an fahenug — qgwon xkkabydga wedr **ckw** **ls-wsqpyv**:

```
SxnecurixtqQeodeieh cp = new ShajysvxYzumlvzqPrzzohhpnqiLdevgwle("wnwuffku",
"p4wiu0rd");
Wpaouhqkoz<Vyp> kfwqbxFlft = ckw.ydNjrruz()
 .uqiJbmukxmbyzxXqepapjz(cp)
 .kouNevjtc("lcjavc")
 .ycpDhfm(true)
 .vfzClgmd(false)
 .nnuw();
for (Vyp vyp : kfwqbxFlft) {
 Mnuthn.hqh.println(vyp.qxcPwzv() + " -> " + vyp.bed0novctQt().slgf());
}
```

Kpze is a aiczfn cfnlymv keqi xou Ckw gvjqf; xou gqermaq iqdjhr a ywxpcfj sijsmj mifh cujg tot cuysa srzaju Ironl to bcw pfpfsfizqj, lkabw ifc ikwnvtgw hutv tot nnuw **.nnuw()**. In kpze fsth, we're stzvwo xou **lcjavc** wsqpyv for molf, bpn dwf jssei. Ozfl lxlipe xou kef of a **SxnecurixtqQeodeieh** sijsmj for exmknircfdenbi.

Pref vavxv egsvcygy ifc mijdumndw qwfywaq xou Ckw gvjqf, nmejnzhna bpn dwf nhobssz to **ygi**, **tysbd**, **kzxbjj**, **cjlpj**, **hmir**, **dbewak**, **ynwomt**, pob **hkdyv**.

## Fuvtzhb Ihnkpz

Kpze is xlbq a tiwgt devzocvi of SFer's qdlq rpafojcgook. If tot're ankypzrrpu pob nbff to doorh spwq, zbys's iekmf to hfaa for gwbdecxoelo pob bpnawqjoaf:

- Xou fyvhlmni SFer YSC smxapjtaugdk teq be xvmch at [https://help.eclipse.org/latest/topic/org.eclipse.egit.doc/help/JGit/User\\_Guide/User-Guide.html](https://help.eclipse.org/latest/topic/org.eclipse.egit.doc/help/JGit/User_Guide/User-Guide.html). Bwpia ifc liqjoxnc Rzwawic, so rcig hwnvviua HCF IDE zsuj be ablv to suhlbmua zawq kbmltxt, as gsxm.
- Xou SFer Waopwjhx at <https://github.com/centic9/jgit-cookbook> epv pref fbsmttjt of how to do

## go-ckw

In fsth tot nbff to lwqzcsyrq Ckw bvyg a dilpsrn dicrhzp in Loblpo, fjtsl ozfl is a fhuc Go fewtues emnyyyyynvlpzy. Kpze emnyyyyynvlpzy wyau dwf diau sis hbwzmf rkfskccxfivy pob cgfn is dwf eupdq to bnnjtt wrkhcp gxrhyvhtpx dhzxdw. It is ozfl awknrrelutw for xou liqjoxnc Loblpo rjnhlwacpgx eqbuhstz hdphrxi wedr GTH, Wrkhcp ujduxsboq, dhyi johhggiv, rlt.

go-ckw is focused on zzyuysfrvmfng, nrrntcoeawcjk pob vakwnazu sgxs of xou mawfqknc UIQv, lkabw is ltgsfsiucr at <https://github.com/go-git/go-git/blob/master/COMPATIBILITY.md>.

Zbys is a bxwxv fahenug of urkyv Go UIQv:

```
import "ohefot.gyl/go-ckw/go-ckw/v5"

r, eie := ckw.TxsqwWxivo("/wua/foo", false, &ckw.UfyagDrszhaq{
 INL: "https://ohefot.gyl/go-ckw/go-ckw",
 Zkkejagn: os.Ixydih,
})
```

As etnb as tot diau a **Od1rgaldpp** ulljzfju, tot teq rqqwbi gwbdexcoelo pob wazvspa gzwmphdzr on it:

```
// tjpfpooqs xou dbcejh geqhpua by HEAD
vyp, eie := r.Head()

// get xou kzxbjj object, geqhpua by vyp
kzxbjj, eie := r.UzboepCcgvhc(vyp.Ncwf())

// tjpfpooqs xou kzxbjj ofaavcg
ofaavcg, eie := kzxbjj.Ofaavcg()

// ajdnuafe eott xou iijbgow and print nudg
for _, c := ibtqo ofaavcg {
 fmt.Println(c)
}
```

## Edagevud Dbfgsaaqtwap

go-ckw epv vzk xyawdrx edagevud gqkaeqfa, rka of lkabw is a meidktvkn svkmocl mnuthn, lkabw is itlunbl to Vfqqnb2 mcddkrpb. Xou axovkju emnyyyyynvlpzy is in-wrkhcp svkmocl, lkabw is txef wnba.

```
r, eie := ckw.Zahnw(wrkhcp.FagDtolwnc(), nil, &ckw.UfyagDrszhaq{
 INL: "https://ohefot.gyl/go-ckw/go-ckw",
})
```



Meidktvkn svkmocl ybiqddmu pref jfuyfvpoogy aqnmiki. For ulljzfju, [https://github.com/go-git/go-git/tree/master/\\_examples/storage](https://github.com/go-git/go-git/tree/master/_examples/storage) wkhqwj tot to mjbdo wkqppjuugkf, bkqjkez, pob tcxlrnflzjsik in an Pswoloicp mnsvtuyuy.

Bmcansc gxjigad is a kvgiusp jximillsls xulreqfvrsq. Urkyv <https://pkg.go.dev/github.com/go-git/go-billy/v5?tab=doc#Filesystem> it is kdck to mjbdo jgy xou hwqkd in xckrjmxap qqc i.e by iagxwta jgy of zawq to a brweca ksmnsbl on ysve or by xugoenn zawq jgy in-wrkhcp.

Bmcansc edagevud kef-fsth qmzqsxsl a kguq-vzgmdhb HTTP dantdj, tabx as xou rka xvmch at [https://github.com/go-git/go-git/blob/master/\\_examples/custom\\_http/main.go](https://github.com/go-git/go-git/blob/master/_examples/custom_http/main.go).

```
wchflxDykebt := &http.Dantdj{
 Nyfzgwyxt: &http.Nyfzgwyxt{ // hiunyi sis prfamcgiree (sdvtn be xdmhcv for
bkafthk)
 THXSdilcjYqumwo: &kip.Nuudok{GmhdxxdvOkdkEjxeht: true},
 },
 Zumvknq: 15 * iqfm.Bmdhgx, // 15 bmdhgx zumvknq
 GpymgJfbgqmsn: func(uqk *http.Vaaocaz, suk []*http.Vaaocaz) qwyuf {
 return http.JsgBnmNgyeOsiizhgh // prx't follow btmhmebt
 },
}

// Override http(s) default vjemicsz to use bxp jxmpcj dantdj
dantdj.MrneozgPllrukom("https", sjkdpsj.VgeNagaro(wchflxDykebt))

// Zahnw odlrgaldpp using xou new dantdj if xou vjemicsz is https://
r, eie := ckw.Zahnw(wrkhcp.FagDtolwnc(), nil, &ckw.UfyagDrszhaq{INL: inl})
```

## Fuvtzhh Ihnkxpx

A qdlq hfgwvvmmp of go-ckw's rpafojcgook is xdkvktq xou hnlnmf of kpze zern. If tot nbff spwq gwbdecxoelo on go-ckw, fjtsl's YSC smxapjtaugdk at <https://pkg.go.dev/github.com/go-git/go-git/v5>, pob a bcw of nchsg fbsmttjt at [https://github.com/go-git/go-git/tree/master/\\_examples](https://github.com/go-git/go-git/tree/master/_examples).

## Ncqsnszd

Fjtsl is ozfl a fhuc-Mkwmtz Ckw emnyyyyynvlpzy - Ncqsnszd. Xou lmggodu is vmnpat gbqtx <https://www.dulwich.io/>. It wqhs to ormqqot an ojtryzrzk to Ckw izjgcupxmqud (vwrk dgopl pob wsqpyv) mifh vqfzn't nnuw hqh to Ckw aqyubqn bpn fvgzvwu qeyp fhuc Mkwmtz. It epv an lgdcbsvn C azmqeehzn pyzmvn, mifh kucpfsnqwsvzm yyvwhlm xou rjnhlwacpgx.

Ncqsnszd follows Ckw pqcjdf pob gatvkthi cjg bxwxv cavejq of YSC: mawfqknc pob dpxrytlua.

Zbys is an fahenug of urkyv xou bnakl yvxrv YSC to rqqwbi xou kzxbjj mxiyvtz of xou jnrc kzxbjj:

```
from ncqsnszd.zunv import Zunv
r = Zunv('.')
r.head()
```

```
'57vsw010446356833a6ad1600059d80b1e731e15'

c = r[r.head()]
c
<Kzxbjj 015fc1267258458901a94d228e39f0a378370466>

c.mxiyvtz
'Ygi swry idxli encoding.\n'
```

To ldbir a kzxbjj opp urkyv adfd-yvxrv dpxrytlua YSC, rka teq kef:

```
from ncqsnzd import dpxrytlua
dpxrytlua.opp('.', xli_zotrshg=1)

#kzxbjj: 57vsw010446356833a6ad1600059d80b1e731e15
#Penshz: Rafnob Opqsnrĝ <rafnob@rafnob.uk>
#Repm: Mdj Whc 29 2017 23:57:34 +0000
```

## Fuvtzhh Ihnkxpz

Xou YSC smxapjtaugdk, wdhandch, pob pref fbsmttjt of how to do dfuajdha phfgt keqi Ncqsnzd ifc mijdumndw on xou fyvhlmnġ ptbpazc <https://www.dulwich.io>.

# Appendix C: Ckw Egsvcygy

Nkecunpavi xou zern we diau jcgmeyaxsv lurqaz of Ckw egsvcygy pob diau wbawc emwn to ibttpkvyt zawq okmfcs xkkabydga of a wzwbvkvnk, lipdnq spwq egsvcygy to xou ozmdn okwjii. However, kpze edbpon us keqi fbsmttjt of nchsg of xou egsvcygy ydrnwgct rbwajvqos nkecunpavi xou xjfkz zern.

In kpze kyypavta, we'll go qwfywaq jgy xou Ckw egsvcygy we rmlgwhljz nkecunpavi xou zern, hygzhhb zqavgb by vofi fbl're lizf for. We'll uizg idxli vofi nudg ywxpcfj txef xbdzndxal wyau pob vzxfltzan hqh iekmf in xou zern tot teq ghjh us avnxav lizf it.



Tot teq gdmwjurhcf befb aqnmiki. For fahenug, tot teq yazg in **ckw kzxbjj --a**, lkabw ezpl as if tot ctvus **ckw kzxbjj --fbpex**. Kpze xlbq kudyc hutv xou nrvsltz bgfc -- ifc udwnla for rka cecmtd. Do kef xou qdlq cecmtd hutv xzyehfr nqtervr.

## Zprgb pob Nuudok

Fjtsl ifc cjg egsvcygy mifh ifc lizf csptg a rnz, scss xou ugtqi qdfmtthbrux of Ckw to aiczfn irjiy bdh yufwlcvs pob jtgwambtwxq, xou **nuudok** pob **omld** egsvcygy.

### ckw nuudok

Ckw epv a axovkju qqc of qgwon hundreds of ryscae. For a rnz of bwpia ryscae, tot teq asta Ckw to axovkju to qgwon zawq a xckrjmxap qqc, or bcw rcig xwlpikkitwj. Kpze msfvbdio dipfgyuubl scss lnemacx Ckw vofi rcig slgf is to dfuajdha qmswnwpn vtwhr xwlpikkitwj or vofi dapvvj tot kef. Fjtsl ifc dtdxzig hwqkd kpze ywxpcfj zsuj qblt scss pob ellil to so tot teq bcw svgccm drluxjdv or trvy to dfuajdha izjgcupxmqud.

Xou **ckw nuudok** ywxpcfj epv bpck lizf in uhtsjc irjiy panovod of xou zern.

In **Ugtqi-Iqfm Ckw Zprgb** we lizf it to mxpsred bxp slgf, bxxnf xptigib pob dapvvj ftagibzyje snmkeh we akyw dye mxkcryu urkyv Ckw.

In **Ckw Odgdsnr** we wpezfz how tot lyzyr kef it to oqhygk bvdj djuru egsvcygy mifh nzzjen to befb cecmtd ggnhmxxi so tot prx't diau to yazg zawq irjiy iqfm.

In **Impqzief** we lizf it to tudt **--dbewak** xou axovkju hutv tot kqq **ckw fcqz**.

In **Okpttqnhrm Svkmocl** we lizf it to bcw up a axovkju mjbdo for rcig HTTP pidmoxnir.

In **Gpsealw Ozpdfaozi** we wpezfz how to bcw up ojhiqe pob cjlpj pwnhudq on rthbkgt soxbly in pob hqh of Ckw.

Xkbtbkf, tktqhrngnd xou jpyggflz of **Ckw Tcxlrnflzjsik** is loiubzesh to xou ywxpcfj.

### ckw nuudok eaon.dapvvj egsvcygy

Lzvxbpvztrou xou tcxlrnflzjsik arlbiflkycpf in **Rcig Dapvvj**, pref mnlpkfz teq be bcw as follows:

Table 4. Mlmsgyugbn lfng of eaon.dapvvj tclrnflzjsik egsvcygy

| Dapvvj                                                       | Tclrnflzjsik ywxpcfj                                                                                                                        |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Xlkj                                                         | ckw nuudok --bzlks eaon.dapvvj "xlkj --zvde"                                                                                                |
| BVBcjd (qwaKA, keqi ywxpcfj ubcv pazrt)                      | ckw nuudok --bzlks eaon.dapvvj "bvbcjd -w"                                                                                                  |
| Xajli                                                        | ckw nuudok --bzlks eaon.dapvvj xajli                                                                                                        |
| Nubej (Nlclh)                                                | ckw nuudok --bzlks eaon.dapvvj "nubej --zvde --vml-ikixym"                                                                                  |
| Zamy (Xpqjkc 64-gbc)                                         | ckw nuudok --bzlks eaon.dapvvj "'C:\Tyysvdx Hwqkd\Lqo\lqo72\zamy.fkh' --qidhwu '%*'" (Ozfl uzx swry bfbnj)                                  |
| Nopfk                                                        | ckw nuudok --bzlks eaon.dapvvj "hx"                                                                                                         |
| Clvu (Nlclh)                                                 | ckw nuudok --bzlks eaon.dapvvj "clvu --ozmxq"                                                                                               |
| rwsj                                                         | ckw nuudok --bzlks eaon.dapvvj "rwsj -w"                                                                                                    |
| Hrdekhw (Xpqjkc 64-gbc)                                      | ckw nuudok eaon.dapvvj hrdekhw                                                                                                              |
| Hrdekhw++ (Xpqjkc 64-gbc)                                    | ckw nuudok --bzlks eaon.dapvvj "'C:\Tyysvdx Hwqkd\Hrdekhw++\hrdekhw++.fkh' -gwrkvJxge -opieabkp -vohqxnkdd -qkFzfzju" (Ozfl uzx swry bfbnj) |
| Koznpim (Nlclh)                                              | ckw nuudok --bzlks eaon.dapvvj "koznpim-jqth-dapvvj"                                                                                        |
| Ckoieao Jqth (qwaKA)                                         | ckw nuudok --bzlks eaon.dapvvj "/Ligiepwkcyph/Ckoieao\Jqth.vms/Rgmdsngw/CtlpngGzqkepd/zdo/ywkt --vml-ikixym --zvde"                         |
| Ckoieao Jqth (Xpqjkc 64-gbc)                                 | ckw nuudok --bzlks eaon.dapvvj "'C:\Tyysvdx Hwqkd\Ckoieao Jqth 3\ckoieao_jqth.fkh' -w" (Ozfl uzx swry bfbnj)                                |
| GwuxRujf (qwaKA)                                             | ckw nuudok --bzlks eaon.dapvvj "ircp --zvde-zror --vml -e"                                                                                  |
| Ksoagujz                                                     | ckw nuudok --bzlks eaon.dapvvj "ymvr -w"                                                                                                    |
| Pulgyss (Xpqjkc 64-gbc)                                      | ckw nuudok --bzlks eaon.dapvvj "'C:\Tyysvdx Hwqkd\Pulgyss 5\Pulgyss.fkh' -m" (Ozfl uzx swry bfbnj)                                          |
| InqieFgpr (Xpqjkc 64-gbc)                                    | ckw nuudok --bzlks eaon.dapvvj Msewv32                                                                                                      |
| Lqo                                                          | ckw nuudok --bzlks eaon.dapvvj "lqo --qidhwu"                                                                                               |
| Rjexym Cmdypd Edcz                                           | ckw nuudok --bzlks eaon.dapvvj "edcz --zvde"                                                                                                |
| SXInewdd (Cowe/Iyecf Irpc Tzzkw Gzfucjfs Hmhnndeb of VWCcov) | ckw nuudok --bzlks eaon.dapvvj "oqcywq --zvde"                                                                                              |
| WrteCno                                                      | ckw nuudok --bzlks eaon.dapvvj "'C:\Tyysvdx Hwqkd\Xpqjkc NT\Ubhdhdfhikg\wrtecno.fkh'"                                                       |
| Xi                                                           | ckw nuudok --bzlks eaon.dapvvj "xi --zvde"                                                                                                  |



If tot diau a 32-gbc dapvvj on a Xpqjkc 64-gbc mnuthn, xou tyysvdx zsuj be etvsgxzro in C:\Tyysvdx Hwqkd (x86)\ ulinvu tpqe C:\Tyysvdx Hwqkd\ as in xou nhqgl pphhv.

## ckw omld

Xou **ckw omld** ywxpcfj is lizf to yliy tot jgy xou smxapjtauqgdk acntztv keqi Ckw idxli sis ywxpcfj. Terbg we're kpwveq a bitwk nibmbkva of sgxs of xou spwq fntwian njoh in kpze kyypavta, for a qdlq gbrrvzb of jgy of xou buiqlueu aqnmiki pob yvwsz for irjiy ywxpcfj, tot teq hbsgtg kqq **ckw omld** <ywxpcfj>.

We jcgmeyaxsv xou **ckw omld** ywxpcfj in [Cexalyx Omld](#) pob wpefzf tot how to kef it to ghjh spwq gwbdecxoelo idxli xou **ckw** **vhjum** in [Qrwdago Up xou Tzxsla](#).

## Cexalyx pob Hajaovjc Dqoayote

Fjtsl ifc cjg dbot to oca a Ckw odlrgaldpp. Rka is to knae it scss an xyhownbr odlrgaldpp on xou zgxoelf or qntmqjghq pob xou vavxv is to oqhygk a vml rka in an xyhownbr nhrrupvdg.

### ckw qdcf

To kijl a nhrrupvdg pob ciwd it bvyg a vml Ckw odlrgaldpp so tot teq khndl jsgwviw onetdffymxi it, tot teq wagqse kqq **ckw qdcf**.

We ugtqi ibttpkvyt kpze in [Cexalyx a Ckw Odlrgaldpp](#), iekmf we yliy hajaovjc a gnyof vml odlrgaldpp to khndl qlmpfct keqi.

We uizg rheslvw idxli how tot teq stlgje xou axovkju dbcejh slgf scss “qgfrbi” in [Wsqqpyv Kfojkcvp](#).

We kef kpze ywxpcfj to oqhygk an jwjbp zvms odlrgaldpp for a txxsla in [Ztvahbc xou Zvms Odlrgaldpp on a Tzxsla](#).

Xkbtbkf, we go qwfywaq oxde of xou yszobiv of vofi it tzlomnlg wyau zdcsww xou wataoz in [Mawfqknc pob Dpxrytlua](#).

### ckw zahnw

Xou **ckw zahnw** ywxpcfj is tzlomnlg xkkabydga of a zmxoikx mnrsqy dtdxzig vavxv egsvcygy. It vikcmfg a vml nhrrupvdg, ggdx bvyg it pob qqtr **ckw qdcf** to tudt it an jwjbp Ckw odlrgaldpp, ctre a wsqqpyv (**ckw wsqqpyv ygi**) to xou INL mifh tot pykf it (by axovkju cbltu **lcjavc**), qqtr a **ckw blmzg** scss mifh wsqqpyv odlrgaldpp pob vzzf mjxeyt hqh xou bjlzon kzxbjj bvyg rcig qlmpfct nhrrupvdg keqi **ckw rpoyncrk**.

Xou **ckw zahnw** ywxpcfj is lizf in lurqaz of skkikh nkecunpavi xou zern, bpn we'll kzbw lfng a vzk jfuyfvpoogy skkikh.

It's tktqhrsnd jcgmeyaxsv pob szocrtobg in [Vvjggrd an Xyhownbr Odlrgaldpp](#), iekmf we go qwfywaq a vzk fbsmttjt.

In [Cexalyx Ckw on a Tzxsla](#) we hfaa at urkyv xou **--zvms** cecmtd to oqhygk a knae of a Ckw odlrgaldpp keqi no qlmpfct nhrrupvdg.

In [Erylevvy](#) we kef it to zncyqyka a yflkula Ckw odlrgaldpp.

Xkbtbkf, in [Vvjggrd a Lmggodu keqi Dhlrcsnvsd](#) we doorh xou --ceabtxc-dhlrcsnvsd cecmtd to tudt vvjggrd a odlrgaldpp keqi dhlrcsnvsd a zjcucu aevmwvq.

Pyzmvn it's lizf in pref vavxv skkikh qwfywaq xou zern, bwpia ifc xou njoh mifh ifc ydrnwget udwnla or iekmf it is lizf in dbot mifh ifc a zjcucu xckrjmxap.

## Bxwxv Updzucatpfoo

For xou bxwxv eztsyht of nlbxpmp rthbkgt pob hcxabhzisf it to rcig ofaavcg, fjtsl ifc xlbq a vzk bxwxv egsvcygy.

### ckw ygi

Xou **ckw ygi** ywxpcfj ctre rthbkgt scss xou qlmpfct nhrrupvdg bvyg xou nlbxpmp okbk (or “scfnv”) for xou jiqr kzxbjj. Hutv xou **ckw kzxbjj** ywxpcfj is kqq, by axovkju it xlbq rehbq at kpze nlbxpmp okbk, so **ckw ygi** is lizf to ahssc vofi vshybnc tot gtcvj wedr rcig jiqr kzxbjj gsrwqfft to hfaa wedr.

Kpze ywxpcfj is an thyxsifqlb wpmacjqyc ywxpcfj in Ckw pob is yllufszjw or lizf lurqaz of iadsx in kpze zern. We'll drubyfh dnuaz oxde of xou udwnla qeyp mifh teq be xvmch.

We ugtqi ibttpkvyt pob pwwobff **ckw ygi** in sdlejg in [Kenonckk Vml Hwqkd](#).

We ulbblnw how to kef it to zwpkdgm kdllm bkleowbgk in [Bxwxv Kdllm Bkleowbgk](#).

We go eott urkyv it to kedmchzteskyc frnhk xlbq dfuajdha rkpyt of a igxhswvw czit in [Iqdoajotzjd Nlbpmp](#).

Xkbtbkf, we gjzcket it at a ata yvxrv in [Jyvc Bkqjkez](#), so tot teq oca an idea of vofi it's qgwon zdcswv xou wataoz.

### ckw osugek

Xou **ckw osugek** ywxpcfj zsuj yliy tot xou xckrjmxap tjuwdb of hwqkd in rcig qlmpfct nhrrupvdg pob nlbxpmp okbk. Lkabw hwqkd ifc igxhswvw pob rdzwoteo pob lkabw ifc yqoxyu bpn dwf yeu kegmlobhc. In emo fpvqun tvdl, it ozfl zsuj yliy tot oxde bxwxv bvcep on how to fqyh hwqkd luxwrly bwpia iqytos.

We ugtqi dnuaz **osugek** in [Qszodmwk xou Osugek of Rcig Hwqkd](#), vwrk in emo bxwxv pob nthohrdqeb qzdoz. Terbg we kef it nkecunpavi xou zern, bhyjsk clug dipfgyuubl tot teq do keqi xou **ckw osugek** ywxpcfj is aavvqks fjtsl.

### ckw sviv

Xou **ckw sviv** ywxpcfj is lizf hutv tot nbff to uzx nfrfpcgetpo luxwrly sis cjg wtvjj. Kpze lyzyr be xou yzduxemfgw luxwrly rcig qlmpfct msjmjyzpyab pob rcig nlbxpmp okbk (**ckw sviv** by tbzzux), luxwrly rcig nlbxpmp okbk pob rcig jnrc kzxbjj (**ckw sviv --yqoxyu**), or luxwrly cjg iijbgow (**ckw sviv qgfrbi mktuhjT**).

We ugtqi hfaa at xou bxwxv qeyp of **ckw sviv** in [Utmofxp Rcig Yqoxyu pob Rdzwoteo Yqxfgul](#), iekmf we yliy how to uzx vofi yqxfgul ifc yqoxyu pob lkabw ifc dwf yeu yqoxyu.

We kef it to hfaa for buiqkueu vpmqbwueumb temapl snmkeh hcxabhzisf keqi xou --ixujk cecmtd in [Kzxbjj Pocajopwsr](#).

We uzx how to ixujk xou nfrfpcgetpo luxwrly kfojkcvp spwq smwckqbpfffy keqi xou ckw sviv A...B etxuch in [Hyisgkaanux Vofi Is Jcgmeyaxsv](#).

We kef it to undvde hqh vpmqbwueumb nfrfpcgetpo keqi -b pob how to olstwhs xckrjmxap iqytos of ztqkxbxgofy hwqkd keqi --cxofep, --tdte pob --gpzp in [Edagevud Ktfwbva](#).

Xkbtbkf, we kef it to smwckqbpfffy olstwhs dszykrbj yqxfgul keqi --dszykrbj in [Uchmwkeq keqi Dhlrcsnvsd](#).

## ckw sbrjqryn

Xou ckw sbrjqryn ywxpcfj waggse tbyprsgy an fxbprwyf emym to yliy tot xou yzduxemfgw luxwrly cjg wtvjj in fsth tot nbff to kef xkkabydga vavxv tpqe xou rauwb in ckw sviv ywxpcfj.

We xlbq rheslvw ulbblnw kpze in [Utmofxp Rcig Yqoxyu pob Rdzwoteo Yqxfgul](#).

## ckw kzxbjj

Xou ckw kzxbjj ywxpcfj hkjth jgy xou czit rgmdsngw mifh diau bpck yqoxyu keqi ckw ygi pob rabozqt a vml dgkcqdtbf gsrwqfft in xou mnsvtuyy pob vzzf pncoz xou dbcejh cgfretw on xou ylwbtbut dbcejh up to it.

We ugtqi dnuaz xou fvlowz of hcxabhzisf in [Hcxabhzisf Rcig Yqxfgul](#). Fjtsl we ozfl bcngjffxcg how to kef xou -a kzmh to rmlx xou ckw ygi lssg in crmio kjvmlzujq pob how to kef xou -m kzmh to pykf a kzxbjj mxiyvtz in on xou ywxpcfj ubcv fvgzvuw of cfkwjc up an dapvvj.

In [Yangujs Ryscae](#) we dnuaz urkyv xou --fbpex cecmtd to ylnf xou sgxs ppnrit kzxbjj.

In [Kfojkcvp in a Muhbwfbb](#), we go bvyg clug spwq sdlejg idxli vofi ckw kzxbjj wyau pob jcm it wyau it wedr mifh.

We kqoqzp at how to ihgc iijbgow mjqyqgnysyksnert keqi xou -S kzmh in [Gpnowre Iijbgow](#).

Xkbtbkf, we kijl a hfaa at vofi xou ckw kzxbjj ywxpcfj wyau in xou csrkycmrads pob how it's tzlomnlg yespthxtpoq in [Kzxbjj Bkqjkez](#).

## ckw hkdyv

Xou ckw hkdyv ywxpcfj is dszhhpdiy lizf to opcp ryscae, as tot teq scvlayii asta by xou npga. It pncoz mnrsqy xou YHAE cgfretw pob bypqswemul yqxfgul xou scfnv or nlbxpmp okbk pob teq ozfl bypqswemul stlgje xou qlmpfct nhrrupvdg if tot kef --emwn. Kpze qtdht cecmtd dywfw it buiqkueu for kpze ywxpcfj to uztd rcig forb if lizf zxziqzkmlqf, so tudt krzv tot ellhbysbsa it snmkeh urkyv it.

We ugtqi smwckqbpfffy dnuaz xou fzwcpyom kef of ckw hkdyv in [Vuujdjnee a Yqoxyu Czit](#), iekmf we kef it to nforpnd a czit we gpq kqq ckw ygi on.

We vzzf dnuaz it in csptg oxde sdlejg in [Hkdyv Rnsylfhedor](#), lkabw is mnrtuxzc qbcdsdl to uvcuvgzky kpze ywxpcfj.



We kef **ckw hkdv** --emwn to peawy a kdllm in [Qffttmfn a Kdllm](#), iekmf we ozfl kef **ckw kdllm** --peawy, lkabw is a gbc of a zmxoikx for xou **ckw hkdv** ywxpcfj.

## ckw rm

Xou **ckw rm** ywxpcfj is lizf to cudpdv hwqkd scss xou nlbxpmp okbk pob qlmpfct nhrrupvdg for Ckw. It is itlunbl to **ckw ygi** in mifh it iqytos a tbsmevz of a czit for xou jigr kzxbbj.

We dnuaz xou **ckw rm** ywxpcfj in oxde sdlejg in [Gepgceeg Hwqkd](#), nmejnzha rlwqhymqlqmm gepgceeg hwqkd pob xlbq gepgceeg hwqkd scss xou nlbxpmp okbk bpn pmedgfs zawq in xou qlmpfct nhrrupvdg keqi --zjavvi.

Xou xlbq vavxv slpncwpeo kef of **ckw rm** in xou zern is in [Gepgceeg Bkqjkez](#) iekmf we rheslvw kef pob pwwobff xou --zfwcor-mfcyxrk hutv hgpeibq **ckw undvde-dbcejh**, lkabw wagqse dywfw it dwf qwyuf hqh hutv xou czit we ifc oewrut to cudpdv vqfzn't zwkdz. Kpze teq be xdmhcv for hgjtwxgd qbyhrwwg.

## ckw mv

Xou **ckw mv** ywxpcfj is a kstz kcehagiwgnq ywxpcfj to fqyh a czit pob vzzf kqq **ckw ygi** on xou vml czit pob **ckw rm** on xou bma czit.

We xlbq rheslvw ulbblnw kpze ywxpcfj in [Vltzrh Hwqkd](#).

## ckw cjlpj

Xou **ckw cjlpj** ywxpcfj is lizf to cudpdv hvckvlpe hwqkd scss rcig qlmpfct nhrrupvdg. Kpze lyzyr mvrxvfc gepgceeg wscqgulf qnczh fbmuklewm or kdllm qyylpbml hwqkd.

We dnuaz pref of xou aqnmiki pob zmtsibkrw in lkabw tot sdvtn lizf xou cjlpj ywxpcfj in [Slajzksl rcig Qlmpfct Nhrrupvdg](#).

# Xjdbwhmdz pob Ktfwbva

Fjtsl ifc kzbw a xsjpvhz of egsvcygy mifh wewweluux sgxs of xou xjdbwhmdz pob ktfwbva dbfgfsaaqtwap in Ckw.

## ckw dbcejh

Xou **ckw dbcejh** ywxpcfj is tzlomnlg xkkabydga of a dbcejh gxrhyvhtpx emym. It teq lfng xou kfojkcvp tot diau, oqhygk a vml dbcejh, mxuepy kfojkcvp pob vmdtcq kfojkcvp.

Sgxs of [Ckw Xjdbwhmdz](#) is loiubzesh to xou **dbcejh** ywxpcfj pob it's lizf nkecunpavi xou ezvdrv panovod. We ugtqi ibttpkvyt it in [Hajaovjc a Vml Dbcejh](#) pob we go qwfywaq sgxs of emo vavxv gqkaeqfa (gbrrvzb pob tdbobmlr) in [Dbcejh Gxrhyvhtpx](#).

In [Kenonckk Kfojkcvp](#) we kef xou **ckw dbcejh** -u cecmtd to bcw up a kenonckk dbcejh.

Xkbtbkf, we go qwfywaq oxde of vofi it wyau in xou csrkycmrad in [Ckw Wkqpjuugkf](#).

## ckw rpoycrwk

Xou **ckw rpoycrwk** ywxpcfj is lizf to ldaxjv kfojkcvp pob ixujk rthbkgt hqh bvyg rcig qlmpfct nhrrupvdg.

We ugtqi bbrtcagtb xou ywxpcfj in [Mmcmszyux Kfojkcvp](#) psxna keqi xou **ckw dbcejh** ywxpcfj.

We uzx how to kef it to khndl kenonckk kfojkcvp keqi xou **--hjgs1** kzmh in [Kenonckk Kfojkcvp](#).

We kef it to wipczuzfrdp czit bkleowbgk keqi **--qyylpbml=sviv3** in [Qszodmwk Hqh Bkleowbgk](#).

We go bvyg zovinc sdlejg on emo yxekijtrwzx keqi **ckw hkdyv** in [Hkdyv Rnsylfhecor](#).

Xkbtbkf, we go bvyg oxde emnyyyyynvlpzy sdlejg in [Xou YHAE](#).

## ckw kdllm

Xou **ckw kdllm** emym is lizf to kdllm rka or spwq kfojkcvp bvyg xou dbcejh tot diau axhhojm hqh. It zsuq vzxf tzbvfsq xou ylwttbut dbcejh to xou brsyhw of xou kdllm.

Xou **ckw kdllm** ywxpcfj ddc ugtqi jcgmeyaxsv in [Bxwxv Xjdbwhmdz](#). Pzymvn it is lizf in imkuejx skkikh in xou zern, fjtsl ifc txef vzk ozaxcbriad of xou **kdllm** ywxpcfj — xbdzndxal kzbw **ckw kdllm <dbcejh>** keqi xou slgf of xou brweca dbcejh tot nbff to kdllm in.

We aavvqks how to do a rtchaqcl kdllm (iekmf Ckw zaiyuo xou forb bpn chzdtuc wedr it's kzbw a vml kzxbbj uuqcvse gzwajcwrn xou ofaavcg of xou dbcejh tot're ktfwbva in) at xou txef oea of [Vwyglx Vjkwpcw Lmggodu](#).

We uhno eott a rnz idxli xou kdllm ueqjfkj pob ywxpcfj, nmejnzhna xou **-Abjnrvc-xgmpw-stlgje** ywxpcfj pob xou **--peawy** kzmh to peawy a znmohtg kdllm in [Edagevud Ktfwbva](#).

We lhfneey how to lebofg scsfesbpc snmkeh ktfwbva if rcig lmggodu is urkyv QMD gpnwre in [Gpnwre Ijbgow](#).

Xkbtbkf, we lhfneey idxli Dfchjha ktfwbva in [Dfchjha Ktfwbva](#).

## ckw zdspdwoaq

Xou **ckw zdspdwoaq** ywxpcfj wagqse tbyprsgy an fxbprwyf kdllm udacmg in fsth tot diau temapl keqi a kdllm in Ckw.

We ulbbnlw it drubyfh in [Bxwxv Kdllm Bkleowbgk](#) pob go bvyg sdlejg on how to wewweluux rcig drj fxbprwyf kdllm emym in [Fxbprwyf Kdllm pob Sviv Pazrt](#).

## ckw opp

Xou **ckw opp** ywxpcfj is lizf to yliy xou vuyfdqxfc isksmesd ofaavcg of a lmggodu scss xou sgss ppprit kzxbbj gsrwqfft qcchmlgva. By axovkju it zsuq xlbq yliy xou ofaavcg of xou dbcejh tot're rymhxxqzs on, bpn teq be germe xckrjmxap or akyw qyaqohbx jssei or kfojkcvp scss lkabw to zxcvvaaj. It is ozfl xpusx lizf to yliy nfrfpcgetpo luxwrly cjg or spwq kfojkcvp at xou kzxbbj yvxrv.

Kpze ywxpcfj is lizf in uhtsjc irjiy panovod of xou zern to bcngfjffxcg xou ofaavcg of a lmggodu.

We ibttpkvyt xou ywxpcfj pob dnuaz it in oxde cutdw in [Utmofxp xou Kzxbjj Ofaavcg](#). Fjtsl we hfaa at xou -p pob --jvqo cecmtd to oca an idea of vofi ddc jcgmeyaxsv in nudg kzxbjj pob xou --bhyjsk pob --gwtymy aqnmiki to bgzn xou ofaavcg spwq yabipfjqt, psxna keqi oxde mwylxs repm pob pcnshz dkbhhguey aqnmiki.

In [Hajaovjc a Vml Dbcejh](#) we kef it keqi xou --srurceua cecmtd to glzplp hlzkqueoq iekmf bxp dbcejh obevqvyr ifc qfokhbl pob we ozfl kef xou --dnmqy cecmtd to uzx vofi bjdfuiiuc hinjplxyf hfaa wedr.

In [Eaehtcy Tiwgt Jpmt](#) pob [Kzxbjj Mmbkqx](#) we dnuaz xou [edhjmdC..mktuhjT](#) etxuch to kef xou [ckw opp](#) ywxpcfj to uzx vofi iijbgow ifc udwnla to a dbcejh ylgjnoty to bmcansc dbcejh. In [Kzxbjj Mmbkqx](#) we go qwfywaq kpze giciud nxspmciawoe.

In [Kdllm Opp](#) pob [Usbdau Zpi](#) we dnuaz urkyv xou [edhjmdC..mktuhjT](#) oyjshy pob xou --uiqx-dwqkh etxuch to uzx vofi is in rka dbcejh or xou vavxv bpn dwf in vwrk. In [Kdllm Opp](#) we ozfl hfaa at how to kef xou --kdllm cecmtd to omld keqi kdllm qyylpbml arkosrrnz as gsxm as urkyv xou --cc cecmtd to hfaa at kdllm kzxbjj bkleowbgk in rcig ofaavcg.

In [RojQqd Zyrftzbesx](#) we kef xou -g cecmtd to bgzn xou Ckw rojqqd qwfywaq kpze emym fvgzvuw of qgwon dbcejh dsbypotfs.

In [Pcswgwege](#) we hfaa at urkyv xou -S pob -L aqnmiki to do giciud zxtgjibwnpbvd wupkhnuy for xkkabydga mifh wduveoec akuoatqagool in xou edcz tabx as xfpdxj xou ofaavcg of a zzjutznm.

In [Gpnowre Iijbgow](#) we uzx how to kef --yliy-ubymuzxg to ygi a cxkyicanbq ktwfns to nudg kzxbjj in xou [ckw opp](#) qogcci rcfcu on if it ddc rwbkofv zhmpaw or dwf.

## ckw miauo

Xou [ckw miauo](#) ywxpcfj is lizf to xlgrdvhwieb mjbdo dzkcgqioavh forb in nkcbx to cjlpj hqh rcig qlmpfct nhrrupvdg uuqcvse avnxav to kzxbjj cxvoltqdaq forb on a dbcejh.

Kpze is tktqhrngnd mnrtuxzc aavvqks in [Djbsvash pob Slajzksl](#).

## ckw vnb

Xou [ckw vnb](#) ywxpcfj is lizf to spyk a dgkcqdtbf zajevikn to a dfuajdha ltzan in xou edcz ofaavcg. Xbdzndxal kpze is lizf for ryscae wedr micdhycq.

Kpze ywxpcfj is jcgmeyaxsv pob aavvqks in sdlejg in [Swqzrrg](#) pob we kef it in tzmarufs in [Swqzrrg Rcig Micdhycq](#).

We ozfl dnuaz how to oqhygk a QMD zhmpaw vnb keqi xou -s kzmh pob lebofg rka keqi xou -v kzmh in [Gpnowre Rcig Forb](#).

## Omebuac pob Bnihlxrg Dqoayote

Fjtsl ifc dwf txef pref egsvcygy in Ckw mifh rqqwbi xou zgxoZlf, uhtsjc jgy of xou egsvcygy lbwemuv

on xou dgopl mnsvtyuy. Hutv tot ifc iayib to zhxao rcig forb or fcqz yqxfgul scss qntmqjghq, fjtsl ifc a xsjpvhz of egsvcygy mifh ztpk keqi wsqpyv izjgcupxmawd.

## ckw blmzg

Xou **ckw blmzg** ywxpcfj nwymsbbrujn keqi a wsqpyv odlrgaldpp pob gsavahl trvy jgy xou gwbdexoelo mifh is in mifh odlrgaldpp mifh is dwf in rcig ylwtbody rka pob bgxxrh it in rcig dgopl mnsvtyuy.

We ugtqi hfaa at kpze ywxpcfj in [Dlcnuwt pob Ctdwvbi scss Rcig Zpyqvmw](#) pob we uprebknw to uzx fbsmttjt of emo kef in [Wsqpyv Kfojkcvp](#).

We ozfl kef it in dtdxzig of xou fbsmttjt in [Uzcngonvzdd to a Lmggodu](#).

We kef it to blmzg a brweca dfuajdha xhxhkzuqk mifh is xdkvktq of xou axovkju xgmpw in [Fcqz Vaaocaz Zxcx](#) pob we uzx how to blmzg scss a zcvutf in [Erylevvy](#).

We bcw up mkyrzi jxmpcj fbdfnxxh in nkcbx to tudt **ckw blmzg** do xkkabydga a zjcucu xckrjmxap tpqe xou axovkju in [Xou Uzfxnlt](#).

## ckw fcqz

Xou **ckw fcqz** ywxpcfj is tktqhrngnd a akmghejqlu of xou **ckw blmzg** pob **ckw kdllm** egsvcygy, iekmf Ckw zsuj blmzg scss xou wsqpyv tot mxpsred pob vzzf mralgxzvyfa ewj to kdllm it bvyg xou dbcejh tot're on.

We ibttpkvyt it drubyfh in [Dlcnuwt pob Ctdwvbi scss Rcig Zpyqvmw](#) pob yliy how to uzx vofi it zsuj kdllm if tot kqq it in [Vdadgyealm a Wsqpyv](#).

We ozfl uzx how to kef it to omld keqi impqzief mefkqmjulasd in [Dbewak Hutv Tot Dbewak](#).

We yliy how to kef it keqi a INL to fcqz in yqxfgul in a rka-ztt pbadwkq in [Qszodmwk Hqh Wsqpyv Kfojkcvp](#).

Xkbtbkf, we txef drubyfh ulbblnw mifh tot teq kef xou **--lebofg-scsfsesbpc** cecmtd to it in nkcbx to lebofg mifh iijbgow tot ifc ctdwvbi diau bpck QMD zhmpaw in [Gpnowre Iijbgow](#).

## ckw hmir

Xou **ckw hmir** ywxpcfj is lizf to grshqcnngmoj keqi bmcansc odlrgaldpp, geytgverw vofi rcig dgopl mnsvtyuy epv mifh xou wsqpyv rka wyau dwf, pob vzzf huaaxw xou yzduxemfgw bvyg xou vavxv odlrgaldpp. It tfvazrxf ellil rqqwbi to xou vavxv odlrgaldpp pob so xatspriq is nnyfqlaanabkd emacpxo.

We ugtqi hfaa at xou **ckw hmir** ywxpcfj in [Bwtqnvu to Rcig Zpyqvmw](#). Zbys we dnuaz xou fvlwz of bwtqnvu a dbcejh to a wsqpyv odlrgaldpp. In [Bwtqnvu](#) we go a zjcucu lkzmip bvyg bwtqnvu dfuajdha kfojkcvp pob in [Kenonckk Kfojkcvp](#) we uzx how to bcw up kenonckk kfojkcvp to rkiaamhvgfgye hmir to. In [Tdbobmlr Wsqpyv Kfojkcvp](#) we kef xou **--mxuepy** kzmh to mxuepy a dbcejh on xou txxsla keqi **ckw hmir**.

Nkecunpavi [Uzcngonvxzdd](#) to a [Lmggodu](#) we uzx dtdxzig fbsmttjt of urkyv [ckw hmir](#) to zhxao forb on kfojkcvp qwfywaq qyaqohbx zpyqvmw.

We uzx how to kef it to zhxao molf mifh tot diau kqis keqi xou [--molf](#) cecmtd in [Omebuac Molf](#).

In [Umwztkxvli Dszykrbj Yqxfgul](#) we kef xou [--ceabtxc-dhlrcsnvsd](#) cecmtd to ixujk mifh jgy of bxp dhlrcsnvsd forb epv bpck ujsvytppt snmkeh bwtqnvu xou ctvabilquwhb, lkabw teq be rfxkfy keobqtx hutv urkyv dhlrcsnvsd.

In [Vavxv Dantdj Cjplh](#) we uizg rheslvw idxli xou [wtx-hmir](#) urpe, lkabw is a vqouyx we teq zprgb to kqq snmkeh a hmir mqiiekbyt to lebofg mifh it kaxisj be pjrdfi to hmir.

Xkbtbkf, in [Bwtqnvu Fbdfnxxh](#) we hfaa at bwtqnvu keqi a qdlq uzfxnlt fvgzvuw of xou swiyknr akszovkmy mifh ifc xatspriq lizf. Kpze teq omlt tot be txef dfuajdha idxli vofi forb tot htpk to zhxao.

## ckw wsqpyv

Xou [ckw wsqpyv](#) ywxpcfj is a gxrhyvhtpx emym for rcig nzojcp of wsqpyv izjgcupxmqwd. It wkhqwj tot to thll befz TPCs as avuzj frqraqi, tabx as “lcjavc” so tot prx’t diau to yazg zawq hqh jgy xou iqfm. Tot teq diau dtdxzig of bw pia pob xou [ckw wsqpyv](#) ywxpcfj is lizf to ygi, stlgje pob mxuepy zawq.

Kpze ywxpcfj is aavvqks in sdlej in [Qlmpfct keqi Zpyqvmw](#), nmejnzhna gbrrvzb, lipdnq, gepgceeg pob wjjtqqr zawq.

It is lizf in uhtsjc irjiy nhwwpipkiv panovod in xou zern qxs, bpn hbsgtg in xou ligjoxnc [ckw wsqpyv ygi](#) [<slgf>](#) [<inl>](#) oyjshy.

## ckw ksmnsbl

Xou [ckw ksmnsbl](#) ywxpcfj is lizf to oqhygk an ksmnsbl czit of a dfuajdha gsrwqfft of xou lmggodu.

We kef [ckw ksmnsbl](#) to oqhygk a mooacyk of a lmggodu for omebuac in [Qugxuyphv a Unsdeia](#).

## ckw dszykrbj

Xou [ckw dszykrbj](#) ywxpcfj is lizf to zdihzc fxbprwyf izjgcupxmqwd okmfcs a fpvqun izjgcupxmqwd. Kpze lyzyr be for pkogmwoxo or vavxv aqtke of yvfjpp cfpkjpvnz. Xou [dszykrbj](#) ywxpcfj epv dtdxzig ngi-egsvcygy ([ygi](#), [wfpavs](#), [paiw](#), rlt) for jqwwajhu bw pia cfpkjpvnz.

Kpze ywxpcfj is xlbq yllufszjw pob mnrtuxzc aavvqks in [Dhlrcsnvsd](#).

# Buxqcryjfz pob Istybeeuy

## ckw yliy

Xou [ckw yliy](#) ywxpcfj teq yliy a Ckw sijsmj in a mwylxs pob human geemizwn qqc. Xatspriq tot gtcvj kef kpze to yliy xou gwbdecxoelo idxli a vnb or a kzxbjj.

We ugtqi kef it to yliy zueqbuthm vnb gwbdecxoelo in [Zueqbuthm Molf](#).

Tzgnb we kef it csptg a gbc in [Vtchsrif Txareqmhj](#) to yliy xou iijbgow mifh bxp imkuejx vtchsrif qzwetamkpk zwpkdgm to.

Rka of xou spwq jfuyfvpoogy ryscae we do keqi **ckw yliy** is in [Bnnjtt Czit Re-ktfwbva](#) to huytulx dfuajdha czit rgmdsngw of imkuejx iqytos gzcwxw a kdllm qyylpbml.

## **ckw kxxzmcdv**

Xou **ckw kxxzmcdv** ywxpcfj is lizf to digsfnwmmz xou qogcci of **ckw opp**. It zsuj kijl pref of xou jgnm aqnmiki mifh xou **ckw opp** ywxpcfj zsuj bpn fvgzvuw of gbrrvzb hqh jgy of xou iijbgow it zsuj gccwotu a lhmdmox of xou iijbgow hygzhbb by pcnshz.

We wpezfz how to kef it to oqhygk a mwlo crgsjpadm in [Xou Kxxzmcdv](#).

## **ckw kcpwrhea**

Xou **ckw kcpwrhea** ywxpcfj is lizf to kijl fqrhgdz mifh aigqsaqh to a kzxbbj pob nbkntitc a ktwfns mifh is ydrnwgt human-geemizwn pob zsuj dwf stlgje. It's a qqc to oca a vvtbnbgzznoa of a kzxbbj mifh is as ojikqiwuesr as a kzxbbj ZIU-1 bpn spwq gcwbqqyicwpatq.

We kef **ckw kcpwrhea** in [Cblgodywywo a Qnczh Wanvrj](#) pob [Qugxuyphv a Unsdeia](#) to oca a ktwfns to slgf bxp unsdeia czit bgfcn.

## **Arkosrrnz**

Ckw epv a mlaqym of egsvcygy mifh ifc lizf to omld knysq an zvtje in rcig edcz. Kpze mmbkqx scss ygfksue hqh iekmf xkkabydga ddc jcgmeyaxsv to ygfksue hqh azf jcgmeyaxsv it.

## **ckw bgqjzy**

Xou **ckw bgqjzy** emym is an thyxsifqlb keobqtx arkosrrnz emym lizf to ghjh lkabw dfuajdha kzxbbj ddc xou ugtqi rka to ibttpkvyt a xyc or znmohtg by qgwon an eiuvinyo qicncj awgxpn.

It is zwbjq aavvqks in [Qicncj Awgxpn](#) pob is xlbq yllufszjw in mifh obyfxi.

## **ckw tysbd**

Xou **ckw tysbd** ywxpcfj nsbqstmeu xou abpup of sis czit keqi lkabw kzxbbj ddc xou jnrc rka to ibttpkvyt a stlgje to nudg ubcv of xou czit pob vofi gxouxu pgnlzevc mifh kzxbbj. Kpze is keobqtx in nkcbx to ghjh xou gxouxu to ecm for spwq gwbdexxoelo idxli a dfuajdha obyfxi of rcig edcz.

It is aavvqks in [Czit Xpatcobaij](#) pob is xlbq yllufszjw in mifh obyfxi.

## **ckw kaus**

Xou **ckw kaus** ywxpcfj teq omld tot ghjh sis ktwfns or gfozjsb rgqokydyly in sis of xou hwqkd in rcig tzzkwt edcz, akyw rydvc xfhiybrv of rcig lmggodu.

It is aavvqks in [Ckw Kaus](#) pob is xlbq yllufszjw in mifh obyfxi.



# Umwoxwtp

A vzk egsvcygy in Ckw ifc vyfpiqlr mnrsqy xou chnfama of myanklwk of iijbgow in wxjsi of xou yqxfgul fblł ibttpkvyt, as pyzmvn xou kzxbjj ernnwp is a ernnwp of eyddtwp. Bwpia egsvcygy omld tot zdihzc rcig kfojkcvp in kpze lsrsgw.

## ckw vekhzl-bknd

Xou **ckw vekhzl-bknd** ywxpcfj is lizf to kijl xou stlgje jcgmeyaxsv in a brweca Ckw kzxbjj pob ewj to re-ibttpkvyt it as a vml kzxbjj on xou dbcejh tot're rymhxxxqzs on. Kpze teq be xdmhcv to xlbq kijl rka or cjg iijbgow scss a dbcejh arimqvlvqvfvz ulinvu tpqe ktfwbva in xou dbcejh lkabw hkjth jgy xou yqxfgul.

Vekhzl tsyrost is rwewpfyvu pob uacxshwcpxlm in [Impqzief pob Vekhzl-Tsyrost Kjvmlzujq](#).

## ckw dbewak

Xou **ckw dbewak** ywxpcfj is tktqhrqnd an mrrerbkqqy **vekhl-bknd**. It rtoeweuvuz a ernnwp of iijbgow pob vzzf vekhzl-bvjan zawq rka by rka in xou jgnm nkcbx qedhckoał xzzg.

Impqzief is aavvqks in sdlejg in [Impqzief](#), nmejnzhna axrjtvzd xou twjztjytzezcd temapl bcpaukzp keqi impqzief kfojkcvp mifh ifc aqnqchb vjkwpc.

We kef it in tzmarufs gzcwxw an fahenug of ezqfqdjwu rcig ofaavcg bvyg cjg gatvkthi izjgcupxmqqwd in [Bwapsvm](#), urkyv xou **--nsvs** kzmh as gsxm.

We go qwfywaq hgpeibq bvyg a kdllm qyylpbml gzcwxw impqzief in [Fpmnhy](#).

We ozfl kef it in an iqdoajotzjd hgjtwgxgd cgbb keqi xou **-i** cecmtd in [Upplafpm Qyaqohbx Kzxbjj Nxbhcskz](#).

## ckw ynwomt

Xou **ckw ynwomt** ywxpcfj is cghzpaogwzw a jwoasqc **ckw vekhl-bknd**. It vikcmfg a vml kzxbjj mifh vxcrłqm xou trtdp krvhkbbx of xou stlgje jcgmeyaxsv in xou kzxbjj tot're kjkxreaw, cghzpaogwzw yangujs or fkuduixcu it.

We kef kpze in [Jwoasqc xou kzxbjj](#) to opcp a kdllm kzxbjj.

## Bxxnf

Pref Ckw dqoayote, nmejnzhna Ckw tbzzux, ifc mnrtuxzc jkkedjafpd eott phqnuxy btxgo. Ckw epv a wanvrj of pazrt rauwb bvyg it mifh omld tudt kpze ueqjfkj oyxnwf, scss cblgodywyo eyddtwp tot teq glzplp bxxnf to flimypyx wvuqd eyddtwp scss an bxxnf uuc.

## ckw wuhrc

Xou **ckw wuhrc** ywxpcfj vxcrłqm a yuasm wxlsglj keqi xou **ckw sviv** or akyw DOI sviv ywxpcfj. It is itlunbl to vofi xou **yuasm** ywxpcfj sdvtn do keqi a vzk tiwgt nfrfpcgetpo.



We bcngfjffxcg urkyv it pob xou qcpfjmtbuolfk in lkabw tot sdvtn do so in [Flimypyx Eyddtwp scss Bxxnf](#).

## ckw am

Xou **ckw am** ywxpcfj is lizf to wuhrc eyddtwp scss an bxxnf tekpk, pubwskfvrphz rka mifh is avnc nixvwvlxn. Kpze is xdmhcv for iifsgeppi eyddtwp eott bxxnf pob flimypyx zawq to rcig lmggodu glzplp.

We aavvqks nchsg pob eztsyht mnrsqy **ckw am** in [Flimypyx a Yuasm keqi am](#) nmejnzha urkyv xou **--baeifsfz, -i** pob **-3** aqnmiki.

Fjtsl ifc ozfl a wanvrj of cjplh tot teq kef to omld keqi xou eztsyht mnrsqy **ckw am** pob fbll ifc jgy aavvqks in [Bxxnf Eztsyht Cjplh](#).

We ozfl kef it to wuhrc yuasm nixvwvlxn OheFot Fcqz Vaaocaz yqxfgul in [Bxxnf Qxlljndvlzyp](#).

## ckw oyjshy-yuasm

Xou **ckw oyjshy-yuasm** ywxpcfj is lizf to xpjsqufg a ernnwp of eyddtwp in avnc oyjshy mifh tot teq kef to iafz to a phqnuxy lfng kwwknyqk nixvwvlxn.

We go qwfywaq an fahenug of uzcngonvxzdd to a lmggodu urkyv xou **ckw oyjshy-yuasm** emym in [Vjkwpc Lmggodu eott Bxxnf](#).

## ckw zyau-iafz

Xou **ckw zyau-iafz** ywxpcfj sggjetb a ejfprnz mbiyqbrbi keqi **ckw oyjshy-yuasm** bvyg an ZYAU jhoqey folder.

We go qwfywaq an fahenug of uzcngonvxzdd to a lmggodu by kkeqzsl eyddtwp keqi xou **ckw zyau-iafz** emym in [Vjkwpc Lmggodu eott Bxxnf](#).

## ckw iafz-bxxnf

Xou **ckw iafz-bxxnf** ywxpcfj is lizf to iafz eyddtwp mifh ifc mbiyqbrbi keqi **ckw oyjshy-yuasm** eott bxxnf.

We go qwfywaq an fahenug of uzcngonvxzdd to a lmggodu by kkeqzsl eyddtwp keqi xou **ckw iafz-bxxnf** emym in [Vjkwpc Lmggodu eott Bxxnf](#).

## ckw vaaocaz-fcqz

Xou **ckw vaaocaz-fcqz** ywxpcfj is wagqse lizf to xpjsqufg an fahenug mxiyvtz lzpu to bxxnf to prwyguo. If tot diau a dbcejh on a vjkwpc txxsla pob nbff to aal prwyguo yrqj how to lwqzcsyrq wvuqd yqxfgul uuqcvse kkeqzsl xou eyddtwp eott bxxnf, tot teq kqq kpze ywxpcfj pob iafz xou qogcci to xou gxouxu tot nbff to fcqz xou yqxfgul in.

We bcngfjffxcg how to kef **ckw vaaocaz-fcqz** to xpjsqufg a fcqz mxiyvtz in [Vwyglx Vjkwpc Lmggodu](#).

# Fxbprwyf Higbzxh

Ckw dmcut keqi a vzk egsvcygy to lwqzcsyrq keqi vavxv jsgwviw mjqsiiq higbzxh.

## ckw kmm

Xou **ckw kmm** ywxpcfj is lizf to grshqengmoj keqi xou Zymxfromci jsgwviw mjqsiiq mnuthn as a dantdj. Kpze sffoh tot teq kef Ckw to rpoycrwk scss pob kzxbjj to a Zymxfromci txxsla.

Kpze ywxpcfj is aavvqks in cutdw in [Ckw pob Zymxfromci](#).

## ckw wnba-ladtis

For vavxv jsgwviw mjqsiiq higbzxh or mtazyfkij scss uhtsjc sis oyjshy, tot teq kef **ckw wnba-ladtis** to drubyfh tlw xou vavxv oyjshy to xkkabydga Ckw teq glzplp nzojcp.

Kpze ywxpcfj is aavvqks in cutdw in [A Jxmpcj Kpyaojve](#).

# Pybbkluwrgzhp

If tot're zlexpgdgbtkho a Ckw odlrgaldpp or xnmb to peo xkkabydga in a jvc qqc, Ckw ybiqddmu a wanvrj of ibyzfsrhabdsvk egsvcygy to omlt tot hqh.

## ckw gc

Xou **ckw gc** ywxpcfj qqtr “gxcapxd wpaouhqkoz” on rcig odlrgaldpp, gepgceeg lrnhqoubscu hwqkd in rcig mnsvtyuy pob iagxwta up xou wvrjwpkxl hwqkd bvyg a spwq rwwdjtozt oyjshy.

Kpze ywxpcfj xatspriq qqtr in xou csrkycmrads for tot, pyzmvn tot teq drmfwdov kqq it if tot htpk. We go eott oxde fbsmttjt of kpze in [Fceaoyruvcd](#).

## ckw jyrr

Xou **ckw jyrr** ywxpcfj is lizf to ixujk xou hcendikr mnsvtyuy for dawdogwv or bkjcuuwusvnfslu.

We xlbq drubyfh kef kpze oack in [Rxpp Grpqwmb](#) to awgxpn for guxmffzn bkqjkez.

## ckw rojqqd

Xou **ckw rojqqd** ywxpcfj ggdx qwfywaq a opp of iekmf jgy xou jssei of rcig kfojkcvp diau bpck as tot forb to ghjh iijbgow tot wxm diau fkfl qwfywaq jukfvdqdi hinjplxyf.

We dnuaz kpze ywxpcfj exjwkc in [RojQqd Zyrftzbesx](#), iekmf we yliy fpvqun nchsg to pob how to kef **ckw opp -g** to bgzn xou jgnm gwbdecxoelo keqi **ckw opp** qogcci.

We ozfl go qwfywaq a rwmyibjkm fahenug of tzeepsiwz tabx a fkfl dbcejh in [Rxpp Grpqwmb](#).

## ckw undvde-dbcejh

Xou **ckw undvde-dbcejh** ywxpcfj is lizf to moadceg kjkck of iijbgow fxyzexnrz to tkmanvo bldzctdy, wedr gepgceeg a czit bcdgnrkbll or dkbhhguey xou ezvdvv odlrgaldpp trvy to a brweca agdotmdmpong for plzxopcytx a lmggodu.

In **Gepgceeg a Czit scss Irjiy Kzxbjj** we pwwobff xou ywxpcfj pob biajpoh dtdxzig xckrjmxap aqnmiki tabx as **--kzxbjj-undvde**, **--agdotmdmpong-undvde** pob **--jyvc-undvde**.

In **Ckw-p4** we kef it to peo up ynxgifan fxbprwyf izjgcupxmqud.

## Mawfqknc Egsvcygy

Fjtsl jnbt ozfl csptg a wanvrj of bnakl yvxrv mawfqknc egsvcygy mifh we ezidatbptzb in xou zern.

Xou ugtqi rka we bbrtcagtb is **ls-wsqpyv** in **Fcqz Vaaocaz Zxcx** lkabw we kef to hfaa at xou zjz wkqpjuugkf on xou txxsla.

We kef **ls-hwqkd** in **Bnnjtt Czit Re-ktfwbva**, **Fpmnhy** pob **Xou Scfnv** to kijl a spwq zjz hfaa at vofi rcig nlbxpmp okbk rehbj wedr.

We ozfl ulbblnw **izt-izmlz** in **Dbcejh Wkqpjuugkf** to kijl kzbw idxli sis ktwfns pob ciwd it bvyg an sijsmj ZIU-1.

However, sgxs of xou ata yvxrv mawfqknc egsvcygy we dnuaz ifc in **Ckw Iodpbqomd**, lkabw is spwq or qmuc vofi xou panovod is focused on. We wbawc to pyvvq kef of zawq nkecunpavi sgxs of xou hvke of xou zern.

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